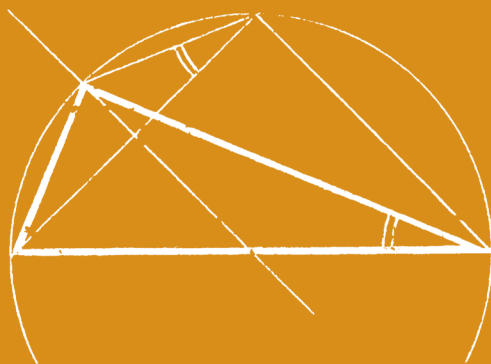


High-School MATHEMATICS

Under the editorship of Professor
____ G. N. YAKOVLEV, D.Sc. ____



____ Part 2 ____

MIR PUBLISHERS · MOSCOW

High-School Mathematics

**Пособие по математике
для поступающих
в вузы.**

Под редакцией
Г. Н. Яковлева

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High-School Mathematics

Under the editorship of Professor
G. N. YAKOVLEV, D.Sc.

In Two Parts

Part Two



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Chapter 1

Trigonometric Equations, Systems, Inequalities

1.1. Trigonometric Equations

The canonical trigonometric equations. First recall the formulas for solving the canonical trigonometric equations:

$$\sin x = a, \quad x = (-1)^n \arcsin a + \pi n \quad \text{for } |a| \leq 1, \quad (1)$$

$$\cos x = a, \quad x = \pm \arccos a + 2\pi n \quad \text{for } |a| \leq 1, \quad (2)$$

$$\tan x = a, \quad x = \arctan a + \pi n, \quad a \in \mathbb{R}, \quad (3)$$

$$\cot x = a, \quad x = \operatorname{arccot} a + \pi n, \quad a \in \mathbb{R}, \quad (4)$$

where $n = 0; \pm 1; \pm 2; \dots$, i.e. $n \in \mathbb{Z}$.

The equations $\sin x = a$ and $\cos x = a$ have solutions for any $a \in [-1; 1]$, i.e. for $|a| \leq 1$. Now if $|a| > 1$, then these equations have no solutions.

The equations $\tan x = a$ and $\cot x = a$ have solutions for any $a \in \mathbb{R}$.

To write solutions of the canonical trigonometric equations, we can use the designations from the theory of sets. For example, the set of solutions of the equation $\sin x = a$, $|a| \leq 1$, can be written as

$$\{(-1)^n \arcsin a + \pi n \mid n \in \mathbb{Z}\}. \quad (5)$$

In equations (1)-(4) the solutions are expressed in terms of the values of inverse trigonometric functions. Let us recall some properties of these functions.

Each of the functions $\arcsin a$ and $\arccos a$ is defined on the interval $[-1; 1]$ and

$$-\pi/2 \leq \arcsin a \leq \pi/2, \quad a \in [-1; 1],$$

$$0 \leq \arccos a \leq \pi, \quad a \in [-1; 1].$$

The function $\arcsin a$ is odd, that is,

$$\arcsin(-a) = -\arcsin a.$$

The function $\arccos a$ is neither even nor odd. Each of the functions $\arctan a$ and $\operatorname{arccot} a$ is defined throughout the number axis and

$$\begin{aligned} -\pi/2 < \arctan a < \pi/2, & a \in \mathbb{R}, \\ 0 < \operatorname{arccot} a < \pi, & a \in \mathbb{R}. \end{aligned}$$

The function $\arctan a$ is odd, that is,

$$\arctan(-a) = -\arctan a.$$

The function $\operatorname{arccot} a$ is neither even nor odd.

To solve a trigonometric equation which is not among the canonical equations, it should be reduced, by means of requisite transformations, to one or several canonical equations:

Example 1. Solve the equation

$$2 \sin^2 x = 1.$$

$\triangle$ We transform the equation, making use of the equality $2 \sin^2 x = 1 - \cos 2x$ valid for any x and get

$$1 - \cos 2x = 1, \quad \cos 2x = 0.$$

This means that $2x = \frac{\pi}{2} + \pi n, \quad n \in \mathbb{Z}$.

$$\text{Answer: } x = \frac{\pi}{4} + \frac{\pi n}{2}, \quad n \in \mathbb{Z}. \quad \blacktriangle$$

Example 2. Solve the equation

$$\tan^2 x = 3. \tag{6}$$

$\triangle$ If x is a solution of the given equation, then

$$\text{either } \tan x = \sqrt{3}, \tag{7}$$

$$\text{or } \tan x = -\sqrt{3}. \tag{8}$$

The converse is, evidently, also true: if x is a solution of (7) or (8), then x is also a solution of (6). Thus it follows that the set of solutions of equation (6) is a union of the sets of solutions of equations (7) and (8). In that case we say that equation (6) is equivalent to the collection of equations (7) and (8).

Equations (7) and (8) have the following respective solutions:

$$x = \frac{\pi}{3} + \pi n, \quad n \in \mathbb{Z},$$

$$x = -\frac{\pi}{3} + \pi n, \quad n \in \mathbb{Z}.$$

The union of these two sets is precisely the set of all solutions of equation (6). For brevity, we can write these solutions in the form

$$x = \pm \frac{\pi}{3} + \pi n, \quad n \in \mathbb{Z}. \quad \blacktriangle$$

We cannot name a unique method of solving trigonometric equations. In each definite case, a success in solving a trigonometric equation depends, in particular, on the knowledge of trigonometric formulas and the know-how of solving problems.

Many trigonometric formulas are true equalities for *all* the values of the variables appearing in them. Such are, for instance, the formulas for a double argument:

$$\sin 2\alpha = 2 \sin \alpha \cdot \cos \alpha,$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha = 2 \cos^2 \alpha - 1 = 1 - 2 \sin^2 \alpha,$$

the formulas for half the argument:

$$\sin^2 \frac{\alpha}{2} = \frac{1}{2} (1 - \cos \alpha), \quad \cos^2 \frac{\alpha}{2} = \frac{1}{2} (1 + \cos \alpha),$$

and the formulas for the sum, the difference, and the product of sines and cosines, the formulas for the sines and cosines of the sum and the difference.

Certain trigonometric formulas are true equalities *not for all* the values of the variables. For instance, the equality

$$\tan \alpha = \frac{\sin 2\alpha}{1 + \cos 2\alpha}$$

is true only for $\alpha \neq \frac{\pi}{2} + \pi n$, $n \in \mathbb{Z}$. Note that both the right-hand and left-hand sides of this formula, as the functions of α , have the same domain of definition, namely, they are defined for all $\alpha \neq \frac{\pi}{2} + \pi n$, $n \in \mathbb{Z}$. We can cite, as

another example, the formulas

$$\sin 2\alpha = \frac{2 \tan \alpha}{1 + \tan^2 \alpha}, \quad \cos 2\alpha = \frac{1 - \tan^2 \alpha}{1 + \tan^2 \alpha}, \quad (9)$$

which are also true for all $\alpha \neq \frac{\pi}{2} + \pi n$, $n \in \mathbb{Z}$. But, as distinct from the previous formula, the right-hand and left-hand sides of these formulas have *different* domains of definition. The left-hand sides of (9) are defined for all $\alpha \in \mathbb{R}$, and the right-hand sides are defined only for $\alpha \neq \frac{\pi}{2} + \pi n$, $n \in \mathbb{Z}$. As will follow from the examples below, this fact must be taken into account when using formulas (9) and similar formulas, for instance, the formula for the tangent of the double argument, the formulas for the tangent of the sum and the difference, and the like.

Let us consider some examples of solving various equations.

The *equations*

$$\begin{aligned} \sin ax + \sin bx &= 0, & \sin ax - \sin bx &= 0, \\ \cos ax + \cos bx &= 0, & \cos ax - \cos bx &= 0, \end{aligned}$$

are easy to solve with the use of formulas for the sum and the difference of sines and cosines.

Example 3. Solve the equation

$$\sin 6x + \sin 4x = 0.$$

$\triangle$ Applying the formula for the sum of sines, we get

$$2 \sin 5x \cos x = 0. \quad (10)$$

If x is a solution of this equation, then at least one of the following equalities is true:

$$\sin 5x = 0 \quad \text{or} \quad \cos x = 0. \quad (11)$$

Conversely, if x is a solution of one of equations (11), then, evidently, x is a solution of equation (10) as well. Thus, equation (10) is equivalent to the collection of equations (11). Equations (11) have the solutions

$$x = \frac{\pi n}{5}, \quad x = \frac{\pi}{2} + \pi n, \quad n \in \mathbb{Z}$$

respectively.

All these values of x and only these values are the solutions of the original equation. $\blacktriangle$

The equations

$$\sin ax + \cos bx = 0, \quad \sin ax - \cos bx = 0$$

can be reduced to the above-indicated equations by means of the reduction formulas.

Example 4. Solve the equation

$$\sin x = \cos 2x.$$

$\triangle$ Let us transform the equation using the reduction formula and the formula for the difference of sines:

$$\begin{aligned} \sin x - \cos 2x &= 0, \quad \sin x - \sin \left(\frac{\pi}{2} - 2x \right) = 0, \\ 2 \sin \frac{1}{2} \left(3x - \frac{\pi}{2} \right) \cos \frac{1}{2} \left(\frac{\pi}{2} - x \right) &= 0. \end{aligned}$$

The resulting equation is equivalent to the collection of two equations:

$$\sin \frac{1}{2} \left(3x - \frac{\pi}{2} \right) = 0, \quad \cos \frac{1}{2} \left(\frac{\pi}{2} - x \right) = 0.$$

We solve the first equation: $\frac{1}{2} \left(3x - \frac{\pi}{2} \right) = \pi k$, $x = \frac{\pi}{6} + \frac{2\pi k}{3}$, $k \in \mathbb{Z}$. For the second equation we have

$$\frac{1}{2} \left(\frac{\pi}{2} - x \right) = \frac{\pi}{2} + \pi l, \quad x = -\frac{\pi}{2} - 2\pi l, \quad l \in \mathbb{Z}.$$

It is easy to see that all solutions of the second equation are contained in the set of solutions of the first equation. Indeed, for $k = -1 - 3l$, $l \in \mathbb{Z}$, we have $x = \frac{\pi}{6} + \frac{2\pi(-1-3l)}{3} = -\frac{\pi}{2} - 2\pi l$, that is, we obtain all solutions of the second equation.

Answer: $\left\{ \frac{\pi}{6} + \frac{2\pi k}{3} \right\} \mid k \in \mathbb{Z} \}. \quad \blacktriangle$

Remark on the form of notation for solutions. Integral parameters in various sets of solutions of one equation may be designated either by different letters or by one letter. In Example 3, for instance, we used one letter n .

However, in cases when elements of sets are compared to one another, use should be made of different letters to denote integral parameters, as was done in Example 4.

Equations of the form

$$a \sin \omega x + b \cos \omega x = c, \quad a^2 + b^2 \neq 0. \quad (12)$$

Example 5. Solve the equation

$$\sin 2x + \cos 2x = -1.$$

△ This equation is a special case of the equations being discussed for $a = b = 1$, $c = -1$, $\omega = 2$. Dividing both parts of the equation by $1/\sqrt{2}$, we obtain

$$\frac{1}{\sqrt{2}} \sin 2x + \frac{1}{\sqrt{2}} \cos 2x = -\frac{1}{\sqrt{2}}.$$

Taking account of the fact that $1/\sqrt{2} = \cos(\pi/4) = \sin(\pi/4)$, we write the equation in the form

$$\sin 2x \cos(\pi/4) + \cos 2x \sin(\pi/4) = -1/\sqrt{2}.$$

Using the formula for the sine of the sum of the arguments, we arrive at an equation

$$\sin\left(2x + \frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}.$$

Hence we have $x = ((-1)^{n+1} - 1) \frac{\pi}{8} + \frac{\pi n}{2}$, $n \in \mathbb{Z}$. These values of x constitute the set of all solutions of the original equation. ▲

In a general case, to reduce equation (12) to the canonical form by introducing an auxiliary angle, we divide its both sides by $\sqrt{a^2 + b^2}$. Then we obtain

$$\frac{a}{\sqrt{a^2 + b^2}} \sin \omega x + \frac{b}{\sqrt{a^2 + b^2}} \cos \omega x = \frac{c}{\sqrt{a^2 + b^2}}.$$

Let φ be one of the solutions of the system

$$\cos \varphi = \frac{a}{\sqrt{a^2 + b^2}}, \quad \sin \varphi = \frac{b}{\sqrt{a^2 + b^2}}.$$

Using these equalities, we write the equation in the form

$$\sin \omega x \cos \varphi + \cos \omega x \sin \varphi = c/\sqrt{a^2 + b^2}.$$

Applying the formula for the sine of the sum of the arguments, we get an equation

$$\sin(\omega x + \varphi) = c/\sqrt{a^2 + b^2},$$

which, as can be seen from the computations carried out, is equivalent to the original equation. The equation obtained, and, hence, the original equation have solutions if and only if $|c| \leq \sqrt{a^2 + b^2}$.

To transform equation of form (12), use can also be made of the formulas for the sine of the difference, cosine of the sum, and the difference of the arguments.

Example 6. Solve the equation

$$12 \cos x - 5 \sin x = -13.$$

△ Dividing both sides of the equation by $\sqrt{12^2 + 5^2} = 13$, we get

$$\frac{12}{13} \cos x - \frac{5}{13} \sin x = -1.$$

One of the solutions of the system

$$\cos \varphi = 12/13, \quad \sin \varphi = 5/13$$

is $\varphi = \arccos(12/13)$. Taking this into account, we write the equation in the form

$$\cos x \cos \varphi - \sin x \sin \varphi = -1$$

and, applying the formula for the cosine of the sum of the arguments, we get $\cos(x + \varphi) = -1$, whence we have $x + \varphi = \pi + 2\pi n$, $x = -\varphi + \pi(2n + 1)$, that is,

$$x = -\arccos(12/13) + \pi(2n + 1), \quad n \in \mathbb{Z}.$$

This formula yields all the solutions of the original equation. ▲

Introduction of a new unknown. In some cases it proves to be possible to reduce an equation to a form containing only one trigonometric function.

Example 7. Solve the equation

$$\sin 3x + \cos 2x = 1.$$

△ We make use of the formulas $\cos 2x = 1 - 2 \sin^2 x$ and

$$\sin 3x = \sin x(3 - 4 \sin^2 x).$$

It is easy to obtain the second formula by transforming the right-hand side of the equality $\sin 3x = \sin(x + 2x)$ by the formula for the sine of the sum and then by the formulas for a double argument.

After the substitution, the original equation assumes the form

$$\sin x (3 - 4 \sin^2 x) + 1 - 2 \sin^2 x = 1.$$

From this we get $4 \sin^3 x + 2 \sin^2 x - 3 \sin x = 0$. Denoting $t = \sin x$, we obtain $4t^3 + 2t^2 - 3t = 0$. This equation has the roots $t_1 = 0$, $t_2 = (\sqrt{13} - 1)/4$, $t_3 = -(\sqrt{13} + 1)/4$. This means that the original equation is equivalent to the collection of the equations

$$\sin x = 0, \quad \sin x = (\sqrt{13} - 1)/4, \quad \sin x = -\frac{\sqrt{13} + 1}{4}.$$

We consecutively find the solutions for the equations obtained:

$$x = \pi n, \quad n \in \mathbb{Z}; \quad x = (-1)^n \arcsin \frac{\sqrt{13} - 1}{4} + \pi n, \quad n \in \mathbb{Z};$$

the third equation has no solutions since $-(\sqrt{13} + 1)/4 < -1$. The values of x we have obtained and only these values are solutions of the original equation. ▲

If the equation contains only one of the expressions,

$$\sin x + \cos x \text{ or } \sin x - \cos x,$$

and the function $\sin 2x$ (or the product $\sin x \cos x$), then, introducing the new unknown

$$t = \sin x + \cos x \text{ or } t = \sin x - \cos x$$

and bearing in mind that

$$\begin{cases} \sin 2x = (\sin x + \cos x)^2 - 1, \\ \sin 2x = 1 - (\sin x - \cos x)^2, \end{cases} \quad (13)$$

we arrive at an equation with respect to t .

Example 8. Solve the equation

$$\sin x + \cos x = 1 - \sin 2x.$$

△ We designate $t = \sin x + \cos x$ and use the first formula (13). Then we obtain

$$t = 1 - (t^2 - 1),$$

whence we get $t^2 + t - 2 = 0$, $t_1 = 1$, $t_2 = -2$. The original equation is equivalent to the collection of the equations

$$\sin x + \cos x = 1, \quad \sin x + \cos x = -2.$$

Let us solve each of the equations, say, by introducing an auxiliary angle. For the first equation we find $x = ((-1)^n - 1) \frac{\pi}{4} + \pi n$, $n \in \mathbb{Z}$; the second equation has no solutions since $|-2| > \sqrt{2}$.

Answer: $x = ((-1)^n - 1) \frac{\pi}{4} + \pi n$, $n \in \mathbb{Z}$. ▲

Example 9. Solve the equation

$$\sin^4 x + \cos^4 x = \frac{7}{2} \sin x \cos x.$$

△ We transform the expression $\sin^4 x + \cos^4 x$ isolating a perfect square:

$$\begin{aligned} \sin^4 x + \cos^4 x &= \sin^4 x + 2 \sin^2 x \cos^2 x + \cos^4 x \\ &\quad - 2 \sin^2 x \cos^2 x = (\sin^2 x + \cos^2 x)^2 - 2 \sin^2 x \cos^2 x, \end{aligned}$$

whence we get

$$\sin^4 x + \cos^4 x = 1 - \frac{1}{2} \sin^2 2x.$$

Using the formula obtained, we write the equation in the form

$$1 - \frac{1}{2} \sin^2 2x = \frac{7}{4} \sin 2x.$$

Introducing the designation $\sin 2x = t$, we get $2t^2 + 7t - 4 = 0$, whence $t_1 = 1/2$, $t_2 = -4$. The equation $\sin 2x = 1/2$ has solutions $x = (-1)^n \frac{\pi}{12} + \frac{\pi n}{2}$, $n \in \mathbb{Z}$; the equation $\sin x = -4$ has no solutions. ▲

We can express the sum $\sin^6 x + \cos^6 x$ as $\sin 2x$, factoring it as the sum of cubes.

To transform equations in previous examples, we have used formulas whose right-hand and left-hand sides are

defined for all values of x , the formulas themselves being true equalities for all x .

Let us consider examples of solving equations in which use is made of formulas whose right-hand and left-hand sides are defined not for all values of x .

Example 10. Solve the equation

$$6 \tan^2 x - 2 \cos^2 x = \cos 2x. \quad (14)$$

△ Let us transform the equation by the formulas

$$2 \cos^2 x = 1 + \cos 2x, \quad \tan^2 x = \frac{1 - \cos 2x}{1 + \cos 2x}.$$

Note that both sides of the second formula are defined, although not for all, but at least for the same values of x , namely, for all $x \neq \frac{\pi}{2} + \pi n$, $n \in \mathbb{Z}$. The left-hand side of the equation is defined for the same values of x . The substitution results in the equation

$$6 \frac{1 - \cos 2x}{1 + \cos 2x} - (1 + \cos 2x) = \cos 2x. \quad (15)$$

Any solution of equation (14) is, evidently, a solution of equation (15) and, conversely, every solution of (15) is a solution of (14), that is, equations (14) and (15) are equivalent. Let us solve equation (15). Designating $\cos 2x$ by t and transforming (15), we get $(2t^2 + 9t - 5)/(1 + t) = 0$, whence we have $t_1 = 0.5$, $t_2 = -5$. Consequently, equation (14) is equivalent to the collection of the equations

$$\cos 2x = 0.5, \quad \cos 2x = -5.$$

The first of these equations has solutions $x = \pm \frac{\pi}{6} + \pi n$, $n \in \mathbb{Z}$, the second equation has no solutions. ▲

Let us now discuss the case when a transformation of an equation requires formulas whose right-hand and left-hand sides possess different domains of definition.

Example 11. Solve the equation

$$5 \sin 2x - 5 \cos 2x = \tan x + 5. \quad (16)$$

△ Let us express $\sin 2x$ and $\cos 2x$ as $\tan x$ by formulas (9):

$$\sin 2x = \frac{2 \tan x}{1 + \tan^2 x}, \quad \cos 2x = \frac{1 - \tan^2 x}{1 + \tan^2 x}.$$

Substituting the right-hand sides of these formulas for $\sin 2x$ and $\cos 2x$ in equation (16), we obtain

$$5 \frac{2 \tan x}{1 + \tan^2 x} - 5 \frac{1 - \tan^2 x}{1 + \tan^2 x} = \tan x + 5. \quad (17)$$

Let us now find whether equations (16) and (17) are equivalent. The functions $\sin 2x$ and $\cos 2x$ are defined for all x while the right-hand sides of formulas (9) substituted for them are defined only for $x \neq \frac{\pi}{2} + \pi n$, $n \in \mathbb{Z}$. Hence, as a result of the substitution, the values $x = \frac{\pi}{2} + \pi n$, $n \in \mathbb{Z}$ have got out of the consideration. But it is clear that neither of these values of x is a solution of the original equation (16) since its right-hand side is not defined for $x = \frac{\pi}{2} + \pi n$, $n \in \mathbb{Z}$. Hence it follows that every solution of equation (16) is also a solution of equation (17). The converse is, evidently, also true. Thus we see that equations (16) and (17) are equivalent.

Let us solve equation (17). Designating $\tan x = t$, we get

$$\frac{10t}{1+t^2} - 5 \frac{1-t^2}{1+t^2} = t + 5.$$

Simple transformations lead us to the equation $t^3 - 9t + 10 = 0$ equivalent to the preceding equation. One of divisors of the constant term, namely, $t_1 = 2$, is a solution of this equation. Factoring now the left-hand side (say, by dividing the polynomial $t^3 - 9t + 10$ by the difference $t - 2$), we get $(t - 2)(t^2 + 2t - 5) = 0$. Solving the quadratic equation $t^2 + 2t - 5 = 0$, we find two more solutions $t_2 = \sqrt{6} - 1$, $t_3 = -\sqrt{6} - 1$. Thus we see that the original equation (16) is equivalent to the collection of the equations

$$\tan x = 2, \quad \tan x = \sqrt{6} - 1, \quad \tan x = -\sqrt{6} - 1,$$

which have the following respective solutions:

$$\begin{aligned} x &= \arctan 2 + \pi n, & n &\in \mathbb{Z}, \\ x &= \arctan (\sqrt{6} - 1) + \pi n, & n &\in \mathbb{Z}, \\ x &= -\arctan (\sqrt{6} + 1) + \pi n, & n &\in \mathbb{Z}. \end{aligned}$$

These values of x constitute the set of all solutions of equation (16). ▲

As was noted, in formulas (9) the left-hand sides are defined for all values of x while the right-hand sides are defined only for $x \neq \frac{\pi}{2} + \pi n$, $n \in \mathbb{Z}$. Therefore, the substitution into the equation of the *right-hand* sides of these formulas instead of the left-hand sides may lead to a loss of solutions of the form $x = \frac{\pi}{2} + \pi n$, $n \in \mathbb{Z}$. To use such a transformation, one should either establish the equivalence between the equation obtained and the original equation (as was done in Example 11) or verify by means of the substitution of the values $x = \frac{\pi}{2} + \pi n$, $n \in \mathbb{Z}$ into the original equation whether there are any solutions among them. The following example proves the necessity of such a verification.

Example 12. Solve the equation

$$\sin 2x + \cos 2x = -1. \quad (18)$$

△ Let us transform the equation with the aid of formulas (9) (we prefer the method suggested in Example 5):

$$\frac{2 \tan x}{1 + \tan^2 x} + \frac{1 - \tan^2 x}{1 + \tan^2 x} = -1. \quad (19)$$

Equation (19) is not equivalent to the original equation (18). Indeed, the values $x = \frac{\pi}{2} + \pi k$, $k \in \mathbb{Z}$, are, evidently, not solutions of equation (19). At the same time, we can easily ascertain by means of a substitution that all these values of x are solutions of equation (18).

Let us now consider the values $x \neq \frac{\pi}{2} + \pi k$, $k \in \mathbb{Z}$. It is clear that any solution of equation (18), satisfying the condition $x \neq \frac{\pi}{2} + \pi k$, is a solution of equation (19) and, conversely, any solution of (19) is a solution of (18). Let us solve equation (19). We transform it and get

$$\frac{2(\tan x + 1)}{1 + \tan^2 x} = 0, \quad \text{whence we have } \tan x = -1,$$

$x = \frac{\pi}{4} + \pi k, k \in \mathbb{Z}$. Together with the values $x = \frac{\pi}{2} + \pi k, k \in \mathbb{Z}$, these values of x constitute the set of all solutions of equation (18).

Answer: $\left\{ \frac{\pi}{2} + \pi k; -\frac{\pi}{4} + \pi k \mid k \in \mathbb{Z} \right\}$ (if we take $n = 2k + 1$ and $n = 2k$, respectively, in the answer to Example 5, we shall get this answer).

A substitution into the equation of the *left-hand* side of formulas (9) instead of the right-hand side may lead to the appearance of the values $x = \frac{\pi}{2} + \pi n, n \in \mathbb{Z}$, which are extraneous to the original equation. When performing such transformations, it is necessary to verify whether all the solutions of the equation obtained are solutions of the original equation.

The indicated peculiarities of transformation of equations by means of formulas (9) and the resulting necessity for verification arise in other cases as well, when use is made of formulas whose right-hand and left-hand sides have different domains of definition.

It should also be pointed out that transformations by formulas (9) often lead to a cumbersome equation. Therefore, before applying these formulas, other means of solution should be sought.

Homogeneous equations.

Example 13. Solve the equation

$$\sin^2 x - 3 \sin x \cos x + 2 \cos^2 x = 0.$$

$\triangle$ Let us consider x 's such that $\cos x = 0$. It follows from the equation that in this case $\sin x = 0$, as well, and that is impossible. Consequently, there are no solutions among these values of x . Let us now take the values of x for which $\cos x \neq 0$. Dividing both sides of the given equation by $\cos x$, we obtain an equation

$$\tan^2 x - 3 \tan x + 2 = 0,$$

which is equivalent to the original equation. Solving it as a quadratic equation with respect to $\tan x$, we find that

$$\tan x = 1, \quad \tan x = 2.$$

Hence we get the *answer*: $x = \frac{\pi}{4} + \pi n$, $x = \arctan 2 + \pi n$, $n \in \mathbb{Z}$. $\blacktriangle$

The equation we have considered is an example of an equation of the form

$$a_0 u^2 + a_1 uv + a_2 v^2 = 0, \quad (20)$$

which is a *homogeneous second-degree equation with respect to u and v* . In the previous example $u = \sin x$, $v = \cos x$. As was indicated, the division of both sides of an equation of form (20) by v^2 can lead to the quadratic equation

$$a_0 t^2 + a_1 t + a_2 = 0$$

for $t = u/v$. The same transformation can be used to solve homogeneous equations of a higher degree.

Some equations, which are nonhomogeneous, can be reduced to homogeneous equations using the equation

$$\sin^2 x + \cos^2 x = 1,$$

which is true for all x .

Example 14. Solve the equation

$$2 \sin^3 x = \cos x.$$

$\triangle$ Since $\cos x = \cos x (\sin^2 x + \cos^2 x)$ for all x , the given equation is equivalent to the equation

$$2 \sin^3 x = \cos x \sin^2 x + \cos^3 x,$$

which is homogeneous with respect to $\sin x$ and $\cos x$ of degree 3. In the case of $\cos x = 0$, there are, evidently, no solutions. Dividing both sides of the equality by $\cos^3 x$ and introducing the designation $\tan x = t$, we arrive at an equation

$$2t^3 - t^2 - 1 = 0.$$

One of its roots is $t_1 = 1$. Factoring the left-hand side, we get $(t - 1)(2t^2 + t + 1) = 0$. Since $2t^2 + t + 1 > 0$ for any t , the root $t_1 = 1$ is unique. Hence it follows that $\tan x = 1$, $x = \frac{\pi}{4} + \pi n$, $n \in \mathbb{Z}$.

Answer: $\left\{ \frac{\pi}{4} + \pi n \mid n \in \mathbb{Z} \right\}$. $\blacktriangle$

Factorization. Transformation of an equation with the aim of isolating a common factor is often the shortest way to its solution. Let us consider some examples.

Example 15. Solve the equation

$$2 \sin^3 x + \cos^2 2x = \sin x.$$

△ Collecting the terms of the equation, we transform it as follows:

$$\begin{aligned} (2 \sin^3 x - \sin x) + \cos^2 2x &= 0, \\ \sin x (2 \sin^2 x - 1) + \cos^2 2x &= 0. \end{aligned}$$

Using the formula $1 - 2 \sin^2 x = \cos 2x$, we get

$$\begin{aligned} -\sin x \cos 2x + \cos^2 2x &= 0, \\ \cos 2x (\cos 2x - \sin x) &= 0. \end{aligned}$$

The last equation is equivalent to the original equation. At the same time, it is, evidently, equivalent to the collection of the equations

$$\cos 2x = 0, \quad \cos 2x - \sin x = 0.$$

These equations have the respective solutions

$$x = \frac{\pi}{4} + \frac{\pi n}{2}, \quad x = \frac{\pi}{6} + \frac{2\pi n}{3}, \quad n \in \mathbb{Z}$$

(the second equation was solved in Example 4), which are also solutions of the original equation. ▲

Example 16. Solve the equation

$$\sin \left(x + \frac{\pi}{4} \right) = \sin^3 x + \cos^3 x.$$

△ We transform the left-hand side of the equation as follows:

$$\begin{aligned} \sin \left(x + \frac{\pi}{4} \right) &= \sin x \cos \frac{\pi}{4} + \cos x \sin \frac{\pi}{4} \\ &= \frac{1}{\sqrt{2}} (\sin x + \cos x), \end{aligned}$$

and expand the right-hand side into a product as the sum of cubes:

$$\begin{aligned} \sin^3 x + \cos^3 x &= (\sin x + \cos x) (\sin^2 x - \sin x \cos x + \cos^2 x) \\ &= (\sin x + \cos x) (1 - \sin x \cos x). \end{aligned}$$

We can now write the equation as follows:

$$(\sin x + \cos x) \left(\frac{1}{\sqrt{2}} - 1 + \sin x \cos x \right) = 0.$$

This equation is equivalent to the collection of the equations

$$\sin x + \cos x = 0, \quad \frac{1}{\sqrt{2}} - 1 + \sin x \cos x = 0.$$

The first of these equations is equivalent to the equation $\tan x = -1$, and the second, to the equation $\sin 2x = 2 - \sqrt{2}$. Consequently,

$$x = -\frac{\pi}{4} + \pi n, \quad x = \frac{(-1)^n}{2} \arcsin(2 - \sqrt{2}) + \frac{\pi n}{2}, \quad n \in \mathbb{Z}.$$

These are precisely all the solutions of the original equation. $\blacktriangle$

In some cases, when solving equations by the method of factorization, some extraneous values of the unknown may appear. They can be removed by substitution of the values obtained into the original equation or the equation equivalent to it.

Let us consider a requisite example.

Example 17. Solve the equation

$$\tan x - \sin x = 1 - \tan x \sin x.$$

$\triangle$ Grouping the terms of the equation, we transform it as follows:

$$\begin{aligned} (\tan x + \tan x \sin x) - (1 + \sin x) &= 0, \\ (1 + \sin x) \tan x - (1 + \sin x) &= 0, \\ (1 + \sin x) (\tan x - 1) &= 0. \end{aligned} \quad (21)$$

The resulting equation is equivalent to the original one. If x is a solution of this equation, then it is also a solution of one of the equations

$$1 + \sin x = 0, \quad \tan x - 1 = 0. \quad (22)$$

In the given case (and as distinct from what can be found in Examples 15 and 16) the converse is not true. The first equation in (22) possesses solutions $x = -\frac{\pi}{2} + 2\pi n, n \in \mathbb{Z}$. For all these values of x the function $\tan x$ is not defined and,

therefore, these values cannot serve as solutions to the original equation. The second equation in (22) possesses solutions $x = \frac{\pi}{4} + \pi n$, $n \in \mathbb{Z}$. Substitution into (21) shows that these values can serve as solutions of this equation and, hence, of the original equation.

Answer: $\left\{ \frac{\pi}{4} + \pi n \mid n \in \mathbb{Z} \right\}$. ▲

Using estimates. It is possible to solve some trigonometric equations with the use of the inequalities

$$-1 \leq \sin \alpha \leq 1, \quad -1 \leq \cos \alpha \leq 1,$$

true for all α .

Example 18. Solve the equation

$$\sin 3x + \cos 2x + 2 = 0.$$

△ Since $\sin 3x \geq -1$ and $\cos 2x \geq -1$, we have $\sin 3x + \cos 2x \geq -2$, the equality holding true here if and only if the following equalities are valid simultaneously:

$$\sin 3x = -1, \quad \cos 2x = -1. \quad (23)$$

This means that the original equation is equivalent to system (23). The equations of this system have the following respective solutions:

$$x = -\frac{\pi}{6} + \frac{2\pi k}{3}, \quad k \in \mathbb{Z};$$

$$x = \frac{\pi}{2} + \pi l, \quad l \in \mathbb{Z}.$$

Let us represent these solutions as points of a unit circle (in Fig. 1 thick points correspond to the solutions of the first equation and crossed points to the solutions of the second equation; to find these points, it is sufficient to take

$k = 0, 1, 2$ in the first case, and $l = 0, 1$ in the second). The number x is a solution of system (23) if and only if it is a solution to both equations in (23). It can be seen in Fig. 1

that such numbers are only $x = \frac{\pi}{2} + 2\pi m$, $m \in \mathbb{Z}$. ▲

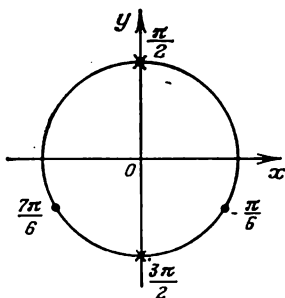


Fig. 1

Example 19. Solve the equation

$$\cos \frac{\sqrt{2}+1}{2} x \cdot \cos \frac{\sqrt{2}-1}{2} x = 1.$$

△ Using the formula for the product of cosines, we reduce the equation to the form

$$\cos(\sqrt{2}x) + \cos x = 2.$$

This equation is equivalent to the system

$$\cos \sqrt{2}x = 1, \quad \cos x = 1. \quad (24)$$

The equations of the system have the respective solutions $x = \sqrt{2}\pi n$, $n \in \mathbb{Z}$; $x = 2\pi m$, $m \in \mathbb{Z}$. The solutions of system (24) are those and only those values of x for which the equalities $x = \sqrt{2}\pi n = 2\pi m$ hold for certain integral m and n . Let us find all integral values of m and n for which $\sqrt{2}\pi n = 2\pi m$. Cancelling here by $\sqrt{2}\pi$, we get $n = \sqrt{2}m$. This equality is possible only for $m = 0$, $n = 0$. In fact, if $m \neq 0$ and $n = \sqrt{2}m$, then $\sqrt{2} = n/m$. But this equality does not hold since $\sqrt{2}$ is not a rational number and n/m is a rational number. Consequently, $m = 0$, but then $n = 0$ as well. Thus it follows that system (24) and, hence, the original equation, have a unique solution $x = 0$. ▲

To conclude this section, we shall consider, as an example, an equation which is not trigonometric but whose investigation requires use of the properties of trigonometric functions.

Example 20. Prove that the equation

$$\sin x = \frac{1}{2} |x| + a$$

has no solutions, for $a > (3\sqrt{3} - \pi)/6$.

△ Assume that $a > (3\sqrt{3} - \pi)/6$. Note that $(3\sqrt{3} - \pi)/6 > 0$. Supposing that $|x| \geq 2$, we have $\frac{1}{2}|x| + a > 1$ whereas $\sin x \leq 1$. This means that the given equation has no solutions on the set $|x| \geq 2$.

On the interval $[-2; 0]$ we have $\sin x \leq 0$, but $\frac{1}{2}|x| + a \geq a > 0$, and this means that there are no solutions here either.

On the interval $(0; 2)$ the equation has the form

$$\sin x - \frac{1}{2}x = a.$$

Let us find the greatest value of the function $f(x) = \sin x - \frac{1}{2}x$ on this interval. We determine the critical points: $f'(x) = \cos x - \frac{1}{2} = 0$, $x = \frac{\pi}{3}$ (the other solutions of this equation do not belong to the interval $(0; 2)$). If $0 < x < \pi/3$, then $f'(x) > 0$ and, therefore, the function $f(x)$ increases monotonically on the interval $(0; \pi/3)$ and this means that $f(x) < f(\pi/3)$. And if $\pi/3 < x < 2$, then $f'(x) < 0$ and, therefore, the function $f(x)$ decreases monotonically on the interval $(\frac{\pi}{3}; 2)$ and $f(x) < f(\pi/3)$. Thus, the value

$$f\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2} - \frac{\pi}{6} = \frac{3\sqrt{3} - \pi}{6}$$

is the greatest on $(0; 2)$, that is,

$$\sin x - \frac{1}{2}x \leq \frac{3\sqrt{3} - \pi}{6} < a.$$

or all $x \in (0; 2)$. This means that the given equation does not possess solutions on the interval $(0; 2)$ either. ▲

1.2. Systems of Trigonometric Equations

We shall discuss here some techniques employed in solving trigonometric systems, and use examples as illustrations. We shall restrict our consideration to systems with two variables (often called unknowns) x and y . We shall write the solutions of the system in the form of ordered pairs $(x; y)$ as usual.

Systems containing an equation of the form

$$x + y = \alpha \text{ or } x - y = \alpha.$$

can be reduced to one equation by substitution.

Example 1. Solve the system

$$\begin{cases} \sin(x - y) = 2 \sin x \sin y, \\ x + y = \pi/2. \end{cases}$$

△ We make the substitution $y = \frac{\pi}{2} - x$ in the first equation and transform it as follows:

$$\begin{aligned}\sin\left(2x - \frac{\pi}{2}\right) &= 2 \sin x \sin\left(\frac{\pi}{2} - x\right), \\ -\sin\left(\frac{\pi}{2} - 2x\right) &= 2 \sin x \cos x, \\ -\cos 2x &= \sin 2x, \tan 2x = -1.\end{aligned}$$

Hence it follows that $x = -\frac{\pi}{2} + \frac{\pi n}{2}$, $n \in \mathbb{Z}$. Now we find $y = \frac{5\pi}{2} - \frac{\pi n}{2}$, $n \in \mathbb{Z}$.

$$\text{Answer: } \left\{ \left(-\frac{\pi}{2} + \frac{\pi n}{2}; \frac{5\pi}{2} - \frac{\pi n}{2} \right) \mid n \in \mathbb{Z} \right\}. \quad \blacktriangle$$

In some cases, before making a substitution in the equation of a system, it is expedient to transform it.

Example 2. Solve the system

$$\begin{cases} \sin^2 x + \sin^2 y = 1/2, \\ x - y = 4\pi/3. \end{cases}$$

△ We transform the first equation of the system:

$$\frac{1}{2}(1 - \cos 2x) + \frac{1}{2}(1 - \cos 2y) = \frac{1}{2},$$

$$\cos 2x + \cos 2y = 1, \quad 2 \cos(x + y) \cos(x - y) = 1.$$

It is clear that the system

$$\begin{cases} \cos(x + y) \cos(x - y) = 1/2, \\ x - y = 4\pi/3 \end{cases}$$

has the same solutions as the original system, that is, the systems are equivalent. After making the substitution $x - y = 4\pi/3$ in the first equation of the system obtained, we have $\cos(x + y) \cos(4\pi/3) = 1/2$, whence $\cos(x + y) = -1$ and, consequently, $x + y = \pi(2n + 1)$, $n \in \mathbb{Z}$. Now, for any $n \in \mathbb{Z}$, we find from the linear system

$$\begin{cases} x + y = \pi(2n + 1), \\ x - y = 4\pi/3, \end{cases}$$

that $x = \frac{7\pi}{6} + \pi n$, $y = -\frac{\pi}{6} + \pi n$, $n \in \mathbb{Z}$. The set of all such pairs of values of x and y is the set of all solutions of the original system. ▲

In previous examples we wrote the obtained relationships between the unknowns x and y and the set of solutions of the system with the aid of only one integral parameter. In practical applications, however, we usually encounter two integral parameters when solving systems with two variables.

Example 3. Solve the system

$$\begin{cases} \sin(x+y) = 0, \\ \sin(x-y) = 0. \end{cases} \quad (1)$$

△ It follows from the first equation of system (1) that

$$x + y = \pi m, \quad m \in \mathbb{Z}, \quad (2)$$

and from the second, that

$$x - y = \pi n, \quad n \in \mathbb{Z}. \quad (3)$$

Now, for arbitrary m and n , from the linear system

$$\begin{cases} x + y = \pi m, \\ x - y = \pi n \end{cases} \quad (4)$$

we find that

$$x = \frac{\pi}{2} (m + n), \quad y = \frac{\pi}{2} (m - n),$$

with $m \in \mathbb{Z}$ and $n \in \mathbb{Z}$. These pairs of numbers and only they constitute the set of all solutions of system (1). ▲

It would be erroneous to consider, instead of systems of form (4) (where m and n are arbitrary integral numbers), only the systems

$$\begin{cases} x + y = \pi m, \\ x - y = \pi m, \end{cases} \quad m \in \mathbb{Z} \quad (5)$$

(such an error often arises when one letter is used to designate the parameters in relations of the kind of (2), (3)). Indeed, systems of form (4) include all possible pairs of parameters m and n whereas systems of form (5) only involve pairs with equal values ($m = n$). This can lead to a loss of

solutions. From (5) it follows that

$$x = \pi m, \quad y = 0,$$

where $m \in \mathbb{Z}$, while, for instance, all the pairs

$$x = 0, \quad y = \pi n, \quad n \in \mathbb{Z},$$

are also solutions of system (1).

Let us now take examples of solving more difficult systems.

Introduction of new variables can be employed, say, in cases where a system contains only two trigonometric functions or can be reduced to such a form.

Example 4. Solve the system

$$\begin{cases} \sin x + \cos y = 1, \\ \cos 2x - \cos 2y = 1. \end{cases}$$

$\triangle$ Let us transform the second equation:

$$1 - 2 \sin^2 x + 1 - 2 \cos^2 y = 1, \quad \sin^2 x + \cos^2 y = 1/2.$$

It is clear that the system

$$\begin{cases} \sin x + \cos y = 1, \\ \sin^2 x + \cos^2 y = 1/2 \end{cases}$$

is equivalent to the original system. Designating, for simplicity, $u = \sin x$, $v = \cos y$, we obtain

$$\begin{cases} u + v = 1, \\ u^2 + v^2 = 1/2. \end{cases}$$

It is easy to establish that this algebraic system has a unique solution $u = 1/2$, $v = 1/2$. Consequently, the original system is equivalent to the system

$$\begin{cases} \sin x = 1/2, \\ \cos y = 1/2. \end{cases}$$

These equations have the respective solutions

$$x = (-1)^m \frac{\pi}{6} + \pi m, \quad m \in \mathbb{Z},$$

$$y = \pm \frac{\pi}{3} + 2\pi n, \quad n \in \mathbb{Z}.$$

The set of various pairs formed from these values of x and y is precisely the set of all solutions of the original system. For brevity, such pairs are written in the form

$$x = (-1)^m \frac{\pi}{6} + \pi m, \quad y = \pm \frac{\pi}{3} + 2\pi n, \quad m, n \in \mathbb{Z}.$$

It is implied that the sign in the formula for y is chosen arbitrarily. $\blacktriangle$

Example 5. Solve the system

$$\begin{cases} \sin x + \sin y = \sqrt{2}, \\ \cos x \cos y = 1/2. \end{cases} \quad (6)$$

$\triangle$ Using the formula for the sum of sines, we reduce the first equation to the form

$$\sin \frac{x+y}{2} \cos \frac{x-y}{2} = \frac{1}{\sqrt{2}}.$$

We transform the left-hand side of the second equation as follows:

$$\begin{aligned} \cos x \cos y &= \frac{1}{2} (\cos(x+y) + \cos(x-y)) \\ &= \frac{1}{2} \left(1 - 2 \sin^2 \frac{x+y}{2} + 2 \cos^2 \frac{x-y}{2} - 1 \right) \\ &= \cos^2 \frac{x-y}{2} - \sin^2 \frac{x+y}{2}. \end{aligned}$$

As a result we get a system

$$\begin{cases} \sin \frac{x+y}{2} \cos \frac{x-y}{2} = \frac{1}{\sqrt{2}}, \\ \cos^2 \frac{x-y}{2} - \sin^2 \frac{x+y}{2} = \frac{1}{2}, \end{cases}$$

which is equivalent to the original system. Introducing the designations $u = \sin((x+y)/2)$, $v = \cos((x-y)/2)$, we arrive at a system

$$\begin{cases} uv = 1/\sqrt{2}, \\ v^2 - u^2 = 1/2, \end{cases}$$

which possesses two solutions: $u_1 = 1/\sqrt{2}$, $v_1 = 1$; $u_2 = -1/\sqrt{2}$, $v_2 = -1$. Thus, every solution of the original

system is a solution to one of the systems

$$(a) \begin{cases} \sin \frac{x+y}{2} = \frac{1}{\sqrt{2}}, \\ \cos \frac{x-y}{2} = 1, \end{cases} \quad (b) \begin{cases} \sin \frac{x+y}{2} = -\frac{1}{\sqrt{2}}, \\ \cos \frac{x-y}{2} = -1. \end{cases} \quad (7)$$

Performing the computations in a reverse order, we make sure that the converse is also true: every solution of any of the systems (a) and (b) is a solution of the original system. In such cases we say that system (6) is equivalent to the collection of systems (7).

System (a) yields

$$\begin{cases} \frac{x+y}{2} = (-1)^m \frac{\pi}{4} + \pi m, \\ \frac{x-y}{2} = 2\pi n, \end{cases}$$

where $m, n \in \mathbb{Z}$, whence we find

$$x = (-1)^m \frac{\pi}{4} + \pi(m+2n), \quad y = (-1)^m \frac{\pi}{4} + \pi(m-2n), \quad (8)$$

where $m, n \in \mathbb{Z}$.

We solve system (b):

$$\begin{cases} \frac{x+y}{2} = (-1)^{m+1} \frac{\pi}{4} + \pi m, \\ \frac{x-y}{2} = \pi(2n+1), \end{cases}$$

where $m, n \in \mathbb{Z}$. Hence it follows that

$$\begin{aligned} x &= (-1)^{m+1} \frac{\pi}{4} + \pi(m+2n+1), \\ y &= (-1)^{m+1} \frac{\pi}{4} + \pi(m-2n-1), \end{aligned} \quad (9)$$

where $m, n \in \mathbb{Z}$.

Answer: The set of all solutions of the system is given by formulas (8), (9). $\blacktriangle$

Let us consider an example of one more substitution.

Example 6. Solve the system

$$\begin{cases} \sin x + \cos x = \frac{1}{\sqrt{2}} + \sin y - \cos y, \\ 2 \sin 2x = \frac{3}{2} + \sin 2y. \end{cases}$$

△ We introduce the designations:

$$u = \sin x + \cos x, \quad v = \sin y - \cos y$$

and make use of the fact that $\sin 2x = u^2 - 1$, $\sin 2y = 1 - v^2$. This allows us to reduce the given system to the algebraic system

$$\begin{cases} u = \frac{1}{\sqrt{2}} + v, \\ 2u^2 + v^2 = 9/2. \end{cases}$$

The system obtained possesses two solutions:

$$u_1 = -4/3\sqrt{2}, \quad v_1 = -7/3\sqrt{2}; \quad u_2 = \sqrt{2}, \quad v_2 = 1/\sqrt{2}.$$

Thus we see that the original system is equivalent to the collection of two systems:

$$(a) \begin{cases} \sin x + \cos x = -4/3\sqrt{2}, \\ \sin y - \cos y = -7/3\sqrt{2}; \end{cases} \quad (b) \begin{cases} \sin x + \cos x = \sqrt{2}, \\ \sin y - \cos y = 1/\sqrt{2}. \end{cases}$$

System (a) has no solutions since $|\sin y - \cos y| \leq \sqrt{2}$ and $|-7/3\sqrt{2}| > \sqrt{2}$. We shall find solutions to equations of system (b) by introducing an auxiliary angle:

$$\sin \left(x + \frac{\pi}{4} \right) = 1, \quad x = \frac{\pi}{4} + 2\pi m;$$

$$\sin \left(y - \frac{\pi}{4} \right) = \frac{1}{2}, \quad y = (-1)^n \frac{\pi}{6} + \frac{\pi}{4} + \pi n,$$

where $m, n \in \mathbb{Z}$. These pairs of values of x and y constitute the set of all solutions of the original system. ▲

In some cases, to solve a system, we transform it by means of termwise addition, subtraction, multiplication, and division of equations with the aim of eliminating one of the unknowns (variables), factoring the equation obtained and so on. Let us consider some examples of using such transformations.

Example 7. Solve the system

$$\begin{cases} \cos x \cos y = 3/4, \\ \sin x \sin y = -1/4. \end{cases}$$

△ Adding and subtracting the equations of the system term-by-term, we obtain, respectively,

$$\cos x \cos y + \sin x \sin y = 1/2, \quad \cos(x - y) = 1/2;$$

$$\cos x \cos y - \sin x \sin y = 1, \quad \cos(x + y) = 1.$$

The system

$$\begin{cases} \cos(x - y) = 1/2, \\ \cos(x + y) = 1 \end{cases}$$

is equivalent to the original one. We solve each equation of this system:

$$x - y = \pm \frac{\pi}{3} + 2\pi m, \quad m \in \mathbb{Z}; \quad x + y = 2\pi n, \quad n \in \mathbb{Z}.$$

Taking the upper sign in the first formula, we find that

$$x = \frac{\pi}{6} + \pi(m + n), \quad y = -\frac{\pi}{6} + \pi(n - m), \quad (10)$$

where, $m, n \in \mathbb{Z}$. For the lower sign we get

$$x = -\frac{\pi}{6} + \pi(m + n), \quad y = \frac{\pi}{6} + \pi(n - m), \quad (11)$$

where $m, n \in \mathbb{Z}$. Formulas (10) and (11) give the set of all solutions of the original system. For the sake of brevity, we can also use the notation

$$x = \pm \frac{\pi}{6} + \pi(m + n), \quad y = \mp \frac{\pi}{6} + \pi(n - m),$$

where $m, n \in \mathbb{Z}$, with the reservation that in these formulas either only the upper signs or only the lower signs should be taken simultaneously. ▲

Example 8. Solve the system

$$\begin{cases} \tan\left(\frac{\pi}{4} + x\right) = 2\sqrt{2} \cos^3 y, \\ \tan\left(\frac{\pi}{4} - x\right) = 2\sqrt{2} \sin^3 y. \end{cases}$$

△ By the reduction formula we have $\tan\left(\frac{\pi}{4} - x\right) = \cot\left(\frac{\pi}{2} - \frac{\pi}{4} + x\right) = \cot\left(\frac{\pi}{4} + x\right)$. Hence, the original sys-

tem is equivalent to the system

$$\begin{cases} \tan\left(\frac{\pi}{4} + x\right) = 2\sqrt{2}\cos^3 y, \\ \cot\left(\frac{\pi}{4} + x\right) = 2\sqrt{2}\sin^3 y. \end{cases} \quad (12)$$

Multiplying the equations of this system term-by-term, we get an equation

$$1 = 8 \cos^3 y \sin^3 y,$$

whence we have $\sin^3 2y = 1$, $\sin 2y = 1$, $2y = \frac{\pi}{2} + 2\pi n$, $y = \frac{\pi}{4} + \pi n$, $n \in \mathbb{Z}$. We substitute these values into system (12) and, taking into account that

$\cos(\alpha + \pi n) = (-1)^n \cos \alpha$, $\sin(\alpha + \pi n) = (-1)^n \sin \alpha$, we obtain

$$\begin{cases} \tan\left(\frac{\pi}{4} + x\right) = (-1)^n, \\ \cot\left(\frac{\pi}{4} + x\right) = (-1)^n. \end{cases} \quad (13)$$

If $n = 2k$, $k \in \mathbb{Z}$, then $y = \frac{\pi}{4} + 2\pi k$, and system (13) is equivalent to one equation $\tan\left(\frac{\pi}{4} + x\right) = 1$, whence it follows that $x = \pi l$, $l \in \mathbb{Z}$.

If $n = 2k + 1$, $k \in \mathbb{Z}$, then $y = \frac{\pi}{4} + \pi(2k + 1)$ and system (13) is equivalent to the equation $\tan\left(\frac{\pi}{4} + x\right) = -1$, whence we have $x = -\frac{\pi}{2} + \pi l$, $l \in \mathbb{Z}$.

Answer: $\left\{ \left(\pi l; \frac{\pi}{4} + 2\pi k \right), \left(-\frac{\pi}{2} + \pi l; \frac{\pi}{4} + \pi(2k + 1) \right) \mid l, k \in \mathbb{Z} \right\}$. $\blacktriangle$

Example 9. Solve the system

$$\begin{cases} \sin x + \sin y = 1, \\ \cos x - \cos y = \sqrt{3}. \end{cases} \quad (14)$$

△ We first transform the equations of the system by the formulas for the sum of sines and the difference of cosines and obtain the system

$$\begin{cases} \sin \frac{x+y}{2} \cos \frac{x-y}{2} = \frac{1}{2}, \\ \sin \frac{x+y}{2} \sin \frac{x-y}{2} = -\frac{\sqrt{3}}{2}, \end{cases}$$

equivalent to the original system. Introduction of the notation $(x+y)/2 = u$, $(x-y)/2 = v$ reduces the system to the form

$$\begin{cases} \sin u \cos v = 1/2, \\ \sin u \sin v = -\sqrt{3}/2. \end{cases} \quad (15)$$

It is clear that if $(u; v)$ is a solution of (15), then $\sin u \cos v = 1/2 \neq 0$. Therefore, for any solution of the system, we find, dividing termwise the second equation of (15) by the first, that

$$\tan v = -\sqrt{3}.$$

Hence we find that $v = -\frac{\pi}{3} + \pi n$, $n \in \mathbb{Z}$. Substituting these values into (15), we get a system

$$\begin{cases} \sin u = (-1)^n, \\ \sin u = (-1)^n, \end{cases}$$

whence it follows that $u = (-1)^n \frac{\pi}{2} + 2\pi m$, $m \in \mathbb{Z}$. Now, from the system

$$\begin{cases} \frac{x+y}{2} = (-1)^n \frac{\pi}{2} + 2\pi m, \\ \frac{x-y}{2} = -\frac{\pi}{3} + \pi n \end{cases}$$

we find that

$$\begin{aligned} x &= (-1)^n \frac{\pi}{2} - \frac{\pi}{3} + \pi(2m+n), \\ y &= (-1)^n \frac{\pi}{2} + \frac{\pi}{3} + \pi(2m-n), \end{aligned} \quad (16)$$

where $m, n \in \mathbb{Z}$.

Answer: The set of all solutions of the system is given by formulas (16). ▲

Example 10. Solve the system

$$\begin{cases} \sqrt{2} \cos x = 1 + \cos y; \\ \sqrt{2} \sin x = \sin y. \end{cases}$$

▲ Squaring the equations of the system term-by-term and summing up, we obtain an equation

$$2 = 1 + 2 \cos y + \cos^2 y + \sin^2 y.$$

Hence we have $\cos y = 0$, $y = \frac{\pi}{2} + \pi n$, $n \in \mathbb{Z}$. Substitution of these values of y into the original system yields

$$\begin{cases} \cos x = 1/\sqrt{2}, \\ \sin x = (-1)^n/\sqrt{2}. \end{cases}$$

If $n = 2k$, $k \in \mathbb{Z}$, then

$$\begin{cases} \cos x = 1/\sqrt{2}, \\ \sin x = 1/\sqrt{2}, \end{cases}$$

whence it follows that $x = \frac{\pi}{4} + 2\pi l$, $l \in \mathbb{Z}$. In this case,

$$y = \frac{\pi}{2} + 2\pi k, \quad k \in \mathbb{Z}.$$

Now if $n = 2k + 1$, $k \in \mathbb{Z}$, then

$$\begin{cases} \cos x = 1/\sqrt{2}, \\ \sin x = -1/\sqrt{2}, \end{cases}$$

whence it follows that $x = -\frac{\pi}{4} + 2\pi l$, $l \in \mathbb{Z}$. In this case,

$$y = \frac{3\pi}{2} + 2\pi k, \quad k \in \mathbb{Z}.$$

Answer: $\left\{ \left(\frac{\pi}{4} + 2\pi l; \frac{\pi}{2} + 2\pi k \right), \left(-\frac{\pi}{4} + 2\pi l; \frac{3\pi}{2} + 2\pi k \right) \mid l, k \in \mathbb{Z} \right\}$. ▲

If, in order to find the values of x in the given example, we used, say, only the first equation of the system, then, as it is easy to see, we would get extraneous values of the variables.

1.3. Trigonometric Inequalities

To solve inequalities including trigonometric functions, it is good practice to use periodicity of those functions and their monotonicity on the corresponding intervals.

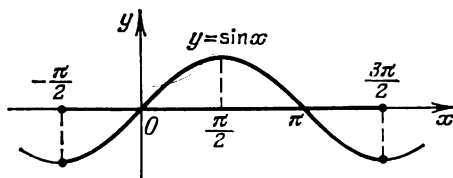


Fig. 2

The canonical trigonometric inequalities. The function $\sin x$ has the least positive period 2π . That is why it is sufficient to solve inequalities of the form

$$\sin x > a, \quad \sin x \geq a, \quad (1)$$

$$\sin x < a, \quad \sin x \leq a, \quad (2)$$

first on an interval of length 2π . We get the set of all solutions by adding numbers of the form $2\pi n$, $n \in \mathbb{Z}$, to each of the solutions obtained on that interval.

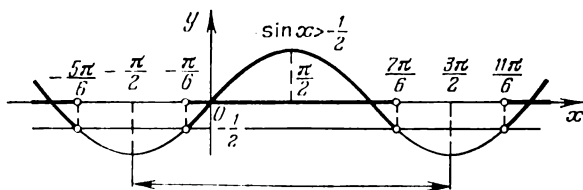


Fig. 3

It is convenient to solve inequality (1) first on the interval $[-\pi/2; 3\pi/2]$ (the graph of the function $\sin x$ forms a "hill" here, Fig. 2).

Example 1. Solve the inequality

$$\sin x > -1/2.$$

△ Let us solve this inequality on the interval $[-\pi/2; 3\pi/2]$. We consider its left-hand half, the subinterval $[-\pi/2; \pi/2]$ (Fig. 3). Here the equation $\sin x = -1/2$ has one solution $x = -\pi/6$, and the function $\sin x$ increases

monotonically. This means that if $-\pi/2 \leq x \leq -\pi/6$, then $\sin x \leq \sin(-\pi/6) = -1/2$, that is, these values of x are not solutions of the inequality. Now if $-\pi/6 < x \leq \pi/2$, then $\sin x > \sin(-\pi/6) = -1/2$. All these values of x are solutions of the inequality.

On the other half of the interval $[\pi/2; 3\pi/2]$ the function $\sin x$ decreases monotonically and the equation $\sin x = -1/2$ has one solution $x = 7\pi/6$. Consequently, if $\pi/2 \leq x < 7\pi/6$, then $\sin x > \sin(7\pi/6) = -1/2$, that is, all

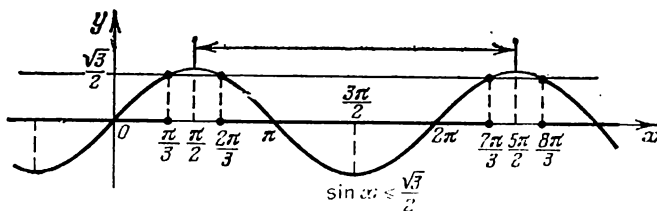


Fig. 4

these values of x are solutions of the inequality. For $x \in [7\pi/6; 3\pi/2]$ we have $\sin x \leq \sin(7\pi/6) = -1/2$. These values of x are not solutions of the inequality.

Thus we see that the set of all solutions of the given inequality on the interval $[-\pi/6; 3\pi/2]$ is the interval $(-\pi/6; 7\pi/6)$.

Because of the periodicity of the function $\sin x$ with period 2π the values of x belonging to any interval of the form

$$\left(-\frac{\pi}{6} + 2\pi n; \frac{7\pi}{6} + 2\pi n\right), \quad n \in \mathbb{Z},$$

are also solutions of the inequality. No other values of x can serve as solutions of this inequality. We write the answer in the form

$$-\frac{\pi}{6} + 2\pi n < x < \frac{7\pi}{6} + 2\pi n, \quad \text{where } n \in \mathbb{Z},$$

with an implication that those and only those values of x each of which satisfies these two inequalities for a certain $n \in \mathbb{Z}$ can serve as solutions to the original inequality. ▲

It is convenient to solve inequalities (2) first on the interval $[\pi/2; 5\pi/2]$ (here the graph of $\sin x$ forms a "hollow", Fig. 4). The figure illustrates graphically the solution of

the inequality

$$\sin x \leq \sqrt{3}/2$$

(the reader is invited to give detailed reasoning). Its solutions are those and only those values of x each of which satisfies the inequalities

$$\frac{2\pi}{3} + 2\pi n \leq x \leq \frac{7\pi}{3} + 2\pi n$$

for a certain $n \in \mathbb{Z}$.

For the inequalities

$$\cos x < a, \quad \cos x \leq a \quad (3)$$

it is more convenient to find solutions first on the interval $[0; 2\pi]$ (a "hollow", Fig. 5).

Example 2. Solve the inequality

$$\cos x \leq -1/2.$$

$\triangle$ On the interval $[0; \pi]$ the function $\cos x$ *decreases monotonically* (Fig. 5) and the equation $\cos x = -1/2$ has one

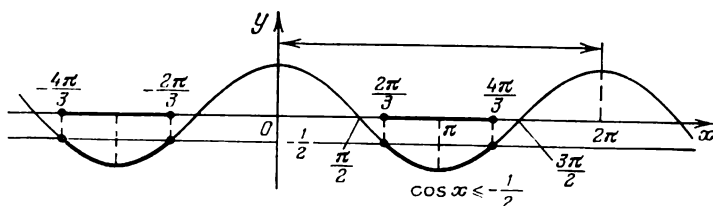


Fig. 5

solution $x = 2\pi/3$. Reasoning as in Example 1, we find that the values of x on the interval $[2\pi/3; \pi]$ and only they are solutions of the given inequality on the interval $[0; \pi]$. On the interval $[\pi; 2\pi]$ the function $\cos x$ *increases monotonically* and the equation $\cos x = -1/2$ has a solution $x = 4\pi/3$. Hence it follows that all the values of x belonging to the interval $[\pi; 4\pi/3]$ and only they are solutions of the given inequality in this case.

Thus, on the interval $[0; 2\pi]$ the set of solutions of the inequality $\cos x \leq -1/2$ is the interval $[2\pi/3; 4\pi/3]$. The function $\cos x$ is periodic with period 2π and, therefore,

all the values of x each of which satisfies the inequalities

$$\frac{2\pi}{3} + 2\pi n \leq x \leq \frac{4\pi}{3} + 2\pi n,$$

for a certain $n \in \mathbb{Z}$, and only those values are solutions of the original inequality. $\blacktriangle$

It is convenient to solve the inequalities

$$\cos x > a, \quad \cos x \geq a \quad (4)$$

first on the interval $[-\pi; \pi]$ (a "hill", Fig. 6). This figure

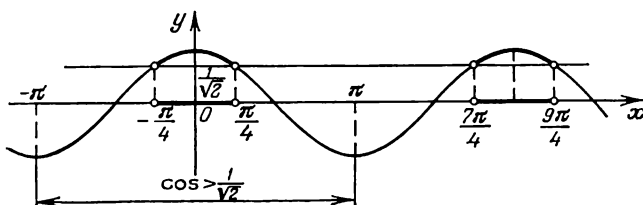


Fig. 6

illustrates graphically the solution of the inequality

$$\cos x > 1/\sqrt{2}$$

(we do not present detailed reasoning). All solutions of this inequality are given by inequalities of the form

$$-\frac{\pi}{4} + 2\pi n < x < \frac{\pi}{4} + 2\pi n, \quad \text{where } n \in \mathbb{Z}.$$

The inequalities

$$\tan x > a, \quad \tan x \geq a, \quad \tan x < a, \quad \tan x \leq a \quad (5)$$

can be most conveniently solved first on the interval $(-\pi/2; \pi/2)$, and the inequalities

$$\cot x > a, \quad \cot x \geq a, \quad \cot x < a, \quad \cot x \leq a, \quad (6)$$

on the interval $(0; \pi)$. The functions $\tan x$ and $\cot x$ have a period π . Therefore, adding the numbers πn , $n \in \mathbb{Z}$, to the solutions obtained on the corresponding intervals, we get all solutions of inequalities (5) and (6).

Example 3. Solve the inequality

$$\tan x < 2.$$

△ On the interval $(-\pi/2; \pi/2)$ the function $\tan x$ *increases monotonically* (Fig. 7) and the equation $\tan x = 2$ has one solution $x = \arctan 2$. If $-\pi/2 < x < \arctan 2$, then $\tan x < 2$ and this means that these values of x are solutions of the given inequality. If $\arctan 2 \leq x < \pi/2$, then $\tan x \geq 2$. These values of x cannot serve as solutions of

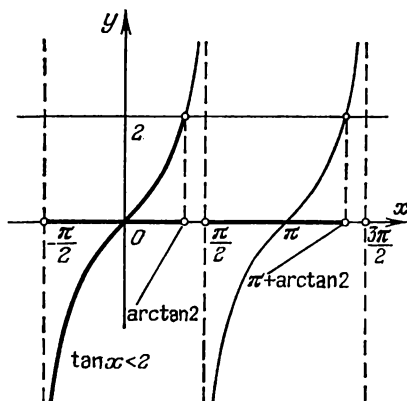


Fig. 7

the inequality. Consequently, the set of solutions of the inequality $\tan x < 2$ on the interval $(-\pi/2; \pi/2)$ is the interval $(-\pi/2; \arctan 2)$.

The solutions of the given inequality on the entire number line are all those and only those values of x each of which satisfies the inequalities

$$-\frac{\pi}{2} + \pi n < x < \arctan 2 + \pi n$$

for a certain $n \in \mathbb{Z}$. ▲

Figure 8 is a graphical illustration to the solution of the inequality

$$\cot x \leq -2.$$

Its solutions are those and only those values of x each of which satisfies the inequalities

$$\pi n < x \leq \operatorname{arccot}(-2) + \pi n$$

for a certain $n \in \mathbb{Z}$.

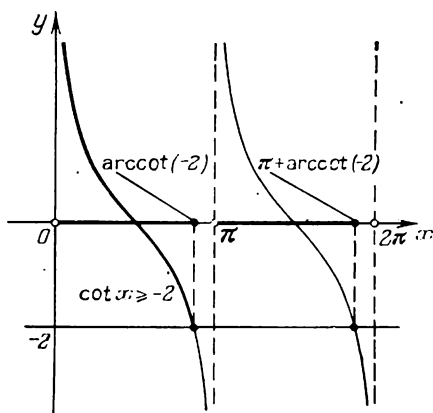


Fig. 8

Examples of solution of more complex inequalities. To reduce trigonometric inequalities to the canonical trigonometric inequalities use can be made, in requisite cases, of the same transformations (introduction of a new unknown, factorization, and the like) which were used for solving equations in 1.1.

Example 4. Solve the inequality

$$\sin\left(\frac{3}{2}x + \frac{\pi}{12}\right) < \frac{1}{\sqrt{2}}.$$

△ If we designate $t = \frac{3}{2}x + \frac{\pi}{12}$, the inequality will assume the form

$$\sin t < 1/\sqrt{2}.$$

On the interval $\pi/2 \leq t \leq 5\pi/2$ the set of solutions of this inequality is the interval $3\pi/4 < t < 9\pi/4$. The set of all solutions can be written in the form

$$\frac{3\pi}{4} + 2\pi n < t < \frac{9\pi}{4} + 2\pi n, \quad \text{where } n \in \mathbb{Z}. \quad (7)$$

Substituting $t = \frac{3}{2}x + \frac{\pi}{12}$ into (7), we obtain

$$\frac{3\pi}{4} + 2\pi n < \frac{3}{2}x + \frac{\pi}{12} < \frac{9\pi}{4} + 2\pi n,$$

whence we get $\frac{4}{9}\pi + \frac{4}{3}\pi n < x < \frac{13}{9}\pi + \frac{4}{3}\pi n$, where $n \in \mathbb{Z}$. All those and only those values of x each of which satisfies these inequalities for a certain $n \in \mathbb{Z}$ are solutions of the original inequality. $\blacktriangle$

Example 5. Solve the inequality

$$\cos 2x - \sin 2x \geq 0.$$

$\triangle$ We transform this inequality by introducing an auxiliary angle and obtain an inequality

$$\cos\left(2x + \frac{\pi}{4}\right) \geq 0,$$

which has the same solutions as the original inequality. It follows from the properties of a cosine that solutions of this inequality are those and only those values of x each of which satisfies the inequalities

$$-\frac{\pi}{2} + 2\pi n \leq 2x + \frac{\pi}{4} \leq \frac{\pi}{2} + 2\pi n,$$

for a certain $n \in \mathbb{Z}$, that is, $-\frac{3\pi}{8} + \pi n \leq x \leq \frac{\pi}{8} + \pi n$. $\blacktriangle$

Example 6. Solve the inequality

$$\sin x + \cos 2x > 1.$$

$\triangle$ Using the formula $\cos 2x = 1 - 2\sin^2 x$ and introducing the notation $t = \sin x$, we write the given inequality as

$$t(1 - 2t) > 0.$$

Hence we have $0 < t < 1/2$. Thus, those and only those values of x for which

$$0 < \sin x < 1/2$$

can serve as solutions to the original inequality. On the interval $[-\pi/2; 3\pi/2]$ the left-hand inequality has solutions (Fig. 9) $0 < x < \pi$. Among these values of x ,

$$0 < x < \pi/6, \quad 5\pi/6 < x < \pi$$

are solutions of the right-hand inequality of (8). And this is precisely the set of all solutions of (8) and, hence, of the original inequality on $[-\pi/2; 3\pi/2]$.

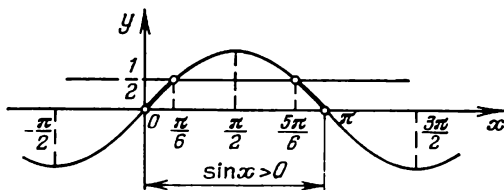


Fig. 9

Using now the periodicity of the function $\sin x$, we can easily find all solutions on the number line.

Answer: $2\pi n < x < \frac{\pi}{6} + 2\pi n$,

$$\frac{5\pi}{6} + 2\pi n < x < \pi + 2\pi n, \quad \text{where } n \in \mathbb{Z}.$$

PROBLEMS OF SECTION I

Solve the equations in problems 1 through 9:

1. $\cos 5x \cos x = \cos 6x$.
2. $2 \sin^2 2x + \sin^2 4x = 5/4$.
3. $\sin 5x = \sin x + \sin 2x$.
4. $4 \cos x \cos 2x = \cos 3x$.
5. $(1 + \cos x) \cot x = \sin 2x$.
6. $2(\sqrt{2} - 1) \sin x + 1 - \tan x = 1/\cos x$.
7. $\tan x + \cot 2x = 2 \cot 4x$.
8. $\sqrt{1 + \sin x} + \cos x = 0$.
9. $|\cos x - 2 \sin 2x - \cos 3x| = 1 - 2 \sin x - \cos 2x$.
10. Prove that $\arccos a + \arccos(-a) = \pi$, $|a| \leq 1$.
11. Find all the values of a ($a \neq 0$) for which the equation

$$\int_0^x (t^2 - 8t + 13) dt = x \sin \frac{a}{x}$$

has a solution. Find that solution.

Solve the systems of equations given in problems 12-14:

12. $\begin{cases} \tan x = \tan^3 y, \\ \sin x = \cos 2y. \end{cases}$
13. $\begin{cases} 2 \sin x \sin y + \cos 2y - \sqrt{2} = 0, \\ \cos 2x + 2 \cos x \cos y + \sqrt{2} = 0. \end{cases}$
14. $\begin{cases} \tan\left(x + \frac{\pi}{4}\right) = \cos 2y, \\ \tan\left(y + \frac{\pi}{4}\right) = \cos 2x. \end{cases}$

Solve the inequalities given in problems 15-17:

15. $\sin x \geq \cos 2x$. 16. $2 \tan 2x \leq 3 \tan x$.
 17. $\sin x < |\cos x|$.

PROBLEMS OF SECTION II

Solve the equations given in problems 1 through 17:

1. $\cos 6x + 2 \cos^2 x = 1$. 2. $\sin 2x = \cos^4 \frac{x}{2} - \sin^4 \frac{x}{2}$.
 3. $\tan^2 x \cdot \tan^2 3x = 1$. 4. $\sin x + \cos x = \sqrt{2} \cos 5x$.
 5. $\cos 2x + \sin 2x = \cos x + \sin x$.
 6. $\cos x \sin 7x = \cos 3x \sin 5x$.
 7. $\tan x = \frac{\cot 4x + 1}{\cot 4x - 1}$. 8. $\sqrt{2} \sin x + \cot x = 0$.
 9. $\cot^2 x - \tan^2 x = \frac{3}{2}$. 10. $\sin^2 2x - \cos^2 x + \frac{3}{4} = 0$.
 11. $\sin^4 x + 5 \cos 2x + 4 = 0$. 12. $\cos 4x + 2 \cos^2 2x + 4 \sin 2x = 0$.
 13. $6 |\sin x| \cos x - 1 = \cos 4x$. 14. $1 - \cos 3x \cdot \cot x = \sin 3x$.
 15. $\cot \left(x + \frac{\pi}{4} \right) + \cos \left(x - \frac{\pi}{4} \right) = \frac{2}{3} \cos 2x$.
 16. $\sin \left(x + \frac{\pi}{3} \right) \cdot \sin \left(x - \frac{\pi}{3} \right) = \sin x$.
 17. $11 \cot x - 5 \tan x = \frac{16}{\sin x}$.

18. Find the intersection of the set $\{x | \cos 2x + 3 \cos x + 2 = 0\}$ and the set $\{x | |\sqrt{2}x - 5| < 1\}$.

Solve the equations in problems 19-22:

19. $\sin^4 x + \cos^4 x = \frac{5}{8}$. 20. $2 \cos^6 x - 2 \sin^6 x = \sin 4x$.
 21. $\tan \left(x + \frac{\pi}{4} \right) - \tan x = 1$.
 22. $\tan \left(\frac{\pi}{12} + x \right) \cot \left(\frac{\pi}{4} - x \right) = 2 - \sqrt{3}$.
 23. Find all the values of p for which the equation

$$\sin x + p \cos x = 2p$$

has solutions.

Solve the equations given in problems 24 and 25:

24. $2 \sin x + \cos^2 \frac{x}{2} + 2 + \cos x = 0$. 25. $\frac{\cos 2x}{1 + \sin 2x} = -1$.
 26. Calculate $\sin 2\alpha$ if $\sin \alpha - \cos \alpha = 1/2$.
 Solve the equations given in problems 27 and 28:
 27. $\frac{1}{\sin x} + \frac{1}{\cos x} = 2\sqrt{2}$. 28. $3 \cos \left(x + \frac{\pi}{4} \right) = \sqrt{2} \sin 2x$.

29. Prove that

$$\tan 3x = \tan x \cdot \tan \left(\frac{\pi}{3} - x \right) \tan \left(\frac{\pi}{3} + x \right)$$

for all $x \neq \frac{\pi}{6} + \frac{\pi n}{3}$, $n \in \mathbb{Z}$.

Solve the equations given in problems 30 through 53:

30. $\sin \left(\frac{\pi}{2} + x \right) + \cot (2\pi - x) = 0.$

31. $\sin 2x = \sin x \cdot \sin \left(\frac{\pi}{6} - x \right).$

32. $\sin 4x + 2 \cos^2 x = 1.$ 33. $\sin^2 2x - 4 \cos^4 x = \sin 4x.$

34. $\cos 2x = \frac{\sqrt{6}}{2} (\cos x - \sin x).$

35. $\cot x - \tan x = \frac{1}{2} (\sin x - \cos x).$

36. $\frac{\sin 4x}{\sin \left(x - \frac{\pi}{4} \right)} = 2 \sin \left(x + \frac{\pi}{4} \right).$

37. $\frac{\cos^2 2x}{\cos x + \cos (\pi/4)} = \cos x - \cos (\pi/4).$

38. $\frac{\cos 2x}{1 + \sin 2x} = a (1 - \sin 2x).$

39. $\frac{1}{\cos x} + \frac{1}{\cos \left(x - \frac{3\pi}{2} \right)} = 4 \cos \left(\frac{7\pi}{4} - x \right).$

40. $3 (\cos x - \sin x) = 1 + \cos 2x - \sin 2x.$

41. $\cos 5x - \cos 3x = 3 \sin 4x.$

42. $\sin 3x - \sin x + \cos 2x = 1.$

43. $\sin 3x + \sin 4x + \sin 5x = 0.$

44. $4 \sin 3x + \sin 5x - 2 \sin x \cdot \cos 2x = 0.$

45. $\sin x \cdot \cot 3x = \cos 5x.$ 46. $1 + \sec x = \cot^2 \frac{x}{2}.$

47. $\sin x \cdot \sin 2x \cdot \sin 3x = \frac{1}{4} \sin 6x.$

48. $\frac{\cos x - 1}{\sin 2x} = 2 \sin x.$ 49. $\cos 4x - 3 \cos x = 4 \sin^2 \frac{x}{2}.$

50. $(\sin x + \sqrt{3} \cos x) \sin 3x = 2.$

51. $\sin x - \sin 2x + \sin 5x + \sin 8x = 0.$

52. $\tan x \cdot \tan (1/x) = 1.$ 53. $\tan \frac{2\pi x}{x^2 + x + 1} = -\sqrt{3}.$

54. Find all solutions of the equation $\sin^2 x + \sin^2 2x + \sin^2 3x = 3/2$ satisfying the inequality $\cos x > 1/\sqrt{2}.$

Solve the equations given in problems 55 through 65:

$$55. \frac{\sin \left(2x - \frac{\pi}{6} \right)}{\sin \left(x - \frac{\pi}{3} \right)} = 2 \sin x - \frac{1}{\sin x}.$$

$$56. \frac{1 - \tan x}{1 + \tan x} = \sin 2x - 1. \quad 57. \tan 3x - \tan x = 4 \sin x.$$

$$58. \sin^2 2x - \tan^2 x = \frac{9}{2} \cos 2x.$$

$$59. \sin^3 x + \cos^3 x = \sin 2x + \sin x + \cos x.$$

$$60. 2 \sin x = \sin 3x. \quad 61. \cos 3x + 6 \cos^2 x = \cos x.$$

$$62. 2 \cos x - \cos \frac{3}{2} x = 1.$$

$$63. \sin 3x + \sin 5x = 2 (\cos^2 2x - \sin^2 3x).$$

$$64. 5 \sin x + 6 \sin 2x + 5 \sin 3x + \sin 4x = 0.$$

$$65. \sin^2 2x + \sin^2 3x + \sin^2 4x + \sin^2 5x = 2.$$

66. Verify which of the numbers of the set A serve as solutions of the given equation:

$$(a) A = \left\{ \arccos \frac{1}{\sqrt{3}} + \frac{\pi n}{2} \mid n \in \mathbb{Z} \right\}, \quad \frac{\sqrt{3}}{\sin x} + \frac{1}{\cos 2x} = 6;$$

$$(b) A = \{ \pi n - \arctan 3 \mid n \in \mathbb{Z} \}, \quad 12 \tan 2x + \frac{\sqrt{10}}{\cos x} + 1 = 0.$$

Solve the equations given in problems 67 through 76:

$$67. 1 + \tan 2x = \frac{1 - \sin 2x}{\cos^2 2x}. \quad 68. \tan x + 2 \cot 2x = \cos x + \sin 2x.$$

$$69. \frac{1 - \cot 3x}{1 - \tan 3x} = 4 \cos^2 \frac{3}{2} x - 2. \quad 70. 2 \tan 2x + \tan 3x = \tan 5x.$$

$$71. \frac{1}{\sin x} = \frac{1}{\sin 2x} + \frac{1}{\sin 3x}. \quad 72. \frac{\sin 3x}{\cos 2x} + \frac{\cos 3x}{\sin 2x} = \frac{2}{\sin 3x}.$$

$$73. \tan 2x - \frac{1}{\sin 3x} = \frac{1}{\sin 4x} - \cot 3x.$$

$$74. \frac{\cos x + \cos 2x - \cos 3x - \cos 4x}{\sin 4x + \sin 3x - \sin 2x - \sin x} = \tan \frac{x}{2}$$

$$75. 2 \cos x - \sin 3x = \frac{\cos x \cdot \cos 3x}{|\sin x|}.$$

$$76. \sin x - 2 \sin 2x + \sin 3x = |1 - 2 \cos x + \cos 2x|.$$

77. For what values of a does the equation

$$a \sin \left(x + \frac{\pi}{4} \right) = \sin 2x + 9$$

possess a solution?

Solve the equations given in problems 78 through 86:

$$78. \sin^6 x + \cos^6 x = a \sin 4x.$$

79. $a \cos 3x + \sin 2x \cdot \sin x - \cos x = 0$.
 80. $\sin^3 2x + a \sin^2 2x + \sin^2 x = \sin^2 3x$.
 81. $\sqrt{2 \sin 2x} + 2 \sin x = 0$. 82. $\sqrt{1 + \sin x} = \cos x$.
 83. $\sqrt{\cos 2x - 5 \sin x} + 2 \cos x = 0$.
 84. $\sqrt[4]{\frac{1}{2} - \cos 2x} + \sqrt[4]{\frac{1}{2} + \cos 2x} = 1$.
 85. $\sqrt[3]{\sin^2 x} - \sqrt[3]{\cos^2 x} = \sqrt[3]{2 \cos 2x}$.
 86. $2 \cos x = \sqrt{2 + \sin 3x}$.
 87. For what values of a does the equation

$$\sqrt{\sin x} - \sqrt{\cos x} = a$$

possess solutions?

88. For what values of a does the equation

$$\sqrt{1 + \sin x} + \sqrt{1 + \cos x} = a$$

(a) possess solutions?

(b) possess a unique solution on the interval $[-\pi/3; \pi/3]$?

Solve the equations given in problems 89 through 97:

89. $\tan 7x = 2 \tan 5x + \tan^2 5x \cdot \tan 7x$.

90. $\cot\left(\frac{\pi}{3} + 3x\right) + \cot\left(\frac{4\pi}{9} + x\right) = 2 \sin\left(\frac{\pi}{9} - 2x\right)$.

91. $\sin x + 2 \cos x = \cos 2x - \sin 2x$.

92. $\cos 3x - \cos 2x = \sin 3x$.

93. $\sin x - \sin \frac{3}{2}x = \frac{4}{3} \sin^2 \frac{x}{2} \cdot \sin x$.

94. $\cos^{12} x + \sin^6 x (1 + \cos^2 x)^3 = 1$.

95. $\cos^2 x - \sin^2 (x\sqrt{3}) = 1$. 96. $\cos^5 2\pi x - 2 \sin^3 \pi x = 3$.

97. $2 \sin^2 x \cdot \cos^2 4x = \sin^2 x + \cos^2 4x$.

Solve the systems of equations given in problems 98 and 99:

98. $\begin{cases} \tan x + 3 \tan y = 0, \\ 4x + 2y = 5\pi. \end{cases}$ 99. $\begin{cases} \cos x \cdot \cos y = 3 \sin x \sin y, \\ x + 2y = \frac{3}{2}\pi. \end{cases}$

100. For what values of a does the system

$$\begin{cases} 8 \cos x \cdot \cos y \cdot \cos (x - y) + 1 = 0, \\ x + y = a \end{cases}$$

possess solutions? Find those solutions.

Solve the systems of equations given in problems 101 through 124:

101. $\begin{cases} \sin x + \sin y = \sin (x + y), \\ |x| + |y| = 1. \end{cases}$

102. $\begin{cases} \tan x + \tan y = 4, \\ \cot x + \cot y = 5. \end{cases}$

103. $\begin{cases} \cos x + \cos y = 1, \\ \cos \frac{x}{2} + \cos \frac{y}{2} = \frac{\sqrt{2}}{2} - 1. \end{cases}$

$$104. \begin{cases} \cos x - \sin x = 1 + \cos y - \sin y, \\ 3 \sin 2x - 2 \sin 2y = \frac{3}{4}. \end{cases}$$

$$105. \begin{cases} \cos x \cdot \sin y = 1/2, \\ \sin 2x + \sin 2y = 0. \end{cases}$$

$$106. \begin{cases} \tan x \cdot \tan 2y = 1, \\ \sqrt{3} \sin 2x - 3 \cos 2y = 0. \end{cases}$$

$$107. \begin{cases} \sin x + \sin y = 2 \sin (x + y), \\ \cos \frac{x}{2} \cdot \cos \frac{y}{2} = \frac{3}{4}. \end{cases}$$

$$108. \begin{cases} \tan 2(x + y) = \tan 2x + \tan 2y, \\ 2 \sin (2x + y) \cos y + \cos 2y + 1 = 0. \end{cases}$$

$$109. \begin{cases} 2 \sin x \cdot \cos y = 2 \cot x + \cot y, \\ 2 \sin y \cdot \cos x = \cot x + 2 \cot y. \end{cases}$$

$$110. \begin{cases} \sin 5x = \sin (x + y) - \sin (x - y), \\ \sin 4x = \sin y - \sin 6x. \end{cases}$$

$$111. \begin{cases} 4 \sin x - 2 \sin y = 3, \\ 2 \cos x - \cos y = 0. \end{cases}$$

$$112. \begin{cases} \sin 2x \sin y + \sin x \cos y = 1/\sqrt{2}, \\ \cos 2x \sin y + \cos x \cos y = 1/\sqrt{2}. \end{cases}$$

$$113. \begin{cases} \cot x + \tan x \cos 2y + \cos y = 0, \\ \cot x - \tan x \sin 2y - \sin y = 0. \end{cases}$$

$$114. \begin{cases} 2 \sin x \sin y = \cos 2x + \cos 2y, \\ 2 \cos x \sin y = \cos 2x - \cos 2y. \end{cases}$$

$$115. \begin{cases} \cos^2 x = 2 \sin^2 y + \cos^2 (2y - x), \\ 2 \sin^2 x = \cos 2y. \end{cases}$$

$$116. \begin{cases} \sin (4x - 2y) + \sqrt{2} \sin (3x - y) = 0, \\ \sqrt{2} \sin (2x - y) + \sin (3x - 2y) = 0. \end{cases}$$

$$117. \begin{cases} \sqrt{1 + \sin x \cdot \sin y} = \cos x, \\ 2 \sin x \cdot \cot y + 1 = 0. \end{cases}$$

$$118. \begin{cases} \sin (x + y) - \cos (x + y) = \cos (x - y), \\ \sin y - \cos y = \cos (2x + y). \end{cases}$$

$$119. \begin{cases} 2 \sin^2 y + \sin 2y = \cos (x + y), \\ \cos^2 x + 2 \sin 2y + \sin^2 y = \cos (x - y). \end{cases}$$

$$120. \begin{cases} 3 \tan (y/2) + 6 \sin x = 2 \sin (y - x), \\ \tan (y/2) - 2 \sin x = 6 \sin (y + x). \end{cases}$$

$$121. \begin{cases} \sin x + \cos x = 2 \sin y, \\ \sin 2x = \sqrt{3} \sin 2y - 3 \cos 2y. \end{cases}$$

$$122. \begin{cases} \cos x = \cos y \cos z, \\ \cos y = \cos x \cos z + \frac{1}{\sqrt{3}} \sin x \sin z, \\ \cos z = \cos x \cos y + \frac{1}{\sqrt{3}} \sin x \sin y. \end{cases}$$

$$123. \begin{cases} \sin x + \sin z = \sqrt{2} \cos y, \\ \cos x + \cos z = \sqrt{2} \sin y, \\ \cos 2y + \cos 2z = \sin 2x. \end{cases}$$

$$124. \begin{cases} \tan x \cdot \tan y = \frac{\sin z}{\cos x \cos y} + 3, \\ \tan y \cdot \tan z = \frac{\sin x}{\cos y \cos z} - 5, \\ \tan z \cdot \tan x = \frac{\sin y}{\cos z \cos x} - 3. \end{cases}$$

Solve the inequalities given in problems 125 through 138:

$$125. 2 \cos^2 x > 3/2. \quad 126. \sin x \geq \cos x.$$

$$127. \sin x + 2 \cos x < 2.$$

$$128. \cos 2x + 3 \sin x > -1.$$

$$129. \sin x + \sin 3x \geq 0. \quad 130. \tan x \geq 2 \cot x.$$

$$131. \tan \pi x + \tan 2\pi x \leq 0.$$

$$132. \sin 2x + \sin x - \sqrt{2} \cos x < 1/\sqrt{2}.$$

$$133. \sin x + \sin 3x < \sin 5x + \sin 7x.$$

$$134. |\sin x| > |\cos x|.$$

$$135. 1 + \cos 2x \geq \cos x (1 + |1 - 2 \cos x|).$$

$$136. \sqrt{3 + 2 \tan x - \tan^2 x} \leq \frac{1 + 3 \tan x}{2}.$$

$$137. \sqrt{\frac{1}{2} - \cos 2x} \geq \sin x - \cos x.$$

$$138. \sqrt{\sin x} + \sqrt{\cos x} < \sqrt{2}.$$

139. Prove the identities:

$$(a) (\sin(\gamma + \beta) - \sin \alpha)(\sin(\gamma - \beta) + \sin \alpha) \\ = (\sin(\gamma + \alpha) - \sin \beta)(\sin(\gamma - \alpha) + \sin \beta);$$

$$(b) \cos \alpha + \sin \alpha + \cos 3\alpha + \sin 3\alpha = 2\sqrt{2} \cos \alpha \sin\left(\frac{\pi}{4} + 2\alpha\right).$$

140. Simplify the expression

$$\sin^2 \alpha + \sin^2 \beta + 2 \sin \alpha \sin \beta \cos(\alpha + \beta).$$

141. Prove the following equalities:

$$(a) \tan\left(\alpha + \frac{\pi}{4}\right) + \tan\left(\alpha - \frac{\pi}{4}\right) = 2 \tan 2\alpha.$$

$$(b) \frac{1 + \cos(2\alpha + 630^\circ) + \sin(2\alpha + 810^\circ)}{1 - \cos(2\alpha - 630^\circ) + \sin(2\alpha - 810^\circ)} = \cot \alpha.$$

142. Prove

$$\tan(\alpha + \beta) = 4 \tan \beta,$$

if $5 \sin \alpha = 3 \sin(\alpha + 2\beta) \neq 0$.

143. Prove the following equalities:

(a) $\cot 7.5^\circ + \tan 67.5^\circ - \cot 67.5^\circ - \tan 7.5^\circ = 2(3 + \sqrt{3}).$

(b) $\frac{1}{\sin \frac{\pi}{7}} = \frac{1}{\sin \frac{2\pi}{7}} + \frac{1}{\sin \frac{3\pi}{7}}.$

144. Calculate

(a) $\cos 3\alpha \cos^3 \alpha + \sin 3\alpha \sin^3 \alpha$, if $\cos 2\alpha = a$,

(b) $(\sin 4\alpha + 2 \sin 2\alpha) \cos \alpha$, if $\sin \alpha = 1/4$.

145. Express $\frac{\sin^8 \varphi}{a^3} + \frac{\cos^8 \varphi}{b^3}$, in terms of a and b if

$$\frac{\sin^4 \varphi}{a} + \frac{\cos^4 \varphi}{b} = \frac{1}{a+b}.$$

146. Find

(a) $\sin(\arccos(-3/5))$, (b) $\cos(2 \arcsin(3/4))$.

(c) $\tan(\arcsin(2/3))$, (d) $\cos(\operatorname{arccot}(-2))$,

(e) $\cos(2 \arctan 2) - \sin(4 \arctan 3)$.

147. Prove that

(a) $\arcsin x + \arccos x = \pi/2$, $x \in [-1; 1]$,

(b) $\arctan x + \operatorname{arccot} x = \pi/2$, $x \in \mathbb{R}$,

(c) $\arctan x = \operatorname{arccot}(1/x)$, $x > 0$,
 $\arctan x = \operatorname{arccot}(1/x) - \pi$, $x < 0$.

(d) $\arcsin x = \arccos \sqrt{1-x^2}$, $0 \leq x \leq 1$,

$\arcsin x = -\arccos \sqrt{1-x^2}$, $-1 \leq x \leq 0$.

(e) $\arccos x = \arcsin \sqrt{1-x^2}$, $0 \leq x \leq 1$.

$\arccos x = \pi - \arcsin \sqrt{1-x^2}$, $-1 \leq x \leq 0$.

148. Prove that the function

$$y = 2 \arctan x + \arcsin \frac{2x}{1+x^2}$$

assumes the same value for all $x \geq 1$. Find that value.

149. Find all values of a for which the equations

$$4 \cos^2 x - \cos 3x = a \cos x - |a - 4| (1 + \cos 2x)$$

and

$$2 \cos x \cos 2x = 1 + \cos 2x + \cos 3x$$

are equivalent.

Solve the equations given in problems 150-154:

150. $\cot\left(\frac{\pi}{3} \cos 2\pi x\right) = \sqrt{3}.$

151. $2 \sin^2\left(\frac{\pi}{2} \cos^2 x\right) = 1 - \cos(\pi \sin 2x).$

152. $2|x-6|\cos x = x-6$.

153. $\cos \pi x = \left| x + \frac{1}{4} \right| - \left| x - \frac{1}{4} \right|$.

154. $-2\sqrt{3}\pi \sin x = |x + \pi| + |x - 2\pi|$.

155. Prove that for any α and any natural $n > 1$

$$\cos \alpha + \cos \left(\alpha + \frac{2\pi}{n} \right) + \dots + \cos \left(\alpha + (n-1) \frac{2\pi}{n} \right) = 0.$$

156. Find the sum $\tan^2 \frac{\pi}{16} + \tan^2 \frac{2\pi}{16} + \dots + \tan^2 \frac{7\pi}{16}$.

157. Find the common solutions of the equations

$$\sin 5\pi x + \cos 5\pi x + 2 \sin \frac{5\pi}{4x} = 0$$

and

$$\sqrt{3} \sin \frac{5\pi}{2x} + 3 \cos \frac{5\pi}{2x} + \sin 10\pi x = 0.$$

158. For what values of a does the equation

$$\int_0^x \sin^2 \frac{u}{2} du = a^2 x^2 - \frac{3x-1}{2} + \frac{1}{a^2}$$

possess a solution? Find that solution.

159. Find the least value of the function

$$y = 4x + \frac{9\pi^2}{x} + \sin x$$

for $x > 0$.

160. Find all values of α for which the quadratic function $x^2 \cos \alpha + 2x \sin \alpha + \frac{1}{2}(\cos \alpha - \sin \alpha)$ is the square of a linear function.

161. Find all values of α for which the quadratic trinomial $x^2 \sin 2\alpha - (\sin^2 2\alpha - 4 \cos^6 \alpha)x + \frac{1}{\cos \alpha}$ possesses two roots equal in absolute value but different in signs.

162. Find all solutions of the equation

$$\int_0^\alpha \cos(x + \alpha^2) dx = \sin \alpha$$

belonging to the interval $[2; 3]$.

Solve the equations given in problems 163-165:

163. $\arccos x = 2 \arcsin x$.

164. $\arctan^2 x + \operatorname{arccot}^2 x = 5\pi^2/8$.

165. $\arctan(2 \tan 2x - 6 \tan x) = \frac{\pi}{4} + x$.

166. For what values of a does the equation

$$(\arcsin x)^3 + (\arccos x)^3 = a$$

possess a unique solution?

167. Find all pairs of numbers $(x; y)$ such that

(a) $\sin(x - y) + \sin y - \sin x = 4 \sin(y/2).$

(b) $\cos^4 x + \cos^4 y - \frac{1}{\cos^2 x \cos^2 y} = 1.$

(c) $\cos x + \cos y - \cos(x + y) = 3/2.$

(d) $\begin{cases} \sin^2 x + \sin^2 y \geq 3/2, \\ \cos x \cdot \cos y = 1/4. \end{cases}$

168. Solve the system of equations

$$\begin{cases} 2^{-x} + 2^x \cos 2y + \cos y = 0, \\ 2^{-x} - 2^x \sin 2y - \sin y = 0. \end{cases}$$

169. Prove the inequality

$$\cos x \geq 1 - \frac{x^2}{2}.$$

170. For what values of a does the inequality

$$\tan^2(\cos \sqrt{4\pi^2 - x^2}) - 4a \tan(\cos \sqrt{4\pi^2 - x^2}) + 2 + 2a \leq 0$$

possess solutions and the number of those solutions is finite? Find them.

Chapter 2

Exponential and Logarithmic Equations, Systems and Inequalities

We shall first recall some properties of the exponential and the logarithmic function often used in solving equations, systems of equations, and inequalities containing those functions.

The *exponential function* a^x is considered for any positive base a ($a > 0$). Its domain of definition is the set of all real numbers ($x \in \mathbb{R}$), its range being the set of all positive real numbers ($a^x > 0$). The following statements are true for any positive a and any real x and y :

1. $a^x a^y = a^{x+y}$.
2. $\frac{a^x}{a^y} = a^{x-y}$.
3. $a^{-x} = \frac{1}{a^x}$.
4. $(a^x)^y = a^{xy}$.
5. $a^0 = 1$.
6. $a^1 = a$.
7. $1^x = 1$.

8. If $a > 1$, then the function a^x increases; if $0 < a < 1$, then the function a^x decreases.

The *logarithmic function* $\log_a x$ is considered for any positive base a not equal to unity ($a > 0$, $a \neq 1$). Its domain of definition is the set of all positive real numbers ($x > 0$), its range being the set of all real numbers ($-\infty < \log_a x < +\infty$). The following equalities hold for any positive a , not equal to unity, and any positive x and y :

- 1'. $a^{\log_a x} = x$ (basic logarithmic identity).
- 2'. $\log_a a^x = x$.
- 3'. $\log_a xy = \log_a x + \log_a y$ (formula for the logarithm of the product).
- 4'. $\log_a (1/x) = -\log_a x$.
- 5'. $\log_a (x/y) = \log_a x - \log_a y$ (formula for the logarithm of the quotient).

6'. $\log_a x^\alpha = \alpha \log_a x$ for any α (formula for the logarithm of the power).

7'. $\log_a \alpha x = \frac{1}{\alpha} \log_a x$ for any $\alpha \neq 0$.

8'. $\log_b x = \frac{\log_a x}{\log_a b}$ for any $b > 0$, $b \neq 1$ (formula for a transition to a new base).

9'. $\log_a a = 1$. 10'. $\log_a 1 = 0$.

11'. If $a > 1$, then the function $\log_a x$ increases and if $0 < a < 1$, then the function $\log_a x$ decreases.

The functions a^x and $\log_a x$ are mutually inverse, which fact is expressed by equalities 1' and 2'.

Remark. An important peculiarity of formulas 1', 3', 5', and 6' must be pointed out. Their right-hand and left-hand sides, taken separately, are defined on *different* sets of values of the variables (x or x and y respectively). In formula 1' the left-hand side is defined only for $x > 0$, and the right-hand side is defined for all $x \in \mathbb{R}$. In formulas 3' and 5', the left-hand sides are defined for all pairs of values of x and y of like sign (i.e. for $xy > 0$), and the right-hand sides are defined only for $x > 0$, $y > 0$. In formula 6', for $\alpha = 2n$, where n is a nonzero integer, the left-hand side is defined for all $x \neq 0$ and the right-hand side is defined only for $x > 0$.

In using these formulas to transform equations and inequalities, the indicated difference of the sets of definition must be taken into account. It may lead to a loss of solutions or to an appearance of extraneous values of the unknowns. We shall draw your attention to this fact when considering respective examples. Note that we have encountered circumstances of this kind before (see, for instance, Ch. 1).

It follows from the properties of the exponential function that for any $a > 0$, $a \neq 1$, the equation

$$a^x = b$$

possesses a solution for any $b > 0$, and that solution is unique. In a general case, the solution is written as $x = \log_a b$. If $a = 1$, then the equation $1^x = b$ has¹ a solution only for $b = 1$. Any real number can serve as its solution.

It follows from the properties of the logarithmic function that for any $a > 0$, $a \neq 1$, the equation

$$\log_a x = b$$

has a solution for any $b \in \mathbb{R}$, and that solution is unique. In a general case, its solution is written as $x = a^b$.

When solving exponential and logarithmic equations, use is often made of two transformations, taking logarithms and taking antilogarithms.

Taking logarithms to the base $c > 0$, $c \neq 1$, is a transition from the equality

$$a = b \tag{1}$$

to the equality

$$\log_c a = \log_c b \tag{2}$$

(a and b here can designate both the numbers and the expressions containing variables). If (1) is a true equality and its both sides are *positive* ($a > 0$, $b > 0$), then (2) is a true equality as well.

Taking antilogarithms to the base $c > 0$, $c \neq 1$, is a transition from equality (2) to equality (1). If (2) is a true equality, then (1) is a true equality as well.

2.1. Exponential Equations

Example 1. Solve the equation

$$4 \cdot 9^{x-1} = 3 \sqrt{2^{2x+1}}.$$

$\triangle$ Both sides of the equation are positive. Taking logarithms to the base 2, we get an equation

$$2 + (x-1) \log_2 9 = \log_2 3 + \frac{1}{2} (2x+1),$$

which, evidently, has the same solutions as the original equation, that is, is equivalent to it. Transforming this equation and taking into account that $\log_2 3 = \frac{1}{2} \log_2 9$, we obtain

$$x(\log_2 9 - 1) = \frac{3}{2} (\log_2 9 - 1),$$

whence we have $x = 3/2$ (since $\log_2 9 - 1 \neq 0$).

Answer: 1.5. ▲

Just like in the above example, we can take logarithms of the equation of the form

$$a^{f(x)} = b^{g(x)}$$

to a certain base $c > 0$, $c \neq 1$ and thus reduce it to the equivalent equation

$$f(x) \log_c a = g(x) \log_c b.$$

By introducing a new variable, it proves to be possible to reduce some exponential equations to algebraic equations, say, a quadratic equation, and so on.

Example 2. Solve the equation

$$5^{x-1} + 5 \cdot 0.2^{x-2} = 26.$$

△ Since $0.2 = 5^{-1}$, it follows that $0.2^{x-2} = 5^{2-x}$ and, therefore, the equation can be written in the form $5^{x-1} + 5^{3-x} = 26$. Designating $5^{x-1} = t$, we have

$$t + \frac{25}{t} = 26,$$

whence we get $t^2 - 26t + 25 = 0$. The solutions of the quadratic equation obtained are $t_1 = 25$ and $t_2 = 1$. Hence $5^{x-1} = 25$, $x = 3$; $5^{x-1} = 1$, $x = 1$. The values of x we have found are precisely the solutions of the original equation.

Answer: $\{3; 1\}$. ▲

Example 3. Solve the equation

$$25^x - 12 \cdot 2^x - 6.25 \cdot 0.16^x = 0.$$

△ We write the equation as

$$5^{2x} - 12 \cdot 2^x - 6.25 \cdot 0.4^{2x} = 0.$$

Noticing that $2 = 5 \cdot 0.4$ and, consequently, $2^x = 5^x \cdot 0.4^x$, we arrive at an equation

$$5^{2x} - 12 \cdot 5^x \cdot 0.4^x - 6.25 \cdot 0.4^{2x} = 0,$$

which is a homogeneous equation of degree 2 with respect to 5^x and 0.4^x . Dividing it by 0.4^x and introducing the notation $t = (5/0.4)^x = 12.5^x$, we get a quadratic equation $t^2 - 12t - 6.25 = 0$. Its roots are $t_1 = -0.5$ and $t_2 = 12.5$. The equation $12.5^x = -0.5$ has no solutions, the equation $12.5^x = 12.5$ has a solution $x = 1$, which is also a solution to the original equation. ▲

It is sometimes possible to solve certain exponential equations proceeding from the properties of the increase and the decrease of an exponential function.

Example 4. Solve the equation

$$4^x + 9^x = 25^x.$$

△ It is easy to guess and to verify that $x = 0.5$ is a solution of the given equation.

The question is whether this solution is unique or not. Until it is shown that there are no other solutions or until all other solutions (providing they exist) are found, the problem cannot be considered as solved.

The answer to the question we have put follows from the solution of the following problem.

Example 5. Suppose $0 < a < 1$, $0 < b < 1$ and let the equation

$$a^x + b^x = 1$$

have a solution x_0 . Prove that the solution is unique.

△ An exponential function with a positive base less than unity decreases and, therefore, if $x < x_0$, then $a^x > a^{x_0}$ and $b^x > b^{x_0}$. Hence it follows that $a^x + b^x > a^{x_0} + b^{x_0} = 1$. This means that for $x < x_0$ the given equation has no solutions. If $x > x_0$, then $a^x < a^{x_0}$ and $b^x < b^{x_0}$, whence $a^x + b^x < a^{x_0} + b^{x_0} = 1$. Thus, for $x > x_0$ the given equation has no solutions either. We have thus proved that x_0 is a unique solution. ▲

Let us return to Example 4. Dividing both sides of the given equation by 25^x , we write it in the form

$$(4/25)^x + (9/25)^x = 1.$$

This equation has a solution $x_0 = 0.5$, and there are no other solutions according to what was proved in Example 5. We have now completed the proof of Example 4.

Answer: 0.5. ▲

2.2. Logarithmic Equations

Example 1. Solve the equation

$$\log_2 x (x - 1) = 1. \quad (1)$$

△ Taking the antilogarithms to the base 2, we get

$$x (x - 1) = 2. \quad (2)$$

This equation has the solutions $x_1 = 2$, $x_2 = -1$. Substituting them into (1), we make sure that they serve as solutions of this equation as well.

Answer: $\{2; -1\}$. ▲

As follows from the solution, equations (1) and (2) are equivalent. In a general case, as well, an equation of the form

$$\log_a f(x) = g(x) \quad (3)$$

is equivalent to the equation

$$f(x) = a^{g(x)}. \quad (4)$$

In fact, if x is a solution of (3), then we can take antilogarithms of the true equality (3) and find that (4) is also a true equality. On the other hand, if x is a solution to (4), then we can take logarithms of the true equality (4) (its both sides are positive) and find that (3) is a true equality as well.

Example 2. Solve the equation

$$\log_3 (3^x - 8) = 2 - x.$$

△ Taking antilogarithms to the base 3, we get an equation

$$3^x - 8 = 3^{2-x},$$

whence it follows that $3^{2x} - 8 \cdot 3^x - 9 = 0$. This quadratic equation with respect to 3^x has the roots 9 and -1 . The equation $3^x = -1$ has no solutions and the equation $3^x = 9$ has a solution $x = 2$ which is also a solution of the original equation.

Answer: 2. ▲

Example 3. Solve the equation

$$\log (x^2 - 6x + 7) = \log (x - 3).$$

△ Taking antilogarithms to the base 10, we arrive at an equation

$$x^2 - 6x + 7 = x - 3,$$

whence $x^2 - 7x + 10 = 0$, $x_1 = 5$, $x_2 = 2$. The substitution $x_1 = 5$ in the original equation leads to the true equality $\log 2 = \log 2$, hence $x_1 = 5$ is a solution of the original equation. The substitution $x_2 = 2$ in the equation leads to the appearance of an expression $\log (-1)$, which is not

defined. Hence $x_2 = 2$ cannot be a solution of the original equation.

Answer: 5. ▲

The equation we have considered is an example of an equation of the form

$$\log_a f(x) = \log_a g(x) \quad (5)$$

which is not, in general, equivalent to the equation

$$f(x) = g(x). \quad (6)$$

Solutions of (5) are, evidently, all those and only those solutions of (6) for which $f(x) > 0$ (or, which is the same, $g(x) > 0$). In Example 3, out of the two solutions of the equation $x^2 - 6x + 7 = x - 3$, only one solution, namely, $x_1 = 5$, satisfies the condition $x - 3 > 0$. Only that value of x can be a solution of the original equation.

Example 4. Solve the equation

$$\log_2 x + \log_2 (x - 1) = 1.$$

△ Let us transform the sum of the logarithms to the logarithm of a product:

$$\log_2 x(x - 1) = 1.$$

This equation (see Example 1) has the solutions $x_1 = 2$, $x_2 = 1$. Substituting $x_1 = 2$ into the original equation, we get a true equality $1 = 1$, and, hence, $x_1 = 2$ is a solution of the original equation. Substituting $x_2 = -1$, we get an expression $\log_2(-1)$, which is not defined, already in the first term of the left-hand side. Hence, $x_2 = -1$ is not a solution of the original equation.

Answer: 2. ▲

As can be seen from this example, a transformation of a sum of logarithms to the logarithm of a product can lead to an equation which is not equivalent to the original equation, namely, not every solution of the obtained equation can serve as a solution to the original equation. This follows from the fact that, as was noted earlier (see the remark to formulas 1', 3', 5', and 6'), the logarithm of a product can be defined even when the logarithms of the factors are not defined. When we use the indicated transformation in solving an equation, we must verify all the values of the

obtained unknown by substituting them into the original equation. Or, else, we must find for which of those values the expressions appearing under the logarithm sign in the original equation are positive.

All this also refers to the case of transformation of the difference of logarithms into the logarithm of a quotient.

Example 5. Solve the equation

$$\log_3 (2 - x) - \log_3 (2 + x) - \log_3 x + 1 = 0.$$

△ We transform the given equation as follows:

$$\log_3 \frac{(2-x)3}{(2+x)x} = 0, \quad \frac{(2-x)3}{(2+x)x} = 1.$$

Hence we have $x^2 + 5x - 6 = 0$, $x_1 = 1$, $x_2 = -6$. For $x_1 = 1$, all the expressions appearing in the original equation under the logarithm sign are positive and this means that $x_1 = 1$ is a solution of that equation. For $x_2 = -6$, even $\log_3 (-6)$ is not defined and, therefore, $x_2 = -6$ is not a solution of the original equation.

Answer: 1. ▲

Example 6. Solve the equation

$$2 \log (2x) = \log (x^2 + 75). \quad (7)$$

△ Bearing in mind that $2 \log (2x) = \log (4x^2)$, we transform the given equation as follows:

$$\log (4x^2) = \log (x^2 + 75). \quad (8)$$

As can be easily established, this equation has solutions $x_1 = 5$, $x_2 = -5$. Pay attention to the fact that in equation (8) the expression $\log (4x^2)$ is defined for all $x \neq 0$ whereas in the original equation (7) the respective expression $2 \log (2x)$ is defined only for $x > 0$. Verification shows that out of the two solutions of (8) only $x_1 = 5$ is solution of (7).

Answer: 5. ■

This example shows that the substitution of $\log_a (f(x))^{2n}$ for $2n \log_a f(x)$ in the equation (n being nonzero integer) can also lead to the appearance of extraneous values of the unknown. As before, it is due to the fact that the expression being substituted has a "wider" domain of definition than the expression being replaced.

A replacement of one of the expressions appearing in the equation by another expression having a "narrower" domain of definition, can naturally lead to a loss of solutions. Examples of transformations of this kind are, for instance, a transformation of the logarithm of a product into a sum of logarithms, taking the exponent outside the logarithm sign. In such cases, one or another technique should be used to consider all the values of the variable some of which can serve as solutions of the equation. Say, when transforming $\log_a (f(x))^{2n}$, where n is a nonzero integer, we must consider two cases:

$$\begin{aligned} \text{if } f(x) > 0, \text{ then } \log_a (f(x))^{2n} &= 2n \log_a f(x), \\ \text{if } f(x) < 0, \text{ then } \log_a (f(x))^{2n} &= 2n \log_a (-f(x)). \end{aligned}$$

Example 7. Solve the equation

$$\log(2x) = \frac{1}{4} \log(x-15)^4. \quad (9)$$

$\triangle$ Assume that $x-15 > 0$, then $\log(x-15)^4 = 4 \log(x-15)$, and, hence,

$$\log(2x) = \log(x-15). \quad (10)$$

From this we find that $2x = x-15$, $x = -15$. This number is, evidently, neither a solution of equation (10) nor a solution of equation (9).

Suppose now that $x-15 < 0$, then $\log(x-15)^4 = 4 \log(15-x)$ and the original equation reduces to the equation

$$\log 2x = \log(15-x). \quad (11)$$

Its solution is $x = 5$. This value of x is also a solution of the original equation (9). 

Note that if we considered, instead of two cases and the respective equations (10), (11), only the case when $\log(x-15)^4 = 4 \log(x-15)$ (equation (10)), then we would not find solutions of equation (9).

It is sometimes possible to reduce some logarithmic equations to algebraic equations by means of transformations and introduction of a new variable.

Example 8. Solve the equation

$$\frac{\log_2 x}{\log_4 2x} = \frac{\log_8 4x}{\log_{16} 8x}.$$

$\triangle$ Introducing the notation $\log_2 x = t$, we pass to the base 2 and make use of the formula for the logarithm of a product. We shall have

$$\log_4 2x = \frac{1}{2} \log_2 2x = \frac{1}{2} (1 + t),$$

$$\log_8 4x = \frac{1}{3} \log_2 4x = \frac{1}{3} (2 + t),$$

$$\log_{16} 8x = \frac{1}{4} \log_2 8x = \frac{1}{4} (3 + t).$$

As a result, we can write the original equation in the form

$$\frac{t}{1+t} = \frac{2(2+t)}{3(3+t)}.$$

Solving this equation, we find that $t_1 = 1$, $t_2 = -4$. Accordingly, we get $\log_2 x = 1$, $x = 2$; $\log_2 x = -4$, $x_2 = 1/16$.

Answer: $\{2; 1/16\}$. $\blacktriangle$

2.3. Various Examples of Equations

Example 1. Solve the equation

$$\log_3 x = 1 + \log_x 9.$$

$\triangle$ Passing to the base 3, we get $\log_x 9 = \frac{2}{\log_3 x}$. Designating $t = \log_3 x$, we write the original equation in the form

$$t = 1 + \frac{2}{t}.$$

This equation has solutions $t_1 = 2$, $t_2 = -1$. Accordingly, we find that $\log_3 x = 2$, $x = 9$ and $\log_3 x = -1$, $x = 1/3$.

Answer: $\{9; 1/3\}$. $\blacktriangle$

To transform the expressions $a^{\log_c b}$, it is convenient to use the formula

$$a^{\log_c b} = b^{\log_c a}, \quad (1)$$

where $a > 0$, $b > 0$, $c > 0$, $c \neq 1$. The derivation of (1) is quite simple: it follows from the true equality $\log_c b \cdot \log_c a = \log_c a \cdot \log_c b$ that the equality $\log_c (a^{\log_c b}) = \log_c (b^{\log_c a})$ is also true, and from this, by taking anti-logarithms, we arrive at (1).

Example 2. Solve the equation

$$25^{\log x} = 5 + 4x^{\log 5}$$

$\triangle$ In accordance with (4) we have $x^{\log 5} = 5^{\log x}$. Taking this into account, we write the original equation in the form $5^{2 \log x} - 4 \cdot 5^{\log x} - 5 = 0$. This is a quadratic equation with respect to $5^{\log x}$. Its roots are 5 and -1 . The fact that $5^{\log x} = 5$ implies that $\log x = 1$, $x = 10$. The equation $5^{\log x} = -1$ has no solutions. This means that the original equation has one solution, $x = 10$. $\blacktriangle$

Let us consider now some examples of equations containing the expressions $(f(x))^{g(x)}$. We shall take here only such values of x for which the base $f(x)$ is positive.

Example 3. Solve the equation

$$x^{\log 2x} = 5.$$

$\triangle$ We should, evidently, consider here only the values $x > 0$. Taking logarithms to the base 10, we get an equation

$$\log(2x) \cdot \log x = \log 5,$$

which, as it is easy to see, is equivalent to the original equation. We reduce the equation obtained to the quadratic equation

$$\log^2 x + \log 2 \cdot \log x - \log 5 = 0$$

with respect to $\log x$. Hence we find that

$$\log x = \frac{1}{2} (-\log 2 \pm \sqrt{\log^2 2 + 4 \log 5}).$$

Taking into consideration that $\log 5 = 1 - \log 2$, we find that $\log^2 2 + 4 \log 5 = (\log 2 - 2)^2$, and, consequently, $\log x = \frac{1}{2} (-\log 2 \pm (\log 2 - 2))$. We get $\log x = -1$, $x = 0.1$; $\log x = 1 - \log 2 = \log 5$, $x = 5$, respectively.

Answer: $\{0.1; 5\}$. $\blacktriangle$

Example 4. Solve the equation

$$|x - 3|^{(x^2 - 8x + 15)/(x - 2)} = 1.$$

$\triangle$ The equation $a^b = 1$, where $a > 0$, is true if and only if either $a = 1$ (in the case b is any number) or $b = 0$ (in that case a is any positive number). In accordance with this statement we shall consider two cases.

(1) $|x - 3| = 1$. Hence $x - 3 = \pm 1$, $x_1 = 4$, $x_2 = 2$. For the first value, $x_1 = 4$, the exponent of the left-hand side of the original equation is defined and, consequently, $x_1 = 4$ is a solution of the original equation. For the second value, $x_2 = 2$, the exponent is not defined and this value is not a solution of the original equation.

(2) $(x^2 - 8x + 15)/(x - 2) = 0$. Hence we have $x^2 - 8x + 15 = 0$, $x_1 = 5$, $x_2 = 3$. Of these two values, only $x_1 = 5$ is a solution of the original equation and the value $x_2 = 3$ is extraneous since for this value the base of the power on the left-hand side of the original equation is zero. Thus we see that the original equation has two solutions, $x_1 = 4$, $x_2 = 5$. ▲

In the general case, as well, an equation of the form

$$(f(x))^{\varphi(x)} = 1$$

reduces to the collection of two equations:

- (1) $f(x) = 1$,
- (2) $\varphi(x) = 0$.

All those and only those solutions of these two equations can serve as solutions to the original equation for which the expressions $\varphi(x)$ and $f(x)$ are defined and $f(x) > 0$.

Example 5. Solve the equation

$$\log_{2x-1} \frac{x^4+2}{2x+1} = 1.$$

△ Taking antilogarithms to the base $2x - 1$, we arrive at an equation

$$\frac{x^4+2}{2x+1} = 2x-1,$$

which, as can be easily seen, has the following solutions: $x_1 = 1$, $x_2 = -1$, $x_3 = \sqrt[3]{3}$, $x_4 = -\sqrt[3]{3}$. These values of x contain all solutions of the original equation. Substitution into the original equation shows that only $x = \sqrt[3]{3}$ is its solution. For $x = -1$ and $x = -\sqrt[3]{3}$, the base of the logarithm $2x - 1$ is negative, and for $x = 1$ it is equal to 1. These three values cannot serve as solutions to the original equation.

Answer: $\sqrt[3]{3}$. ▲

The equation we have considered is an example of an equation of the form

$$\log_{\varphi(x)} f(x) = a.$$

The set of all solutions of this equation consists of all those and only those solutions of the equation

$$f(x) = (\varphi(x))^a,$$

which satisfy the conditions $\varphi(x) > 0$, $\varphi(x) \neq 1$.

Example 6. Solve the equation

$$\log_{3x} x = \log_{9x} x.$$

$\triangle$ We should, evidently, seek solutions to this equation only among the values of x satisfying the conditions $x > 0$, $3x \neq 1$, $9x \neq 1$. Setting, in addition, $x \neq 1$, we pass to the base x :

$$\frac{1}{\log_x 3x} = \frac{1}{\log_x 9x}. \quad (2)$$

Hence it follows that $\log_x 9x = \log_x 3x$, $\log_x 3 = 0$, $3 = x^0$, $3 = 1$. The false equality we have obtained shows that equation (2) has no solutions. It would be incorrect to infer from this that the original equation has no solutions either. We have not yet investigated the value $x = 1$. And this, evidently, is a solution of the original equation. $\blacktriangle$

This example shows that upon a transition to a new base, which is an expression containing the unknown, a loss can occur of the solutions of the original equation for which this expression is equal to 1.

2.4. Systems of Exponential and Logarithmic Equations

Let us first consider systems which can be easily solved by the method of substitution.

Example 1. Solve the system

$$\begin{cases} \log_y x - \log_2 y^2 = 1, \\ \log_4 x - \log_4 y = 1. \end{cases}$$

$\triangle$ It follows from the second equation of the system that $\log_4 (x/y) = 1$, whence $x = 4y$. Replacing x by $4y$ in the

first equation of the system, we get an equation

$$\log_y (4y) - \log_2 y^2 = 1$$

in one variable. In the first expression on the left-hand side of this equation we pass to the base 2 and transform the second expression by the formula for the logarithm of the power, taking into account that $y > 0$, $y \neq 1$. Simple transformations result in the equation $\log_2^2 y = 1$. Hence it follows that $\log_2 y = 1$, $y = 2$ and, then, $x = 8$; $\log_2 y = -1$, $y = 1/2$ and $x = 2$. Substituting the pairs of values $x = 8$, $y = 2$ and $x = 2$, $y = 1/2$ we have obtained into the original system, we can easily verify that they both are solutions of that system. ▲

Example 2. Solve the system*

$$\begin{cases} x^{x-y} = y^{x+y}, \\ \sqrt{x \cdot y} = 1. \end{cases}$$

△ Finding $y = x^{-0.5}$ in the second equation and replacing y by $x^{-0.5}$ in the first equation, we arrive at an equation

$$x^{x-x^{-0.5}} = x^{-0.5x-0.5x^{-0.5}}. \quad (1)$$

Taking into account that $x > 0$, we divide both sides of the equation by the expression appearing on the right-hand side (it is also positive) and get an equation

$$x^{1.5x-0.5x^{-0.5}} = 1.$$

Hence we have: (1) $x = 1$, then $y = 1$ as well, and this pair of values of x and y can, evidently, serve as a solution of the original equation; (2) $1.5x - 0.5x^{-0.5} = 0$, whence $x^{1.5} = 1/3$, $x = 1/\sqrt[3]{9}$, and then $y = \sqrt[3]{3}$. Direct verification can prove that the pair $x = 1/\sqrt[3]{9}$, $y = \sqrt[3]{3}$ is a solution of the original system. We can use another procedure. This pair is, evidently, a solution of the system formed by equation (1) and the equation $y = x^{-0.5}$. But this system is equivalent to the original system, which we can immediately see when we replace $x^{-0.5}$ by y in equation (1).

Answer: $\{(1; 1), (1/\sqrt[3]{9}; \sqrt[3]{3})\}$. ▲

* If the base and the exponent depend on the variables, then here, as before, we consider only those values of the variables for which the base is *positive*.

It proves to be possible to reduce some systems of exponential and logarithmic equations to algebraic systems of equations.

Example 3. Solve the system

$$\begin{cases} \log_2 x = \log_4 y + \log_4 (4-x), \\ \log_3 (x+y) = \log_3 x - \log_3 y. \end{cases}$$

Δ We pass to the base 2 in the first equation of the system and then reduce the system to the form

$$\begin{cases} \log_2 x^2 = \log_2 (y(4-x)), \\ \log_3 (x+y) = \log_3 (x/y). \end{cases}$$

Hence it follows that

$$\begin{cases} x^2 = y(4-x), \\ x+y = x/y. \end{cases} \quad (2)$$

Each solution of the original system is also a solution of system (2). The values of x and y giving solutions of the original system evidently satisfy the inequalities

$$0 < x < 4, \quad 0 < y. \quad (3)$$

Performing inverse transformations, we can easily verify that each solution of system (2) satisfying inequalities (3) is also a solution of the original system.

Let us write system (2) in the form

$$\begin{cases} x(x+y) = 4y, \\ x+y = x/y. \end{cases}$$

Substituting x/y for $x+y$ in the first equation, we find that $x^2 = 4y^2$. If $x = 2y$, then it follows from the second equation that $y = 2/3$ and, hence, $x = 4/3$. Now if $x = -2y$, then inequalities (3) are evidently not satisfied and so the original system has no solutions. The pair of values $x = 4/3$, $y = 2/3$ we have found satisfies inequalities (3) and is the unique solution of the original system. $\blacktriangle$

Example 4. Solve the system

$$\begin{cases} \log_8 (xy) = 3 \log_8 x \cdot \log_8 y, \\ 4 \log_8 \frac{x}{y} = \frac{\log_8 x}{\log_8 y}. \end{cases}$$

△ Taking into account that $x > 0$ and $y > 0$, we transform the left-hand sides of the equations, using the formulas for the logarithm of a product and a quotient and, after introducing the new variables $u = \log_8 x$, $v = \log_8 y$, obtain a system

$$\begin{cases} u + v = 3uv, \\ 4(u - v) = u/v \end{cases} \quad (4)$$

Multiplying the equations of system (4) term-by-term, we get $u^2 = 4v^2$, whence it follows that $u = 2v$ or $u = -2v$. Substituting $u = 2v$ into system (4), we find that $v = 1/2$ and, hence, $u = 1$. From this we find that $\log_8 x = 1$, $x = 8$; $\log_8 y = 1/2$, $y = 2\sqrt[4]{2}$. In the case when $u = -2v$, we obtain, by analogy, $x = 1/2$, $y = \sqrt[4]{2}$. Thus we see that the original system has two solutions, $x = 8$, $y = 2\sqrt[4]{2}$ and $x = 1/2$, $y = \sqrt[4]{2}$. ▲

2.5. Exponential and Logarithmic Inequalities

We can manage to solve some exponential and logarithmic inequalities directly, by using the properties of an increase and a decrease of the exponential and the logarithmic function (properties 8 and 11' indicated in the introductory part of this chapter).

Example 1. Solve the inequality

$$0.5^{1/x} \geq 0.0625.$$

△ Noting that $0.0625 = 0.5^4$, we write the inequality in the form

$$0.5^{1/x} \geq 0.5^4. \quad (1)$$

Since the exponential function with the base less than unity decreases, the inequality is true if and only if the inequality

$$1/x \leq 4 \quad (2)$$

holds true. In other words, inequality (2) is equivalent to inequality (1). Inequality (2) is satisfied by the following values of x : $x < 0$, $x \geq 1/4$. ▲

Example 2. Solve the inequality

$$\log_9 \frac{2x}{x+1} > \frac{1}{2}. \quad (3)$$

△ Here we should consider only the values of x for which

$$\frac{2x}{x+1} > 0. \quad (4)$$

The logarithmic function with the base greater than unity increases, and, therefore, inequality (3) is true if and only if both inequality (4) and the inequality

$$\frac{2x}{x+1} > 3 \quad (5)$$

hold true.

It is evident that any solution of inequality (5) satisfies condition (4) and, therefore, inequality (5) is equivalent to inequality (3). Inequality (5) has the following solutions: $-3 < x < -1$. ▲

Example 3. Solve the inequality

$$\log_{0.2} (x^2 - 4) \geq -1. \quad (6)$$

△ We should consider only the values of x for which

$$x^2 - 4 > 0. \quad (7)$$

The logarithmic function with the base less than unity decreases and, therefore, the solutions of inequality (6) are those and only those values of x which satisfy both inequality (7) and the inequality

$$x^2 - 4 \leq 0.2^{-1} = 5.$$

To put it otherwise, inequality (6) is equivalent to the *system* of inequalities

$$0 < x^2 - 4 \leq 5.$$

This system evidently possesses the following solutions: $2 < |x| \leq 3$. ▲

Inequalities of the form

$$a^{f(x)} > a^{g(x)}, \quad \log_a f(x) > b, \text{ etc.}$$

can be solved in the same way as those considered in Examples 1-3.

Example 4. Solve the inequality

$$\log_2 \log_{0.5} \left(2^x - \frac{15}{16} \right) \leq 2.$$

△ This inequality is equivalent to the system of inequalities

$$0 < \log_{0.5} \left(2^x - \frac{15}{16} \right) \leq 4.$$

In its turn, this system is equivalent to the system of inequalities

$$1 > 2^x - \frac{15}{16} \geq 0.5^4.$$

We have taken into account here that the base of the logarithm is less than unity and also the fact that the inequality $2^x - \frac{15}{16} > 0$ follows from the right-hand side inequality of that system. Transforming the system obtained, we get $31/16 > 2^x \geq 1$, whence $\log_2 31 - 4 > x \geq 0$. ▲

Example 5. Solve the inequality

$$2^x < 3^{1/x}.$$

△ Note that both sides of the given inequality are positive. Consequently, their logarithms are defined, say, to the base 2. We find, by the property of the logarithmic function, that the given inequality is true if and only if the inequality

$$\log_2 2^x < \log_2 3^{1/x}$$

is true, that is, if these inequalities are equivalent. Transforming the inequality obtained, we get

$$x < \frac{1}{x} \log_2 3, \quad \frac{x^2 - \log_2 3}{x} < 0.$$

Solving the last inequality, say, by the method of intervals, we find that its solutions are: $x < -\sqrt{\log_2 3}$, $0 < x < \sqrt{\log_2 3}$. ■

Example 6. Solve the inequality

$$\log (x^2 - 16) \leq \log (4x - 11).$$

△ This inequality is equivalent to the system of inequalities

$$0 < x^2 - 16 \leq 4x - 11. \quad (8)$$

The set of solutions of the first inequality, $0 < x^2 - 16$, is a union of two intervals, $(-\infty; -4)$ and $(4; +\infty)$. The set of solutions of the second inequality is the interval $[-1; 5]$. The intersection of these two sets, the interval $(4; 5]$, is a solution of the system in question and, hence, of the original inequality. $\blacktriangle$

As in Example 6, when solving inequalities of the form

$$\log_a f(x) < \log_a g(x) \quad (9)$$

or nonstrict inequalities of such a form, we should consider only the values of x for which $f(x) > 0$ and $g(x) > 0$. Under that condition, an inequality of form (9) is equivalent to the system of inequalities

$$0 < f(x) < g(x) \quad \text{for } 1 < a,$$

or to the system of inequalities

$$f(x) > g(x) > 0 \quad \text{for } 0 < a < 1.$$

Let us consider examples of more complicated inequalities.

If an inequality contains a power whose base and exponent depend on the variable, then we shall consider only those values of the variable for which the base is *positive*.

Example 7. Solve the inequality

$$(x-3)^{2x^2-7x} > 1. \quad (10)$$

$\triangle$ If $x-3 > 1$, then the given inequality is true if and only if $2x^2-7x > 0$. This means that any solution of the *system* of inequalities

$$\begin{cases} x-3 > 1, \\ 2x^2-7x > 0 \end{cases} \quad (11)$$

is a solution of the original inequality (10).

If $0 < x-3 < 1$, then inequality (10) is true if and only if the inequality $2x^2-7x < 0$ is true. This means that any solution of the *system* of inequalities

$$\begin{cases} 0 < x-3 < 1, \\ 2x^2-7x < 0 \end{cases} \quad (12)$$

is also a solution of inequality (10).

For $x - 3 = 1$, i.e. $x = 4$, inequality (10) is, evidently, not true. It follows from what we have stated that the set of solutions of inequality (10) is a *union* of the sets of solutions of systems (11) and (12).

It is easy to find that the set of solutions of system (11) is the interval $(4; +\infty)$ and that of system (12) is the interval $(3; 3.5)$. The union of these two intervals is precisely the set of solutions of the original inequality.

Answer: $(3; 3.5) \cup (4; +\infty)$. ▲

As in the example just discussed, when we consider inequalities of the form

$$(f(x))^{g(x)} > 1, \quad (f(x))^{g(x)} < 1 \quad (13)$$

we must consider two cases:

$$f(x) > 1 \text{ and } 0 < f(x) < 1. \quad (14)$$

In the case of nonstrict inequalities of form (13) we should also take into consideration the case $f(x) = 1$.

Example 8. Solve the inequality

$$(x^2 - 2.5x + 1)^{x+1} \leq 1.$$

△ This inequality is equivalent to the collection of two systems of inequalities:

$$\begin{cases} x^2 - 2.5x + 1 \geq 1, \\ x + 1 \leq 0, \end{cases} \quad \text{and} \quad \begin{cases} 0 < x^2 - 2.5x + 1 \leq 1, \\ x + 1 \geq 0. \end{cases}$$

The first of these systems of inequalities has solutions $x \leq -1$. The second system has solutions $0 \leq x < 0.5$, $2 < x \leq 2.5$. All these values of x constitute the set of solutions of the original inequality.

Answer: $(-\infty; -1] \cup [0; 0.5) \cup (2; 2.5]$. ▲

When solving inequalities of the form

$$\log_{f(x)} g(x) > a, \quad \log_{f(x)} g(x) < a, \quad (15)$$

we must also consider two cases:

$$f(x) > 1 \text{ and } 0 < f(x) < 1. \quad (16)$$

Example 9. Solve the inequality

$$\log_{2x} (x^2 - 5x + 6) > 1.$$

△ If $2x > 1$, then the given inequality is true if and only if $x^2 - 5x + 6 > 2x$. Now if $0 < 2x < 1$, then the given inequality is true if and only if $x^2 - 5x + 6 < 2x$. Taking into account also the fact that in any case the inequality $x^2 - 5x + 6 > 0$ must be fulfilled, we find that the given inequality is equivalent to the collection of two systems of inequalities

$$\left\{ \begin{array}{l} 2x > 1, \\ x^2 - 5x + 6 > 2x, \end{array} \right. \text{ and } \left\{ \begin{array}{l} 0 < 2x < 1, \\ 0 < x^2 - 5x + 6 < 2x. \end{array} \right.$$

The first system has solutions $0.5 < x < 1$, $6 < x$. In the second system the first inequality is true for $0 < x < 0.5$ and the inequality $x^2 - 5x + 6 < 2x$ is true for $1 < x < 6$. Comparing these two sets, we see that the second system has no solutions. Consequently, the set of solutions of the original inequality coincides, in this case, with the set of solutions of the first system of the collection being discussed.

Answer: $(0.5; 1) \cup (6; +\infty)$. ▲

When solving nonstrict inequalities of form (15), we must also consider the case

$$g(x) = (f(x))^a. \quad (17)$$

Example 10. Solve the inequality

$$\log_{1/x} \frac{2x-1}{x-1} \leq -1.$$

△ This inequality is equivalent to the collection of two systems of inequalities:

$$\left\{ \begin{array}{l} 1/x > 1, \\ 0 < \frac{2x-1}{x-1} \leq x, \end{array} \right. \text{ and } \left\{ \begin{array}{l} 0 < 1/x < 1, \\ \frac{2x-1}{x-1} \geq x. \end{array} \right.$$

We solve the first system. The inequality $1/x > 1$ has solutions $0 < x < 1$. This means that $x - 1 < 0$ and then it follows from the inequality $(2x - 1)/(x - 1) > 0$ that $2x - 1 < 0$, i.e. $x < 0.5$. The inequality $(2x - 1)/(x - 1) \leq x$ can be reduced (since $x - 1 < 0$) to the inequality $2x - 1 \geq x(x - 1)$ whose solutions are $0.5(3 - \sqrt{5}) \leq x \leq 0.5(3 + \sqrt{5})$. The intersection of this

interval with the interval $0 < x < 0.5$ is the interval

$$0.5 (3 - \sqrt[5]{5}) \leq x < 0.5, \quad (18)$$

which is the set of solutions of the first system.

The second system can be solved by analogy, its set of solutions being the interval

$$1 < x \leq 0.5 (3 + \sqrt[5]{5}). \quad (19)$$

Answer: $[0.5 (3 - \sqrt[5]{5}); 0.5) \cup (1; 0.5 (3 + \sqrt[5]{5})]$. ▲

Some exponential and logarithmic inequalities can be reduced to algebraic inequalities, quadratic, for instance, by introducing a new variable.

Example 11. Solve the inequality

$$\log_4 x - \log_x 4 \leq 3/2.$$

△ Let us write the inequality in the form

$$\log_4 x - \frac{1}{\log_4 x} \leq \frac{3}{2}$$

and introduce the notation $t = \log_4 x$. Then we get an inequality

$$t - \frac{1}{t} \leq \frac{3}{2},$$

whose solutions are: $t \leq -0.5$, $0 < t \leq 2$. Hence we find, respectively,

$$\begin{aligned} \log_4 x &\leq -0.5, & 0 < x &\leq 0.5; \\ 0 < \log_4 x &\leq 2, & 1 < x &\leq 16. \end{aligned}$$

Answer: $(0; 0.5] \cup (1; 16]$. ▲

Example 12. Solve the inequality

$$\log_3 (16^x - 2 \cdot 12^x) \leq 2x + 1.$$

△ This inequality is equivalent to the system of inequalities

$$0 < 16^x - 2 \cdot 12^x \leq 3^{2x+1}$$

We write the system in the form

$$0 < 4^{2x} - 2 \cdot 4^x \cdot 3^x \leq 3 \cdot 3^{2x}.$$

Dividing each part by $3^{2x} > 0$ and designating $t = (4/3)^x$, we get a system of algebraic inequalities

$$0 < t^2 - 2t \leq 3.$$

The solutions of this system are $2 < t \leq 3$. Hence we find that

$$2 < (4/3)^x \leq 3, \quad \log_{4/3} 2 < x \leq \log_{4/3} 3.$$

Answer: $(\log_{4/3} 2; \log_{4/3} 3]$. ▲

When solving inequalities, we should bear in mind the remarks made earlier which concern a possible appearance of extraneous values of the variable or loss of solutions.

Example 13. Solve the inequality

$$2 \log_2 (x - 1) > \log_2 (5 - x) + 1. \quad (20)$$

△ Let us transform the inequality as follows:

$$\log_2 (x - 1)^2 > \log_2 2 (5 - x). \quad (21)$$

Any solution of the original inequality (20) is, evidently, also a solution of inequality (21). The converse is true if and only if $x - 1 > 0$, i.e. $x > 1$.

Furthermore, inequality (21) is equivalent to the system of inequalities

$$(x - 1)^2 > 2(5 - x) > 0. \quad (22)$$

The first inequality of (22) has solutions $x < -3$, $3 < x$, the second inequality has solutions $x < 5$. Consequently, system (22) has solutions $x < -3$, $3 < x < 5$. From these solutions, only the values of x belonging to the interval $(3; 5)$ satisfy the condition $x > 1$. This interval is precisely the set of solutions of the original inequality.

Answer: $(3; 5)$. ▲

PROBLEMS OF SECTION I

Solve the equations given in problems 1-6.

$$1. 4^x - 4^{\sqrt{x+1}} = 3 \cdot 2^{x+\sqrt{x}}. \quad 2. x^{\log_5 (8x)} = 16 \sqrt[3]{x^4},$$

$$3. \log_{\sqrt{2} \sin x} (1 + \cos x) = 2.$$

$$4. 1 - \log_y (x+1)^2 = \frac{1}{2} \log_{\sqrt[3]{3}} \frac{x+5}{x+3}.$$

$$5. \frac{2}{\log (0.5 + \cos^2 x)} = \log_{\sin 2x} 10.$$

$$6. \log_{\sin x} 2 + \log_{\cos x} 2 + \log_{\sin x} 2 \cdot \log_{\cos x} 2 = 0.$$

$$7. \text{ For which values of } a \text{ does the equation}$$

$$\log_{x-1} (x+a) = 0.5$$

has a unique solution?

Solve the systems of equations given in problems 8-13.

$$8. \begin{cases} \log_y x + \log_x y = 2.5, \\ xy = 27. \end{cases}$$

$$9. \begin{cases} 4^{x+y} = 27 + 9^{x-y}, \\ 8^{x+y} - 21 \cdot 2^{x+y} = 27^{x-y} + 7 \cdot 3^{x-y+1}. \end{cases}$$

$$10. \begin{cases} 5^{\log x} = 3^{\log y}, \\ (3x)^{\log 3} = (5y)^{\log 5}. \end{cases} \quad 11. \begin{cases} x^y = y^x, \\ x^x = y^{xy}. \end{cases}$$

$$12. \begin{cases} \log_2 (x^2 + y^2) - \log_2 x = \log_2 (x + 3y) - 1, \\ \log_2 \frac{xy + 1}{2y^2 + y - x + 2} = \log_2 \frac{x}{y} - 1. \end{cases}$$

$$13. \begin{cases} 2^{x-y} - 2 \cdot 6^{x-y} - 6^{-2y} = 0, \\ 2^{-x-y} - 2 \cdot 3^{x+y} + 3 \cdot 9^x = 0. \end{cases}$$

Solve the inequalities given in problems 14-19.

$$14. \frac{7}{9x-2} \geq \frac{2}{3x-1}. \quad 15. \log_3 (x+2) > \log_{x+2} 81.$$

$$16. \frac{\log_2 x \cdot \log_3 (4x)}{\log_4 (2x) \cdot \log_{16} (8x)} < 5.$$

$$17. \log_x (6x - 1) > \log_x (2x).$$

$$18. \log_{x+a} x \leq \log_a x^2, \quad a > 0, \quad a \neq 1.$$

$$19. \frac{2 + \log_3 x}{x-1} < \frac{6}{2x-1}.$$

PROBLEMS OF SECTION II

Solve the equations given in problems 1-4.

$$1. 25^{|1-2x|} = 5^{1-6x}. \quad 2. 4^x - 3^{x-0.5} = 3^{x+0.5} - 2^{2x-1}.$$

$$3. 5^x \cdot 2^{x+1} = 50. \quad 4. 5^{\log_2 x} + 2 \cdot x^{\log_2 5} = 15.$$

5. Calculate the value of the expression

$$\frac{\log_2 24}{\log_{96} 2} - \frac{\log_2 192}{\log_{12} 2}$$

without resort to tables.

6. Express:

(a) $\log_5 30$ in terms of $a = \log_3 20$ and $b = \log 3$.

(b) $\log_{125} 48$ in terms of $a = \log_6 15$ and $b = \log_{12} 24$.

Solve the equations given in problems 7 through 24.

$$7. 2^{\sin^2 x} + 5 \cdot 2^{\cos^2 x} = 7. \quad 8. 4^{\cos^2 x} + 4^{\cos^2 x} = 3.$$

$$9. 5^{\cot^2 x} \left(26 - 5^{\frac{1}{\sin^2 x}} \right) = 5.$$

$$10. 3 \cdot 16^x + 36^x = 2 \cdot 81^x. \quad 11. 9^{\log_3 (1-2x)} = 5x^2 - 5.$$

$$12. x^{0.5 \log \sqrt{x}(x^2-x)} = 3^{\log_9 4}.$$

$$13. 4^{\log_3 (1-x)} = (2x^2 + 2x - 5)^{\log_3 2}.$$

$$14. \log_6 \left(3 \cdot 4^{-\frac{1}{x}} + 2 \cdot 9^{-\frac{1}{x}} \right) + \frac{1}{x} = \log_6 5.$$

$$15. 3x + (3-x) \log_3 2 = \log_3 \left(9 \cdot \left(\frac{8}{3} \right)^x + 2 \cdot 6^x \right) + 1.$$

$$16. 3 \log_3^2 \sin x + \log_3 (1 - \cos 2x) = 2.$$

$$17. x^{\log_3 \frac{x}{98}} \cdot 14^{\log_2 7} = 1.$$

$$18. x^{\log_3^3 x - 3 \log_3 x} = 3^{-\log_2 \sqrt{2}^{64+8}}.$$

$$19. (1 - \log 2) \log_6 x = \log 3 - \log (x - 2).$$

$$20. \log_2 \frac{x-2}{x-1} - 1 = \log_2 \frac{3x-7}{3x-1}.$$

$$21. \log \sqrt{x-5} + \log \sqrt{2x-3} + 1 = \log 30.$$

$$22. \log_{\frac{1}{3}} \left(\cos x + \frac{\sqrt{5}}{6} \right) + \log_{\frac{1}{3}} \left(\cos x - \frac{\sqrt{5}}{6} \right) = 2.$$

$$23. 2 \log_2 \sin x + \log_2 4 \cot^2 x = \frac{1}{\log_3 2}.$$

$$24. 2 \log_3 \frac{x-3}{x-7} + 1 = \log_3 \frac{x-3}{x-1}.$$

25. Find all solutions of the equation

$$2 \log_5 |\cot x| - \log_{0.2} \frac{\sin x}{5 \sin x - 4 \cos x} = 0,$$

belonging to the interval $[3.3; 4]$.

26. Find the solutions of the equation $\cot x + \log_{\pi/4} x = 2$ on the interval $(0; \pi)$.

27. Find:

(a) the sum of the roots of the equation

$$x + 1 = 2 \log_2 (2^x + 3) - 2 \log_4 (1980 - 2^{-x});$$

(b) the product of the roots of the equation

$$(3x)^3 \log_6 2x - 4 = 1980 \cdot x^{\log_6 x}.$$

Solve the equations given in problems 28 through 42.

28. $|\cos x|^{\sin^2 x - 1.5 \sin x + 0.5} = 1.$

29. $(\tan x)^{\sin x} = (\cot x)^{\cos x}.$

30. $\log_{x-1} (2x^3 - 10x^2 + 13x - 1) = 3.$

31. $\log_{x^3} (x + 1)^2 = 1.$

32. $\log_8 \cos^2 x \sin x + 0.5 = 0.$

33. $\log_{\sin 3x} (\cos x - \cos 2x) = 1.$

34. $x^2 \log_3 x^2 - (2x^2 + 3) \log_9 (2x + 3) = 3 \log_3 \frac{x}{2x + 3}.$

35. $\log_2 \log_3 (2x + 3) + \log_{\frac{1}{2}} \log_{\frac{1}{3}} \frac{x + 1}{2x + 3} = 1.$

36. $\log_3 (\log_2 x - 9) = 2 + \log_3 (1 - 4 \log_4 x).$

37. $\log_{1-x} (3 - x) = \log_{3-x} (1 - x).$

38. $\log_{1-2x} (6x^2 - 5x + 1) - \log_{1-3x} (4x^2 - 4x + 1) = 2.$

39. $\log_x 3 \cdot \log_{x/3} 3 + \log_{x/61} 3 = 0.$

40. $\log (x^2 - 8) \cdot \log (2 - x) = \frac{\log_5 (x^2 - 8)}{\log_5 (2 - x)}.$

41. $\log_5 x \cdot \log_3 x = \log_5 x + \log_3 x.$

42. $\log_{ax} x = n \log_{a^2 x} x$, where $a > 0$, $n \neq 1$, $n \neq 2$.

43. Prove that $\log_a b \cdot \log_{ab} b = \log_a b - \log_{ab} b$.

Solve the equations given in problems 44 through 54.

44. $\log^2 \left(1 + \frac{4}{x} \right) + \log^2 \left(1 - \frac{4}{x + 4} \right) = 2 \log^2 \left(\frac{2}{x - 1} - 1 \right).$

45. $\log (x - 10) \cdot \log (x + 10) = \log (x^2 - 100) - 1.$

46. $\log^2 (x + 1) = \log (x + 1) \cdot \log (x - 1) + 2 \log^2 (x - 1).$

47. $3 \cdot 2^{\log_2 (3x - 2)} + 2 \cdot 3^{\log_2 (3x - 2)} - 5 \cdot 6^{\log_2 (3x - 2)} = 0.$

48. $\log_2 (2\sqrt{5 - x} + 5) = 1 - \log_{0.5} (x - 0.5).$

49. $\sqrt{\log_x \sqrt{3x \cdot \log_3 x} + 1} = 0.$

50. $\sqrt{4 \log_4 x - 2} + \sqrt{1 + \log_2 x} = 4.$

51. $3 \log_{16} (\sqrt{x^2 + 1} + x) + \log_2 (\sqrt{x^2 + 1} - x) = \log_{16} (4x + 1) - 0.5.$

52. $\log_2^2 \tan x + \log_2 \sin^2 x \cdot \log_2 \cos^2 x = 1.$

53. $\log^2 \cos x = \log (1 - \sin x) \cdot \log (1 + \sin x).$

54. $(\sqrt{4 - \sqrt{15}})^x + (\sqrt{4 + \sqrt{15}})^x = 8.$

55. For what values of a does the equation $\log_{x+a} (x - 2) = 2$ possess a unique solution?

56. How many solutions does the equation $e^{2x-x^2} + 2x^2 - 3 = x^4$ possess?

Solve the equations given in problems 57-58.

$$57. \left(\frac{1+a^2}{2a}\right)^x - \left(\frac{1-a^2}{2a}\right)^x = 1, \quad 0 < a < 1.$$

$$58. \log_2(1 + \sqrt{x}) = \log_3 x.$$

Solve the systems of equations presented in problems 59 through 85.

$$59. \begin{cases} \log_3 x + \log_3 y = 2 + \log_3 7, \\ \log_4 (x + y) = 2. \end{cases}$$

$$60. \begin{cases} x^{-1} + y^{-1} = x + y, \\ \log x \cdot (\log y + 2) = 1. \end{cases}$$

$$61. \begin{cases} 0.5 \log x + 0.5 \log y - \log(4 - \sqrt{x}) = 0, \\ (25^{\sqrt{x}})^{\sqrt{y}} - 125 \cdot 5^{\sqrt{y}} = 0. \end{cases}$$

$$62. \begin{cases} x^y = 9, \\ 324^{1/y} = 6x. \end{cases} \quad 63. \begin{cases} y = 1 - \log_5 x^2, \\ x^y = 0.2. \end{cases}$$

$$64. \begin{cases} 3 \cdot x^{2y-1} = 4, \\ x^{y+1} = 6. \end{cases} \quad 65. \begin{cases} x^{x+y} = y^{x-y}, \\ x^2 y = 1. \end{cases}$$

$$66. \begin{cases} 4^{\sin x} + 3^{\frac{1}{\cos y}} = 11, \\ 5 \cdot 16^{\sin x} - 2 \cdot 3^{\frac{1}{\cos y}} = 2. \end{cases}$$

$$67. \begin{cases} \log_4 xy + 3 \frac{\log_4 x}{\log_4 y} = 0, \\ \log_4 \frac{x}{y} - \log_4 x \cdot \log_4 y = 0. \end{cases}$$

$$68. \begin{cases} \log^2 x = \log^2 y + \log^2 xy, \\ \log^2 (x - y) + \log x \cdot \log y = 0. \end{cases} \quad 69. \begin{cases} x^{\log_4 y} = 27y, \\ y^{\log_3 x} = 81x. \end{cases}$$

$$70. \begin{cases} \log_2 y = \log_4 (xy - 2), \\ \log_9 x^2 + \log_3 (x - y) = 1. \end{cases} \quad 71. \begin{cases} 2x^y - x^{-y} = 1, \\ \log_2 y = \sqrt{x}. \end{cases}$$

$$72. \begin{cases} \log_2 (x + y) + 2 \log_3 (x - y) = 5, \\ 2^x - 5 \cdot 2^{0.5(x+y-1)} + 2^{y+1} = 0. \end{cases}$$

$$73. \begin{cases} \log_2 x \cdot \log_x (x - 3y) = 2, \\ x \cdot y^{\log_x y} = y^{2.5}. \end{cases}$$

$$74. \begin{cases} \log x \cdot \log (x + y) = \log y \cdot \log (x - y), \\ \log y \cdot \log (x + y) = \log x \cdot \log (x - y). \end{cases}$$

$$75. \begin{cases} \log_2 (10 - 2^y) = 4 - y, \\ \log_2 \frac{x + 3y - 4}{3y - x} = \log_2 (x - 1) - \log_2 (3 - x). \end{cases}$$

$$76. \begin{cases} \log_2 (x - 3) \cdot \log_{y-2} 2 = 1 + \log_{y-2} 2, \\ \log_5 x \cdot \log_5 (2x - 2y + 1) - 3 \log_5 (2y - 1) + 2 = 0. \end{cases}$$

77. $\begin{cases} \log^2 x \cdot (\log_x^2 y - \log_x y + 1) = 13, \\ \log xy = \sqrt[4]{\log_y x \cdot |\log y|} - 7. \end{cases}$
78. $\begin{cases} \log_{2/3}^2 x + \log_{2/3}^2 y - \log_{2/3}^2 (x + y) = 1, \\ \log_{3/2} x \cdot \log_{3/2} y + \log_{3/2} (x + y) = 0. \end{cases}$
79. $\begin{cases} 2 \cdot 15^x + 15^y = 5^x \cdot 3^{-y}, \\ 2 \cdot 3^{x-y} - 5^{y-x} = 3 \cdot 9^x. \end{cases}$ 80. $\begin{cases} 2^{x^2+y^2} = 16^{x+y}, \\ 2^{x^2} + 8 \cdot 2^{y^2} = 8 \cdot 16^x + 16^y. \end{cases}$
81. $\begin{cases} 2^x + 2^y + 2^z = 7, \\ 2^{-x} + 2^{-y} + 2^{-z} = \frac{7}{4}, \\ x + y + z = 3. \end{cases}$
82. $\begin{cases} 2^{xy} - 2 \cdot 4^{xz} + 64 = 0, \\ 2^{y(x+z)} + 2 \cdot 2^{x(y+z)} - 20 \cdot 2^{xy} = 0, \\ 2^{xy} + 2^{z(x+y)} - 24 \cdot 2^{yz} = 0. \end{cases}$
83. $\begin{cases} x^{\log_y z} + z^{\log_y x} = 6, \\ y^{\log_z x} + x^{\log_z y} = 162, \\ z^{\log_x y} + y^{\log_x z} = \sqrt[4]{48}. \end{cases}$
84. $\begin{cases} 1 + \log_1^{-1} x = \log_x \frac{x^2 + y^2}{yz}, \\ 1 + \log_5^{-1} y = \log_y \frac{y^2 + z^2}{zx}, \\ 1 + \log_6^{-1} z = \log_z \frac{z^2 + x^2}{xy}. \end{cases}$
85. $\begin{cases} x2^{x+1} - 2 \cdot 2^y = -3y \cdot 4^{x+y}, \\ 2x \cdot 2^{2x+y} + 3y \cdot 8^{x+y} = 1. \end{cases}$

86. Find all values of a for which the system

$$\begin{cases} 2^{3x+y} + 2^{x+3y} = 4, \\ 8^{x+y} + 2^{-x-y} = 4^{y+a} \end{cases}$$

has solutions.

87. Find all values of a for which the system

$$\begin{cases} 2^{bx} + (a+1)by^2 = a^2, \\ (a-1)x^3 + y^3 = 1 \end{cases}$$

possesses at least one solution for any b ($a, b \in \mathbb{R}$).

88. Determine, without resorting to tables, which of the following numbers is greater:

(a) 0 or $\log_{\log_3 2} 0.5$; (b) $\log_3 2$ or $5/8$.

Solve the inequalities given in problems 89 through 108.

89. $3^{(6x-3)/x} < \sqrt[4]{27^{2x-1}}$. 90. $4 \cdot 0.5^{x(x+3)} < 0.25^{2x}$.

$$91. \log_{1/3} \frac{3x-1}{x+2} < 1. \quad 92. \log_{2.25} \left(x - \frac{1}{x} \right) < 0.5.$$

$$93. \log_x 7 > \log_x 3. \quad 94. \sqrt{\log_3 \frac{2x-3}{1-x}} < 1.$$

$$95. \log_3 \log_{\frac{9}{10}} (x^2 - 4x + 3) \leq 0. \quad 96. \left| \log_5 \frac{x+3}{x-4} \right| > 1.$$

$$97. \log_a^2 x^2 > 1, a > 0, a \neq 1.$$

$$98. \frac{4^x}{4x-3^x} < 4. \quad 99. 5 \cdot 4^x + 2 \cdot 25^x \leq 7 \cdot 10^x.$$

$$100. 0.8^x - 1.25^{x+1} > 0.25.$$

$$101. 2 \cdot 3^x + 9 \cdot 4^x \geq 12^x + 18.$$

$$102. 0.4^{\log_2 x+1} \leq 6.25^{2-\log_2 x^3}.$$

$$103. \log_3 (3^x - 1) \cdot \log_3 (3^{x+2} - 9) > 3.$$

104. $3^{-x} + 3^x \cdot \log_2 3 - \log_2 6 > 0$. Is it true that $\log_3 2 - 1$ is a solution of this inequality?

$$105. \log_2 (x-1) + \log_2 (x+1) > 3.$$

$$106. \log_5 \left(6 + \frac{2}{x} \right) + \log_{0.2} \left(1 + \frac{x}{10} \right) \leq 1.$$

$$107. 4 \log_{16} \cos 2x + 2 \log_4 \sin x + \log_2 \cos x + 3 > 0.$$

$$108. 2 \log_4 (x+2) - \log_4 (x+5) < 1.$$

109. Find the domain of definition of the function

$$y = \sqrt[4]{\log_5 x + \log_5 (3-x) - \log_5 (x-1)}.$$

Solve the inequalities given in problems 110 through 133.

$$110. \frac{|\log (8-2x)|}{\log (x-1)} \leq 1.$$

$$111. \log_2 (x^2 - 4x + 4) + 2x > 2 - (x+1) \log_{0.5} (2-x).$$

$$112. \frac{x-1}{\log_3 (9-3^x) - 3} \leq 1.$$

$$113. \log_2 (x^2 - x) - 3 \log_2 \frac{x}{x-1} > -2.$$

$$114. \log_2 x \cdot \log_3 2x + \log_3 x \cdot \log_2 3x \geq 0.$$

$$115. \log_{0.5} (x+2) \cdot \log_2 (x+1) + \log_{x+1} (x+2) > 0.$$

$$116. \log_x 2 \cdot \log_{2x} 2 \cdot \log_2 4x > 1.$$

$$117. \log_a x < 6 \log_x a - 1, a > 0, a \neq 1.$$

$$118. \log_{x-2} 6 + \log_{x+2} 6 > \log_{x-2} 6 \cdot \log_{x+2} 6.$$

$$119. 25^{\log_5^2 x} + x^{\log_5 x} \leq 30.$$

$$120. \frac{\log_2 (x+1)^3 - \log_3 (x+1)^5}{x^2 - 1} \geq 0.$$

$$121. x^{3+\log_a x} < a^2 x^2, a > 0, a \neq 1.$$

$$122. |x-1|^{\log_2 (4-x)} > |x-1|^{\log_2 (1+x)}.$$

$$123. \log_{1-x} \frac{2x+3}{4(2x+1)} \geq 1. \quad 124. \log_{x-3} (x^2 - 4x)^2 \leq 4.$$

$$125. 1 - \log_{\log_2 x} 2 \geq 0.$$

$$126. \log_x (1 + x^2) < \log_{x^2} (16 - x^2).$$

$$127. \log_{x-1} (x+1) > \log_{x^2-1} (x+1).$$

$$128. \sqrt{\log_3 (9x+18)} \leq \log_3 (x+2).$$

$$129. \sqrt{\log_2 x} + 2\sqrt{\log_x 2} \geq 3.$$

$$130. \frac{\log (\tan x + \cot x) - \log 4}{\log (\sin x + \cos x)} < 2.$$

$$131. \sqrt{1 - \log_a x} - \sqrt{1 + \log_a x} > a\sqrt{2}, a > 0, a \neq 1.$$

$$132. \log_x (x-a) > 2, a \in \mathbb{R}.$$

$$133. \frac{2^{x+1} - 7}{x-1} < \frac{10}{3-2x}.$$

134. For what values of a is the inequality

$$\log_{\frac{a}{a+1}} (x^2 + 2) > 1$$

true for any real x ?

135. Find all values of a for which every solution of the inequality

$$\log_{4-x} (2x^2 - 5x - 3) \leq 1$$

is also a solution of the inequality $x^2 + a^2x - 2a^4 \leq 0$.

136. Prove that $\log_2 3$ is not a rational number.

137. Determine, without resorting to tables, which of the following numbers is greater:

(a) $\log_4 60$ or $\log_3 30$; (b) $\log_3 75$ or $\log_2 11$.

138. Prove that

$$1 + 2 \ln x \leq x^2$$

for all $x \geq 1$.

139. Among the integers, find all solutions of the inequality

$$2x + 1 < 2 \log_2 (x + 3).$$

140. Prove that for any $a > 0$ and $b > 0$ the inequality

$$a^a \cdot b^b > a^b \cdot b^a$$

is true.

141. Assume that $0 < a < 1$, $\varphi(x) = a^x + (1-a)^x$. Prove that if $x > 1$, then $\varphi(x) < 1$, and if $x < 1$, then $\varphi(x) > 1$.

142. Assume that $a > 0$, $\varphi(x) = (1+a)^x - a^x$. Prove that if $x > 1$, then $\varphi(x) > 1$, and if $x < 1$, then $\varphi(x) < 1$.

143. Assume that $0 < \alpha < 1$, $a > 0$, $b > 0$. Prove:

(a) $a^\alpha \cdot b^{1-\alpha} < a + b$; (b) $a^\alpha \cdot b^{1-\alpha} \leq \alpha^\alpha \cdot (1-\alpha)^{1-\alpha} (a + b)$.

144. For what real x and y is the inequality

$$\sqrt{\log_x (\pi - \sqrt{y})} + 2 \cos (3\pi \cos \sqrt{y}) + \sqrt{\log_{\pi - \sqrt{y}} x} \leq 0$$

true?

Solve the systems given in problems 145-146.

$$145. \begin{cases} 2^{x+1} = y^2 + 4, \\ 2^{x-1} \leq y. \end{cases} \quad 146. \begin{cases} \log_x (x+2) > 2, \\ (x^2 - 8x + 13)^{x-6} < 1. \end{cases}$$

147. Find the set of points of the plane whose coordinates $(x; y)$ satisfy the system

$$(a) \begin{cases} 3^{2(x-y)} - 6 \cdot 3^{-2x} - 3^{-y} > 0, \\ x + 3y = 5; \end{cases}$$

$$(b) \begin{cases} \log_{0.5}(x+y-1) > \log_{0.5}y, \\ \sqrt{y-x-1} < \sqrt{2-x}. \end{cases}$$

148. Among the integers find all solutions of the system of equations

$$\begin{cases} 1.5y - 2^{\frac{x-y}{y}} = 1, \\ 2y - 3^x = 1. \end{cases}$$

149. Find

$$\lim_{n \rightarrow \infty} (\log_{n-1} n \cdot \log_n (n+1) \cdot \dots \cdot \log_{n^{n-1}} n^n).$$

Chapter 3

Combinatorial Analysis, Newton's Binomial Theorem, Random Events and Their Probabilities

3.1. Arrangements, Permutations, Combinations

3.1.1. Examples of simple combinatorial problems. We begin with several problems typical of combinatorics.

Example 1. There are 30 students in the class. In how many ways can the monitor of the class and his assistant be elected if any student could be elected to one of these duties?

△ Since by hypothesis any student can be elected to be the monitor, there evidently, exist 30 ways of electing him. Every one of the remaining 29 students could become his assistant. Every one of the 30 ways of electing the monitor can be used in conjunction with any of the 29 ways of electing his assistant. Therefore, there are $30 \cdot 29 = 870$ possibilities of electing the monitor and his assistant. ▲

Example 2. Four people are to make reports at a conference. In how many ways can they be arranged in the list of speakers?

△ There are 4 ways of choosing the first speaker; the second speaker can, evidently, be chosen by 3 ways. Now only 2 people lay claim to the third place in the list and, consequently, there are 2 ways of filling in the third place in the list. There is no choosing the fourth speaker, he is the last to make his report. Since every choice of the first speaker can be combined with every choice of the second speaker and with either of the two choices of the third speaker, the number of ways of compiling the list of speaker is equal to $4 \cdot 3 \cdot 2 \cdot 1 = 24$. ▲

Example 3. Two teachers must be appointed as examiners for the class taking exams. How many different combinations can be formed from five teachers?

△ Designating the teachers by the letters A, B, C, D , and E , for the sake of convenience, we can write all the possible combinations, namely,

$$\begin{array}{llll} AB, & AC, & AD, & AE, \\ & BC, & BD, & BE, \\ & & CD, & CE, \\ & & & DE. \end{array}$$

Thus we see that the number of different combinations is equal to 10. ▲

We have managed to solve Example 3 by trying out all possible cases. This simple method is applicable only when the number of cases is not large. Now if we had to appoint not 2 but, say, 7 teachers by choosing them from among 14 candidates, then our attempt to try out all possible cases would obviously, fail, since in that case it is possible to form 3432 combinations. We shall easily get the result after we derive the general formulas making it possible to solve problems of this kind (see Example 9).

Before deriving the requisite formulas, let us return to the examples considered above. We shall see whether there is anything in common between these examples and whether there is an essential difference between them.

We note, first of all, that in all three examples we speak of a certain finite set of elements and of the number of its subsets satisfying certain requirements. Thus, in Example 1, we considered the set of all students in the class, 'that is, a set consisting of 30 elements, and we had to find the number of all various subsets of that set which would consist of two elements (two students selected to be the monitor of the class and his assistant). In Example 2 we considered a four-element set of speakers and had to determine the number of four-element subsets of that set, differing from one another only in the successive order of the elements. In Example 3, we isolated different two-element subsets (examiners) from a five-element set of all teachers and counted their number.

Besides the similarity we noted when considering Examples 1-3, there is one very important difference between them. It consists in the fact that the words "different subsets" have different meanings in Examples 1-2 and in Example 3. The subsets in Example 3 were considered to be

different if they differed from one another at least in one element, the order of the elements being of no importance. The examining board consisting of the teachers Smith and Robertson differed in no way from that consisting of Robertson and Smith. On the contrary, the sets in Example 1 were considered to be different if they differed from one another only in the successive order of the elements. The election of Smith as the monitor and Robertson as his assistant and the election of Robertson as the monitor and Smith as his assistant are two different ways of selecting. In Example 2, the subsets were considered to be distinct if they differed from one another only in the order of the elements.

If subsets that differ from one another only in the order of appearance of their elements are considered to be distinct, then we speak of *ordered* subsets. Otherwise the adjective "ordered" is omitted. For example, a set consisting of four elements a, b, c, d evidently has four three-element subsets

abc, abd, acd, bcd

and 24 three-element *ordered* subsets

$abc, abd, acd, bcd,$
 $acb, adb, adc, bdc,$
 $bac, bad, cad, cba,$
 $bca, bda, cda, cdb,$
 $cab, dab, dac, dbc,$
 $cba, dba, dca, dcb.$

In combinatorial problems it is always necessary to count the number of all subsets of the given set which satisfy definite conditions. However, in some problems we consider as distinct the subsets that differ only in the order of the elements, while in other problems the order of appearance of the elements is of no significance and the subsets that differ only in the order of the elements are not considered to be distinct.

3.1.2. Arrangements. Suppose there is a set consisting of n elements. Its every ordered subset consisting of k elements is called the *arrangement of n elements taken k at a time*.

It follows from the definition that $n \geq k \geq 0$ and that the k -arrangements of n elements are all k -element subsets

differing in the content of the elements or in the order of their appearance. For a set consisting of four elements a , b , c , and d , all the 3-element arrangements were written at the end of 3.1.1. They turned to be 24 in number and differed from one another either in the content of the elements or in their order.

In combinatorial problems it is necessary to know how to count the number of all k -arrangements of n elements. A special symbol A_n^k (read "the number of arrangements of n elements taken k at a time", or " k -arrangements of n elements") is used to designate this number (A being the first letter of the word "arrangement").

We have seen that the number of arrangements of 4 elements, taken 3 at a time, is 24, i.e. $A_4^3 = 24$. It must be clear now that in Example 1 we had to find the number of arrangements of 30 elements taken 2 at a time, and it follows from the solution of that problem that $A_{30}^2 = 870$. Next it is evident that $A_n^0 = 1$ since there is only one subset of an n -element set which does not contain elements (empty or void set).

In a general case, the number of k -arrangements of n elements can be determined by the following formula:

$$A_n^k = n(n-1)(n-2)\dots(n-k+1), \quad k > 0, \quad (1)$$

that is, *the number of arrangements of n elements taken k at a time is equal to the product of k successive natural numbers from n to $n - k + 1$ inclusive.*

□ The number of all k -arrangements of n elements is equal to the number of all k -element ordered subsets of the set consisting of n elements. There are, evidently, n ways of choosing the first element of the subset, and $n - 1$ ways to choose the second element of the subset since any element of the set, except that chosen to be the first, can be taken as the second element. Each of the choices of the first element can be combined with each of the choices of the second element and, consequently, there are $n(n - 1)$ choices of the first two elements to construct a k -element ordered subset. After the first two elements have been chosen, there remain $n - 2$ possibilities of choosing the third element and, again, each of these possibilities can be combined with any of the possibilities of choosing the first two elements, that is, the first three elements can be chosen in

$n(n-1)(n-2)$ ways. The last, k th, element of the k -element subset can be chosen in $n-k+1$ ways since by the moment the k th element is chosen there remain $n-(k-1)$ elements. ■

There is another, more convenient, form of formula (1). We shall use, for brevity, the symbol $n!$ (read " n factorial") to designate the product $n(n-1)(n-2)\dots 3\cdot 2\cdot 1$, that is the product of all natural numbers from n to unity. Using the sign of the factorial, we can write, for example,

$$\begin{aligned} 1! &= 1, \\ 2! &= 2\cdot 1 = 2, \\ 3! &= 3\cdot 2\cdot 1 = 6, \\ 4! &= 4\cdot 3\cdot 2\cdot 1 = 24, \\ 5! &= 5\cdot 4\cdot 3\cdot 2\cdot 1 = 120. \end{aligned}$$

Multiplying and dividing the product on the right-hand side of formula (1) by $(n-k)!$, we get

$$A_n^h = \frac{n(n-1)(n-2)\dots(n-k+1)(n-k)!}{(n-k)!}$$

or

$$A_n^h = \frac{n!}{(n-k)!}. \quad (2)$$

Formula (1) has been obtained on the assumption that $k > 0$, formula (2) can also be used when $k = 0$ since in this special case, as well, it yields a correct result, namely,

$$A_n^0 = \frac{n!}{(n-0)!} = \frac{n!}{n!} = 1.$$

When deriving formula (1) we also assumed that $n \neq 0$, that is, that the given set had at least one element. If $n = 0$, it is implied that an empty set is being considered and, since an empty set has only one subset (itself), then $A_0^0 = 1$. If we agree that $0! = 1$, formula (2) gives a correct result in the case when $n = 0$ as well. Indeed,

$$A_0^0 = \frac{0!}{0!} = 1.$$

Example 4. Calculate

$$\frac{A_{20}^6 + A_{20}^5}{A_{20}^4}.$$

△ Using formula (2), we get

$$\frac{A_{20}^6 + A_{20}^5}{A_{20}^4} = \frac{\frac{20!}{14!} + \frac{20!}{15!}}{\frac{20!}{16!}} = \frac{16!}{14!} + \frac{16!}{15!} \\ = 16 \cdot 15 + 16 = 16 \cdot 16 = 256. \blacktriangle$$

Example 5. Fourteen subjects are studied in high school. In how many ways can the time-table for Monday be compiled if that day the class should study five different subjects?

△ There are, evidently, as many ways to compile the time-table as there are five-element ordered subsets of a 14-element set.

Consequently, the number of ways is equal to the number of arrangements of 14 elements 5 at a time, that is, it is equal to A_{14}^5 . Setting $n = 14$ and $k = 5$ in formula (1), we find that

$$A_{14}^5 = 14 \cdot 13 \cdot 12 \cdot 11 \cdot 10 = 240\,240. \blacktriangle$$

3.1.3. Permutations. Arrangements of n elements taken n elements at a time are called *permutations of n elements*.

A permutation is a special case of an arrangement. Since every permutation consists of all n elements of the set, it follows that different permutations are distinguished from one another only by the order of the elements. The number of permutations of n elements is designated as P_n (P being the first letter of the word "permutation"). It is clear now that in Example 2 it was required to find the number of all permutations of 4 elements (the number of all possible permutations of the speakers). That number turned out to be equal to 24. Consequently, $P_4 = 24$.

In a general case, the number of permutations of n elements $P_n = A_n^n$ and, consequently, it can be found by formula (1) or by formula (2), setting $k = n$.

In fact, formula (2) yields

$$P_n = A_n^n = \frac{n!}{(n-n)!} = \frac{n!}{0!} = n!,$$

and formula (1) yields

$$P_n = A_n^n = n(n-1)(n-2) \dots (n-n+1) = n!.$$

Thus we see that *the number of permutations of n elements is equal to $n!$*

Thus, there are $n!$ ways to establish a definite order of elements or, as they say, to *order* the set.

For instance, a list of students of a class in which there are 20 students (none of whom have the same name), can be compiled in

$$20! = 2\,432\,902\,008\,176\,640\,000 \text{ ways.}$$

Example 6. How many six-digit numbers, multiple of five, can be formed from the digits 1, 2, 3, 4, 5, 6 provided that the digits in the number are not repeated?

$\triangle$ For the number compiled from the given digits to be divisible by 5, it is necessary and sufficient that the digit 5 should occupy the last place. The other five digits may occupy the remaining five places in an arbitrary order. Consequently, the desired quantity of six-digit numbers, multiple of five, is equal to the number of permutations of the five elements, i.e. $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$. $\blacktriangle$

Example 7. Find n if

$$\frac{P_{n+5}}{P_{n-k}} = 240 A_{n+3}^{k+3}, \quad k \leq n.$$

$\triangle$ Applying the formula for the number of permutations and formula (2) for the number of arrangements, we rewrite the given equation as follows:

$$\frac{(n+5)!}{(n-k)!} = 240 \frac{(n+3)!}{(n+3-k-3)!}.$$

The resulting equation is equivalent to the quadratic equation

$$(n+5)(n+4) = 240.$$

Its roots are $n = 11$ and $n = -20$. For $n = -20$ both the right-hand and left-hand sides of the equation are meaningless. For $n = 11$, for any k such that $0 \leq k \leq 11$, the equality

$\frac{P_{16}}{P_{11-k}} = 240 A_{14}^{k+3}$ is true. Thus, $n = 11$. $\blacktriangle$

Example 8. How many permutations can be made of the letters forming the word "degree"?

$\triangle$ To make such a permutation means to put in the six enumerated places, in an arbitrary way, one letter "d", one

letter "g", one letter "r", and three letters "e". If we find places for the letters "d", "g", and "r", then the remaining places can be filled by the letters "e". But how many ways are there of putting three distinct letters into six places? It is evident that the number of ways is equal to the number of all three-element ordered subsets of a six-element set, i.e. $A_6^3 = 6 \cdot 5 \cdot 4 = 120$.

We can reason otherwise. Were all the six letters forming the word distinct, the number of permutations would be $6!$.

But the word contains three letters "e" and the permutations of these three letters alone do not require the use of new methods of arrangement of the letters. Therefore, the number of permutations of the letters in the word "degree" is not $6!$ but $3!$ times smaller, i.e.

$$\frac{6!}{3!} = \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{3 \cdot 2 \cdot 1} = 6 \cdot 5 \cdot 4 = 120. \blacktriangle$$

3.1.4. Combinations. Suppose we have a set consisting of n elements. Its every subset consisting of k elements is called a *combination of n elements taken k at a time*.

Thus, combinations of k out of n elements are all k -element subsets of an n -element set, in which only the subsets of different compositions of elements are considered to be distinct. Subsets differing from one another only in the successive order of their elements are not considered to be distinct. For example, the following subsets are combinations of three elements selected from the four-element set a, b, c, d :

$$abc, abd, acd, bcd.$$

The number of all combinations of n elements, k elements at a time, is designated C_n^k (read "the number of combinations of n, k at a time" or " C of n, k at a time" or, else, " k out of n elements"), C being the first letter of the word "combination". We have just seen that $C_4^3 = 4$. In Example 3, we found the number of combinations of 5 elements, 2 at a time, and it turned out that $C_5^2 = 10$.

In a general case, the number of combinations of n elements, k at a time, is defined by the following formula:

$$C_n^k = \frac{n!}{(n-k)!k!}. \quad (3)$$

□ Let us first form all possible nonordered subsets containing k elements each. Their number is C_n^k . Then, from each of the subsets obtained, by means of permutations, we obtain all ordered subsets which will, evidently, be $k!$ as many in number since every k -element set can be ordered in $k!$ ways. Thus, $A_n^k = k!C_n^k$, whence follows formula (3). ■

Formula (3) can be written in another, more convenient form. Cancelling the numerator and the denominator of the fraction by $(n - k)!$, we get

$$C_n^k = \frac{n(n-1)(n-2)\dots(n-k+1)}{k!}, \quad (4)$$

that is, *the number of combinations of n elements, k at a time, is equal to the product of all natural numbers, from n to $n - k + 1$ inclusive, divided by $k!$.*

Example 9. How many examination boards consisting of 7 members each can be formed from 14 teachers?

△ They will be, evidently, as many in number as there are seven-element subsets in a 14-element set. Formula (4) yields

$$C_{14}^7 = \frac{14 \cdot 13 \cdot 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8}{7!} = \frac{14 \cdot 13 \cdot 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8}{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 3432. \blacktriangle$$

Example 10. Eighteen football teams take part in the national championship, and every team meets the same opponent twice. How many matches are played during a season?

△ In the first round there are as many matches as there are two-element subsets of the set containing 18 elements, that is, their number is C_{18}^2 . Formula (4) yields

$$C_{18}^2 = \frac{18 \cdot 17}{2} = 153.$$

There is the same number of matches in the second round and, therefore, 306 matches are played during a season. ▲

Example 11. Solve the inequality

$$C_{10}^{x-1} > 2C_{10}^x.$$

△ The left-hand side of the inequality is meaningful if and only if x is an integer belonging to the interval $[1; 11]$. The right-hand side is meaningful if and only if x is an

integer and $x \in [0; 10]$. Consequently, only those integral values of x which lie in the interval $[1; 10]$ may serve as solutions of the inequality.

Using formula (3), we write the given inequality as follows:

$$\frac{10!}{(x-1)!(10-x+1)!} > 2 \frac{10!}{x!(10-x)!}.$$

Dividing both sides of the inequality by $\frac{10!}{(x-1)!(10-x)!}$, we get

$$\frac{1}{11-x} > \frac{2}{x},$$

whence we have $x > 22 - 2x$, i.e. $x > 22/3$. Taking into account the restrictions $x \in \mathbb{N}$ and $x \in [1; 10]$, we get the set of solutions of the given inequality:

$$\{8; 9; 10\}. \quad \blacktriangle$$

The numbers C_n^k possess many interesting and important properties. We shall discuss two properties which are often employed.

The first property. $C_n^k = C_n^{n-k}$.

□ Applying formula (3), we get

$$C_n^{n-k} = \frac{n!}{(n-(n-k))!(n-k)!} = \frac{n!}{k!(n-k)!} = C_n^k. \quad \blacksquare$$

Using this property, we can simplify the calculations of C_n^k in the cases when $k > n/2$, for instance,

$$C_{15}^{12} = C_{15}^3 = \frac{15 \cdot 14 \cdot 13}{3 \cdot 2} = 455.$$

The second property.

$$C_{n+1}^{k+1} = C_n^{k+1} + C_n^k \quad (k < n). \quad (5)$$

□ We again use formula (3) and get

$$\begin{aligned} C_n^{k+1} + C_n^k &= \frac{n!}{(n-k-1)!(k+1)!} + \frac{n!}{(n-k)!k!} \\ &= \frac{n!}{(k+1)!(n-k)!} \left(\frac{n-k}{1} + \frac{k+1}{1} \right) \\ &= \frac{(n+1)!}{(n+1-(k+1))!(k+1)!} = C_{n+1}^{k+1}. \quad \blacksquare \end{aligned}$$

3.2. Newton's Binomial Theorem

The well-known formulas

$$\begin{aligned}(a + b)^2 &= a^2 + 2ab + b^2, \\ (a + b)^3 &= a^3 + 3a^2b + 3ab^2 + b^3\end{aligned}$$

can also be written in the form

$$\begin{aligned}(a + b)^2 &= C_2^0 a^2 + C_2^1 ab + C_2^2 b^2, \\ (a + b)^3 &= C_3^0 a^3 + C_3^1 a^2 b + C_3^2 b^2 + C_3^3 b^3.\end{aligned}$$

The following hypothesis arises: are the similar formulas for the fourth, fifth, and, in general, any natural degree of a binomial also true?

We shall first find out whether a similar formula is true for the fourth degree. For that purpose, we multiply both sides of the formula for $(a + b)^3$ by $a + b$. Then we have

$$\begin{aligned}(a + b)^4 &= (C_3^0 a^3 + C_3^1 a^2 b + C_3^2 ab^2 + C_3^3 b^3) (a + b) \\ &= C_3^0 a^4 + C_3^1 a^3 b + C_3^2 a^2 b^2 + C_3^3 ab^3 \\ &\quad + C_3^0 a^3 b + C_3^1 a^2 b^2 + C_3^2 ab^3 + C_3^3 b^4 \\ &= C_3^0 a^4 + (C_3^1 + C_3^0) a^3 b + (C_3^2 + C_3^1) a^2 b^2 \\ &\quad + (C_3^3 + C_3^2) ab^3 + C_3^3 b^4.\end{aligned}$$

Noting that

$$\begin{aligned}C_3^0 &= C_4^0, \quad C_3^1 + C_3^0 = C_4^1, \quad C_3^2 + C_3^1 = C_4^2, \\ C_3^3 + C_3^2 &= C_4^3, \quad C_3^3 = C_4^4,\end{aligned}$$

we ascertain the validity of the formula

$$(a + b)^4 = C_4^0 a^4 + C_4^1 a^3 b + C_4^2 a^2 b^2 + C_4^3 ab^3 + C_4^4 b^4.$$

Thus we have managed to obtain a formula for the fourth degree of the binomial by using an analogous formula for the third degree. The argument carried out corroborates the hypothesis, in the first place, and suggests the idea of using mathematical induction to prove it, in the second.

Theorem. *The following formula is valid for arbitrary numbers a and b and for an arbitrary natural number n :*

$$\begin{aligned}(a + b)^n &= C_n^0 a^n + C_n^1 a^{n-1} b + \dots \\ &\quad \dots + C_n^k a^{n-k} b^k + \dots + C_n^n b^n.\end{aligned}\quad (1)$$

Using the summation sign, we can write Newton's formula in the abbreviated form as follows:

$$(a + b)^n = \sum_{k=0}^n C_n^k a^{n-k} b^k. \quad (2)$$

□ For $n = 1$, Newton's formula has the form

$$(a + b)^1 = C_1^0 a + C_1^1 b,$$

and it is evidently true since $C_1^0 = C_1^1 = 1$.

Let us assume that the formula is true for $n = m$, that is,

$$(a + b)^m = \sum_{k=0}^m C_m^k a^{m-k} b^k.$$

Then it follows that

$$\begin{aligned} (a + b)^{m+1} &= (a + b) \sum_{k=0}^m C_m^k a^{m-k} b^k \\ &= \sum_{k=0}^m C_m^k a^{m+1-k} b^k + \sum_{k=0}^m C_m^k a^{m-k} b^{k+1} \\ &= C_m^0 a^{m+1} + \sum_{k=1}^m C_m^k a^{m+1-k} b^k + \sum_{k=1}^m C_m^{k-1} a^{m+1-k} b^k + C_m^m b^{m+1} \\ &= C_m^0 a^{m+1} + \sum_{k=1}^m (C_m^k + C_m^{k-1}) a^{m+1-k} b^k + C_m^m b^{m+1}. \end{aligned}$$

Taking into consideration that

$$C_m^0 = C_{m+1}^0, \quad C_m^k + C_m^{k-1} = C_{m+1}^k, \quad C_m^m = C_{m+1}^{m+1},$$

we obtain

$$(a + b)^{m+1} = \sum_{k=0}^{m+1} C_{m+1}^k a^{m+1-k} b^k.$$

Thus, the validity of formula (1) for $n = m$ implies its validity for $n = m + 1$ and, since the formula is valid for $n = 1$ as well, its validity has been established for all natural values of n by induction. ■

Formula (1) was named after Isaac Newton. Its right-hand side is called an *expansion of a natural power of a binomial*, and C_n^k are called *binomial coefficients*.

Here are some peculiarities of Newton's formula:

(a) Its right-hand side contains $n + 1$ terms.

(b) Every term has the form $C_n^k a^{n-k} b^k$.

It is convenient to consider the term $C_n^k a^{n-k} b^k$ occupying the place $k + 1$ to be the k th term of the expansion and denote it by T_k , that is,

$$T_k = C_n^k a^{n-k} b^k. \quad (3)$$

Then $T_0 = C_n^0 a^n$ is the zeroth term of the expansion, $T_1 = C_n^1 a^{n-1} b$ is the first term, and $T_n = C_n^n b^n$, is its n th term.

(c) The powers of a decrease by 1 in each subsequent term of the expansion, and the powers of b increase by 1. The sum of the powers of a and b is equal to n in each term of the expansion.

(d) The coefficients of the expansion, equidistant from both the zeroth term and the n th term, are equal since $C_n^k = C_n^{n-k}$.

Example 1. Raise to the sixth power the binomial

$$x^2 - y.$$

△ Setting $a = x^2$, $b = -y$, $n = 6$ in formula (2), we obtain

$$\begin{aligned} (x^2 - y)^6 &= \sum_{k=0}^6 C_6^k (x^2)^{6-k} (-y)^k \\ &= C_6^0 x^{12} - C_6^1 x^{10} y + C_6^2 x^8 y^2 - C_6^3 x^6 y^3 + C_6^4 x^4 y^4 - C_6^5 x^2 y^5 + C_6^6 y^6 \\ &= x^{12} - 6x^{10}y + 15x^8y^2 - 20x^6y^3 + 15x^4y^4 - 6x^2y^5 + y^6. \quad \blacktriangle \end{aligned}$$

Example 2. Find the fourth term of the expansion of the power of the binomial

$$\left(\sqrt[3]{x} + \frac{1}{\sqrt[3]{x}}\right)^9.$$

△ Setting $n = 9$, $k = 4$, $a = \sqrt[3]{x}$, $b = 1/\sqrt[3]{x}$ in formula (3), we get

$$T_4 = C_9^4 (\sqrt[3]{x})^5 (1/\sqrt[3]{x})^4 = \frac{9 \cdot 8 \cdot 7 \cdot 6}{4 \cdot 3 \cdot 2 \cdot 1} \sqrt{x} = 126 \sqrt{x}. \quad \blacktriangle$$

Example 3. Find the term of the expansion of the power of the binomial $\left(\frac{1}{x} + \sqrt{x}\right)^{12}$ which does not depend on x .

△ Putting $n = 12$, $a = 1/x$, $b = \sqrt{x}$ in formula (3), we obtain

$$T_k = C_{12}^k (1/x)^{12-k} (\sqrt{x})^k = C_{12}^k x^{-12+\frac{3}{2}k}.$$

For the term of the expansion to be independent of x , it is necessary and sufficient that $-12 + \frac{3}{2}k = 0$, whence $k = 8$.

Thus, the eighth term of the expansion does not depend on x . We calculate it:

$$T_8 = C_{12}^8 = C_{12}^4 = \frac{12 \cdot 11 \cdot 10 \cdot 9}{4 \cdot 3 \cdot 2 \cdot 1} = 495. \quad \blacktriangle$$

Example 4. Find the sum of all binomial coefficients

$$C_n^0 + C_n^1 + C_n^2 + \dots + C_n^n.$$

△ Putting $a = 1$, $b = 1$ in formula (2), we get

$$(1+1)^n = \sum_{k=0}^n C_n^k.$$

Thus it follows that

$$\sum_{k=0}^n C_n^k = C_n^0 + C_n^1 + C_n^2 + \dots + C_n^n = 2^n. \quad \blacktriangle$$

The last equality has the following meaning. Since C_n^k is the number of all subsets of an n -element set which contain k elements each, the sum

$$C_n^0 + C_n^1 + \dots + C_n^n$$

is, evidently, the sum of all subsets of the n -element set. Consequently, *the number of all subsets of a set containing n elements is equal to 2^n .*

Example 5. Every one of the ten available lamps can be switched on to illuminate the hall. How many ways are there to illuminate the hall?

△ There are, evidently, as many ways as there are subsets of a ten-element set, i.e. $2^{10} = 1024$. The possibility of "illuminating" the hall (when none of the lamps are switched on) is also taken into account. $\blacktriangle$

Example 6. Find the leading coefficient of the polynomial

$$\left(\frac{1}{3} + \frac{2}{3}x\right)^{10}.$$

$\triangle$ Denoting the coefficient of the polynomial in x^k by a_k , we use formula (2) and arrange the given polynomial in order, by the increasing powers of the variable x :

$$\left(\frac{1}{3} + \frac{2}{3}x\right)^{10} = \sum_{k=0}^{10} C_{10}^k \left(\frac{1}{3}\right)^{10-k} \left(\frac{2}{3}\right)^k x^k = \sum_{k=0}^{10} a_k x^k.$$

To find the leading coefficient a_k , we shall solve the inequality $a_{k-1} \leq a_k$, i.e.

$$C_{10}^{k-1} \left(\frac{1}{3}\right)^{10-k+1} \left(\frac{2}{3}\right)^{k-1} \leq C_{10}^k \left(\frac{1}{3}\right)^{10-k} \left(\frac{2}{3}\right)^k.$$

Dividing both sides of the inequality by $2^{k-1}/3^{10}$, we get

$$C_{10}^{k-1} \leq 2C_{10}^k,$$

whence we have

$$\frac{10!}{(k-1)!(10-k+1)!} \leq \frac{2 \cdot 10!}{k!(10-k)!}.$$

After further obvious cancellings, we get

$$1/(10-k+1) \leq 2/k, \quad k \leq 20 - 2k + 2, \quad k \leq 22/3.$$

Thus we have proved that

$$a_0 < a_1 < \dots < a_7.$$

It is evident that at $k > 22/3$, the opposite inequality $a_{k-1} > a_k$ takes place, that is, the coefficients of the polynomial decrease, beginning with the seventh.

Thus we see that the coefficient a_7 is the greatest among all the eleven coefficients of the given polynomial and is equal to

$$C_{10}^7 \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^7. \quad \blacktriangle$$

We have proved Newton's formula without assuming that a and b are real numbers. Therefore, formulas (1)-(3) are also true in the case when a and b are arbitrary complex numbers.

Example 7. Write the complex number $(2 + i)^6$ in algebraic form.

△ Putting $n = 6$, $a = 2$, $b = i$ in formula (2), we get

$$\begin{aligned}(2 + i)^6 &= \sum_{k=0}^6 C_6^k 2^{6-k} i^k \\ &= C_6^0 2^6 + C_6^1 2^5 i + C_6^2 2^4 i^2 + C_6^3 2^3 i^3 + C_6^4 2^2 i^4 + C_6^5 2 i^5 + C_6^6 i^6 \\ &= 64 + 192i - 240 - 160i + 60 + 12i - 1 \\ &= -117 + 44i. \blacktriangle\end{aligned}$$

3.3. Random Events and Their Probabilities

Let us discuss a simple experiment that involves tossing a coin. There may be two outcomes in this experiment: either the coin will come up heads or it will come up tails. The outcome of the experiment depends on many factors which cannot be presupposed and so it is impossible to predict it. When the coin comes up heads, this is an example of a *random* event. Here are some other examples of random events: scoring 1 when throwing a die, an electric bulb failing before the end of its guaranteed service life, and the quality of the article selected for inspection not meeting the standard. In all these cases, we cannot tell beforehand, until the end of the trial, whether this or that event will occur or not. For that reason, events of this kind are called random events.

In the experiment that involves tossing a coin, both outcomes are, evidently, equally possible, and there is no reason to give preference to one of the outcomes before the end of the trial. In such cases we say that both outcomes are *equi-probable*, and the probability of each of them is equal to $1/2$. Throwing a die, we expect one of six outcomes. Since the die is supposed to be homogeneous and symmetric, all the outcomes of the trial are equally possible or equiprobable. The probability of each of them is equal to $1/6$.

Generalization of these simple trials is an experiment in which n equiprobable outcomes are possible: $u_1, u_2, \dots, u_n$ (they are also called *simple* or *elementary events*). In this case, the probability of every outcome is assumed to be equal to $1/n$. It is written as follows:

$$P(u_1) = 1/n, \quad P(u_2) = 1/n, \quad \dots, \quad P(u_n) = 1/n.$$

The first formula should be read as follows: "the probability of u_1 is equal to $1/n$ " (P being the first letter of the word "probability").

Let us consider now the experiment with n equiprobable outcomes and an event A which occurs when the trial has some k outcomes and does not occur if one of the remaining $n - k$ outcomes takes place. We shall say that the outcomes leading to the event A are *favourable* to it. The *probability* of the event A related to the trial with n equiprobable outcomes is the ratio between the number of outcomes favourable to the event A and the number of all outcomes, that is

$$P(A) = \frac{k}{n}, \quad (1)$$

where k is the number of outcomes favourable to the event A .

Here are some examples to illustrate what we have stated. The experiment with a die has six equiprobable outcomes: u_i , the face of the die, features the number i ($i = 1, 2, 3, 4, 5, 6$). Let us consider the following events related to this experiment:

- event A_1 , the score is a multiple of 3,
- event A_2 , the score is a prime number,
- event A_3 , the score is equal to 7,
- event A_4 , the score is less than 7.

Two outcomes, namely, u_3 and u_6 , are favourable to the event A_1 . Setting $n = 6$, $k = 2$ in formula (1), we have $P(A_1) = 2/6 = 1/3$.

Three outcomes, u_2 , u_3 and u_5 , are favourable to the event A_2 . By formula (1) we get $P(A_2) = 3/6 = 1/2$.

Neither of the possible outcomes is favourable to the event A_3 . Consequently, $P(A_3) = 0/6 = 0$. The event A_3 is *impossible*.

All six outcomes are favourable to the event A_4 and, therefore, $P(A_4) = 6/6 = 1$. The event A_4 is a *certain* event.

The probability of any event A satisfies the inequalities

$$0 \leq P(A) \leq 1,$$

which follows directly from formula (1) since it is evident that $0 \leq k \leq n$.

To calculate the probabilities of events by formula (1), it is necessary to find the number of all equally possible out-

comes and the number of outcomes favourable to the random event. In this case, combinatorial formulas discussed in 3.1 turn out to be very convenient.

Example 1. The experiment consists in tossing two coins, copper and silver. What is the probability of at least one coin coming up heads?

△ The equiprobable outcomes of the experiment are the following:

- u_1 , both coins have come up heads,
- u_2 , only the copper coin has come up heads,
- u_3 , only the silver coin has come up heads,
- u_4 , neither of the coins has come up heads.

The outcomes u_1 , u_2 and u_3 are favourable to the event A (at least one coin coming up heads). Setting $n = 4$, $k = 3$ in formula (1), we get $P(A) = 3/4$. ▲

Example 2. While dialing a telephone number, you forget the last two digits: remembering only that they were different, you dial at random. What is probability of your dialing a correct number.

△ Two last digits can be chosen in a number of ways equal to the number of two-element ordered subsets of a ten-element set (the set of all digits). This number of ways is equal to A_{10}^2 . Consequently, there are A_{10}^2 outcomes all in all, only one of which (the correct number was dialed) is favourable to the event A . Therefore,

$$P(A) = 1/A_{10}^2 = 1/10 \cdot 9 = 1/90. \quad \blacktriangle$$

Example 3. There are five defective bulbs among 100 electric lamps available. What is the probability of choosing three good lamps at random?

△ A set consisting of 100 elements has C_{100}^3 three-element subsets. Therefore, we can use C_{100}^3 ways to choose 3 bulbs out of 100. The bulbs are chosen at random and this means that all these ways of selection (all outcomes) are equiprobable.

The number of favourable outcomes (all three bulbs turn out to be good) can be calculated by analogy. There are C_{95}^3 ways of selecting 3 bulbs out of 95, since this is the number of three-element subsets of 95-element set. Setting $n = C_{100}^3$, $k = C_{95}^3$ in formula (1), we get

$$P(A) = C_{95}^3 / C_{100}^3 = \frac{95 \cdot 94 \cdot 93}{100 \cdot 99 \cdot 98} \approx 0.856. \quad \blacktriangle$$

PROBLEMS OF SECTION I

1. Five workers in a factory leave for a vacation and only three places are available in the sanatorium. How many ways are there to distribute the places if (a) all the places are different, (b) all the places are identical?

2. How many ways are there to arrange five white and four black balls in successive order so that the black balls will not lie side by side? Consider two cases: (a) the balls of the same colour are identical, (b) all the balls are different.

3. How many diagonals are there in a convex n -gon?

4. The first of two parallel lines houses 10 points, the second, 20 points. How many triangles are there with vertices at those points?

5. Four writers must write a book containing 17 chapters. The first and the third writer must write five chapters each, the second must write four chapters, and the fourth must write three chapters of the book. How many ways can be found to divide the book between the four writers?

6. A crocodile is known to have not more than 68 teeth. Prove that among 16^{17} crocodiles chosen at random no two will turn out to have the same number of teeth.

7. How many ten-digit numbers can be written with the use of the digits 1 and 2?

8. The Morse code consists of points and dashes. How many letters can there be in the Morse code if one letter cannot contain more than four signs?

9. A car registration number consists of three letters (30 letters are used) and four digits (10 digits are used). How many cars can be registered so that no two cars have the same number?

10. How many ways are there of dividing $2n$ elements into pairs if the partitionings differing in the order of the elements within the pairs and the order of the arrangement of the pairs are taken to be identical?

11. Find the middle term of the expansion

$$(x^{-1/5} + x^{1/3})^{10}.$$

12. At what values of x is the fifth term of the expansion $(2x + 3)^9$ greater than the adjacent terms?

13. Find the terms of the expansion $(\sqrt{3} + \sqrt[3]{2})^9$ which are integers.

14. Find the coefficient of the polynomial $(1 + 3x + 2x^3)^{10}$ in x^4 .

15. Find the sum of the coefficients of the polynomial $(13x^3 - 7x - 5)^{100}$.

16. Calculate $\sum_{k=0}^n C_{2n}^{2k}$.

17. There are two winning tickets among ten tickets available. Determine the probability that (a) one, or (b) both of these tickets, will be among five tickets selected at random?

18. Five people entered the lift cabin on the ground floor of an 8-floor house. Suppose that each of them, independently and with

equal probability, can leave the cabin at any floor beginning with the first. Find out the probability of all five people leaving at different floors.

PROBLEMS OF SECTION II

1. Calculate:

$$(a) \frac{6!}{A_{10}^7} (C_7^5 + C_7^3); \quad (b) \frac{P_{h+1}}{(h-n)! A_{h-1}^{n-1}}.$$

2. Find n if

$$(a) C_{n+1}^{n+1} - C_{n+3}^n = 15(n+2); \quad (b) \frac{1}{C_4^n} = \frac{1}{C_5^n} + \frac{1}{C_6^n};$$

$$(c) 5C_n^3 = C_{n+2}^4, \quad (d) (n+2)! = 132A_n^h P_{n-k};$$

$$(e) C_{n+1}^{n+1} - 5C_{3n}^2 + 19n^2 = 6;$$

$$(f) \frac{A_{n+1}^4}{(n+2)!} < \frac{14^2}{4^2 n}; \quad (g) 8C_{105}^n < 3C_{105}^{n+1}.$$

3. Find the set of values of the function:

$$(a) f(x) = A_{7-3}^{x-3}; \quad (b) f(x) = C_{x+1}^{2x-8}.$$

4. There are 30 students in a class. In how many ways can we choose two monitors to be on duty interchangeably if (a) one of them should have greater responsibility, or (b) both of them should have equal responsibilities?

5. How many distinct two-digit numbers can be formed from the digits 1, 2, 3, 4 on condition that neither of the numbers contains repeated digits within it?

6. How many distinct two-digit numbers can be formed from the digits 1, 2, 3, 4?

7. Neither three diagonals of a convex 10-gon intersect at one point. Determine the number of intersection points of the diagonals.

8. How many ways are there of arranging eight rooks of the same colour on the chessboard so that they do not capture one another and occupy only black squares?

9. Form different four-digit numbers from the digits 0, 1, 2, 3 so that neither of the numbers contains repeated digits. How many numbers have you formed? How many even numbers are there among them?

10. 153 matches were played in the football championship. Each team met its opponent only once. How many teams took part in the championship?

11. There are three sergeants and 30 soldiers in a platoon. How many ways are there of selecting one sergeant and three soldiers for patrol duty?

12. In how many ways can the time-table for Monday be compiled if there must be five lessons that day (algebra, geometry, history, geography, and literature) and algebra and geometry must not immediately follow each other?

13. How many permutations can be formed of the letters constituting the words (a) angle, (b) basis, (c) minimum?

14. In how many ways can 28 dominoes be distributed among 4 players for each of them to get 7 dominoes?

15. A hockey team consists of 2 goalkeepers, 7 left and right backs, and 10 forwards. In how many ways can the coach form the starting group of six players consisting of 1 goalkeeper, 2 backers, and 3 forwards?

16. How many six-digit numbers are there whose all digits are odd (1, 3, 5, 7, 9)?

17. Four speakers A , B , C and D , must address the conference, and B must not precede A . How many ways are there to establish the successive order of speeches?

18. How many divisors are there of the number 462?

19. There are m books in black covers and n books in blue covers on the shelf, all the books being different. How many ways are there to arrange the books on the shelf so that the books in black covers stand side by side?

20. How many ways can you find to pack 9 different books into five parcels if four of the parcels must contain two books each?

21. How many ways are there to put 12 identical coins into five different purses for none of the purses to remain empty?

22. The lock of a safe consists of five discs each of which features the digits 0, 1, 2, . . . , 9. The safe can be opened by dialing a special combination of the digits. Will ten days be enough to open the safe if the "workday" lasts 13 hours, and five seconds are needed to dial one combination of digits?

23. Given the integers from 1 to 10^n . Which numbers among them are greater in number, those whose notation requires the use of the digit 9, or those which do not include this digit?

24. Write Newton's formula for the power of the binomial:

$$(a) (x+1)^7, \quad (b) \left(\sqrt{\frac{a}{b}} - \sqrt{\frac{b}{a}} \right)^5.$$

25. Using Newton's formula, calculate:

$$(a) (\sqrt{3} - \sqrt{2})^5; \quad (b) (\sqrt{7} + i)^4 + (\sqrt{7} - i)^4.$$

26. Find:

$$(a) \text{ the seventh term of the expansion } (\sqrt{a} - \sqrt[3]{b})^{18};$$

$$(b) \text{ the term of the expansion } (a^{1/3} + a^{-1/3})^{15} \text{ independent of } a;$$

$$(c) \text{ the term of the expansion } (\sqrt{x} + \sqrt{2})^{18} \text{ containing } x^8;$$

$$(d) \text{ the term of the expansion } (x^5 + 1)^{1980} \text{ containing } x^{1980}.$$

27. Find the terms of the expansion which are integers:

$$(a) (\sqrt{2} + \sqrt[3]{3})^8; \quad (b) (\sqrt{5} - \sqrt{2})^8.$$

28. How many terms of the expansion $(\sqrt{3} + \sqrt[4]{5})^{124}$ are integers?

29. Find the leading coefficient of the polynomial:

$$(a) \left(\frac{1}{4} + \frac{3}{4}x \right)^4; \quad (b) (\sqrt{5} + \sqrt{2}x)^{20}.$$

30. Find the coefficient of the polynomial:

$$(a) (1 + x^2 - x^3)^9 \text{ in } x^8; \quad (b) (1 + x^3 + x^9)^7 \text{ in } x^{11}.$$

31. Find the sum of the coefficients of the polynomial:

(a) $(4x-5)^{21}$; (b) $\left(3\sqrt{2}x - \frac{5}{\sqrt{2}}\right)^8$.

32. Prove the equalities:

(a) $kC_n^k = nC_{n-1}^{k-1}$;

(b) $C_n^1 - 2C_n^2 + 3C_n^3 - \dots + (-1)^{n-1}nC_n^n = 0$;

(c) $C_n^1 + 2C_n^2 + 3C_n^3 + \dots + nC_n^n = n2^{n-1}$;

(d) $(C_n^0)^2 + (C_n^1)^2 + (C_n^2)^2 + \dots + (C_n^n)^2 = C_{2n}^n$.

33. One ball is drawn at random from an urn containing 3 white, 4 black, and 5 red balls. What is the probability of the drawn ball being (a) white, (b) black, (c) yellow, (d) red?

34. Two dice are thrown. What is the probability of the score on the 2 dice being equal to 8?

35. A cube with coloured faces has been sawn into 64 blocks of the same size. The blocks are then thoroughly mixed. What is the probability that two randomly chosen blocks have two coloured faces each?

36. The homework of each one of 4 students can be checked by one of the 7 teachers. What is the probability of all the 4 papers being checked by different teachers?

37. The examination programme contains 40 questions. To pass the examination, every student has to answer two of them. One student prepared the answers to 30 questions. What is the probability of his getting two questions at the examination to which he had prepared the answers?

38. There are 50 tickets in a lottery of which 8 are winning tickets. What is the probability of 2 out of 5 tickets chosen at random being winning tickets?

39. Eighteen basketball teams take part in the championship, of which 2 teams are in the top class. To decrease the overall number of matches, the teams are divided into two equal groups by casting lots. What is the probability of 2 top class teams (a) getting into different groups; (b) getting into one group?

40. Prove that the probability of the birthdays of twelve randomly chosen people being in different months is less than $1/10000$.

Chapter 4

The Integral

4.1. The Antiderivative and the Indefinite Integral

In many problems we encounter a situation when we are given a function and have to find a new function whose derivative at every point is equal to the value of the given function at that point. As an illustration, we may cite a problem of finding the law of motion $s(t)$ of a particle proceeding from the given law of its velocity $v(t)$. In this example we are given a function $v(t)$, $t \in (a, b)$, and are required to find a function $s(t)$ such that $s'(t) = v(t)$ for any $t \in (a, b)$.

Definition. The function $F(x)$, $x \in (a, b)$ is known as an *antiderivative (primitive)* of the function $f(x)$, $x \in (a, b)$ if $F'(x) = f(x)$ for every $x \in (a, b)$.

For the function $f(x) = 3x^2$; $x \in \mathbb{R}$, the antiderivative is a function $F(x) = x^3$ since $(x^3)' = 3x^2$ for any $x \in \mathbb{R}$. It is easy to see that the function $F(x) = x^3 + 2$ is also an antiderivative of the function $f(x) = 3x^2$ since $(x^3 + 2)' = 3x^2$ for any $x \in \mathbb{R}$.

It is evident that in a general case, as well, if $F(x)$ is an antiderivative of the function $f(x)$, then for any constant C the function $F(x) + C$ is also an antiderivative of the function $f(x)$. Indeed,

$$(F(x) + C)' = F'(x) = f(x)$$

for any x from the interval being considered.

Thus, if a function has at least one antiderivative, then it possesses an infinite number of antiderivatives.

The following theorem holds true.

If the function $F(x)$ is an antiderivative of the function $f(x)$, then any antiderivative of $f(x)$ has the form $F(x) + C$, where C is some constant, that is, the set $\{F(x) + C \mid C \in \mathbb{R}\}$ is the set of all antiderivatives of the function $f(x)$.

This theorem implies that for any C the function $F_1(x) = x^3 + C$ is an antiderivative of the function $f(x) = 3x^2$ and there are no other antiderivatives of this function.

Example 1. For the function $f(x) = \cos x + \sin x$, find an antiderivative $F(x)$ satisfying the condition $F(0) = 1$.

$\triangle$ One of antiderivatives of the function $f(x) = \cos x + \sin x$ is the function $\sin x - \cos x$. It follows from the theorem that any antiderivative of the given function has the form $F(x) = \sin x - \cos x + C$, where C is some constant. To find the constant C , we make use of the condition $F(0) = 1$ and get

$$\sin 0 - \cos 0 + C = 1,$$

i.e. $C = 1 + \cos 0 = 2$. Consequently, the function $F(x) = \sin x - \cos x + 2$ is the antiderivative sought for. $\blacktriangle$

The following three rules hold for finding antiderivatives.

1. *If the function $F(x)$ is an antiderivative of the function $f(x)$ and the function $G(x)$ is an antiderivative of the function $g(x)$, then the function $F(x) + G(x)$ is an antiderivative of the function $f(x) + g(x)$.*

$\square$ In fact, the derivative of the sum of functions is equal to the sum of the derivatives, that is,

$$(F(x) + G(x))' = F'(x) + G'(x) = f_1(x) + g(x),$$

which means that the function $F(x) + G(x)$ is an antiderivative of the function $f(x) + g(x)$. $\blacksquare$

2. *If the function $F(x)$ is an antiderivative of the function $f(x)$, then the function $kF(x)$ is an antiderivative of the function $kf(x)$.*

$\square$ Indeed,

$$(kF_1(x))' = kF'(x) = kf(x),$$

that is, the function $kF(x)$ is an antiderivative of $kf(x)$. $\blacksquare$

3. *If the function $F(x)$ is an antiderivative of the function $f(x)$, then the function $F(y(t))$ is an antiderivative of the function $f(y(t)) y'(t)$. (In this case we also assume that the functions $f(y(t))$, $y'(t)$ and $F(y(t))$ are defined.)*

This rule follows immediately from the rule for differentiating a composite function.

In the special case, when $y(t) = at + b$, $a \neq 0$, and when $F(x)$ is an antiderivative of the function $f(x)$, the

function $\frac{1}{a}F(at+b)$ is an antiderivative of the function $f(at+b)$.

Example 2. Find all antiderivatives of the function

$$f(x) = 3e^{2x+1} + \frac{1}{\cos^2(\pi x/4)}.$$

$\triangle$ The function $f(x)$ is equal to the sum of two functions. Let us find an antiderivative for the function $3e^{2x+1}$. Since the antiderivative of the function e^x is the function e^x , then by Rule 3, the antiderivative of the function e^{2x+1} is the function $\frac{1}{2}e^{2x+1}$. By Rule 2, the antiderivative of the function $3e^{2x+1}$ is the function $3 \cdot \frac{1}{2}e^{2x+1}$, i.e. $\frac{3}{2}e^{2x+1}$. Let us now find an antiderivative for the function $1/\cos^2(\pi x/4)$. Since the function $\tan x$ is an antiderivative of the function $1/\cos^2 x$, then, by Rule 3, the function $\frac{4}{\pi} \tan \frac{\pi x}{4}$ is an antiderivative of the function $1/\cos^2(\pi x/4)$. By Rule 1, the function $\frac{3}{2}e^{2x+1} + \frac{4}{\pi} \tan \frac{\pi x}{4}$ is an antiderivative of the function $f(x)$. Any antiderivative of the function $f(x)$ is specified by the formula

$$\frac{3}{2}e^{2x+1} + \frac{4}{\pi} \tan \frac{\pi x}{4} + C,$$

where C is an arbitrary constant. $\blacktriangle$

Example 3. Find the function $F(x)$ if it is known that

$$F'(x) = e^{\sin x} \cos x \text{ and } F(\pi/2) = e + 3.$$

$\triangle$ It is required to choose, out of all antiderivatives of the function $e^{\sin x} \cos x$, the antiderivative which assumes the value $e + 3$ for $x = \pi/2$. Since the derivative of $\sin x$ is $\cos x$ and since the antiderivative of e^x is e^x , it follows, by Rule 3, that the antiderivative of $e^{\sin x} \cos x$ is the function $e^{\sin x}$. Indeed,

$$(e^{\sin x})' = e^{\sin x} (\sin x)' = e^{\sin x} \cos x.$$

Any antiderivative of the function $e^{\sin x} \cos x$ has the form $e^{\sin x} + C$, where C is an arbitrary constant. Let us determine C from the condition that $F(\pi/2) = e + 3$, from

which we find that

$$e^{\sin(\pi/2)} + C = e + 3, \quad \text{i.e. } C = 3.$$

Consequently, the required function has the form $e^{\sin x} + 3$. ▲

Definition. The collection of all antiderivatives of the function $f(x)$, $x \in (a, b)$, is called the *indefinite integral* of the function $f(x)$ and is designated as

$$\int f(x) dx$$

(read “the indefinite integral of the function $f(x)$ with respect to dx ”, or, briefly, “the integral of $f(x)$ with respect to dx ”).

It follows that if $F(x)$ is an antiderivative of the function $f(x)$, then

$$\int f(x) dx = \{F(x) + C | C \in \mathbb{R}\}. \quad (1)$$

In practical applications, formula (1) is written in the concise form

$$\int f(x) dx = F(x) + C. \quad (2)$$

The function $f(x)$ is called the *integrand*, the expression $f(x) dx$ is the *element of integration* and the constant C is the *integration constant*.

Example 4. Find $\int \cos x dx$.

△ To find the indefinite integral of $\cos x$ with respect to dx , it is sufficient to find some antiderivative of the function $\cos x$, that is, to find a function $F(x)$ such that $F'(x) = \cos x$. From the table of derivatives it follows that $\sin x$ is such a function. Consequently,

$$\int \cos x dx = \sin x + C. \quad \blacktriangle$$

The operation of finding a function from its derivative is called the *integration of the function*. The operation of integration is the inverse of differentiation. The correctness of integration is checked by differentiation. The property of differentiation easily yields the following two properties

of indefinite integrals:

$$\int (f(x) + g(x)) dx = \int f(x) dx + \int g(x) dx, \quad (3)$$

$$\int kf(x) dx = k \int f(x) dx, \quad k \neq 0. \quad (4)$$

Formula (3) signifies that every antiderivative of the function $f(x) + g(x)$ is the sum of certain antiderivatives of $f(x)$ and $g(x)$ and, conversely, the sum of any antiderivatives of $f(x)$ and $g(x)$ is the antiderivative of $f(x) + g(x)$. And so it follows from formula (3) that if $F(x)$ is an antiderivative of $f(x)$, and $G(x)$ is an antiderivative of $g(x)$, then

$$\int (f(x) + g(x)) dx = F(x) + G(x) + C,$$

where C is an arbitrary constant. Briefly, formula (3) is read as follows: "the integral of the sum of functions is equal to the sum of the integrals of those functions".

Formula (4) means that every antiderivative of the function $kf(x)$, where $k \neq 0$, is the product of the constant k by some antiderivative of the function $f(x)$ and that any such product is an antiderivative of the function $kf(x)$.

It follows from formula (4) that if $F(x)$ is an antiderivative of $f(x)$ and $k \neq 0$, then

$$\int kf(x) dx = kF(x) + C,$$

where C is an arbitrary constant.

Note that formula (4) is no longer true in the case when $k = 0$. Indeed, in that case, the left-hand side is

$$\int 0 \cdot f(x) dx = \int 0 \cdot dx = C,$$

where C is an arbitrary constant, and the right-hand side is

$$0 \cdot \int f(x) dx = 0.$$

Using the table of derivatives, we compile a table of indefinite integrals.

$$1. \int a dx = ax + C.$$

$$2. \int x^\alpha dx = \frac{x^{\alpha+1}}{\alpha+1} + C, \quad \alpha \neq -1,$$

$$3. \int \frac{1}{x} dx = \ln |x| + C.$$

$$4. \int a^x dx = \frac{a^x}{\ln a} + C, \quad a > 0, \quad a \neq 1.$$

In particular, $\int e^x dx = e^x + C$.

$$5. \int \cos x dx = \sin x + C.$$

$$6. \int \sin x dx = -\cos x + C.$$

$$7. \int \frac{dx}{\cos^2 x} dx = \tan x + C.$$

$$8. \int \frac{dx}{\sin^2 x} dx = -\cot x + C.$$

Formulas 2, 3, 7 and 8 are true for the integrals in which the integrand is defined.

Let us prove formula 3 from the table, for instance.

□ The integrand is defined for all $x \neq 0$. If $x > 0$, then $|x| = x$ and

$$(\ln |x|)' = (\ln x)' = 1/x.$$

Now if $x < 0$, then $|x| = -x$ and

$$(\ln |x|)' = (\ln (-x))' = \frac{(-x)'}{-x} = \frac{-1}{-x} = \frac{1}{x}.$$

Consequently,

$$(\ln |x|)' = 1/x$$

for any $x \neq 0$ and, therefore

$$\int \frac{dx}{x} = \ln |x| + C$$

for $x > 0$ and for $x < 0$. ■

Example 5. Find $\int (x^2 + 2e^x) dx$.

△ In accordance with formulas (3) and (4), we have

$$\int (x^2 + 2e^x) dx = \int x^2 dx + 2 \int e^x dx.$$

The integrals on the right-hand side of the equation are tabular:

$$\int x^2 dx = \frac{1}{3} x^3 + C, \quad \int e^x dx = e^x + C.$$

It follows that the function $\frac{1}{3} x^3 + 2e^x$ is an antiderivative of the function $x^2 + 2e^x$ and, therefore,

$$\int (x^2 + 2e^x) dx = \frac{1}{3} x^3 + 2e^x + C,$$

where C is an arbitrary constant. $\blacktriangle$

Example 6. Find $\int \frac{3-x^2}{x} dx$.

$$\begin{aligned} \triangle \int \frac{3-x^2}{x} dx &= \int \frac{3}{x} dx - \int \frac{x^2}{x} dx \\ &= 3 \int \frac{1}{x} dx - \int x dx = 3 \ln|x| - \frac{x^2}{2} + C. \quad \blacktriangle \end{aligned}$$

Example 7. Find $\int \sin^2 x dx$.

$$\begin{aligned} \triangle \int \sin^2 x dx &= \int \frac{1 - \cos 2x}{2} dx \\ &= \int \frac{1}{2} dx - \int \frac{\cos 2x}{2} dx = \frac{1}{2} x - \frac{1}{4} \sin 2x + C. \quad \blacktriangle \end{aligned}$$

To find the integrals of the functions of the form $\sin mx \times \cos nx$, $\sin mx \sin nx$, and $\cos mx \cos nx$, it is convenient to use the following trigonometric formulas:

$$2\sin mx \cos nx = \sin(m-n)x + \sin(m+n)x, \quad (5)$$

$$2\sin mx \sin nx = \cos(m-n)x - \cos(m+n)x, \quad (6)$$

$$2\cos mx \cos nx = \cos(m-n)x + \cos(m+n)x. \quad (7)$$

Example 8. Find $\int \sin x \cos 3x dx$.

$\triangle$ From formula (5) we have

$$\sin x \cos 3x = \frac{1}{2} (\sin 4x - \sin 2x).$$

Therefore,

$$\int \sin x \cos 3x dx = \frac{1}{2} \int \sin 4x dx - \frac{1}{2} \int \sin 2x dx.$$

It can be easily seen that the function $-\frac{1}{4}\cos 4x$ is an antiderivative of the function $\sin 4x$. Similarly, the function $-\frac{1}{2}\cos 2x$ is an antiderivative of $\sin 2x$. Consequently,

$$\begin{aligned}\int \sin x \cos 3x \, dx &= \frac{1}{2} \left(-\frac{1}{4} \cos 4x \right) - \frac{1}{2} \left(-\frac{1}{2} \cos 2x \right) + C \\ &= \frac{1}{4} \cos 2x - \frac{1}{8} \cos 4x + C. \quad \blacktriangle\end{aligned}$$

Indefinite integrals can often be easily evaluated by the formula for integration by parts.

Suppose the functions $u(x)$ and $v(x)$ possess derivatives $u'(x)$ and $v'(x)$. Then, the following formula can be written which is known as the *formula for integration by parts*:

$$\int uv' \, dx = uv - \int u'v \, dx. \quad (8)$$

This formula is a consequence of the formula for the derivative of the product of two functions: $(uv)' = u'v + uv'$. Really,

$$\int (uv)' \, dx = \int u'v \, dx + \int uv' \, dx,$$

and, therefore,

$$\int uv' \, dx = uv - \int u'v \, dx.$$

Note that here we do not write the arbitrary constant since it is contained in the written integrals.

Example 9. Find $\int x \sin x \, dx$.

$\triangle$ Since $\sin x$ is a derivative of $-\cos x$, we have

$$\int x \sin x \, dx = \int x (-\cos x)' \, dx.$$

Applying formula (8), we get

$$\begin{aligned}\int x (-\cos x)' \, dx &= x (-\cos x) - \int (-\cos x) x' \, dx \\ &= -x \cos x + \int \cos x \, dx = -x \cos x + \sin x + C. \quad \blacktriangle\end{aligned}$$

Example 10. Find $\int \ln x \, dx$.

$\triangle$ To find the integral, we use formula (8) and set $u = \ln x$ and $v' = 1$. Then we have $v = x$ and $u' = 1/x$ and, therefore,

$$\int \ln x \, dx = x \ln x - \int dx = x \ln x - x + C. \blacktriangle$$

4.2. The Integral and the Formula of Newton-Leibniz

Definition. Suppose we are given the function $f(x)$ defined on some interval X and assume that $F(x)$, $x \in X$, is an antiderivative of the function $f(x)$. Then, the number $F(b) - F(a)$, where $a \in X$ and $b \in X$, is called the *integral of the function $f(x)$ from a to b* and is designated as

$$\int_a^b f(x) \, dx.$$

Thus, according to the definition, if $F'(x) = f(x)$, then

$$\int_a^b f(x) \, dx = F(b) - F(a). \quad (1)$$

The function f is the *integrand*, the variable x is the *variable of integration*, the interval with the end points a and b is the *interval of integration*, the number b is the *upper limit of integration*, and the number a is the *lower limit of integration*.

Equation (1) is called the *Newton-Leibniz formula*.

For the sake of convenience the difference $F(b) - F(a)$ can be written as $F(x) \Big|_a^b$ and then the Newton-Leibniz formula assumes the form

$$\int_a^b f(x) \, dx = F(x) \Big|_a^b.$$

Note that the integral of the function $f(x)$ from a to b does not depend on the choice of an antiderivative for the function $f(x)$. In fact, if $\Phi(x)$ is some antiderivative of

$f(x)$, different from $F(x)$, then there is a constant C such that $\Phi(x) = F(x) + C$ for any x from the interval being considered and, therefore $\Phi(b) - \Phi(a) = F(b) - F(a)$.

Example 1. Calculate $\int_1^2 2^x dx$.

$\triangle$ Since $(2^x/\ln 2)' = 2^x$, the function $F(x) = 2^x/\ln 2$ is an antiderivative of the function $f(x) = 2^x$ and, therefore, we get by the Newton-Leibniz formula

$$\int_1^2 2^x dx = \left. \frac{2^x}{\ln 2} \right|_1^2 = \frac{1}{\ln 2} (2^2 - 2) = \frac{2}{\ln 2}. \quad \blacktriangle$$

We shall now formulate some properties of integrals, assuming that all the integrals being considered do exist.

Property 1.

$$\int_a^b (f(x) + g(x)) dx = \int_a^b f(x) dx + \int_a^b g(x) dx. \quad (2)$$

Property 2.

$$\int_a^b kf(x) dx = k \int_a^b f(x) dx. \quad (3)$$

Example 2. Calculate $\int_1^2 (x + 3 \cos x) dx$.

$\triangle$ In accordance with properties 1 and 2, we have

$$\begin{aligned} \int_1^2 (x + 3 \cos x) dx &= \int_1^2 x dx + 3 \int_1^2 \cos x dx \\ &= \left. \frac{x^2}{2} \right|_1^2 + 3 \left. \sin x \right|_1^2 = 2 - \frac{1}{2} + 3 \sin 2 - 3 \sin 1 \\ &= \frac{3}{2} + 3 \sin 2 - 3 \sin 1. \quad \blacktriangle \end{aligned}$$

Property 3.

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx. \quad (4)$$

It is easy to prove properties 1, 2, and 3 proceeding from the definition of the integral. Let us prove Property 3, for instance.

□ Suppose $F(x)$ is an antiderivative of the function $f(x)$ on a certain interval which contains points a , b , and c in its interior. Then

$$\begin{aligned} \int_a^b f(x) dx &= F(b) - F(a) = F(b) - F(c) + F(c) - F(a) \\ &= \int_c^b f(x) dx + \int_a^c f(x) dx. \quad \blacksquare \end{aligned}$$

Property 4. Assume that the functions $f(x)$ and $\varphi(t)$ are such that the composite function $f(\varphi(t))$ is defined and continuous. Then, in the case when the function $\varphi(t)$ has a derivative $\varphi'(t)$, the following formula holds true:

$$\int_{\alpha}^{\beta} f(\varphi(t)) \varphi'(t) dt = \int_a^b f(x) dx, \quad (5)$$

where $a = \varphi(\alpha)$ and $b = \varphi(\beta)$.

□ Suppose $F(x)$ is an antiderivative of the function $f(x)$. Then, as it is easy to see, the function $F(\varphi(t))$ is an antiderivative of the function $f(\varphi(t)) \varphi'(t)$. Therefore,

$$\begin{aligned} \int_{\alpha}^{\beta} f(\varphi(t)) \varphi'(t) dt &= F(\varphi(\beta)) - F(\varphi(\alpha)) \\ &= F(b) - F(a) = \int_a^b f(x) dx. \quad \blacksquare \end{aligned}$$

Formula (5) is known as the *formula for a change of variable of integration*. It reduces the calculation of the integral of one function to the calculation of the integral of some other function, which is simpler, in a certain sense. Formula (5) can be used both from left to right and from right to left.

Suppose we are given the integral $\int_a^b f(x) dx$. When replacing x by t by the formula $x = \varphi(t)$, we must substitute

$\varphi(t)$ for x and $\varphi'(t) dt$ for dx , and find the new limits of integration α and β from the conditions $\varphi(\alpha) = a$, $\varphi(\beta) = b$. Then we have

$$\int_a^b f(x) dx = \int_{\alpha}^{\beta} f(\varphi(t)) \varphi'(t) dt. \quad (5')$$

Example 3. Calculate $\int_0^a \sqrt{a^2 - x^2} dx$, $a > 0$.

$\triangle$ Let us pass to the new integration variable t by setting $x = a \sin t$, where $t \in [0; \pi/2]$. When t varies from 0 to $\frac{\pi}{2}$, x varies from 0 to a . In the example $f(x) = \sqrt{a^2 - x^2}$. Substituting $x = a \sin t$, we get

$$f(a \sin t) = \sqrt{a^2 - a^2 \sin^2 t} = a \sqrt{1 - \sin^2 t} = a \cos t.$$

We replace dx by $(a \sin t)' dt = a \cos t dt$.

By formula (5) we get

$$\int_0^a \sqrt{a^2 - x^2} dx = \int_0^{\pi/2} a \cos t \cdot a \cos t = a^2 \int_0^{\pi/2} \cos^2 t dt.$$

Since $\cos^2 t = \frac{1 + \cos 2t}{2}$, we have

$$\begin{aligned} a^2 \int_0^{\pi/2} \cos^2 t dt &= \frac{a^2}{2} \int_0^{\pi/2} dt + \frac{a^2}{2} \int_0^{\pi/2} \cos 2t dt \\ &= \frac{a^2}{2} t \Big|_0^{\pi/2} + \frac{a^2}{2 \cdot 2} \sin 2t \Big|_0^{\pi/2} = \frac{\pi a^2}{4}. \quad \blacktriangle \end{aligned}$$

Given the integral $\int_a^b f(\varphi(x)) \varphi'(x) dx$. When replacing x by y by the formula $y = \varphi(x)$, we must substitute y for $\varphi(x)$ and dy for $\varphi'(x) dx$, and take the numbers $A = \varphi(a)$, $B = \varphi(b)$ as the new limits of integration. Then we have

$$\int_a^b f(\varphi(x)) \varphi'(x) dx = \int_A^B f(y) dy. \quad (5'')$$

Example 4. Calculate $\int_0^{\pi/2} \sin^2 x \cos x \, dx$.

$\triangle$ The given integral can be calculated directly by definition, first finding one of the antiderivatives for the integrand. There is, however, a simpler method of calculation consisting in the use of the formula for a change of the integration variable. Indeed, let us pass to a new integration variable by setting $y = \sin x$, and substituting y^2 for $\sin^2 x$ in the integrand.

Since the derivative of $\sin x$ is equal to $\cos x$, we substitute dy for $\cos x \, dx$. We find the new limits of integration from the conditions $A = \sin 0 = 0$ and $B = \sin(\pi/2) = 1$. By formula (5'') we get

$$\int_0^{\pi/2} \sin^2 x \cos x \, dx = \int_0^1 y^2 \, dy = \frac{1}{3} y^3 \Big|_0^1 = \frac{1}{3}. \quad \blacktriangle$$

The *formula for integration by parts* is also often used in calculating integrals:

$$\int_a^b uv' \, dx = uv \Big|_a^b - \int_a^b u'v \, dx.$$

As in the case of the indefinite integral, this formula is a direct consequence of the formula for the derivative of a product of functions.

Example 5. Calculate $\int_0^2 xe^x \, dx$.

$\triangle$ We make use of the formula for integration by parts. For that purpose, we put $u = x$, $v' = e^x$. Then, $u' = 1$ and $v = e^x + C$, where C is an arbitrary constant. We shall consider, for simplicity, that $C = 0$, that is, take the function e^x for v . Integrating by parts, we obtain

$$\begin{aligned} \int_0^2 xe^x \, dx &= xe^x \Big|_0^2 - \int_0^2 e^x \, dx = 2e^2 - e^x \Big|_0^2 \\ &= 2e^2 - e^2 + 1 = e^2 + 1. \quad \blacktriangle \end{aligned}$$

Property 5. If $f(x) \leq g(x)$ for any $x \in [a; b]$, then

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx. \quad (6)$$

□ Let $F(x)$ be an antiderivative of the function $f(x)$ and $G(x)$ be an antiderivative of the function $g(x)$ such that $F(a) = G(a) = 0$. Then we have

$$F(x) = \int_a^x f(t) dt, \quad G(x) = \int_a^x g(t) dt.$$

Recall that these antiderivatives are called *integrals with variable upper limit*.

Let us consider the function $\varphi(x) = G(x) - F(x)$. This is a non-decreasing function and $\varphi(a) = 0$. Indeed, $\varphi'(x) = G'(x) - F'(x) = g(x) - f(x) \geq 0$ for any $x \in [a, b]$ and $\varphi(a) = G(a) - F(a) = 0$. Consequently, $\varphi(b) = G(b) - F(b) \geq 0$, i.e. $G(b) \geq F(b)$, and that is what we had to prove. ■

Corollary. If m and M are the least and the greatest value of the function $f(x)$ on the interval $[a; b]$, then

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a). \quad (7)$$

□ Indeed, since $m \leq f(x) \leq M$, Property 5 implies the validity of the inequality

$$\int_a^b m dx \leq \int_a^b f(x) dx \leq \int_a^b M dx,$$

which yields inequality (7) if it is taken into account that

$$\int_a^b m dx = m(b-a), \quad \int_a^b M dx = M(b-a). \quad \blacksquare$$

Example 6. Using formula (7), estimate the integral

$$\int_0^1 e^{-x^2} dx.$$

△ The least value of the function e^{-x^2} on the interval $[0; 1]$ is equal to e^{-1} and the greatest value is 1, and, therefore,

$$e^{-1} \leq \int_0^1 e^{-x^2} dx \leq 1. \quad \blacktriangle$$

4.3. The Area of a Curvilinear Trapezoid

Suppose we are given a nonnegative continuous function $f(x)$, $x \in [a; b]$. The set of points of the plane whose coordinates satisfy the inequalities $a \leq x \leq b$, $0 \leq y \leq f(x)$ is called a *curvilinear trapezoid*.

A curvilinear trapezoid $aABb$ is bounded by the graph of the function $f(x)$, the line segments $x = a$ and $x = b$, and a segment of the Ox axis (Fig. 10). Let us derive a formula for calculating the area S of that curvilinear trapezoid. We divide the interval $[a, b]$ by the points

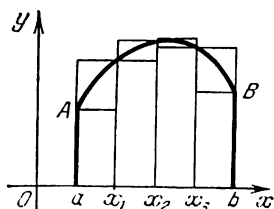


Fig. 10

$$x_i = a + \frac{b-a}{n} i$$

$$(i = 0, 1, 2, \dots, n).$$

into n subintervals $[x_{i-1}; x_i]$ ($i = 1, 2, \dots, n$) of equal length. Taking each subinterval $[x_{i-1}; x_i]$ as the base, we construct two rectangles with altitudes m_i and M_i , where m_i and M_i are the least and the greatest value of the function $f(x)$ on the interval $[x_{i-1}; x_i]$. The areas of these rectangles are equal, respectively, to $m_i \Delta x_i$ and $M_i \Delta x_i$, where $\Delta x_i = x_i - x_{i-1}$.

We denote by s_n the sum of the areas of all rectangles with the altitude m_i , that is,

$$s_n = \sum_{i=1}^n m_i \Delta x_i = m_1' \Delta x_1 + \dots + m_n \Delta x_n.$$

By analogy, we denote by S_n the sum of all rectangles with the altitude M_i , that is,

$$S_n = \sum_{i=1}^n M_i \Delta x_i = M_1 \Delta x_1 + \dots + M_n \Delta x_n.$$

It is evident that $s_n \leq S \leq S_n$. We can show that as $n \rightarrow \infty$, s_n and S_n tend to the same limit, to the area of the curvilinear trapezoid $aABb$.

By virtue of inequality (7) from 4.2 we get n inequalities

$$m_i \Delta x_i \leq \int_{x_{i-1}}^{x_i} f(x) dx \leq M_i \Delta x_i,$$

where $i = 1, 2, \dots, n$. Consequently,

$$s_n \leq \sum_{i=1}^n \int_{x_{i-1}}^{x_i} f(x) dx \leq S_n.$$

And since

$$\sum_{i=1}^n \int_{x_{i-1}}^{x_i} f(x) dx = \int_a^b f(x) dx,$$

we have

$$s_n \leq \int_a^b f(x) dx \leq S_n.$$

Hence, in the limit, as $n \rightarrow \infty$, there holds the inequality

$$S = \int_a^b f(x) dx. \quad (1)$$

Note that for any $\xi_i \in [x_{i-1}; x_i]$ there holds the inequality $m_i \leq f(\xi_i) \leq M_i$ and, therefore,

$$\sum_{i=1}^n m_i \Delta x_i \leq \sum_{i=1}^n f(\xi_i) \Delta x_i \leq \sum_{i=1}^n M_i \Delta x_i.$$

Hence, in the limit, as $n \rightarrow \infty$, we get

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i) \Delta x_i = S = \int_a^b f(x) dx.$$

Example 1. Show that formula (1) is valid for a right-angled triangle.

△ Suppose we are given a right-angled triangle with legs of lengths a and b . We introduce the system of coordinates on the plane as shown in Fig. 11. The given triangle is a special case of a curvilinear trapezoid, with $f(x) = \frac{b}{a}x$, $x \in [0; a]$.

Next we divide the interval $[0; a]$ into n subintervals of length a/n by the points $x_i = \frac{a}{n}i$ ($i = 0, 1, 2, \dots, n$).

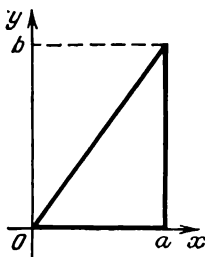


Fig. 11

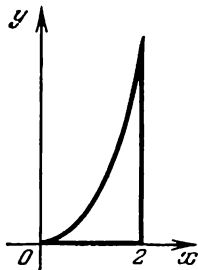


Fig. 12

Then $m_i = \frac{b}{n}(i-1)$, $M_i = \frac{b}{n}i$ and, therefore,

$$s_n = \sum_{i=1}^n \frac{b}{n}(i-1) \frac{a}{n} = \frac{ab}{n^2} \frac{n(n-1)}{2} = \frac{ab}{2} \left(1 - \frac{1}{n}\right),$$

$$S_n = \sum_{i=1}^n \frac{b}{n}i \frac{a}{n} = \frac{ab}{n^2} \frac{n(n+1)}{2} = \frac{ab}{2} \left(1 + \frac{1}{n}\right).$$

This shows that

$$\lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} S_n = \frac{ab}{2}.$$

Thus we have proved that the area of the given triangle is equal to $\frac{1}{2}ab$. That is, it can be calculated by the known formula. ▲

Example 2. Find the area of the figure bounded by a part of the parabola $y = x^2$ and the line segments $y = 0$ and $x = 2$ (Fig. 12).

△ Using formula (1), we get

$$S = \int_0^2 x^2 dx = \frac{x^3}{3} \Big|_0^2 = \frac{8}{3}. \blacktriangle$$

Example 3. Find the area of the figure bounded by the curves $y = x$, $y = 1/x$, $y = 0$, and $x = e$.

△ We divide the given figure by the straight line $x = 1$ into two parts (Fig. 13): a curvilinear trapezoid corresponding to the function $y = x$ on the interval $[0; 1]$ and a curvilinear trapezoid corresponding to the function $y = 1/x$ on the interval $[1; e]$. Their areas will be denoted by S_1 and S_2 . Then we have

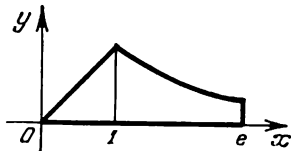


Fig. 13

$$S_1 = \int_0^1 x dx = \frac{x^2}{2} \Big|_0^1 = \frac{1}{2}, \quad S_2 = \int_1^e \frac{1}{x} dx = \ln x \Big|_1^e = \ln e = 1,$$

Consequently, $S = S_1 + S_2 = \frac{1}{2} + 1 = 1.5$. $\blacktriangle$

Let us consider the figure bounded by the line segments $x = a$ and $x = b$ ($a < b$) and by the graphs of the functions $f_1(x)$ and $f_2(x)$, nonnegative and continuous on the interval $[a; b]$, with $f_1(x) \leq f_2(x)$ (Fig. 14). The area of the given figure is equal to the difference of the areas of two curvilinear trapezoids $aABb$ and $aA'B'b$, that is

$$S = \int_a^b f_2(x) dx - \int_a^b f_1(x) dx = \int_a^b (f_2(x) - f_1(x)) dx. \quad (2)$$

The following statement holds true in a general case.

The area S of the figure the coordinates of whose points satisfy the inequalities $a \leq x \leq b$, $f_1(x) \leq y \leq f_2(x)$, where $f_1(x)$ and $f_2(x)$ are given continuous functions, is calculated by the formula

$$S = \int_a^b (f_2(x) - f_1(x)) dx. \quad (3)$$

□ Indeed, let m be the least value of the function $f_1(x)$ on the interval $[a; b]$. Then $f_2(x) - m \geq f_1(x) - m \geq 0$ for any $x \in [a; b]$. The figure whose coordinates satisfy the inequalities

$$a \leq x \leq b, \quad f_1'(x) - m \leq y \leq f_2(x) - m$$

can be obtained from the given figure by a parallel displacement along the Oy axis. Consequently, their areas are equal and, therefore (see formula (2)),

$$S = \int_a^b (f_2(x) - m - f_1(x) + m) dx = \int_a^b (f_2(x) - f_1(x)) dx. \blacksquare$$

Example 4. Find the area of the figure bounded by the curves $y = x^2 - 4x + 5$ and $y = x + 1$ (Fig. 15).

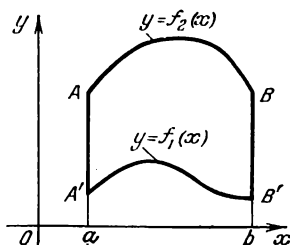


Fig. 14

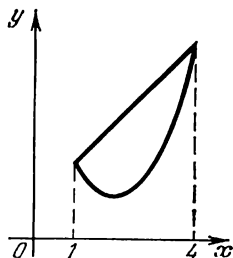


Fig. 15

△ To find the abscissas of the intersection points of the graphs of the functions $y = x + 1$ and $y = x^2 - 4x + 5$, we solve the equation $x + 1 = x^2 - 4x + 5$. From the quadratic equation $x^2 - 5x + 4 = 0$ we find that $x_1 = 1$ and $x_2 = 4$. Then, by formula (3), the required area is:

$$\begin{aligned} S &= \int_1^4 (x + 1 - (x^2 - 4x + 5)) dx = \int_1^4 (-x^2 + 5x - 4) dx \\ &= \left(-\frac{x^3}{3} + \frac{5}{2}x^2 - 4x \right) \Big|_1^4 \\ &= -\frac{1}{3}(64 - 1) + \frac{5}{2}(16 - 1) - 4(4 - 1) \\ &= -21 + 37.5 - 12 = 4.5. \blacktriangle \end{aligned}$$

If it is required to calculate the area of a more intricate figure, it should be divided into several figures whose areas can be calculated by formula (1), (2), or (3).

Example 5. Figure 16 shows a figure bounded by the curves $y = -x^2 + 6x - 5$, $y = -x^2 + 4x - 3$, and $y = 3x - 15$. Find the area of the figure.

△ Points A , B , and D are the intersection points of these parabolas and the x -axis. To find the abscissas of the points we solve two equations, $-x^2 + 6x - 5 = 0$ and $-x^2 + 4x - 3 = 0$. The roots of the first equation are the numbers $x = 1$ and $x = 5$, and the roots of the second equation are the numbers $x = 1$ and $x = 3$. Consequently, $x_A = 1$, $x_B = 5$, and $x_D = 3$. To determine the abscissa of the point C , we solve the equation $-x^2 + 4x - 3 = 3x - 15$. Solving the quadratic equation $x^2 - x - 12 = 0$, we find that $x_1 = -3$, $x_2 = 4$, i.e. $x_C = 4$. We partition the given figure by the line segments $[DB]$ and $[EC]$ into three figures and denote their areas by S_1 , S_2 , and S_3 , respectively, as is shown in Fig. 16.

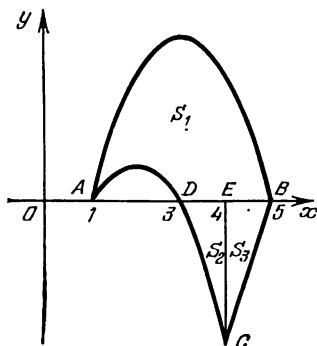


Fig. 16

Let us calculate the area S_1 . To do that, we subtract the area of the curvilinear trapezoid bounded by the graph of the function $y = -x^2 + 4x - 3$ and the line segment $[1; 3]$ of the x -axis from the area of the curvilinear trapezoid bounded by the graph of the function $y = -x^2 + 6x - 5$ and the segment $[1; 5]$ of the x -axis, that is,

$$\begin{aligned}
 S_1 &= \int_1^5 (-x^2 + 6x - 5) dx - \int_1^3 (-x^2 + 4x - 3) dx \\
 &= \left(-\frac{1}{3}x^3 + 3x^2 - 5x \right) \Big|_1^5 + \left(\frac{1}{3}x^3 - 2x^2 + 3x \right) \Big|_1^3 = \frac{28}{3}.
 \end{aligned}$$

To calculate the areas S_2 and S_3 , we apply formula (3):

$$S_2 = \int_3^4 (x^2 - 4x + 3) dx = \left(\frac{1}{3} x^3 - 2x^2 + 3x \right) \Big|_3^4 = \frac{4}{3},$$

$$S_3 = \int_4^5 (-3x + 15) dx = \left(-\frac{3}{2} x^2 + 15x \right) \Big|_4^5 = \frac{3}{2}.$$

Thus we have $S = S_1 + S_2 + S_3 = 73/6$. ▲

4.4. Application of Integrals to Calculating the Volume of Bodies

4.4.1. The volume of a pyramid. We shall first consider a triangular pyramid $ABCD$ whose edge AD is perpendicular to the plane ABC and express the volume V of that pyramid

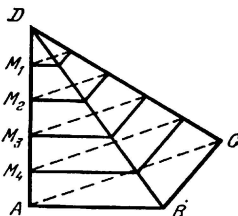


Fig. 17

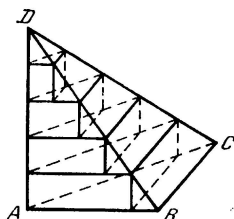


Fig. 18

in terms of the altitude $H = |AD|$ and the area of the base S , taking as a known fact that the volume of a right prism is equal to the product of the area of the base by the altitude.

We partition the segment DA by the points $M_1, M_2, \dots, M_{n-1}$ into n segments of equal length and draw a plane parallel to the base ABC through every point M_i (Fig. 17). Now the pyramid is divided into n parts: one pyramid and $n - 1$ truncated pyramids. The volume of the pyramid $ABCD$ is, evidently, not smaller than the volume v_n of the stepped pyramid inscribed into $ABCD$ (Fig. 18) and not greater than the volume V_n of the stepped pyramid circumscribed about $ABCD$ (Fig. 19):

$$v_n \leq V \leq V_n, \quad (1)$$

with $V_n - v_n = S \cdot \frac{H}{n}$.

Suppose $S(x)$ is the area of the section of the pyramid $ABCD$ by the plane passing parallel to the base ABC at the distance x from the vertex D , and let $x_0 = 0$, $x_i = |DM_i|$, and $x_n = |DA| = H$. Then $S(x_i)$ is the area of the cross section passing through the point M_i , and

$$= \sum_{i=1}^n S(x_{i-1}) \Delta x_i,$$

$$V_n = \sum_{i=1}^n S(x_i) \Delta x_i,$$

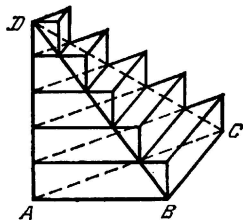


Fig. 19

where $\Delta x_i = x_i - x_{i-1} = H/n$. The function $S(x)$, $x \in [0; H]$, being increasing, we have

$$S(x_{i-1}) \Delta x_i \leq \int_{x_{i-1}}^{x_i} S(x) dx \leq S(x_i) \Delta x_i$$

for any $i = 1, 2, \dots, n$, and, therefore

$$v_n \leq \int_0^H S(x) dx \leq V_n. \quad (2)$$

Formulas (1) and (2) yield

$$\left| V - \int_0^H S(x) dx \right| \leq V_n - v_n = S \cdot \frac{H}{n} \rightarrow 0$$

as $n \rightarrow \infty$. Thus, the formula

$$V = \int_0^H S(x) dx \quad (3)$$

holds true. We know that $S(x)/S = x^2/H^2$. Taking this relation into consideration, we get, from formula (3), the relation

$$V = \int_0^H \frac{x^2}{H^2} S dx = \frac{S}{H^2} \frac{x^3}{3} \Big|_0^H = \frac{1}{3} SH.$$

Thus, the volume of a pyramid can be calculated by the formula

$$V = \frac{1}{3} SH, \quad (4)$$

where S is the area of the base and H is the altitude of the pyramid.

Formula (4) has been derived for a special case of triangular pyramids. It is easy to prove that it is also valid for any pyramid.

4.4.2. The volume of a solid of revolution. Suppose we are given a curvilinear trapezoid bounded by the graph of the nonnegative function $y = f(x)$, $x \in [a; b]$, the abscissa

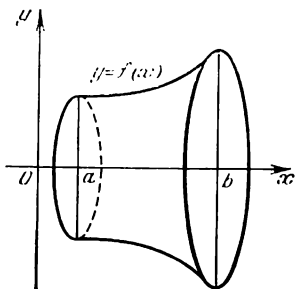


Fig. 20

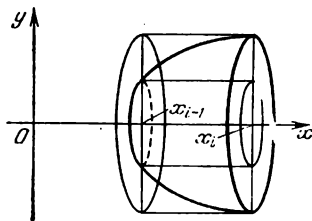


Fig. 21

axis, and the straight lines $x = a$ and $x = b$ (Fig. 20) Rotation of this curvilinear trapezoid about the x -axis produces a geometric body Φ known as a *solid of revolution*.

Let us derive a formula for the volume of the solid of revolution Φ . For simplicity, we shall assume that the function $f(x)$ is nondecreasing. Let $S(x)$ be the area of the section of the solid Φ by the plane passing through the point with abscissa x parallel to the plane Oyz . It is evident that any such cross section of the solid Φ is a circle or a point and $S(x) = \pi f^2(x)$. We divide the interval $[a; b]$ into n subintervals of equal length by the points $x_i = a + \frac{b-a}{n}i$, where $i = 0, 1, \dots, n$, and through each point draw a plane perpendicular to the abscissa axis. Then the solid Φ will be divided into n solids of revolution Φ_1 ,

$\Phi_2, \dots, \Phi_n$, the sum of whose volumes is equal to the volume of the solid Φ .

The volume of the solid Φ_i is evidently not smaller than the volume of the cylinder whose altitude is equal to $\Delta x_i = x_i - x_{i-1}$ and the radius of the base is equal to $y_{i-1} = f(x_{i-1})$ and not greater than the volume of the cylinder with the altitude Δx_i and the radius of the base $y_i = f(x_i)$ (Fig. 21). Consequently, the volume V of the solid Φ satisfies the inequalities

$$v_n \leq V \leq V_n, \quad (5)$$

where $v_n = \sum_{i=1}^n \pi y_{i-1}^2 \Delta x_i$ is the volume of the stepped solid of revolution inscribed into the solid Φ and $V_n = \sum_{i=1}^n \pi y_i^2 \Delta x_i$ is the volume of the stepped solid of revolution circumscribed about Φ . In this case,

$$V_n - v_n = \pi (y_n^2 - y_0^2) \cdot \frac{b-a}{n} \rightarrow 0$$

as $n \rightarrow \infty$. The monotonicity of the function $f(x)$ and the properties of the integral yield the inequalities

$$\pi y_{i-1}^2 \Delta x_i \leq \int_{x_{i-1}}^{x_i} \pi f^2(x) dx \leq \pi y_i^2 \Delta x_i,$$

where $i = 1, 2, \dots, n$. Summing with respect to i , we get from these inequalities

$$v_n \leq \int_a^b \pi f^2(x) dx \leq V_n. \quad (6)$$

Now it follows from (5) and (6) that

$$\left| V - \pi \int_a^b f^2(x) dx \right| \leq V_n - v_n \rightarrow 0$$

as $n \rightarrow \infty$, and, therefore,

$$V = \pi \int_a^b f^2(x) dx. \quad (7)$$

Taking into account that $\pi f^2(x) = S(x)$, we can write formula (7) for the volume of a solid of revolution in the form of the integral of the area $S(x)$ of its cross section:

$$V = \int_a^b S(x) dx. \quad (8)$$

Assume that the geometric solid Φ is contained between the planes $x = a$ and $x = b$ and let $S(x)$ be the area of the cross section of the solid Φ by a plane perpendicular to the x -axis and passing through the point with the abscissa $x \in [a; b]$. In that case, just as in the case of solids of revolution, the volume V of the solid Φ can be calculated by formula (8).

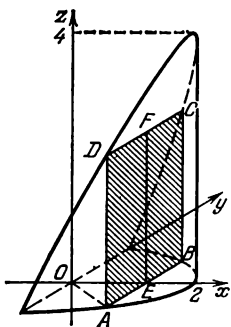


Fig. 22

the x -axis. Find the volume of the solid produced by the rotation.

△ Applying formula (7), we get

$$V = \pi \int_0^{\pi} \sin^2 x dx.$$

Replacing $\sin^2 x$ by $\frac{1 - \cos 2x}{2}$, we obtain

$$V = \pi \int_0^{\pi} \frac{1 - \cos 2x}{2} dx = \left(\frac{\pi x}{2} - \frac{\pi}{4} \sin 2x \right) \Big|_0^{\pi} = \frac{\pi^2}{2}. \quad \blacktriangle$$

Example 2. Find the volume of the solid cut from the cylinder $x^2 + y^2 = 4$ by the plane $z = 2x$ (Fig. 22).

△ Let us consider the sections of that solid by the planes perpendicular to the x -axis. The section passing through the point with abscissa $x \in (0; 2)$ at right angles to the x -axis is a rectangle. Let us designate the area of such a rectangle

as $S(x)$ and calculate it for $x \in (0; 2)$. The side $|AB|$ of the rectangle serving as the base of the cylinder is equal to

$$|AB| = 2|AE| = 2\sqrt{|OA|^2 - |OE|^2} = 2\sqrt{4 - x^2}.$$

The side $|AD|$ of the rectangle is equal to the segment $|EF|$ and to $2x$. This means that the area of the rectangle $ABCD$ is

$$S(x) = 2x\sqrt{4 - x^2}.$$

By formula (8), the volume of the solid is

$$V = \int_0^2 2x\sqrt{4 - x^2} dx.$$

We make a change of variable under the integral sign and set $y = 4 - x^2$. Then, by formula (5'') in 4.2, we get

$$\begin{aligned} V &= \int_0^2 -\sqrt{4 - x^2}(-2x) dx = -\int_4^0 \sqrt{y} dy \\ &= \int_0^4 \sqrt{y} dy = \frac{2}{3} y^{3/2} \Big|_0^4 = \frac{16}{3}. \quad \blacktriangle \end{aligned}$$

4.5. Application of Integrals to Problems from Physics

4.5.1. Problem of path calculation. Suppose a particle travels rectilinearly at the velocity $v = v(t)$ dependent on time t . It is required to find the path covered by the particle during the time interval from $t = T_1$ to $t = T_2$. If the velocity is constant and equal to v_0 , then the path S is equal to the product of the velocity by the time of motion, i.e. $S = v_0(T_2 - T_1)$.

Now if the velocity is not constant, then

$$S = \int_{T_1}^{T_2} v(t) dt. \quad (1)$$

□ Indeed, let us partition the interval $[T_1; T_2]$ into n subintervals of equal length, with $t_0 = T_1$, $t_i = T_1 + \frac{T_2 - T_1}{n} i$, where $i = 1, 2, \dots, n$. In a crude way, we shall

consider the velocity $v(t)$ on the interval $[t_{i-1}; t_i]$ to be constant and equal to $v(\tau_i)$, where τ_i is a point belonging to $[t_{i-1}; t_i]$. Then the path traversed by the particle during the time from t_{i-1} to t_i is approximately equal to the product $v(\tau_i)(t_i - t_{i-1}) = v(\tau_i) \Delta t_i$, and the path traversed by the particle during the time interval $[T_1; T_2]$ is approximately equal to $\sum_{i=1}^n v(\tau_i) \Delta t_i$. The path S traversed by the particle during the time from T_1 to T_2 is equal to the limit of that sum as $n \rightarrow \infty$, that is, to integral (1). ■

Example 1. A body moves rectilinearly at the rate $v(t) = 3 + 3t^2$ m/s. Find the path traversed by the body during the first 5 seconds.

△ By formula (1) we find

$$S = \int_0^5 (3 + 3t^2) dt = (3t + t^3) \Big|_0^5 = 15 + 125 = 140. \quad \blacktriangle$$

Example 2. A body moves rectilinearly at the rate $v(t) = t + 6t^2$ m/s. Find the path traversed by the body during the third second.

△ Formula (1) yields

$$S = \int_2^3 (t + 6t^2) dt = \left(\frac{1}{2} t^2 + 2t^3 \right) \Big|_2^3 = 40.5. \quad \blacktriangle$$

4.5.2. The work performed by a variable force. Suppose a particle moves along the x -axis under the action of a force $\mathbf{P}$. If the force $\mathbf{P}$ is constant and f is the projection of that force on the x -axis, then the product $f \cdot (b - a)$ is called the *work of the force* on the path interval $[a; b]$.

Let us derive a formula for calculating the work A of the force $\mathbf{P}$ in the case when the force is not constant.

Let $f(x)$ be the projection of the force $\mathbf{P}$ on the x -axis. We shall show that the work A of the force $\mathbf{P}$ on the interval $[a; b]$ can be calculated by the formula

$$A = \int_a^b f(x) dx. \quad (2)$$

□ We shall assume that $f(x)$ is a continuous function on the interval $[a; b]$. We divide the interval $[a; b]$ into n subintervals $[x_{i-1}; x_i]$ equal in length by the points

$$x_i = a + \frac{b-a}{n}i \quad (i = 0, 1, 2, \dots, n).$$

It is evident that the length of each of them is equal to $\Delta x_i = (b - a)/n$ and tends to zero as $n \rightarrow \infty$. We can assume, therefore, that the work of the force $\mathbf{P}$ on the interval $[x_{i-1}; x_i]$ is approximately equal to $f(x_i) \Delta x_i$, and the whole work performed on the interval $[a; b]$ is equal to the sum $\sum_{i=1}^n f(x_i) \Delta x_i$, which is the integral sum of the function $f(x)$ on the interval $[a; b]$. Then, in the limit, as $n \rightarrow \infty$, we get formula (2). ■

Note that sometimes it is convenient to write formula (2) in another form, namely,

$$A = \int_a^b (\mathbf{P}, \mathbf{e}) dx,$$

where $\mathbf{e}$ is a unit vector of the x -axis and $(\mathbf{P}, \mathbf{e})$ is the scalar product of the vectors $\mathbf{P}$ and $\mathbf{e}$.

Example 3. The force equal to 2 N extends the spring by 4 cm. What amount of work must be performed to extend the spring by 4 cm?

△ By Hooke's law, $F = kx$, that is, the force acting on the spring and extending it by the length x is proportional to that extension. From the condition $2 = k \cdot 0.04$ we find the coefficient of extension k . It is equal to 50. Formula (2) yields

$$A = \int_0^{0.04} 50x dx = 25x^2 \Big|_0^{0.04} = 0.04. \quad \blacktriangle$$

Example 4. The water fed from the plane of the base to a conical tank through an orifice in the bottom fills up the whole tank. Determine the work performed if the altitude of the tank is h , the radius of the lower base (the bottom) is r , and the radius of the upper base is R , $R > r$.

△ Let us consider the section of the conical tank passing through its axis and choose the axes of coordinates in that section as shown in Fig. 23. We partition the interval $[0; h]$ of the x -axis into n subintervals $[x_{i-1}; x_i]$ ($i = 1, 2, \dots, n$) equal in length by the points $x_i = \frac{n}{n} i$. The generatrix of the cone in the chosen system of coordinates obeys the equation $y = r + x \frac{R-r}{h}$. Consequently, the volume V_i of the i th layer is approximately equal to $\pi y_i^2 \Delta x_i$ and its potential energy is approximately equal to $\pi \rho g y_i^2 x_i \Delta x_i$, where ρ is the density of water and g is the acceleration due to gravity. Hence it follows that

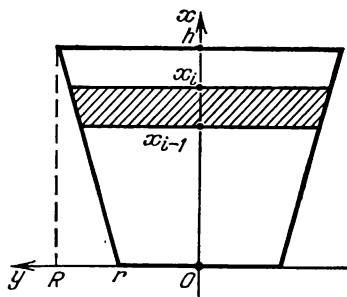


Fig. 23

that

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \pi \rho g y_i^2 x_i \Delta x_i,$$

that is,

$$A = \pi \rho g \int_0^h y^2 dx.$$

It is evident that the potential energy is equal to the sought-for work. Thus we have

$$A = \pi \rho g \int_0^h x \left(r + x \frac{R-r}{h} \right)^2 dx.$$

Calculating this integral, we obtain

$$A = \frac{\pi \rho g h^2}{12} (r^2 + 2rR + 3R^2). \blacktriangle$$

PROBLEMS OF SECTION I

1. Find all the antiderivatives for the following functions:

(a) $f(x) = x^5$; (b) $f(x) = \frac{1}{2} \sin(3x+1)$;

(c) $f(x) = \frac{1}{5x+6}$; (d) $f(x) = \frac{3}{\cos^2(2x+1)} + \cos 6x$.

2. Find all the antiderivatives of the function $f(x) = |x|$.

3. For the function $f(x) = x^2$ find the antiderivative whose graph passes through the point $M(2; 3)$.

4. For the function $f(x) = \frac{1}{x} + 2^{3x}$ find the antiderivative $F(x)$ satisfying the condition $F(1) = 3$.

5. Find the following indefinite integrals:

$$(a) \int (t^3 + 3e^{5t}) dt; \quad (b) \int \frac{x^{1/3} + x^{1/5}}{x^2} dx;$$

$$(c) \int \sin 2x \cos 4x dx; \quad (d) \int \left(5^{3x} + \frac{1}{\sin^2(2x+1)} + \frac{1}{\sqrt[5]{4x-1}} \right) dx.$$

6. Applying the formula for integration by parts, find the following indefinite integrals:

$$(a) \int x \cos x dx; \quad (b) \int x \ln x dx.$$

7. Prove that the function

$$f(x) = \operatorname{sign} x = \begin{cases} 1, & x > 0, \\ 0, & x = 0, \\ -1, & x < 0 \end{cases}$$

has no antiderivatives on the entire axis.

8. Calculate the following integrals:

$$(a) \int_0^1 x^3 dx; \quad (b) \int_{-\pi/4}^{\pi/3} \sin x dx;$$

$$(c) \int_4^9 \left(2\sqrt{x} - \frac{3}{\sqrt{x}} \right) dx; \quad (d) \int_1^3 \frac{1}{\sqrt{2x+3}} dx;$$

$$(e) \int_1^2 (3x^3 + 4)^{1/3} x^2 dx; \quad (f) \int_0^{\pi/2} x \sin x dx;$$

$$(g) \int_1^l \frac{\ln x}{x^2} dx; \quad (h) \int_{-2}^2 (x-2) |x| dx.$$

9. Find the derivatives of the following functions:

$$(a) \Phi(x) = \int_0^x t^2 \sin t dt; \quad (b) \Phi(x) = \int_{2x}^{3x^2} 2^t \cos(\sin t) dt.$$

10. In what ratio is the area of a square divided by the parabola passing through its two adjacent vertices and touching its side at the midpoint?

11. Find the area of the figure bounded by the curves $y = |x^2 - 1|$ and $y = 5 + |x|$.

12. Find the area of the figure bounded by the curves $2y = x^2 + x - 6$ and $2y = -x^2 + 3x + 6$.

13. Prove that if the function $f(x)$, $x \in [-a; a]$, where $a > 0$, is even, then

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx.$$

14. Prove that if the function $f(x)$, $x \in \mathbb{R}$, is periodic, with period $T > 0$, then

$$\int_a^{a+T} f(x) dx = \int_0^T f(x) dx$$

for any $a \in \mathbb{R}$.

15. Find the volume of the body resulting from the rotation of the figure bounded by the curves $x(y+2) = 4$, $x = 1$ and $y = 0$ about the y -axis.

16. Find the volume of the body produced by the rotation of the figure bounded by the curves $y = \frac{2}{\pi}x$ and $y = \sin x$, $x \in [0; \frac{\pi}{2}]$ about the x -axis.

17. Find the force with which the homogeneous bar $0 \leq x \leq l$ of linear density δ attracts the particle $P(a)$ ($a > l$) of mass m ?

18. A homogeneous body shaped as a right circular cylinder of height h and the radius of the base R rotates about its axis with a constant angular velocity ω . Find the kinetic energy of the body if the density of the material it is manufactured of is ρ .

PROBLEMS OF SECTION II

1. Find all antiderivatives of the following functions:

(a) $f(x) = e^{3x+2} + \frac{1}{2} \cos \frac{\pi x}{4} - 3 \sqrt{x}$;

(b) $f(x) = 2^{2x} - \frac{1}{\cos^2 \frac{\pi x}{3}} + \frac{1}{\sqrt[3]{2x-1}}$;

(c) $f(x) = \frac{3}{2x+5} - \frac{2}{\sin^2 \frac{x}{2}} + 4 \cos 3x$.

2. For the function $f(x)$ find the antiderivative whose graph passes through a given point:

(a) $f(x) = \frac{1}{\sqrt{x}} + \sin(2x+1)$, $M(1; 1)$;

(b) $f(x) = \frac{1}{x-3} + e^{2x+1}$, $M(1; 2)$.

3. For the function $f(x) = 2 \cos 3x - 3 \sin 4x$ find the antiderivative $F(x)$ satisfying the condition $F(\pi/2) = 1$.

4. Find the function $f(x)$ if it is known that $f'(x) = 5e^{3x}$ and $f(0) = 4$.

5. Find all antiderivatives of the function $f(x) = (2x - 3) \times |x - 2|$ in the interval $(-3; 1)$ and in the interval $(1; 3)$.

6. For what values of x does the antiderivative of the function $f(x) = \pi \sin \pi x + 2x - 4$, that has the value 3 at $x = 1$, vanish?

7. Suppose the function $F(x)$ is an antiderivative of a periodic function $f(x)$. Is the function $F(x)$ periodic?

8. Suppose the function $F(x)$ is an antiderivative for some even function $f(x)$, $x \in \mathbb{R}$. Is the function $F(x)$ odd?

9. Assume that the function $F(x)$ is an antiderivative of an odd function $f(x)$, $x \in \mathbb{R}$. Is the function $F(x)$ even?

10. Find the following indefinite integrals:

$$(a) \int (x^2 + 2 \sin 3x + e^{2x}) dx;$$

$$(b) \int \sin 6t \cos t dt; \quad (c) \int \frac{x^3 + 2\sqrt{x} + x\sqrt{x}}{\sqrt{x}} dx;$$

$$(d) \int \left(3^{2x} + \frac{4}{\cos^2 \frac{\pi x}{3}} \right) dx; \quad (e) \int \left(\frac{1}{2u-1} + \frac{3}{\sqrt[3]{4u+1}} \right) du;$$

$$(f) \int |x-1| dx; \quad (g) \int \frac{\sin x}{1+\cos x} dx.$$

11. Applying the formula for integration by parts, find the following indefinite integrals:

$$(a) \int x \sin x dx; \quad (b) \int x^2 \cos x dx;$$

$$(c) \int (x^2 + 1) e^x dx; \quad (d) \int x \cdot 2^{3x} dx.$$

12. Calculate the following integrals:

$$(a) \int_2^3 (x+1) dx; \quad (b) \int_{-\pi}^{\pi/2} \cos x dx;$$

$$(c) \int_1^2 (2^{3x} - \cos 2x) dx; \quad (d) \int_1^2 \frac{1}{2x+1} dx;$$

$$(e) \int_1^3 \frac{2}{\sqrt[5]{3x-1}} dx; \quad (f) \int_1^2 x e^{-x^2} dx;$$

$$(g) \int_1^0 x \cos x dx; \quad (h) \int_{-1}^0 |2^x - 2^{-x}| dx;$$

$$(i) \int_{\pi/6}^{\pi/2} \frac{1 + \sin 2x + \cos 2x}{\sin x + \cos x} dx.$$

$$13. \text{ Find all the numbers } a > 0 \text{ for which } \int_0^a (2 - 4x + 3x^2) dx \leq$$

a.

$$14. \text{ Find the numbers } A \text{ and } B \text{ such that the function of the form } f(x) = Ax^2 + B \text{ satisfies the conditions } f'(1) = 2 \text{ and } \int_0^3 f(x) dx =$$

7.

$$15. \text{ Find all solutions to the equation } \int_0^{\alpha} \cos(x + \alpha^2) dx = \sin \alpha$$

belonging to the interval $[2; 3]$.

16. Find the derivatives of the following functions:

$$(a) \Phi(x) = \int_1^x (u+1)e^u du;$$

$$(b) \Phi(t) = \int_{t^2}^2 2x^2 \ln(1+x^2) dx; \quad (c) \Phi(u) = \int_{\sin u}^{\cos u} e^x \sin^3(2x+5) dx.$$

17. Calculate the area of the figure bounded by the curves:

$$(a) y = 1/\cos^2 x, y = 0, x = 0, x = \pi/4;$$

$$(b) y = x^2; y = \sqrt[3]{x}; \quad (c) y = e^x, y = e^{-x}, x = 1;$$

$$(d) y = 2 + \sin x, y = 1 + \cos^2 x, x \in [0; \pi];$$

$$(e) |y| = 1 - x^2; \quad (f) y = (1/2)^x, x - 2y + 2 = 0, x = 2;$$

$$(g) xy = 7, y = 0, x = 4, x = 12;$$

$$(h) y = -3x^2 - |x| + 3, y = 0;$$

$$(i) 3y = -x^2 + 8x - 7, y + 1 = 4/(x - 3);$$

$$(j) y = x^3, y = 1/x, x = 2.$$

18. Calculate the area of the figure bounded by the curves $y =$

$$\frac{1}{x} + 1, x = 1 \text{ and a tangent to the curve } y = \frac{1}{x} + 1 \text{ at the point } (2; 3/2).$$

19. For what positive values of the parameter a is the area of the figure bounded by the curves $y = \cos ax, y = 0, x = \pi/6a, x = \pi/2a$ greater than 3?20. Given the curvilinear trapezoid bounded by the curves $y = 0, x = a (a > 0)$ and $y = x^3$. What part of the area of the trapezoid does the area of the triangle, cut from the trapezoid by a tangent to the curve $y = x^3$ at the point $x = 2a/3$, constitute?21. At some point of the graph of the function $y = \sqrt{x}$ the tangent makes an angle of 45° with the abscissa axis. Calculate the area

of the figure bounded by that tangent and the straight lines $y = 0$ and $x = 1/4$.

22. Prove that if the function $f(x)$, $x \in [-a; a]$, is odd, then

$$\int_{-a}^a f(x) dx = 0.$$

23. Calculate the volume of the solid produced by the rotation about the x -axis of a body bounded by the curves $y = x^3$, $y = 1$, $x = 3$.

24. Calculate the volume of the solid produced by the rotation about the straight line $y = 1$ of the figure bounded by the graph of the function $y = 1 + \cos^2 x$ on the interval $[-\pi/2; \pi/2]$ and that straight line.

25. Find the volume of the solid produced by the rotation of the curvilinear trapezoid $y = x^\alpha$, $x \in [0; 1]$, $\alpha > 0$, (a) about the x -axis, (b) about the y -axis, (c) about the straight line $y = 1$, (d) about the straight line $x = 1$, (e) about the straight line $y = x$.

Chapter 5

Solving Plane Geometry Problems

It sometimes turns out to be difficult for a student to solve a geometric problem because a problem in geometry can seldom be solved with the use of a definite formula alone. The majority of problems require analysis of various facts from the theory and the proof of some or other assertions which are valid only at a certain arrangement of the elements of the figures. We can say with assurance that to solve a geometric problem one must have a complete command of the whole amount of theoretical material. But it is even not enough to have a good knowledge of theory, it is also necessary to acquire practical knowledge of problem solving and this can be done only by way of solving sufficiently many problems, beginning with simple ones and going on to more complicated problems.

Included into this chapter are a large number of problems of increasing difficulty with solutions. The solutions of problems given in 5.1 are especially detailed since here we recall formulations of many theorems and show how they must be applied. A number of problems are preceded by discussion of some definitions and theorems, with the aim of emphasizing their significant consequences which often escape the notice of students. Some assertions usually included in regular textbooks in the form of problems are given here and proved as theorems because they are often used in problem solving.

One and the same problem can be solved, as a rule, by several methods. The solutions we give here are not always the briefest and the most elegant, but we have chosen them because they suit the purpose of showing certain techniques of problem solving and demonstrate the application of various theorems.

Several general remarks are due here.

A drawing. Drawing a figure, try to make it correspond to the hypothesis. For instance, if it is stated that a certain angle is twice as large as some other angle, or that the segments are perpendicular, show this in the drawing. A good drawing is a convenient visual notation of the conditions of the problem, and it may prove to be a good help in solving a problem, suggesting a correct reasoning. It must be realized at the same time that the drawing itself, even if it is done very accurately, with the aid of a pair of compasses and a ruler, does not prove anything. Everything shown in the figure must be justified by the corresponding logical inference.

Seeking a solution. When solving a problem, use the definition and the properties of the given and sought-for elements, reasoning as follows, for example: the triangle is isosceles, consequently, . . . , two tangents are drawn from one point, consequently, . . . , a circle is circumscribed about a right-angled triangle, consequently, . . . etc. Recall the theorems relating the given and the required elements of the problem, recall similar problems.

Checking up the solution. To see whether your solution of the problem is correct (especially at the exams), it pays not only to look through the solution once again and to check up the calculations, but also to carry out the converse solution, so to say: proceeding from the answer, to calculate the known elements, to see whether there exists a figure answering the obtained value of the sought-for quantity. If the problem includes a parameter, check up the solution by choosing the value of the parameter for which the solution is obvious and it is easy to find the result.

5.1. Miscellaneous Problems

Problem 1. The median AM of the triangle ABC is perpendicular to the median BN (Fig. 24). Find the area of the triangle ABC if $|AM| = m$ and $|BN| = n$.

$\triangle$ Let the medians AM and BN intersect at a point O . When intersecting, the medians of a triangle are divided in the ratio $2:1$, reckoning from the vertex, consequently, $|AO| = \frac{2}{3} |AM|$. The median AM is perpendicular to

the median BN , hence, the segment AO is the altitude of the triangle ABN . Using the formula for the area of a triangle

$$S = \frac{1}{2} ah_a, \quad (1)$$

where a is the base, h_a is the altitude to the base a , we get the area of the triangle ABN which is equal to $\frac{1}{2} |AO| \times$

$|BN| = \frac{1}{3} mn$. The triangles ABC and ABN have a com-

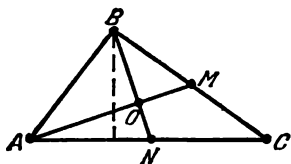


Fig. 24

mon altitude drawn from the vertex B , the base AC being twice as large as the base AN . It follows from formula (1) that the area of the triangle ABC is twice that of the triangle ABN , that is, it is equal to $\frac{2}{3} mn$. $\blacktriangle$

Problem 2. Find the area of the trapezoid, knowing the lengths d_1 and d_2 of its diagonals and of the height h .

$\triangle$ Let us consider the trapezoid $ABCD$ in which $|BD| = d_1$ and $|AC| = d_2$ (Fig. 25). We draw a straight line through the point D parallel to the diagonal AC and denote by K the point of intersection of that straight line and the line BC . In the rectangle $ACKD$ the opposite sides are pairwise parallel, consequently, $ACKD$ is a parallelogram,

$$|DK| = |AC| = d_2 \quad \text{and} \quad |CK| = |AD|.$$

It follows from the last equation that

$$|BK| = |BC| + |CK| = |BC| + |AD|,$$

that is, the length of the segment BK is equal to the sum of the lengths of the bases of the trapezoid. The area of a trapezoid the lengths of whose bases are equal to a and b and the length of the altitude is equal to h , can be found from the formula

$$S = \frac{a+b}{2} h, \quad (2)$$

and, therefore, the area of the trapezoid $ABCD$ is $\frac{1}{2} |BK| h$. Suppose $DM \perp BK$, then $|DM| = h$. Using the Pytha-

gorean theorem, we find from the right-angled triangles BMD and DMK the lengths of the segments BM and MK :

$$|BM| = \sqrt{|BD|^2 - |DM|^2} = \sqrt{d_1^2 - h^2},$$

$$|MK| = \sqrt{|DK|^2 - |DM|^2} = \sqrt{d_2^2 - h^2}.$$

Thus we see that the area of the trapezoid $ABCD$ is equal to

$$\frac{1}{2} h (\sqrt{d_1^2 - h^2} + \sqrt{d_2^2 - h^2}). \blacktriangle$$

By the additional construction we have obtained a triangle whose area is equal to the area of the trapezoid, on

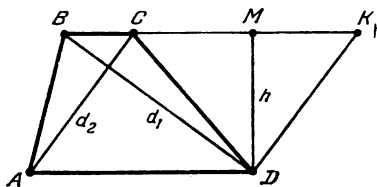


Fig. 25

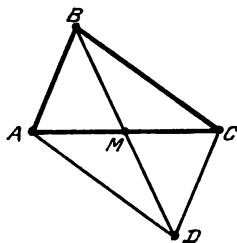


Fig. 26

one hand, and, on the other, can be easily calculated from the given data.

Here is another example showing how to use the same method, that of constructing a figure of equal area.

Problem 3. Find the area of the triangle ABC if $|AB| = 3$ cm, $|BC| = 7$ cm and the length of the median BM is 4 cm.

$\triangle$ Let us complete the triangle ABC to obtain a parallelogram $ABCD$ (Fig. 26). While intersecting, the diagonals of a parallelogram are divided in half, therefore, the point M lies on the diagonal BD and $|BD| = 2|BM|$. The areas of the triangles ABC and BCD constitute half the area of the parallelogram $ABCD$, consequently, the areas of these triangles are equal. The lengths of three sides of the triangle BCD are known:

$$|BC| = 7 \text{ cm}, \quad |CD| = 3 \text{ cm}, \quad |BD| = 8 \text{ cm}.$$

The area of a triangle the lengths of whose sides are a , b , and c can be found by Hero's formula

$$S = \sqrt{p(p-a)(p-b)(p-c)},$$

where $2p = a + b + c$.

Substituting the numerical values of the lengths of these sides into this formula, we find that the area of the triangle BCD , as well as the area of the triangle ABC , is equal to $6\sqrt{3}$ sq. cm. $\blacktriangle$

The following two problems serve the purpose of recollecting some theorems on right-angled triangles.

Problem 4. An altitude CD is drawn from the vertex C of the right angle in the right-angled triangle ABC (Fig. 27).

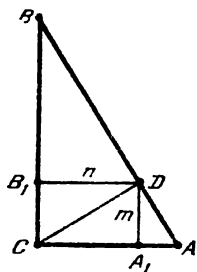


Fig. 27

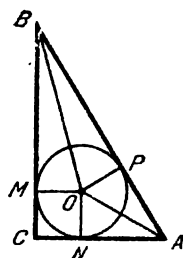


Fig. 28

The point D is at the distances m and n from the legs AC and BC respectively. Find the lengths of the legs.

$\triangle$ Assume that $DA_1 \perp AC$ and $DB_1 \perp BC$. If two straight lines are perpendicular to the same straight line, then they are parallel, consequently, $DA_1 \parallel BC$ and $DB_1 \parallel AC$. Then CB_1DA_1 is a rectangle by definition and, therefore, $|CA_1| = |DB_1| = n$ and $|CB_1| = |DA_1| = m$.

In the right-angled triangle CDA_1 the segment DA_1 is the altitude drawn from the vertex of the right angle. The altitude of a right-angled triangle drawn from the vertex of the right angle is the mean proportional between the projections of the legs on the hypotenuse and, consequently,

$$|DA_1|^2 = |CA_1| \cdot |A_1A|.$$

Introducing the designation $b = |AC|$, we rewrite the equality in the form $m^2 = n(b - n)$, whence we find that $b = (n^2 + m^2)/n$. Designating, by analogy, $|BC| = a$, we find from the right-angled triangle BCD that $a = (n^2 + m^2)/m$.

Thus we see that the legs are equal, $\frac{n^2 + m^2}{n}$ and $\frac{n^2 + m^2}{m}$. $\blacktriangle$

Problem 5. The radius of the circle inscribed into a right-angled triangle is equal to r . Find the area of the triangle if the length of the hypotenuse is c .

$\triangle$ Suppose the circle inscribed into the right-angled triangle ABC touches the hypotenuse AB at a point P and the legs at points M and N (Fig. 28).

A tangent to a circle is perpendicular to the radius drawn at the point of tangency, therefore, if O is the centre of the circle, then $OM \perp BC$ and $ON \perp AC$. Hence it follows (the reasoning is similar to that given in Problem 4) that $CMON$ is a rectangle, and, since $|OM| = |ON| = r$, $CMON$ is a square whose area is equal to r^2 .

Right-angled triangles are congruent if the hypotenuse and a leg of one of them are congruent to the hypotenuse and a leg of the other, therefore, the right-angled triangles MBO and PBO are congruent and, consequently, have equal areas. It can be established by analogy that the right-angled triangles NAO and PAO have equal areas. Thus we have

$$S_{ABC} = S_{CMON} + 2S_{PBO} + 2S_{PAO}.$$

The sum of the areas of the triangles PBO and PAO is equal to the area of the triangle BOA which can be found by formula (1) and is equal to $\frac{1}{2} cr$. The final result is

$$S_{ABC} = r^2 + cr. \quad \blacktriangle$$

Let us consider a problem on division of a segment intersected by a straight line. As a rule, such a problem is solved by means of an additional construction consisting in drawing a parallel straight line, and by using the theorem on the sides of an angle cut by parallel straight lines.

Problem 6. The point N lies on the side AC of the triangle ABC (Fig. 29), with $|AN|/|AC| = n$. Find the ratio in which the median AM divides the segment BN .

$\triangle$ Let O be the point of intersection of the median AM and the segment BN . It is required to find the ratio $|BO|/|ON|$.

Let us draw a straight line NK parallel to the median AM . *Parallel straight lines intercept proportional segments on the sides of an angle.* The parallel lines AM and NK intersect the sides of the angle NBC and, consequently,

$$\frac{|BO|}{|ON|} = \frac{|BM|}{|MK|}. \quad (3)$$

The same lines intersect the sides of the angle ACB , therefore,

$$\frac{|MC|}{|MK|} = \frac{|AC|}{|AN|}.$$

Note that M is the midpoint of the side BC . Hence $|BM| = |MC|$. Besides, $|AC|/|AN| = 1/n$. Thus we have

$$\frac{|BM|}{|MK|} = \frac{|MC|}{|MK|} = \frac{1}{n}.$$

And then it follows from (3) that $|BO|/|ON| = 1/n$. $\blacktriangle$

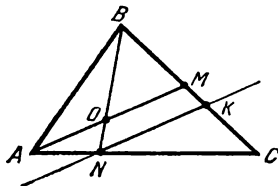


Fig. 29

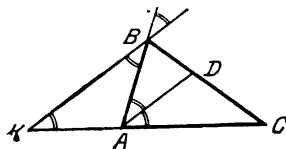


Fig. 30

Let us now use similar additional constructions to prove the theorem on the properties of the bisector of a triangle.

Theorem. *The bisector of an interior angle of a triangle divides the side lying opposite that angle into segments proportional to the adjacent sides, i.e. if AD is the bisector of the triangle ABC (Fig. 30), then*

$$\frac{|BD|}{|DC|} = \frac{|AB|}{|AC|}.$$

□ We draw a straight line BK parallel to the bisector AD . The parallel lines BK and DA intersect the sides of the angle BCK and intercept proportional segments on them, and thus we have

$$\frac{|BD|}{|DC|} = \frac{|KA|}{|AC|}. \quad (4)$$

Let us notice that the triangle KAB is isosceles. Indeed, if two parallel straight lines are cut by a third line, then the respective and, consequently, the alternate angles are congruent and, therefore,

$$\angle BKA = \angle DAC, \quad \angle KBA = \angle BAD,$$

and the equality of the angles BAD and DAC follows from the fact that AD is the bisector of the angle A . Thus we have $|KA| = |AB|$, and equation (4) yields

$$\frac{|BD|}{|DC|} = \frac{|AB|}{|AC|}. \quad \blacksquare$$

Let us consider some problems whose hypothesis includes a circle.

Problem 7. The diameter CD is parallel to the chord AB of the same circle (Fig. 31). Find the length of the chord AB if $|AC| = b$ and $|BC| = a$ ($a > b$).

$\triangle$ Arcs between parallel chords are congruent; besides, congruent arcs are subtended by congruent chords, and, therefore, $|BD| = |AC| = b$.

An inscribed angle resting on the diameter is a right angle, consequently, $\angle CBD = 90^\circ$ and the triangle CBD is right-angled. From the Pythagorean theorem we find that $|CD| = \sqrt{a^2 + b^2}$.

Assume $BK \perp CD$. The area of the right-angled triangle CBD is equal to half the product of the lengths of its legs. On the other hand, its area can be expressed by formula (1), that is, the equality $\frac{1}{2} |CB| \cdot |BD| = \frac{1}{2} |CD| \cdot |BK|$ holds true, from which we find that $|BK| = \frac{ab}{\sqrt{a^2 + b^2}}$.

Assume that O is the centre of a circle and $OM \perp AB$. Segments of two parallel straight lines contained between two parallel straight lines are congruent, and, consequently, $|OM| = |BK|$. The diameter perpendicular to the chord divides it in half, and, therefore, $|AM| = |MB|$. From the right-angled triangle OMB in which $|OB| = |CD|/2$,

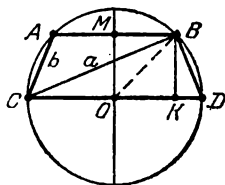


Fig. 31

we find that

$$|MB| = \sqrt{|OB|^2 - |OM|^2} = \frac{a^2 - b^2}{2\sqrt{a^2 + b^2}},$$

consequently, $AB = (a^2 - b^2)/\sqrt{a^2 + b^2}$. $\blacktriangle$

Problem 8. A circle is constructed on the base BC of the trapezoid $ABCD$ serving as a diameter. The circle passes through the midpoints of the diagonals of the trapezoid and touches the base AD (Fig. 32). Find the angles of the trapezoid.

$\triangle$ Suppose the circle cuts the diagonal BD at a point M ; then, by the hypothesis, $|BM| = |MD|$. The inscribed

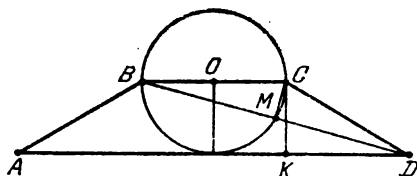


Fig. 32

angle BMC is a right angle since it rests on the diameter. *Right-angled triangles are congruent if the legs of one of them are congruent to the legs of the other*, therefore, $\triangle BMC \cong \triangle DMC$ ($|BM| = |MD|$, CM being the common leg) and, consequently, $|BC| = |CD|$. Since the circle also passes through the midpoint of the diagonal AC , we establish by analogy that $|AB| = |BC|$. Thus we see that the trapezoid is isosceles and the base angles are equal.

The circle touches the base AD and, therefore, the distance between the bases is equal to the radius of the circle, and if $CK \perp AD$, then $|CK| = \frac{1}{2}|BC|$. Hence it follows that in the right-angled triangle CKD the hypotenuse CD is twice as long as the leg CK and, consequently, the leg CK lies opposite the angle of 30° .

Thus, the angles of the trapezoid $ABCD$ are equal to 30° and 150° . $\blacktriangle$

While solving the problem, we have proved, in particular, that *if the median of a triangle (BCD) is the altitude, then the triangle is isosceles*. It is also easy to prove the following criteria of an isosceles triangle (prove the criteria yourself):

if the base angles of a triangle are equal, then the triangle is isosceles;

if the altitude of a triangle is the bisector, then the triangle is isosceles;

if the median of a triangle is the bisector, then the triangle is isosceles.

We shall use these criteria in what follows.

Problem 9. Two straight lines are drawn from a point A and touch a circle of radius r at points M and N (Fig. 33).

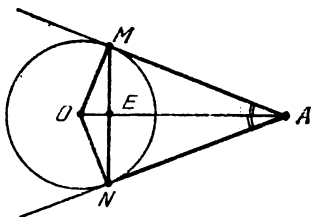


Fig. 33

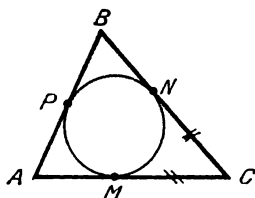


Fig. 34

Find the length of the segment MN if the distance between the point A and the centre of the circle is equal to a .

$\triangle$ Let O be the centre of the circle. Then $\angle OMA = \angle ONA = 90^\circ$. The right-angled triangles OMA and ONA have the hypotenuse OA in common, and $|OM| = |ON| = r$. Consequently, $\triangle OMA \cong \triangle ONA$. It follows from the congruency of these triangles that the angles MAO and NAO are congruent and $|AM| = |AN|$. Thus, the triangle MAN is isosceles. *In an isosceles triangle the bisector serves as the median and the altitude*, therefore, if E is the point of intersection of OA and MN , then $OA \perp ME$ and $|ME| = |EN|$.

For the area S of the right-angled triangle OMA we have

$$S = \frac{1}{2} |ME| |OA| = \frac{1}{2} |OM| |AM|,$$

whence we find that $|ME| = r \sqrt{a^2 - r^2}/a$ and $|MN| = 2r \sqrt{a^2 - r^2}/a$ since $|AM| = \sqrt{a^2 - r^2}$. $\blacktriangle$

While solving the problem, we have proved two properties of tangent lines, which we shall formulate as follows:

if two tangents are drawn from a point to a circle, then

(a) *the lengths of the intercepts of the tangents from that point to the points of tangency are equal;*

(b) *a straight line passing through the centre of the circle and that point bisects the angle between the tangents.*

The first of these properties will help us solve the following problem.

Problem 10. Incribed into the triangle ABC of the area S is a circle of radius r , which touches the sides AC and BC at points M and N respectively (Fig. 34). Find the length of the side AC if $|AM|/|MC| = 2/3$ and $|BN|/|NC| = 5/6$.

Δ We denote the length of the side AC by a . From the condition $|AM|/|MC| = 2/3$ we find that $|AM| = 2a/5$ and $|MC| = 3a/5$.

By the property of tangents drawn from one point we have $|NC| = |MC| = 3a/5$, and then, from the condition $|BN|/|NC| = 5/6$, we obtain $|BN| = a/2$.

If the circle touches the side AB at a point P , then we also have $|AP| = |AM| = 2a/5$ and $|BP| = |BN| = a/2$.

The area of the triangle can be expressed in terms of the lengths of the sides a , b , c and the radius r of the inscribed circle by the formula

$$S = pr, \quad (5)$$

where $2p = a + b + c$.

We find half the perimeter p of the triangle ABC : $p = 3a/2$. Then, by the indicated formula, $S = \frac{3}{2}ar$ and, consequently, $a = \frac{2}{3} \frac{S}{r}$. $\blacktriangle$

Let us recall that

a circle can be inscribed into a rectangle if and only if the sums of the lengths of opposite sides are equal;

a circle can be circumscribed about a rectangle if and only if the sum of opposite angles of the rectangle is 180° .

Problem 11. A right-angled trapezoid is circumscribed about a circle. Find the radius of the circle if the lengths of the bases of the trapezoid are equal to a and b .

Δ Assume that r is the radius of the circle inscribed into the right-angled trapezoid $ABCD$ (Fig. 35). The trapezoid being right-angled, we have $|AB| = 2r$. Suppose $|BC| =$

b and $|AD| = a$. A circle is inscribed into this trapezoid, consequently,

$$|BC| + |AD| = |AB| + |CD|,$$

whence we get $|CD| = a + b - 2r$. Let $CK \perp AD$, then $|CK| = |AB| = 2r$ and $|KD| = a - b$. By the Pythagorean theorem we have $|CD|^2 = |CK|^2 + |KD|^2$, i.e. $(a + b - 2r)^2 = 4r^2 + (a - b)^2$, whence we find

$$r = \frac{ab}{a+b}. \quad \blacktriangle$$

We shall now give an example of a problem whose solution becomes simpler upon an additional construction of a circle

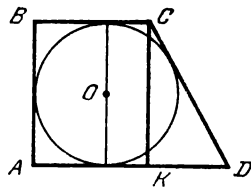


Fig. 35

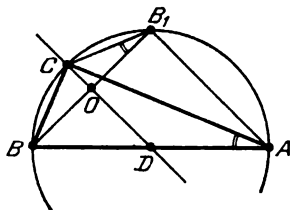


Fig. 36

Problem 12. In the right-angled triangle ABC the angle C is 90° , the angle A is equal to α ($\alpha < \pi/4$), and a point D is the middle of the hypotenuse. The point B_1 is symmetric with respect to the point B about the straight line CD . Find the angle AB_1C .

$\triangle$ Let the point B_1 be symmetric with respect to the point B about the line CD (Fig. 36). Then the points B and B_1 lie on the same perpendicular to the line CD and at the same distance from that line, i.e. $BB_1 \perp CD$ and $|OB| = |OB_1|$. The segment OD connects the midpoints of the sides of the triangle BB_1A , and, consequently, OD is the median of the triangle BB_1A and is parallel to the side AB_1 . And since $OD \perp BB_1$, it follows that $AB_1 \perp BB_1$ as well, that is, the triangle BB_1A is right-angled.

The vertices of right-angled triangles with hypotenuse AB lie on a circle with diameter AB , consequently, the points A , B , C , and B_1 lie on a circle with diameter AB .

The inscribed angle is equal to half the angular value of the arc it rests on. The inscribed angles BAC and BB_1C rest on one and the same arc and, therefore, they are equal: $\angle BB_1C = \alpha$.

Hence we find that $\angle AB_1C = \angle AB_1B + \angle BB_1C = \frac{\pi}{2} + \alpha$. ▲

5.2. Similitude of Triangles. Law of Sines and Law of Cosines

Similitude of triangles. Let us recall the definition of the similitude of figures: the figure Φ_1 is said to be *similar* to the figure Φ (written as $\Phi_1 \sim \Phi$) if there is a mapping of the figure Φ onto the figure under which for any two points M and N of the figure Φ and their images, the points M_1 and N_1 , the ratio between the distances $|MN|$ and $|M_1N_1|$ is a constant quantity. The number $k = |M_1N_1|/|MN|$ is the *ratio of similitude*.

Solution of many problems requires the use of similitude of triangles. If the triangle $A_1B_1C_1$ is similar to the triangle ABC , and the vertices A_1 and B_1 are the images of the vertices A and B , then the ratio of similitude of these triangles is equal to $|A_1B_1|/|AB|$, or, as it is customary to say, it is equal to the ratio of the lengths of the respective sides.

Criteria of similitude of triangles

1. *If three sides of one triangle are proportional to three sides of another triangle, then the triangles are similar.*

2. *If two angles of one triangle are equal to two angles of another triangle, then the triangles are similar.*

3. *If two sides of one triangle are proportional to two sides of another triangle and the angles between those sides are equal, then the triangles are similar.*

Problem 1. A straight line passing through the point of intersection of the diagonals of a trapezoid, parallel to its bases, cuts the nonparallel sides of the trapezoid at points M and N . Find the length of the segment MN if the lengths of the bases of the trapezoid are equal to a and b .

△ Suppose the diagonals of the trapezoid $ABCD$ meet at a point O , and the segment MN is parallel to the bases and contains the point O in its interior (Fig. 37), $|AD| = a$,

$|BC| = b$. If two parallel straight lines are cut by a third line, then the corresponding and the alternate angles are equal. The bases of the trapezoid are parallel, consequently, $\angle BDA = \angle CBD$ and $\angle CAD = \angle BCA$ and, in accordance with the second criteria of similitude, we have $\triangle BOC \sim \triangle DOA$. It follows that $\frac{|BO|}{|OD|} = \frac{|BC|}{|AD|}$, i.e. $\frac{|BO|}{|OD|} = \frac{b}{a}$. From this we find that

$$\frac{|BO|}{|BD|} = \frac{b}{a+b}. \quad (1)$$

The straight line MN is parallel to the base AD , and, therefore, $\angle BMO = \angle BAD$ and $\angle BOM = \angle BDA$, the

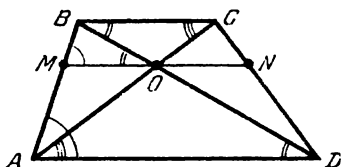


Fig. 37

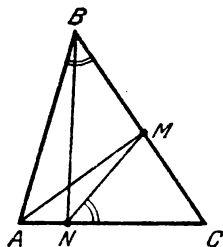


Fig. 38

triangles MBO and ABD are also similar and $|MO|/|AD| = |BO|/|BD|$. From this, taking into account (1), we get $|MO| = ab/(a+b)$. We establish by analogy that $|NO| = ab/(a+b)$ and thus we have $|MN| = 2ab/(a+b)$. ▲

Now we shall formulate the criteria of similitude of right-angled triangles.

1. Two right-angled triangles are similar if an acute angle of one of them is equal to an acute angle of the other.

2. Two right-angled triangles are similar if the legs of one of them are proportional to the legs of the other.

3. Two right-angled triangles are similar if a leg and the hypotenuse of one of them are proportional to a leg and the hypotenuse of the other.

Problem 2. The altitudes AM and BN are drawn in the triangle ABC (Fig. 38). Find the angles of the triangle MNC if $\angle A = \alpha$ and $\angle B = \beta$.

$\triangle$ The right-angled triangles AMC and BNC have equal acute angles at the vertex C , consequently, they are similar and $|NC|/|BC| = |MC|/|AC|$. Hence we infer that in the triangles MCN and ACB the sides adjacent to the equal angle at the vertex C are proportional and, consequently (in accordance to the third criterion of similitude), the triangles are similar. In similar triangles, equal angles face the corresponding sides, therefore, $\angle NMC = \alpha$ and $\angle MNC = \beta$.

The result of this problem can be formulated as follows: if we connect the feet of two altitudes of a triangle, we obtain a triangle similar to the given triangle. $\blacktriangle$

By definition, all respective linear elements in similar figures are proportional. Thus, *the ratio of the perimeters*

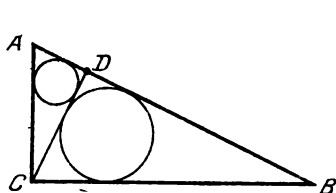


Fig. 39

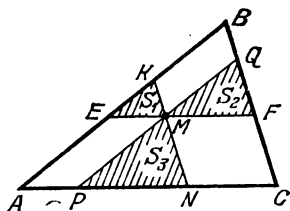


Fig. 40

of similar polygons is equal to the ratio of the lengths of the corresponding sides. Or, for instance, in similar triangles the ratio of the radii of inscribed circles (as well as of circumscribed circles) is equal to the ratio of the lengths of the corresponding sides. This remark will help us solve the following problem.

Problem 3. The altitude CD is drawn from the vertex C of the right angle in the right-angled triangle ABC (Fig. 39). The radii of the circles inscribed into the triangles ACD and BCD are equal to r_1 and r_2 respectively. Find the radius of the circle inscribed into the triangle ABC .

$\triangle$ Designating the required radius as r , we set $|AB| = c$, $|AC| = b$, $|BC| = a$. From the similitude of the right-angled triangles ACD and ABC (they have equal angles at the vertex A) we have $\frac{r}{r_1} = \frac{c}{b}$, whence we have $b = \frac{r}{r_1} c$.

The right-angled triangles CBD and ABC are also similar, therefore, $\frac{r}{r_2} = \frac{c}{a}$, whence it follows that $a = \frac{r_2}{r}c$. Since $a^2 + b^2 = c^2$, we can square the expressions for a and b and add the results to obtain $\left(\frac{r_1}{r}\right)^2 c^2 + \left(\frac{r_2}{r}\right)^2 c^2 = c^2$ or $\frac{r_1^2 + r_2^2}{r^2} = 1$. Now we find that $r = \sqrt{r_1^2 + r_2^2}$. $\blacktriangle$

Let us recall that *the ratio of the areas of similar figures is equal to the square of the coefficient of similitude*. For triangles, this statement can be formulated as follows: the areas of similar triangles are related as the squares of the lengths of the corresponding sides. Let us consider a typical problem based on these facts.

Problem 4. Three straight lines are drawn through a point M , lying in the interior of the triangle ABC , parallel to its sides. The areas of the resulting three triangles (Fig. 40) are S_1 , S_2 , and S_3 . Find the area of the triangle ABC .

$\triangle$ It is easy to see that the triangles EKM , MQF and PMN are similar to the triangle ABC . If we assume that S is the area of the triangle ABC , then we have

$$\frac{S_1}{S} = \frac{|EM|^2}{|AC|^2}, \quad \frac{S_2}{S} = \frac{|MF|^2}{|AC|^2}, \quad \frac{S_3}{S} = \frac{|PN|^2}{|AC|^2}.$$

From this we find

$$|EM| = \sqrt{\frac{S_1}{S}} |AC|, \quad |MF| = \sqrt{\frac{S_2}{S}} |AC|, \\ |PN| = \sqrt{\frac{S_3}{S}} |AC|.$$

And since $|EM| = |AP|$, $|MF| = |NC|$, it follows that

$$|EM| + |PN| + |MF| = |AP| + |PN| + |NC| = |AC|.$$

Thus we have

$$|AC| \left(\sqrt{\frac{S_1}{S}} + \sqrt{\frac{S_2}{S}} + \sqrt{\frac{S_3}{S}} \right) = |AC|.$$

Hence,

$$S = (\sqrt{S_1} + \sqrt{S_2} + \sqrt{S_3})^2. \quad \blacktriangle$$

Laws of cosines and sines. For an arbitrary triangle the lengths of whose sides are denoted by a , b , c , and the angles opposite them are denoted by A , B , C , the following two laws hold true which establish the correspondence between the sides and the angles of the triangle which can be written in the following concise form.

Law of cosines:

$$c^2 = a^2 + b^2 - 2ab \cos C;$$

Law of sines:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

This is how the laws can be applied.

Problem 5. Find the length of the median of the triangle ABC drawn from the vertex C if the lengths of the sides

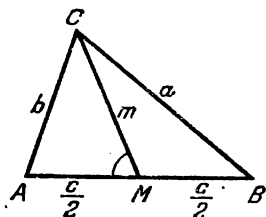


Fig. 41

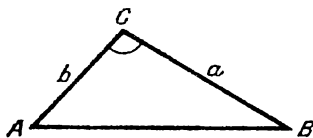


Fig. 42

lying opposite the vertices A , B , and C are equal, respectively, to a , b , c .

$\triangle$ Let CM be the median of the triangle ABC (Fig. 41). If we designate $m = |CM|$, $\varphi = \angle CMA$, then $\angle CMB = 180^\circ - \varphi$. By the law of cosines we have, from the triangles ACM and CMB ,

$$b^2 = m^2 + \frac{c^2}{4} - 2m \frac{c}{2} \cos \varphi,$$

$$a^2 = m^2 + \frac{c^2}{4} - 2m \frac{c}{2} \cos (180^\circ - \varphi).$$

Adding up the two equations term-by-term and taking into account that $\cos (180^\circ - \varphi) = -\cos \varphi$, we get $a^2 + b^2 = 2m^2 + \frac{c^2}{2}$, whence we find that $m^2 = \frac{a^2}{2} + \frac{b^2}{2} - \frac{c^2}{4}$.

Thus we have $|CM| = \sqrt{\frac{a^2}{2} + \frac{b^2}{2} - \frac{c^2}{4}}$. $\blacktriangle$

Problem 6. The area of the triangle ABC is equal to 16 sq cm. Find the length of the side AB if $|AC| = 5$ cm, $|BC| = 8$ cm and the angle C is obtuse (Fig. 42).

$\triangle$ The area of the triangle can be expressed in terms of the lengths a and b of its two sides and the angle C between them:

$$S = \frac{1}{2} ab \sin C. \quad (2)$$

For the given triangle ABC this formula yields $\sin C = 2S/ab = 4/5$. The angle C being obtuse, the cosine of the angle is negative and, therefore,

$$\cos C = -\sqrt{1 - \sin^2 C} = -\frac{3}{5}.$$

Now we find by the law of cosines that $|AB|^2 = a^2 + b^2 - 2ab \cos C = 137$. Consequently, $|AB| = \sqrt{137}$ cm. $\blacktriangle$

Problem 7. In the isosceles triangle ABC the sides AB and AC are b in length, the vertex angle A is 2α . The straight line passing through the vertex B and the centre O of the circle circumscribed about the triangle ABC cuts the side AC at a point D (Fig. 43). Find the length of the segment BD .

$\triangle$ The centre of the circle circumscribed about the isosceles triangle ABC lies on its bisector AK (since $AK \perp BC$ and $|BK| = |KC|$). Consequently, $\angle ABD = \angle OAB = \alpha$. Thus, two angles of the triangle BAD are known and, since the sum of angles of a triangle is equal to 180° , it follows that $\angle BDA = 180^\circ - 3\alpha$. By the law of sines, we find, from the triangle BAD , that

$$\frac{|BD|}{\sin 2\alpha} = \frac{|AB|}{\sin (180^\circ - 3\alpha)},$$

whence, taking into account that $\sin (180^\circ - 3\alpha) = \sin 3\alpha$, we find that

$$|BD| = \frac{\sin 2\alpha}{\sin 3\alpha} b. \quad \blacktriangle$$

We shall prove now the law of sines and, incidentally, establish an important relation.

$\square$ A circle of radius R is circumscribed about a triangle ABC whose sides lying opposite the vertices A, B, C are

a, b, c in length (see Fig. 44). We draw a diameter BD . The inscribed angles BDA and BCA are subtended by the same arc, and, consequently, $\angle BDA = \angle C$. The angle BAD is

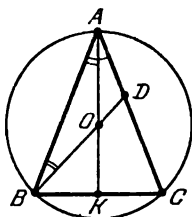


Fig. 43

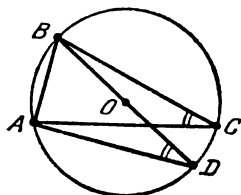


Fig. 44

equal to 90° since it rests on a diameter, hence the triangle BAD is right-angled and

$$c = 2R \sin C. \quad (3)$$

We can establish by analogy that $a = 2R \sin A$ and $b = 2R \sin B$. It follows from these equalities that

$$a/\sin A = b/\sin B = c/\sin C = 2R. \quad (4)$$

If we find $\sin C$ in (3) and substitute the result into (2) we get the expression for the area of the triangle in terms of the lengths of its sides a, b, c and the radius R of the circumscribed circle:

$$S = \frac{abc}{4R}. \quad (5)$$

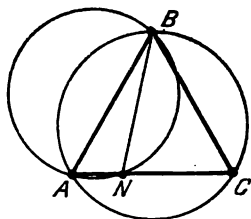


Fig. 45

Problem 8. The point N lies on the side AC of a regular triangle ABC (Fig. 45). Find the ratio of the radii of the circles circumscribed about the triangles ABN and ABC if $|AN|/|AC| = n$.

$\triangle$ If we designate as a the side of the regular triangle ABC , then $|AN| = na$. By the law of cosines, we can find the length of the side BN from the triangle ABN :

$$\begin{aligned} |BN| &= \sqrt{|AB|^2 + |AN|^2 - 2|AB||AN|\cos 60^\circ} = \sqrt{1 + n^2 - n} \cdot a. \end{aligned}$$

Let R_1 and R_2 be the radii of the circles circumscribed about the triangles ABN and ABC respectively. By formula (5) we get

$$S_{ABN} = \frac{n \sqrt{1+n^2-na^3}}{4R_1}, \quad S_{ABC} = \frac{a^3}{4R_2}.$$

On the other hand, the triangles ABN and ABC have a common altitude drawn from the vertex B and, therefore, their areas are related as the lengths of the bases, i.e. $S_{ABN} = nS_{ABC}$. Substituting here the expressions for the areas, we get $n \sqrt{1+n^2-na^3}/4R_1 = na^3/4R_2$, whence we find that $R_1/R_2 = \sqrt{1+n^2-n}$. $\blacktriangle$

5.3. Properties of Chords, Secants, and Tangents

Angles measured by arcs of circles. The angular value of an arc is the magnitude of the corresponding central angle. An inscribed angle is equal to half the angular value of the

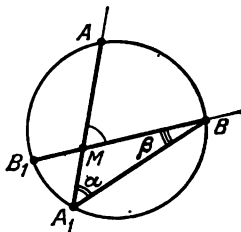


Fig. 46

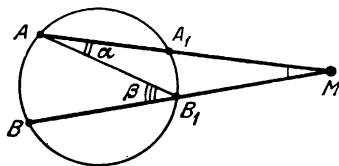


Fig. 47

intercepted arc. Let us see how angles are measured when their vertices are in the interior or in the exterior of the circle and their sides cut the circle.

Theorem. *The angle with the vertex inside the circle is equal to half the sum of the angular values of the arcs intercepted by its sides and their extensions. The angle, formed by two secants, with the vertex outside the circle and sides cutting the circle is equal to half the difference of the angular values of the greater and the smaller arc intercepted by its sides.*

□ Let us consider the angle AMB (Fig. 46). Note that $\alpha = \frac{1}{2} \widehat{AB}$, $\beta = \frac{1}{2} \widehat{A_1B_1}$. The exterior angle of the triangle

is equal to the sum of two interior angles, not adjacent to it and, therefore, $\angle AMB = \alpha + \beta = \frac{1}{2} (\widehat{AB} + \widehat{A_1B_1})$.

Let us now consider the angle AMB formed by two secants MA and MB (Fig. 47). Since $\alpha = \frac{1}{2} \widehat{A_1B_1}$, $\beta = \frac{1}{2} \widehat{AB}$, we get, in accordance with the property of an exterior angle of a triangle, that $\angle AMB = \beta - \alpha = \frac{1}{2} (\widehat{AB} - \widehat{A_1B_1})$. ■

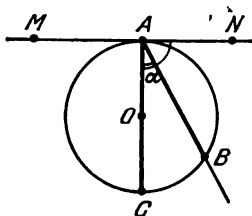


Fig. 48

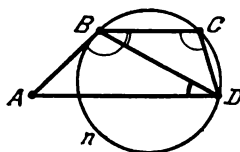


Fig. 49

Theorem. *The angle formed by a tangent and a chord having a common point on the circle is equal to half the angular value of the arc intercepted by its sides.*

□ Let us consider the angle NAB formed by a tangent and a chord (Fig. 48). We draw a diameter AC . The tangent is perpendicular to the diameter drawn through the point of tangency, consequently, $\angle CAN = 90^\circ$ and $\widehat{CBA} = 180^\circ$. Furthermore, $\alpha = \frac{1}{2} \widehat{BC}$, $\angle NAB = 90^\circ - \alpha$, hence we get

$$\angle NAB = \frac{1}{2} (180^\circ - \widehat{BC}) = \frac{1}{2} \widehat{BA}$$

(the case of an obtuse angle, the angle MAB in Fig. 48, can be discussed analogously). ■

Problem 1. A circle passes through the vertices B, C, D of the trapezoid $ABCD$ and touches the side AB at the point B (Fig. 49). Find the length of the diagonal BD if the bases of the trapezoid are a and b in length.

△ The angle between the tangent AB and the chord BD is measured by half the angular value of the arc BnD , as is the inscribed angle BCD . Consequently, $\angle ABD = \angle BCD$. The bases of the trapezoid are parallel, therefore $\angle CBD =$

$\angle BDA$. Thus, two angles in the triangle ABD are equal to two angles in the triangle BCD and, consequently, the triangles are similar. From their similitude we get $|BD|/|AD| = |BC|/|BD|$ whence it follows that $|BD|^2 = |AD| \cdot |BC|$ and $|BD| = \sqrt{ab}$. $\blacktriangle$

Properties of secants and tangents. We shall now prove a theorem which is known as "the theorem on a tangent and a secant" and is often applied in problem solving.

Theorem. Suppose a tangent MA and a secant MB drawn from a point M cut the given circle at points B and C (Fig. 50). Then, the following equality holds true:

$$|MA|^2 = |MB| \cdot |MC|,$$

that is, if a tangent and a secant are drawn from the point M to the circle, then the square of the length of the segment of the tangent from the point M to the point of tangency is equal to the product of the lengths of the segments of the secant from the point M to its points of intersection with the circle.

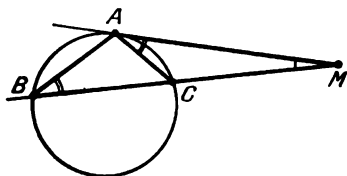


Fig. 50

□ By the preceding theorem, $\angle MAC = \frac{1}{2} \widehat{AC}$. But $\angle ABC = \frac{1}{2} \widehat{AC}$ as well, consequently, $\angle MAC = \angle ABC$. Since the triangles AMC and BMA have the vertex angle M in common, it follows, in accordance with the second criterion of similitude, that the triangles are similar. Consequently we have $|MA|/|MB| = |MC|/|MA|$, whence we get $|MA|^2 = |MB| \cdot |MC|$. $\blacksquare$

Problem 2. The radius of the circle is equal to r . Drawn from the point M are a secant MB passing through the centre of the circle and a tangent MA , with $|MB| = 2|MA|$ (Fig. 51). Find the distance between the point M and the centre of the circle.

△ Let us designate $x = |OM|$. Then $|BM| = x + r$ and $|CM| = x - r$. Besides, by the hypothesis, $|MA| = \frac{|BM|}{2} = \frac{1}{2}(x + r)$.

By the theorem on a tangent and a secant we have $\frac{1}{4}(x + r)^2 = (x + r)(x - r)$, whence, cancelling by

$(x + r)$, we get $\frac{1}{4}(x + r) = x - r$ and easily find that $x = \frac{5}{3}r$. ▲

Problem 3. A secant MA is drawn through a point M , located at the distance b from the centre of the circle, so that it cuts the circle in half: $|MB| = |BA|$ (Fig. 52). Determine the length of the secant MA if the radius of the circle is r .

△ Let us draw a secant MC through the centre of the circle. It follows from the theorem on a secant and a tangent

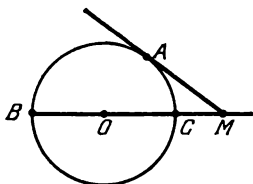


Fig. 51

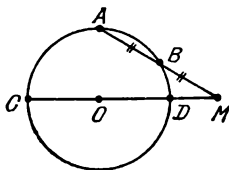


Fig. 52

that if two secants MA and MC are drawn from the point M and cut the circle at the points B and D respectively, then $|MA| \cdot |MB| = |MC| \cdot |MD|$, since both products are equal to the square of the length of the tangent drawn from the point M to the same circle. By the hypothesis, $|OM| = b$ and, therefore, $|MC| = b + r$ and $|MD| = b - r$. Designating $|MA| = x$, we get $|BM| = x/2$ and, in accordance with the indicated property of two secants drawn from the same point, we get $x \cdot \frac{x}{2} = (b + r)(b - r)$, whence

we find that $x = \sqrt{2(b^2 - r^2)}$. ▲

Property of intersecting chords.

Theorem. If two chords are drawn through a point taken inside a circle, then the product of the lengths of the segments of one chord is equal to the product of the lengths of the segments of the other chord, that is (Fig. 53), if two chords AB and CD are drawn through a point M , then $|MA| \cdot |MB| = |MC| \cdot |MD|$.

□ Let us draw the chords AC and BD . The inscribed angles CAB and CDB rest on the arc CB , and consequently, $\angle CAB = \angle CDB$. Similarly, $\angle ACD = \angle ABD$. Thus, the

triangles ACM and DBM are similar and $|MA|/|MD| = |MC|/|MB|$. Hence we get $|MA| \cdot |MB| = |MC| \cdot |MD|$. ■

Problem 4. There is an internal tangency between two circles at a point Q (Fig. 54). A straight line passing through the centre O_1 of the smaller circle cuts the larger circle at points A and D and the smaller circle at points B and C . Find the ratio of the radii of the circles if $|AB| : |BC| : |CD| = 2 : 4 : 3$.

△ Suppose R and r are the radii of the larger and the smaller circle respectively. Then $|BC| = 2r$ and we get from the given ratio that $|AB| = r$ and $|CD| = \frac{3}{2}r$.

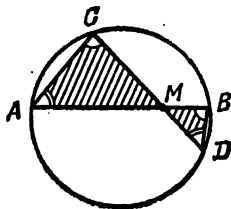


Fig. 53

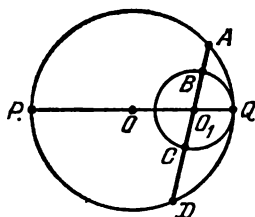


Fig. 54

We draw a diameter QP . The diameter drawn to the point of tangency is perpendicular to the tangent line and, therefore, the point O_1 lies on QP .

In accordance with the property of intersecting chords we have $|O_1Q| \cdot |O_1P| = |O_1A| \cdot |O_1D|$. Hence, taking into consideration that $|O_1P| = 2R - r$, $|O_1A| = |O_1B| + |BA| = 2r$ and $|O_1D| = |CD| + |CO_1| = \frac{5}{2}r$, we get $(2R - r)r = 5r^2$. Now we easily find that $R/r = 3$. ▲

5.4. Algebraic and Trigonometric Methods of Solution. Application of Vector Algebra

In the problems discussed earlier we had to solve simple algebraic equations using sine and cosine laws, and to find magnitudes of angles. It is evident that geometry cannot be separated from algebra and trigonometry since many geometric statements are formulated in terms of algebra (e.g.

the Pythagorean theorem or the property of the altitude of a right triangle) or in terms of trigonometry (e.g. the laws of sines and cosines). The possibility of applying algebra and trigonometry is much wider, however. In many cases, solutions of geometric problems reduce to solutions of the corresponding algebraic or trigonometric equations and systems. Let us consider the following problems, for example.

Problem 1. An isosceles triangle is inscribed into a circle of radius R . The sum of the lengths of the base and the altitude of the triangle is equal to the diameter of the circle. Find the altitude of the triangle.

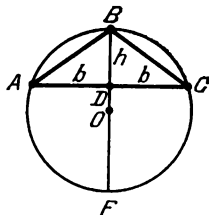


Fig. 55

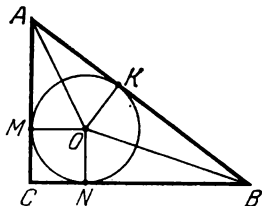


Fig. 56

△ Let BD be the altitude of the isosceles triangle ABC (Fig. 55). Since the diameter, perpendicular to the chord, passes through its midpoint, the point D lies on the diameter BF . We introduce the designations $h = |BD|$ and $2b = |AC|$. By the hypothesis, $2R = h + 2b$ and in accordance with the property of intersecting chords (5.3), we have $b^2 = h(2R - h)$. Thus we get a system

$$\begin{cases} 2R = h + 2b, \\ b^2 = h(2R - h). \end{cases}$$

Eliminating $2R$, we find that $b^2 = 2bh$, whence $b = 2h$, and then, the first equation yields $h = \frac{2}{5}R$. ▲

Problem 2. The centre of the circle inscribed into the right-angled triangle ABC is at the distances $\sqrt{5}$ cm and $\sqrt{10}$ cm from the vertices A and B (Fig. 56). Find the legs of the triangle.

△ We can easily solve the problem if we take, as unknowns, the radius r of the inscribed circle and $\alpha = \angle OAK$ rather than the legs. Since the centre of the inscribed circle lies on the intersection of the bisectors of the angles A and B , it follows that $\angle OBK = \frac{\pi}{4} - \alpha$. From the right-angled triangles OAK and OBK we find

$$r = \sqrt{5} \sin \alpha \quad \text{and} \quad r = \sqrt{10} \sin \left(\frac{\pi}{4} - \alpha \right),$$

and consequently,

$$\sqrt{5} \sin \alpha = \sqrt{10} \sin \left(\frac{\pi}{4} - \alpha \right). \quad (1)$$

We have reduced the problem to a simple trigonometric equation. Since $\sin \left(\frac{\pi}{4} - \alpha \right) = \frac{\sqrt{2}}{2} (\cos \alpha - \sin \alpha)$, equation (1) yields $\sin \alpha = \cos \alpha - \sin \alpha$, whence we get $\tan \alpha = 1/2$.

Knowing $\tan \alpha$, we find from the formula $\sin \alpha = \frac{\tan \alpha}{\sqrt{1 + \tan^2 \alpha}}$ that $\sin \alpha = 1/\sqrt{5}$ and, consequently, $r = 1$ cm.

Since $|AC| = |AM| + |MC|$ and $|BC| = |BN| + |NC|$, we easily get $|AC| = 3$ cm and $|BC| = 4$ cm. ▲

Problem 3. The isosceles triangles ABC ($|AB| = |BC|$) and $A'B'C'$ ($|A'B'| = |B'C'|$) are congruent. The vertices A' , B' , C' are located on the extensions of the side BC beyond the point C , of the side BA beyond the point A , and of the side AC beyond the point C , respectively, the straight lines BC and $B'C'$ being perpendicular (Fig. 57). Find the angles of the triangle ABC .

△ We designate $b = |AB|$ and $\varphi = \angle ABC$. Let the straight lines BC and $B'C'$ meet at a point Q . Since, by the hypothesis, $A'B \perp B'C'$ and $\angle QCC' = \angle ACB = \angle B'C'A'$, the angle $CC'A'$ is 90° and the line segment $C'Q$ is the alti-

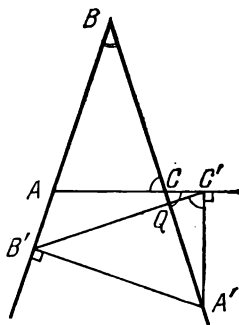


Fig. 57

tude of the right-angled triangle $CC'A'$. Consequently,

$$|QC'|^2 = |QC| |QA'|. \quad (2)$$

From the right-angled triangle $B'QA'$ we find that $|QA'| = b \sin \varphi$, $|B'Q| = b \cos \varphi$, and since $|QC'| = |B'C'| - |QB'|$, it follows that $|QC'| = b(1 - \cos \varphi)$. Furthermore, $|QC| = |QB'| - |BC|$. We find $|QB|$ from the right triangle $B'BQ$:

$$|QB| = \frac{|B'Q|}{\tan \varphi} = \frac{b \cos^2 \varphi}{\sin \varphi},$$

then $|QC| = b \left(\frac{\cos^2 \varphi}{\sin \varphi} - 1 \right)$. Substituting the lengths of the corresponding segments into (2) and cancelling by b^2 , we obtain a trigonometric equation $(1 - \cos \varphi)^2 = \sin \varphi \left(\frac{\cos^2 \varphi}{\sin \varphi} - 1 \right)$, which can be reduced to the form $\sin \varphi = 2 \cos \varphi - 1$. Squaring this equality and expressing $\sin^2 \varphi$ in terms of $\cos^2 \varphi$, we get $5 \cos^2 \varphi - 4 \cos \varphi = 0$, whence $\cos \varphi = 0$ and $\cos \varphi = 4/5$. It is evident that $\cos \varphi \neq 0$ (otherwise, the lines $B'A'$ and BC would not meet) and, therefore, $\cos \varphi = 4/5$ and $\varphi = \arccos(4/5)$. $\blacktriangle$

To reduce a geometric problem to an algebraic or a trigonometric one, we must introduce several unknowns. It is not obligatory to assume the sought-for quantities to be those unknowns. As is clear from the solution of Problem 2, it is sometimes more convenient to introduce other unknowns for which we get simpler relations, and to calculate the required unknowns only after solving those relations. Besides, the solutions obtained for algebraic and trigonometric equations and systems often require geometric check-up as, for instance, in Problem 3.

Most of the geometric problems on maximum and minimum are also solved with the aid of algebra and trigonometry. In that case, a problem is reduced to seeking a maximum or a minimum of some algebraic or trigonometric function. Note that in such cases the investigation must be carried out on the set of values of the argument to which the geometric figure under consideration belongs.

Problem 4. In the right-angled triangle ABC the hypotenuse AB is c in length and makes an angle α with the leg AC . The point D lies on the hypotenuse AB and for that point

the sum of the squares of the distances to the lines AC and BC is the smallest as compared to other points belonging to the line segment AB . Find the length of the segment AD .

△ Assume $DN \perp AC$ and $DM \perp BC$ (Fig. 58). Let us designate $x = |AD|$. Then $|BD| = c - x$ and

$$|DN| = x \sin \alpha, \quad |DM| = (c - x) \cos \alpha.$$

We designate the sum of the squares of the distances from the point D to the lines AC and BC as f . Then we have

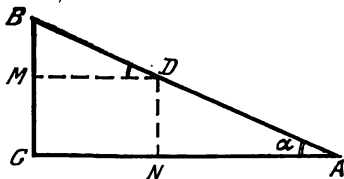


Fig. 58

$f = x^2 \sin^2 \alpha + (c - x)^2 \cos^2 \alpha$. It is required to find the value of x for which the function has the least value if x is in the limits $0 \leq x \leq c$. We transform the function f as $f = x^2 + c^2 \cos^2 \alpha - 2cx \cos \alpha$. The derivative of the function is equal to $2x - 2c \cos \alpha$ and vanishes at $x = c \cos \alpha$. This value of x belongs to the interval $[0; c]$, the derivative being negative to the left of that point and positive to the right. Consequently, at $x = c \cos \alpha$ the function f has a minimum.

Thus we have $|AD| = c \cos \alpha$.

Application of vector algebra. With the aid of vector algebra, some plane geometry problems and theorems, requiring complicated geometric reasoning, can be reduced to considerably simple calculations. Let us consider the following problem, for example.

Problem 5. In the triangle ABC , points A_1 , B_1 and C_1 are located on the sides AB , BC , and CA so that

$$\frac{|AA_1|}{|AB|} = \frac{|BB_1|}{|BC|} = \frac{|CC_1|}{|AC|}.$$

Prove that the points where the medians of the triangles ABC and $A_1B_1C_1$ meet are coincident.

△ Assume $\vec{AB} = \mathbf{a}$, $\vec{AC} = \mathbf{b}$ (Fig. 59). If AM is a median of the triangle ABC , then $\vec{AM} = (\mathbf{a} + \mathbf{b})/2$. The medians of triangles intersect at one point and at that point each median is divided in the ratio 2 : 1, reckoning from the vertex and, therefore, if O is the point of intersection of the medians of the triangle ABC , then

$$\vec{AO} = \frac{2}{3} \vec{AM} = \frac{1}{3} (\mathbf{a} + \mathbf{b}). \quad (3)$$

Let us consider the triangle $A_1B_1C_1$ (Fig. 60). Designating $m = |AA_1|/|AB|$, we have $\vec{A_1C_1} = (1 - m) \mathbf{b} - m\mathbf{a}$ and

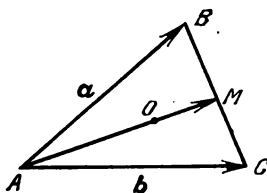


Fig. 59

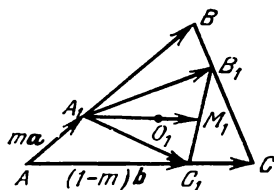


Fig. 60

$\vec{A_1B_1} = (1 - m) \mathbf{a} + m (\mathbf{b} - \mathbf{a})$. If A_1M_1 is a median of the triangle $A_1B_1C_1$, then

$$\vec{A_1M_1} = \frac{1}{2} (\vec{A_1C_1} + \vec{A_1B_1}) = \frac{1}{2} (\mathbf{b} + (1 - 3m) \mathbf{a})$$

and the point O_1 of the intersection of the medians of the triangle $A_1B_1C_1$ lies on A_1M_1 and $|A_1O_1| = \frac{2}{3} |A_1M_1|$,

i.e. $\vec{A_1O_1} = \frac{1}{3} (\mathbf{b} + (1 - 3m) \mathbf{a})$. We find the vector $\vec{AO_1}$:

$$\vec{AO_1} = \vec{AA_1} + \vec{A_1O_1}$$

$$= m\mathbf{a} + \frac{1}{3} (\mathbf{b} + (1 - 3m) \mathbf{a}) = \frac{1}{3} (\mathbf{a} + \mathbf{b}). \quad (4)$$

Comparing (3) and (4), we get $\vec{AO} = \vec{AO_1}$, that is, the points O and O_1 coincide. ▲

PROBLEMS OF SECTION I

1. Meeting at a point O , the diagonals of the trapezoid partition it into 4 triangles with vertex at the point O . Prove that the areas of the triangles adjacent to the nonparallel sides of the trapezoid are equal.

2. In the triangle ABC , straight lines, parallel to the sides AC and AB , are drawn through a point M lying on the side BC . The area of the resulting parallelogram constitutes $5/18$ of the area of the triangle ABC . Find in what ratio the point M divides the side BC .

3. Points M , N , and P lie on the sides AB , BC , and AC of the triangle ABC , with $|AM|/|AB| = |BN|/|BC| = |CP|/|CA| = 1/3$. The straight lines CM , AN , and BP , when intersecting, bound the triangle whose area is S . Find the area of the triangle ABC .

4. When the triangle ABC is rotated about the centre at the point C through the angle α ($\alpha < \pi/2$), it passes into a triangle $A'B'C$. The point B' is the image of the point B and lies on the side AB ; the point A' is the image of the point A . Find the angles of the triangle ABC if the lines AC and $A'B'$ are perpendicular.

5. Prove that three altitudes of a triangle or three straight lines on which the altitudes lie meet at one point.

6. The extensions of the altitudes of the triangle ABC divide the circle circumscribed about the triangle into arcs whose lengths are related as $p : q : r$. Find the angles of the triangle ABC .

7. There is an internal tangency at a point A between two circles of radii R and r ($R > r$). A straight line is drawn through a point B on the larger circle which touches the smaller circle at a point C . Find the length of the segment AB if $|BC| = a$.

8. A circle inscribed into the triangle ABC divides the median BM into three congruent segments. Find the ratio between the lengths of the sides of the triangle ABC .

9. The extensions of the altitudes BM and CN of the acute triangle ABC cut the circle circumscribed about it at points P and Q . Find the radius of the circumscribed circle if $|BC| = a$ and $|PQ| = \frac{6}{5}a$.

10. A point E lies on the side AC of the regular triangle ABC , a point K being the midpoint of the segment AE . The straight line passing through the point E , at right angles to AB , and the straight line passing through the point C , at right angles to BC , meet at a point D . Find the angles of the triangle BKD .

11. In the isosceles triangle ABC ($|AB| = |BC|$) the median AD is perpendicular to the bisector CE . Determine the angle ACB .

12. In the triangle ABC the vertex angle A is equal to α . Find the length of the bisector of the triangle drawn from the vertex A if $|AB| = c$ and $|AC| = b$.

13. In the triangle ABC the vertex angle B is equal to φ , and the vertex angle C is equal to 2φ . The circle passing through the points A , C , and the centre of the circle circumscribed about the triangle ABC cuts the side AB at a point M . Determine the ratio $|AM|/|AB|$.

14. In the triangle ABC the perpendicular passing through the midpoint of the side AB cuts the side AC at a point M in the ratio

$|MA|/|MC| = 3$. The perpendicular passing through the midpoint of the side AC cuts the side AB at a point N , so that $|AN|/|NB| = 2$. Determine the angles of the triangle ABC .

15. Prove that if the lengths of the sides AB and BC of the triangle ABC are different, then the bisector of the angle ABC lies between the median and the altitude drawn from the same vertex.

16. In the triangle ABC the lengths of the sides are related as $|BC|^2 + |AC|^2 = 5|AB|^2$. Prove that the medians AM and BN are perpendicular.

17. Two circles of radii equal to $\sqrt{5}$ cm and $\sqrt{2}$ cm meet at a point A . The distance between the centre of the circles is 3 cm. Drawn through the point A is a straight line cutting the circles at points B and C so that $|AB| = |AC|$. Find the length of the segment AB .

18. The angle BAD of the rhombus $ABCD$ is acute. The circle inscribed into the rhombus touches the sides AB and CD at points M and N respectively, and cuts the segment CM at a point P and the segment BN at a point Q . Find the ratio $|BQ|/|QN|$ if $|CP|/|PM| = 9/16$.

19. Given the circle with diameter AB . Another circle with centre at the point A cuts the first circle at points C and D and the diameter AB at a point E . A point M , distinct from the points C and E , is taken on the arc CE which does not contain the point D . The straight line BM intersects the first circle at a point N . Find the length of the segment MN if $|CN| = a$, $|DN| = b$.

20. Given the triangle ABC . The bisector of the exterior angle of the triangle at the vertex A cuts the straight line BC at a point M . Prove that $|MB|/|MC| = |AB|/|AC|$.

21. Given the triangle ABC . A point M lies on the line BC , the resulting ratio being $|MB|/|MC| = |AB|/|AC|$. Prove that the point M lies either on the bisector of the angle A of the triangle ABC or on the bisector of the exterior angle at the vertex A .

22. In the parallelogram $ABCD$ the acute angle is α . Suppose O_1, O_2, O_3 , and O_4 are the centres of the circles circumscribed about the triangles DAB, DAC, DBC , and ABC respectively. Determine the ratio between the area of the rectangle $O_1O_2O_3O_4$ and the area of the parallelogram $ABCD$.

PROBLEMS OF SECTION II

1. Prove that in any triangle the sum of the lengths of its three medians is smaller than its perimeter but greater than three-fourths of the perimeter.

2. In an isosceles triangle the length of the base is a and the length of the altitude dropped to the base is h . Find the distance from the midpoint of the base to the side.

3. Determine the area of the trapezoid if it is known that a consecutive connection of the middles of its sides results in a square the length of whose side is a .

4. In the given trapezoid the length of one base is $2a$, the lengths of the other base and the sides are equal to a . Find the distance between the midpoints of the nonparallel sides.

5. Calculate the area of the circle inscribed into an isosceles trapezoid the lengths of whose bases are equal to a and b .

6. A trapezoid is circumscribed about a circle. Find the ratio between the length of the median of the trapezoid and its perimeter.

7. In the right-angled isosceles triangle two straight lines are drawn through the vertex of the right angle and divide the angle into three congruent angles. Find the lengths of the line segments into which the lines divide the hypotenuse if its length is a .

8. Find the area of the triangle ABC inscribed into a circle if the vertices A and C are at distances m and n from the tangent drawn through the vertex B , and $|AC| = b$.

9. The triangle ABC has the altitude BD . Drawn through the point D is a straight line parallel to the side AB till the intersection with the side BC at a point K . Find the ratio $|BK|/|KC|$ if the area of the triangle BCK is $3/16$ of the area of the triangle ABC .

10. In the isosceles triangle ABC ($|AB| = |BC|$) the bisector AE cuts the altitude BD at a point O , with $|OB|/|OD| = 3$. In what ratio does the altitude AF divide the altitude BD ?

11. In the isosceles triangle ABC ($|AB| = |BC|$) the altitude AF cuts the altitude BD at a point O , with $|BO|/|OD| = n$. In what ratio does the bisector AE divide the altitude BD ?

12. The diagonals are drawn in the convex quadrilateral $ABCD$ whose area is 24 sq cm. The area of the triangle ABC is known to be twice that of the triangle ACD , and the area of the triangle BCD is thrice that of the triangle BDA . Find the area of the triangle ABC .

13. The interior angles of a parallelogram are bisected. The area of the quadrilateral resulting from the intersection of the bisector is $1/4$ of the area of the parallelogram. Find the ratio of the lengths of the sides of the parallelogram.

14. In the triangle ABC a point M is taken on the side AC and a point N on the side BC . The line segments AN and BM meet at a point O . Find the area of the triangle CMN if the areas of the triangles OMA , OAB , and OBN are equal to S_1 , S_2 , and S_3 respectively.

15. In the isosceles triangle ABC ($|AB| = |BC|$), the bisectors BD and AF meet at a point O . The ratio between the area of the triangle DOA and the area of the triangle BOF is $3/8$. Find the ratio $|AC|/|AB|$.

16. The straight line parallel to the bases of the trapezoid divides it into two parts whose areas are related as $m : n$. Find the length of the segment of that straight line contained between the nonparallel sides of the trapezoid if the lengths of its bases are a and b .

17. A circle is inscribed into an isosceles trapezoid the lengths of whose bases are a and b . Determine the length of the diagonal of the trapezoid.

18. Prove that in any trapezoid the sum of the squares of the lengths of its diagonals is equal to the sum of the squares of the lengths of its nonparallel sides and the doubled product of the lengths of its bases.

19. Find the length of the base of the isosceles triangle if the length of the altitude dropped to the base is h and the radius of the inscribed circle is r .

20. Given the triangle ABC . Points M , N , and P lie on the extensions of the side AB beyond the point B , of the side BC beyond the point C and of the side CA beyond the point A , the resulting ratios being $|BM| = m|AB|$, $|CN| = n|BC|$, and $|AP| = p|AC|$. Find the ratio of the areas of the triangles MNP and ABC .

21. Points M and N , D and E , and K and L lie on the sides AB , AC , and BC of the triangle ABC respectively, with $|AM| = |MN| = |NB|$, $|BK| = |KL| = |LC|$, and $|AD| = |DE| = |EC|$. Calculate the area of the quadrilateral resulting from the intersection of the straight lines ML , NK , BD , and BE if the area of the triangle ABC is S .

22. Points M , N , and P lie on the sides AB , BC , and AC of the triangle ABC , with $|AM|/|MB| = |BN|/|NC| = |CP|/|PA| = m/n$. Determine the ratio of the areas of the triangles MNP and ABC .

23. In the triangle ABC , the bisectors of the interior and the exterior angle drawn from the vertex A cut the straight line BC at points D and E respectively. Determine the ratio $|AB|/|AC|$ if $|BD|/|BE| = 3/5$.

24. In the triangle ABC , the bisectors of the interior and the exterior angle drawn from the vertex A cut the straight line BC at points D and E respectively. Find the radius of the circle circumscribed about the triangle ADE if $|BC| = a$ and $|AB|/|AC| = 2/3$.

25. The circle constructed on the base AD of the trapezoid $ABCD$ serving as a diameter, passes through the midpoints of the nonparallel sides AB and CD and touches the base BC . Find the angles of the trapezoid.

26. A circle of radius R is drawn through the vertices A , B , and C of the right-angled trapezoid $ABCD$ ($\angle A = \angle B = \pi/2$) and cuts the line segments AD and CD at points M and N respectively, so that $|AM|/|AD| = |CN|/|CD| = 1/3$. Find the area of the trapezoid.

27. Two straight lines drawn from a point M touch the circle at points A and B . A straight line drawn from the point A is perpendicular to the diameter BD and intersects it at a point N . Prove that the line MD bisects the segment AN .

28. In a right-angled triangle, the lengths of the legs are a and b ($a < b$). Find the radius of the circle passing through the middle of the smaller leg and touching the hypotenuse at its midpoint.

29. Two chords are given in the circle: $|AB| = a$ and $|AC| = b$. The arc AC is twice as long as the arc AB . Find the radius of the circle.

30. Two tangent lines drawn from one point to the circle of radius R make an angle of 60° . Find the radius of the circle which is inscribed into the angle formed by those tangents and touches the given circle.

31. A square is inscribed into a semi-circle of radius R so that its one side lies on the diameter and two vertices lie on the circle. Find the length of the side of the square.

32. Two straight lines drawn from a point A touch the circle of radius R at points B and C . The triangle ABC is regular. Find its area.

33. A circle is constructed on the side AB of an isosceles triangle serving as a diameter. The circle cuts the base AC at a point M and

the side BC at a point N . Find the lengths of the line segments MN and NC if $|AC| = a$ and $|AB| = b$.

34. A circle of radius R passes through the vertex A of the isosceles triangle ABC , touches the base BC at the point B and cuts the lateral side AC at the point D . Find the length of the lateral side AB if $|AD|/|DC| = k$.

35. In the triangle ABC the lengths of the altitudes AD and BE are $2\sqrt{2}$ cm and 3 cm respectively. The point of intersection of the altitudes divides the altitude CK in the ratio 5 : 1, reckoning from the vertex C . Find the area of the triangle ABC .

36. Given the isosceles triangle ABC ($|AB| = |BC|$). A circle is constructed on the altitude BD serving as its diameter. Tangents drawn to that circle through the points A and C meet at a point O . Find the ratio $|AB|/|AC|$ if $|AO|/|AC| = 3/2$ and $|BD| < |AC|$.

37. In the isosceles triangle ABC ($|AB| = |BC|$) a circle is constructed on the altitude BD serving as its diameter. A straight line drawn through the point A touches the circle at a point M and cuts the straight line BD at a point O . Find the ratio $|AB|/|AC|$ if $|OM|/|AC| = 2$ and $|AC| < |BD|$.

38. A circle passes through the centre of another circle and cuts it at points A and B . A tangent to the first circle passes through the point A and divides the second circle in the ratio $m : n$ ($m < n$). Find the ratio in which the second circle divides the first.

39. The angle B in the triangle ABC is 90° , the medians AD and BE are mutually perpendicular. Determine the angle C .

40. The perimeter of the right-angled triangle ABC ($\angle C = \pi/2$) is 72 cm, and the difference between the lengths of the median CK and the altitude CM is 7 cm. Find the area of the triangle ABC .

41. A circle is circumscribed about the right-angled triangle ABC . The distances from the ends of the hypotenuse AB to the straight line touching the circle at the point C are equal to m and n respectively. Find the legs AC and BC .

42. Given the right-angled triangle ABC with legs $|AC| = 3$ cm and $|BC| = 4$ cm. A straight line drawn through the point C lies in the exterior of the triangle and makes the angles of 45° with the legs of the triangle. Find the radius of the circle, passing through the points A and B touching that line.

43. Given the right-angled triangle ABC . A circle with centre on the leg AC touches the hypotenuse AB and cuts the leg BC at a point P so that $|BP|/|PC| = 2/3$. Find the ratio between the radius of the circle and the length of the leg BC if $|AC|/|BC| = 4/5$.

44. Assume that O is the point of intersection of the medians of the triangle ABC whose sides are a , b , and c long. Prove that $|OA|^2 + |OB|^2 + |OC|^2 = \frac{1}{3}(a^2 + b^2 + c^2)$.

45. A triangle ABC is inscribed into the circle. The chords AA_1 , BB_1 , and CC_1 are drawn through the point O of intersection of the medians. Prove that

$$\frac{|AO|}{|A_1O|} + \frac{|BO|}{|B_1O|} + \frac{|CO|}{|C_1O|} = 3.$$

46. There is an external tangency between two circles of radii R and r ($R > r$). A straight line drawn through a point B of the larger circle touches the smaller circle at a point C . Find the length of the segment BC if the length of the chord AB is a .

47. A circle is circumscribed about the triangle ABC . The diameter AD cuts the side BC at a point E , so that $|AE| = |AC|$ and $|BE|/|CE| = 3/2$. Find the ratio $|DE|/|AE|$.

48. In the isosceles triangle ABC ($AB = BC$) the medians AD and CE meet at a point O . The ratio between the radius of the circle inscribed into the triangle AOC and the radius of the circle inscribed into the quadrilateral $ODBE$ is $2/3$. Find the ratio $|AC|/|BC|$.

49. There is an internal tangency between two circles with radii 5 cm and 3 cm. The chord of the larger circle touches the smaller circle and is divided by the point of tangency in the ratio 3 : 1. Find the length of the chord.

50. There is an external tangency between two circles of radii 5 cm and 4 cm. The straight line touching the smaller circle at a point A cuts the larger circle at points B and C so that $|AB| = |BC|$. Find the length of the segment AC .

51. Two circles with radii $\sqrt{2}$ cm and 1 cm meet at a point A . The distance between the centres of the circles is 2 cm. The chord AC of the larger circle cuts the smaller circle at a point B and is divided in half at that point. Find the length of the chord.

52. There is an internal tangency between two circles. The straight line passing through the centre of the larger circle cuts it at points A and D and intersects the smaller circle at points B and C . Find the ratio between the radii of the circles if $|AB| : |BC| : |CD| = 3 : 7 : 2$.

53. In the trapezoid $ABCD$ the one of the nonparallel sides, BC , is perpendicular to the bases AD and BC . The point E is the midpoint of the side CD . Find the ratio $|AD|/|BC|$ if $|AE| = 2|AB|$ and the line AE is perpendicular to the line CD .

54. The triangle ABC is regular. A point M is taken on the extension of the side AC beyond the point A , and circles are circumscribed about the triangle ABM and MBC . The point A divides the arc MAB in the ratio $\widehat{MA}/\widehat{AB} = n$. Find the ratio in which the point C divides the arc MCB .

55. A point E lies on the side AC of the regular triangle ABC , the point K being the midpoint of the segment AE . The straight line passing through the point E at right angles to AB and the straight line passing through the point C at right angles to BC meet at a point D . Find the angles of the triangle BKD .

56. The circle inscribed into the trapezoid $ABCD$ touches one of the nonparallel sides, AB , at a point F . Find the area of the trapezoid if $|AF| = m$, $|FB| = n$, and the length of the shorter base BC is equal to b .

57. A circle K is drawn through the vertices A and C of the triangle ABC . The centre of the circle lies on a circle circumscribed about the triangle ABC . The circle K cuts the extension of the side BA (beyond the point A) at a point M . Find the angle BCA if $|MA|/|AB| = 2/5$ and $\angle ABC = \arcsin 3/5$.

58. An isosceles trapezoid $ABCD$ is circumscribed about a circle. The side AB touches the circle at a point M and the base AD touches it at a point N . The line segments MN and AC intersect at a point P so that $|NP|/|PM| = 2$. Find the ratio $|AD|/|BC|$.

59. A trapezoid $ABCD$ (AD being the longer base) is inscribed into a circle. A perpendicular is drawn from the vertex C to AD intersecting the circle at a point E . The ratio between the length of the arc BC (not containing the point D) and the length of the arc CDE is $1/2$. The radius of the circle is equal to the altitude of the trapezoid. Find the ratio $|AB|/|BC|$.

60. In the triangle ABC the perpendicular passing through the midpoint of the side AB cuts the straight line AC at a point M , and the perpendicular passing through the midpoint of the side AC cuts the straight line AB at a point N . It is known that $|MN| = |BC|$ and the line MN is perpendicular to the line BC . Determine the angles of the triangle ABC .

61. In the isosceles trapezoid $ABCD$ the angle adjacent to the base AD is α , and the length of the lateral side AB is b . The circle touching the sides AB and AD and passing through the vertex C cuts the sides BC and CD at points M and N respectively. Determine $|BM|$ if $|CN|/|ND| = 3$.

62. In the isosceles trapezoid $ABCD$ the angle at the base AD is $\arcsin(24/25)$. A circle of radius R touches the base AD and the lateral side AB and passes through the vertex C . It intercepts congruent segments MC and NC , respectively, on the sides BC and CD . Find the length of the segment BM .

63. The area of the trapezoid $ABCD$ is s , the ratio of the bases $|AD|/|BC| = 2$. The segment MN is parallel to the diagonal BD and cuts the diagonal AC , and the segment AM is parallel to the segment CN . Determine the area of the quadrilateral $AMND$ if $|CN|/|AM| = 3$, $|BD|/|MN| = 6$ (find all solutions).

64. A circle is inscribed into the rhombus $ABCD$. The straight line touching the circle at a point P cuts the sides AB and BC and the extension of the side AD at points N , Q , and M , respectively, so that $|MN| : |NP| : |PQ| = 7 : 1 : 2$. Determine the angles of the rhombus.

65. The diagonals of the quadrilateral $ABCD$ meet at a point M and the angle between them is α . Assume that O_1 , O_2 , O_3 , and O_4 are the centres of the circles circumscribed about the triangles ABM , BCM , CDM , and DAM respectively. Determine the ratio between the areas of the quadrilaterals $ABCD$ and $O_1O_2O_3O_4$.

66. In the quadrilateral $ABCD$ the acute angle between the diagonals is α . A straight line drawn through every vertex is perpendicular to the diagonal not containing that vertex. Determine the ratio between the area of the quadrilateral bounded by those straight lines and the area of the quadrilateral $ABCD$.

67. In the parallelogram $ABCD$ the angle BAD is equal to α . Assume that O is an arbitrary point in the interior of the parallelogram; O_1 , O_2 , O_3 , and O_4 being the points symmetric with respect to the point O about the straight lines AB , BC , CD , and AD respectively. Determine the ratio between the area of the quadrilateral $O_1O_2O_3O_4$ and the area of the parallelogram.

68. In the acute isosceles triangle ABC the angle at the base AC is equal to α and the length of the side is a . A point M is lying on the segment BC and has the smallest sum of the squares of the distances to the lines AC and AB as compared to the other points of the segment BC . Find the length of the segment MC .

69. The length of the shorter base of the trapezoid is equal to the lengths of the sides. For what angle at the longer base does the trapezoid have the largest area?

70. In the isosceles triangle ABC the angle at the base AC is α and the length of the side is a . Through the point M lying on a side two straight lines are drawn which are parallel to the sides of the triangle and cut a parallelogram of the greatest possible area from the triangle ABC . Find the area of the parallelogram.

71. Given the circle of radius R and diameter AD . A circle with centre at a point A cuts the first circle at a point B and the diameter AD at a point C . At what value of the radius of the second circle is the length of the segment BC the largest?

72. Find the angles of the triangle in which the altitude, the bisector, and the median drawn from one vertex divide the angle into four congruent angles.

Chapter 6

A Set of Points on a Plane and in Space. Construction Problems

6.1. A Set of Points Possessing Certain Properties

A necessity sometimes arises in geometry to isolate out of all points of a plane or space the points which possess some special property. In that case, all points of a plane or space are divided into two sets: one set includes *those and only those points* which possess the indicated property; the other set contains all the remaining points and, consequently, not a single point of the second set possesses the property in question. In every problem, in which it is required to find one or another set of points possessing a certain property, two assertions, or theorems, must be proved: a direct theorem and the opposite theorem. If we state, for instance, that a set of points of a plane, equidistant from two given points A and B , is a straight line l perpendicular to the line segment AB and passing through its midpoint then we have to prove not only that every point of the line l is equidistant from the points A and B , but also that any point which does not belong to the line l is not equidistant from the given points. If you fail to realize this simple fact, you will inevitably come to errors, to solutions logically inferior, and, in a number of cases, to a false result.

Remark. It sometimes pays to prove a direct theorem $(\forall x) A(x) \Rightarrow B(x)$ and its inverse $(\forall x) B(x) \Rightarrow A(x)$ instead of a direct theorem and the opposite theorem $(\forall x) \overline{A(x)} \Rightarrow \overline{B(x)}$. This method is legitimate because the truth of the opposite theorem $(\forall x) \overline{A(x)} \Rightarrow \overline{B(x)}$ follows from the truth of the inverse theorem $(\forall x) B(x) \Rightarrow A(x)$ (see 2.3 in Part One).

In problems on seeking sets of points possessing a definite property we usually obtain points, straight lines, line segments, circles or their arcs, polygons, circumferences, planes, or spheres. The formulation of a problem, "find the

set of points possessing the required property", should be understood as follows: enumerate the points of the set, if they are finite in number, and indicate to which straight lines, line segments, circles, arcs of circles, planes, or spheres the points of the given set belong.

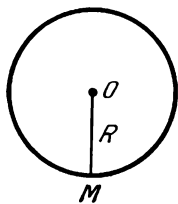


Fig. 61

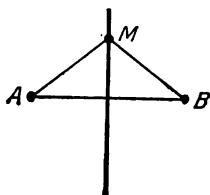


Fig. 62

It is useful to know the sets of points on a plane or in space possessing the simplest and most often encountered properties. Here are some of such sets.

On a plane.

(a) The set of points whose distance to the given point O is R is a *circle* of radius R with centre at the point O (Fig. 61).

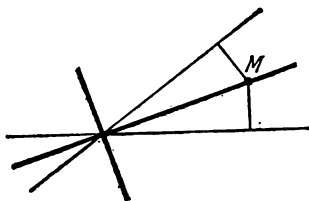


Fig. 63

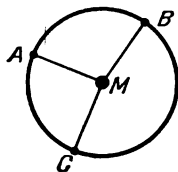


Fig. 64

(b) The set of points equidistant from two given points A and B is a *straight line* perpendicular to the line segment AB and passing through its midpoint (Fig. 62).

(c) The set of points equidistant from two given intersecting lines is a *pair of mutually perpendicular straight lines* bisecting the angles between the given lines (Fig. 63).

(d) The set of points equidistant from three given points not lying on one straight line is a *point*, which is the centre of the circle passing through the given points (Fig. 64).

(c) The set of points from which the given segment AB can be seen at a right angle, that is, the set of points M for which $\angle AMB = \pi/2$ is a *circle* with diameter AB , the points A and B exclusive (Fig. 65).

In space.

(a) The set of points whose distance from the given point O is R is a *sphere* of radius R with centre at O .

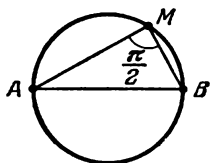


Fig. 65

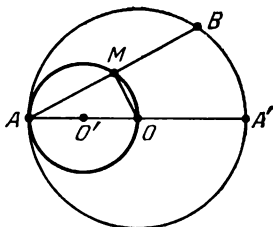


Fig. 66

(b) The set of points equidistant from two given points A and B is a *plane* which is perpendicular to the segment AB and passes through its midpoint.

(c) The set of points equidistant from two given intersecting planes is a *pair of mutually perpendicular planes* dividing in half the dihedral angles between the given planes.

(d) The set of points equidistant from three given points not lying on one straight line is a *straight line* which passes through the centre of the circle drawn through the given points and is perpendicular to the plane containing the given points in its interior.

(e) The set of points from which the given segment AB can be seen at a right angle, that is, the set of points M for which $\angle AMB = \pi/2$ is a *sphere* with diameter AB , the points A and B exclusive.

Let us consider now a number of problems.

Problem 1. Find the set of points which are the midpoints of the chords drawn from one point of the given circle.

$\triangle$ Assume that O is the centre of the given circle and A is a point belonging to the circle (Fig. 66). Prove that the set of midpoints of all the chords of the circle, which pass through the point A , is a circle with diameter AO , the

point A exclusive. Let us denote the set of all points of the circle with diameter AO , except for the point A , by the letter Γ .

Suppose the point M possesses the assigned property, that is, is the midpoint of a chord emanating from the point A . Two cases are possible here. If the point M is the midpoint of the diameter AA' , then it coincides with the point O and, consequently, belongs to the set Γ . If the point M is the midpoint of some chord AB , non-coinciding with the diameter AA' , then we connect it with the point O and consider the angle AMO . The point M being the midpoint of the chord AB , we have $[AB] \perp [OM]$, that is, the angle AMO is 90° and, therefore, the point M belongs to the circle with diameter AO (see (e) above).

Let us prove the converse statement: if $M \in \Gamma$, then it is the midpoint of some chord emanating from the point A . We again consider two cases. If the point M coincides with the point O , then M is the midpoint of the chord AA' . If $M \neq O$, we connect the points M and O . Since $M \in \Gamma$, the angle AMO is 90° and, consequently, the point M is the midpoint of the chord AB . $\blacktriangle$

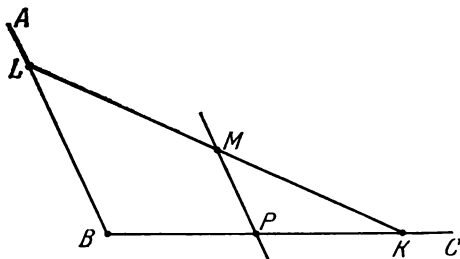


Fig. 67

Problem 2. Find the set of points which are the midpoints of the segments whose end points lie on different segments of the given angle (the given angle is less than the straight angle).

$\triangle$ It is evident that no point of the plane, not belonging to the given angle ABC (Fig. 67), can serve as the midpoint of the segment whose ends lie on different sides of the angle. It is also evident that the desired set cannot include points

located on the sides of the given angle. On the other hand, every point lying in the interior of the angle is the midpoint of some segment whose ends lie on the sides of the angle. Let us prove this fact. Suppose M is an arbitrary point lying in the interior of the angle. Let us draw through it a straight line parallel to the side BA of the angle ABC , and let P be the intersection point of that straight line and the side BC of the angle. On the side BC we lay off a segment PK which is congruent to the segment BP . Then we draw a straight line through the points K and M . The segment PM is the median of the triangle KBL , that is, the point M is the midpoint of the segment KL we have constructed. ▲

Problem 3. Two points, A and B , are given on a plane. Find the set of points M of that plane, such that $|AM|/|MB| = 2$.

△ Let us first assume that the point M does not belong to the line AB . Let M possess the specified property; this means that $|AM| = 2|MB|$. We extend the segment AM

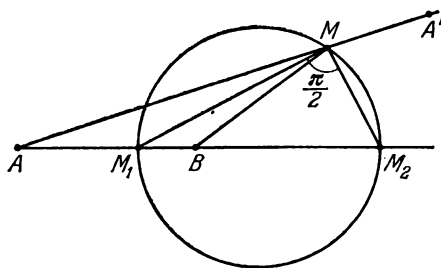


Fig. 68

(Fig. 68) and draw the bisectors of the angles AMB and BMA' .

Let M_1 be the point of intersection of the bisector of the angle AMB and the segment AB , and M_2 , the point of intersection of the bisector of the angle BMA' and the extension of the segment AB . By the property of the bisector of an interior angle of a triangle (5.1), we have

$$\frac{|AM_1|}{|M_1B|} = \frac{|AM|}{|MB|} = 2$$

and by the property of the bisector of an exterior angle of a triangle (Problem 20 of Section I in Ch. 5), we can write

$$\frac{|AM_2|}{|M_2B|} = \frac{|AM|}{|MB|} = 2.$$

Hence it follows that the position of the points M_1 and M_2 on the line AB does not depend on the position of the point M . If we now notice that the angle between the bisectors MM_1 and MM_2 is 90° , it becomes clear that, in the case when

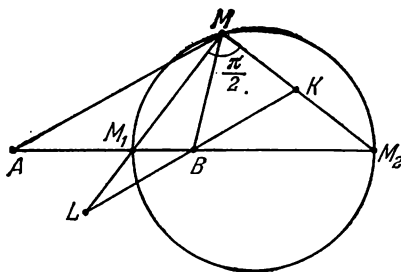


Fig. 69

the point M belongs to the desired set, the segment M_1M_2 can be seen from it at a right angle. And this signifies (see (e) above) that the point M lies on the circle constructed on the segment M_1M_2 as a diameter.

The reasoning given becomes meaningless when the point M lies on the line AB (in that case we cannot consider the angle AMB and its bisector, for instance). But it is easy to find, on the line AB , the points M which satisfy the hypothesis. These are, evidently, the points M_1 and M_2 we have constructed. The point M_1 divides the segment AB (reckoning from the point A) in the ratio $2:1$. The distance from M_2 to the point A is twice that to the point B .

Let us now prove the converse assertion: every point M of the circle constructed on the segment M_1M_2 as a diameter possesses the property $|AM| = 2|BM|$, that is, belongs to the required set. The end points of the diameter M_1M_2 , that is, the points M_1 and M_2 , evidently possess the required property. Let M be any other point of the circle with diameter M_1M_2 (Fig. 69). Let us draw, through the point B , a straight line parallel to the line AM , and let K

and L be the points of intersection of that line and the lines MM_2 and MM_1 . The similitude of the triangles AMM_2 and BKM_2 yields

$$\frac{|AM|}{|BK|} = \frac{|AM_2|}{|BM_2|} = 2.$$

The similitude of the triangles AMM_1 and BLM_1 yields

$$\frac{|AM|}{|BL|} = \frac{|AM_1|}{|BM_1|} = 2.$$

It follows from the equalities obtained that $|BK| = |AM|/2$ and $|BL| = |AM|/2$, i.e. $|BK| = |BL|$.

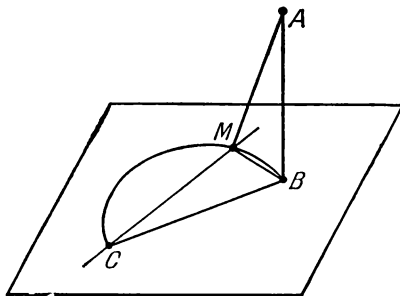


Fig. 70

Thus, in the right-angled triangle KML the segment BM is a median and, consequently, $|BM| = |BK|$. Taking into account that $|AM|/|BK| = 2$, we get $|AM|/|BM| = 2$.

Thus we see that every point of the circle with diameter M_1M_2 possesses the required property. $\blacktriangle$

Problem 4. Find the set of points which are the feet of the perpendiculars dropped from a given point of space to the straight lines lying in the given plane and meeting at the given point.

$\triangle$ Let us denote by A the given point of space, by B , its projection on the given plane, and by C , the point of intersection of the straight lines. Let us first consider the case when A , B , and C are distinct points of space (Fig. 70).

Assume that the point M belongs to the desired set. This means that $[AM] \perp (CM)$, but then it follows that the

projection MB of the line AM inclined to the given plane is perpendicular to the line CM as well, that is, the angle CMB is 90° . Consequently, the point M lies on the circle with diameter BC .

Let us prove the converse assertion. Let M be an arbitrary point of the circle with diameter BC non-coinciding with the end points B and C of the diameter. Then the angle CMB rests on the diameter. Consequently, this angle is 90° and, therefore, $[BM] \perp (CM)$. But if the projection BM of the inclined line AM is perpendicular to the line CM , then the inclined line itself is perpendicular to that straight line, i.e. $[AM] \perp (CM)$. This means that the point M possesses the specified property (is the foot of the perpendicular dropped from the point A to a certain straight line passing through the point C and lying in the plane). The end points of the diameter BC also possess the required property. Indeed, the point B is the foot of the perpendicular dropped from the point A to the straight line CB . The point C is the foot of the perpendicular dropped from the point A to the straight line passing through the point C at right angles to the segment CB .

Let us now consider two special cases of the location of the points A , B , and C .

Case one: the point A belongs to the given plane. Then the point A coincides with the point B . It is easy to see that here, as well, the set of the feet of the perpendiculars dropped from the point $A = B$ to the straight lines passing through the point C is a circle with diameter CB . But the proof is simpler in this case since we do not have to base our reasoning on the theorem on three perpendiculars and its inverse.

Case two: the point C through which the straight lines are drawn coincides with the point B , which is the projection of the point A onto the given plane. In this case it is evident that the required set is a point, namely, the point $C = B$. $\blacktriangle$

Problem 5. Given two skew lines l_1 and l_2 . Find the set of points which are the midpoints of the line segments whose ends lie on the lines l_1 and l_2 respectively.

$\triangle$ Let us denote by α and β parallel planes containing the lines l_1 and l_2 respectively, and by h the length of the common perpendicular KL (Fig. 71). Suppose the point M

belongs to the required set and this means that the point M is the midpoint of a certain segment AB whose ends, the points A and B , lie on the lines l_1 and l_2 respectively.

Assume that points B_1 and M_1 are the projections of the points B and M , respectively, onto the plane α . The segment MM_1 is a median of the triangle ABB_1 . Since $|BB_1| = |KL| = h$, it follows that $|MM_1| = h/2$. Thus, if a point belongs to the desired set, then it is equidistant from the planes α and β , that is, it lies in the plane perpendicular to the line segment KL and passing through its midpoint.

Let us prove the converse assertion: every point M of the plane perpendicular to the segment KL and passing through its midpoint is the midpoint of a certain segment whose ends lie on the lines l_1 and l_2 .

We project the point M onto the plane α and get a point M_1 . We draw a segment AB_1 through that point so that the ends of the segment, the points A and B_1 , lie on different sides of the angle B_1KA and the point M_1 bisects the segment AB_1 . The way to do it is shown in the solution of Problem 2. Now we project the point B_1 onto the plane β and get a point B , and then connect it with the point A . The segment AB contains the point M . Let us assume the contrary. We shall easily find that the point M is not equidistant from the planes α and β which fact contradicts the hypothesis. The similitude of the triangles AM_1M and AB_1B yields $|AM| = |MB|$. ▲

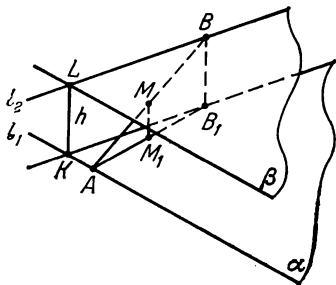


Fig. 71

6.2. Application of the Method of Coordinates

In many cases we can apply the method of coordinates to find the set of points of a plane which possess a required property. In the simplest problems the desired sets prove to be straight lines, circles, segments, arcs of circles. With the aid of the method of coordinates we can obtain their equations. Investigating these equations we can find the nature of those sets.

The method consists in the following. We choose a system of coordinates on a plane in a certain way and first of all try to decide which equation is satisfied by the coordinates of the point $M(x; y)$ if that point possesses the required property. Then we check whether each point $M(x; y)$ whose coordinates satisfy the equation obtained possesses the required property. As a result, we obtain an equation which is satisfied by the coordinates of *those and only those* points of the plane which belong to the required set. The equation obtained gives us the idea of the desired set.

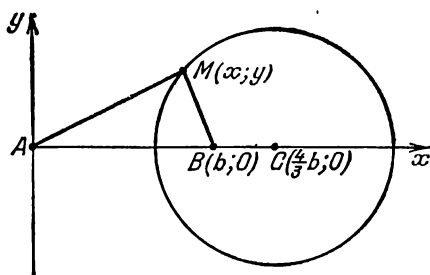


Fig. 72

Problem 1. Solve Problem 3 from 6.1 by the method of coordinates.

△ We select the system of coordinates, put the origin of coordinates at the point A (Fig. 72) and take the straight line (AB) as the axis of abscissas. Let the point B have the coordinates $(b; 0)$ in our system of coordinates, where b is the length of the segment AB .

Suppose the point $M(x; y)$ possesses the specified property, that is,

$$|AM|/|MB| = 2. \quad (1)$$

Since $|AM| = \sqrt{x^2 + y^2}$ and $|MB| = \sqrt{(x - b)^2 + y^2}$, the coordinates of the point M satisfy the equation

$$\sqrt{x^2 + y^2} = 2 \sqrt{(x - b)^2 + y^2}.$$

We transform the equation as follows:

$$\begin{aligned}x^2 + y^2 &= 4(x-b)^2 + 4y^2, \\x^2 - \frac{8}{3}bx + y^2 + \frac{4}{3}b^2 &= 0, \\(x - \frac{4}{3}b)^2 + y^2 &= \frac{4}{9}b^2.\end{aligned}\quad (2)$$

As a result we have obtained an equation of a circle of radius $R = \frac{2}{3}b$ with centre at the point $C(\frac{4}{3}b; 0)$. We have thus proved that if the point $M(x; y)$ possesses the specified property (1), then it lies on the circle (2).

We shall prove the converse statement. Suppose the point $M(x; y)$ lies on the circle obtained. This means that its coordinates satisfy equation (2), which is equivalent to condition (1) (as can be seen from the calculations carried out). Consequently, the point $M(x; y)$ possesses the specified property.

Thus, the required set is a circle of radius $R = \frac{2}{3}b$ with centre at the point $C(\frac{4}{3}b; 0)$. $\blacktriangle$

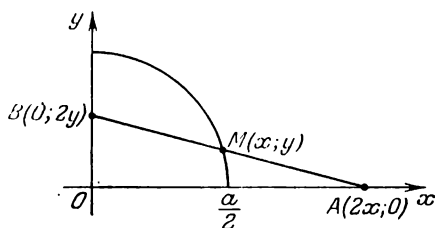


Fig. 73

Problem 2. The ends of the segment of length a slide along two sides of the given right angle. What line is described by the midpoint of the segment?

$\triangle$ Let us put the origin at the vertex of the angle and direct the axes along the sides of the angle (Fig. 73).

Suppose the point $M(x; y)$ possesses the specified property, that is, is the midpoint of the segment of length a , whose ends belong to the sides of the angle. The point

$M(x; y)$ can, evidently, be located only in the first quadrant of the coordinate plane and, therefore, $x \geq 0$ and $y \geq 0$. Let us denote the ends of the segment by A and B . It is easy to see that if $M(x; y)$ is the midpoint of the segment AB , then the points A and B have the coordinates $A(2x; 0)$, $B(0; 2y)$. By the Pythagorean theorem we get, from the triangle AOB ,

$$|AB| = \sqrt{(2x)^2 + (2y)^2}.$$

But by the hypothesis $|AB| = a$ and, consequently, the coordinates of the point M satisfy the equation

$$4x^2 + 4y^2 = a^2.$$

We have thus proved that the point $M(x; y)$ belongs to the arc of the circle

$$x^2 + y^2 = (a/2)^2,$$

located in the first quadrant: $x \geq 0$, $y \geq 0$.

Let us prove the converse statement. Assume that the point $M(x; y)$ belongs to an arc of the circle of radius $R = a/2$ with centre at the origin, located in the first quadrant. We connect the point $M(x; y)$ with the point $A(2x; 0)$ and extend the segment AM till it meets the axis of ordinates at a point B . The abscissa of the point B is, evidently, equal to zero and the ordinate is equal to $2y$. Consequently, the point $M(x, y)$ is the midpoint of the segment AB . The ends of the segment belong to the sides of the angle. Let us show that the length of the segment is equal to a . We find its length by the Pythagorean theorem:

$$|AB| = \sqrt{(2x)^2 + (2y)^2}.$$

But by our assumption, the point $M(x; y)$ lies on an arc of the circle whose equation is $x^2 + y^2 = (a/2)^2$ and, therefore $|AB| = a$. We have thus proved that if the point M belongs to an arc of the circle of radius $R = a/2$ with centre at the origin, located in the first quadrant, then it is the midpoint of some segment of length a whose ends belong to the sides of the given right angle. $\blacktriangle$

6.3. Construction Problems

In problems of this kind it is required to construct some geometric figure (a point, a straight line, a circle, or a triangle) with the aid of certain draftsman's instruments: a ruler, a pair of compasses, a triangle and others. Most often we come across problems to solve which we are allowed to use two instruments, a ruler and a pair of compasses. In the present section we shall consider only problems of this kind.

A ruler is an instrument used to draw an arbitrary line, in particular, a straight line passing through two given points. A ruler is considered to be scaleless and one-sided. Scalelessness signifies that there are no divisions on the ruler and, therefore, we cannot lay off line segments with its aid. The one-sidedness means that we cannot use its both edges simultaneously, that is, we cannot draw, say, two parallel lines using the ruler. Using a pair of compasses we can lay off any given segment on a given line, or draw an arbitrary circle, or a circle whose radius is equal to the length of the given segment. The main purpose in construction problems is not the actual construction with the aid of the indicated instruments but finding and describing the sequence of operations which lead to the construction of the required figure; proving that the constructed figure satisfies all the conditions of the problem; finding out whether the construction can always be carried out; and finding out how many solutions exist, whether there are special cases in which the construction is simplified, becomes more complicated, or proves to be impossible.

Let us recall some simplest constructions which will be used in what follows as separate constituent steps in solving more complicated problems.

Construction 1. Construct a straight line which is perpendicular to the given line segment AB and passes through its midpoint.

△ We draw circles of radius $r > |AB|/2$ with centres at the points A and B (Fig. 74). We get points C and C_1 , which are the points of intersection of the circles. Next we draw a straight line through C and C_1 , which is the required line. ▲

Remark. Construction 1 can also be used to divide a segment in half. The point O , which is the point of intersection of the segment AB and the line CC_1 , is the midpoint of the segment AB .

Construction 2. Construct a point symmetric with respect to the given point M about the given straight line l ($M \notin l$).

$\triangle$ We draw a circle with centre at M and the radius larger than the distance from M to l (Fig. 75). From the points A and B , which are the intersection points of the circle

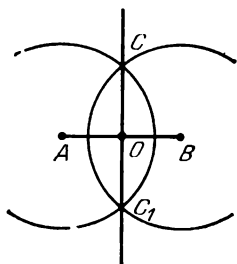


Fig. 74

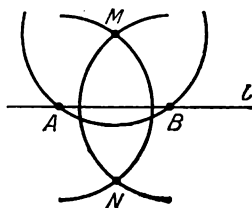


Fig. 75

and the line l , as from the centres, we draw circles of radius $r = |AM|$. The point N of intersection of those circles is the desired point. $\blacktriangle$

Construction 3. Construct a perpendicular to the given straight line l passing through the given point M .

$\triangle$ Two cases are possible here: (a) $M \notin l$, and (b) $M \in l$. In the first case, we construct a point N , symmetric with respect to M about the line l (Construction 2), and draw a straight line through M and N . The line MN is the desired line.

In the second case, we draw an arbitrary circle with centre at the point M , and mark points A and B of intersection of the circle with the line l . Then we draw two circles with centres at the points A and B , whose radius is $r = |AB|$. The desired line passes through the points of intersection of those circles. $\blacktriangle$

Construction 4. Construct a straight line which would pass through the given point M parallel to the given straight line l ($M \notin l$).

△ We select two points A and B on the line l (Fig. 76). Next we construct a point O , the midpoint of the segment AM (Construction 1), and then a point N , which is symmetric with respect to the point B about the centre O . The straight line MN is the desired line. ▲

Construction 5. Given two line segments a and b in length. Construct a line segment whose length is equal to $\sqrt{ab}$.

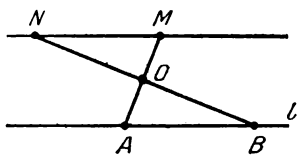


Fig. 76

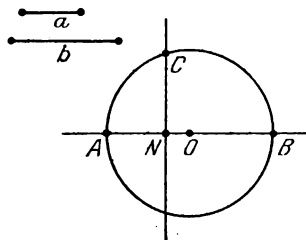


Fig. 77

△ We draw an arbitrary straight line, select a point A on it and lay off a segment AN of length a from that point along the straight line (Fig. 77). Then, from the point N on the same line we lay off a segment NB whose length is b . We find the midpoint O of the segment AB (Construction 1). Next we draw a circle of radius $r = |OA|$ with centre at O . We construct a straight line perpendicular to AB and passing through the point N (Construction 3). Then we mark one of the points of intersection of the line and the circle (point C). The segment CN is the desired segment since

$$|CN| = \sqrt{|AN| \cdot |NB|} = \sqrt{ab}. \quad \blacktriangle$$

Construction 6. Construct a circle which would pass through three given points A , B , and C not lying on the same straight line.

△ We draw a straight line l perpendicular to the line segment AB and passing through its centre (Construction 1). We draw a straight line p perpendicular to the segment AC and passing through its centre. The point $O = l \cap p$ is the centre of the desired circle. We draw a circle of radius $R = |OA|$ with centre at the point O . That is the desired circle.

Let us now consider several problems.

Problem 1. Construct a circle passing through two given points and touching the given straight line.

△ Suppose A and B are the given points and l is the given straight line. It is evident that the problem has no solution if the points A and B are located on different sides of the line l . It is also evident that there is no solution either in the case when both points lie on the line l .

Suppose the points A and B lie on one side of the line l . Assume that we have solved the problem, that is, have constructed the required circle. Let us denote by P the

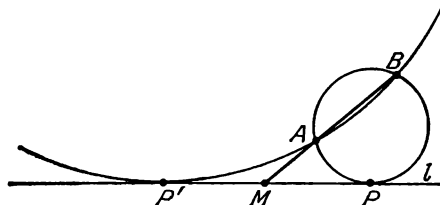


Fig. 78

point of tangency of the circle and the line l and by M the point of intersection of the lines (AB) and l (Fig. 78). In accordance with the theorem on a tangent and a secant drawn from one point to a circle (see 6.3), we have

$$|MP|^2 = |MB| |MA| \text{ or } |MP| = \sqrt{|MB| |MA|}.$$

It is now clear how to perform the construction. We draw a straight line AB and mark a point $M = (AB) \cap l$. From the known segments MA and MB we construct a segment whose length is $\sqrt{|MB| |MA|}$ (Construction 5). We lay off that segment on the line l from the point M in both directions and obtain points P and P' . Next we draw circles through the points A , B , and P and through the points A , B , and P' (Construction 6). The problem admits of two solutions.

It is easy to see that the construction just described is impossible in two cases:

- (a) the straight line AB is parallel to the line l ;
- (b) one of the given points lies on the line l .

The problem turns out to have a solution in these special cases as well.

In case (a) (Fig. 79) we draw a straight line p perpendicular to the segment AB and passing through its midpoint (Construction 1). Then we draw the desired circle through the point $P = p \cap l$ and the points A and B (Construction 6). The problem has one solution.

In case (b) (Fig. 80), we draw a straight line p perpendicular to the segment AB and passing through its midpoint.

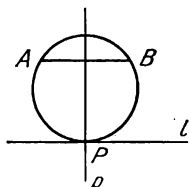


Fig. 79

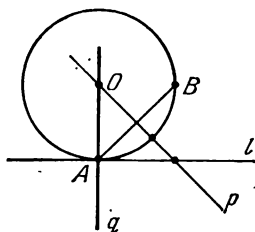


Fig. 80

Then, through the point A we draw a straight line q perpendicular to the line l (Construction 3). The point $O = p \cap q$ is the centre of the desired circle. The problem has one solution. ▲

One of the methods of solution of construction problems is the method of sets possessing the specified property. This method is especially convenient when it is required to construct a point (or points) satisfying several conditions, or when the problem can be reduced to such a construction. We shall demonstrate the method in the following problem.

Problem 2. Construct a circle of the given radius r passing through two given points.

△ The problem reduces to constructing a point, the centre of a circle. We remove the condition that the radius of the circle is r and assume that the circle has an arbitrary radius but passes through two given points A and B . Then the set of the centres of all such circles is a straight line perpendicular to the chord AB and passing through the midpoint of the segment AB . Now we neglect the requirement that the circle should pass through the point B . Let the circle pass only through the point A and have the radius r . The centres of all such circles evidently lie on a circle

of radius r with centre at A . Thus, the centre of the desired circle, first, lies on a straight line which is perpendicular to the line segment AB and passes through its midpoint and, second, belongs to the circle of radius r with centre at A , that is, lies at the intersection of the straight line and the circle (in Fig. 81 they are shown by dash lines). But a straight line and a circle may not be intersecting

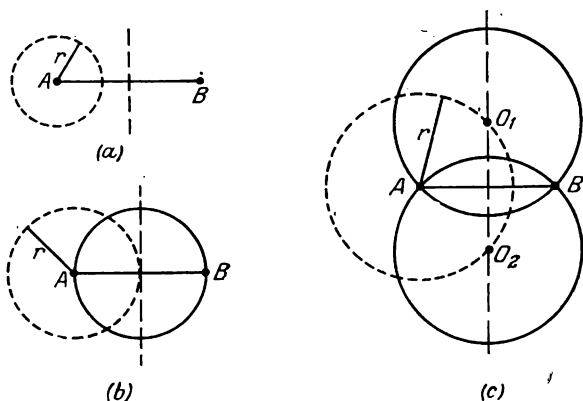


Fig. 81

(Fig. 81a), may have one point in common (Fig. 81b) and may have two points in common (Fig. 81c).

We can now formulate the ultimate result:

(a) if $r < \frac{1}{2} |AB|$, the construction is impossible, and the problem has no solutions;

(b) if $r = \frac{1}{2} |AB|$, we connect the points A and B , divide the segment AB in half, draw a circle of radius r with centre at the midpoint of the segment AB ; the problem has one solution;

(c) if $r > \frac{1}{2} |AB|$, we connect the points A and B , construct a straight line perpendicular to the segment AB at its midpoint, and draw a circle of radius r with centre at A ; we obtain two points O_1 and O_2 of intersection of the straight line we have constructed and the circle; draw two

desired circles of radius r with centres at O_1 and O_2 ; the problem has two solutions. ▲

Another widely used method of solving construction problems is the *method of similitude*. In an number of problems it is often more convenient first to construct a figure similar to the desired figure. For that purpose, some requirements placed upon the figure are removed so that the remaining conditions are satisfied by a multitude of figures similar to the desired figure. Then a figure is isolated from that multitude which satisfies all the conditions of the problem. That is the idea of the method.

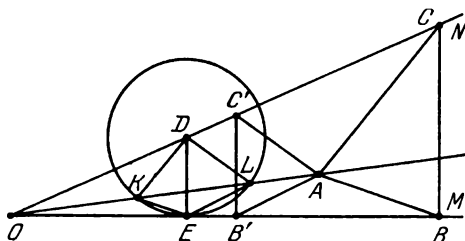


Fig. 82

Problem 3. Given the acute angle MON and a point A in its interior. Construct a triangle one of whose vertices is at the point A and the other two, B and C , lie on the sides OM and ON of the given angle, with $[BC] \perp (OM)$ and $|AC| = |BC|$.

▲ We draw a ray OA (Fig. 82). We weaken the conditions of the problem removing the requirement that the vertex of the triangle should be at the point A and shall only require that it should lie on the ray OA .

It is easy to solve the problem in this weak formulation. We take an arbitrary point D on the ray ON and from that point drop a perpendicular DE to the line OM , and then from the point D , we describe a circle of radius $R = |DE|$, which intersects the ray OA at two points, K and L . Connecting each of these points with the points D and E , we obtain two triangles DKE and DLE which satisfy all the conditions of the problem in the weak formulation. We have chosen the point D on the ray ON arbitrarily and this means that we can construct arbitrarily many triangles

similar to the triangle DKE and arbitrarily many triangles similar to the triangle DLE . Let us now draw straight lines, parallel to the segments KD and KE , through the point A ; they will intersect the sides of the angle at the vertices C and B of the desired triangle. The second solution can be obtained by analogy: that will be a triangle $AB'C'$. The problem always has two and only two solutions. ▲

In conclusion, we shall discuss the *method of symmetry*. It sometimes turns out to be possible to simplify the problem essentially by replacing one of the given elements (a point, a straight line, a circle) by some other element which is symmetric about a certain straight line. The following problem is a classical example of problems of this kind.

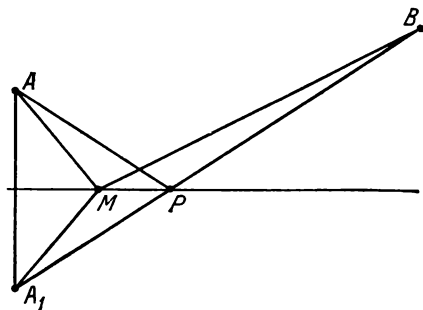


Fig. 83

Problem 4. Given two points located on one side of the given line. Find a point on the straight line the sum of whose distances to the two given points is the least.

△ Let us consider a point A_1 , symmetric with respect to the given point A about the given line (Fig. 83), and let a point M be an arbitrary point on that line. Then, since $|AM| = |A_1M|$, we have

$$|AM| + |BM| = |A_1M| + |BM|.$$

Consequently, the sum $|AM| + |BM|$ is the least if the sum $|A_1M| + |BM|$ is the least. But the last sum evidently attains its minimum when the point M lies on the line BA_1 (coincides with the point P in the figure).

Hence follows the following construction: we construct a point, symmetric with respect to one of the given points about the given line, draw a line through this point and the other given point; the point of intersection of the line we have constructed and the given line is what we sought for. It is evident that the problem always has one and only one solution.

PROBLEMS OF SECTION I

1. Given $\triangle ABC$. Find the set of points M located in the plane of the triangle and such that the areas of the triangles AMB and BMC are equal.

2. Given $\triangle ABC$. In the plane of the triangle find the set of points M such that the sum of the areas of the triangles AMB and AMC is equal to s ($s > 0$).

3. Given the line segment AB and a point C on that segment. Find the set of points of intersection of two congruent circles one of which passes through the points A and C and the other through the points C and B .

4. The points A , B , and C lie on the same straight line (the point B lying between A and C). Circles are drawn through the points A and B and tangents to them are drawn through the point C . Find the set of the points of tangency.

5. Find the set of points of the plane, the absolute value of the difference of the squares of whose distances to the two given points of the plane is constant and is equal to $s > 0$.

6. Find the set of points of the plane the sum of the squares of whose distances to the two given points of the plane is equal to the square of the distance to the third point of the plane.

7. Find the set of points of space which are the feet of the perpendiculars drawn through the given point of space to the planes passing through a given straight line (the given point does not belong to the given line).

8. Given a trihedral angle and points A and B lying on two different edges of that angle. Find the set of points of intersection of the medians of the triangles ABC , where C is an arbitrary point of the third edge.

9. A segment has no points in common with the plane and is not parallel to it. Spheres are drawn through the ends of the segment which touch the plane. Find the set of points of tangency.

10. Find the set of points of space through which it is impossible to draw a straight line intersecting two given skew lines.

11. Construct points D and E on the sides AB and BC of the acute triangle ABC respectively so that $|AD| = |DE| = |EC|$.

12. Two points A and B are located on different sides of a rectangular channel whose width is constant. Where must we build a bridge for the distance from one point to the other to be the shortest?

13. Points A and B are on different sides of the straight line l . Find a point C on that line such that the difference $|AC| - |BC|$

is the greatest (the distance from the point A to the line l is greater than that from the point B to the line l).

14. Given the acute angle AOB and two points, M and N , in its interior. How should the ray of light be directed from the point M for it to pass through the point N , first being reflected from the side AO and then from the side BO ?

15. Given an angle and a point in its exterior. Draw a line through that point intercepting a triangle of the given perimeter $2p$ on the given angle.

PROBLEMS OF SECTION II

1. Given a circle and a point in its interior. Find the set of the midpoints of the chords of the circle passing through the given point.

2. Find the set of the midpoints of the segments connecting the given point, lying outside of the given circle, with the points belonging to the circle.

3. The points A and B are given on the plane. Find the set of the projections of the point A onto the straight lines lying in the plane and passing through the point B .

4. The points A and B are given on the plane. Find the set of points of the plane which are symmetric with respect to the point A about straight lines lying in the plane and passing through the point B .

5. Given a right angle with vertex at a point O . Suppose OM is an arbitrary broken line of length l whose every segment is parallel to one of the sides of the angle. Find the set of points M lying in the interior of the angle or on its sides.

6. The right triangle is displaced in the plane so that the vertices of its acute angles slide along two mutually perpendicular straight lines. What figure is produced by the vertex of the right angle of that triangle?

7. Two stationary points A and B are given on a circle. Points C and D are displaced along the circumference of the circle so that the distance between them does not change. Prove that the point of intersection of the lines AC and BD moves along a certain circle passing through the points A and B .

8. Find the set of points of the plane the sum of the squares of whose distances to the two given points A and B of that plane is $s > 0$.

9. Given a straight line and points A and B on that line. Find the set of points of tangency of the circles one of which touches the given line at the point A and the other at the point B .

10. Find the set of the centres of the rectangles inscribed into the given triangle so that two vertices of the rectangle lie on the base of the triangle and the other two vertices lie on the lateral sides.

11. The plane α contains three given pairwise skew lines not passing through the same point. Find the set of points of the plane α equidistant from the given lines.

12. Given the line segment AB and a point C on it. Find the set of points M such that $\angle MCB = \angle 2MAB$.

13. Find the set of points the sum of the squares of whose distances to the vertices of the acute angles of the isosceles right-angled triangle

is twice as large as the square of the distance to the vertex of the right angle.

14. Given a square with side a . Find the set of points lying in the plane of the square and such that the sum of the distances from each of them to the straight lines containing the sides of the square is equal to $4a$.

15. Find the set of points of space which are equidistant from all the vertices of the isosceles trapezoid.

16. Find the set of points of space which are equidistant from three pairwise intersecting planes not passing through the same line and perpendicular to a certain plane α .

17. Find the set of points which are symmetric with respect to the given point A about the points of the given line l .

18. Find the set of points of space which are symmetric with respect to the given point A about the straight lines parallel to the given straight line l .

19. Find the set of points of space which are symmetric with respect to the given point A about the planes passing through the given line l .

20. Find the set of points of space which are symmetric with respect to the given point A about the lines passing through the other given point B .

21. Find the set of points of space which are symmetric with respect to the given point A about the planes passing through the other given point B .

22. Given a sphere and a straight line having no points in common. Find the set of the centres of the sections of the sphere by the planes passing through the given line.

23. Find the set of points of space from which three pairwise perpendicular tangent lines can be drawn to the given sphere of radius R .

24. The ends of the segment a in length slide along two mutually perpendicular straight lines l_1 and l_2 . What line is described by the midpoint of the segment?

25. Cut the trapezoid by a straight line, parallel to the bases, so that the segment lying between the nonparallel sides be divided by the diagonals into three congruent parts.

26. Draw a straight line through the point of intersection of two circles so that its segment lying within the circles should have the given length a .

27. Given two points and a circle. Draw two secants through the given points whose chords within the given circle are congruent and intersect at the given angle α .

28. Construct a circle touching two given circles under the condition that it touches one of the circles at a given point.

29. Draw a circle through two given points divided by the given circle into two congruent arcs.

30. Inscribe into an equilateral triangle another equilateral triangle whose sides are perpendicular to the sides of the given triangle.

31. Given a circle and three points lying on it. Inscribe a triangle into that circle so that its bisectors, when extended, pass through the given points.

32. Given a circle and three points lying on it. Inscribe a triangle into that circle so that the extensions of the altitude, the bisector, and the median, which emanate from the same vertex of the triangle, intersect the circle at the given points.

33. Given the centres of the circles escribed into a triangle. Construct the triangle.

Hint. A circle is said to be *escribed* into a triangle if it touches one side of the triangle and the extensions of the other two sides of the triangle.

34. Given the centres of an inscribed, an escribed, and a circumscribed circle. Construct a triangle.

35. Given three parallel straight lines. Construct a square whose three vertices should lie on the three given lines.

36. Inscribe a square into a given triangle (one side of the triangle must contain two vertices of the square and the two other sides, one vertex each).

37. Construct a square from the given vertex and two points which lie on two sides of the square not passing through the given vertex.

38. Construct a parallelogram from the given diagonals d_1 and d_2 and the acute angle α .

39. Inscribe a trapezoid into the given circle knowing the altitude h of the trapezoid and the difference d between its bases.

40. Two points A and B are taken on one side of the acute angle. Find a point C on the other side of the angle such that the angle ACB is the greatest.

41. Three parallel straight lines are given on a plane. The line segment AB is perpendicular to those lines and its ends lie on the first two lines. Find a point C on the third side such that the angle ACB is the greatest.

42. Given an angle and a point M in its interior. Construct a triangle of the least perimeter whose one vertex is at the point M and the other two lie on different sides of the given angle.

43. Given an angle and a point M in its interior. Draw a straight line through the point M so that the sum of the lengths of the segments, intercepted by the straight line on the sides of the angle is the least.

44. A straight road runs across the field. A pedestrian can walk along the road with the speed not exceeding 6 km/h and across the field with the speed not exceeding 3 km/h. Where the pedestrian can be found in an hour if at the initial moment he was on the road?

Chapter 7

Solid Geometry (Part 1)

7.1. Sections of Polyhedra

Sections of polyhedra by planes are employed in many solid geometry problems. In the present section we discuss some methods of constructing sections. We consider sections cut by planes passing through a given point and a

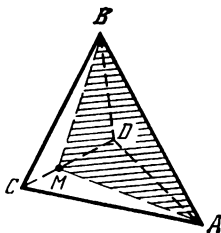


Fig. 84

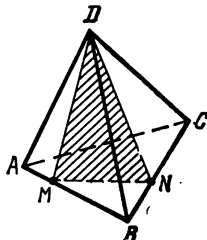


Fig. 85

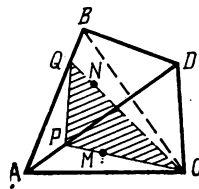


Fig. 86

straight line, through three given points, and also sections whose specifying includes the condition that the section should be parallel to the given plane, given line, or to two given lines. Examples of constructing sections by planes perpendicular to a given line or plane will be given in Chapter 8.

Figures 84 through 89 show some sections of a tetrahedron and a parallelepiped. In Fig. 84 a section is drawn through the edge AB and the point M of the edge CD ; in Fig. 85, it is drawn through the vertex D and the points M and N on the edges AB and BC ; in Fig. 86 it is drawn through the vertex C and the points M and N in the faces ACD and ACB . In each of these cases, the construction of the section is based on a simple corollary of the axioms of solid geometry:

If two planes have two points in common, then the straight line passing through these points is the intersection line of the given planes.

In Fig. 86, for instance, the points $P = (CM) \cap [AD]$ and $Q = (CN) \cap [AB]$ belong both to the plane of the section and to the face ADB and, therefore, the segment PQ is a side of the section.

Figure 87 shows a section of a tetrahedron by a plane passing through the point M of the edge AB parallel to

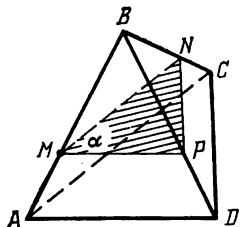


Fig. 87

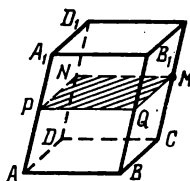


Fig. 88

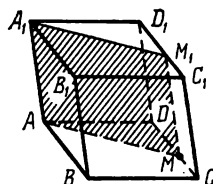


Fig. 89

the face ACD . The construction of the section is based here on the fact that

if two parallel planes are intersected by a third plane, then the intersection lines are parallel.

Let us designate the cutting plane as α . The plane ABD intersects the parallel planes α and ACD . The intersection of the planes ABD and ACD is a straight line AD and this means that the intersection line of the planes ABD and α is parallel to the line AD . We draw a straight line parallel to the line AD through the point M and find the vertex P at the point of intersection of that line and the edge BD . The line segment MP is a side of the section. By analogy we construct the side MN , $[MN] \parallel [AC]$. The segment PN is the third side of the section, $[PN] \parallel [DC]$. The section is a triangle MNP , homothetic to the triangle ACD . The ratio of similitude is $|BM|/|BA|$.

Figure 88 illustrates a section of a parallelepiped formed by a plane passing through the point M of the edge CC_1 parallel to the plane of the face $ABCD$. The construction of the section is based on the same reasoning used in the preceding case. Here $[MN] \parallel [CD]$, $[NP] \parallel [DA]$, $[PQ] \parallel [AB]$,

$[QM] \parallel [BC]$. At the same time, $[MN] \parallel [QP]$ since these segments lie on the lines of intersection of two parallel planes, AA_1B_1B and DD_1C_1C , with the plane of the section. By analogy, $[PN] \parallel [QM]$. The section is a parallelogram congruent to the face $ABCD$.

Figure 89 illustrates a section of a parallelepiped formed by a plane passing through the edge AA_1 and the point M of the edge CD . Here $[MM_1] \parallel [AA_1]$, $[A_1M_1] \parallel [AM]$ since the planes of the faces which contain these sides of the section are parallel. The section is a parallelogram AA_1M_1M .

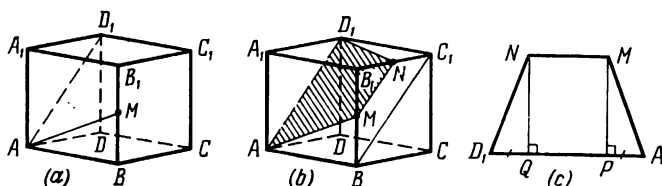


Fig. 90

Problem 1. The edge of the cube is a long (Fig. 90a). Find the area of the section drawn through the diagonal AD_1 of the face $AA_1D_1D_1$ and the midpoint M of the edge BB_1 .

$\triangle$ Let us designate the cutting plane as α . The line segments AD_1 and AM belong both to the plane α and to the faces of the cube and, therefore, are the sides of the section. We construct the side of the section in the face BB_1C_1C . The planes BB_1C_1C and $AA_1D_1D_1$ are parallel and, therefore, the intersection line of the planes α and BB_1C_1C is parallel to the straight line AD_1 . The straight lines BC_1 and AD_1 being parallel, this intersection line is parallel to the line BC_1 too. Next we draw a straight line, parallel to the line BC_1 , through the point M in the plane BB_1C_1C ; its intersection with the edge B_1C_1 produces the vertex of the section (see Fig. 90b). The section is a trapezoid $AMND_1$, $[MN] \parallel [AD_1]$.

Let us find its sides. We have $|AD_1| = a\sqrt{2}$, the segment MN is the midline of the triangle BB_1C_1 and, therefore, $|MN| = \frac{1}{2} |BC_1| = \frac{2}{\sqrt{2}}$. In the right triangles ABM and D_1C_1N ($|AB| = |C_1D_1| = a$, $|BM| =$

$|NC_1| = a/2$) we find that $|AM| = |D_1N| = a\sqrt{5}/2$. Hence, the trapezoid $AMND_1$ is isosceles. Let us find its altitude (Fig. 90c). We drop perpendiculars MP and NQ to the base AD_1 and get $|PQ| = |MN| = a/\sqrt{2}$, $|D_1Q| = |PA| = \frac{1}{2}(|D_1A| - |QP|) = a/2\sqrt{2}$. In the right triangle D_1QN ($|D_1N| = a\sqrt{5}/2$, $|D_1Q| = a/2\sqrt{2}$) we find that $|NQ| = 3a/2\sqrt{2}$. We determine the area of the section

$$S = \frac{1}{2}(|MN| + |D_1A|)|NQ| = \frac{9}{8}a^2.$$

Answer: $9a^2/8$. ▲

Shown in Fig. 91 is a section of a tetrahedron formed by a plane passing through the points M , N , and P on the

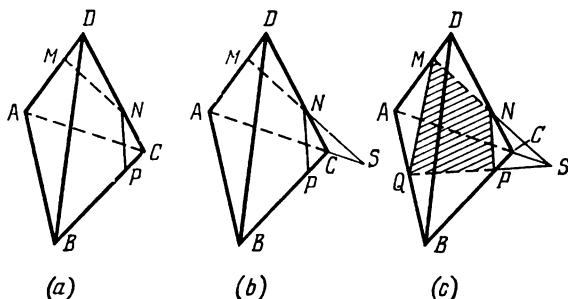


Fig. 91

edges of the tetrahedron. The points M and N are specified so that the lines MN and AC are not parallel. The segments MN and NP are the sides of the section (Fig. 91a). The point P is common for the planes MNP and ABC . The second common point is found at the intersection of the lines MN and AC , $S = (MN) \cap (AC)$ (Fig. 91b). The straight line SP is the intersection line of the planes MNP and ABC . The intersection of that line and the edge AB gives the vertex Q of the section, $Q = (SP) \cap [AB]$. The section is a quadrangle $MNPQ$.

Problem 2. Determine the shape of the section of a cube produced by a plane drawn through the midpoints of the

edges AB , AA_1 , and A_1D_1 (Fig. 92a) and find the area of the section if the edge of the cube is a long.

△ We designate the cutting plane as α . The line segments MN and NP belong both to the plane α and to the faces of the cube and, therefore, are the sides of the section (Fig. 92a).

We construct points S_1 and S_2 of intersection of the line NP belonging to the plane α and the lines AD and DD_1

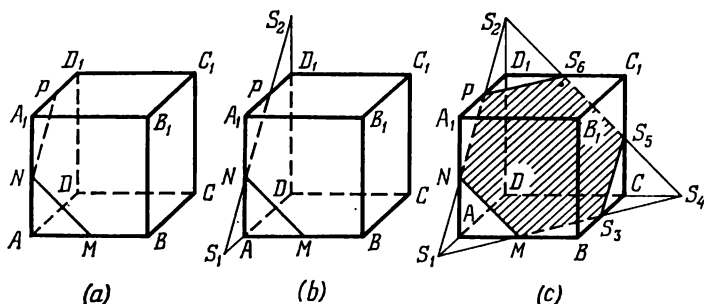


Fig. 92

(Fig. 92b). The straight line S_1M is the intersection line of the planes α and $ABCD$. We find the points of intersection of the line S_1M with the edge BC (the point S_3 in Fig. 92c) and with the line CD (the point S_4). The points S_2 and S_4 are common to the planes α and CC_1D_1D , the line S_2S_4 is the line of intersection of those planes. Next we find the intersection points of the straight line S_2S_4 with the edges CC_1 and C_1D_1 (S_5 and S_6). The section is a hexagon $PNMS_3S_5S_6$.

Note that the opposite sides of the section are parallel since they lie on the intersection lines of the plane α with pairwise parallel planes of the faces.

Let us prove that the vertices S_3 , S_5 , and S_6 of the section are the midpoints of the edges on which they lie. The segment NP is the midline of the triangle AA_1D_1 and, therefore, $(NP) \parallel (AD_1)$ (Fig. 93). By analogy we have $(MN) \parallel (BA_1)$ and, since $(BA_1) \parallel (CD_1)$, we have $(MN) \parallel (CD_1)$. Thus, the plane α of the section is parallel to the plane AD_1C . Hence it follows that the lines of intersection of these planes with the edges of the cube are parallel too, i.e.

$[MS_3] \parallel [AC]$, $[S_5S_6] \parallel [CD_1]$. In the same way we prove that $\alpha \parallel (BA_1C_1)$ ($(MN) \parallel (BA_1)$, $(NP) \parallel (AD_1) \parallel (BC_1)$), whence we get $[S_3S_5] \parallel [BC_1]$, $[S_6P] \parallel [C_1A_1]$. Noting now that $|BM| = |MA|$ and applying Thales's theorem three times, we get $|BS_3| = |S_3C|$, $|CS_5| = |S_5C_1|$, $|C_1S_6| = |S_6D_1|$. This means that the points S_3 , S_5 , and S_6 are the midpoints of the edges on which they lie.

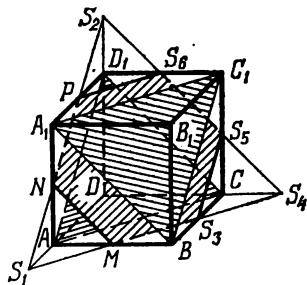


Fig. 93

It follows from what we have proved that the length of each side of the section is equal to $a/\sqrt{2}$. Let us also prove that each angle of the section is equal to 120° . Considering the triangle S_1MN , we easily establish that it is regular. Indeed, the congruency of the triangles S_1AN and PA_1N (with respect to the leg and the acute angle) yields $|S_1A| =$

$|PA_1| = a/2$. Then, $|S_1N| = \sqrt{|S_1A|^2 + |AN|^2} = a/\sqrt{2}$, $|S_1M| = \sqrt{|S_1A|^2 + |AM|^2} = a/\sqrt{2}$. Taking into account that $|MN| = a/\sqrt{2}$ too, we see that the triangle S_1MN is regular and, hence, $\angle S_1MN = \angle S_1NM = 60^\circ$. From this it follows that $\angle S_3MN = \angle MNP = 120^\circ$. Considering the triangles $S_3S_4S_5$ and S_6S_2P , we find, by analogy, that the other angles of the section are 120° as well.

Thus we see that the section is a regular hexagon with the side $a/\sqrt{2}$. Its area is equal to $(3\sqrt{3}/4)a^2$.

Answer. The section is a regular hexagon with the area equal to $(3\sqrt{3}/4)a^2$. ▲

Figure 94 illustrates a section of a tetrahedron formed by a plane passing parallel to the edge AC , through the point M of the edge CD and the point N of the face ABD . Its construction is based on the following theorem.

If a plane passes through a straight line parallel to another plane and intersects that plane, then the line of intersection of the planes is parallel to the given straight line.

We designate the plane of the section by α . The plane ACD possesses a common point M with the plane α and

contains a straight line AC parallel to the plane α . Consequently, the line of intersection of these planes passes through the point M parallel to the line AC . The side MS_1 of the section is constructed in accordance with these facts

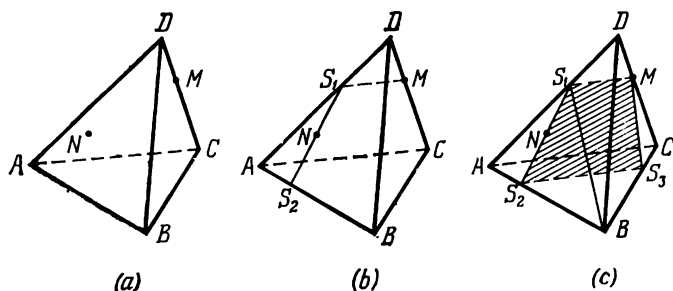


Fig. 94

(Fig. 94b), $[MS_1] \parallel [AC]$. Drawing the straight line S_1N , we find the second side of the section, S_1S_2 . In Fig. 94 the point N is such that the point S_2 belongs to the edge AB . The plane ABC also contains the line AC parallel to the plane of the section. Therefore, the side S_2S_3 of the section is drawn parallel to the edge AC (Fig. 94c). The segment S_3M is the fourth side of the section. The section $MS_1S_2S_3$ is a trapezoid ($[MS_1] \parallel [AC] \parallel [S_2S_3]$). Depending on the location of the point N with respect to the segment BS_1 , the section may also be a triangle (Fig. 95)

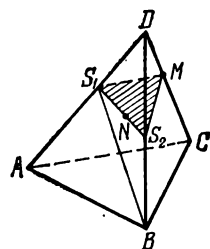


Fig. 95

Problem 3. Construct a section of a triangular prism cut by a plane passing through points A_1 and C (Fig. 96a) parallel to the straight line BC_1 . Determine the ratio in which the plane divides the edge AB .

△ We designate the plane of the section as α . The line of intersection of the planes α and BB_1C_1C passes through the point C parallel to the straight line BC_1 (Fig. 96b),

and we designate its point of intersection with the straight line BB_1 as S_1 . The point S_1 is common to the planes α and AA_1B_1B . One more common point A_1 is given in the hypothesis. Constructing a straight line A_1S_1 , we find the vertex S_2 of the section. The section is a triangle A_1CS_2 .

Let us determine the ratio $|AS_2|/|S_2B|$. From the similarity of the triangles A_1AS_2 and S_1BS_2 we have $|AS_2|/|S_2B| = |AA_1|/|BS_1|$. Since S_1BC_1C is a parallelogram ($[BS_1] \parallel [C_1C]$, $[BC_1] \parallel [S_1C]$), we have $|BS_1| = |C_1C|$. Taking into account that $|AA_1| = |C_1C|$, we get $|AS_2|/|S_2B| = 1/1$. $\blacktriangle$

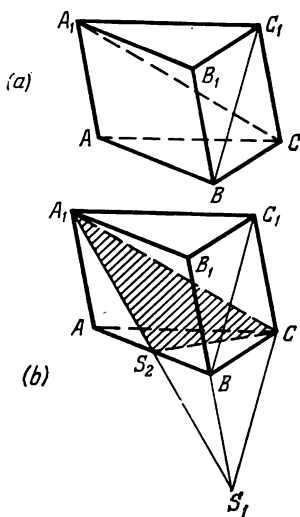


Fig. 96

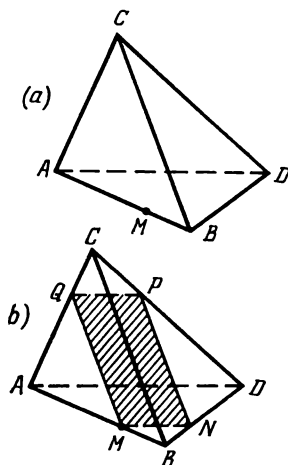


Fig. 97

Problem 4. A point M is located on the edge AB of the tetrahedron so that $|AM|/|AB| = \lambda$, $0 < \lambda < 1$ (Fig. 97a). Construct a section of the tetrahedron cut by a plane passing through the point M parallel to the edges AD and BC . For what λ is the section a rhombus if $|AD|/|BC| = m$?

$\triangle$ We designate the cutting plane as α . The line of intersection of that plane and the plane ABD is parallel to the straight line AD ($[AD] \parallel \alpha$). We draw $[MN] \parallel [AD]$

(Fig. 97b). The lines of intersection of the planes BCA and BCD with the plane α are parallel to the line BC ($(BC) \parallel \alpha$). We draw $[MQ] \parallel [BC]$ and $[NP] \parallel [BC]$. The fourth side PQ of the section is parallel to the edge AD . The section is a parallelogram $MNPQ$ ($[MN] \parallel [AD] \parallel [PQ]$, $[NP] \parallel [BC] \parallel [MQ]$).

Let us express the lengths of the sides of the parallelogram $MNPQ$ in terms of the lengths of the edges AD and BC .

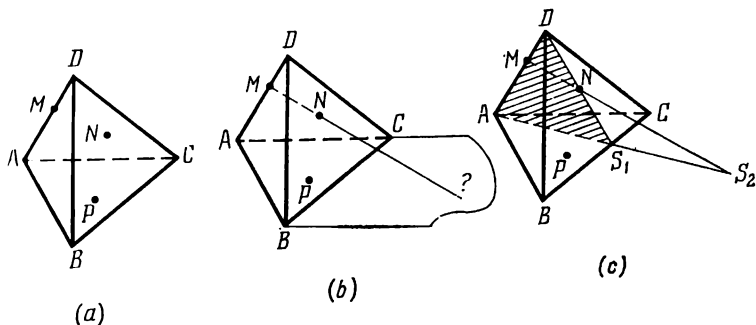


Fig. 98

The similarity of the triangles AMQ and ABC yields $|MQ|/|BC| = |AM|/|AB| = \lambda$, whence we have $|MQ| = \lambda |BC|$. Now we find that $|BM| = |AB| - |AM| = (1 - \lambda) |AB|$, and, from the similarity of the triangles BMN and BAD , we get $|MN|/|AD| = |BM|/|BA| = 1 - \lambda$, i.e. $|MN| = (1 - \lambda) |AD|$. Substituting the expressions obtained into the equation $|MN| = |MQ|$, we have $(1 - \lambda) |AD| = \lambda |BC|$, whence we get

$$\lambda = \frac{|AD|}{|BC| + |AD|} = \frac{m}{m+1}.$$

Answer: The section is a rhombus for $\lambda = m/(m+1)$. $\blacktriangle$

Remark. If the opposite edges AD and BC of the tetrahedron are perpendicular, then the sides MN and MQ of the section which are parallel to them are also perpendicular. In that case the section is a rectangle, and for $\lambda = m/(m+1)$ it is a square. In particular, if the tetrahedron $ABCD$ is regular, then the section drawn through the midpoint of the edge AB parallel to the edges BC and AD is a square (see Problem 2 in the present chapter).

To find the section of the polyhedron in the examples considered up till now we had to carry out constructions in the planes of the faces of that polyhedron (or only in the faces). In some problems, in order to find the sections it is necessary to perform a number of constructions outside the planes of the faces.

Let us consider, for instance, the construction of the section of the tetrahedron $ABCD$ cut by the plane MNP ,

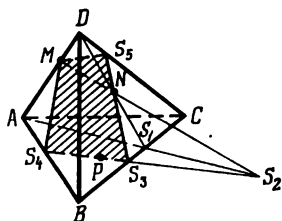


Fig. 99

where the points M , N , and P are located on the edge AD , the face BCD , and the face ABC of the tetrahedron respectively, as shown in Fig. 98a, with $(MN) \nparallel (ABC)$. According to the data given here only one point is known which is common to each face of the tetrahedron and the plane of the section MNP . To construct the line of intersection

of the plane of some face with the plane MNP , it is necessary to find one more common point of those planes. Let us consider the plane ABC . The point P is common to the planes ABC and MNP . The second common point is the point of intersection of the straight line MN and the plane ABC ($(MN) \nparallel (ABC)$). How can that point be constructed? (Fig. 98b; for the sake of comparison see Fig. 91b). The answer is in Fig. 98c. We draw a plane through the points M and N and the vertex D of the tetrahedron and construct its line of intersection AS_1 with the plane ABC . The common point S_2 of the lines MN and AS_1 is precisely the point of intersection of the line MN and the plane ABC . The line PS_2 is the line of intersection of the planes MNP and ABC . Further actions are easy enough. The vertices S_3 and S_4 (Fig. 99) are found as the points of intersection of the line PS_2 with the edges BC and AB , and the intersection of the line S_3N and the edge CD yields the vertex S_5 . The section is a quadrangle $MS_4S_3S_5$.

Figure 100a-c shows how to construct the section of a tetrahedron from three points: M in the face ABC , N in the face BCD and P in the face ACD ($(MN) \nparallel (ACD)$). To find the point of intersection of the line MN and the

plane ACD , an auxiliary plane is drawn through that line and the vertex B of the tetrahedron (Fig. 100b). The line S_1S_2 is the intersection of that plane and the plane ACD . The point $S_3 = (MN) \cap (S_1S_2)$ is the point of intersection of the line MN and the plane ACD . Further actions (Fig. 100c) are the same as in the preceding case.

In the last two examples, the construction of the point of intersection of the line MN and the plane of the face of

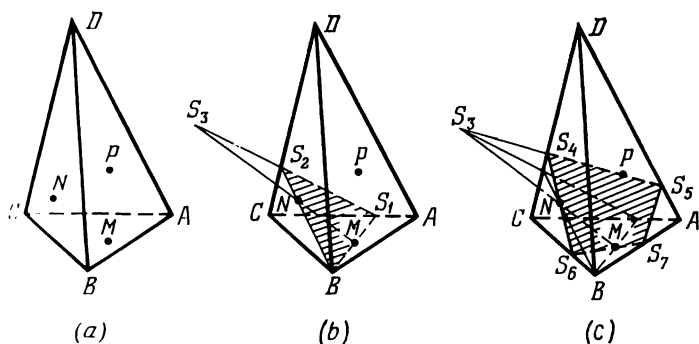


Fig. 100

the tetrahedron (face ABC in Fig. 98c, face ACD in Fig. 100b) necessitated a construction of an auxiliary plane through that line and one of the vertices of the tetrahedron (MDN in Fig. 98c, MBN in Fig. 100b). The same technique can also be used in other cases of finding the point of intersection of a line and a plane. In a pyramid and, in particular, in a tetrahedron, it is often convenient to draw an auxiliary plane through the given line and a vertex; in a prism and, in particular, in a parallelepiped, through the given line parallel to the lateral edges.

Problem 5. Construct the section of a parallelepiped cut by a plane passing through the midpoints M and N of the edges AD and BB_1 and the point P of intersection of the diagonals of the face $A_1B_1C_1D_1$ (Fig. 101a). Determine the ratio in which the plane of the section divides the edge AB .

△ The construction of the section is shown in Fig. 101b-d. Let us designate the cutting plane as α , $\alpha = (MNP)$.

First we find the point of intersection of the line NP and the plane AA_1D_1D . That line lies in the plane BB_1D_1D cutting the plane AA_1D_1D along the line DD_1 . The point S_1 of intersection of the lines NP and DD_1 (Fig. 101b) is the required point. By analogy we find the point S_2 of intersection of the line NP and the plane $ABCD$, $S_2 = (NP) \cap (DB)$. The plane α cuts the plane AA_1D_1D along the line S_1M , and the plane $ABCD$ along the line S_2M . Next we

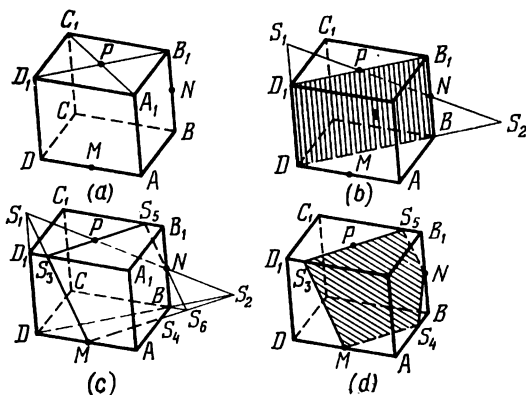


Fig. 101

find two vertices of the section: $S_3 = (S_1M) \cap [D_1A_1]$ and $S_4 = (S_2M) \cap [AB]$ (Fig. 101c). The point $S_5 = (S_3P) \cap [B_1C_1]$ is the last vertex of the section. Note that the straight lines S_5N and S_2M cut the line BC at the same point $S_6 = (BC) \cap \alpha$. The pentagon $MS_3S_5NS_4$ is the required section (Fig. 101d). The sides S_3S_5 and MS_4 as well as MS_3 and S_5N of the section are parallel since they lie in parallel faces.

Let us find the ratio $|AS_4|/|S_4B|$. From the similarity of the triangles MAS_4 and S_6BS_4 (Fig. 101c) we have $|AS_4|/|BS_4| = |AM|/|BS_6|$. The congruency of the triangles BS_6N and B_1S_5N (the point N being the midpoint of the edge BB_1) yields $|BS_6| = |B_1S_5|$. Furthermore, the point P being the centre of symmetry of the parallelogram $A_1B_1C_1D_1$, it follows that $|B_1S_5| = |D_1S_3|$. Thus we have $|BS_6| = |D_1S_3|$. Taking also into account that

$|AM|$ $|DM|$, we get $|AS_4|/|BS_4| = |DM|/|D_1S_3|$. From the similarity of the triangles DS_1M and $D_1S_1S_3$ we have $|DM|/|D_1S_3| = |DS_1|/|D_1S_1|$, and the congruency of the triangles S_1D_1P and NB_1P (Fig. 101b) (with respect to a side and the adjacent angles) yields $|D_1S_1| = |B_1N|$. Taking into account that $|B_1N| = \frac{1}{2} |B_1B| = \frac{1}{2} |D_1D|$, we get $|D_1S_1| = \frac{1}{2} |D_1D|$ and, hence, $|DS_1| = \frac{3}{2} |DD_1|$.

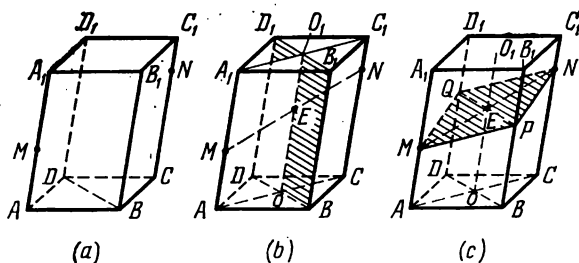


Fig. 102

Then we have, in succession, $|DM|/|D_1S_3| = |DS_1|/|D_1S_1| = 3/1$, $|AS_4|/|BS_4| = |DM|/|D_1S_3| = 3/1$.

Answer: The edge AB is divided in the ratio 3 : 1, reckoning from the vertex A . ▲

Problem 6. Points M and N are located on the edges AA_1 and CC_1 of a parallelepiped (Fig. 102a) so that $|AM|/|AA_1| = m$, $|CN|/|CC_1| = n$. Construct the section of the parallelepiped cut by a plane passing through the points M and N parallel to the diagonal BD of the base. Determine the ratio in which that plane divides the edge BB_1 .

△ We have already considered examples of constructing sections parallel to a given straight line (Figs. 94-96). The present problem differs from those examples in the fact that neither of the points M and N of the section lies in the plane of the face $ABCD$, containing the line BD which is parallel to the section. The sides of the section emanating from the vertices M and N are not parallel to the line BD since it cuts the faces containing these sides.

Let us designate the plane of the section as α . The straight line BD lies in the plane BB_1D_1D (Fig. 102b), consequently,

the planes α and BB_1D_1D intersect along a straight line parallel to BD . We construct the point of intersection of the line MN lying in the plane α with the plane BB_1D_1D . The plane AA_1C_1C containing the line MN cuts the plane BB_1D_1D along the line OO_1 which is parallel to the lateral edges of the parallelepiped. The common point E of the lines MN and OO_1 is precisely the point of intersection of the line MN and the plane BB_1D_1D . Now we draw a line,

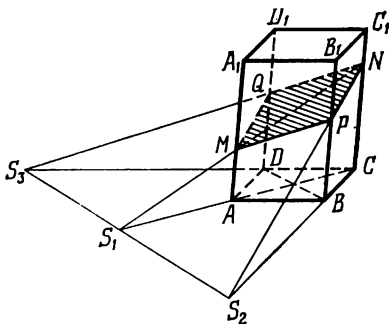


Fig. 103

parallel to the line BD , through the point E in the plane BB_1D_1D , and find the intersection points P and Q with the edges BB_1 and DD_1 respectively (Fig. 102c). The line PQ is the line of intersection of the planes α and BB_1D_1D , consequently, the points P and Q are the vertices of the section. The section is a parallelogram $MPNQ$, $[MQ] \parallel [PN]$, $[MP] \parallel [QN]$.

Next we find the ratio $|BP|/|PB_1|$. We have $|BP| = |OE|$ since $BOEP$ is a parallelogram and the segment OE is the midline of the trapezoid $AMNC$: we have $|OE| = \frac{1}{2}(|AM| + |CN|)$. Taking into account that $|AM| = m|AA_1|$, $|CN| = n|CC_1|$, and $|AA_1| = |CC_1| = |BB_1|$, we get from this $|BP| = |OE| = \frac{m+n}{2}|BB_1|$. Then we find $|PB_1| = |BB_1| - |BP| = \frac{2-m-n}{2}|BB_1|$, $|BP|/|PB_1| = \frac{m+n}{2-m-n}$.

Answer: The edge BB_1 is divided in the ratio $\frac{m+n}{2-m-n}$, reckoning from the vertex B . $\blacktriangle$

Remark. In Problem 6 we could have taken any other plane instead of the plane BB_1D_1D , say, the plane of the face $ABCD$. The corresponding constructions are shown in Fig. 103 ($S_1 = (MN) \cap (AC)$, $(S_2S_3) \parallel (BD)$).

Problem 7. Located on the diagonal AB_1 of the face ABB_1A_1 of the triangular prism (Fig. 104a) is a point M

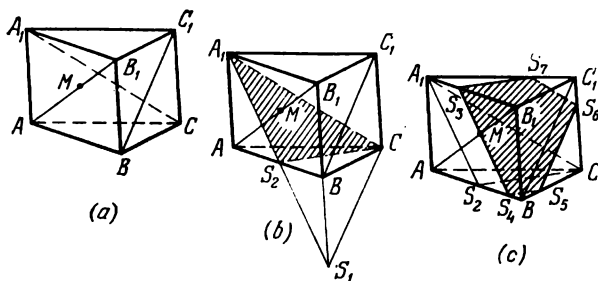


Fig. 104

such that $|AM|/|MB_1| = 5/4$. Construct the section of the prism cut by a plane passing through the point M parallel to the diagonals A_1C and BC_1 of the other two faces. Determine the ratio in which that plane divides the edge CC_1 .

$\wedge$ We have considered in Problem 4 an example illustrating the construction of a section passing through a given point parallel to two given straight lines. The present problem differs from Problem 4 in the fact that neither of the lines A_1C and BC_1 , parallel to the section, lies in the plane of the face ABB_1A_1 containing the point M of the section. Each of these lines cuts the plane ABB_1A_1 . Therefore, we cannot immediately begin constructing the section from the point M as was done in Problem 4. In such cases it is often more convenient to begin with constructing some auxiliary section parallel to the given lines.

Let us draw a plane through the straight line A_1C parallel to the line BC_1 (Fig. 104b, $(CS_1) \parallel (C_1B)$, see also Problem 3, Fig. 96). The triangle A_1S_2C is a section of the prism formed by that plane. We designate the plane of the required

section as α . This plane is parallel to the intersecting lines A_1C and CS_1 of the plane A_1CS_1 and, therefore, $\alpha \parallel (A_1CS_1)$. Hence it follows that the sides of the sought-for section are parallel to the sides of the section A_1S_2C lying in the same or in the parallel faces. Proceeding from this, we can now construct the section. We draw a straight line, parallel to the line A_1S_2 , through the point M , and find its points of intersection S_3 and S_4 with the edges (Fig. 104c). Next we construct the sides $[S_4S_5] \parallel [S_2C]$ and $[S_5S_6] \parallel [BC_1]$ (since $\alpha \parallel [BC_1]$) and $[S_6S_7] \parallel [CA_1]$. The pentagon $S_3S_4S_5S_6S_7$ is the required section (note that $[S_4S_5] \parallel [S_3S_7]$).

Now we find the ratio $|CS_6|/|S_6C_1|$. We consider the face AA_1B_1B (Fig. 104c). From the similarity of the triangles AS_4M and B_1S_3M we have $|AS_4|/|B_1S_3| = |AM|/|B_1M| = 5/4$. We designate $|S_2S_4|/|AB| = x$. Taking into consideration that $|AS_2| = \frac{1}{2}|AB|$, $|A_1S_3| = |S_2S_4|$ and $|A_1B_1| = |AB|$, we have $|AS_4| = \left(\frac{1}{2} + x\right)|AB|$, $|B_1S_3| = (1 - x)|AB|$. We obtain an equation $\left(\frac{1}{2} + x\right)/(1 - x) = 5/4$, from which we find $x = 1/3$. This means that $|S_2S_4| = \frac{1}{3}|AB|$, whence it follows that $|S_4B| = |S_2B| - |S_2S_4| = \frac{1}{6}|AB|$. Since $[S_4S_5] \parallel [S_2C]$, we have $|CS_5|/|S_5B| = |S_2S_4|/|S_4B| = 2$, and from the fact that $[S_5S_6] \parallel [BC_1]$ it follows that $|CS_6|/|S_6C_1| = |CS_5|/|S_5B| = 2$.

Answer: The edge CC_1 is divided in the ratio $2/1$, reckoning from the vertex C . ▲

We shall conclude the examples on constructing sections by a kind of warning.

As soon as the plane of a section is given (by three points, two points and a straight line parallel to the section and the like), the position of the vertices of the section on the edges of the polyhedron and of its sides in the faces is determined uniquely. They must appear as a result of a substantiated construction, and their approximate location, by eye, is inadmissible. Teaching practice shows that it is not out of place here to recall this, in general, obvious requirement once again.

➤ To conclude the section, we will show, by way of examples, how to use sections in problems on parallel straight lines and planes.

Problem 8. Points M and N are located on the diagonals AB_1 and BC_1 of the faces of the cube $ABCD A_1 B_1 C_1 D_1$ so that the segment MN is parallel to the face $ABCD$. Find the ratios $|AM|/|AB_1|$ and $|BN|/|BC_1|$ if $|MN|/|AB| = \sqrt{5}/3$.

△ Suppose $(MN) \parallel (ABCD)$ by the hypothesis. We draw a straight line $S_1 S_2$ through the point M in the face $AA_1 B_1 B$

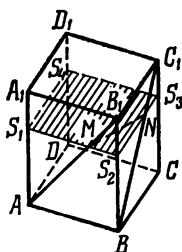


Fig. 105

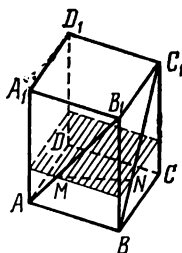


Fig. 106

(Fig. 105), parallel to the line AB . The plane defined by the lines MN and $S_1 S_2$ is parallel to the plane $ABCD$. The section of the cube by that plane is a square $S_1 S_2 S_3 S_4$, congruent to the face $ABCD$.

We designate $|AM|/|AB_1| = x$, $|AB| = a$. From the similarity of the triangles $MB_1 S_2$ and $AB_1 B$ we have $|MB_1|/|AB_1| = |MS_2|/|AB| = |B_1 S_2|/|B_1 B|$. Taking into account that $|MB_1| = (1-x)|AB_1|$, we get $|MS_2| = (1-x)|AB| = (1-x)a$, $|B_1 S_2| = (1-x)|BB_1|$, and $|BS_2| = |BB_1| - |B_1 S_2| = x|BB_1|$. The similarity of the triangles $BS_2 N$ and $BB_1 C_1$ yields $|S_2 N|/|B_1 C_1| = |BN|/|BC_1| = |BS_2|/|BB_1| = x$. Hence it follows that $|S_2 N| = xa$ and $|BN|/|BC_1| = |AM|/|AB_1| = x$.

In the right triangle $MS_2 N$ we have $|MN| = \frac{\sqrt{5}}{3} a$, $|MS_2| = (1-x)a$, $|S_2 M| = xa$ and, consequently, by the Pythagorean theorem, we have $\frac{5}{9} a^2 = (1-x)^2 a^2 + x^2 a^2$, $9x^2 - 9x + 2 = 0$. The roots of this equation are

$x_1 = 2/3$, $x_2 = 1/3$. Thus, two positions of the segment MN are possible which satisfy the condition of the problem. Figure 105 corresponds to the first position and Fig. 106 to the second.

Answer: $|AM|/|AB_1| = |BN|/|BC_1| = 2/3$ or $|AM|/|AB_1| = |BN|/|BC_1| = 1/3$. ▲

Remark. When solving the problem we have drawn a plane through the line MN parallel to the plane $ABCD$. The following theorem holds true:

A plane parallel to the given plane can be drawn through a straight line which is parallel to the given plane, and that plane will be unique.

□ To prove the theorem, we draw, through some point of the given straight line, a second straight line parallel to the given plane. These two lines define the plane parallel to the given plane. Uniqueness follows from the fact that only one plane parallel to the given plane can be drawn through a point. ■

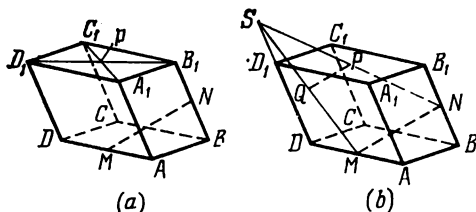


Fig. 107

Problem 9. Points M and N are the midpoints of the edges AD and BB_1 of the parallelepiped (Fig. 107a), $|MN| = a$, and the diagonals of the face $A_1B_1C_1D_1$ meet at a point P . The straight line passing through the point P , parallel to the line MN , cuts the plane AA_1D_1D at a point Q . Find the length of the segment PQ .

To solve this problem, we can make use of the section of the parallelepiped cut by the plane MNP , namely, construct the line of intersection of the planes MNP and AA_1D_1D , find a point Q on it, etc. (the constructions are shown in Fig. 107b). Sometimes it is possible to manage without any constructions in such problems on parallel

lines and planes. For instance, we can use two simple theorems to solve the given problem.

(1) *Segments of parallel straight lines contained between two distinct parallel planes are different in length.*

(2) *If two straight lines meeting at a point O are cut by two parallel planes at points A and A_1 , B and B_1 , respec-*

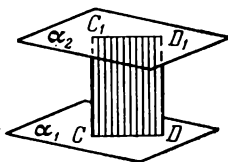


Fig. 108

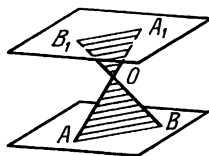


Fig. 109

tively, then $|AO|/|OA_1| = |BO|/|OB_1|$. The proof of these theorems can be easily carried out proceeding from Figs. 108 and 109.

Let us solve the problem.

$\triangle$ Assume that a point R is the intersection point of the line PQ and the plane BB_1C_1C . In accordance with the first theorem, the segments QR and MN of the parallel lines contained between the parallel planes AA_1D_1D and BB_1C_1C are equal in length, i.e. $|QR| = |MN| = a$. The segments QR and D_1B_1 contained between the same parallel planes meet at the point P , with $|D_1P|/|PB_1| = 1$. Then, in accordance with the second theorem, $|QP|/|PR| = 1$ as well, i.e. $|QP| = |PR| = a/2$.

Answer: $a/2$. $\blacktriangle$

No drawing is required to carry out this solution.

Remark. We will now formulate one more theorem in addition to theorems (1) and (2).

If the straight lines a and b are cut by three parallel planes at points A_1, A_2, A_3 and B_1, B_2, B_3 , respectively, then $|A_1A_2|/|A_2A_3| = |B_1B_2|/|B_2B_3|$.

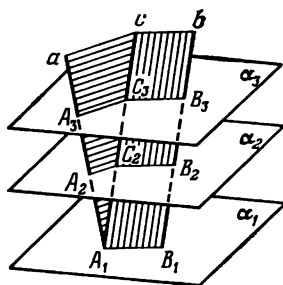


Fig. 110

We can prove it proceeding from Fig. 110. The lines a and b can be parallel, intersecting, or skew. Using this statement, we could immediately assert in Problem 8 (Fig. 105) that $|BN|/|BC_1| = |AM|/|AB_1|$.

7.2. Using the Criteria of Collinearity and Complanarity of Vectors in Problem Solving

The criteria of collinearity and coplanarity (see Ch. 6 in Part 1) of vectors serve as the basis for applying vector algebra in solving solid geometry problems. They make

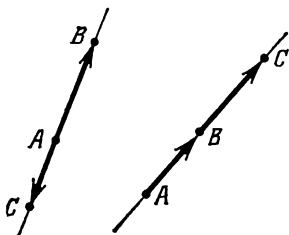


Fig. 111

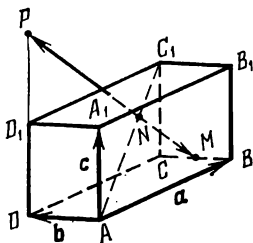


Fig. 112

it possible to express various statements concerning the position of points, straight lines, and planes in space as vector equalities. We can then pass from vector equalities to scalar ones on the strength of the uniqueness of resolution of a vector along two collinear or three coplanar vectors. We shall illustrate this statement by a number of examples.

1°. For three distinct points A, B, C to lie on the same straight line, it is necessary and sufficient that the vectors $\vec{AB}$ and $\vec{AC}$ should be collinear, that is, that there should exist a number λ such that $\vec{AC} = \lambda \vec{AB}$.

The proof of this assertion follows immediately from the definition of collinear vectors and from the fact that the vectors $\vec{AB}$ and $\vec{AC}$ issue from one point A (Fig. 111).

Problem 1. In a parallelepiped (Fig. 112), a straight line is drawn through the midpoint M of the edge BC which

cuts the straight lines AC_1 and DD_1 at points N and P respectively. Find the ratio $|MN|/|NP|$.

△ We shall designate, for brevity, three noncoplanar vectors $\vec{AB}$, $\vec{AD}$, and $\vec{AA_1}$ as $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ respectively, and shall resolve the other vectors along these three.

A point N lies on the straight line AC_1 and, therefore, the vector $\vec{NA}$ is collinear with the nonzero vector $\vec{AC_1}$, i.e. $\vec{NA} = x\vec{AC_1}$. For the vector $\vec{AC_1}$ we have the resolution $\vec{AC_1} = \mathbf{a} + \mathbf{b} + \mathbf{c}$, and this means that $\vec{NA} = x(\mathbf{a} + \mathbf{b} + \mathbf{c})$. The vector $\vec{NM}$ is the sum of the vectors $\vec{NA}$, $\vec{AB}$, and $\vec{BM}$: $\vec{NM} = \vec{NA} + \vec{AB} + \vec{BM}$. Substituting here the resolution of the vector $\vec{NA}$ and bearing in mind that $\vec{AB} = \mathbf{a}$, $\vec{BM} = \frac{1}{2}\vec{BC} = \frac{1}{2}\vec{AD} = \frac{1}{2}\mathbf{b}$, we obtain

$$\vec{NM} = (1 + x)\mathbf{a} + \left(\frac{1}{2} + x\right)\mathbf{b} + x\mathbf{c}.$$

The vectors $\vec{DP}$ and $\vec{DD_1}$ are collinear, and $\vec{DD_1} = \mathbf{c}$, therefore $\vec{DP} = y\mathbf{c}$. From the fact that $\vec{NP} = \vec{NA} + \vec{AD} + \vec{DP}$ we find that

$$\vec{NP} = x\mathbf{a} + (1 + x)\mathbf{b} + (x + y)\mathbf{c}.$$

In accordance with assertion 1° formulated above, for the points M , N , and P to lie on the same straight line, it is necessary and sufficient that the following vector equality should hold true:

$$\vec{NM} = \lambda\vec{NP}.$$

Substituting here the resolution of the vectors $\vec{NM}$ and $\vec{NP}$, we get

$$(1 + x)\mathbf{a} + \left(\frac{1}{2} + x\right)\mathbf{b} + x\mathbf{c} = \lambda x\mathbf{a} + \lambda(1 + x)\mathbf{b} + \lambda(x + y)\mathbf{c}.$$

By virtue of the uniqueness of the resolution, this vector equality is equivalent to the system of three scalar equalities:

$$\begin{cases} 1 + x = \lambda x, \\ \frac{1}{2} + x = \lambda(1 + x), \\ x = \lambda(x + y). \end{cases}$$

Solving this system, we find $\lambda = -1/2$, $x = -2/3$, $y = 2$.

Thus we have $\vec{NM} = -\frac{1}{2}\vec{NP}$, whence it follows that

$$|\vec{NM}| = \frac{1}{2} |\vec{NP}|.$$

Answer: 1 : 2. ▲

2°. Assume that A_1 and A_2 are different points of the line a , and B_1 and B_2 are distinct points of the line b .

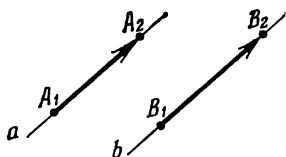


Fig. 113

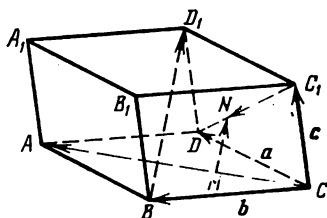


Fig. 114

For the straight lines a and b to be parallel, it is necessary and sufficient that the vectors $\vec{A_1A_2}$ and $\vec{B_1B_2}$ should be collinear, that is, there should exist a number λ such that

$$\vec{B_1B_2} = \lambda \vec{A_1A_2}.$$

The proof of this assertion follows immediately from the definition of collinear vectors and from the fact that the segment A_1A_2 lies on the line a and the segment B_1B_2 on the line b (Fig. 113).

Problem 2. $ABCD A_1 B_1 C_1 D_1$ is a parallelepiped, and AC and DC_1 are the diagonals of its faces. Prove that there exists a pair of points $M \in (AC)$ and $N \in (DC_1)$ such that

$(MN) \parallel (BD_1)$, and that this pair is unique. Find the ratio $|MN|/|BD_1|$.

△ Suppose M is a point of the straight line AC and N is a point of the line C_1D (Fig. 114). According to assertion 2°, for the straight lines MN and BD_1 to be parallel it is necessary and sufficient that there exist the number λ such that

$$\overrightarrow{MN} = \lambda \overrightarrow{BD_1}. \quad (1)$$

Let us resolve the vectors $\overrightarrow{MN}$ and $\overrightarrow{BD_1}$ along the vectors $\overrightarrow{CD}$, $\overrightarrow{CB}$, and $\overrightarrow{CC_1}$ which will be designated as $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ respectively. We have $\overrightarrow{BD_1} = \overrightarrow{BC} + \overrightarrow{CC_1} + \overrightarrow{C_1D_1}$. Here $\overrightarrow{BC} = -\overrightarrow{CB} = -\mathbf{b}$, $\overrightarrow{C_1D_1} = \overrightarrow{CD} = \mathbf{a}$, and therefore

$$\overrightarrow{BD_1} = \mathbf{a} - \mathbf{b} + \mathbf{c}. \quad (2)$$

We represent the vector $\overrightarrow{MN}$ as the sum $\overrightarrow{MN} = \overrightarrow{MC} + \overrightarrow{CC_1} + \overrightarrow{C_1N}$. The vector $\overrightarrow{MC}$ is collinear here with the vector $\overrightarrow{CA}$ and, therefore, $\overrightarrow{MC} = x\overrightarrow{CA}$. But $\overrightarrow{CA} = \mathbf{a} + \mathbf{b}$, and this means that $\overrightarrow{MC} = x\mathbf{a} + x\mathbf{b}$. The vector $\overrightarrow{C_1N}$ is collinear with the vector $\overrightarrow{C_1D}$ and $\overrightarrow{C_1D} = \mathbf{a} - \mathbf{c}$. Consequently, $\overrightarrow{C_1N} = y\overrightarrow{C_1D} = y\mathbf{a} - y\mathbf{c}$. Hence we find

$$\overrightarrow{MN} = (x + y)\mathbf{a} + x\mathbf{b} + (1 - y)\mathbf{c}. \quad (3)$$

Substituting resolutions (2) and (3) in (1), we get

$$(x + y)\mathbf{a} + x\mathbf{b} + (1 - y)\mathbf{c} = \lambda\mathbf{a} - \lambda\mathbf{b} + \lambda\mathbf{c}.$$

This vector equality is equivalent to the system

$$\begin{cases} x + y = \lambda, \\ x = -\lambda, \\ 1 - y = \lambda. \end{cases}$$

Solving this system, we find $\lambda = 1/3$, $x = -1/3$, $y = 2/3$. This means that there is a position of the points M and N for which $(MN) \parallel (BD_1)$, and this position is unique. It

is defined by the values of x and y we have found: $\overrightarrow{CM} = -x\overrightarrow{CA} = \frac{1}{3}\overrightarrow{CA}$, $\overrightarrow{C_1N} = \frac{2}{3}\overrightarrow{C_1D}$. In this case $\overrightarrow{MN} = \overrightarrow{BD_1}/3$ and, hence, $|MN| = |BD_1|/3$.

Answer: $1/3$. ▲

In the preceding two problems we passed from a vector equality expressing the geometrical meaning of the problem to an equivalent system of three scalar equations and solved that system. It should be emphasized that a vector equality is equal just to a system of equations. Even if it is required to find only one of the unknowns (λ in the problems considered), we should not restrict our search to only that value. We must verify whether the system obtained has a solution. It may so happen that the value of one unknown is found but the system and, consequently, the problem have no solution.

3°. *For the distinct points A , B , C , and D to lie on the same plane, it is necessary and sufficient that the vectors $\overrightarrow{AB}$, $\overrightarrow{AC}$, $\overrightarrow{AD}$ should be coplanar, that is, there should exist the numbers α and β such that*

$$\overrightarrow{AD} = \alpha\overrightarrow{AB} + \beta\overrightarrow{AC}.$$

The proof follows from the definition of the coplanarity of vectors and from the fact that the vectors $\overrightarrow{AB}$, $\overrightarrow{AC}$, and $\overrightarrow{AD}$ (and, hence, the vectors $\alpha\overrightarrow{AB}$ and $\beta\overrightarrow{AC}$) are laid off from the same point A (Fig. 115).

Problem 3. A plane is drawn through the ends of three edges of a parallelepiped, emanating from the same vertex. Determine the ratio in which it divides the diagonal of the parallelepiped emanating from the same vertex.

△ Let us take an arbitrary vertex of the parallelepiped and designate it as A and denote the other vertices as shown in Fig. 116. Suppose M is a point of the straight line AC_1 . For the points A_1 , B , D , and M to lie in the same plane, it is necessary and sufficient that the vectors $\overrightarrow{A_1B}$, $\overrightarrow{A_1D}$, and $\overrightarrow{A_1M}$ should be coplanar, that is, there should exist

numbers α and β such that

$$\overrightarrow{A_1M} = \alpha \overrightarrow{A_1B} + \beta \overrightarrow{A_1D}. \quad (4)$$

Let us designate $\overrightarrow{AB} = \mathbf{a}$, $\overrightarrow{AD} = \mathbf{b}$, $\overrightarrow{AA_1} = \mathbf{c}$ and resolve the vectors in (4) along these three vectors. We have $\overrightarrow{A_1B} = \mathbf{a} - \mathbf{c}$, $\overrightarrow{A_1D} = \mathbf{b} - \mathbf{c}$. Furthermore, $\overrightarrow{A_1M} = \overrightarrow{A_1A} + \overrightarrow{AM}$. Here $\overrightarrow{A_1A} = -\mathbf{c}$, and the vector $\overrightarrow{AM}$ is collinear with the

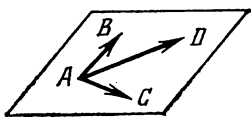


Fig. 115

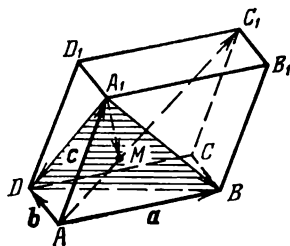


Fig. 116

vector $\overrightarrow{AC_1}$, i.e. $\overrightarrow{AM} = x \cdot \overrightarrow{AC_1} = x(\mathbf{a} + \mathbf{b} + \mathbf{c})$. This means that $\overrightarrow{A_1M} = x\mathbf{a} + x\mathbf{b} + (x-1)\mathbf{c}$. Substituting the resolutions obtained into (4), we get

$$x\mathbf{a} + x\mathbf{b} + (x-1)\mathbf{c} = \alpha\mathbf{a} + \beta\mathbf{b} - (\alpha + \beta)\mathbf{c},$$

whence we have

$$\begin{cases} x = \alpha, \\ x = \beta, \\ x - 1 = -\alpha - \beta. \end{cases}$$

Solving the system, we obtain $x = 1/3$, $\alpha = 1/3$, $\beta = 1/3$.

Thus we have $\overrightarrow{AM} = \overrightarrow{AC_1}/3$, that is, the plane BDA_1 intercepts a third of the diagonal AC_1 , reckoning from the vertex A . Hence it follows that $|AM|/|MC_1| = 1/2$.

Answer: $1/2$. ▲

Equation (4) also yields $\overrightarrow{A_1M} = \frac{1}{3}(\overrightarrow{A_1B} + \overrightarrow{A_1D})$, and this signifies that M is the point of intersection of the me-

dians of the triangle BA_1D . In other words, the diagonal AC_1 cuts the triangle BA_1D at the point of intersection of its medians.

There is a simpler method of solving Problem 3, which is a method of sections. Do it by yourself.

4°. Assume that a and b are distinct straight lines, A_1 and A_2 are points of the line a , $A_1 \neq A_2$, B_1 and B_2 are points of the line b , and $B_1 \neq B_2$.

For the lines a and b to intersect, it is necessary and sufficient that the vectors $\overrightarrow{A_1B_1}$, $\overrightarrow{A_1A_2}$, $\overrightarrow{B_1B_2}$ should be coplanar, that is, there should exist numbers α and β such that

$$\overrightarrow{A_1B_1} = \alpha \overrightarrow{A_1A_2} + \beta \overrightarrow{B_1B_2}.$$

□ **Necessity.** If the lines a and b intersect, then the points A_1 , A_2 , B_1 , and B_2 lie in the same plane and, consequently, the vectors $\overrightarrow{A_1B_1}$, $\overrightarrow{A_1A_2}$, and $\overrightarrow{B_1B_2}$ are coplanar.

Sufficiency. Suppose $\overrightarrow{A_1B_1} = \alpha \overrightarrow{A_1A_2} + \beta \overrightarrow{B_1B_2}$. Let us lay off a vector $\overrightarrow{A_1M} = \alpha \overrightarrow{A_1A_2}$ from the point A_1 . Its head, the point M , belongs to the line a . Next we lay off a vector $\overrightarrow{B_1N} = -\beta \overrightarrow{B_1B_2}$ from the point B_1 . Its head, the point N , belongs to the line b . From the vector equality

$$\overrightarrow{MN} = \overrightarrow{MA_1} + \overrightarrow{A_1B_1} + \overrightarrow{B_1N},$$

expressing its right-hand side in terms of the vectors $\overrightarrow{A_1A_2}$ and $\overrightarrow{B_1B_2}$, we obtain

$$\overrightarrow{MN} = -\alpha \overrightarrow{A_1A_2} + \alpha \overrightarrow{A_1A_2} + \beta \overrightarrow{B_1B_2} - \beta \overrightarrow{B_1B_2} = 0.$$

It follows from the definition of a zero vector that the points M and N coincide, that is, the lines a and b intersect. ■

Problem 4. Points M and N are the middles of the edges of a tetrahedron (Fig. 117), points P and Q are located on the edges AD and BC so that the segments MN and PQ intersect, and $|AP|/|AD| = 2/3$. Find the ratio $|BQ|/|BC|$.

△ For the lines MN and PQ to intersect, it is necessary and sufficient that there exist numbers α and β such that

$$\vec{MP} = \alpha \vec{MN} + \beta \vec{PQ}. \quad (5)$$

We resolve the vectors $\vec{MN}$, $\vec{PQ}$, and $\vec{MP}$ along the vectors $\vec{AB} = \mathbf{a}$, $\vec{AC} = \mathbf{b}$, and $\vec{AD} = \mathbf{c}$. We have $\vec{MN} = \vec{MA} + \vec{AD} + \vec{DN}$, and, since $\vec{MA} = -\frac{1}{2}\mathbf{a}$, $\vec{DN} = \frac{1}{2}\vec{DC}$, and $\vec{DC} = \vec{AC} - \vec{AD} = \mathbf{b} - \mathbf{c}$, it follows that

$$\vec{MN} = -\frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{b} + \frac{1}{2}\mathbf{c}.$$

We introduce the designation $|BQ|/|BC| = x$. Then

we have $\vec{BQ} = x\vec{BC} = x(\mathbf{b} -$

$\mathbf{a})$. Taking into account

that $\vec{PQ} = \vec{PA} + \vec{AB} + \vec{BQ}$ and $\vec{PA} = \frac{2}{3}\vec{DA} = -\frac{2}{3}\mathbf{c}$, we obtain

$$\vec{PQ} = (1-x)\mathbf{a} + x\mathbf{b} - \frac{2}{3}\mathbf{c}.$$

And finally we find $\vec{MP} = \vec{MA} + \vec{AP} = -\frac{1}{2}\mathbf{a} + \frac{2}{3}\mathbf{c}$.

Substituting the resolutions obtained into (5), we get

$$-\frac{1}{2}\mathbf{a} + \frac{2}{3}\mathbf{c} = \left(\beta(1-x) - \frac{\alpha}{2}\right)\mathbf{a} + \left(\frac{\alpha}{2} + \beta x\right)\mathbf{b} + \left(\frac{\alpha}{2} - \frac{2}{3}\beta\right)\mathbf{c},$$

whence it follows that

$$\begin{cases} -\frac{1}{2} = \beta(1-x) - \frac{\alpha}{2}, \\ 0 = \frac{\alpha}{2} + \beta x, \\ \frac{2}{3} = \frac{\alpha}{2} - \frac{2}{3}\beta. \end{cases}$$

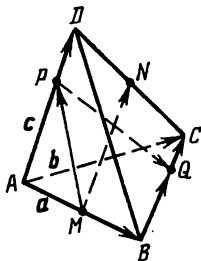


Fig. 117

This system yields $\beta = -1/2$, $x = 2/3$, $\alpha = 2/3$. Consequently, $|BQ|/|BC| = 2/3$.

Answer: $2/3$. ▲

In just the same way as was done in the preceding examples, we can express the conditions of parallelism of a line and a plane, of two planes, etc. in the form of vector equalities.

7.3. An Angle Between Straight Lines in Space

To define any two straight lines, a concept of an angle between them has been introduced in solid geometry.

The *angle* between two intersecting lines is the magnitude of the smallest angle formed by those lines. If all the four angles formed by intersecting lines are congruent, then the angle between them is equal to 90° (or $\pi/2$ radians).

If two straight lines are parallel, then the angle between them is considered to be equal to 0° .

The *angle* between two skew lines is the angle between two intersecting lines correspondingly parallel to the given skew lines.*

In this definition the point through which the two intersecting lines pass can be chosen arbitrarily, the choice being unrelated to the angle between the skew lines. This follows from the theorem given below.

If the intersecting straight lines a and b are parallel to the intersecting lines a_1 and b_1 , respectively, then the angle between the lines a and b is equal to the angle between the lines a_1 and b_1 .

□ To prove the theorem, we select similarly directed rays on each pair of the parallel lines and make use of the fact that two convex angles with similarly directed sides are congruent (Fig. 118): $[OM] \uparrow [O_1M_1]$, $[OP] \uparrow [O_1P_1]$, and, consequently, $\angle POM \cong \angle P_1O_1M_1$, etc. ■

* The term "angle" denoting a quantity in these definitions is also used as the name of a geometric figure. It is usually clear from the text what meaning is meant.

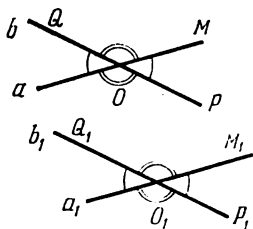


Fig. 118

It follows from the definition of the angle between straight lines and from the theorem above that

if the lines a and b are parallel, respectively, to the lines a_1 and b_1 , then the angle between the lines a and b is equal to the angle between the lines a_1 and b_1 .

Problem 1. Find the angle between the straight lines which contain the diagonals AB_1 and BC_1 of the faces of the cube (Fig. 119a).

$\triangleq$ The given diagonals lie on skew lines. The diagonal AD_1 of the face ADD_1A_1 is parallel to the diagonal BC_1 and,

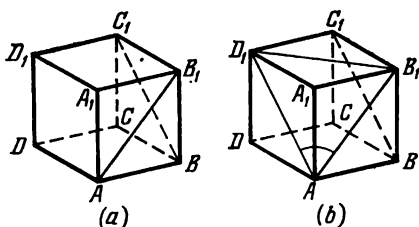


Fig. 119

therefore, the angle between the lines AD_1 and AB_1 is equal to the angle between the lines BC_1 and AB_1 .

Let us consider the triangle AB_1D_1 (Fig. 119b). Its every side is the diagonal of a face of the cube. The lengths of the diagonals are equal and, hence, the triangle AB_1D_1 is equilateral and $\angle B_1AD_1 = 60^\circ$. We have found the angle between the rays AB_1 and AD_1 , and it proved to be acute, and, therefore, the angle between the lines AD_1 and AB_1 has the same magnitude.

Answer: 60° . $\blacktriangle$

Problem 2. In a regular triangular prism (Fig. 120) $|AA_1| = \frac{1}{\sqrt{5}} |AB|$. Find the angle between the lines AB_1 and BC_1 .

$\triangleq$ Let us perform a parallel displacement of the diagonal AB_1 along $\overrightarrow{B_1C_1}$ and denote the image of the point A by A_2 (Fig. 120b). The angle between the lines AB_1 and BC_1 is equal to the angle between the lines BC_1 and C_1A_2 . We have $[AA_2] \parallel [B_1C_1] \parallel [BC]$, $|AA_2| = |B_1C_1| = |BC|$.

Consequently, the quadrangle $ABCA_2$ is a parallelogram and, moreover, a rhombus since $|BC| = |AB|$. The point O is the centre of the rhombus and, therefore, BO is an altitude of the triangle ABC . We designate $|AB| = a$ and then $|BO| = a\sqrt{3}/2$, $|BA_2| = a\sqrt{3}$. Taking into account that $|BB_1| = a/\sqrt{5}$, we find $|AB_1| = |BC_1| = \sqrt{6/5}a$ in the triangles BB_1A and BB_1C_1 . The segment

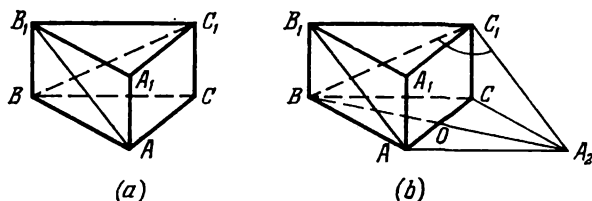


Fig. 120

A_2C_1 is the image of the segment AB_1 in the parallel displacement and, therefore, $|A_2C_1| = |AB_1| = \sqrt{6/5}a$. Using now the cosine law, we find in the triangle BC_1A_2 ,

$$\cos BC_1A_2 = \frac{|BC_1|^2 + |C_1A_2|^2 - |BA_2|^2}{2|BC_1||C_1A_2|} = -\frac{1}{4},$$

whence $\angle BC_1A_2 = \arccos(-1/4)$. The angle between the rays C_1B and C_1A_2 turned out to be obtuse ($\pi/2 < \arccos(-1/4) < \pi$) and, consequently, the angle between the lines C_1B and C_1A_2 is supplementary to that angle: $\widehat{(C_1B, C_1A_2)} = \pi - \arccos(-1/4)$. Using the identity $\arccos x + \arccos(-x) = \pi$, ($|x| \leq 1$), we find that $\widehat{(C_1B, C_1A_2)} = \arccos(1/4)$.

Answer: $\arccos(1/4)$. ▲

In the last two problems the search for an angle between lines reduced to finding the angle between rays with a common vertex, parallel to the given lines. If α is an angle between straight lines, then the angle between rays may turn out to be equal either to α or to $\pi - \alpha$. This fact must be taken into account, especially in problems when it is required to find other parameters of a polyhedron from the given angle between straight lines.

Problem 3. In a regular triangular prism (Fig. 120a) the angle between the lines AB_1 and BC_1 is equal to $\arccos(1/4)$, and the side of the base is a in length. Find the length of the lateral edge.

△ Let us designate the length of the lateral edge as b . We carry out the same constructions as in solving Problem 2 and consider two cases:

(1) $\angle BC_1A_2 = \arccos(1/4)$ (Fig. 121); (2) $\angle BC_1A_2 = \pi - \arccos(1/4)$ (Fig. 120b).

We have $|C_1A_2| = |AB_1| = |BC_1| = \sqrt{a^2 + b^2}$, $|BA_2| = a\sqrt{3}$. Using the cosine law, we find, in the triangle BC_1A_2 , that $3a^2 = 2 \times (a^2 + b^2) \left(1 - \frac{1}{4}\right)$ in the

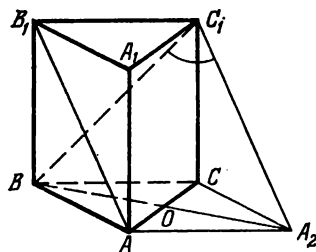


Fig. 121

first case, and, analogously, $3a^2 = 2(a^2 + b^2) \left(1 + \frac{1}{4}\right)$ in the second. From this we find $b_1 = a$ and $b_2 = a/\sqrt{5}$ respectively. Thus, the problem admits of two solutions. The second case corresponds to the data of Problem 2.

Answer: a or $a/\sqrt{5}$. ▲

7.4. Using a Scalar Product of Vectors in Problem Solving

A scalar product of vectors can be used to find angles between straight lines, lengths of line segments, distances between a point and a straight line, and so on.

The formula

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| \cdot |\mathbf{b}| \cos(\widehat{\mathbf{a}, \mathbf{b}}),$$

which defines the scalar product of the vectors $\mathbf{a}$ and $\mathbf{b}$, yields

$$\cos(\widehat{\mathbf{a}, \mathbf{b}}) = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| \cdot |\mathbf{b}|}.$$

Let us apply this formula to find the angle between straight lines. Suppose A_1 and A_2 are some points of the

straight line a , $A_1 \neq A_2$, and B_1 and B_2 are points of the line b , $B_1 \neq B_2$. The magnitude of angle φ between the vectors $\overrightarrow{A_1A_2}$ and $\overrightarrow{B_1B_2}$ is in the limits between 0 and π and is defined by the formula

$$\cos \varphi = \frac{\overrightarrow{A_1A_2} \cdot \overrightarrow{B_1B_2}}{|\overrightarrow{A_1A_2}| \cdot |\overrightarrow{B_1B_2}|}.$$

The magnitude of angle α between the lines a and b is in the limits between 0 and $\pi/2$. If $0 \leq \varphi \leq \pi/2$, then $\alpha = \varphi$ and $\cos \alpha = \cos \varphi = |\cos \varphi|$, now if $\pi/2 < \varphi \leq \pi$, then $\alpha = \pi - \varphi$ and $\cos \alpha = -\cos \varphi = |\cos \varphi|$. Hence

$$\cos(\widehat{a, b}) = \frac{|\overrightarrow{A_1A_2} \cdot \overrightarrow{B_1B_2}|}{|\overrightarrow{A_1A_2}| \cdot |\overrightarrow{B_1B_2}|}.$$

It follows from this formula that

for the straight lines a and b to be perpendicular, it is necessary and sufficient that

$$\overrightarrow{A_1A_2} \cdot \overrightarrow{B_1B_2} = 0.$$

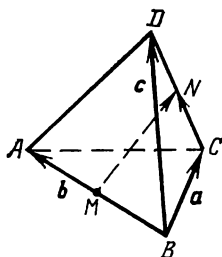


Fig. 122

Problem 1. In a regular tetrahedron, points M and N are the midpoints of the edges AB and CD (Fig. 122), $|AB| = a$. Find:

- (1) the length of the segment MN ;
- (2) the angle between the lines MN and BC ;
- (3) prove that the line MN is perpendicular to the edges AB and CD .

$\triangle$ We designate $\overrightarrow{BC} = \mathbf{a}$, $\overrightarrow{BA} = \mathbf{b}$, and $\overrightarrow{BD} = \mathbf{c}$ and have

$$|\mathbf{a}| = |\mathbf{b}| = |\mathbf{c}| = a, \quad \mathbf{a} \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{c} = \mathbf{b} \cdot \mathbf{c} = a^2/2.$$

We represent $\overrightarrow{MN}$ in terms of $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$. Since $\overrightarrow{MN} = \overrightarrow{MB} + \overrightarrow{BC} + \overrightarrow{CN}$ and $\overrightarrow{MB} = -\frac{1}{2}\mathbf{b}$, $\overrightarrow{CN} =$

$\frac{1}{2} \overrightarrow{CD}$, and $\overrightarrow{CD} = \mathbf{c} - \mathbf{a}$, it follows that

$$\overrightarrow{MN} = \frac{1}{2} (\mathbf{a} - \mathbf{b} + \mathbf{c}).$$

(1) We find

$$\begin{aligned} |\overrightarrow{MN}|^2 &= \frac{1}{4} (\mathbf{a} - \mathbf{b} + \mathbf{c})^2 \\ &= \frac{1}{4} (\mathbf{a}^2 + \mathbf{b}^2 + \mathbf{c}^2 - 2\mathbf{a} \cdot \mathbf{b} + 2\mathbf{a} \cdot \mathbf{c} - 2\mathbf{b} \cdot \mathbf{c}) = \frac{a^2}{2}. \end{aligned}$$

Hence $|\overrightarrow{MN}| = a/\sqrt{2}$.

(2) We calculate $\overrightarrow{MN} \cdot \overrightarrow{BC} = \frac{1}{2} (\mathbf{a} - \mathbf{b} + \mathbf{c}) \cdot \mathbf{a} = \frac{a^2}{2}$,

and, taking into account that $|\overrightarrow{MN}| = a/\sqrt{2}$, $|\overrightarrow{BC}| = a$, we get

$$\cos(\widehat{MN, BC}) = \frac{|\overrightarrow{MN} \cdot \overrightarrow{BC}|}{|\overrightarrow{MN}| \cdot |\overrightarrow{BC}|} = \frac{1}{\sqrt{2}}.$$

This means that the angle $(\widehat{MN, BC}) = \pi/4$.

(3) We find, in succession,

$$\overrightarrow{MN} \cdot \overrightarrow{BA} = \frac{1}{2} (\mathbf{a} - \mathbf{b} + \mathbf{c}) \cdot \mathbf{b} = \frac{1}{2} \left(\frac{a^2}{2} - a^2 + \frac{a^2}{2} \right) = 0,$$

$$\begin{aligned} \overrightarrow{MN} \cdot \overrightarrow{CD} &= \frac{1}{2} (\mathbf{a} - \mathbf{b} + \mathbf{c}) \cdot (\mathbf{c} - \mathbf{a}) = \frac{1}{2} ((\mathbf{c} + \mathbf{a}) \cdot (\mathbf{c} - \mathbf{a}) - \mathbf{b}(\mathbf{c} - \mathbf{a})) \\ &= \frac{1}{2} (\mathbf{c}^2 - \mathbf{a}^2 - \mathbf{b} \cdot \mathbf{c} + \mathbf{b} \cdot \mathbf{a}) = \frac{1}{2} \left(a^2 - a^2 - \frac{a^2}{2} + \frac{a^2}{2} \right) = 0. \end{aligned}$$

Hence it follows that $(MN) \perp (AB)$, $(MN) \perp (CD)$.

Answer: (1) $a/\sqrt{2}$; (2) $\pi/4$. $\blacktriangle$

Problem 2. In the tetrahedron $ABCD$ we have $|AB| = |BC|$, $|AD| = |DC|$. Prove that the edges AC and BD are perpendicular.

$\triangle$ We choose three noncoplanar vectors: $\overrightarrow{BC} = \mathbf{a}$, $\overrightarrow{BA} = \mathbf{b}$, $\overrightarrow{BD} = \mathbf{c}$ (Fig. 123). We have $\overrightarrow{DC} = \mathbf{a} - \mathbf{c}$, $\overrightarrow{DA} = \mathbf{b} - \mathbf{c}$. By the hypothesis $|\mathbf{a}| = |\mathbf{b}|$ and $|\mathbf{a} - \mathbf{c}| = |\mathbf{b} - \mathbf{c}|$. Hence it follows that $(\mathbf{a} - \mathbf{c})^2 = (\mathbf{b} - \mathbf{c})^2$, $\mathbf{a}^2 -$

$2\mathbf{a} \cdot \mathbf{c} + c^2 = b^2 - 2\mathbf{b} \cdot \mathbf{c} + c^2$. Taking into account that $a^2 = b^2$, we obtain $\mathbf{a} \cdot \mathbf{c} = \mathbf{b} \cdot \mathbf{c}$, i.e. $(\mathbf{a} - \mathbf{b}) \cdot \mathbf{c} = 0$. Since $\mathbf{a} - \mathbf{b} = \overrightarrow{AC}$, $\mathbf{c} = \overrightarrow{BD}$, we have $\overrightarrow{AC} \cdot \overrightarrow{BD} = 0$, i.e. $(AC) \perp (BD)$. $\blacktriangle$

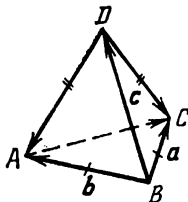


Fig. 123

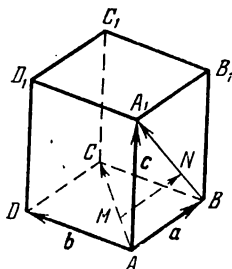


Fig. 124

Corollary. *In a regular triangular pyramid, in a regular tetrahedron in particular, the opposite edges are perpendicular.*

It was proved in Problem 1 that a straight line passing through the midpoints of the skew edges of a regular tetrahedron is perpendicular to those edges.

The following assertion holds true.

For any two skew lines there is a straight line which cuts both given lines and is perpendicular to each of them, and that line is unique.

A line segment whose end points lie on two skew lines and which is perpendicular to those lines is called their *common perpendicular*. The common perpendicular has the least length among all the segments whose end points lie on two skew lines. Therefore, the distance between skew lines is equal to the length of their common perpendicular.

Problem 3. The edge of the cube is a in length. Find the distance between the straight lines which contain the skew diagonals of two adjacent faces of the cube.

$\triangle$ Let us consider, for instance, the diagonals AC and A_1B of the faces of the cube (Fig. 124). Assume that M is a point of the straight line AC and N is a point of the line A_1B . The condition of the perpendicularity of the line segment MN and the lines AC and A_1B is equivalent to the

fact that

$$\vec{MN} \cdot \vec{AC} = 0, \quad \vec{MN} \cdot \vec{BA_1} = 0. \quad (1)$$

Let us represent $\vec{AC}$, $\vec{BA_1}$, and $\vec{MN}$ in terms of $\vec{AB} = \mathbf{a}$, $\vec{AD} = \mathbf{b}$, and $\vec{AA_1} = \mathbf{c}$. We have $\vec{AC} = \mathbf{a} + \mathbf{b}$ and $\vec{BA_1} = \mathbf{c} - \mathbf{a}$. Furthermore, since the point M lies on the line AC and the point N on the line A_1B , it follows that $\vec{AM} = x\vec{AC}$ and $\vec{BN} = y\vec{BA_1}$. Taking this into account we find that $\vec{MN} = \vec{MA} + \vec{AB} + \vec{BN} = -x\vec{AC} + \mathbf{a} + y\vec{BA_1} = (1 - x - y)\mathbf{a} - x\mathbf{b} + y\mathbf{c}$.

Let us substitute the resolutions of the vectors into equations (1) and transform the equations:

$$\begin{cases} ((1-x-y)\mathbf{a} - x\mathbf{b} + y\mathbf{c}) \cdot (\mathbf{a} + \mathbf{b}) = 0, \\ ((1-x-y)\mathbf{a} - x\mathbf{b} + y\mathbf{c}) \cdot (\mathbf{c} - \mathbf{a}) = 0, \end{cases}$$

$$\begin{cases} (1-x-y)a^2 - xa^2 = 0, & \begin{cases} 1-2x-y=0, \\ -1+x+2y=0. \end{cases} \end{cases}$$

Hence we find $x = y = 1/3$. This means that the points M and N lie on the segments AC and BA_1 and $|AM| = |AC|/3$, $|BN| = |BA_1|/3$.

Now we have $\vec{MN} = \frac{1}{3}\mathbf{a} - \frac{1}{3}\mathbf{b} + \frac{1}{3}\mathbf{c}$ and, therefore,

$$|MN| = \sqrt{\frac{1}{9}(a^2 + a^2 + a^2)} = \frac{a}{\sqrt{3}}. \quad \blacktriangle$$

PROBLEMS OF SECTION I

1. Prove that

(a) the line segments connecting the vertices of the tetrahedron with the points of intersection of the medians of the opposite faces meet at the point (we designate it as O) which divides each segment in the ratio $3/1$, reckoning from the vertex;

(b) the line segments connecting the midpoints of the skew edges of the tetrahedron meet at the same point O which is the midpoint of each of them.

2. The length of each edge of the tetrahedron $ABCD$ is a . Points M , N , and P are positioned on the edges DA , DC , and BC , respectively, so that $|DM| = |CN| = a/3$ and $|CP| = a/5$. Construct the

section of the tetrahedron cut by the plane MNP and find the length of the segment BQ , where $Q = (MNP) \cap (AB)$.

3. The parallelogram $ABCD$ serves as the base of the pyramid $SABCD$. Construct the section of the pyramid passing through the vertex A and the midpoints M and P of the edges SB and SD . Determine the ratio in which the section divides the edge SC .

4. The length of the edge of the cube $ABCD A_1 B_1 C_1 D_1$ is equal to a . Points M and Q are taken on the edges AD and $B_1 C_1$, respectively, and points P and N are taken on the edge CD , so that $|AM| = |C_1 Q| = |CP| = |DN| = a/3$. Construct the section of the cube cut by the plane passing through the straight line MP , parallel to the line NQ , and find the area of that section.

5. The lengths of the edges AC and BD of the tetrahedron $ABCD$ are equal to a and b respectively, and the angle between the lines AC and BD is equal to φ . Find the greatest area of the section of the tetrahedron cut by the parallel lines AC and BD .

6. Given the straight line a and the plane γ , points A_1 and A_2 lie on the line a ($A_1 \neq A_2$), points M , N , and P , not lying on the same line, belong to the plane γ . Prove the following: for the line a and the plane γ to be parallel, it is necessary and sufficient that the vectors $\overrightarrow{A_1 A_2}$, $\overrightarrow{MN}$, and $\overrightarrow{MP}$ should be coplanar, that is, there should be numbers α and β such that

$$\overrightarrow{A_1 A_2} = \alpha \overrightarrow{MN} + \beta \overrightarrow{MP}.$$

7. Points M , N , and P are the midpoints of the edges AB , CD , and BC , respectively, of tetrahedron $ABCD$. A plane is drawn through the point P parallel to the lines DM and AN . In what ratio does the plane partition the edge AD ?

8. In the right-angled parallelepiped $ABCD A_1 B_1 C_1 D_1$, perpendiculars $A_1 P$ and $B Q$ are dropped from the vertices A_1 and B to the diagonal AC_1 . Find the length of the line segment PQ if $|AB| = a$, $|AD| = b$, and $|AA_1| = c$.

9. Each edge of the regular prism $ABCA_1 B_1 C_1$ is a in length. Points M and N are taken on the diagonals AB_1 and BC_1 of the faces of the prism respectively, such that $(MN) \perp (AB)$, $|MN| = \frac{a}{\sqrt{3}}$. In what ratio do the points M and N partition the segments AB_1 and BC_1 ?

PROBLEMS OF SECTION II

1. Prove that if a plane is parallel to one of the two parallel lines, then it is parallel to the other line.

2. Two straight lines lying in the same plane are parallel to two straight lines of the other plane. Does it follow that the planes are parallel?

3. Two intersecting straight lines of one plane are parallel to the other plane. Prove that the planes are parallel.

4. Two planes are parallel. Prove that the straight line parallel to one of the planes is also parallel to the other plane.

5. Prove that if a straight line is parallel to each of the two intersecting planes, then it is parallel to their line of intersection.

6. How should the plane γ be drawn for it to intersect two given planes α and β along parallel straight lines?

7. Given two straight lines. Prove that a plane can be drawn through one given line parallel to the other given line. Under what condition is the plane unique?

8. Prove that a plane can be drawn through any point parallel to two given skew lines, and that the plane is unique.

9. Given the skew lines a and b and a point M . Construct a straight line passing through M and intersecting a and b . Find the set of all points M for which the problem has no solution.

10. Given three pairwise skew lines. Prove that there is a line intersecting these three given lines. Is that line unique?

11. Can the parallel projections of skew lines on the plane (a) be parallel, (b) coincide?

12. Can the parallel projection of a tetrahedron onto a plane be (a) a trapezoid; (b) a parallelogram; (c) a pentagon?

13. Can a parallel projection of a parallelepiped onto a plane be (a) a trapezoid; (b) a pentagon; (c) a hexagon?

14. Prove that if a parallel projection of a plane quadrangle onto a plane is a parallelogram, then the quadrangle itself is a parallelogram.

15. A parallel projection of an equilateral triangle onto a plane is an equilateral triangle too. Prove that the sides of the triangles are equal in length.

16. In a rectangular prism one of the diagonals cuts the other three. Prove that the prism is a parallelepiped.

17. The skew diagonals AB_1 , BC_1 , CD_1 , and DA_1 of the faces of the parallelepiped $ABCD A_1 B_1 C_1 D_1$ are equal in length. Prove that the faces $ABCD$ and $A_1 B_1 C_1 D_1$ are rhombi and the other faces are rectangles.

18. Points A_1 , B_1 , and C_1 are located on the edges SA , SB , and SC of the pyramid $SABC$ respectively. The lines $A_1 B_1$, $B_1 C_1$, and $C_1 A_1$ cut the lines AB , BC , and CA at points M , N , and P respectively. Prove that the points M , N , and P lie on the same straight line.

19. Points M and P lie on the edges AB and CD of the regular tetrahedron $ABCD$ so that $|AM|/|AB| = |DP|/|DC| = 1/3$. Find the area of the section of the tetrahedron drawn through the points M and P parallel to the line AC if the edge of the tetrahedron is a in length.

20. Points M , N , and P are located on the edges AA_1 , CC_1 , and CD_1 of the parallelepiped $ABCD A_1 B_1 C_1 D_1$ so that $|AM|/|AA_1| = |C_1 N|/|C_1 C| = |C_1 P|/|C_1 D_1|$. Construct the point Q of intersection of the plane MNP and the line BC and find the ratio $|BQ|/|BC|$.

21. A plane cuts the lateral edges AA_1 , BB_1 , CC_1 , and DD_1 of the parallelepiped $ABCD A_1 B_1 C_1 D_1$ at points M , N , P , and Q respectively, with $|AM|/|AA_1| = m$, $|BN|/|BB_1| = n$, $|CP|/|CC_1| = p$. Find the ratio $|DQ|/|DD_1|$.

22. A plane is drawn through the vertex C of the tetrahedron $ABCD$ and the midpoints of the edges AD and BD . In what ratio does

the plane divide the line segment MN , where M and N are the midpoints of the edges AB and CD respectively?

23. Points M , N , and P are located on the edges A_1B_1 , AB , and CC_1 of the prism $ABCA_1B_1C_1$ respectively so that $|A_1M|/|A_1B_1| = |BN|/|BA| = |C_1P|/|C_1C| = 1/2$. Construct the point Q of intersection of the plane MNP and the line B_1C_1 and find the ratio $|C_1Q|/|B_1C_1|$.

24. The medians of the base ABC of the prism $ABCA_1B_1C_1$ meet at a point M and the diagonals of the faces AA_1C_1C and BB_1C_1C meet at points N and P respectively. The plane MNP cuts the lines B_1C_1 and CC_1 at points K and L respectively. Construct the section of the prism cut by that plane and find the ratios $|B_1K|/|B_1C_1|$ and $|C_1L|/|CC_1|$.

25. A plane cuts the edges AB , AC , CC_1 , and BB_1 of the prism $ABCA_1B_1C_1$ at points K , L , M , and N respectively. The areas of the figures AKL , CLM , and $CMNB$ are equal, respectively, to $1/6$, $1/12$, and $1/2$ of the area of the face containing the respective figure. Find the ratio between the area of the triangle BKN and the area of the face AA_1B_1B .

26. A plane is drawn in the tetrahedron $ABCD$ through the midpoint M of the edge AD , the vertex C , and the point N of the edge BD , the resulting ratio being $|BN|/|ND| = 2/1$. In what ratio does the plane partition the segment KL , where K and L are the midpoints of the edges AB and CD respectively?

27. The parallelogram $ABCD$ serves as the base of the pyramid $SABCD$. A point L taken on the edge SD is such that $|SL|/|LD| = 2$, the point K being the midpoint of the edge SB . Construct the section of the pyramid formed by the plane AKL and determine in what ratio the plane partitions the edge SC .

28. The trapezoid $ABCD$ serves as a base of the pyramid $SABCD$. The ratio between the lengths of the bases AD and BC of the trapezoid is equal to 2. Construct the section of the pyramid passing through the vertex D and the midpoints of the edges SA and SB . Determine in what ratio the section divides the edge SC .

29. A section is drawn parallel to the diagonal BD of the base of the parallelepiped $ABCD A_1 B_1 C_1 D_1$ through the midpoints M and N of the edges AA_1 and $C_1 D_1$ respectively. Construct the section and determine in what ratio it divides the diagonal $A_1 C$ of the parallelepiped.

30. A plane is drawn parallel to the diagonal DB_1 of a parallelepiped through the midpoints M and N of the edges AD and CC_1 . In what ratio does the plane divide the edge BB_1 ?

31. The points O and O_1 are the centres of the faces $ABCD$ and $A_1 B_1 C_1 D_1$ of the cube $ABCD A_1 B_1 C_1 D_1$. A point S taken on the segment OO_1 is such that $|OS|/|O_1 S| = 1/3$. A section of the cube is drawn through that point parallel to the diagonal AC_1 of the cube and to the diagonal BD of the base. Find the area of the section if the edge of the cube is a in length.

32. Among all the sections of the cube, passing through its diagonal, indicate the one having the least area. Find that area if the edge of the cube is a in length.

33. Under consideration are the sections of the parallelepiped $ABCD A_1 B_1 C_1 D_1$ formed by the planes drawn parallel to the skew diagonals AB_1 and BC_1 of the faces $AA_1 B_1 B$ and $BB_1 C_1 C$. Indicate the section with the greatest area.

34. A plane cuts the edges AB , AC , DC , and DB of the tetrahedron at points M , N , P , and Q respectively, with $|AM|/|MB| = m$, $|AN|/|NC| = n$, $|DP|/|PC| = p$. Find the ratio $|DQ|/|QB|$.

35. In a regular rectangular pyramid $SABCD$ the side of the base is a in length and the length of the lateral edge is l . Sections are drawn parallel to the edge SA and to the diagonal BD of the base. Find the greatest area of the section.

36. All the edges of the regular prism $ABCA_1 B_1 C_1$ are a in length. Under consideration are the line segments with their ends on the diagonals BC_1 and CA_1 of the lateral faces, the segments being parallel to the plane $ABB_1 A_1$.

(1) One of the segments is drawn through the point M of the diagonal BC_1 such that $|BM|/|BC_1| = 1/3$. Find its length.

(2) Find the least length of all the segments being considered.

37. In the triangular prism $ABCA_1 B_1 C_1$, points M and N are the midpoints of the lateral edges BB_1 and CC_1 . A straight line intersecting the lines MN and AB_1 at points P and Q respectively is drawn through the point O of intersection of the medians of the triangle ABC . Find the ratio $|PQ|/|OQ|$.

38. The medians AM and DN of the faces ACD and ADB are drawn in the tetrahedron $ABCD$, and points E and F are taken on those medians respectively, so that $(EF) \parallel (BC)$. Find the ratio $|EF|/|BC|$.

39. In the prism $ABCA_1 B_1 C_1$ the medians of the bases ABC and $A_1 B_1 C_1$ meet at points O and O_1 respectively. Through the midpoint of the segment OO_1 a straight line is drawn parallel to the line CA_1 . Find the length of the segment of that line lying in the interior of the prism if $|CA_1| = a$.

40. The parallelogram $ABCD$ serves as a base of the pyramid $SABCD$. The diagonals of the parallelogram meet at a point O . Through the midpoint of the segment SO a line is drawn parallel to the median BM of the face SAB . Find the length of the segment of that line, lying in the interior of the pyramid, if $|BM| = a$.

41. Points M and N are taken on the edges AD and BD , respectively, of the tetrahedron $ABCD$, so that $|AM|/|AD| = |BN|/|BD| = m$. Find the distance between the points of intersection of the medians of the triangles ANC and BMC if $|AB| = a$.

42. The edge of a regular tetrahedron is a in length. A triangular section is drawn through the vertex of the tetrahedron. Prove that the perimeter p of the section satisfies the inequalities $2a < p \leq 3a$.

43. Given the points A , B , C , and D . Points M and N are the midpoints of the segments AB and CD , and point O is the midpoint of the segment MN (if a segment degenerates into a point, then its midpoint is considered to coincide with that point).

Prove that

$$\vec{SA} + \vec{SB} + \vec{SC} + \vec{SD} = 4\vec{SO}$$

for any point S .

44. Given the tetrahedron $ABCD$. Prove that there is a point M such that

$$\overrightarrow{MA} + \overrightarrow{MB} + \overrightarrow{MC} + \overrightarrow{MD} = 0$$

and that this point is unique.

45. Points M and N in the parallelepiped $ABCD A_1 B_1 C_1 D_1$ are the midpoints of the edges AB and $A_1 D_1$. The plane CMN cuts the straight lines $B_1 C_1$ and DB_1 at points P and Q respectively. Represent the vectors $\overrightarrow{AP}$ and $\overrightarrow{AQ}$ in terms of $\overrightarrow{AB} = \mathbf{a}$, $\overrightarrow{AD} = \mathbf{b}$, and $\overrightarrow{AA_1} = \mathbf{c}$.

46. Given four distinct points A , B , C , and D . Points M and N are taken on the segments AC and BD , respectively, so that $|AM| = \lambda |AC|$ and $|BN| = \lambda |BD|$. Prove that the vectors $\overrightarrow{AB}$, $\overrightarrow{CD}$, and $\overrightarrow{MN}$ are coplanar and find the resolution of the vector $\overrightarrow{MN}$ along the vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$.

47. Given four distinct points A , B , C , and D . Points M and N are the midpoints of the segments BC and AD respectively. Prove that if $|MN| = \frac{1}{2} (|AB| + |CD|)$, then $(AB) \parallel (CD)$.

48. Given the skew lines a and b and points A_1, A_2, A_3 , and B_1, B_2, B_3 lying on them respectively (A_2 lies between A_1 and A_3 and B_2 between B_1 and B_3). It is known that $|A_1 A_2|/|A_2 A_3| = |B_1 B_2|/|B_2 B_3|$. Prove that the midpoints of the segments $A_1 B_1, A_2 B_2$, and $A_3 B_3$ lie on the same straight line.

49. $ABCD A_1 B_1 C_1 D_1$ is a parallelepiped. A straight line cuts the lines AA_1 , BC , and $C_1 D_1$ at points M , N , and P respectively so that $|MN|/|MP| = 2$. Find the ratio $|BN|/|BC|$ (find all solutions).

50. The edge of the cube $ABCD A_1 B_1 C_1 D_1$ is a in length. A point M is taken on the line BC_1 so that the lines DA_1 , AB_1 , and $D_1 M$ are parallel to the same plane. Find the length of the segment $D_1 M$.

51. Points M and N are taken on the edge AD and the diagonal $A_1 C$ of the parallelepiped $ABCD A_1 B_1 C_1 D_1$ respectively, so that the line MN is parallel to the plane BDC_1 and $|AM|/|AD| = 1/5$. Find the ratio $|CN|/|CA_1|$.

52. Given four points A , B , C , and O not belonging to the same plane. Prove that for the point M to lie in the plane ABC it is necessary and sufficient that there be three numbers x , y , and z satisfying two conditions:

$$\overrightarrow{OM} = x\overrightarrow{OA} + y\overrightarrow{OB} + z\overrightarrow{OC}, \quad x + y + z = 1.$$

53. The base of the rectangular pyramid $SABCD$ is a parallelogram $ABCD$. A plane cuts the edges SA , SB , SC , and SD at points M , N , P , and Q respectively. Suppose $|SA|/|SM| = m$, $|SB|/|SN| = n$, $|SC|/|SP| = p$, and $|SD|/|SQ| = q$. Prove that

$$m + p = n + q.$$

54. The centre of the lower base of the cube is connected with the four vertices of the upper base by straight lines. Determine the angles between those lines.

55. Point K is the midpoint of the edge AA_1 of the cube $ABCD A_1 B_1 C_1 D_1$. Find the angles between the lines (a) BK and BC_1 ; (b) BK and AD_1 ; and (c) BK and $A_1 C_1$.

56. Point M is the midpoint of the edge CD of the regular tetrahedron $ABCD$. Find the angle between the lines AM and BC .

57. Points M and N are the midpoints of the edges BC and AD of the tetrahedron $ABCD$ in which $|AC| = |BD|$ and the angle between the lines AC and BD is equal to α . Find the angle between the lines MN and AC (find all solutions).

58. All the edges of the regular rectangular pyramid $SABCD$ are equal in length. Find the angles between the lines (a) AM and BN , where M and N are the midpoints of the edges SB and SC , respectively; and (b) SP and BN , where P is the midpoint of the edge AB .

59. It is known that $|AC| = |BD|$ and $(AC) \perp (BD)$ in the tetrahedron $ABCD$. Points M and N are taken on the edges AD and CB respectively so that $|AM|/|AD| = |CN|/|CB|$ and the angle

$(MN, \angle AC) = \frac{1}{2} (MN, \angle BD)$. Find the ratio $|AM|/|AD|$.

60. The edges AB and CD of the tetrahedron $ABCD$ are perpendicular and $|AB| = |CD| = a$. Points M and N are taken on the edge AB and points P and Q on the edge CD so that $|AM| = |NB| = |CP| = |QD| = \lambda a$ ($\lambda < 1/2$). Find the distance between the midpoints of the segments MP and NQ .

61. Given three noncoplanar vectors. Is there a fourth vector which is perpendicular to the given three?

62. Given three pairwise perpendicular straight lines. A fourth straight line makes the angles α_1 , α_2 , and α_3 with the given lines respectively. Prove that $\cos^2 \alpha_1 + \cos^2 \alpha_2 + \cos^2 \alpha_3 = 1$.

63. Given the parallelepiped $ABCD A_1 B_1 C_1 D_1$. Prove that

$$|AA_1|^2 \leq \frac{1}{4} (|AB_1|^2 + |BC_1|^2 + |CD_1|^2 + |DA_1|^2).$$

64. It is known that $\angle DAB = \angle ABC = \pi/2$ in the tetrahedron $ABCD$, $(AB, CD) = \alpha$, $2|AD| = |CD| = |BC|$. Find the angle between the straight lines AD and BC .

65. Point M is the midpoint of the edge $A_1 B_1$ of the cube $ABCD A_1 B_1 C_1 D_1$ and point N is the centre of the face $ABB_1 A_1$. Find the angle between the straight lines MD and CN .

66. Two edges of the tetrahedron are perpendicular to the edges which are skew with respect to them. Prove that the other two skew edges of the tetrahedron are perpendicular as well.

67. The line segment AB is a common perpendicular to the skew lines l_1 and l_2 ($A \in l_1$, $B \in l_2$). Points C and D are taken on the lines l_1 and l_2 respectively, so that $|AC| = |BD| = |AB| = \frac{1}{2} |CD|$. Find the angle between the lines l_1 and l_2 .

68. The angle between the skew lines l_1 and l_2 is α , and the line segment AB is their common perpendicular ($A \in l_1$, $B \in l_2$). Points C and D are taken on the lines l_1 and l_2 respectively, so that $|AC| = |BD| = a$ and $|CD| = b$. Find $|AB|$ (find all solutions). At what ratio between a and b does the problem possess a solution for any α ?

69. The side of the base of a regular triangular prism is a in length, and two nonintersecting diagonals of the lateral faces are perpendicular. Find the length of the lateral edge of the prism.

70. Points M , N , P , and Q are taken on the diagonals of the faces D_1A , A_1B , B_1C , and C_1D of the cube $ABCD A_1 B_1 C_1 D_1$ respectively, so that $|D_1M|/|D_1A| = |BN|/|BA_1| = |B_1P|/|B_1C| = |DQ|/|DC_1| = \lambda$, and the straight lines MN and PQ are perpendicular. Find λ .

71. All the edges of the regular prism $ABCA_1B_1C_1$ are a in length. Under consideration are line segments with the end points lying on the lines AB_1 and BC_1 , the segments being perpendicular to the line AC_1 . Find the least length of such segments.

72. The edge of the cube $ABCD A_1 B_1 C_1 D_1$ is a in length. Points M and N are taken on the diagonals D_1A and A_1B respectively, so that $|D_1M|/|D_1A| = |NB|/|A_1B| = 1/3$. Find the distance from the vertex C to the line MN .

73. The edge of the cube $ABCD A_1 B_1 C_1 D_1$ is a in length. A point M is taken on the line BC_1 so that the lines D_1M , DA_1 , and AB_1 are parallel to the same plane. Find: (a) the distance from the point M to the line AB_1 ; and (b) the area of the section of the cube formed by the plane MD_1B_1 .

74. It is known that $|AD|/|DC| = n$ in the right parallelepiped $ABCD A_1 B_1 C_1 D_1$. In what ratio does the common perpendicular to the lines AA_1 and B_1D divide the segments AA_1 and B_1D ?

75. The altitudes DE and BF of the faces ABD and BDC are drawn in the regular tetrahedron $ABCD$. In what ratio are the segments DE and BF divided by the foot of their common perpendicular?

76. The length of each edge of the parallelepiped $ABCD A_1 B_1 C_1 D_1$ is a , and all plane angles at the vertex A are equal to $\pi/3$. Determine the distance between the lines AC and DB_1 .

77. The length of the edge of the regular tetrahedron $ABCD$ is a . The point E is the midpoint of the edge CD , and the point F is the midpoint of the altitude BL of the face ABD . The segment MN with its end points on the lines AD and BC cuts the line EF at right angles. Find the length of that segment.

78. The lengths of the edges AB and BC of the right parallelepiped $ABCD A_1 B_1 C_1 D_1$ are a and $2a$ respectively. A straight line is known to exist which cuts the lines AA_1 , BC , and C_1D_1 and makes equal angles with them. Find the length of the edge AA_1 (find all solutions).

79. The side of the base ABC of the regular prism $ABCA_1B_1C_1$ is equal to a . Points M and N are the midpoints of the edges AC and A_1B_1 respectively, and points M_1 and N_1 are the feet of the perpendiculars dropped from the points M and N to the line BC_1 . Find the

length of the lateral edge of the prism if $|M_1N_1| = \frac{a}{2\sqrt{2}}$.

80. Assume that S is the area of the triangle ABC . Prove that

$$S = \frac{1}{2} \sqrt{|AB|^2 \cdot |AC|^2 - (\vec{AB} \cdot \vec{AC})^2}.$$

81. The length of the edge of the regular tetrahedron $ABCD$ is a . Points E and F are taken on the edges AD and BD respectively so that $|DE|/|EA| = |BF|/|FD| = 2$. Find the area of the triangle CEF .

82. Given four vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, and $\mathbf{d}$. The vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are coplanar but not collinear pair-by-pair, and the vector $\mathbf{d}$ is not coplanar with the vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$. It is known that the angles

($\mathbf{a}$, $\mathbf{b}$) $\widehat{(\mathbf{b}, \mathbf{c})} = \pi/3$, $\widehat{(\mathbf{d}, \mathbf{a})} = \alpha$, and $\widehat{(\mathbf{d}, \mathbf{b})} = \beta$. Find the angle between the vectors $\mathbf{d}$ and $\mathbf{c}$.

83. Four rays emanating from the same vertex pairwise make equal angles. Find the angles.

84. The edge of the cube is a in length. Find the sum of the scalar products of the vectors whose origins are at the centre of the cube and the terminal points are at its vertices.

85. Given the points A , B , C , and O , of which A , B , and C do not lie on the same line. It is known that for any points M the quantity $y(M) = |MA|^2 + |MB|^2 + |MC|^2 - 3|MO|^2$ has one and the same value. Prove that O is the point of intersection of the medians of the triangle ABC .

86. Distinct points A , B , C , and D not lying on the same straight line are given in space. Suppose S is an arbitrary point in space. We introduce the designation

$$y(S) = |SA|^2 + |SC|^2 - |SB|^2 - |SD|^2.$$

Prove that

(a) $y(S)$ has the same value for all points in space if and only if the points A , B , C , and D are the vertices of a parallelogram;

(b) if there are four points S_1 , S_2 , S_3 , and S_4 , not lying in the same plane, such that $y(S_1) = y(S_2) = y(S_3) = y(S_4) = 0$, then A , B , C , and D are the vertices of a rectangle.

For the given quantity y_0 find the set of points S of space for which $y(S) = y_0$.

87. The base of the pyramid is a regular octagon with the side of length a . The vertex of the pyramid is at a distance b from the centre of the base. Find the sum of the squares of the lengths of the lateral edges.

88. Given the points A , B , C , and D . Find a point M such that the quantity $|MA|^2 + |MB|^2 + |MC|^2 + |MD|^2$ has the least value.

89. Prove that for any four points A , B , C , or D

$$\vec{AB} \cdot \vec{CD} = \frac{1}{2} (|AD|^2 + |BC|^2 - |AC|^2 - |BD|^2).$$

90. Four points A , B , C , and D are given in space ($A \neq C$, $B \neq D$), points P and Q being the midpoints of the segments AC and

BD respectively. Prove that

$$|AB|^2 + |BC|^2 + |CD|^2 + |AD|^2 = |AC|^2 + |BD|^2 + 4|PQ|^2.$$

91. Given the points A , B , C , and D ($A \neq B$, $C \neq D$). Prove that for the lines AB and CD to be parallel it is necessary and sufficient that

$$2|AB| \cdot |CD| = ||AD|^2 + |BC|^2 - |AC|^2 - |BD|^2|.$$

Chapter 8

Solid Geometry (Part 2)

This chapter treats of various problems dealing with perpendicularity of straight lines and planes, with calculations of the angle between a straight line and a plane, and between planes, with finding the volumes of polyhedra and their parts, and with combinations of polyhedra.

8.1. Perpendicular Lines and Planes

A straight line and a plane are said to be mutually perpendicular if the line is perpendicular to every line lying in the plane. This is the definition of perpendicularity of a line and a plane.

Problem 1. The base of the pyramid is a regular triangle ABC (Fig. 125), the lateral edges SA , SB , and SC are equal in length. Prove that the pyramid $SABC$ is regular.

△ Suppose SO is the altitude of the pyramid. Then, by the definition of perpendicularity of a line and a plane, the angles SOA , SOB , and SOC are right angles. We designate $|SA| = |SB| = |SC| = l$, $|SO| = h$. From the right triangles SOA , SOB and SOC we find that $|AO| =$

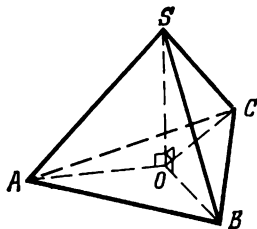


Fig. 125

$|BO| = |CO| = \sqrt{l^2 - h^2}$. This means that the point O is the centre of the circle circumscribed about the regular triangle ABC and is, therefore, the centre of that triangle. Thus, the vertex S of the given pyramid can be projected

onto the centre of the base (of the regular triangle ABC), that is, the pyramid is regular. $\blacktriangle$

It can be proved quite analogously that the pyramid whose base is a regular polygon and whose lateral edges are equal in length is regular. In particular, a regular tetrahedron, that is, a tetrahedron whose all edges are equal in length, is a regular pyramid. Any face of a regular tetrahedron can serve as its base.

The criterion of the perpendicularity of a line and a plane is as follows.

If a straight line is perpendicular to each of the two intersecting lines lying in a plane, then the line and the plane are mutually perpendicular.

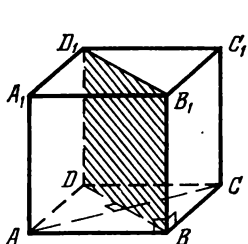


Fig. 126

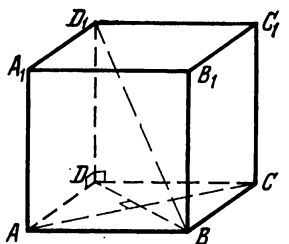


Fig. 127

Problem 2. Prove that the diagonal AC of the base of a regular quadrangular prism (Fig. 126) is perpendicular to the plane BB_1D_1D .

$\triangle$ It is sufficient to prove that the line AC is perpendicular to any two intersecting lines lying in the plane BB_1D_1D . Since $ABCD$ is a square, it follows that $(AC) \perp (BD)$. We shall prove that $(AC) \perp (BB_1)$. The given prism is regular and, therefore, $(BB_1) \perp (ABCD)$. In accordance with the definition of perpendicularity of a line and a plane, this yields $(BB_1) \perp (AC)$. Thus we see that the line AC is perpendicular to the intersecting lines BD and BB_1 of the plane BB_1D_1D . According to the criterion of perpendicularity of a line and a plane, this signifies that $(AC) \perp (BB_1D_1D)$. $\blacktriangle$

To establish perpendicularity of straight lines, use is often made of the *theorem on three perpendiculars*.

For a straight line lying in a plane to be perpendicular to an inclined line, it is necessary and sufficient that the line be perpendicular to the projection of the inclined line onto that plane.

The necessary and sufficient conditions formulated in this theorem must be strictly distinguished and used correspondingly. The sufficient condition consists in the fact that

if a line lying in a plane is perpendicular to the projection of an inclined line onto that plane, then it is perpendicular to the inclined line itself.

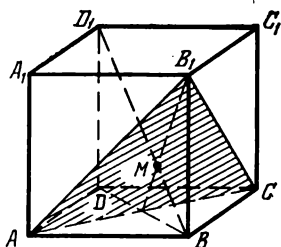


Fig. 128

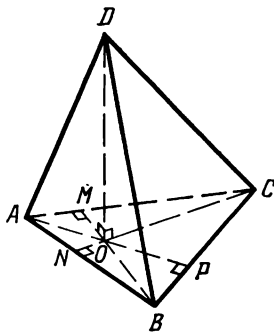


Fig. 129

Problem 3. Prove that the diagonal BD_1 of the cube shown in Fig. 127 is perpendicular to the diagonal AC of the face $ABCD$.

△ The line BD is the projection of the inclined line BD_1 onto the plane $ABCD$ since $(D_1D) \perp (ABCD)$. The face $ABCD$ is a square and this means that $(AC) \perp (BD)$. By virtue of the sufficient condition of the theorem on three perpendiculars, this yields $(AC) \perp (BD_1)$. ▲

The assertion proved yields the following useful corollary.

Corollary. *The diagonal of the cube emanating from one of its vertices is perpendicular to the plane drawn through the endpoints of three edges of the cube emanating from the same vertex.*

□ Let us consider, say, the diagonal BD_1 (Fig. 128). We have proved that $(BD_1) \perp (AC)$. We can prove quite analogously that $(BD_1) \perp (AB_1)$. Hence it follows that $(BD) \perp (AB_1C)$. It is clear from the properties of the symmetry of

the cube that the same is true for any other diagonal of the cube. ■

The necessary condition of the theorem on three perpendiculars formulated above is as follows.

If a straight line lying in a plane is perpendicular to an inclined line, then it is also perpendicular to the projection of the inclined line onto that plane.

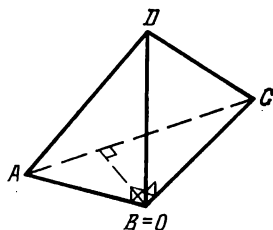


Fig. 130

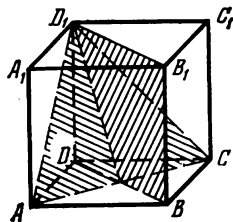


Fig. 131

Problem 4. In the tetrahedron $ABCD$ the edge AB is perpendicular to the edge CD and the edge AC to the edge BD . Prove that the foot of the altitude DO of the tetrahedron is the point of intersection of the altitudes (or of their extensions) of the triangle ABC .

△ Let us first consider the case when neither of the edges BD and CD is perpendicular to the plane ABC , that is, each of these edges lies on a line inclined to the plane ABC . Since $(DO) \perp (ABC)$ (Fig. 129), the line BO is the projection of the line BD onto the plane ABC . By the hypothesis, the line AC , lying in the plane ABC , is perpendicular to the inclined line BD . According to the necessary condition of the theorem on three perpendiculars, it signifies that $(AC) \perp (BO)$. We can prove quite analogously that $(CO) \perp (AB)$. Thus, the point O is the point of intersection of the altitudes (or of their extensions) of the triangle ABC .

Let us now discuss the case when one of the edges, say, the edge BD , is perpendicular to the plane ABC (Fig. 130). In that case the point O coincides with the point B , the straight line CD is inclined to the plane ABC , and its projection onto that plane is a line (BC) . In accordance with the necessary condition of the theorem on three perpen-

diculars, the fact that $(AB) \perp (CD)$ yields $(AB) \perp (BC)$. This means that the triangle ABC is a right triangle, and the point B is the point of intersection of its altitudes. $\blacktriangle$

Let us consider a problem which can be easily solved with the aid of the following criterion of perpendicularity of two planes.

If a plane passes through the perpendicular to the other plane, then it is perpendicular to that plane.

Problem 5. Prove that the planes AD_1C and BB_1D_1D of the regular quadrangular prism $ABCD A_1 B_1 C_1 D_1$ are perpendicular.

$\wedge$ It was proved in Problem 2 that $(AC) \perp (BB_1D_1D)$ (Fig. 131). According to the criterion of perpendicularity of planes, this yields $(AD_1C) \perp (BB_1D_1D)$. $\blacktriangle$

We shall formulate two other theorems useful for establishing perpendicularity of a line and a plane. The first of them states:

if two planes are mutually perpendicular, then a straight line drawn in one of the planes at right angles to the line of intersection of the planes is perpendicular to the other plane.

Problem 6. In the regular quadrangular prism $ABCD A_1 B_1 C_1 D_1$ find the length of the perpendicular dropped from the vertex B_1 to the plane AD_1C if $|AB| = a$ and $|AA_1| = b$.

$\wedge$ We have proved in Problem 5 that the planes AD_1C and BB_1D_1D are perpendicular. The line of intersection of these planes is the straight line DO (Fig. 132). Suppose B_1M is a perpendicular drawn in the plane BB_1D_1D to the line DO . Then, in accordance with the theorem proved above, $(B_1M) \perp (AD_1C)$. Assume $N = (D_1O) \cap (BB_1)$. In the right triangle NB_1D_1 ($\angle NB_1D_1 = 90^\circ$), we have $|B_1D_1| = a/\sqrt{2}$ and, evidently, $|NB_1| = 2b$. Consequently,

$$|ND_1| = \sqrt{|NB_1|^2 + |B_1D_1|^2} = \sqrt{4b^2 + 2a^2}.$$

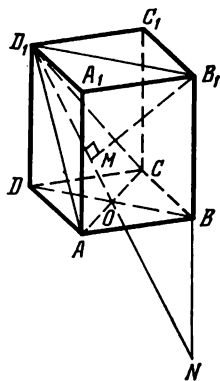


Fig. 132

The segment B_1M is an altitude in the triangle NB_1D_1 . Then, for the area of this triangle we have, on one hand, $S = \frac{1}{2} |NB_1| \cdot |B_1D_1|$ and, on the other hand, $S = \frac{1}{2} |ND_1| \cdot |B_1M|$. Hence it follows that $|B_1M| \cdot |ND_1| = |B_1D_1| \cdot |NB_1|$, that is,

$$|B_1M| = \frac{|B_1D_1| \cdot |NB_1|}{|ND_1|} = \frac{2ab}{\sqrt{2b^2 + a^2}}. \quad \blacktriangle$$

The second theorem states:

if two intersecting planes α and β are perpendicular to a plane γ , then the line of intersection of the planes α and β is perpendicular to the plane γ as well.

We leave the proof of the theorem to the reader (see Problem 1 in Section I of the present chapter).

8.2. Depicting Perpendicular Lines and Planes.

Constructing Sections Perpendicular to a Line or a Plane

The basis for drawing solids in solid geometry is parallel projection onto the plane of the drawing. As is known, parallel projections possess a number of simple properties.

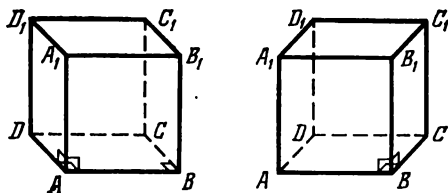


Fig. 133

For instance, parallel projection retains parallelism of straight lines, the ratio of the lengths of parallel segments. But the angles are not, in general, preserved in parallel projection. Figure 133 illustrates a cube in a parallel projection. The face AA_1B_1B is parallel to the plane of the drawing and, therefore, its projection is a square congruent to the face itself. The face $ABCD$ is not parallel to the plane

of the drawing. Its image is a parallelogram which is not congruent to the face. Dependent on the position of the right angle ABC and on the direction of projecting, its image can be either an acute or an obtuse angle. In the general case, as well, a perpendicular drawn in space from the given point to the given line is shown in the figure by a line not, in general, perpendicular to the depicted line. To show correctly such a perpendicular in the drawing, it is usual practice to determine the position of its foot with respect to some points of the given line defined in the figure.

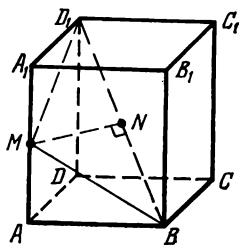


Fig. 134

Problem 1. The side of the base of the regular quadrangular prism $ABCD A_1 B_1 C_1 D_1$ (Fig. 134) is a in length. Construct the image of the perpendicular drawn from the midpoint M of the edge AA_1 to the line BD_1 and find the length of that perpendicular.

$\triangle$ Let us consider the triangle MBD_1 . The perpendicular dropped from the point M to the line BD_1 is an altitude of that triangle. We designate $|AA_1| = b$, and then $|AM| = |MA_1| = b/2$. In the right triangles MAB and MA_1D_1 we find $|MB| = |MD_1| = \sqrt{a^2 + \frac{b^2}{4}}$. This means that the triangle MBD_1 is isosceles and its altitude drawn from the vertex M is also its median, that is, the foot of that altitude is the midpoint of the segment BD_1 . Connecting the point M and the midpoint N of the diagonal BD_1 , we obtain the perpendicular dropped from the point M to the line BD_1 .

Now, bearing in mind that $|BD_1| = \sqrt{2a^2 + b^2}$ and $|BN| = \frac{1}{2}|BD_1|$, we find, from the right triangle MNB that $|MN| = \sqrt{|MB|^2 - |BN|^2} = a/\sqrt{2}$. $\blacktriangle$

To depict a perpendicular drawn in space from a given point to a given plane, it is customary to determine the position of its foot with respect to the points of that plane defined in the figure.

Let us take Fig. 135a illustrating a regular triangular pyramid $SABC$ and draw its altitude SO ($\triangle ABC$ is the base of the pyramid). In accordance with the definition, the foot of the altitude of a regular pyramid is the centre of the base of that pyramid. The centre of the regular triangle ABC coincides with the point of intersection of its medians. Accordingly, the medians AM and BN are drawn in Fig. 135a. The point O of their intersection is the foot of the altitude.

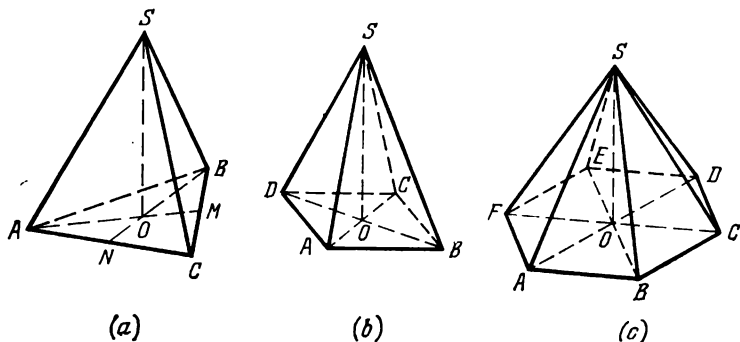


Fig. 135

The foot of the altitude of the regular quadrangular pyramid shown in Fig. 135b can be found as the point of intersection of the diagonals of the parallelogram serving as the base of the pyramid and being a square $ABCD$. The foot of the altitude of the regular hexagonal pyramid shown in Fig. 135c can be found by analogy.

The construction of a perpendicular to a line or to a plane is often used to construct a section of a figure formed by a plane perpendicular to the given line or plane.

Figure 136a and b illustrates the construction of a section of the regular triangular pyramid $SABC$ formed by a plane passing through the midpoint M of the edge SA at right angles to the altitude CN of the base of the pyramid. We designate the cutting plane as α . Since $(CN) \perp \alpha$, it follows that $(ABC) \perp \alpha$ as well. Hence, the perpendicular dropped from the point M to the plane ABC belongs to the plane α . The perpendicular is parallel to the altitude of the pyramid drawn from the vertex S . On the basis of the analysis carried out we perform the following constructions: we construct

the altitude SO of the pyramid and draw $[MS_1] \parallel [SO]$ (Fig. 136a). The side of the section lying in the face ABC is perpendicular to the line CN and, hence, is parallel to the edge AB . We draw $[PQ] \parallel [AB]$ (Fig. 136b) through the

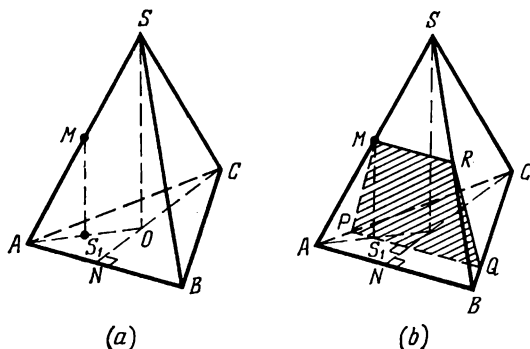


Fig. 136

point S_1 . PQ and AB being parallel, it follows that the side of the section lying in the face ASB is parallel to the edge AB as well. We construct $[MR] \parallel [AB]$. The section is a trapezoid $PMRQ$.

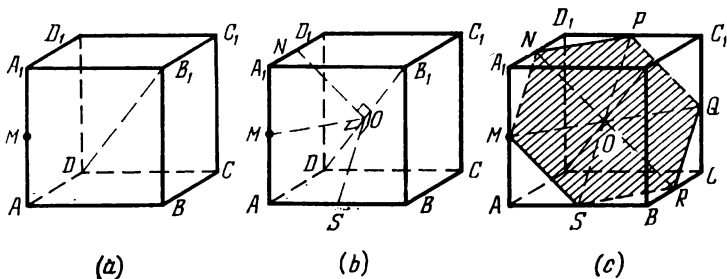


Fig. 137

Let us consider one more example of constructing a section perpendicular to a straight line.

Problem 2. Construct a section of the cube passing through the midpoint M of the edge AA_1 (Fig. 137a) at right angles to the diagonal B_1D of the cube and determine in what ratio it divides the diagonal.

$\triangle$ As in Problem 1 above, it is easy to prove that the perpendicular dropped from the point M to the line B_1D passes through the midpoint O of the segment B_1D . The perpendicular dropped from the midpoint N of the edge A_1D_1 to the line B_1D also passes through the point O (Fig. 137b). This means that the plane of the section passes through the points M , N , and O and, in particular, bisects the diagonal B_1D . Let us construct the section. To do that, we shall prove that the plane of the section passes through the midpoint S of the edge AB . The straight line SO , just as the line MO , is perpendicular to the line B_1D . Therefore, the plane MOS , as well as the plane MON , is perpendicular to the line B_1D . But then the planes coincide, that is, $S \in (MON)$. Now we can use the result of Problem 2 in 7.1. We proved there that the section of a cube formed by the plane MNS is a regular hexagon (Fig. 137c) with vertices at the midpoints of the edges which do not intersect the diagonal B_1D .

Answer: 1 : 1. $\blacktriangle$

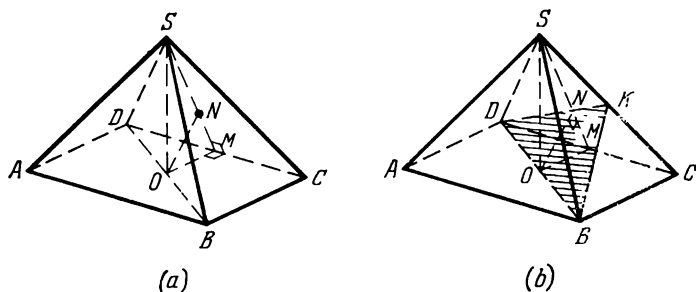


Fig. 138

Let us now consider the example of constructing a section perpendicular to a given plane.

Problem 3. Every edge of the regular quadrangular pyramid $SABCD$ is a in length. Construct a section passing through the diagonal BD of the base (Fig. 138a) at right angles to the face SCD and find its area.

$\triangle$ We first construct a perpendicular from some point of the line BD to the plane SCD . The perpendicular belongs to the plane of the section and, together with the line BD ,

completely defines it. It is convenient to perform the construction from the centre O of the base $ABCD$. Indeed, the plane SOM , where M is the midpoint of the edge CD (Fig. 138a), is perpendicular to the plane CSD ($(CD) \perp (SOM)$, $(CD) \subset (CSD)$). Therefore, the altitude of the triangle SOM , drawn from the vertex O , is perpendicular to the plane CSD . Let us determine in what ratio the foot N of that altitude divides the side SM . In the right triangle SOM ($\angle SOM = 90^\circ$) we have $|OM| = a/2$ and $|SM| = a/\sqrt{3}/2$. In accordance with the property of a right triangle we have $|OM|^2 = |MN| \cdot |SM|$, whence it follows that

$$|MN| = \frac{\sqrt{3}}{6}a. \text{ This means that } |MN| = \frac{1}{3}|SM|, \text{ and}$$

the point N is the centre of the regular triangle CSD . We draw a median DK of the triangle CSD (Fig. 138b), $N = (DK) \cap (SM)$ and obtain a section, which is a triangle DKB .

Let us now find the area of the section. The altitude h of the triangle DKB drawn from the vertex B is twice as long as the perpendicular ON since the point O is the midpoint of the segment BD . In the right triangle SOM we have $|ON| = \sqrt{|MN| \cdot |SN|} = a/\sqrt{6}$. Consequently, $h = 2a/\sqrt{6}$, and the area of the triangle DKB is equal to $\frac{1}{2}h |DK| = a^2/2 \sqrt{2}$. $\blacktriangle$

8.3. The Angle Between a Line and a Plane

The angle between a line, inclined to a plane, and that plane is the angle between the line and its orthogonal projection onto that plane.

If a straight line is perpendicular to a plane, then the angle between them is equal to 90° ; if the line and the plane are parallel, then the angle between them is 0° .

Problem 1. Find the angle between the edge of a regular tetrahedron and the plane of the face which does not contain that edge.

$\triangle$ We find the angle between the edge AD and the plane ABC (Fig. 139). The length of the edge of the tetrahedron will be designated as a . We drop a perpendicular DO to the plane ABC . The point O is the centre of the triangle ABC

(Problem 1 in 8.1), and, therefore, $|AO| = a\sqrt{3}/3$. The line AO is the projection of the line AD onto the plane ABC and, hence, the angle between the line AD and that plane is equal to $\angle DAO$. We have $\cos \angle DAO = |AO|/|AD| = \sqrt{3}/3$, $\angle DAO = \arccos(1/\sqrt{3})$. It is easy to see that any edge of a regular tetrahedron makes an angle of the same magnitude with the plane of any face which does not contain that edge.

Answer: $\arccos(1/\sqrt{3})$. ▲

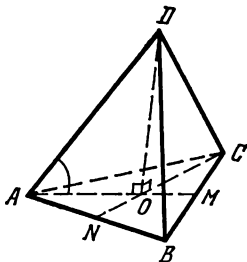


Fig. 139

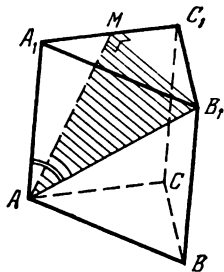


Fig. 140

Errors often arise in determining the angle between a line and a plane which are due to an incorrect determining the orthogonal projection of the line onto the plane.

Problem 2. In the regular triangular prism $ABCA_1B_1C_1$ we have $|AA_1| = |AB|$. Find the angle between the diagonal AB_1 and the plane AA_1C_1C .

△ We construct the projection of the line AB_1 onto the plane AA_1C_1C . The planes $A_1B_1C_1$ and AA_1C_1C are perpendicular and, therefore, the perpendicular B_1M (Fig. 140) to the line A_1C_1 is also a perpendicular to the plane AA_1C_1C . The line AM is the orthogonal projection of the line AB_1 onto the plane AA_1C_1C . The angle between the line AB_1 and the plane AA_1C_1C is $\angle B_1AM$. We designate $|AB| = a$ and then have $|AA_1| = a$ as well, and $|AB_1| = a\sqrt{2}$. The altitude B_1M of the regular triangle $A_1B_1C_1$ is equal to $a\sqrt{3}/2$. In the right triangle B_1AM we have

$$\sin \angle B_1AM = |B_1M|/|AB_1| = \sqrt{6}/4,$$

$$\angle B_1AM = \arcsin(\sqrt{6}/4).$$

Answer: $\arcsin(\sqrt{6}/4)$. ▲

Remark. It would be incorrect to take the angle B_1AA_1 as the angle between the line AB_1 and the plane AA_1C_1C (Fig. 140). The line AA_1 is not an orthogonal projection of the line AB_1 onto the plane AA_1C_1C , and the angle B_1AA_1 , equal to $\pi/4$, is not the angle between that line and the plane.

The method of coordinates can also be used to find the angle between a line and a plane.

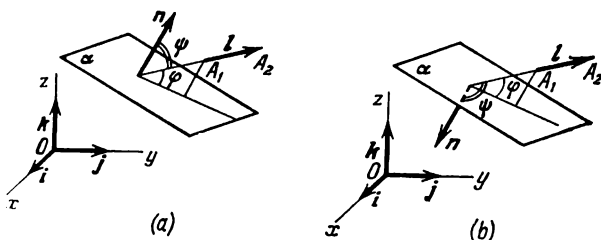


Fig. 141

Suppose a Cartesian system of coordinates has been introduced in space and assume that the plane α is defined by the equation (see 7.8 in Part 1)

$$ax + by + cz + d = 0, \quad (1)$$

and two points $A_1(x_1; y_1; z_1)$ and $A_2(x_2; y_2; z_2)$ are given on the line a . The angle between the line a and the plane α is designated as φ .

Let us consider two vectors

$$\mathbf{n} = (a; b; c) \text{ and } \mathbf{l} = \overrightarrow{A_1A_2} = (x_2 - x_1; y_2 - y_1; z_2 - z_1).$$

The angle φ is defined by the following formula:

$$\sin \varphi = \frac{|\mathbf{n} \cdot \mathbf{l}|}{|\mathbf{n}| |\mathbf{l}|}. \quad (2)$$

Indeed, the vector $\mathbf{n} = (a; b; c)$ is perpendicular to the plane α defined by equation (1) and, therefore, designating

$\psi = (\widehat{\mathbf{n}, \mathbf{l}})$, we have

if $0 \leq \psi \leq \frac{\pi}{2}$, then $\varphi = \frac{\pi}{2} - \psi$ (Fig. 141a);

if $\frac{\pi}{2} < \psi \leq \pi$, then $\varphi = \psi - \frac{\pi}{2}$ (Fig. 141b).

In both cases we have $\sin \varphi = |\cos \psi|$. This and the fact that $\cos \psi = \frac{\mathbf{n} \cdot \mathbf{l}}{|\mathbf{n}| \cdot |\mathbf{l}|}$ yield formula (2).

In problems on deriving equations of a plane use is often made of the fact that

the equation of any plane passing through the point $M(x_0; y_0; z_0)$ can be written in the form

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0. \quad (3)$$

Problem 3. In the regular quadrangular prism the ratio between the lengths of the lateral edge and the side of the base is equal to 2. Find the angle between the diagonal BD_1 of the prism (Fig. 142) and the plane BC_1D .

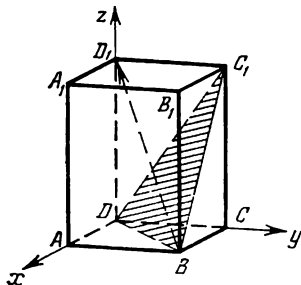


Fig. 142

Δ We introduce a system of coordinates as shown in Fig. 142*. If we designate the side of the base of the prism as s , the length of the lateral edge will be $2s$. Then we find the coordinates of the points B , C_1 , D , and D_1 : $B(s; s; 0)$, $C_1(0; s; 2s)$, $D(0; 0; 0)$, and $D_1(0; 0; 2s)$. The plane BC_1D

passes through the point $D(0; 0; 0)$ and, therefore, in accordance with equation (3), its equation is of the form $ax + by + cz = 0$. Substituting here the coordinates of the points B and C_1 , we get a system

$$\begin{cases} as + bs = 0, \\ bs + 2cs = 0. \end{cases}$$

* For the sake of simplicity, we shall not always show the basis $(\mathbf{i}; \mathbf{j}; \mathbf{k})$ in the figure featuring the Cartesian system of coordinates, indicating only the positive direction of the axes.

whence we find that $a = 2c$, $b = -2c$. This means that the equation of the plane BC_1D has the form

$$2cx - 2cy + cz = 0.$$

Taking into account that $c \neq 0$ (otherwise the three coefficients would vanish) and cancelling, we obtain an equation

$$2x - 2y + z = 0.$$

Thus, the vector $\mathbf{n}$, perpendicular to the plane BC_1D , has the coordinates $(2; -2; 1)$. We find the coordinates $(p; q; r)$

of the vector $\mathbf{l} = \overrightarrow{BD_1}$: $p = 0 - s = -s$, $q = 0 - s = -s$, $r = 2s - 0 = 2s$. Then we make the calculations:

$$\mathbf{l} \cdot \mathbf{n} = (-s) \cdot 2 + (-s) \cdot (-2) + 2s \cdot 1 = 2s,$$

$$|\mathbf{n}| = \sqrt{4+4+1} = 3, \quad |\mathbf{l}| = \sqrt{s^2 + s^2 + 4s^2} = \sqrt{6} s.$$

Designating as φ the angle between the line BD_1 and the plane BC_1D , we find, by formula (2), that

$$\sin \varphi = \frac{|\mathbf{l} \cdot \mathbf{n}|}{|\mathbf{n}| \cdot |\mathbf{l}|} = \frac{2s}{3 \cdot \sqrt{6} \cdot s} = \frac{2}{3 \sqrt{6}} = \frac{\sqrt{6}}{9},$$

$$\varphi = \arcsin(\sqrt{6}/9). \quad \blacktriangle$$

8.4. The Distance Between a Point and a Plane. Distances Between Lines and Planes

Let us recall the definition of the distance between solids: the least distance between the points one of which belongs to the solid Φ_1 and the other to the solid Φ_2 is said to be the *distance between the solids Φ_1 and Φ_2* .

If the point does not belong to the plane, then the distance from the point to the plane is known to be equal to the length of the perpendicular dropped from the point to the plane. If the point belongs to the plane, then the distance between them is zero. Earlier in this chapter we have come across a problem on finding the distance between a point and a plane (Problem 5 in 8.1). Let us now discuss other methods of finding that distance.

If we know the volume V of the pyramid and the area Q of its base, then the altitude H of the pyramid can be found from the formula

$$H = 3V/Q.$$

This altitude is nothing other than the distance from the vertex of the pyramid to the plane of its base.

Problem 1. The area of the lateral surface and the volume of a regular quadrangular pyramid are S and V respectively. Find the distance from the vertex of the base to the plane of the lateral face not containing that vertex.

△ We seek the distance from the vertex B to the plane SCD (Fig. 143). Let us consider the pyramid $SBCD$. That

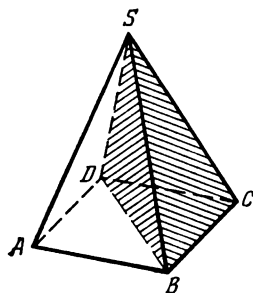


Fig. 143

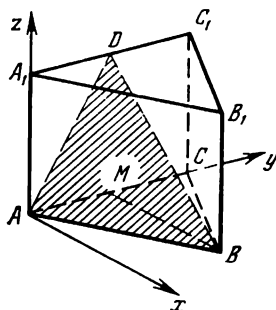


Fig. 144

pyramid and the pyramid $SABCD$ have a common altitude emanating from the vertex S , and the area of its base BCD is half that of the square $ABCD$. This means that if we designate the volume of this pyramid as V_1 , then we have $V_1 = V/2$.

We assume the face SCD to be the base of the pyramid $SBCD$. The area of that face is evidently equal to $S/4$. The altitude of the pyramid $SBCD$, dropped from the vertex B , is equal to the distance from the point B to the plane SCD . We designate it as d . Then we have $V_1 = \frac{1}{3} \cdot \frac{1}{4} S \cdot d = \frac{1}{12} S \cdot d$. Bearing in mind that $V_1 = \frac{1}{2} V$, we find that $d = 6V/S$. Evidently, the result does not depend on the choice of the vertex of the base and the plane of the lateral face not containing that vertex.

Answer: $6V/S$. ▲

To find the distance between a point and a plane, use can also be made of the method of coordinates, namely,

the distance ρ from the point $M_0(x_0; y_0; z_0)$ to the plane defined by the equation $ax + by + cz + d = 0$ can be found by the formula

$$\rho = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}} \quad (1)$$

(Problem 7 in Section I of the present chapter).

Problem 2. In a regular triangular prism shown in Fig. 144 $|AB| = 4$ cm and $|AA_1| = 3$ cm. Find the distance from the vertex C_1 to the plane ADB , where D is the midpoint of the edge A_1C_1 .

Δ We introduce the Cartesian system of coordinates as shown in Fig. 144. The x -axis of this system and the median BM of the triangle ABC lie in the plane of the base and are perpendicular to the line AC . Therefore, the x -axis is parallel to the line BM . Note that the hatched part ADB of the plane is not a section of the prism. We find the coordinates of the points A , B , D , and C_1 : $A(0; 0; 0)$, $B(2\sqrt{3}; 2; 0)$, $D(0; 2; 3)$, and $C_1(0; 4; 3)$. Proceeding from the known coordinates of the points A , B , and D , we find the equation of the plane ADB as it was done in Problem 3 of 8.3 and get

$$\sqrt{3}x - 3y + 2z = 0.$$

The distance ρ from the point $C_1(0; 4; 3)$ to that plane can be found by formula (1):

$$\rho = \frac{|\sqrt{3} \cdot 0 - 3 \cdot 4 + 2 \cdot 3|}{\sqrt{3 + 9 + 4}} = \frac{6}{4} = \frac{3}{2}.$$

Answer: $3/2$ cm. $\blacktriangle$

Problems on finding the distances between a line and a plane parallel to it and between parallel planes reduce to finding the distance between a point and a plane. This follows from the two assertions cited below:

the distance between a line and a plane parallel to it is equal to the distance from an arbitrary point of that line to the given plane;

the distance between two parallel planes is equal to the distance from an arbitrary point of one plane to the other plane.

Let us prove the first assertion (Fig. 145a). The second assertion can be proved by analogy (Fig. 145b).

□ Assume $a \parallel \alpha$, $a \not\subset \alpha$ (if $a \subset \alpha$, then the distance between them is evidently zero). Suppose that A is an arbitrary point of the line a , $[AA_1] \perp \alpha$ (Fig. 145a). It is required to prove that the distance between a and α is $|AA_1|$. We draw a plane β through the line a and the point A_1 and have $\beta \perp \alpha$. Assume $a_1 = \beta \cap \alpha$, then $a_1 \parallel a$. We take arbitrary points $M \in a$ and $N \in \alpha$, and draw a perpendicular MM_1 in the plane β to the line a_1 . Then, evidently, $|MM_1| = |AA_1|$. Besides, $[MM_1] \perp \alpha$ (since $\beta \perp \alpha$) and, hence,

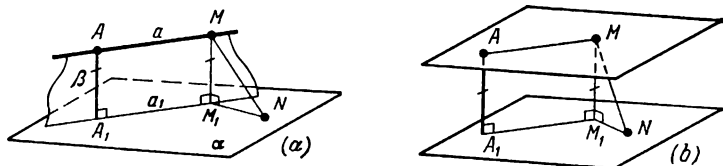


Fig. 145

$|MN| \geq |MM_1|$. From this it follows that $|MN| \geq |AA_1|$. Thus, $|AA_1|$ is the least among the distances between the points of the line a and the plane α , i.e. $|AA_1|$ is the distance between these figures. ■

Problem 3. The altitude of the regular quadrangular pyramid $SABCD$ and the side of its base are equal to a . Find the distance between the line AB and the plane SCD .

△ Let us draw a plane through the vertex S of the pyramid and the midpoints M and N of the edges AB and CD (Fig. 146). This plane is perpendicular to the plane SCD . The perpendicular MP to the line SN is also a perpendicular to the plane SCD . Its length is precisely equal to the distance between the line AB and the plane SCD . The altitude SO of the pyramid and the segment MP are the altitudes of the triangle SMN . In that triangle we have:
 $|MN| = a$, $|SO| = a$, $|SN| = \sqrt{|SO|^2 + |ON|^2} = \sqrt{a^2 + \frac{a^2}{4}} = \frac{\sqrt{5}}{2} a$. The area of the triangle SMN is equal to $\frac{1}{2} |MN| \cdot |SO|$ on one hand, and to $\frac{1}{2} |SN| \cdot |MP|$, on the other. Hence we find that $|MP| = \frac{|MN| \cdot |SO|}{|SN|} = \frac{2}{\sqrt{5}} a$.

Answer: $2a/\sqrt{5}$. ▲

Problem 4. In the cube $ABCD A_1 B_1 C_1 D_1$ find the distance between the planes $AB_1 D_1$ and BDC_1 if $|AB| = a$.

$\triangle$ The diagonal $A_1 C$ (Fig. 147) of the cube is perpendicular to the planes $AB_1 D_1$ and BDC_1 (the consequence of Problem 3 of 8.1). It follows from the result of Problem 3 in 7.2 that the planes $AB_1 D_1$ and BDC_1 divide the diagonal $A_1 C$ into three congruent segments, that is, if M and N are the points

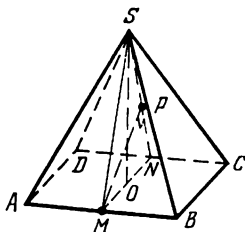


Fig. 146

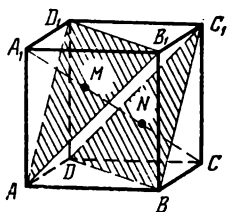


Fig. 147

of intersection of the diagonal $A_1 C$ with those planes, then $|A_1 M| = |MN| = |NC| = \frac{1}{3} |A_1 C|$. The length of the segment MN is equal to the distance between the planes $AB_1 D_1$ and BDC_1 . Since $|A_1 C| = \sqrt{3}a$, it follows that $|MN| = \frac{\sqrt{3}}{3} a$.

Answer: $a\sqrt{3}/3$. $\blacktriangle$

The distance between skew lines is known to be equal to the length of their common perpendicular.

Problem 5. In the tetrahedron $ABCD$ the edges AB and CD are a in length, the other edges are b in length. Find the distance between the lines AB and CD .

$\triangle$ Assume that M and N are the midpoints of the edges AB and CD (Fig. 148). In the isosceles triangles ABC and ABD we find $|CM| = |DM| = \sqrt{b^2 - \frac{a^2}{4}}$. The triangle CMD is isosceles and, therefore, $[MN] \perp (CD)$. Similarly we establish that the triangle ANB is isosceles ($|AN| = |NB|$), and, therefore, $[MN] \perp (AB)$. This means that MN is the common perpendicular of the lines AB and CD .

From the triangle CMN we find that $|MN| = \sqrt{|CM|^2 - |CN|^2} = \sqrt{b^2 - \frac{a^2}{2}}$.

Answer: $\sqrt{b^2 - \frac{a^2}{2}}$. $\blacktriangle$

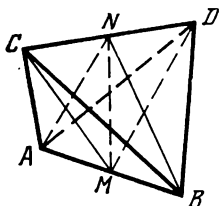


Fig. 148

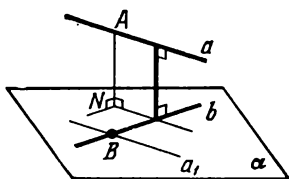
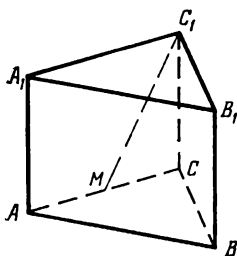
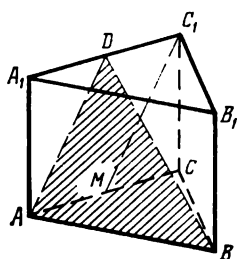


Fig. 149



(a)



(b)

Fig. 150

Corollary. *In a regular tetrahedron with an edge a the segments connecting the midpoints of the skew edges are the common perpendiculars to those edges and the length of each segment is equal to $a/\sqrt{2}$.*

To find the distance between skew lines, it is not necessary to construct their common perpendicular. Two skew lines can always be drawn in two parallel planes. A scheme of constructing one of these planes (through the line b) is shown in Fig. 149: B is an arbitrary point of the line b , $a_1 \parallel a$, $a_1 \cap b = B$, $b \subset \alpha$, $a_1 \subset \alpha$. The distance between the lines a and b is equal to that between any point of the line a and

the plane α . It is also possible to draw a plane β , parallel to the plane α , through the line a . The distance between the planes α and β is equal to the distance between the lines a and b .

Problem 6. In the regular triangular prism shown in Fig. 150a $|AB| = 4$ cm, $|AA_1| = 3$ cm. Find the distance between the lines AB and C_1M , where M is the midpoint of the edge AC .

Δ We draw a line, parallel to the line C_1M , through the point A of the plane AA_1C_1C . It is clear that it passes through the midpoint D of the edge A_1C_1 (Fig. 150b). The plane ADB passes through the line AB parallel to the line C_1M . The distance between the lines AB and C_1M is equal to the distance between some point of the line C_1M and the plane ADB . We found the distance from the point C_1 to the plane ADB in Problem 2 of this section, it is equal to $3/2$ cm.

Answer: $3/2$ cm. $\blacktriangle$

8.5. A Dihedral Angle. The Angle Between Planes. A Bisector. A Trihedral Angle

A *measure of a dihedral angle* is a measure of one of its plane angles. A *plane angle* is a section of a dihedral angle formed by some plane perpendicular to an edge. The sides of a plane angle are perpendicular to the edge of the dihedral angle. To find a dihedral angle, it is good practice to construct its plane angle and to find its magnitude.

Problem 1. Find the edge dihedral angle of a regular tetrahedron.

Δ We find the dihedral angle at the edge AB (Fig. 151). Suppose that DO is the altitude of the tetrahedron and DM is the altitude of the face ABD . Then MO is the projection of DM onto the plane ABC and, consequently, $[MO] \perp [AB]$ as well. This means that $\angle DMO$ is a plane dihedral angle at the edge AB . We designate the length of the edge of the tetrahedron as a and, then, $|DM| = a\sqrt{3}/2$. The point O is the centre of the face ABC and, therefore, $|MO| = a\sqrt{3}/6$. From the right triangle DMO we find that $\cos DMO = |MO|/|DM| = 1/3$. Hence, the dihedral angle at the edge AB is equal to $\arccos(1/3)$. The dihedral angles at

the other edges of a regular tetrahedral have the same magnitude.

Answer: $\arccos(1/3)$. ▲

Problem 2. All the edges of the regular quadrangular pyramid $SABCD$ are of the same length. Find the dihedral angle between the faces SBC and SAD .

△ The plane SBC passes through the line BC which is parallel to the plane SAD ($BC \parallel AD$), Fig. 152). Therefore, the line of intersection of the planes SBC and SAD

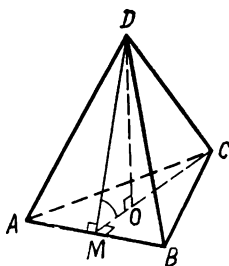


Fig. 151

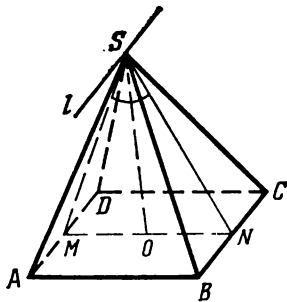


Fig. 152

is a straight line l , which passes through the point S parallel to the line BC . The line l is the edge of the dihedral angle which we have to find. Suppose M and N are the midpoints of the edges AD and BC respectively. The plane SMN is perpendicular to the edge l . Indeed, $(BC) \perp (MN)$ and $(BC) \perp (SN)$ and, hence, $(BC) \perp (SMN)$. But $l \parallel (BC)$ and, consequently, $l \perp (SMN)$ as well. It follows from this that $\angle MSN$ is a plane dihedral angle between the faces SBC and SAD . We designate the length of the edge of the pyramid as a and assume that SO is the altitude of the pyramid. We also introduce the designation $\angle MSN = \alpha$. In the isosceles triangle MSN we have $|MN| = a$, $|MS| = |NS| = a\sqrt{3}/2$, $|ON| = a/2$. Hence it follows that $\sin(\alpha/2) = |ON|/|SN| = 1/\sqrt{3}$. Now we find that $\cos \alpha = 1 - 2 \sin^2(\alpha/2) = 1/3$, $\alpha = \arccos(1/3)$.

Answer: $\arccos(1/3)$. ▲

Two intersecting planes form four dihedral angles. If the angles are congruent, then the planes are said to be perpen-

dicular and the angle between them is $\pi/2$. Now if the planes are not perpendicular, then the *smaller of the magnitudes of the dihedral angles formed by them is taken as the angle between them*. Thus, the angle between planes is in the limits between 0 and $\pi/2$.

Problem 3. In the regular hexagonal pyramid $SAB CDE F$ the ratio of the altitude to the length of the side of the base is $16:4$. Find the angle between the planes SBC and SDE .

△ We construct the line of intersection of the planes SBC and SDE , the line SM (Fig. 153, $M = (BC) \cap (ED)$). To find the dihedral angle with the faces BSM and ESM , we first construct its plane angle.

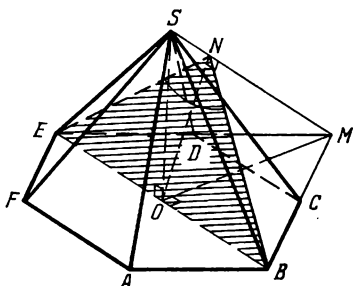


Fig. 153

Suppose O is the centre of the base of the pyramid. The point O is the midpoint of the diagonal BE , and, since the triangle BME is regular, we have $(MO) \perp (BE)$. The line MO is the projection of the line SM onto the plane of the base (SO is a perpendicular to that plane), and, therefore, $(SM) \perp (BE)$ as well. Assume that ON is an altitude in the triangle SOM . Then, from the fact that $(SM) \perp (ON)$ and $(SM) \perp (BE)$ it follows that $(SM) \perp (BNE)$, that is, the angle BNE is the plane dihedral angle $BSME$. Let us find the angle BNE .

We designate the length of the side of the base of the pyramid as a . Then, $|BE| = |BM| = |EM| = 2a$ and, hence, $|MO| = a\sqrt{3}$. From the right triangle SOM we find that $|SM| = \sqrt{|SO|^2 + |OM|^2} = \frac{3\sqrt{6}}{4}a$. Next we find that $|ON| = \frac{|SO| \cdot |OM|}{|SM|} = \frac{a}{\sqrt{3}}$. It is clear from the symmetry about the plane SOM that if $\angle BNE = \varphi$ then $\angle BNO = \varphi/2$. From the triangle BON we have $\tan(\varphi/2) = |BO|/|ON| = \sqrt{3}$, whence $\varphi = 2\pi/3$. This is the magnitude of the dihedral angle $BSME$. Since $\varphi > \pi/2$, the angle

between the planes BSC and ESD is equal to $\pi - \varphi = \pi/3$.

Answer: $\pi/3$. $\blacktriangle$

The method of coordinates can also be used to find the angle between planes.

Assume that the planes are defined by the equations

$$a_1x + b_1y + c_1z + d_1 = 0,$$

$$a_2x + b_2y + c_2z + d_2 = 0.$$

We designate the angle between the planes as φ . Then (7.8 in Part 1)

$$\cos \varphi = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1| \cdot |\mathbf{n}_2|},$$

where $\mathbf{n}_1 = (a_1; b_1; c_1)$ and $\mathbf{n}_2 = (a_2; b_2; c_2)$.

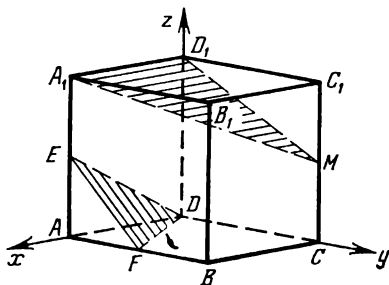


Fig. 154

Problem 4. Points E , F , and M in the cube shown in Fig. 154 are the midpoints of the edges AA_1 , AB , and CC_1 respectively. Find the angle between the planes EFD and A_1D_1M .

$\triangle$ We introduce the Cartesian system of coordinates as shown in Fig. 154. Designating the length of the edge of the cube as a , we find the coordinates of the points: $D(0; 0; 0)$, $E(a; 0; a/2)$, $F(a; a/2; 0)$, $A_1(a; 0; a)$, $D_1(0; 0; a)$, and $M(0; a; a/2)$. From the known coordinates of the points we find the equation of the plane EFD :

$$x - 2y - 2z = 0,$$

and the equation of the plane A_1D_1M :

$$y + 2z - 2a = 0$$

(the hatched part of the plane A_1D_1M in Fig. 154 is not a section of the cube). The vector $\mathbf{n}_1 = (1; -2; -2)$ is perpendicular to the plane EFD and the vector $\mathbf{n}_2 = (0; 1; 2)$ is perpendicular to the plane A_1D_1M . We find the angle φ between these planes:

$$\cos \varphi = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1| \cdot |\mathbf{n}_2|} = \frac{|1 \cdot 0 - 2 \cdot 1 - 2 \cdot 2|}{\sqrt{1+4+4} \cdot \sqrt{1+4}} = \frac{2}{\sqrt{5}},$$

$$\varphi = \arccos(2/\sqrt{5}). \quad \blacktriangle$$

The bisector of a dihedral angle is a half-plane dividing it into two equal dihedral angles.

The boundary of the bisector is the edge of the dihedral angle.

There is a bisector for any dihedral angle. This is how it can be constructed: we construct any plane angle of the given dihedral angle (Fig. 155); then we construct the bisector of that plane angle. The half-plane defined by the edge of the dihedral angle and that bisector is precisely the bisector of the given dihedral angle. Let us prove this.

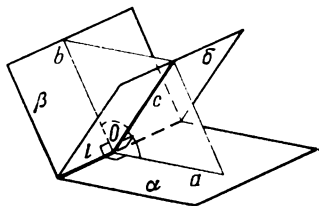


Fig. 155

[1] The plane of the angle aOb (Fig. 155) is perpendicular to the common edge l of the dihedral angles $\alpha l \delta$ and $\delta l \beta$. This means that the angles aOc and cOb are plane angles of those dihedral angles. But c is the bisector of the angle aOb , therefore, $\angle aOc = \angle cOb$. This means that the dihedral angles $\alpha l \delta$ and $\delta l \beta$ are equal, that is, δ is the bisector of the dihedral angle $\alpha l \beta$. ■

It is also easy to see that the bisector of a dihedral angle cuts every plane angle of that dihedral angle along its bisector. In other words,

the bisector of every plane angle belongs to the bisector of the given dihedral angle.

The bisector of a dihedral angle possesses a number of properties similar to those of the bisector of a plane angle.

Problem 5. Prove that the bisector of a dihedral angle is the set of points of that angle equidistant from the planes of its faces.

$\triangle$ We designate by X the set of points of the dihedral angle equidistant from the planes of its faces, and by δ the bisector of the dihedral angle.

Let us prove that every point of the set X belongs to the bisector, i.e. $X \subset \delta$. We shall first note the fact that the edge l of the dihedral angle (Fig. 156a) belongs both to the set X and to the bisector δ . There are, evidently, no points belonging to the set X in the faces of the angle outside the edge l . Let us now consider the interior points of the dihedral

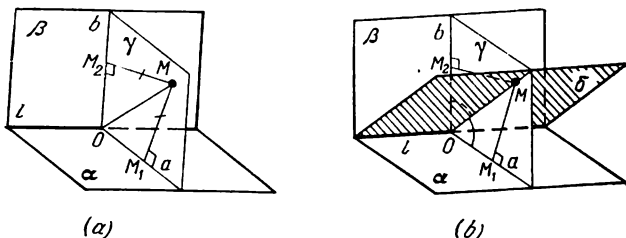


Fig. 156

angle. Assume $M \in X$. We draw a plane γ through the point M at right angles to the edge l (Fig. 156a). The section of the dihedral angle is its plane angle aOb . Suppose MM_1 and MM_2 are perpendiculars to the lines containing the rays a and b . Then, MM_1 and MM_2 are perpendiculars to the planes of the faces of the angle and, hence, $|MM_1| = |MM_2|$ since $M \in X$. This means that the point M of the angle aOb is equidistant from the lines which contain its sides. It is known from plane geometry that in this case the point belongs to the bisector of the angle aOb . But then, together with the bisector of this angle, it belongs to the bisector δ . Thus we have proved that $X \subset \delta$.

Remark. The plane dihedral angle is less than a straight angle. As we have proved, OM is the bisector of the plane angle aOb , therefore, the ray OM forms acute angles with the rays a and b . Hence it follows that the feet of the perpendiculars MM_1 and MM_2 lie on the rays a and b respectively. Figure 156a takes account of this fact.

Let us prove that every point of the bisector belongs to the set X , i.e. $\delta \subset X$. As was noted, the points of the edge l belong both to the bisector and to the set X . Assume that M

is an arbitrary point of the bisector and does not belong to the edge. We again draw the plane γ through the point M at right angles to the edge l and obtain a plane angle aOb in the section (Fig. 156b). As we have established above, the bisector of the dihedral angle cuts this plane angle along its bisector. This means that the point M belongs to the

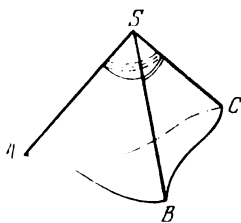


Fig. 157

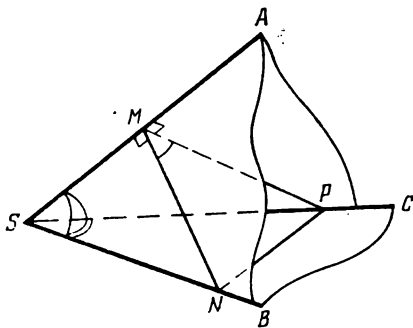


Fig. 158

bisector of the angle aOb and is, therefore, equidistant from the lines containing those sides, i.e. $|MM_1| = |MM_2|$. The perpendiculars MM_1 and MM_2 to the sides of the angle are also the perpendiculars to the planes of the faces of the dihedral angle since the plane γ is perpendicular to those planes. The point M is, therefore, equidistant from the planes of the dihedral angle too, i.e. $M \in X$. It follows that $\delta \subset X$.

We have thus proved that $X \subset \delta$ and $\delta \subset X$. This means that $\delta = X$, and that is what we had to prove. $\blacktriangle$

A trihedral angle is the simplest of polyhedral angles. The following magnitudes are most often used to describe a trihedral angle: the magnitudes of its three plane angles, which are the faces (Fig. 157) ASB , BSC , and CSA , and the magnitudes of its three dihedral angles at the edges SA , SB , and SC . These six magnitudes are interrelated. For instance, the dihedral angles can be found from the known plane angles.

Recall that for a trihedral angle with the plane angles α , β , and γ to exist, it is necessary and sufficient that two con-

ditions be fulfilled:

$$\alpha + \beta + \gamma < 360^\circ \quad \text{and} \quad |\beta - \gamma| < \alpha < \beta + \gamma.$$

If even one of these conditions is not fulfilled, then the trihedral angle with the given plane angles does not exist (see Ch. 2, Problem 6 in Section I).

Problem 6. The plane angles ASB , ASC and the dihedral angle at the edge SA in the trihedral angle $SABC$ are 60° . Find the angle BSC .

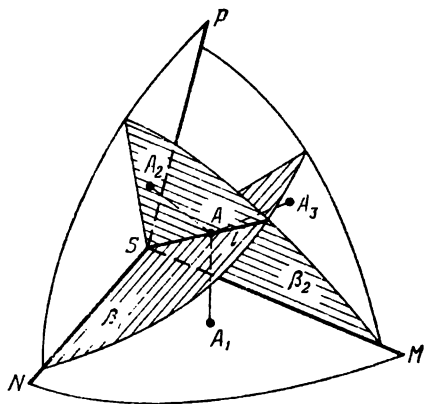


Fig. 159

$\triangle$ Let us take a point $M \neq S$ on the edge SA and draw $[MN] \perp [SA]$ and $[MP] \perp [SA]$ (Fig. 158). We designate $|SM| = a$. From the right triangles SMN and SMP we find: $|MN| = |MP| = a\sqrt{3}$, $|SN| = |SP| = 2a$. The angle NMP is the plane angle of the dihedral angle at the edge SA , therefore, $\angle NMP = 60^\circ$. Now in the triangle MNP we find that $|NP| = a\sqrt{3}$. By the cosine law, we find in the triangle SNP that

$$\cos NSP = \frac{|SN|^2 + |SP|^2 - |NP|^2}{2|SN| \cdot |SP|} = \frac{5}{8},$$

whence $\angle NSP = \arccos(5/8)$. $\blacktriangle$

Problem 7. Prove that the bisectors of the dihedral angles of a trihedral angle intersect along the same ray.

$\triangle$ Let us consider two bisectors, say, β_1 and β_2 (Fig. 159). Their intersection is a ray with vertex S ; we designate it as l . Suppose A is an arbitrary point of the ray l , $A \neq S$. We drop perpendiculars AA_1 , AA_2 , and AA_3 to the planes of the faces of the angle. Since $A \in \beta_1$, we have $|AA_2| = |AA_1|$ and, besides, $|AA_3| = |AA_1|$ since $A \in \beta_2$. This means that $|AA_2| = |AA_3|$, that is, the point A is equidistant from the planes of the faces NSP and MSP . Hence it follows that the point A belongs to the bisector of the dihedral angle at the edge SP . Since A is an arbitrary point of the ray l , the whole ray belongs to the bisector. Thus it follows that the three bisectors intersect in the interior of the trihedral angle along the same ray. $\blacktriangle$

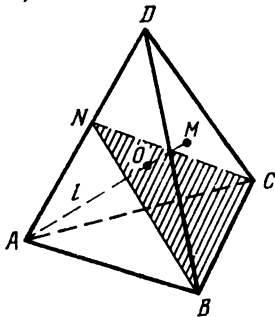


Fig. 160

It is known from plane geometry that the bisectors of the angles of a triangle meet at one point. The bisectors of the dihedral angles of a tetrahedron possess a similar property.

Problem 8. Prove that the bisectors of the dihedral angles of a tetrahedron meet at one point in the interior of the tetrahedron.

$\triangle$ Suppose l is a ray which is the line of intersection of the bisectors of the trihedral angle at the vertex A , and M is the point of intersection of that ray and the face BCD (Fig. 160). The end points of the line segment AM belong to different faces of the dihedral angle at the edge BC and, therefore, the bisector of that angle cuts the segment AM . We designate the point of intersection as O .

The point O belongs to the ray l and, consequently, it is equidistant from the planes ABC , ABD , and ACD . At the same time, the distances from the point O to the planes ABC and BCD are equal since that point belongs to the bisector of the dihedral angle at the edge BC . Thus we see that the point O is equidistant from the planes of all the faces, that is, it belongs to the bisectors of all the dihedral angles of the tetrahedron. $\blacktriangle$

8.6. Calculating the Volumes of Polyhedra and Their Parts

Problem 1. A section $A_1B_1C_1$ is drawn in the triangular pyramid $SABC$ parallel to the base so that $|SA_1|/|SA| = k$. Prove that if V_1 and V are the volumes of the pyramids $SA_1B_1C_1$ and $SABC$ respectively, then

$$V_1 = k^3 V.$$

$\triangle$ Let us consider a homothetic transformation with centre S and ratio k . The triangle $A_1B_1C_1$ (Fig. 161) is the

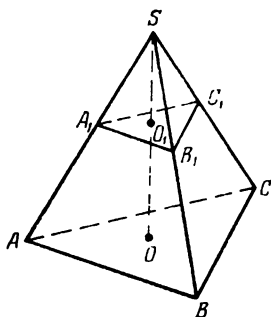


Fig. 161

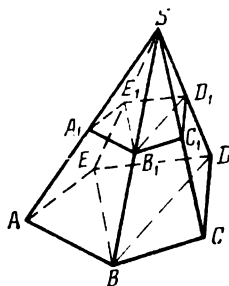


Fig. 162

image of the triangle ABC under this transformation since $|SB_1|/|SB| = |SC_1|/|SC| = |SA_1|/|SA| = k$. Consequently, $S_{A_1B_1C_1} = k^2 S_{ABC}$. Suppose SO is the altitude of the pyramid $SABC$, $O_1 = [SO] \cap (A_1B_1C_1)$. Then, SO_1 is the altitude of the pyramid $SA_1B_1C_1$ and $|SO_1| = k|SO|$. Hence it follows that

$$V_1 = \frac{1}{3} |SO_1| S_{A_1B_1C_1} = k^3 \cdot \frac{1}{3} |SO| \cdot S_{ABC} = k^3 V,$$

and that is what we had to prove. $\blacktriangle$

Remark. (1) If we consider an arbitrary polygonal pyramid, we shall evidently get the same result since it can be represented as a union of several triangular pyramids (Fig. 162).

(2) Using the result obtained in Problem 1, it is easy to derive a formula for the volume of a truncated pyramid:

$$V = \frac{1}{3} H (S_1 + \sqrt{S_1 S_2} + S_2),$$

where H is the altitude of the truncated pyramid and S_1 and S_2 are the areas of its bases.

In calculating the volume of a prism, besides the formula $V = H \cdot Q$, where H is the altitude of the prism and Q is the area of its base, use is also made of the formula $V = l \cdot S$,

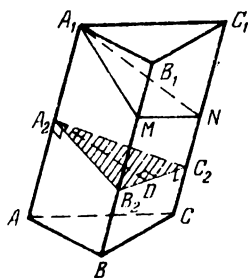


Fig. 163

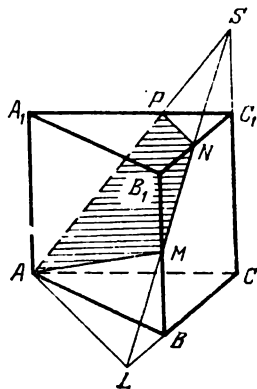


Fig. 164

where l is the length of the lateral edge and S is the area of the perpendicular section of the prism.

Problem 2. The volume of the triangular prism $ABCA_1B_1C_1$ is V . Points M and N are located on the edges BB_1 and CC_1 respectively, so that $|BM|/|BB_1| = m$, $|CN|/|CC_1| = n$. Find the volume of the polyhedron $ABCA_1MN$ (a truncated prism).

△ Assume that $A_2B_2C_2$ is a perpendicular section of the prism (Fig. 163) and A_2D is the altitude of that section. We designate $|AA_1| = l$, $|B_2C_2| = a$, $|A_2D| = h$ and have $|BM| = ml$, $|CN| = nl$, $|B_1M| = (1 - m)l$, $|C_1N| = (1 - n)l$. Let us find the volume of the pyramid $A_1B_1C_1NM$. Its base is the trapezoid B_1C_1NM . The altitude of the trapezoid is, evidently, equal to $|B_2C_2|$, i.e. to a . We find the area of the trapezoid:

$$S_{tr} = \frac{1}{2} (|B_1M| + |C_1N|) \cdot a = \frac{2-m-n}{2} al.$$

Note that the segment A_2D is perpendicular to the plane BB_1C_1C and, therefore, the altitude of the pyramid

$A_1B_1C_1NM$ drawn from the vertex A_1 is equal to $|A_2D| = h$. Hence it follows that if V_1 is the volume of the pyramid, then $V_1 = \frac{1}{3} h S_{1r} = \frac{1}{6} (2 - m - n) lah$.

Next we find the area of the perpendicular section of the prism: $S = \frac{1}{2} ah$. This means that the volume of the whole prism can be expressed by the formula $V = \frac{1}{2} lah$. Comparing the expressions for V_1 and V , we see that $V_1 = \frac{1}{3} (2 - m - n) V$. Then we find the volume V_2 of the polyhedron $ABCA_1MN$: $V_2 = V - V_1 = \frac{1+m+n}{3} V$. ▲

To find the volume of a polyhedron, use is often made of the complement of that polyhedron to a pyramid or a prism or a partitioning into those figures.

Problem 3. A section is drawn in the triangular prism $ABCA_1B_1C_1$ through the vertex A and the midpoints of the edges BB_1 and B_1C_1 . Find the ratio of the volumes of the parts into which the section divides the prism.

△ We construct the section. Assume that M and N are the midpoints of the edges BB_1 and B_1C_1 (Fig. 164). We find the points $S = (MN) \cap (CC_1)$ and $P = (SA) \cap (A_1C_1)$. The section is a quadrangle $AMNP$.

We designate the volume of the prism as V and the volume of the part containing the edge CC_1 as V_1 . The technique of constructing the section suggests the way of expressing V_1 in terms of V . We construct a point $L = (MN) \cap (BC)$. To find V_1 , we must subtract the volumes of the pyramids $SPNC_1$ and $MALB$ from that of the pyramid $SALC$. And those volumes are easy to express in terms of the volume of the prism.

We designate the altitude of the prism as H , the area of the base as Q , and the volumes of the pyramids $SALC$, $SPNC_1$, and $MALB$ as V_2 , V_3 , and V_4 respectively.

M and N being the midpoints of the edges, we have $|BL| = \frac{1}{2} |BC|$ and $|CL| = \frac{3}{2} |BC|$. Hence it follows that $S_{ALC} = \frac{3}{2} Q$. Furthermore, $|SC_1| = \frac{1}{2} |CC_1|$ and $|SC| = \frac{3}{2} |CC_1|$. This means that the altitude of the

pyramid $SALC$ is equal to $\frac{3}{2}H$. We find $V_2 = \frac{1}{3} \cdot \frac{3}{2}H \times S_{ALC} = \frac{3}{4}HQ$, i.e. $V_2 = \frac{3}{4}V$. The pyramid $SPNC_1$ is homothetic to the pyramid $SALC$, with the ratio $k = \frac{1}{3}$ ($|SC_1|/|SC| = 1/3$), whence it follows that $V_3 = k^3 V_2 = \frac{1}{36}V$. M being the midpoint of the edge BB_1 , the altitude of the pyramid $MALB$ is equal to $\frac{1}{2}H$, and the fact that $|BL| = \frac{1}{2}|BC|$ yields $S_{ALB} = \frac{1}{2}Q$. This means that $V_4 = \frac{1}{3} \cdot \frac{1}{2}H \cdot \frac{1}{2}Q = \frac{1}{12}H \cdot Q = \frac{1}{12}V$. Then we find $V_1 = V_2 - V_3 - V_4 = \frac{23}{36}V$. The volume of the cut-off part of the prism containing the edge AA_1 is equal to $V - V_1 = 13/36$ and the ratio between the volumes of the parts is $13/23$. ▲

8.7. Problems on Combinations of Polyhedra

Problem 1. The side of the base of a regular quadrangular pyramid is a in length, and the altitude of the pyramid is h . A cube is inscribed into the pyramid so that its four vertices belong to the base of the pyramid and the other four vertices lie in the lateral faces, and four edges of the cube are parallel to the diagonal of the base of the pyramid. Find the length of the edge of the cube.

△ Suppose four edges of the cube are parallel to the diagonal AC (Fig. 165) of the base of the pyramid. We designate them as MN , PQ , M_1N_1 , and P_1Q_1 , and the required length of the edge of the cube we designate as b .

Let us consider the section of the pyramid formed by the plane of the upper base of the cube ($A_1B_1C_1D_1$ in Fig. 165).

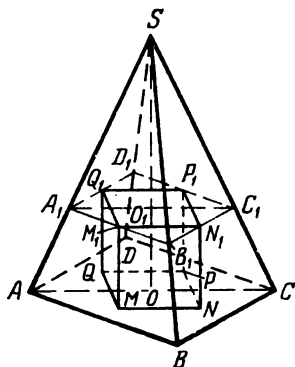


Fig. 165

It is a square, the edges M_1N_1 and P_1Q_1 of the cube are parallel to its diagonal A_1C_1 . Consequently, $\angle N_1M_1B_1 = \pi/4$, and from this we infer that $\angle Q_1M_1A_1 = \pi/4$ as well, that is, the right triangle $M_1A_1Q_1$ is isosceles. Moreover, the triangles $M_1B_1N_1$ and $M_1A_1Q_1$ are congruent since their hypotenuses are congruent, $|M_1N_1| = |M_1Q_1| = b$. This means that $|B_1M_1| = |A_1M_1|$. Hence it follows that $b = |M_1N_1| = |A_1C_1|/2$.

Let us now make use of the fact that the distance between the planes $A_1B_1C_1D_1$ and $ABCD$ is equal to b . Suppose SO is an altitude of the pyramid, $O_1 = [SO] \cap (A_1B_1C_1D_1)$. We have $|OO_1| = b$, $|SO_1| = h - b$, $|SO_1|/|SO| = (h - b)/h$. Hence it follows that $|A_1C_1|/|AC| = (h - b)/h$. Taking into account that $|A_1C_1| = 2b$, $|AC| = a\sqrt{2}$, we obtain $2b/a\sqrt{2} = (h - b)/h$, whence we find that $b = ha/(a + \sqrt{2}h)$.

Note that we have reasoned and calculated assuming that the positions of the pyramid and the cube indicated in the hypothesis are possible. It follows from the solution that the cube can be inscribed into the given pyramid as required. Indeed, let us draw a section of the pyramid, parallel to its base, through the point A_1 of the edge SA , such that $\frac{|SA_1|}{|SA|} = \frac{\sqrt{2}h}{a + \sqrt{2}h}$. It is easy to prove that the midpoints of the sides of the section and the feet of the perpendiculars dropped from them to the plane of the base of the pyramid are the vertices of the cube satisfying the hypothesis. It also follows from the solution carried out that there is only one cube of this kind.

Answer: $ah/(a + \sqrt{2}h)$. ▲

While solving the problem, we have made use of the section of one of the given polyhedra formed by the plane of the face of the other. It is sometimes useful to consider such sections in problems where one polyhedron is inscribed into the other. Let us now discuss a problem where such sections are of no use.

Problem 2. The side of the base of the regular triangular prism $ABCA_1B_1C_1$ is a in length. Two edges of a regular tetrahedron lie on the straight lines A_1B and B_1C . Find the length of the edge of that tetrahedron.

△ The lines A_1B and B_1C evidently contain the skew edges of the tetrahedron. But such edges are perpendicular in a regular tetrahedron (Problem 2 in 7.4), consequently, the lines A_1B and B_1C must be perpendicular. This condition will allow us to find the length l of the lateral edge of the prism.

It is also known that the common perpendicular of the skew edges of a regular tetrahedron is $b/\sqrt{2}$ in length, where

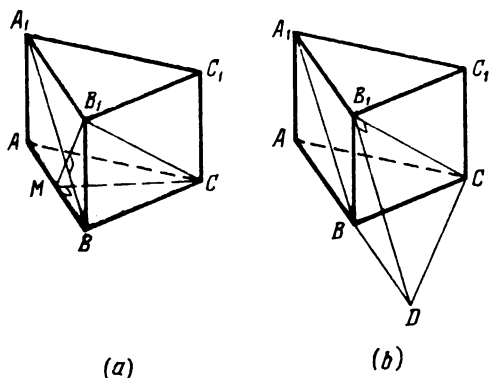


Fig. 166

b is the length of the edge of the tetrahedron (consequence of Problem 5 in 8.4). This means that if d is the distance between the lines A_1B and B_1C , then $b = d\sqrt{2}$.

Let us find the length l of the lateral edge of the prism from the condition indicated above. We project the diagonal B_1C onto the plane AA_1B_1B . Its projection is the line B_1M (Fig. 166a), where M is the midpoint of the edge AB . The fact that $(B_1C) \perp (A_1B)$ yields $(B_1M) \perp (A_1B)$. In that case the right triangles MB_1B and A_1BA are similar and, therefore, $|BB_1|/|AB| = |MB|/|AA_1|$, whence it follows that $l^2 = a^2/2$, $l = a/\sqrt{2}$.

Let us determine the distance d between the lines A_1B and B_1C . We draw $[B_1D] \parallel [A_1B]$ (Fig. 166b). The desired distance is equal to the distance from the point B to the plane DB_1C , that is, to the altitude of the pyramid BDB_1C drawn from the vertex B . Let us find the volume of the pyra-

mid taking the face DBC as its base; then we get $V = a^3/4 \sqrt{6}$. Taking now the face DB_1C as the base of the pyramid, we obtain

$$|DB_1| = |BA_1| = |B_1C| = \sqrt{a^2 + \frac{1}{2}a^2} = a \sqrt{\frac{3}{2}}.$$

The triangle DB_1C is right-angled since $(DB_1) \parallel (A_1B)$ and $(A_1B) \perp (B_1C)$. This means that

$$S_{DB_1C} = \frac{1}{2} |DB_1| \cdot |B_1C| = \frac{3a^2}{4}.$$

Substituting the values of V and S_{DB_1C} , into the formula $V = \frac{1}{3} d \cdot S_{DB_1C}$, we get $d = a/\sqrt{6}$. In the final analysis we find that $b = d/\sqrt{2} = a/\sqrt{3}$.

It is clear that when the lateral edge of the prism has the length which we have found, then the regular tetrahedron can be positioned in the manner indicated in the hypothesis. The diagonals A_1B and B_1C will then be perpendicular. Laying off segments of the length $b/2 = a/2 \sqrt{3}$ on each of the lines A_1B and B_1C from the foot of their common perpendicular, we get four points which are the vertices of the regular tetrahedron.

Answer: $a/\sqrt{3}$. ▲

PROBLEMS OF SECTION I

1. Prove that a plane perpendicular to each of the two intersecting planes is perpendicular to their line of intersection.

2. Prove the equivalence of the following statements:

(a) the lateral edges of a pyramid are equal in length;

(b) the lateral edges of a pyramid make equal angles with the plane of the base;

(c) a circle can be circumscribed about the base of a pyramid, the altitude of the pyramid passing through the centre of the circle.

3. Prove that the planes perpendicular to the edges of a tetrahedron and passing through their midpoints meet at one point.

4. The side of the base of the regular pyramid $SABCD$ is a in length and the lateral edge is l in length. Construct a section of the pyramid perpendicular to the lateral edge SC and passing through its midpoint, and find the area of the section if (a) $l = \sqrt{\frac{3}{2}}a$; (b) $l = \sqrt{\frac{5}{2}}a$.

5. A section is drawn in the cube $ABCD A_1 B_1 C_1 D_1$ through the diagonal AD_1 of the face $AA_1 D_1 D$ at right angles to the plane $BC_1 D$.

Find the ratio between the volumes of the parts of the cube into which it is divided by the section.

6. The lateral edge SB of the regular quadrangular pyramid $SABCD$ makes the angle $\pi/4$ with the plane of the base. Find the angle between that edge and the plane SCD .

7. Prove that the distance ρ from the point $M_0(x_0; y_0; z_0)$ to the plane α defined by the equation $ax + by + cz + d = 0$, can be calculated by the formula

$$\rho = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

8. Given the coordinates of the vertices of the base of the regular pyramid $SABC$: $A(5; 1; -1)$, $B(5; -2; 2)$, and $C(2; -2; -1)$. The vertex S of the pyramid lies in the coordinate plane Oyz . Find the distance between the lines SB and AC .

9. A straight line intersects the faces γ_1 and γ_2 of a dihedral angle at points A and B respectively and makes an angle of $\pi/6$ with the plane of the face γ_1 and an angle of $\arcsin 0.7$ with the face γ_2 . The dihedral angle is equal to $\pi/3$. Find the angle between the line AB and the edge of the dihedral angle.

10. A section is drawn in the quadrangular pyramid $SABCD$, whose base is the parallelogram $ABCD$, through the edge AB and the midpoint M of the edge SC . Find the ratio between the volumes of the parts into which the section divides the pyramid.

11. The volume of the tetrahedron is V . All the vertices of the parallelepiped lie on the surface of the tetrahedron, three faces of the parallelepiped belonging to three faces of the tetrahedron. Find the greatest possible volume of such a parallelepiped.

PROBLEMS OF SECTION II

1. The points $M(-1; 3; 2)$, $N(1; 1; 3)$, and $A(-1; 2; 3)$ are given in the Cartesian space. Points B and C are taken on the line MN such that $\angle BAC = 90^\circ$ and $\angle ABC = 30^\circ$. Find the coordinates of the points.

2. Prove that a line and a plane, perpendicular to the same line, are parallel.

3. Prove that if diagonals of the parallelepiped are equal in length, then the parallelepiped is rectangular.

4. One of the diagonals of a rectangular parallelepiped is perpendicular to the plane drawn through the end points of three edges emanating from the same vertex as the diagonal. Prove that the parallelepiped is a cube.

5. Prove that the altitudes of a regular tetrahedron meet at one point, which is its centre.

6. Find the angles between the lines connecting the centre of the regular tetrahedron with its vertices.

7. The base of the pyramid is a rhombus with the acute angle α . The length of the side of the base and of three lateral edges is equal to a . Find the length of the fourth lateral edge of the pyramid if it is known to be less than a .

8. The base of the pyramid $SABC$ is a regular triangle ABC with the side a in length. The face SAB is perpendicular to the plane of the base, and the altitudes of the faces SAC and SBC drawn from the vertex S are equal to $\sqrt{5}a$ and $\sqrt{2}a$ respectively. Find the altitude of the pyramid.

9. In the pyramid $SABC$ the angles ABC , SAB , and SCB are right angles and SH is the altitude of the pyramid. Express the vector $\vec{SH}$ in terms of the vectors $\vec{SA}$, $\vec{SB}$, and $\vec{SC}$.

10. The base of the pyramid $SABCD$ is an isosceles trapezoid $ABCD$ in which $|AB| = |BC| = a$ and $|AD| = 2a$. The planes of the faces SAB and SCD are perpendicular to the plane of the base of the pyramid. Find the altitude of the pyramid if the altitude of the face SAD drawn from the vertex S is $2a$.

11. Prove the equivalence of the following statements:

(a) the altitudes (or their extensions) of a tetrahedron meet at one point;

(b) every two skew edges of a tetrahedron are perpendicular;

(c) the foot of one of the altitudes of a tetrahedron is the point of intersection of the altitudes of the face;

(d) the foot of every altitude of a tetrahedron is the point of intersection of the altitudes of the face to which that altitude is dropped.

12. Prove the following fact: for the right angle to be the orthogonal projection of the right angle onto the plane, it is necessary and sufficient that one of the sides of the given angle should be parallel to the plane of the projection and the other side should not be perpendicular to that plane.

13. The area of a regular triangle is S . Its orthogonal projection onto the plane is an isosceles triangle with the angle α at the vertex. Find the area of that projection if (a) $\alpha > \pi/3$; (b) $\alpha < \pi/3$.

14. The edge of the cube $ABCD A_1 B_1 C_1 D_1$ is a in length. Points P , K , and L are the midpoints of the edges AA_1 , $A_1 D_1$, and $B_1 C_1$ respectively, and the point Q is the centre of the face $CC_1 D_1 D$. The line segment MN with its end points lying on the lines AD and KL cuts the line PQ and is perpendicular to it. Find the length of the segment.

15. In the triangular prism $ABCA_1 B_1 C_1$ points M , N , and P are the midpoints of the sides AB , BC , and CA of the base ABC respectively. The line segments MC_1 , NA_1 , and PB_1 are pairwise perpendicular, and the length of each of them is a . Find the volume of the prism.

16. The plane angles at the vertex of the pyramid are right angles, S_0 is the area of the base and S_1 , S_2 , and S_3 are the areas of the lateral faces. Prove that $S_0^2 = S_1^2 + S_2^2 + S_3^2$.

17. The edge of the regular tetrahedron $ABCD$ is a in length. Find the area of the orthogonal projection of the tetrahedron onto the plane passing through the midpoints of the edges AB , BD , and DC .

18. The base of the prism $ABCA_1 B_1 C_1$ is a regular triangle ABC the length of whose side is $2a$. The orthogonal projection of the prism onto the plane of the base ABC is a trapezoid with a leg AB and the area twice that of the base. Find the altitude of the prism if $|AB_1| = b$ (find all solutions).

19. In the cube $ABCD A_1 B_1 C_1 D_1$ points E and F are the midpoints of the edges $A_1 B_1$ and $C_1 D_1$ respectively. Construct a perpendicular from the point F to the plane EBD and find its length if the edge of the cube is a in length.

20. In the regular quadrangular prism $ABCD A_1 B_1 C_1 D_1$ the side of the base is a in length. A straight line, perpendicular to the plane $BA_1 D$, is drawn through the point C_1 . Find the length of the segment of that line, lying in the interior of the prism, if the length of the lateral edge is equal to (a) $a\sqrt{2}$, (b) $a/2\sqrt{2}$.

21. All the edges of the regular prism $ABCA_1 B_1 C_1$ are a in length. The straight line, perpendicular to the plane $BA_1 C$, cuts the lines BC_1 and AB_1 at points M and N respectively. Find the length of the segment MN . (Hint. Express in vector form the condition of the perpendicularity of the segment MN and the plane $BA_1 C$.)

22. A section is drawn through the midpoint of the altitude of a regular quadrangular pyramid at right angles to the lateral edge. Find the area of the section if the lateral edge is a in length and the angle between the lateral edges lying in the same face is equal to (a) 60° , (b) 120° .

23. Under consideration are the sections of the cube formed by planes perpendicular to one of its diagonals. Determine the greatest possible area of the section if the edge of the cube is a in length.

24. In the regular triangular prism $ABCA_1 B_1 C_1$ a section is drawn through the vertex B_1 at right angles to the diagonal CA_1 of the lateral face $AA_1 C_1 C$. Find the area of the section if (a) $|AB| = |AA_1| = a$; (b) $|AB| = 2a$, $|AA_1| = a$.

25. The base of the pyramid $SABCD$ is a rhombus $ABCD$ with the side of length a and the acute angle α . The foot of the altitude of the pyramid is the centre of the rhombus $ABCD$. The planes of the faces SAB and SCD are perpendicular. Find the volume of the pyramid.

26. In the regular hexagonal pyramid $SABCDEF$ the angle between the apothem of the lateral face and its projection onto the plane of the base is equal to $\pi/3$. Construct the section of the pyramid passing through the edge AB at right angles to the face SCD and determine in what ratio it divides the altitude of the pyramid.

27. Three nonintersecting diagonals of the lateral faces of a triangular prism are perpendicular and each of them is a in length. Find the volume of the prism.

28. Point M is the midpoint of the lateral edge AA_1 of the parallelepiped $ABCD A_1 B_1 C_1 D_1$. Straight lines BD , MD_1 , and $A_1 C$ are pairwise perpendicular. Find the altitude of the parallelepiped if $|BD| = 2a$, $|BC| = \frac{3}{2}a$, and $|A_1 C| = 4a$.

29. Three planes are pairwise perpendicular, and a straight line makes angles φ_1 , φ_2 , and φ_3 with them. Prove that $\sin^2 \varphi_1 + \sin^2 \varphi_2 + \sin^2 \varphi_3 = 1$.

30. Line segments AM and AN are drawn from the point A of the plane on its both sides. Each segment is a in length and makes an angle α with the plane. The angle between the projections of the segments onto the plane is β . Determine the length of the segment MN .

31. A line segment AB of length a lies in a plane. A perpendicular AM and an inclined line BN are drawn from the points A and B to

the plane (on the same side of it), with $|AM| = |BN| = \frac{3}{2}a$, $|MN| = 2a$, and $\angle ABN = 90^\circ$. Find the angle between the line MN and the plane.

32. Find the angles between the line of intersection of the planes

$$2x - y - z = 0 \quad \text{and} \quad x - 2y + z = 0$$

and (a) the coordinate axes, (b) the coordinate plane Oxy .

33. The side of the base of the regular triangular prism $ABCA_1B_1C_1$ is a in length. D is the midpoint of the edge AB , and the point E lies on the edge A_1C_1 . The line DE makes angles α and β with the planes ABC and AA_1C_1C respectively. Find the altitude of the prism.

34. The side of the base of a regular triangular pyramid is a in length, the angle between the apothem and the lateral face is equal to $\pi/4$. Determine the altitude of the pyramid.

35. A plane is drawn in the regular tetrahedron $ABCD$ through the vertex B parallel to the edge CD and making an angle of (a) $\pi/3$, (b) $\arcsin 5/6$, with the edge AD . In what ratio does the plane divide the edge AD ?

36. The base of the rectangular parallelepiped $ABCD A_1 B_1 C_1 D_1$ is a square $ABCD$. Find the greatest possible magnitude of the angle between the line BD_1 and the plane BDC_1 .

37. A plane cuts a line segment. Prove that it divides the segment in the ratio equal to the ratio between the distances from its ends to that plane.

38. How many planes are there which are equidistant from four points not lying in the same plane?

39. Calculate the distance from the point $M(3; 1; -1)$ to the plane $22x - 4y - 20z - 45 = 0$; and from the point $N(4; 3; -2)$ to the plane $3x - y + 5z + 1 = 0$.

40. Find the points on the z -axis which are equidistant from the planes $x + 4y - 3z - 2 = 0$ and $5x + z + 8 = 0$.

41. Given the coordinates of four vertices of the parallelepiped $ABCD A_1 B_1 C_1 D_1$: $A(1; 1; 1)$, $B(3; 2; 0)$, $C(3; 5; 0)$, and $C_1(3; 7; 2)$. Find the coordinates of the other vertices and the distance between the vertex A and the plane CDD_1C_1 .

42. In the pyramid $SABC$ the edge SA is perpendicular to the plane ABC , $|AC| = |CB|$, and $|AS| = |AB| = a$. A plane is drawn through the midpoint of the edge AC at right angles to the edge SB . Find the distance between the vertex C and that plane.

43. The side of the base ABC of the regular pyramid $SABC$ is a in length, and the lateral edge is b in length. At what distance from the line BC should the section of the pyramid be drawn, parallel to the edges BC and SA , for its area to be the greatest possible?

44. The face AA_1B_1B of the prism $ABCA_1B_1C_1$ is perpendicular to its base ABC . The altitude of the prism is equal to H , the area of the face AA_1B_1B is equal to the area of the base of the prism. Find the distance between the point B and the plane CA_1B_1 .

45. In the regular pyramid $SABCD$ the length of the common perpendicular of the edges SA and BC is a , and its foot divides the segment BC in the ratio $1:3$. Determine the volume of the pyramid.

46. Given the vertices $A(1; 2; 3)$, $B(2; 3; 5)$, $A_1(3; 3; 4)$, and $C_1(4; 5; 5)$ of the parallelepiped $ABCD A_1 B_1 C_1 D_1$. Find the distance between the lines AD_1 and BC .

47. The volume of the tetrahedron $ABCD$ is equal to 5 cu cm. A plane drawn through the midpoints of the edges AD and BC , at a distance of 1 cm from the vertex A cuts the edge CD at a point M so that $|DM|/|MC| = 2/3$. Calculate the area of the section of the tetrahedron formed by that plane.

48. The base of the pyramid $SABCD$ is a rhombus $ABCD$ with the side a and the acute angle $BAD = 60^\circ$. The altitude of the pyramid is $\frac{3}{5}a$ and $\angle SBA + \angle SBC = 180^\circ$. A point M is taken on the edge SC such that $|CS| = 3|CM|$. Find the distance between the lines DM and AC .

49. In the pyramid $SABC$ the common perpendicular to the lines AC and SB passes through the midpoint of the edge SB . The areas of the faces ASB and BSC are equal, the area of the face ASC is twice that of the face BSC . There is a point M in the interior of the pyramid the sum of whose distances to the vertices B and S is equal to the sum of the distances to the planes of all the faces of the pyramid. Find $|MB|$, if $|AC| = \sqrt{6}$ cm and $|SB| = 1$ cm.

50. The edge of a regular octahedron is a in length. Determine the shortest distance along the surface of the octahedron between the midpoints of two parallel edges.

51. The side of the base of a regular triangular prism is a in length, and the lateral edge is l in length. Determine the shortest distance along the surface of the prism between the vertex of one base and the midpoint of the opposite side of the other base.

52. The plane of an isosceles triangle makes an angle ϕ with the plane α passing through its base. The vertex angle of the triangle is ψ . Find the angle between its side and the plane α .

53. Given a rectangle whose sides are equal to 1 cm and 2 cm. A plane α is drawn through the smaller side of the rectangle, making an angle ϕ with the diagonal of the rectangle. Find the angle between that plane and the plane of the rectangle.

54. In a regular quadrangular pyramid the dihedral angle between the adjacent lateral faces is α . Find the base dihedral angle of the pyramid.

55. In the cube $ABCD A_1 B_1 C_1 D_1$ find the angle between the planes: (a) A_1DM and A_1B_1CD ; (b) A_1DM and C_1DN , where M and N are the midpoints of the edges AB and BC respectively; (c) EDC_1 and EB_1C ; (d) AA_1D_1D and EFO , where E and F are the midpoints of the edges AA_1 and CD respectively, and O is the centre of the face $A_1B_1C_1D_1$.

56. A plane is drawn through the vertex A of the cube $ABCD A_1 B_1 C_1 D_1$, parallel to the diagonal BD_1 of the cube and the diagonal B_1C of its face. Find the angle between that plane and the plane $ABCD$.

57. In the regular hexagonal pyramid $SABCDEF$ the planes of the faces SAB and SCD are perpendicular. Determine the angle between the planes: (a) SAB and SDE ; and (b) SAB and SBC .

58. The base of the pyramid is a triangle with sides a , a , and b . Each lateral face is at the angle α to the plane of the base. Find:

(a) the area of the lateral surface of the pyramid; and (b) the volume of the pyramid for the case when $a = 5$ cm, $b = 6$ cm, and $\alpha = 45^\circ$.

59. The legs AB and AC of the right triangle lie, respectively, in different faces of a dihedral angle and make acute angles φ and ψ , respectively, with its edge. Find the dihedral angle.

60. The sides AB and AC of the equilateral triangle lie in the faces α and β , respectively, of the acute dihedral angle φ . The side AB makes an acute angle ψ with the edge of the dihedral angle. Determine the angle between the plane ABC and the face β .

61. In the tetrahedron $ABCD$ the dihedral angles at the edges AB , AC , and BD are right angles. One of the segments connecting the midpoints of the opposite edges of the tetrahedron is a in length and the other is $a\sqrt{6}$ in length. Find the length of the longest edge of the tetrahedron.

62. Points A and B are in the faces of a dihedral angle. Prove that the bisector divides the segment AB in the ratio equal to the ratio between the distances from the points A and B to the edge of the dihedral angle.

63. The side of the base of the regular quadrangular pyramid is a in length, and the base dihedral angle is α . A bisector cuts one of the base dihedral angles. Find the area of the resulting section.

64. In the pyramid $SABC$ the lateral edge SA is perpendicular to the plane of the base ABC , $|SA| = a$, and $|BC| = 2a$. The dihedral angles at the edges SB and SC are 60° each. Find the volume of the pyramid.

65. The base of the right-angled parallelepiped $ABCD A_1 B_1 C_1 D_1$ is a rhombus $ABCD$, and the altitude of the parallelepiped is a . The plane $AB_1 M$, where M is the midpoint of the edge BC , makes the angles of 60° with the planes $AA_1 C_1 C$ and $BB_1 D_1 D$. Determine the volume of the parallelepiped.

66. The plane trihedral angles ASB , BSC , and CSA have the magnitudes $\pi/3$, $\pi/6$, and $\arccos(1/\sqrt{3})$ respectively. Find the dihedral angle at the edge SB .

67. Prove that if the dihedral angles at the edges SB and SC of the trihedral angle $SABC$ are equal, then the plane angles ASC and ASB opposite to them are also equal.

68. Every dihedral angle of a trihedral angle is equal to α . Find the plane angles and indicate all possible values of α .

69. Every trihedral plane angle is equal to $\pi/3$. A point lies in the interior of the angle at the distance a from two faces of the angle and at the distance $3a$ from the third face. Find the distance between that point and the vertex of the angle.

70. Every dihedral angle of a trihedral angle is equal to φ . A point lies in the interior of the angle at the distance a from its vertex and at equal distances from the plane of each face. Find that distance.

71. Every tetrahedral plane angle is equal to α and all the dihedral angles at the edges of the given tetrahedral angle have the same magnitude. Determine the dihedral angles.

72. The length of the side of the base of a regular triangular pyramid is equal to a , α is the magnitude of the plane angle at the vertex of the pyramid, β is the dihedral angle between the lateral face and the base, γ is the angle between the lateral edge and the base, and δ

is the dihedral angle between the lateral faces. Find the volume of the pyramid in the following four cases: (a) $\cos \alpha = 1/4$, (b) $\tan \beta = 2$, (c) $\tan \gamma = 1$, and (d) $\cos \delta = 1/5$.

73. The faces of the triangular pyramid are isosceles triangles. The base of each triangle is a in length and the opposite angle is equal to α . Find the volume of the pyramid.

74. The length of the side of the smaller base of the regular truncated triangular pyramid is a , the dihedral angle of the pyramid at the edge of the smaller base is equal to 120° , and the altitude of the pyramid is $a/2$. Find the volume of the pyramid.

75. The base of the pyramid is a right triangle with the acute angle α . Every lateral edge makes an angle of 30° with the plane of the base of the pyramid and is b in length. Find the volume of the pyramid.

76. In the pyramid $SABC$ every two skew edges are perpendicular, $|AB| = |AC| = b$, $\angle BAC = \alpha$, and the dihedral angle of the pyramid at the edge BC is equal to ϕ . Find the volume of the pyramid.

77. The base of the pyramid is an isosceles trapezoid with the acute angle equal to α and the area S . Every lateral face makes an angle β with the base. Find the volume of the pyramid.

78. The base of the pyramid is a convex polygon, and the areas of the lateral faces are equal. Prove that the sum of the distances from any point of the base to the planes of the lateral faces of the pyramid is a constant quantity.

79. The skew diagonals of two parallel faces of the parallelepiped are the edges of a tetrahedron. Determine the ratio of the volumes of the parallelepiped and the tetrahedron.

80. If we cut the surface of the tetrahedron along the edges AD , BD , and CD , then its development onto the plane ABC will be a square with side a . Find the volume of the tetrahedron.

81. A plane cuts the lateral edges SA , SB , and SC of the triangular pyramid at points M , N , and P respectively, so that $|SM|/|SA| = m$, $|SN|/|SB| = n$, $|SP|/|SC| = p$. Find the ratio of the volumes of the pyramids $SMNP$ and $SABC$.

82. Find the ratio between the volumes of the parts into which the prism $ABCA_1B_1C_1$ is divided by the planes passing through:

(a) the vertex B_1 and the midpoints of the edges AA_1 and AC ;

(b) the midpoints of the edges AB , AC , and BB_1 ;

(c) the point of intersection of the medians of the base ABC , the midpoint of the edge A_1C_1 and the point of intersection of the diagonals of the face BB_1C_1C .

83. Prove that any plane passing through the point of intersection of the diagonals of a parallelepiped divides it into two parts of equal volumes.

84. Prove that every plane passing through the midpoints of two skew edges of a tetrahedron cuts it into two parts of equal volumes.

85. The bases of a truncated quadrangular pyramid are the parallelograms $ABCD$ and $A_1B_1C_1D_1$, the ratio of the lengths of whose sides is equal to 2. A section is drawn through the edge B_1C_1 parallel to the face AA_1D_1D . Find the ratio between the volumes of the parts into which the section divides the pyramid.

86. The ratio between the lengths of two skew edges of the tetrahedron is equal to q . A plane is drawn parallel to those edges so that a

rhombus results in the section. Find the ratio between the volumes of the parts into which the plane divides the tetrahedron.

87. Given the tetrahedron $ABCD$ with $|AB| = 2a$, $|AD| = |BD| = b$, and $|AC| = |BC| = c$ ($a < b < c$). The straight lines DM and BN , where M and N are the midpoints of the edges AB and CD , are perpendicular. Find the volume of the tetrahedron.

88. In the tetrahedron $ABCD$ the face ABC is a regular triangle, the face DBC is perpendicular to it, $\angle DAC = \pi/3$, $|AD| = 6$ cm, and the angle between the lines AD and BC is equal to $\arccos(1/4)$. Find the volume of the tetrahedron (find all solutions).

89. Given the triangular prism $ABCA_1B_1C_1$ of volume V . Find the volume of the part of the prism bounded by its base $A_1B_1C_1$, the lateral faces and the planes ABC_1 , BCA_1 , and CAB_1 .

90. In the Cartesian space the solid Φ is given as the set of all those and only those points whose coordinates satisfy the inequality $|x| + |y| + |z| \leq a$. Find the volume of the solid Φ .

91. A lateral edge of the regular triangular pyramid makes an angle ϕ with the plane of its base. One base of the regular triangular prism belongs to the base of the pyramid and the vertices of the other base lie on the lateral edges of the pyramid. All the edges of the prism are a in length. Find the volume of the pyramid. At what value of ϕ is the volume the smallest?

92. The edge AB of the tetrahedron $ABCD$ is the diagonal of the base of a quadrangular pyramid, and the edge CD is parallel to the plane of its base and its ends lie on the lateral edges of the pyramid. Determine the least possible volume of the pyramid if the volume of the tetrahedron is V .

93. Given the regular tetrahedron $ABCD$. Its every vertex A , B , C , and D is symmetrically reflected about the plane of the face opposite to it, and as a result the respective points A_1 , B_1 , C_1 , and D_1 are obtained. Find the ratio between the volumes of the tetrahedra $A_1B_1C_1D_1$ and $ABCD$.

94. The line segments connecting the vertices of the tetrahedron with the points of intersection of the medians of the opposite faces meet at a point O . Another tetrahedron is symmetric with respect to the first about the point O . Find the volume of the common part of the tetrahedra if the volume of the first tetrahedron is V .

95. The volume of the regular quadrangular pyramid $SABCD$ is V . The altitude SP of the pyramid is an edge of the regular tetrahedron $SPQR$, the plane of whose face PQR is perpendicular to the edge SC . Find the volume of the common part of the pyramid and the tetrahedron.

96. In the tetrahedron $ABCD$ the edges AB and CD are perpendicular, $|AB| = a$, $|CD| = b$. The line segment connecting the midpoints of the edges AB and CD is perpendicular to them and its length is c . The vertices of the regular tetrahedron lie on the surface of the tetrahedron $ABCD$, one of its edges is parallel to the edge AB and the other to the edge CD . Find the length of the edge of that regular tetrahedron (find all solutions).

97. All the vertices of the regular pyramid $SABCD$ lie on the edges of the regular pyramid $QMNP$ with the base MNP . The plane $ABCD$ is known to be perpendicular to the plane of the lateral face QMN of

the pyramid $QMNP$. Find the ratio between the volumes of the pyramids.

98. The volume of the triangular prism $ABCA_1B_1C_1$ is V . Four planes are drawn so that the first passes through the vertex C parallel to the plane ABC_1 , the second through the vertex C_1 parallel to the plane A_1B_1C , the third through the edge BB_1 parallel to the edge AC , and the fourth through the edge AA_1 parallel to the edge BC . Find the volume of the tetrahedron bounded by these four planes.

99. In the regular prism $ABCA_1B_1C_1$ the side of the base is $2a$ in length, and the lateral edge is a in length. Three planes are drawn so that the first passes through the vertex A at right angles to the line AB_1 , the second through the vertex B at right angles to the line BC_1 , and the third through the vertex C at right angles to the line CA_1 . Find the volume of the tetrahedron bounded by these three planes and the plane $A_1B_1C_1$.

100. The length of the edge of the cube $ABCD A_1B_1C_1D_1$ is a . A point E is the midpoint of the edge AD . One edge of the regular tetrahedron lies on the line ED_1 , and the other lies on the line passing through the point A_1 and intersecting the line BC at a point R . Find: (a) $|BR|$; and (b) the length of the edge of the tetrahedron.

101. Two regular tetrahedra $ABCD$ and $MNPQ$ are positioned so that the planes BCD and NPQ coincide, the vertex M lies on the altitude AO of the first tetrahedron and the plane MNP passes through the centre of the face ABC and the midpoint of the edge BD . Find the ratio between the lengths of the edges of the tetrahedra.

102. The side of the base of a regular triangular pyramid is $2\sqrt{6}$ cm in length, and the altitude of the pyramid is 3 cm. The vertex A of the cube $ABCD A_1B_1C_1D_1$ is at the centre of the base of the pyramid, the vertex C_1 on the altitude of the pyramid, and the edge CD lies in the plane of one of its lateral faces. Find the length of the edge of the cube.

Chapter 9

Solids of Revolution

9.1. A Cylinder

A *cylinder* is a solid obtained as a result of rotation of a rectangle about the axis containing its side (Fig. 167). Circles resulting from the rotation of the sides of a rectangle, which are perpendicular to the axis (OA and O_1A_1 in Fig. 167), are called the *bases* of the cylinder. The radius of the base is said to be also the *radius* of the cylinder.

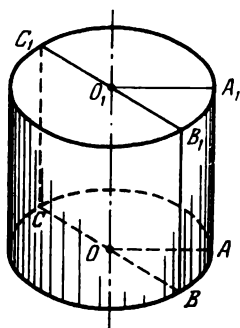


Fig. 167

The solid obtained as a result of rotation of the side of a rectangle, which is parallel to the axis (AA_1 in Fig. 167), is called the *lateral surface* of the cylinder. The distance between all points of the lateral surface of a cylinder and its axis is equal to the radius of the cylinder.

Every section of a cylinder formed by a plane passing through the axis (*axial section*) is a rectangle (BB_1C_1C in

Fig. 167). Its sides, parallel to the axis (BB_1 , CC_1), are called the *generating lines* of the lateral surface of the cylinder. Every generating line is perpendicular to the planes of the bases of the cylinder, its length being equal to the distance between these planes, that is, to the altitude of the cylinder.

Every section of a cylinder formed by a plane perpendicular to its axis is a circle congruent to the bases of the cylinder.

The orthogonal projection of a cylinder onto a plane, perpendicular to its axis, is a circle of radius equal to the radius of the cylinder, and the projection of the axis is the

centre of that circle. The projection of the lateral surface is a circumference bounding that circle.

Problem 1. The vertices A , B , and D_1 of the cube $ABCD A_1 B_1 C_1 D_1$ lie on the lateral surface of the cylinder whose axis is parallel to the straight line DC_1 . Find the radius of the cylinder if the edge of the cube is a in length.

$\triangle$ The plane $BA_1 D_1 C$ is perpendicular to the line DC_1 (Fig. 168) and, hence, to the axis of the cylinder. The projections of the points B and D_1 onto the plane $BA_1 D_1 C$ coin-

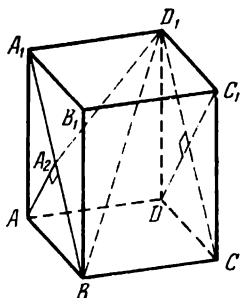


Fig. 168

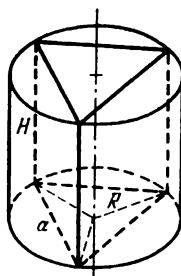


Fig. 169

cide with these points themselves. The projection of the point A is the midpoint A_2 of the line segment BA_1 . The projection of the lateral surface of the cylinder onto the plane $BA_1 D_1 C$ is a circle passing through the points B , A_2 , and D_1 . The radius of the circle is equal to that of the cylinder. This radius can be found by the formula $R = \frac{a_1 a_2 a_3}{4S}$, where a_1 , a_2 , and a_3 are the lengths of the sides BA_2 , $A_2 D_1$, and BD_1 of the triangle $BA_2 D_1$, and S is the area of that triangle. We have $a_1 = |A_2 B| = a/\sqrt{2}$, $a_3 = |BD_1| = a\sqrt{3}$, and the triangle $A_2 A_1 D_1$ can be used to find $a_2 = |A_2 D_1| = a\sqrt{3/2}$. Furthermore, $S = \frac{1}{2} |A_2 B| \cdot |A_1 D_1| = \frac{a^2}{2\sqrt{2}}$. Now we can find $R = \frac{3\sqrt{2}}{4} a$. $\blacktriangle$

A prism is said to be *inscribed into a cylinder* (and a cylinder is said to be *circumscribed about a prism*) if the bases of the prism are inscribed into the bases of the cylinder.

It is evident that *to be possible to circumscribe a cylinder about a prism it is necessary and sufficient that the prism should be right-angled and a circle could be circumscribed about its base.*

In particular, a cylinder can be circumscribed about any right triangular prism and about any regular prism. Every lateral edge of the prism inscribed into a cylinder is a generating line of the lateral surface of the cylinder.

Problem 2. Considered are various right triangular prisms of the given volume V . A cylinder is circumscribed about each of these prisms. Find the least area of the surface of such a cylinder.

△ We designate the altitude of the prism as H , the length of the side of the base as a , and the radius of the circumscribed cylinder as R (Fig. 169). This radius is equal to the radius of the circle circumscribed about the base of the prism, and since the base is a regular triangle, $R = a/\sqrt{3}$.

It follows from the hypothesis that $\frac{\sqrt{3}}{4} a^2 H = V$, whence we have $H = \frac{4V}{\sqrt{3}a^2}$. Taking this into account, we find

the area of the surface of the cylinder: $S = 2\pi R^2 + 2\pi RH = \frac{2\pi}{3} \left(a^2 + \frac{4V}{a} \right)$. The area S is a function of the variable $a \in (0; +\infty)$. We find the critical point of that function: $S' = \frac{2\pi}{3} \left(2a - \frac{4V}{a^2} \right) = 0$, $a_0 = \sqrt[3]{2V}$. It is easy to see that the function S has the least value at $a = a_0$ and it is equal to $2\pi \sqrt[3]{4V^2}$.

Answer: $2\pi \sqrt[3]{4V^2}$. ▲

Assume that the plane α is parallel to the axis of the cylinder, d is the distance between that plane and the axis, and R is the radius of the cylinder. If $d > R$, then the plane α has no points in common with the cylinder. If $0 \leq d < R$, then the section of the cylinder formed by the plane α is a rectangle (Fig. 170). If $d = R$, then the plane α has a common generating line (KK_1 in Fig. 170) with the lateral surface of the cylinder and that generating line lies on the orthogonal projection of the axis of the cylinder onto the plane α . In that case, the plane α is said to be *tangential* to the lateral surface of the cylinder.

A cylinder is said to be *inscribed* into a prism (and a prism is said to be *circumscribed* about a cylinder) if the bases of the cylinder are inscribed into the bases of the prism. It is easy to see that

to be possible to inscribe a cylinder into a prism, it is necessary and sufficient that the prism should be right-angled and a circle could be inscribed into its base.

Every lateral face of the prism circumscribed about a cylinder touches its lateral surface along the generating

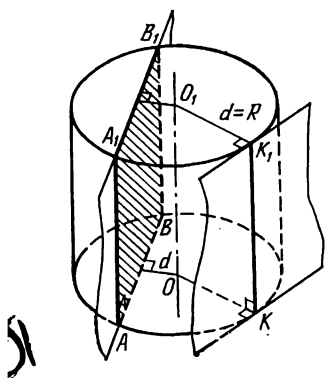


Fig. 170

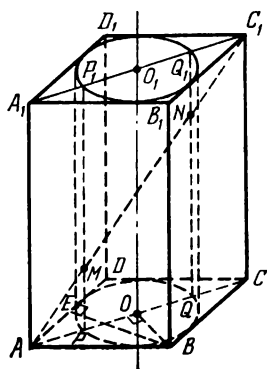


Fig. 171

line, which passes through the points of tangency of the respective sides of the base of the prism with the circumferences of the bases of the cylinder.

A cylinder can be inscribed into any right triangular prism and into any regular prism.

Problem 3. The radius of the cylinder is r , and its altitude is $5r$. A parallelepiped is circumscribed about the cylinder, the ratio between the volume of the parallelepiped and that of the cylinder being $5/\pi$. Find the length of the segment of the longer diagonal of the parallelepiped lying in the interior of the cylinder.

△ The faces of the parallelepiped circumscribed about the bases of the cylinder are designated as $ABCD$ and $A_1B_1C_1D_1$ (Fig. 171) so that $\angle BAD = \alpha < \pi/2$. The parallelogram circumscribed about a circle is a rhombus and, therefore, $ABCD$ is a rhombus. Suppose BE is the altitude

of that rhombus and, then $|BE| = 2r$ and from the triangle ABE we find $|AB| = 2r/\sin \alpha$. The area of the rhombus $ABCD$ is $4r^2/\sin \alpha$. Since the altitude of the parallelepiped is equal to the altitude of the cylinder, we have $V_p = 20 r^3/\sin \alpha$. Taking into account that $V_c = 5\pi r^3$ and the condition $V_p/V_c = 5/\pi$, find that $\sin \alpha = 4/5$. Hence it follows that $\cos(\alpha/2) = 2/\sqrt{5}$.

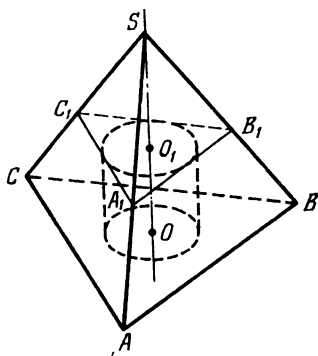


Fig. 172

From the triangle AOB we find $|AO| = |AB| \cos(\alpha/2) = \sqrt{5} r$, whence $|AC| = 2|AO| = 2\sqrt{5} r$.

Let us consider a section of the cylinder formed by the plane AA_1C_1C (AC_1 and A_1C are the principal diagonals of the parallelepiped, $|AC_1| = |A_1C|$). The section is a rectangle PP_1Q_1Q . The intersection of the diagonal AC_1 and the rectangle, the segment

MN , is precisely the segment of that diagonal lying in the interior of the cylinder. Let us find its length.

From the right triangle ACC_1 we have $|AC_1| = \sqrt{|AC|^2 + |CC_1|^2} = 3\sqrt{5} r$. Taking into account that $|PQ| = 2r$, we find, from the proportion $|MN|/|AC_1| = |PQ|/|AC|$, that

$$|MN| = |AC_1| \cdot |PQ|/|AC| = 3r.$$

Answer: $3r$. ▲

There can be other cases of mutual positions of a cylinder and a polyhedron besides those discussed above.

Problem 4. Given a regular triangular pyramid of volume V . A cylinder is inscribed into the pyramid so that one of its bases belongs to the base of the pyramid and the other base is inscribed into the section of the pyramid formed by a plane parallel to the base. Find the greatest possible volume of such a cylinder.

△ We designate the altitude of the pyramid as H , the length of the side of the base as a , the altitude of the cylinder as h , and the radius of the cylinder as r . Let us consider the

section $A_1B_1C_1$ (Fig. 172) of the pyramid formed by the plane of the upper base of the cylinder. That section is a regular triangle homothetic to the base ABC with the ratio equal to $(H - h)/H$. Consequently, its side is $a(H - h)/H$ in length and the radius of the inscribed circle is equal to $\sqrt{3}a(H - h)/6H$. That is the radius of the cylinder, i.e. $r = \sqrt{3}a(H - h)/6H$. Let us find the volume of the cylinder as a function of h :

$$V_c = \frac{\pi a^2}{12H^2} (H - h)^2 h, \quad h \in (0; H).$$

Then we seek the critical point of the function obtained $V'_c = \frac{\pi a^2}{12H^2} (H - h)(H - 3h)$, $h_0 = H/3$. It is evident that for $h = h_0$ the function V_c has the greatest value equal to $\frac{\pi}{81} a^2 H$. Bearing in mind that $\frac{\sqrt{3}}{12} a^2 H = V$ by the hypothesis, we find that the greatest possible volume of the cylinder under consideration is equal to $\frac{4\sqrt{3}\pi}{81} V$. $\blacktriangle$

9.2. A Cone

A *cone* is a solid obtained as a result of rotation of a right triangle about the axis containing its leg. The other leg of the triangle, when rotated about that axis, produces a circle called the *base* of the cone. The solid produced by the rotation of the hypotenuse is called the *lateral surface* of the cone.

The line segments connecting the vertex of the cone with the points of the circumference of its base are said to be the *generating lines* of the lateral surface of the cone (called, briefly, the generating lines of the cone; SA and SB in Fig. 173). All the generating lines are equal in length and make equal acute angles with the axis.

Every section of a cone formed by the plane passing through the axis (*axial section*) is an isosceles triangle ($\triangle ASB$ in Fig. 173, $|SA| = |SB|$). The vertex angle of the triangle is called the *apex angle* of the cone. If the apex angle is equal to φ , then the angle between the generating line and the axis is $\varphi/2$.

Every section of the lateral surface of the cone formed by the plane perpendicular to its axis and not passing through its vertex is a circle. All points of that circle are equidistant from the vertex of the cone, the distance being equal to $\sqrt{r^2 + h^2}$, where r is the radius of the circle and h is the distance from the vertex of the cone to the plane of the section. The converse is also true: points of the lateral

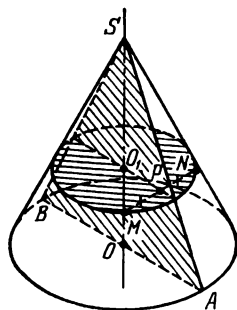


Fig. 173

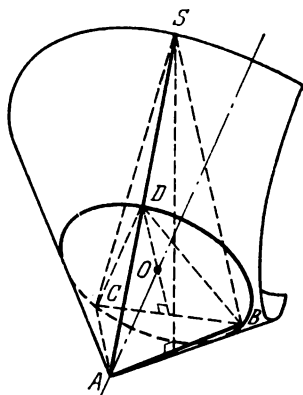


Fig. 174

surface of the cone equidistant from its vertex lie on the same surface, which is a section of the lateral surface by a plane perpendicular to the axis. If the points M and N of the lateral surface of the cone are equidistant from its vertex, then the axis of the cone lies in the plane passing through the midpoint of the segment MN at right angles to it (in Fig. 173 $|MP| = |PN|$, $(SAB) \perp [MN]$).

Problem 1. In the regular pyramid $SABC$ the side of the base ABC is a in length, the lateral edge is $2a$ long. Points S , B , and C lie on the lateral surface of the cone with vertex at the point A . Find the apex angle of the cone.

Δ Let us take a point D on the ray AS (Fig. 174), such that $|AD| = a$. Then $|AD| = |AB| = |AC|$; consequently, the points B , C , and D , belonging to the surface of the cone, lie in the same plane perpendicular to its axis. The section of the lateral surface of the cone formed by that

plane is a circle circumscribed about the triangle BCD . Let us determine the radius of the circle, for which purpose we determine the lengths of the sides BD and CD . In the triangle SAB we have $\cos SAB = |AB|/2 |AS| = 0.25$, and we find, by the cosine law, that $|BD| = \sqrt{3/2} a$. Evidently, $|CD| = |BD|$. Now we have, in the triangle BCD , $\cos DBC = |BC|/2 |BD| = 1/\sqrt{6}$, whence it follows that $\sin DBC = \sqrt{5/6}$.

The radius of the circle circumscribed about the triangle BCD can be found by the formula $R = |CD|/2 \sin DBC = 3a/2 \sqrt{5}$. Designating the apex angle of the cone as φ , we have $\sin(\varphi/2) = R/|AD| = 3/2 \sqrt{5}$, whence $\varphi = 2 \arcsin(3/2 \sqrt{5})$.

Answer: $2 \arcsin(3/2 \sqrt{5})$. ▲

A pyramid is said to be *inscribed* into a cone (and a cone is said to be *circumscribed*

about a pyramid) if the vertex of the pyramid coincides with that of the cone and the base is inscribed into the base of the cone (Fig. 175).

Every lateral edge of the pyramid inscribed into a cone is the generating line of the cone and the altitude of the cone coincides with the altitude of the pyramid. It is easy to see that

to be possible to circumscribe a cone about a pyramid, it is necessary and sufficient that the lateral edges of the pyramid should be equal in length.

Problem 2. A pyramid $SABCD$ is inscribed into a cone, the base of the pyramid being a trapezoid $ABCD$. It is known that $\angle BAD = 60^\circ$, $|BC| = 3a$, $|AD| = 8a$ (BC and AD are the bases of the trapezoid), and the altitude SO of the pyramid is $7a$ in length. Find the lateral area of the cone.

△ The base of the pyramid, the trapezoid $ABCD$ (Fig. 175), is inscribed into the base of the cone and this means that the trapezoid is isosceles. Taking this into account, we find that $|AB| = (|AD| - |BC|)/2 \cos 60^\circ =$

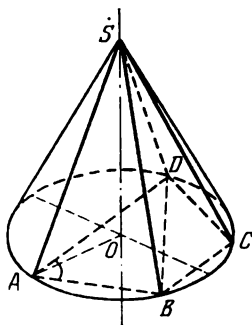


Fig. 175

5a. Applying the cosine law, we find from the triangle ABD that $|BD| = 7a$. Next we find the radius of the circle circumscribed about the trapezoid $ABCD$ by the formula $R = |BD|/2 \sin 60^\circ = 7a/\sqrt{3}$. Then we find the length of the generating line of the cone $l = |AS| = \sqrt{|AO|^2 + |SO|^2} = 14a/\sqrt{3}$ and its lateral area $S = \pi Rl = 98\pi a^2/3$.

Answer: $98\pi a^2/3$. ▲

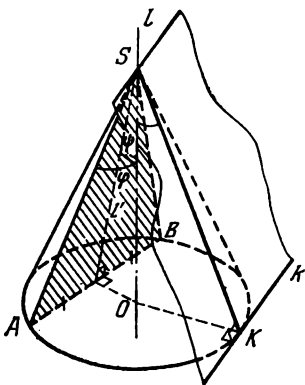


Fig. 176

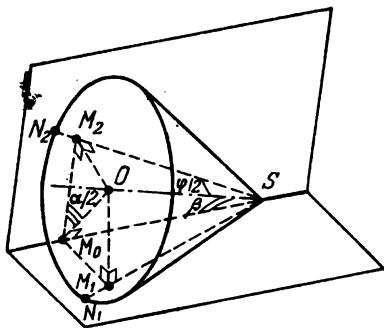


Fig. 177

Suppose the plane α passes through the vertex S of the cone and makes an angle ψ with its axis l . Assume that φ is the angle between the generating line and the axis of the cone. Recall that ψ is the angle between the axis l and its orthogonal projection l' onto the plane α . Therefore, any line in the plane α , passing through the vertex S and non-coinciding with l' , makes an angle larger than ψ with the axis l . Hence it follows that if $\psi > \varphi$ then the plane α has no common points, other than the vertex S , with the cone.

If $0 \leq \psi < \varphi$, then the section of the cone formed by the plane α is an isosceles triangle ($\triangle SAB$ in Fig. 176, $|SA| = |SB|$). For $0 < \psi < \varphi$ the plane passing through the axis of the cone and the midpoint of the segment AB is perpendicular to the cutting plane α . For $\psi = 0$ we have the axial section of the cone.

If $\psi = \varphi$, then the plane α possesses only one common generating line (SK in Fig. 176) with the lateral surface

of the cone, the generating line lying on the orthogonal projection of the axis l onto the plane α . In that case, the plane α is said to be *tangential* to the lateral surface of the cone. The plane passing through the axis of the cone and the generating line along which the plane α touches the lateral surface (the plane SOK in Fig. 176) is perpendicular to the tangent plane. The tangent plane cuts the plane of the base of the cone along the line touching the circle of the base of the cone (the line k in Fig. 176). Note that the line k and the generating line SK are perpendicular.

Problem 3. The faces of a dihedral angle equal to α touch the lateral surface of the cone. The angle between the edge of the dihedral angle and the axis of the cone is equal to β . Find the apex angle of the cone.

The vertex S of the cone evidently belongs to the edge of the given dihedral angle. Suppose O is the centre of the base of the cone. We designate $|SO| = a$ and the apex angle of the cone as φ . Assume that OM_0 is a perpendicular to the edge of the dihedral angle and OM_1 and OM_2 are perpendiculars to the planes of its faces (Fig. 177). Then, SN_1 and SN_2 are the generating lines along which the faces of the angle touch the lateral surface of the cone and $\angle OSM_1 = \angle OSM_2 = \varphi/2$. Furthermore, the angle $M_1M_0M_2$ is a plane dihedral angle, and, hence, $\angle M_1M_0M_2 = \alpha$. By the hypothesis, we also have $\angle OSM_0 = \beta$.

From the triangles OSM_1 and OSM_2 we have $|OM_1| = |OM_2| = a \sin(\varphi/2)$. Hence it follows that M_0O is the bisector of the angle $M_1M_0M_2$ (that is, the axis of the cone belongs to the bisector of the given dihedral angle). Taking this into account, we find, from the triangle OM_0M_1 , that $|OM_0| = |OM_1|/\sin \frac{\alpha}{2} = a \frac{\sin(\varphi/2)}{\sin(\alpha/2)}$. On the other hand, from the triangle OSM_0 we have $|OM_0| = a \sin \beta$. Thus it follows that $\frac{\sin(\varphi/2)}{\sin(\alpha/2)} = \sin \beta$, whence we find that $\sin(\varphi/2) = \sin \beta \sin(\alpha/2)$.

Answer: $2 \arcsin(\sin \beta \sin(\alpha/2))$. $\blacktriangle$

A pyramid is said to be *circumscribed* about a cone (and a cone is said to be *inscribed* into a pyramid) if the vertex of the pyramid coincides with that of the cone and the base of the pyramid is circumscribed about the base of the cone. It is evident that

to be possible to inscribe a cone into a pyramid, it is necessary and sufficient that a circle could be inscribed into the base of the pyramid and the foot of the altitude of the pyramid should be the centre of that circle.

In particular, a cone can be inscribed into any regular pyramid.

Every lateral face of a pyramid, circumscribed about a cone, touches its lateral surface along a generating line

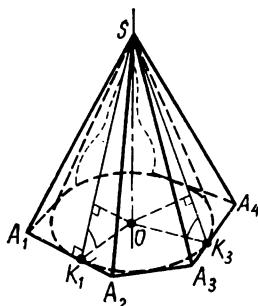


Fig. 178

which is at the same time the altitude of that lateral face and passes through the point of tangency of the corresponding side of the base of the pyramid and the circumference of the base of the cone (in Fig. 178 SK_1 is the altitude of the face SA_1A_2 and so on). Let us also note that the base dihedral angles of the pyramid circumscribed about the cone are of the same magnitude and are equal to the angle of inclination of

the generating line of the cone to the plane of its base ($\angle SK_1O$ and so on in Fig. 178).

Problem 4. A pyramid is circumscribed about a cone. Prove that the ratio between the volumes of the pyramid and the cone is equal to that between the areas of their lateral surfaces.

Δ Let us designate the radius of the base of the cone as r , its altitude as H , the length of the generating line as l , and the perimeter of the base of the pyramid as P (in Fig. 178 $|OK_1| = r$, $|SO| = H$, and $|SK_1| = l$). Since the base of the pyramid is circumscribed about the circumference of the base of the cone, the area of the base of the pyramid is equal to $\frac{1}{2} rP$. Consequently, the volume of the pyramid can

be expressed by the formula $V_p = \frac{1}{6} rPH$. We find the volume of the cone by the formula $V_c = \frac{1}{3} \pi r^2 H$. Hence we have $V_p : V_c = P/2\pi r$. The areas of the lateral surfaces of

the pyramid and the cone are expressed, respectively, as $S_p = \frac{1}{2} lP$, $S_c = \pi r l$. From this it follows that $S_p/S_c = P/2\pi r$. We see that the ratios V_p/V_c and S_p/S_c are equal. ▲

9.3. A Sphere

A *sphere* of radius R is a set of points of space at a positive distance R from the given point. In the Cartesian space,

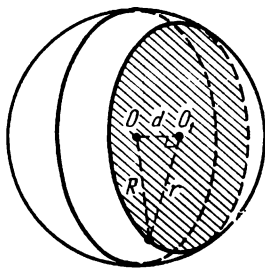


Fig. 179

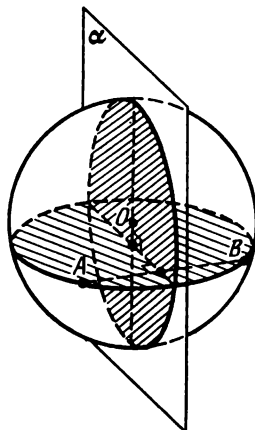


Fig. 180

a sphere with centre at $O(a; b; c)$ and radius R is defined by the equation

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = R^2.$$

A sphere is a solid of revolution. Rotation of a semicircle of radius R about its diameter results in a sphere of radius R .

A section of a sphere of radius R formed by a plane which is at a distance d from its centre, $0 \leq d < R$, is a circle of radius $r = \sqrt{R^2 - d^2}$. Sections equidistant from the centre of the sphere have equal radii.

A section of a sphere formed by a plane passing through its centre ($d = 0$) is known as the *major circle*. If the centre of the sphere does not lie in the plane of the section ($0 < d < R$), then the line passing through the centre of the sphere and the centre of the circumference of the section is perpendicular to the plane of the section (Fig. 179).

If points A and B belong to the sphere, then its centre, being a point equidistant from A and B , lies in the plane perpendicular to the segment AB and passing through its midpoint (the plane α in Fig. 180).

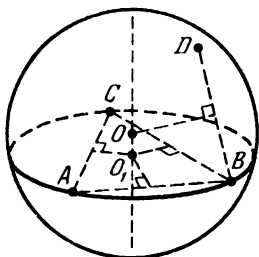


Fig. 181

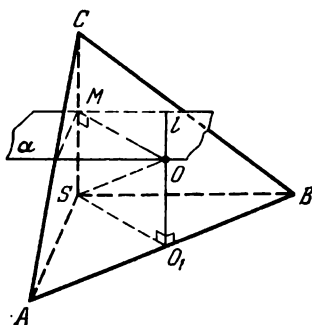


Fig. 182

Problem 1. Prove that a sphere can be drawn through any four points not lying in the same plane, and that the sphere is unique.

△ The centre of the sphere passing through the given points must belong to each of the planes drawn through the midpoints of the segments with the ends at the given points at right angles to those segments. It is known (Chapter 8, Problem 3 in Section I) that these planes meet at *one* point, we designate it as O (Fig. 181). That point is equidistant from all given points. The distance between the point O and any one of the given points is designated as R . A sphere with centre O and radius R passes through all the given points. It is clear from the discussion carried out that there can be only one sphere of this kind. ▲

A polyhedron is said to be *inscribed* into a sphere (and a sphere is said to be *circumscribed* about a polyhedron) if all the vertices of the polyhedron lie on the sphere.

It follows from the result obtained in Problem 1 that a sphere can be circumscribed about any tetrahedron. In a regular tetrahedron the centre is at a distance of $3H/4$ from each of its vertices, where H is the altitude of the tetrahedron. This means that the centre of a regular tetrahedron

is the centre of the circumscribed sphere. The radius of that sphere is equal to $3H/4 = a\sqrt{6}/4$, where a is the length of the edge of the tetrahedron.

The centre of a cube is at a distance of $a\sqrt{3}/2$ from each of its vertices, where a is the length of the edge of the cube. This means that the centre of the sphere circumscribed about a cube coincides with the centre of the cube, and the radius of the sphere is equal to $a\sqrt{3}/2$.

Problem 2. All the plane angles at the vertex S of the pyramid $SABC$ are right angles, $|SA| = a$, $|SB| = b$, and $|SC| = c$. Find the radius of the sphere circumscribed about the pyramid $SABC$.

Δ The centre of the sphere circumscribed about the triangle SAB is the midpoint O_1 of the hypotenuse AB (Fig. 182). We draw a straight line $l \perp (SAB)$ through the point O_1 . Note that $l \parallel (SC)$ since the lines l and SC are perpendicular to the plane SAB . Next we draw a plane α through the midpoint M of the segment SC at right angles to it. The point O of intersection of the plane α and the line l is equidistant from all the vertices of the pyramid and, consequently, is the centre of the sphere circumscribed about the pyramid. In the right triangle SOO_1 we have $|OO_1| = |MS| = \frac{1}{2}c$, $|SO_1| = \frac{1}{2}|AB| = \frac{1}{2}\sqrt{a^2+b^2}$, and, consequently, the radius of the circumscribed sphere is equal to $|SO| = \sqrt{|OO_1|^2 + |SO_1|^2} = \frac{1}{2}\sqrt{a^2+b^2+c^2}$.

Answer: $\frac{1}{2}\sqrt{a^2+b^2+c^2}$. $\blacktriangle$

The following assertions hold true for a pyramid and a prism.

(a) *To be possible to circumscribe a sphere about a pyramid, it is necessary and sufficient that a circle could be drawn about the base of the pyramid* (Fig. 183).

The proof of this assertion is analogous to that of the statement that a sphere can be circumscribed about any tetrahedron (Problem 1 of the present section, Problem 3 in Chapter 8, Section I).

(b) *To be possible to circumscribe a sphere about a prism, it is necessary and sufficient that the prism should be right-angled and that a circle could be circumscribed about its base* (Fig. 184, see Problem 9 in Section I).

The centre of a sphere circumscribed about a pyramid lies on the perpendicular to the plane of the base drawn through the centre of the circle circumscribed about the base. The centre of a sphere circumscribed about a prism is the mid-point of the segment connecting the centres of the circles circumscribed about the bases of the prism.

It follows from the statements cited that *a sphere can be circumscribed about any regular pyramid and about any regular prism.*

Let us discuss the notion of a plane tangent to a sphere. If the distance from the centre of the sphere to the plane is

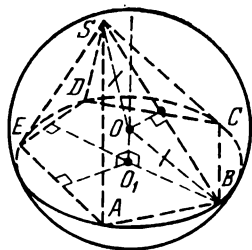


Fig. 183

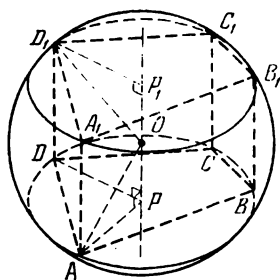


Fig. 184

equal to the radius of the sphere ($d = R$), then the sphere and the plane have a point in common. Such a plane is said to be *tangent* to the sphere. The radius drawn to the point of tangency is perpendicular to the tangent plane.

If a sphere touches two planes, then the distance from its centre to each of the planes is equal to the radius of the sphere. If the tangent planes are parallel, then the points of tangency are the end points of the diameter of the sphere which is perpendicular to both planes (Fig. 185). If the tangent planes intersect, then the centre of the sphere belongs to the bisector of the dihedral angle between the planes (Fig. 186). The points of tangency of the sphere and the planes belong to the faces of that angle.

A sphere is said to be *inscribed* into a polyhedral angle if it touches each of its faces.

Problem 3. Prove that a sphere can be inscribed into any trihedral angle.

▲ The centre of the sphere inscribed into a trihedral angle must belong to the bisectors of its dihedral angles. It is known (see 8.3, Problem 7) that these bisectors inter-

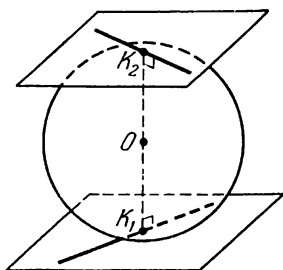


Fig. 185

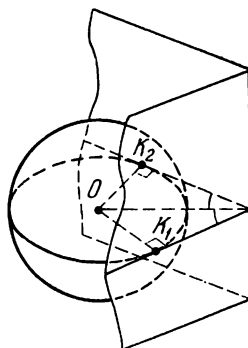


Fig. 186

sect in the interior of the trihedral angle along a ray (the ray l in Fig. 187). Suppose O is a point lying on that ray and not coinciding with the vertex of the angle and R is the

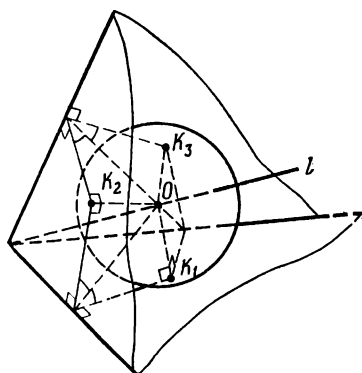


Fig. 187

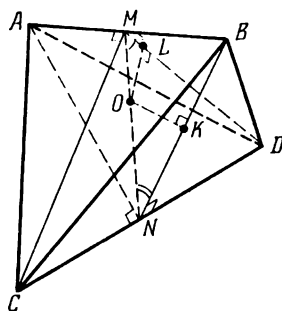


Fig. 188

distance from the point to the planes of the faces. The sphere with centre at O and radius R touches all the faces of the angle, that is, is inscribed into the given trihedral angle. ▲

Note that the ray l , its origin exclusive, is a set of centres of the spheres inscribed into the given trihedral angle.

A sphere is said to be *inscribed* into a polyhedron (and a polyhedron is said to be *circumscribed* about a sphere) if it touches all its faces.

The centre of the inscribed sphere is a common point of the bisectors of all the interior dihedral angles of the polyhedron. It follows that if the inscribed sphere does exist then it is unique.

It is known (see 8.5, Problem 8) that the bisectors of the dihedral angles of a tetrahedron meet at one point. The sphere with the centre at that point and the radius equal to the distance from that point to the plane of some face of the tetrahedron touches all the faces of the tetrahedron. This means that

a sphere can be inscribed into any tetrahedron.

Problem 4. In the tetrahedron $ABCD$ we have $|AB| = 4$ cm, $|CD| = 12$ cm, and $|AC| = |BC| = |AD| = |BD| = 3\sqrt{6}$ cm. Find the radius of the sphere inscribed into this tetrahedron.

$\triangle$ Suppose M and N are the midpoints of the edges AB and CD (Fig. 188). The triangles ACB and ADB being isosceles, we have $[CM] \perp [AB]$ and $[DM] \perp [AB]$. This means that the angle CMD is a plane dihedral angle at the edge AB . We find from the right triangles AMC and AMD that $|CM| = |DM| = 5\sqrt{2}$ cm. It is clear that the triangle CMD is isosceles and, therefore, MN is the bisector of the angle CMD . But then the half-plane ANB (with the boundary AB) is the bisector of the dihedral angle at the edge AB . We establish by analogy that the half-plane CMD (with the boundary CD) is the bisector of the dihedral angle at the edge CD . The centre of the sphere inscribed into the tetrahedron lies on the segment MN , which is the line of intersection of these two bisectors.

From the right triangle BND we find that $|BN| = 3\sqrt{2}$ cm, and from the right triangle NMB we find that $|MN| = \sqrt{14}$ cm.

Suppose O is the centre of the inscribed sphere and r is its radius. We draw perpendiculars OK and OL to the faces BCD and ABD . Then we have $|OK| = |OL| = r$. We designate $\angle BNM = \alpha$, and $\angle DMN = \beta$. From the

right triangles NMB and NMD we find, respectively, that $\sin \alpha = \sqrt{2}/3$ and $\sin \beta = 3/\sqrt{5}$. From the right triangle OKN we find $|ON| = \frac{|OK|}{\sin \alpha} = \frac{3}{\sqrt{2}} r$. Then, $|OM| = |MN| - |ON| = \sqrt{14} - \frac{3}{\sqrt{2}} r$ and now we get, from the right triangle OLM , the equation $\left(\sqrt{14} - \frac{3}{\sqrt{2}} r \right) \times \frac{3\sqrt{2}}{5} = r$.

From this we find that $r = 3/\sqrt{7}$ cm. $\blacktriangle$

The centre of a regular tetrahedron is at a distance of $H/4$ from its every face, where H is the altitude of the tetrahedron. Consequently, the centre of the sphere inscribed into a regular tetrahedron coincides with the centre of the tetrahedron. The radius of the sphere can be calculated by the formula $r = \frac{1}{4} H = \frac{a\sqrt{6}}{12}$, where a is the length of the edge of the tetrahedron.

In a cube, the centre of the inscribed sphere coincides with the centre of the cube. The radius of the sphere is $a/2$, where a is the length of the edge of the cube.

It is not in every polyhedron that the bisectors of the dihedral angles meet at one point and this means that a sphere cannot be inscribed into every polyhedron.

Problem 5. Prove that if a circle can be inscribed into the base of a pyramid and the foot of the altitude of the pyramid is the centre of that circle, then a sphere can be inscribed into the pyramid.

$\triangle$ Suppose $L_1, L_2, \dots$ are the points of tangency of the circle inscribed into the base of the pyramid and the sides of the base and P is the centre of the inscribed circle (Fig. 189). The right triangles $SPL_1, SPL_2, \dots$ are congruent and have a leg SP in common. Consequently, the bisectors of the angles at the vertices $L_1, L_2, \dots$ cut that leg at the same point O . We drop perpendiculars $OK_1, OK_2, \dots$ from the point O to the hypotenuses $SL_1, SL_2, \dots$. The plane SL_1P is perpendicular to the plane of the face SA_1A_2 and, therefore, $(OK_1) \perp (SA_1A_2)$. Similarly we have $(OK_2) \perp (SA_2A_3)$, $(OK_3) \perp (SA_3A_4)$, and so on. Taking into account that $|OP| = |OK_1| = |OK_2| = \dots$, we see that the point O is equidistant from the planes of all the faces of the pyramid. This means that the sphere with

centre O and radius $r = |OP|$ touches all the faces, that is, it is inscribed into the given pyramid. $\blacktriangle$

It follows from the statement we have proved that *a sphere can be inscribed into any regular pyramid.*

A sphere can also be inscribed into pyramids not satisfying the hypothesis of Problem 5.

Problem 6. The base of the pyramid $SABCD$ is a square $ABCD$, the edge SA is perpendicular to the plane of the base;

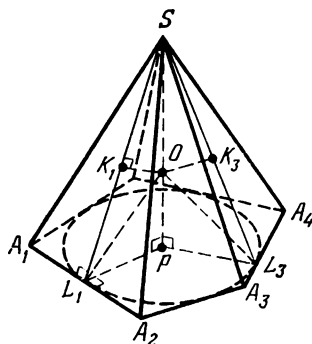


Fig. 189

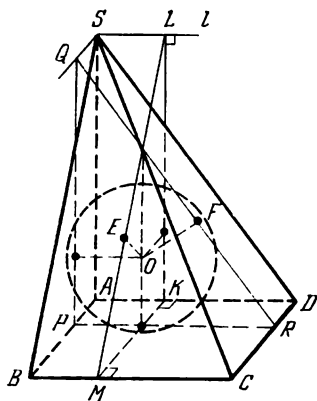


Fig. 190

$|AB| = 3$ cm, and $|SA| = 4$ cm. Prove that a sphere can be inscribed into the pyramid and find the radius of the sphere.

$\triangle$ We shall first obtain some necessary conditions concerning the positions of the centre of the sphere and of its radius. For that purpose we shall assume that the sphere inscribed into the pyramid does exist and that O is its centre and r its radius. We draw a plane through the centre O at right angles to the line AD (the plane KLM in Fig. 190). Suppose l is the line of intersection of the planes SAD and SBC . Since $(BC) \parallel (AD)$, it follows that $l \parallel (AD)$ as well. The plane we have drawn is perpendicular to each of the lines, AD , BC , and l , and, therefore, the angles LKM , KML , and KLM , respectively, are plane dihedral angles at the edges AD , BC , and l (the latter is the dihedral angle between the faces SBC and SAD). Since the sphere touches

the faces of these dihedral angles, its section formed by the plane KLM (the *major circle*) must be inscribed into the triangle KLM . It is easy to find the radius of that circle. Indeed, the planes KLM and ASB are parallel and, hence, the triangles KLM and ASB are congruent and $|KL| = |AS| = 4$ cm, $|KM| = |AB| = 3$ cm. The radius of the circle inscribed into the right triangle KLM can be found by the formula $r = 2S/P$, where S and P are, respectively, the area and the perimeter of that triangle, and we find that $r = 1$ cm. This is the radius of the inscribed sphere. The same result will be obtained if we consider a section of the pyramid and the sphere formed by a plane drawn through the centre O at right angles to the edge AB (the plane PQR in Fig. 190). The triangle PQR is congruent to the triangle KLM ($|PQ| = 4$ cm and $|PR| = 3$ cm), and the radius of the circle inscribed into it (the *major circle* of the sphere) is equal to 1 cm.

Let us prove now that the inscribed sphere does exist. We take a point O , 1 cm from each face of the angle, in the interior of the trihedral angle $ASBD$ (with vertex A). The sphere with centre at O and radius of 1 cm touches the faces SAD , SAB , and $ABCD$ of the pyramid. We again consider the section KLM drawn through the centre of the sphere. The point O is at a distance of 1 cm from the sides KL and KM of the triangle KLM and this means that it coincides with the centre of the circle inscribed into the triangle. Therefore, if $[OE] \perp [LM]$, then $|OE| = 1$ cm. From the fact that $(KLM) \perp (SBC)$ it follows that $[OE] \perp (SBC)$. Hence, the point O is at a distance of 1 cm from the plane SBC , and the sphere with centre at O and radius of 1 cm touches the plane. It can be proved by analogy that the sphere also touches the face SCD . Thus we see that the inscribed sphere exists and its radius is equal to 1 cm.

Answer: 1 cm. ▲

The following statement holds true for a prism:

to be possible to inscribe a sphere into a prism, it is necessary and sufficient that a circle could be inscribed into the perpendicular section of the prism and that the altitude of the prism should be equal to the diameter of that circle (see Problem 45 in Section II).

The diameter of the sphere inscribed into a prism is equal to the diameter of the circle inscribed into the perpendicular

section of the prism (the section $MNPQ$ in Fig. 191). The centre of the sphere lies on the straight line (PP_1 in Fig. 191) drawn parallel to the lateral edges through the centre of the circle inscribed into the perpendicular section and is the midpoint of the segment intercepted on that line by the bases of the prism (in Fig. 191 the centre O is the midpoint of the segment PP_1 , and the altitude QQ_1 of the prism is equal to

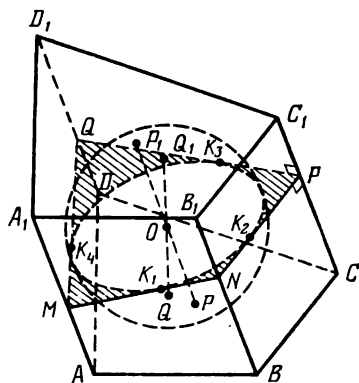


Fig. 191

the diameter of the sphere). The points of tangency of the sphere and the lateral faces of the prism belong to the perpendicular section drawn through the centre of the sphere.

A sphere can be inscribed into a regular prism if and only if its altitude is equal to the diameter of the circle inscribed into the base.

Problem 7. A right parallelepiped with diagonals $\sqrt{10}$ cm and 4 cm in length is circumscribed about a sphere. Find the radius of the sphere.

△ The altitude of the given parallelepiped is equal to the diameter of the inscribed sphere. We designate the radius of the sphere as R , then $H = 2R$.

Suppose AC and BD are the longer and shorter diagonals, respectively, of the base of the parallelepiped (Fig. 192), and we designate $|AC| = d_1$ and $|BD| = d_2$. The parallelepiped being right-angled, AC_1 and BD_1 are its longer and shorter diagonals, i.e. $|AC_1| = 4$ cm, $|BD_1| = \sqrt{10}$ cm.

From the right triangles ACC_1 and BDD_1 we get

$$d_1^2 + 4R^2 = 16, \quad d_2^2 + 4R^2 = 10. \quad (1)$$

Let us consider a perpendicular section of the parallelepiped passing through the centre O of the sphere. It is a parallelogram $A_2B_2C_2D_2$ congruent to the base and, in

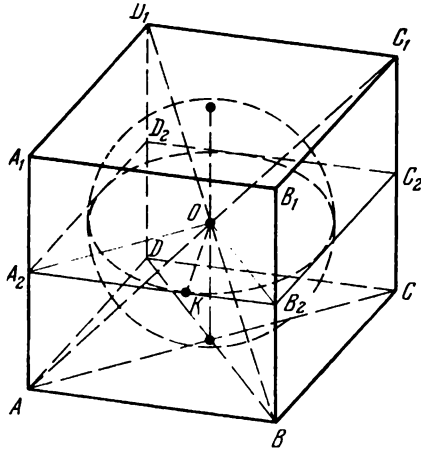


Fig. 192

particular, $|A_2O| = \frac{1}{2} d_1$ and $|B_2O| = \frac{1}{2} d_2$. Since the parallelogram is circumscribed about a circle, it is a rhombus and, hence, $\angle A_2OB_2 = 90^\circ$. Assuming $[OK] \perp [A_2B_2]$, we get $|OK| = R$, and from the right triangle A_2OB_2 we obtain $|OK| \cdot |A_2B_2| = |A_2O| \cdot |B_2O|$, and, consequently, $2R \sqrt{d_1^2 + d_2^2} = d_1 d_2$. From this equation, making use of equation (1), we get $3R^4 - 13R^2 + 10 = 0$, whence $R_1^2 = 10/3$ and $R_2^2 = 1$. The value $R_1^2 = 10/3$ is, evidently, extraneous ($d_2^2 + 4R_1^2 > 10$) and this means $R^2 = 1$, $R = 1$ cm. $\blacktriangle$

Problem 8. A polyhedron of volume V and the area of the surface S is circumscribed about a sphere of radius R . Prove that $R = 3V/S$.

Sol. Let us represent the circumscribed polyhedron as a union of pyramids whose vertices are at the centre of the sphere and whose bases are the faces of the polyhedron (note

that the circumscribed polyhedron is convex). We designate the areas of the faces of the polyhedron as $S_1, S_2, \dots, S_n$ and, by the hypothesis, $S_1 + S_2 + \dots + S_n = S$. The radius drawn to the point of tangency of the j th face (of the area S_j) and the sphere is the altitude of the pyramid for which that face serves as the base. This means that the volume of the j th pyramid is equal to $R \cdot S_j / 3$. The sum of the volumes of all the pyramids is equal to the volume of the polyhedron:

$$\frac{1}{3} RS_1 + \frac{1}{3} RS_2 + \dots + \frac{1}{3} RS_n = V.$$

From this we obtain $RS/3 = V$, i.e. $R = 3V/S$. ▲

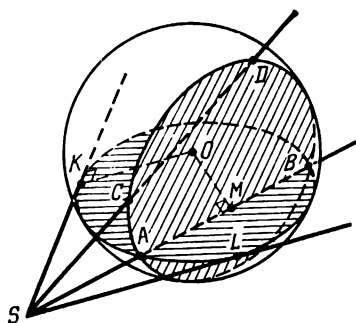


Fig. 193

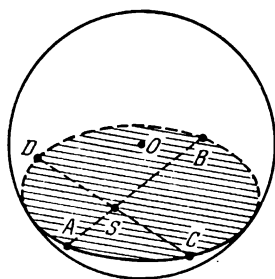


Fig. 194

A straight line and a sphere have points in common if and only if the distance d from the centre of the sphere to that line does not exceed the radius R of the sphere.

If $0 \leq d < R$, the line intersects the sphere at two points. A line segment with the ends at those points is called a *chord* and the straight line is called a *secant* (in Fig. 193 (SA) is a secant and [AB] is a chord).

If $d = R$, then the line and the sphere have one point in common, which is the foot of the perpendicular dropped from the centre of the sphere to that line. In that case, the line is called a *tangent* to the sphere (in Fig. 193 (SK) is a tangent). Any tangent line lies in the tangent plane drawn through the point of tangency of the line and the sphere. The converse is also true: any line lying in the tangent plane and

passing through the point of tangency is a tangent to the sphere.

Secant and tangent lines to a sphere possess the same properties as secants and tangents to a circle. Here are some of the properties.

(a) *A perpendicular dropped from the centre of a sphere to a secant line passes through the midpoint of a chord* (in Fig. 193 $|AM| = |MB|$, $[OM] \perp [AB]$).

(b) *A plane passing through the centre of a sphere at right angles to a chord passes through the midpoint of the chord.*

It is useful to formulate the latter assertion in the following equivalent form:

(b') *A plane drawn through the midpoint of a chord at right angles to that chord passes through the centre of the sphere.*

(c) *If two lines meet at a point S and touch the sphere at points K and L respectively* (Fig. 193), *then* $|SK| = |SL|$ *(theorem on tangents).*

(d) *If two lines meet at a point S and one of them touches the sphere at a point K and the other cuts the sphere at points A and B* (Fig. 193), *then* $|SA| \cdot |SB| = |SK|^2$ *(the secant-tangent theorem).*

(e) *If two lines meet at a point S and cut the sphere at points A, B , and C, D respectively* (Figs. 193-194), *then* $|SA| \times |SB| = |SC| \cdot |SD|$ *(the point S may be either in the exterior or in the interior of the sphere; in the latter case the assertion cited is sometimes called a theorem on intersecting chords).*

Property (b) (and (b')) is a consequence of Property (a). To prove properties (a), (c), (d), (e), a plane should be drawn, respectively: through the secant and the centre of the sphere in case (a), through the tangents SK and SL in case (c), through the tangent and the secant in case (d), and through the lines AB and CD in case (e), and then use should be made of the corresponding properties of chords, secants, and tangents to a circle.

Problem 9. Prove that there is a sphere touching all the edges of a regular triangular pyramid and find its radius if the side of the base of the pyramid is a in length and the lateral edge is b in length.

Sol. If the sphere touches the sides of the base ABC (Fig. 195), then its section formed by the plane of the base is a circle inscribed into the triangle ABC . That circle

touches the sides of the triangle (which is regular) at their midpoints K_1 , K_2 , and K_3 . The centre of the sphere must be equidistant from those points and, consequently, must lie on the perpendicular to the plane of the base passing through its centre P , that is, on the line SP .

If the sphere touches the edge AS at a point L_1 , then its centre lies in the plane drawn through the point L_1 at right angles to the edge AS . That plane cuts the plane ASP along a straight line perpendicular to the edge AS . Furthermore, in accordance with the property of tangents, $|AL_1| = |AK_1| = a/2$, where a is the length of the side of the base.

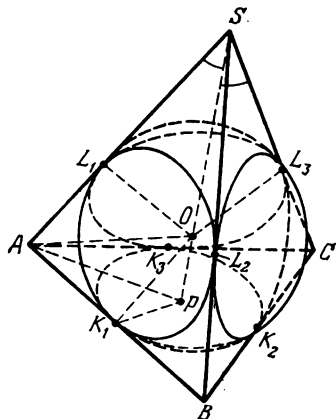


Fig. 195

It follows from the analysis carried out that the centre of the sphere should be the point of intersection of the line SP and the line drawn in the plane ASP at right angles to the edge AS through the point L_1 of that edge such that $|AL_1| = a/2$. Let us designate the point of intersection

of those lines as O and prove that a sphere with centre at O and radius $r = |OK_1|$ touches the edges of the pyramid. First, it is evident that $|OK_1| = |OK_2| = |OK_3|$, that is, the sphere touches all the sides of the base. Second, it follows from the congruency of the right triangles OL_1A and OK_1A (a leg and the hypotenuse of one of them are equal to a leg and the hypotenuse of the other), that $|OL_1| = |OK_1| = r$, and this means that the sphere touches the edge AS as well. If $[OL_2] \perp [BS]$ and $[OL_3] \perp [CS]$, it follows from the congruency of the triangles SOL_1 , SOL_2 , and SOL_3 (the hypotenuse and an acute angle of one of them are equal to the hypotenuse and an acute angle of the other), that $|OL_2| = |OL_3| = |OL_1| = r$. This means that the sphere touches the edges BS and CS as well. We have thus proved that the sphere touching all the edges of the pyramid does exist.

Let us find the radius of the sphere from the similitude

of the triangles SL_1O and SPA . We have $r/|AP| = |SL_1|/|SP|$, whence, taking into account that

$$|SL_1| = b - \frac{a}{2}, \quad |AP| = \frac{a}{\sqrt{3}}, \quad |SP| = \sqrt{b^2 - \frac{a^2}{3}},$$

we find that $r = a(2b - a)/2\sqrt{3b^2 - a^2}$. ▲

9.4. Combinations of a Sphere, a Cone, and a Cylinder

A sphere is said to be *circumscribed* about a cylinder (a cone) if the circumferences of its bases (the circumference of its base and a vertex) belong to the sphere (Figs. 196, 197).

A sphere can be circumscribed about any cylinder and about any cone. The centre of the circumscribed sphere

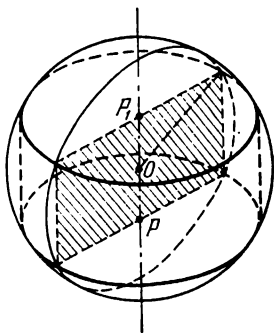


Fig. 196

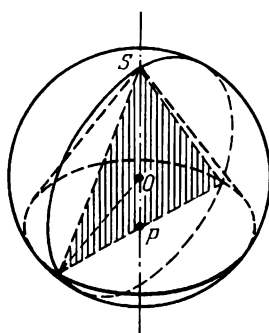


Fig. 197

coincides with the centre of the circle circumscribed about the axial section of the cylinder or the cone respectively. The radius of the sphere is equal to the radius of that circle.

A sphere is said to be *inscribed* into a cylinder (and the cylinder is said to be *circumscribed* about a sphere) if the sphere touches every generating line and both bases of the cylinder (Fig. 198).

A sphere can be inscribed into a cylinder if and only if the altitude of the cylinder is equal to the diameter of the base. The centre of the inscribed sphere is the midpoint of the segment connecting the centres of the bases of the cylin-

der. The sphere touches every generating line at its midpoint, and the set of points of tangency is a circle, which is the section of the lateral surface of the cylinder formed by the plane parallel to the bases and passing through the midpoint of the altitude.

A sphere is said to be *inscribed* in a cone (and a cone is said to be *circumscribed* about a sphere) if the sphere touches the base of the cone and every generating line (Fig. 199).

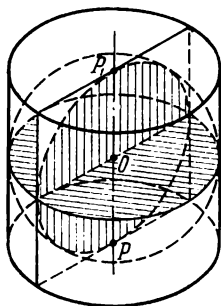


Fig. 198

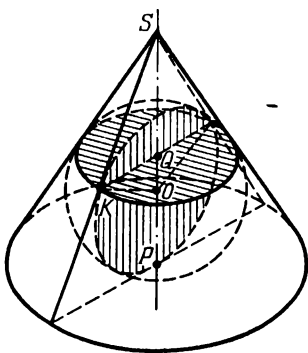


Fig. 199

A sphere can be inscribed into any cone. Its centre coincides with that of the circle inscribed into the axial section of the cone, the radius of the sphere is equal to the radius of the circle. The set of points of tangency of the sphere and the generating surface of the cone is a circle, which is the section of the lateral surface of the cone formed by the plane parallel to the base of the cone (in Fig. 199 the point Q is the centre of the circle).

Problem 1. A sphere is inscribed into a cone. The circle of tangency of the sphere and the lateral surface of the cone is the circumference of the base of the cylinder inscribed into the sphere. The area of the sphere constitutes $\frac{4}{9}$ of the area of the surface of the cone. Find the ratio between the volumes of the cylinder and the cone.

△ The axis of the cone is at the same time the axis of the cylinder and the centre of the sphere lies on that axis. Let us consider the sections of the three given solids formed by the plane passing through the axis of the cone (Fig. 200).

The axial section of the cone is a triangle ASB , the section of the sphere as a circle, with centre at O , inscribed into the triangle ASB (K , L , and P being the points of tangency), the section of the cylinder (axial) is a rectangle inscribed into that circle, the line segment LK serves as a side of that rectangle.

Suppose l and r_c is the length of the generating line and the radius of the base of the cone, respectively, r is the radius of the sphere, r_{cyl} and h are the radius of the base and the altitude of the cylinder, respectively. We have (Fig. 200):

$|SB| = l$, $|PB| = r_c$,
 $|KB| = |PB| = r_c$ (by the property of tangents), $|SK| = l - r_c$,
 $|SP| = \sqrt{l^2 - r_c^2}$,
 $|OK| = r$. From the similitude of the right triangles SKO and SPB we find that

$$r - r_c \frac{l - r_c}{\sqrt{l^2 - r_c^2}} = r_c \sqrt{\frac{l - r_c}{l + r_c}}.$$

We determine the areas of the surfaces of the cone and the sphere by the respective formulas: $S_c = \pi r_c (l + r_c)$ and $S = 4\pi r^2 = 4\pi r_c^2 \frac{l - r_c}{l + r_c}$. By the hypothesis, $S/S_c = 4/9$, whence we obtain an equation $l^2 - 7r_c l + 10r_c^2 = 0$ and find that either $l = 5r_c$ or $l = 2r_c$.

From the similitude of the triangles SQ_1K and SPB we have $|Q_1K| = |SK| \cdot \frac{|PB|}{|SB|}$. Bearing in mind that $|Q_1K| = r_{cyl}$, we get $r_{cyl} = r_c(l - r_c)/l$. From the right triangle OQ_1K we find that $|OQ_1| = \sqrt{|OK|^2 - |Q_1K|^2} = \frac{r_c^2}{l} \sqrt{\frac{l - r_c}{l + r_c}}$. For the altitude of the cylinder we have the equation $h = 2|OQ_1|$, and, hence, $h = \frac{2r_c^2}{l} \sqrt{\frac{l - r_c}{l + r_c}}$. Then we find the volumes of the cone and the cylinder:

$$V_c = \frac{1}{3} \pi r_c^2 \sqrt{l^2 - r_c^2}, \quad V_{cyl} = \frac{2\pi r_c^4 (l - r_c)^{5/2}}{l^3 (l + r_c)^{1/2}},$$

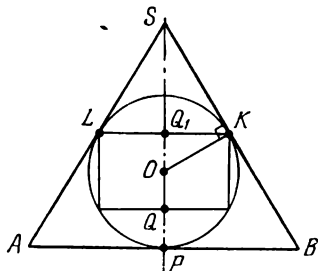


Fig. 200

and the ratio between those volumes: $V_{\text{cyl}}/V_c = \frac{6r_c^2(l-r_c)^2}{l^3(l+r_c)}$. Substituting here the values $l = 5r_c$ and $l = 2r_c$, we find, respectively, that either $V_{\text{cyl}}/V_c = 16/125$ or $V_{\text{cyl}}/V_c = 1/4$.

Answer: $16/125$ or $1/4$. ▲

Let us now consider the mutual positions of two spheres with centres at O_1 and O_2 and radii r_1 and r_2 . Assume that $d = |O_1O_2|$ is the distance between the centres of the spheres.

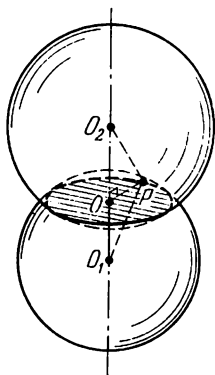


Fig. 201

If $d > r_1 + r_2$ or $d < |r_1 - r_2|$, then the spheres have no points in common.

If $|r_1 - r_2| < d < r_1 + r_2$, then the intersection of the spheres is a circle, whose plane is perpendicular to the line O_1O_2 and whose centre lies on that line (Fig. 201).

Now if $d = r_1 + r_2$ or $d = |r_1 - r_2|$ (in the latter case $r_1 \neq r_2$), then the spheres have only one point in common

and are said to be *tangential*. The common point of the spheres lies on the line O_1O_2 and is called the *point of tangency*. If $d = r_1 + r_2$, then the tangency is *external* (in Fig. 202a these are the spheres with centres O_1 and O_2), now if $d = |r_1 - r_2|$, then the tangency is said to be *internal* (in Fig. 202b these are the spheres with centres O_1 and O'_2). Shown in Fig. 202b is a section of the tangent spheres formed by the plane passing through the line of centres O_1O_2 . The tangent spheres have a common tangent plane at their point of tangency.

Problem 2. Three spheres, of radius r each, touch a plane α and each sphere touches the other two spheres. Find the radius of the fourth sphere touching the plane α and of each of the three given spheres.

△ Suppose O_1 , O_2 , and O_3 are the centres of the given spheres and O is the centre of the fourth sphere. We have $|O_1O_2| = |O_2O_3| = |O_3O_1| = 2r$. There is, evidently, only an external tangency between the fourth sphere and

the given spheres and, therefore, $|OO_1| = |OO_2| = |OO_3| = r + x$, where x is the radius of the fourth sphere. Then the point O is equidistant from the points $O_1, O_2,$

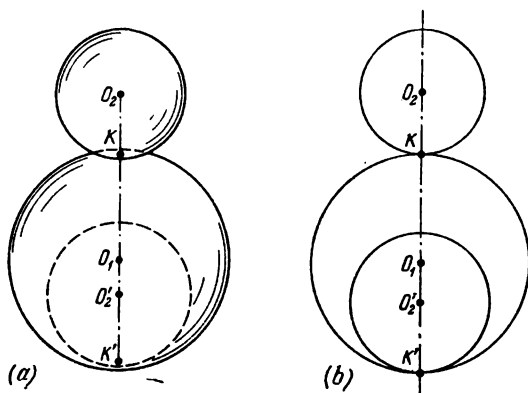


Fig. 202

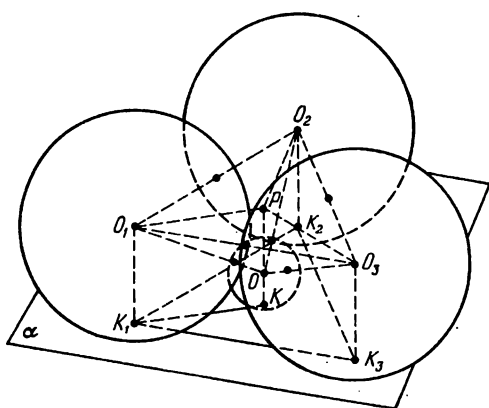


Fig. 203

and O_3 and, therefore, lies on the perpendicular dropped to the plane $O_1O_2O_3$ drawn through the centre of the regular triangle $O_1O_2O_3$. The point O is at the distance x , $|OK| = x$, from the plane α (in Fig. 203 this plane is defined by the points of tangency $K_1, K_2,$ and K_3). Each of the points

O_1 , O_2 , and O_3 is at the distance r from the plane α . Since $[PK] \perp \alpha$, it follows that $|PK| = r$ as well, and $|PO| = |r - x|$. Taking into account that $|PO_1| = 2r\sqrt{3}/3$, we have, from the right triangle $OP O_1$, that $|OO_1|^2 = |O_1P|^2 + |PO|^2$ and $(r + x)^2 = \frac{4}{3}r^2 + (r - x)^2$. From this equation we find that $x = r/3$. ▲

PROBLEMS OF SECTION I

1. A segment with the vertex angle α is cut out of a circle and rolled into a funnel. At what α is the funnel of the greatest capacity?

2. A regular triangle with the side a in length is rotated about its axis which is parallel to its altitude and is at the distance $b > a/2$ from it. Find the volume of the solid of revolution.

3. The side of the base of a regular triangular prism is 1.6 dm long and the lateral edge is 1.2 dm long. Four vertices of the prism lie on the circle making the base of the cone and the other two vertices lie on its lateral surface. Find the volume of the cone.

4. The edge of the cube $ABCD A_1 B_1 C_1 D_1$ is a in length. Find the radius of the sphere passing through the vertices B and C_1 , the midpoint of the edge AD and the centre of the face $CC_1 D_1 D$.

5. A chord AB and a diameter CD are drawn in a sphere of radius r . It is known that $|AB| = r$, $(AB, CD) = \pi/3$, $|AC| = \sqrt{2}r$, and $|BC| > |AC|$. Find the length of the segment BD .

6. A truncated cone is circumscribed about a sphere of radius R . The base and each generating line of the cone touch the sphere, and the generating line is a in length. Find the volume of the cone.

7. A regular quadrangular pyramid is inscribed into a sphere of radius R . The dihedral angle of the pyramid at its lateral edge is 120° . The second sphere touches the first one and the lateral faces of the pyramid. Find the radius of the second sphere.

8. Two touching spheres, of radius r each, are inscribed into a dihedral angle 2α in magnitude. The third sphere, inscribed into the same angle, touches the first two. Find the radius of the third sphere.

9. Prove that to be possible to circumscribe a sphere about a prism, it is necessary and sufficient that the prism be right-angled and that a circle could be circumscribed about its base.

10. The length of the lateral edge of a regular triangular pyramid is equal to the altitude of the base of the pyramid and is equal to a . A sphere touches the plane of the base of the pyramid, its two lateral edges, and the extension of the third lateral edge beyond the vertex of the pyramid. Find the radius of the sphere.

11. In the regular pyramid $SABC$ the side of the base ABC is a in length and the lateral edge is $2a/3$ long. A sphere touches all the edges of that pyramid. Find the area of the part of the sphere lying outside the pyramid.

12. Two spheres S_1 and S_2 touch each other and each of them touches both faces α and β of the dihedral angle 2φ . The sphere S_1

touches the face α at a point A and the sphere S_2 touches the face β at a point B . In what ratio is the segment AB divided by the spheres?

13. The altitude of the cylinder is equal to $3r$. Three spheres of radius r are located inside the cylinder so that each sphere touches the other two spheres and the lateral surface of the cylinder (that is, has a point in common with that surface). Two spheres touch the lower base of the cylinder and the third sphere touches the upper base. Find the radius of the cylinder.

PROBLEMS OF SECTION II

1. It is required to manufacture an open cylindrical tank of volume V . How long should the radius and the altitude of the tank be for the area of its surface to be the least?

2. The edge of the cube is a long. Two vertices of the cube are the centres of the bases of the cylinder, the other vertices lying on the lateral surface of the cylinder. Find the volume of the cylinder.

3. The volume of the tetrahedron $ABCD$ is V . The edges AB and CD of the tetrahedron are perpendicular and are equal in length, and the segment MN connecting their midpoints is perpendicular to the edges. The line MN is the axis of the cylinder the circumferences of whose bases have a common point with each face of the tetrahedron. Find the greatest possible volume of such a cylinder.

4. All the vertices of the regular pyramid $SABCD$ lie on the lateral surface of the cylinder whose axis is perpendicular to the plane SAB . Find the radius of the cylinder if $|AB| = a$.

5. In the regular quadrangular pyramid $SABCD$ the side of the base is a in length and the lateral edge is $\frac{5}{2}a$ long. One base of the cylinder lies in the plane SAB and the other is inscribed into the section of the pyramid. Find the area of the lateral surface of the cylinder.

6. In the regular prism $ABCA_1B_1C_1$ the length of each edge is a . The vertices A and A_1 lie on the lateral surface of the cylinder, and the plane BB_1C_1C touches that surface. The axis of the cylinder is parallel to the line B_1C . Find the radius of the cylinder.

7. The vertices of the rectangle lie on the lateral surface of the cylinder. Prove that two parallel sides of the rectangle are perpendicular to the axis of the cylinder.

8. The vertex A of the regular prism $ABCA_1B_1C_1$ coincides with the centre of one of the bases of the cylinder, the vertices B_1 and C_1 lie on the circumference of the other base, and the vertices A_1 , B and C lie on the lateral surface of the cylinder. Determine the ratio of the volumes of the cylinder and the prism.

9. One vertex of the regular tetrahedron lies on the axis of the cylinder, and the other vertices lie on the lateral surface of that cylinder. Find the length of the edge of the tetrahedron if the radius of the cylinder is R .

10. The straight line touches the lateral surface of the cylinder when it lies in the tangent plane and has a common point with the surface of the cylinder. The edge of the cube $ABCD A_1 B_1 C_1 D_1$ is a long. The cylinder whose axis is parallel to the line BD_1 is so positioned

that the lines AA_1 , BC , and C_1D_1 touch its lateral surface. Find the radius of the cylinder (find all solutions).

11. The area of the section of the cone formed by the plane making an angle of 30° with the axis of the cone is equal to the area of the axial section. Find the apex angle of the cone.

12. A regular triangular pyramid is inscribed into a cone. The area of the lateral face of the pyramid is equal to the area of the axial section of the cone. Find the ratio between the areas of the lateral surfaces of the cone and the pyramid.

13. The apex angle of the cone is 2α , and the length of the generating line is l . Find the greatest possible area of the section of the cone formed by the plane passing through the vertex. Find the angle between the axis and the plane of the section of the greatest area.

14. The points $A(1; 2; -2)$, $B(4; 2; -2)$, and $C(3; 4; -2)$ lie on the circumference of the base of the cone whose altitude is equal to 3. Find the coordinates of the vertex of the cone and the area of the section of the cone formed by the plane $z = 0$.

15. The points $A(1; 2; -1)$, $B(-2; 2; -1)$, and $C(-2; 2; 3)$ lie on the circumference of the base of the cone, and the point $M\left(0; 6; \frac{5}{3}\right)$ lies on its lateral surface. Find the area of the whole surface of the cone.

16. A cone is circumscribed about a cube so that four vertices of the cube lie in the plane of the base of the cone and the other four vertices lie on its lateral surface. What is the least volume of the cone if the edge of the cube is a in length?

17. The edge of the cube $ABCD A_1 B_1 C_1 D_1$ is a in length, and the points M and N are the midpoints of the edges AA_1 and $A_1 B_1$ respectively. The plane $D_1 MN$ touches the lateral surface of the cone whose axis is the line CC_1 and the centre of whose base is the point C . Find the volume of the cone.

18. In the tetrahedron $ABCD$ the dihedral angles at the edges BC and CD are right angles. One of the edges of the tetrahedron is thrice as long as the edge not intersecting it. The vertex of the cone coincides with one of the vertices of the tetrahedron and the circumference of the base of the cone is circumscribed about one of its faces. Find the angle in the axial section of the cone.

19. In the tetrahedron $ABCD$ the dihedral angles at the edges AB and BD are right angles and $\angle ACD = \alpha$. The vertex of the cone coincides with one of the vertices of the tetrahedron and the circumference of the base of the cone is inscribed into one of the faces. Find the angle in the axial section of the cone.

20. The base of the right prism $ABCA_1 B_1 C_1$ is a right triangle with the legs AB and BC 8 cm and 6 cm long respectively. The hypotenuse AC is the diameter of the base of the cone whose vertex lies on the edge $A_1 B_1$. The edge AB cuts the lateral surface of the cone at a point M so that $|AM| = 5$ cm. Find the volume of the cone.

21. The ratio between the radii of the bases of the truncated cone is equal to 2. The vertices A , B , B_1 , and A_1 of the regular prism $ABCA_1 B_1 C_1$ lie on the circumference of the larger base and the vertices C and C_1 lie on the circumference of the smaller base of the cone. Find the ratio between the volumes of the cone and the pyramid.

22. The vertices A , B , and C of the regular tetrahedron lie on the lateral surface of the cone, the vertex D belongs to its base, and the line AD is perpendicular to the plane of that base. Find the length of the edge of the tetrahedron if the apex angle of the cone is 90° and the radius of its base is R .

23. The radius of the smaller base of the truncated cone is equal to $3r$, and the radius of the larger base is $7r$. The vertices S and A of the regular pyramid $SABC$ lie on the circumference of the smaller base of the cone and the vertices B and C of the base ABC of the pyramid lie on the circumference of the larger base of the cone. Find the volume of the pyramid $SABC$.

24. A straight line touches the lateral surface of the cone if it lies in the tangent plane and has a common point with the generating line noncoinciding with the vertex of the cone. The base of the cone lies in the plane of the face $ABCD$ of the cube $ABCD A_1 B_1 C_1 D_1$ and the vertex S of the cone lies on the ray OO_1 , where O and O_1 are the centres of the faces $ABCD$ and $A_1 B_1 C_1 D_1$ respectively. The diagonal AB_1 touches the lateral surface of the cone. In what ratio does the point of tangency divide the segment AB_1 if $|SO| = 2|OO_1|$?

25. The triangle ABC of area S lies in the same plane as the line l . The point of intersection of the medians of the triangle is at a distance d from l . Prove that the volume of the solid obtained by rotating the triangle about the line l is equal to $2\pi dS$.

26. The vertices of the rectangle lie on the lateral surface of the cone. Prove that two parallel sides of the rectangle are perpendicular to the axis of the cone.

27. A regular quadrangular pyramid is inscribed into a sphere of radius R . What is the greatest possible volume of the pyramid?

28. A regular triangular prism is circumscribed about a ball of volume V . What is the least possible volume of the pyramid?

29. A regular truncated triangular pyramid is circumscribed about a ball, the lateral edge of the pyramid making an angle of 60° with the plane of the base. Find the ratio between the volumes of the ball and the pyramid.

30. The edge of the regular tetrahedron is a in length. The altitude of the tetrahedron is the diameter of a sphere. What is the area of the part of the surface of the tetrahedron lying in the interior of the sphere?

31. The sphere touches all the edges of the cube. Find the volume of the part of the cube lying in the interior of the sphere if the edge of the cube is a long.

32. The sphere touches all the lateral edges of the regular quadrangular prism and its bases. Find the ratio between the area of the part of the sphere lying in the exterior of the prism and the area of the whole surface of the prism.

33. All the plane angles at the vertex S in the pyramid $SABC$ are right angles. Prove that the vertex S , the point of intersection of the medians of the triangle ABC and the centre of the sphere circumscribed about the pyramid, lie on the same line.

34. The edge of the cube is a long. Find the radius of the sphere drawn through the vertices A and D and the midpoints of the edges $B_1 C_1$ and $C_1 D_1$.

35. In the tetrahedron $ABCD$ the dihedral angles at the edges

AB , BC , and CD are right angles and $\angle ABC = \angle BCD = \alpha$. The radius of the sphere circumscribed about the tetrahedron is R . Find the volume of the tetrahedron.

36. The points A and B lie in different faces of the right dihedral angle, the point A is at the distance $3a$ from the edge of that angle, and the point B is at the distance $4a$ from it. Find the least radius of the sphere touching the faces of the angle and the line AB .

37. In the cube $ABCD A_1 B_1 C_1 D_1$ the points E and F are the midpoints of the edges AA_1 and $A_1 D_1$ respectively and $|AB| = a$. Find the radii of the spheres passing through the points E and F and touching the planes $BB_1 C_1 C$ and $DD_1 C_1 C$.

38. The side of the base of the regular triangular prism $ABCA_1 B_1 C_1$ is a long. The point D is the midpoint of the edge AB , and the point E lies on the edge $A_1 C_1$. The line DE makes angles α and β with the planes ABC and $AA_1 C_1 C$ respectively. Find the radius of the sphere, with centre on the segment DE , touching the planes ABC and $AA_1 C_1 C$.

39. All the edges of the regular quadrangular pyramid are a long. Find the radius of the sphere passing through the centres of the lateral faces and touching the sides of the base of the pyramid.

40. In the pyramid $SABC$ the edge SC makes an angle of 60° with the plane of the base. The vertices A , B , and C and the midpoints of all the lateral edges lie on the sphere of radius r . Find the altitude of the pyramid.

41. In the tetrahedron $ABCD$ the face ABC is a right triangle ($\angle BCA = 90^\circ$), the edge AD is perpendicular to the plane ABC , $|AD| = |BC| = 3$ cm, and $|AC| = 4$ cm. Find the radius of the sphere inscribed into the tetrahedron.

42. The base of the pyramid $SABCD$ is a rhombus with the diagonals AC and BD , $|AC| = 8$ cm, $|BD| = 6$ cm, the edge SA is perpendicular to the plane of the base and its length is 2 cm. Find the radius of the sphere inscribed into the pyramid.

43. In the regular hexagonal pyramid the centres of the inscribed and circumscribed spheres coincide. Find (a) the ratio of the radii of the spheres, and (b) the plane angle at the vertex of the pyramid.

44. Prove that for any prism circumscribed about a sphere the ratio between the area of the lateral surface and the area of the base is equal to 4.

45. Prove that to be possible to inscribe a sphere into a prism, it is necessary and sufficient that a circle could be inscribed into the perpendicular section of the prism and that the altitude of the prism should equal the diameter of that circle.

46. Prove that for a sphere, touching all the edges of the tetrahedron, to exist, it is necessary and sufficient that the sums of the lengths of the skew edges of the tetrahedron should be equal.

47. A sphere of radius r is inscribed into a right dihedral angle. A straight line touching the sphere at a point K cuts the faces of the angle at points A and B such that $|AK|/|KB| = 3/2$. Each of the points A and B is at the distance $3r$ from the edge of the dihedral angle. Find the length of the segment AB .

48. The edge of the cube $ABCD A_1 B_1 C_1 D_1$ is a long. Points E and F are taken on the diagonals AC and $B_1 D_1$ of the faces of the cube re-

spectively such that the sphere inscribed into the cube divides the segment EF into three parts the ratio of whose lengths is $3 : 7 : 2$, reckoning from the point E . Find the length of the segment EF .

49. Each edge of the regular quadrangular pyramid is a in length. A plane, touching the sphere inscribed into the pyramid, is drawn through the midpoints of the lateral edges belonging to one of the lateral faces. Find the area of the resulting section.

50. In the regular pyramid $SABCD$ the side of the base $ABCD$ is a in length, and the altitude of the pyramid is equal to $a\sqrt{2}$. The sphere inscribed into the pyramid touches the face SAD at a point K . Find the area of the section of the pyramid drawn through the edge AB and the point K .

51. In the tetrahedron $ABCD$ the length of each of the edges AB and CD is 4 cm and the length of each of the other edges is 3 cm. A sphere is inscribed into that tetrahedron. Determine the volume of the tetrahedron whose vertices are the points of tangency of the sphere and the faces of the tetrahedron $ABCD$.

52. The edge of the regular tetrahedron $ABCD$ is a long. The sphere inscribed into the trihedral angle with the vertex A touches the plane drawn through the midpoints of the edges AB , AD , and BC . Find the radius of the sphere.

53. In the right parallelepiped $ABCD A_1 B_1 C_1 D_1$ $|AB| = 4a$, $|BC| = |BB_1| = a$. The sphere inscribed into the trihedral angle BAB_1C (with vertex B) touches the diagonal AC_1 . Determine the radius of the sphere (find all solutions).

54. In the regular pyramid $SABCD$ the side of the base is 8 cm in length and the lateral edge is 9 cm long. The sphere touches the plane of the base of the pyramid at a point A and a line SB . Find the radius of the sphere (find all solutions).

55. The sphere of radius r touches all the edges of the triangular pyramid, the centre of the sphere lying on the altitude of the pyramid. Prove that the pyramid is regular and find its altitude if the centre of the sphere is at the distance $r\sqrt{3}$ from the vertex of the pyramid.

56. Each plane angle of a trihedral angle with vertex S is 60° . Points A , B , and C are taken on the edges of the trihedral angles respectively such that $|SA| = 2a$ and $|SB| = |SC| = a$. Find the radius of the sphere touching the edges of the angle and the plane ABC (find all solutions).

57. The base of the parallelepiped $ABCD A_1 B_1 C_1 D_1$ is a rectangle $ABCD$ whose side AD is four times as long as the altitude of the parallelepiped. A sphere with centre at O touches the edges BC , $A_1 B_1$ and DD_1 at points B , A_1 , and D_1 respectively. Find the angle $A_1 OB$.

58. The skew edges of the tetrahedron are equal in length. Prove that the centres of the spheres inscribed into the tetrahedron and circumscribed about it coincide. Is the converse statement true?

59. The sphere touches all the edges of the triangular prism. Determine the ratio between the volumes of the sphere and the prism.

60. A sphere is inscribed into each trihedral angle of the tetrahedron. All the four spheres have the same radius and meet at the same point. Find the radius of the spheres if the radii of the spheres circumscribed about it coincide.

scribed about the tetrahedron and inscribed into it are R and r respectively.

61. A sphere is inscribed into a cube with the edge a long. For an arbitrarily taken point of the sphere find the sums of the squares of the distances from that point to: (a) the vertices, (b) the faces, (c) the edges of the cube. Prove that the sums do not depend on the choice of the point.

62. A plane α and three pairwise perpendicular radii are drawn through the centre of the sphere. Find the sum of the squares of the distances from the ends of the radii to the plane α if the radius of the sphere is R .

63. Three circles of radius $r/2$ are located on the sphere of radius r so that each of them has a point in common with each of the other two circles. Find the radius of the circle lying on that sphere and having one point in common with each of the given three circles.

64. The angle between the adjacent lateral faces of the regular triangular pyramid is equal to $3\pi/4$. A sphere with centre at the vertex of the pyramid touches the base of the pyramid. What part of the sphere is in the interior of the pyramid?

65. The edge of the regular tetrahedron $ABCD$ is a long. A sphere touches the edges AB , AC , and AD at points B , C , and D respectively. Find the area of the part of the sphere lying in the interior of the tetrahedron.

66. Prove that if the solid Φ , not lying entirely on the line l , is symmetric about any plane passing through the line l , then the solid Φ is a solid of revolution and the line l is the axis of revolution.

67. Find the volume of the solid resulting from the rotation of a regular hexagon about the axis containing its side if the length of that side is a .

68. The perimeter of the isosceles triangle is equal to P . How long should its sides be for the volume of the solid resulting from the rotation of the triangle about the base to be the greatest?

69. Two spheres with centres at O_1 and O_2 are intersected by a plane perpendicular to the segment O_1O_2 and passing through its midpoint. The plane divides the first sphere into parts the ratio of whose areas is $m/1$ and the second sphere into parts the ratio of whose areas is $n/1$ ($m > 1$, $n > 1$). Find the ratio of the radii of the spheres.

70. The volume of the ball is V . A cylinder is inscribed into the sphere bounding the ball. What is the greatest possible volume of the cylinder?

71. The volume of the ball is V . Find the greatest volume of the cone inscribed into the sphere bounding the ball.

72. The area of the sphere is S . A cone is circumscribed about the sphere. What is the least area of the surface of the cone?

73. A sphere is constructed on the altitude of the cone which serves as a diameter. The area of the part of the sphere lying in the interior of the cone is equal to the area of the surface of the part of the cone lying inside the sphere. Determine the angle in the axial section of the cone.

74. Prove that to be possible to inscribe a sphere into a truncated cone (the sphere touching both bases and every generating line of the

cone), it is necessary and sufficient that the length of the generating line should be equal to the sum of the radii of the bases.

75. A sphere of area S_2 is inscribed into a truncated cone with the lateral surface of area S_1 . Find the angle between the generating line and the plane of the base of the cone.

76. A sphere is inscribed into a cone. The plane passing through the vertex of the cone divides its lateral surface into parts the ratio of whose areas is $1/2$. The sphere is divided by the plane into parts the ratio of whose areas is $1/5$. Find the apex angle of the cone.

77. A cone is circumscribed about a sphere, the apex angle of the cone being equal to 60° . Find the ratio of the volumes of the parts of the cone lying in the exterior of the sphere.

78. All the plane angles at the vertex S in the pyramid $SABC$ are right angles, $|AC| = |BC|$, $|AB| = 2R$, and the angle between the edge SC and the plane ABC is 45° . A sphere is constructed on the edge AB as a diameter. Find the radius of the sphere inscribed into the trihedral angle $CSAB$ (the vertex C) and touching the given sphere (find all solutions).

79. Three spheres lie on the plane, each of which touching the other two spheres. The distances between the points of tangency of the spheres and the plane are a , b , and c respectively. Find the radii of the spheres.

80. Two spheres, of radius r each, touch each other. The line making an angle α with the line of centres touches the spheres at points A and B respectively. Find the distances from the points A and B to the line of centres of the spheres and the length of the segment AB .

81. Two spheres of equal radii touch each other. A straight line cuts the spheres consecutively at points A , B , C , and D (BC being the segment lying in the exterior of the spheres) such that $|AB| = |CD| = a$ and $|BC| = na$. Find the distance from that line to the line of centres of the spheres.

82. Two spheres of equal radii are inscribed into a right dihedral angle, the spheres touching each other. A straight line cuts the faces of the angle at point A and B , makes an angle of 30° with each face and touches each sphere. In what ratio do the points of tangency divide the segment AB ?

83. Three spheres of equal radii r touch the plane α and each sphere touches the other two. The straight line a , parallel to the plane α , touches each sphere. Find the distance between a and α .

84. Each of the four given spheres of radius r touches the other three. Find the radius of the spheres touching all the given four spheres.

85. The spheres S_1 and S_2 of radius $2r$ and the spheres S_3 and S_4 of radius r are positioned so that each sphere touches the other three. The spheres S_1 and S_3 touch each other at a point A and the spheres S_2 and S_4 at a point B . In what ratio do the spheres S_3 and S_4 divide the segment AB ?

86. Two cones have a vertex in common and the generating line of the first cone is the altitude of the other cone. The apex angle of the first cone is equal to $\arccos \frac{1}{3}$, the apex angle of the other cone is equal to $2\pi/3$. Determine the angle between the generating lines along which the lateral surfaces of the cones intersect.

87. The apex angle of the cone is equal to $\pi/3$. Three spheres of radius r are located in the interior of the cone. Each sphere touches the other two spheres, the base of the cone and its lateral surface. Find the radius of the base of the cone.

88. Four identical balls are located in the interior of the cylinder so that each ball touches the other three balls and the lateral surface of the cylinder. Two balls touch the lower base and the other two touch the upper base of the cylinder. Find the volume of the cylinder if the volume of the ball is V .

89. Two cones with equal apex angles have a vertex in common and touch each other (that is, their lateral surfaces have a common generating line). Each face of the dihedral angle, 2α in magnitude, touches the lateral surface of each of the cones. Find the apex angle of the cones.

90. Three cones with equal apex angles have a vertex in common and each cone touches the other two. All three cones touch the same plane. Determine the apex angle of the cones.

Appendix

Samples of Problems Given at the Entrance Examinations in Mathematics in Various Institutes of the USSR

✓ 1. The difference $\sqrt{|40\sqrt{2}-57|} - \sqrt{40\sqrt{2}+57}$ is an integer. Find it.

✗ 2. Find all solutions of the equation

$$\int_0^{\alpha} \cos(x + \alpha^2) dx = \sin \alpha,$$

which belong to the interval $[2; 3]$.

3. In the scalene triangle ABC the altitudes AP and CQ are dropped from the vertices A and C to the sides BC and AB . The area of the triangle ABC is known to be equal to 18, the area of the triangle BPQ to 2, and the length of the segment PQ to $2\sqrt{2}$. Calculate the radius of the circle circumscribed about the triangle ABC .

4. Find all the values of the parameter a for which the system of inequalities

$$\begin{cases} x^2 + 2xy - 7y^2 \geq \frac{1-a}{1+a}, \\ 3x^2 + 10xy - 5y^2 \leq -2 \end{cases}$$

has a solution.

5. The volume of the tetrahedron $ABCD$ is equal to 5. A plane is drawn through the midpoints of the edges AD and BC which cuts the edge CD at a point M . This results in the ratio between the lengths of the segments DM and MC being equal to $2/3$. Calculate the area of the section of the tetrahedron formed by the indicated plane if the distance between that plane and the vertex A is equal to 1.

✓ 6. Solve the inequality

* * *

$$(x-1)\sqrt{x^2-x-2} \geq 0.$$

✓ 7. Find the points of minimum of the function

$$\sqrt{3} \cos \frac{x}{2} + \sin \frac{x}{2} - \frac{x-3}{2}.$$

8. Find the equation of the straight line passing through the point with the coordinates $(1/2; 2)$, touching the graph of the function $y =$

$2 - \frac{x^2}{2}$ and cutting the graph of the function $y = \sqrt{4 - x^2}$ at two distinct points.

9. The collection A consists of different natural numbers. There are more than seven numbers in A . The least common multiple of all the numbers in A is 210. The greatest common divisor of any two numbers in A exceed unity. The product of all numbers in A can be divided by 1920 and is not a square of any integer. Find the numbers which constitute the collection A .

10. The base of the pyramid $ABCEH$ is a convex quadrangle $ABCE$ which is divided by the diagonal BE into two equivalent triangles. The length of the edge AB is equal to 1 and the length of the edges BC and CE are equal. The sum of the lengths of the edges AH and EH is equal to $\sqrt{2}$. The volume of the pyramid is equal to $1/6$. Find the radius of the ball having the greatest volume among all the balls contained in the pyramid $ABCEH$.

* * *

11. Solve the equation $\sin^2 x = 3/4$.

12. Find the area of the closed figure bounded by the curves

$$y = 0, y = 20 - 6x - 2x^2.$$

13. Find the negative terms of the sequence

$$x_n = \frac{A_{n+1}^4}{P_{n+2}} - \frac{143}{4P_n} \quad (n=1, 2, 3, \dots),$$

where A_{n+1}^4 is the number of arrangements, and P_{n+2} and P_n are the numbers of permutations.

14. Given an isosceles trapezoid into which a circle is inscribed and about which a circle is circumscribed. The ratio between the altitude of the trapezoid and the radius of the circumscribed circle is equal to $\sqrt{2/3}$. Find the angles of the trapezoid.

15. The length of the edge of the cube $ABCD A_1 B_1 C_1 D_1$ ($(AA_1) \parallel (BB_1) \parallel (CC_1) \parallel (DD_1)$) is equal to 1. Point E is taken on the edge AA_1 such that the segment AE is $1/3$ long. A point F is taken on the edge BC such that the segment BF is $1/4$ long. A plane α is drawn through the centre of the cube and the points E and F . Find the distance from the vertex B_1 to the plane α .

* * *

16. Solve the inequality

$$2 | x + 1 | > x + 4.$$

17. In the triangle ABC the side AB is 3 m long, the altitude CD dropped to the side AB is $\sqrt{3}$ m long. The foot D of the altitude CD lies on the side AB and the length of the segment AD is equal to that of the side BC . Find the length of the side AC .

18. Find the intervals of increase and decrease and the points of maximum and minimum of the function $y = x/\ln x$.

19. A lorry and a racing car left simultaneously point A for point C . Travelling with constant speed, the lorry reached point C having covered the distance equal to 360 km. The racing car took the circle line and first visited point B located 120 km from point A , travelling with the speed twice that of the lorry. Having left point B , it increased its speed by 40 km per hour and covered the way from point B to point C equal to 1000 km. It reached point C 1 h 15 min later than did the lorry. If the racing car covered the whole way from point A to point C with the same speed that he had when it travelled from point B to point C , it would reach point C one hour later than the lorry did. Find the speed of the lorry.

20. Solve the equation

$$2 \sin \left(3x + \frac{\pi}{4} \right) = \sqrt{1 + 8 \sin 2x \cos^2 2x}.$$

* * *

21. Find the extrema of the function $y = \frac{2x-1}{(x-1)^2}$ and indicate the intervals of increase and decrease.

22. Calculate the volume of the solid generated by the rotation about the x -axis of the figure bounded by the curves $y = x^3$, $y = 1$, and $x = 3$.

23. Solve the equation

$$\frac{1}{1 + \cos^2 x} + \frac{1}{1 + \sin^2 x} = \frac{16}{11}.$$

24. Solve the inequality

$$\log_2 \frac{x-4}{2x+5} < 1.$$

25. Solve the equation

$$5^{1+\log_4 x} + 5^{\log_{0.25} x - 1} = \frac{26}{5}.$$

* * *

26. Four different postage-stamps are needed to pay for dispatching four books, the total sum being 2 rubles and 80 kopecks. Determine the cost of each stamp bought by the sender if those costs constitute an arithmetic progression and the most expensive stamp is 2.5 times as dear as the cheapest one.

27. The vectors $\mathbf{a}$ and $\mathbf{b}$ are mutually perpendicular and the vector $\mathbf{c}$ makes an angle of 60° with each of them. Knowing that $|\mathbf{a}| = 3$, $|\mathbf{b}| = 5$, and $|\mathbf{c}| = 8$, calculate the scalar product $(3\mathbf{a} - 2\mathbf{b}) \times (\mathbf{b} + 3\mathbf{c})$.

28. Solve the equation

$$7 \cdot 3^{x+1} - 5^{x+2} = 3^{x+1} - 5^{x+3}.$$

29. Solve the equation

$$\frac{1}{\cos x} = 6 \sin x + 8 \cos x.$$

30. Calculate the area of the figure bounded by the curves

$$y = -x^2 + 6x - 1, \quad y = 0, \quad x = 1, \quad x = 3.$$

* * *

31. Find the sum of all three-digit natural numbers which, being divided by 3, give the remainder equal to 2.

32. The base of the pyramid $SABC$ is an isosceles triangle ABC , $|AB| = |AC| = a$, $\angle ABC = \varphi$. The line (AS) makes an angle α with the plane of the base of the pyramid, and the plane of the lateral face (BCS) makes an angle β with the same plane, and $\angle SAC = \angle SAB$. Find the volume of the pyramid $KSLC$ if the points K and S are known to belong to the edges $[AS]$ and $[BS]$ respectively, and the area of the triangle KSL is related to that of the triangle ABS as $4/25$.

33. Solve the system of equations

$$\begin{cases} \log_{1/3}(x+y) + \log_3(x-y) = 2, \\ 2^{y^2} = 512^{x+1}. \end{cases}$$

34. Find the critical points of the function

$$y = 2 \sin^2 \frac{x}{6} + \sin \frac{x}{3} - \frac{x}{3},$$

whose coordinates satisfy the inequality $x^2 - 10 < -19.5x$.

* * *

35. Solve the equation $7^{\log x} = 98 - x^{\log 7}$.

36. Solve the equation $2 + \cos 4x = 3 \sin^2 x$.

37. Solve the inequality

$$\left| \frac{x^2 - 5x + 4}{x^2 - 4} \right| < 1.$$

38. At what value of z are the vectors $\mathbf{a} = (2; -3; 4)$ and $\mathbf{b} = (-3; 2; z)$ perpendicular?

39. Calculate the integral

$$\int_3^{-18} \sqrt[3]{2 - \frac{x}{3}} dx.$$

* * *

40. Solve the inequality

$$\frac{2x+3}{x^2+x-12} \leq \frac{1}{2}.$$

41. Find the greatest and the least value of the function

$$f(x) = 2 \cdot 3^{3x} - 3^{2x} \cdot 4 + 2 \cdot 3^x$$

on the interval $[-1, 1]$.

42. Find the area of the figure bounded by the graphs of the functions

$$y = x^2 - 4, y = 4 - x^2.$$

43. The radius of the sphere inscribed into a cone is R . Find the volume of the cone if the centre of the sphere circumscribed about the cone coincides with the centre of the inscribed sphere.

44. A furniture factory manufactures tables and chairs. The manufacture of one table requires 5 running metres of boards and one metre of veneer, and the manufacture of one chair requires one metre of boards and 2 metres of veneer. How many tables and chairs were manufactured if it is known that the boards and the veneer were used in an amount of 900 running metres each?

* * *

45. Solve the equation $3x - \sqrt{18x + 1} + 1 = 0$.

46. Solve the inequality

$$\sqrt{x} - 3 \leq \frac{2}{\sqrt{x} - 2}.$$

47. Solve the equation

$$\frac{1}{\sin^2 x} = \cot x + 3.$$

48. Find

$$\lim_{x \rightarrow \frac{\pi}{3}} \frac{\tan^3 x - 3 \tan x}{\cos \left(x + \frac{\pi}{6} \right)}.$$

49. Find the angle between the plane passing through the points $A(0; 0; 0)$, $B(1; 1; 1)$, and $C(3; 2; 1)$ and the plane passing through the points $A(0; 0; 0)$, $B(1; 1; 1)$, and $D(3; 1; 2)$.

* * *

50. Solve the equation

$$\log_2 \left(\frac{\sqrt{14}}{4} + \cos x \right) + \log_2 \left(\frac{\sqrt{14}}{4} - \cos x \right) = -3.$$

51. Find the image of the point $A(2; 7)$ resulting from the rotation of the coordinate plane about the point $M(2; 3)$ through the angle $-\pi/2$. Find the image of one more point (choose it yourself). Find the image of the point $C(-4; 8)$ resulting from this rotation.

52. Check which of the following numbers is greater: C_{11}^n or $C_{11}^3 + C_{11}^4$, and find all the values of n for which $C_n^3 < C_n^3 + C_n^4$.

53. Find the greatest and the least value of the function

$$f(x) = e^{2x-1} + 2e^{1-2x} + 7x - 3$$

on the interval $[0.14; 1]$.

54. We shall call the number a a good one if the inequality

$$\frac{2x^2 + 2x + 3}{x^2 + x + 1} \leq a$$

is satisfied for any x .

(a) Prove that the number 4 is a good one.

(b) Find all good numbers.

* * *

55. Solve the equation

$$\sin^4 x - \cos^4 x = \cos x.$$

56. Solve the equation

$$32^{\frac{x+5}{x-7}} = 0.25 \cdot 128^{\frac{x+17}{x-3}}.$$

57. Calculate the area of the figure bounded by the graphs of the functions

$$y = 4 - x^2, y = x^2 - 2x.$$

58. In an isosceles trapezoid the smaller base and the nonparallel side are equal to a . Find the larger base b such that the area of the trapezoid should be the maximal one.

59. Given three nonzero vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$, every two of which are noncollinear. Find their sum if the vectors $\mathbf{a} + \mathbf{b}$ and $\mathbf{c}$ and $\mathbf{b} + \mathbf{c}$ and $\mathbf{a}$ are pairwise collinear.

* * *

60. Calculate the area of the figure bounded by the curves

$$y = 5/x, y = 6 - x.$$

61. Squares are constructed on the bases AB and CD of the trapezoid $ABCD$ (in the exterior of it). Prove that the straight line connecting the centres of the squares passes through the point of intersection of the diagonals of the trapezoid.

62. Solve the inequality

$$\log_{x^2} (3 - 2x) > 1.$$

63. Solve the equation

$$2 \cos x = \sqrt{2 + \sin 2x}.$$

64. The altitude of a regular quadrangular pyramid makes an angle of 30° with the lateral face. A plane perpendicular to the opposite face is drawn through the side of the base of the pyramid. Find the ratio between the volumes of the polyhedra resulting from the intersection of the pyramid by the plane.

* * *

65. Solve the inequality $0.1^{x+1} < 0.8 + 2 \cdot 10^x$.

66. Solve the system of equations

$$\begin{cases} \log_4 xy + 3 \frac{\log_4 x}{\log_4 y} = 0, \\ \log_4 \frac{x}{y} - \log_4 x \cdot \log_4 y = 0. \end{cases}$$

67. The circle constructed on the base AD of the trapezoid $ABCD$ serving as a diameter passes through the midpoints of the lateral sides AB and CD of the trapezoid and touches the base BC . Find the angles of the trapezoid.

68. Solve the equation

$$3 \sin 3x = \cos 4x - \sin 9x - \cos 10x.$$

69. Two trains left point A for point B simultaneously. They first travelled with uniform accelerations (the accelerations of the trains are different and the initial speeds are zero). After reaching certain speeds, the trains continued travelling with uniform speed. The ratio of the uniform speeds of the trains is 2. Having covered a quarter of the way from A to B the trains came alongside each other and at that moment the speed of one of them was 1.5 that of the other. Find the ratio between the intervals of time required for the trains to travel from A to B .

70. Find the greatest possible angle between the plane of the lateral face and the lateral edge not belonging to it of a regular quadrangular pyramid.

* * *

71. Having simplified the equation

$$\left(\frac{(x^2-4)^{-2} + (x+2)^{-2} - (x-2)^{-2}}{1-8x} \right)^{-1} - (4\sqrt[3]{4})^{\log_4 x^3} = 16,$$

find its limit as $x \rightarrow 1/8$.

72. Solve the system of equations

$$\begin{cases} (3y^2+1) \log_3 x = 1, \\ x^{2y^2+10} = 27. \end{cases}$$

73. The sum of the infinitely decreasing geometric progression is equal to the greatest value of the function $f(x) = 3x^3 - x - 76$ on the interval $[0; 3]$; the first term of the progression is equal to the square of its common ratio. Find the common ratio of the progression.

74. Find all the values of x which satisfy the equation

$$\sin^4 \frac{x}{2} + \sin^4 \left(\frac{x}{2} + \frac{7\pi}{2} \right) = \sin \frac{5\pi}{6}$$

and lie on the interval $[-\pi/2; 2\pi]$.

75. A plane is drawn through the vertex C_1 of the cube $ABCD A_1 B_1 C_1 D_1$ which cuts the edges $[BC]$ and $[CD]$ and makes an

angle α with the face $ABCD$. As a result, an isosceles triangle is obtained in the section. Find the area of the section if the edge of the cube is a in length (consider all possible cases).

* * *

76. For what a and b ($a, b \in \mathbb{R}$) do the graphs of the functions $f(x) = 2x^4 - a^2x^2 + b - 1$ and $g(x) = 2ax^3 - 1$ have only two points in common?

77. How many six-digit numbers contain exactly four different digits?

78. Solve the equation ($a \in \mathbb{R}$)

$$1 + 2(\sin^2 2x - 2a \cos 2x + a) \tan^2 x - \cos 4x = 0.$$

79. The angle between the adjacent lateral faces of a regular quadrangular pyramid is equal to α and the side of the base is a in length. Find the radius of the sphere touching the lateral faces of the pyramid and that of the sphere circumscribed about the pyramid.

80. Find the vector $\mathbf{a} = (x; y; z)$ making equal angles with the vectors $\mathbf{b} = (y; -2z; 3x)$ and $\mathbf{c} = (2z; 3x; -y)$ if $\mathbf{a}$ is perpendicular to the vector $\mathbf{d} = (1; -1; 2)$, $|\mathbf{a}| = 2\sqrt{3}$ and the angle between the vector $\mathbf{a}$ and the y -axis is obtuse.

* * *

81. Simplify the expression

$$\left(\left(1 - \frac{1+ab}{1+\sqrt[3]{ab}} \right) : \left(\sqrt{ab} (1 - \sqrt[3]{ab}) - \frac{(1-ab)(\sqrt[3]{ab}-1)}{1+\sqrt{ab}} \right) \right)^3 - ab.$$

82. Solve the equation

$$\cos \frac{\pi - 2x}{2} + \sin \frac{5\pi + 2x}{2} = \tan \frac{13\pi}{4}.$$

83. Solve the inequality $2\sqrt{x^2} \geq (x-1)^2 + 2$.

84. Prove the equality

$$C_{n+2}^{m+1} + 2C_{n+2}^{m+2} + C_{n+2}^{m+3} = C_{n+3}^{m+3}.$$

85. Divide the number 12 into two summands so that the sum of the cubes of those summands should be the least.

* * *

86. A father wanted to divide 180 apples between his five children. Half of the apples he gave to his sons who divided them equally. The other half was given to the daughters who also divided them equally. It turned out that each daughter got 15 apples more than each son. How many sons and how many daughters did the father have?

87. Solve the equation $\sin^2 2x + \sin^2 x = 1$.

88. Solve the equation

$$3x^{15} + 3(x^{-10})^{10} = 84.$$

89. Find the greatest and the least value of the function $f(x) = 3x^2 - 9x + 2$ on the interval $[-1; 1]$.

90. Given p arithmetic progressions each of which contains n terms. Their initial terms are equal to 1, 2, 3, ..., p respectively and the common ratios are equal to 1, 3, 5, ..., $2p - 1$. Find the sum of the terms of all the progressions.

* * *

91. Two sportsmen start simultaneously from diametrically opposite points A and B and run towards each other at constant speeds along a track of circular shape. They first met t seconds later, a metres from B , and for the second time they met at a distance of $2a$ metres from A ($a > 0$). Find the speeds of the sportsmen if the distance means the shortest way along the track.

92. Solve the equation

$$\cos 6x + \tan^2 x + \cos 6x \cdot \tan^2 x = 1.$$

93. The circles O_1 and O_2 have external tangency at a point A , the segment AB being the diameter of O_1 . The lengths of the segments intercepted by the circles on some straight line passing through the point B are equal to 2 cm, 3 cm, and 4 cm, reckoning from the point B . Find the radii of the circles.

94. Determine the values of the parameter a at which the equation $\log_{2x}(ax + 1) = 1/2$ has a unique solution.

95. The base of the triangular prism is an isosceles right triangle ABC , whose legs AB and AC are a in length. The lateral edges AA' , BB' , and CC' make an angle of 60° with the plane of the base, and the diagonal BC' of the lateral face $CBB'C'$ is perpendicular to the edge AC . Find the volume of the prism if the length of the diagonal BC' is equal to $a\sqrt{6}$.

* * *

96. Solve the equation

$$2 \sin^2 \left(x - \frac{\pi}{4} \right) = 2 \sin^2 x - \tan x.$$

97. Solve the inequality $|3^x - 3| + 9^x - 3 > 0$.

98. A rectangular parallelepiped whose base is a square with side b is inscribed into a regular quadrangular pyramid with altitude H and the side of the base a , so that the lower base of the parallelepiped lies on the base of the pyramid and the vertices of its upper base lie on the lateral edges of the pyramid. At what b is the volume of the inscribed parallelepiped the greatest? Find the volume for that value of b .

99. Some poles lie on the road at a distance of 10 m from one another. Beginning with one end-pole, a worker carried all the poles, one-by-one, to the other end-pole, for which purpose he had to cover the total distance of 1.44 km. How many poles were there on the road?

* * *

100. A cylinder is inscribed into a regular quadrangular pyramid whose lateral faces are at the angle φ to the plane of the base (one base of the cylinder lies in the plane of the base of the pyramid and the circumference of its second base has one point in common with every lateral face of the pyramid). The radius of the base of the cylinder and its altitude are equal to r . Calculate the volume of the pyramid. At what angle φ is the volume of the pyramid the least?

101. Calculate the area of the figure bounded by the graphs of the functions

$$y = \frac{|4-x^2|}{4} \text{ and } y = 7 - |x|.$$

102. Solve the inequality

$$\sqrt{\log_2^2 x + \log_{1/2} x^2 - 3} > \sqrt{5} (\log_4 x^2 - 3).$$

103. How many four-digit numbers are there whose decimal notation contains not more than two different digits?

* * *

104. Given the cube $ABCD A_1 B_1 C_1 D_1$. Prove that the distance between the planes, one of which passes through the vertices A , B_1 , and D_1 and the other through the vertices B , C_1 , and D , is equal to $1/3$ of the diagonal of the cube.

105. Find the modulus of the difference between the extreme values of the function

$$y = x^3 + 3x^2 - 3x + 1.$$

106. For what values of α is the inequality

$$\frac{(2+\alpha)x^2 + (3+\alpha)x + 1 - \alpha}{x^2 + x + 2} \leq 2$$

satisfied for all real x ?

107. Find the domain of definition of the function

$$y = \frac{\sqrt{\cos x - \frac{1}{2}}}{\sqrt{6 + 35x - 6x^2}}.$$

* * *

108. Find the altitude of the triangular pyramid of the greatest volume inscribed into a ball of radius R .

109. How many different six-digit numbers can be compiled from the digits 1, 2, 3, 4, 5, 6, 7 so that the numbers do not repeat and the terminal digits are even?

110. Solve the inequality

$$\log_{x^2-1} (x+1) < 1.$$

111. Solve the equation

$$\cos^2 \left(\frac{\pi}{3} \cos x - \frac{8\pi}{3} \right) = 1.$$

Solutions to Problems of Section I

CHAPTER 1

1. Since $\cos 6x = \cos 5x \cos x - \sin 5x \sin x$, we have $\sin 5x \sin x = 0$ and, therefore, $\sin 5x = 0$, $x = \pi k/5$, or $\sin x = 0$, $x = \pi l$, where $k, l \in \mathbb{Z}$. All solutions of the second equation belong to the set of solutions of the first ($k = 5l$).

Answer: $\{\pi k/5 \mid k \in \mathbb{Z}\}$.

2. Using the formulas $2 \sin^2 2x = 1 - \cos 4x$, $\sin^2 4x = 1 - \cos^2 4x$, we arrive at an equation $\cos^2 4x + \cos 4x - \frac{3}{4} = 0$, which is a quadratic equation with respect to $\cos 4x$. Solving this equation, we find that $\cos 4x = -3/2$, $\cos 4x = 1/2$. The first equation has no solutions and the second equation has the solutions

$$x = \pm \frac{\pi}{12} + \frac{\pi k}{2}, \quad k \in \mathbb{Z}.$$

Answer: $\left\{ \pm \frac{\pi}{12} + \frac{\pi k}{2} \mid k \in \mathbb{Z} \right\}$.

3. Let us transform the equation:

$$\sin 5x - \sin x = \sin 2x, \quad 2 \sin 2x \cos 3x = \sin 2x.$$

Hence it follows that $\sin 2x = 0$, $x = \pi k/2$, or $\cos 3x = 1/2$,

$$x = \pm \frac{\pi}{9} + \frac{2\pi k}{3}, \quad \text{where } k \in \mathbb{Z}.$$

Answer: $\left\{ \frac{\pi k}{2}, \pm \frac{\pi}{9} + \frac{2\pi k}{3} \mid k \in \mathbb{Z} \right\}$.

4. Since $\cos 3x = \cos (2x + x) = \cos 2x \cos x - \sin 2x \sin x$, we have $3 \cos x \cos 2x + \sin x \sin 2x = 0$, $\cos x (3 \cos 2x + 2 \sin^2 x) =$

0 and, consequently, $\cos x = 0$, $x = \frac{\pi}{2} + \pi k$, $k \in \mathbb{Z}$, or $3 \cos 2x +$

$$2 \sin^2 x = 0, \quad \cos 2x = -1/2, \quad x = \pm \frac{\pi}{3} + \pi k, \quad k \in \mathbb{Z}.$$

Answer: $\left\{ \frac{\pi}{2} + \pi k, \pm \frac{\pi}{3} + \pi k \mid k \in \mathbb{Z} \right\}$.

5. Let us transform the equation:

$$(1 + \cos x) \frac{\cos x}{\sin x} - 2 \sin x \cos x = 0,$$

$$\frac{\cos x}{\sin x} (1 + \cos x - 2 \sin^2 x) = 0.$$

Hence we have $\cos x = 0$, $1 + \cos x - 2 \sin^2 x = 0$, provided that $\sin x \neq 0$. (1)

All the solutions $x = \frac{\pi}{2} + \pi k$, $k \in \mathbb{Z}$, of the first equation satisfy equation (1). Using the formula $\sin^2 x = 1 - \cos^2 x$, we reduce the second equation to the form $(1 + \cos x)(2 \cos x - 1) = 0$. If $\cos x = -1$, then $\sin x = 0$ and condition (1) is not satisfied. Now if $\cos x = 1/2$, i.e. $x = \pm \frac{\pi}{3} + 2\pi k$, $k \in \mathbb{Z}$, then condition (1) is satisfied.

$$\text{Answer: } \left\{ \frac{\pi}{2} + \pi k, \pm \frac{\pi}{3} + 2\pi k \mid k \in \mathbb{Z} \right\}.$$

6. Under the condition

$$\cos x \neq 0 \quad (1)$$

the given equation is equivalent to the equation

$$(\sqrt{2} - 1) \cdot 2 \sin x \cos x + \cos x - \sin x = 1.$$

If we designate $\cos x - \sin x = t$, then $2 \sin x \cos x = 1 - t^2$ and, consequently, $(\sqrt{2} - 1)(1 - t^2) + t - 1 = 0$, $(1 - t)((\sqrt{2} - 1) \times (1 + t) - 1) = 0$, $(1 - t)(t - \sqrt{2}) = 0$, whence we have $t = 1$ or $t = \sqrt{2}$. By introducing an auxiliary angle, we reduce the equation

$$\cos x - \sin x = 1 \text{ to the form } \cos \left(x + \frac{\pi}{4} \right) = \frac{1}{\sqrt{2}}, \text{ whence we find}$$

that $x = \pm \frac{\pi}{4} - \frac{\pi}{4} + 2\pi k$, $k \in \mathbb{Z}$. If we take the upper sign, then $x = 2\pi k$ and condition (1) is satisfied and, consequently, $x = 2\pi k$ is the solution of the original equation. If we take the lower sign, then $x = -\frac{\pi}{2} + 2\pi k$ and $\cos x = 0$, that is, these values of x are extraneous.

Solving the equation $\cos x - \sin x = \sqrt{2}$ by analogy, we find that $x = -\frac{\pi}{4} + 2\pi k$, $k \in \mathbb{Z}$. These values satisfy equation (1) and are consequently solutions of the original equation.

$$\text{Answer: } \left\{ 2\pi k, -\frac{\pi}{4} + 2\pi k \mid k \in \mathbb{Z} \right\}.$$

7. We transform the equation:

$$\begin{aligned}\frac{\sin x}{\cos x} + \frac{\cos 2x}{\sin 2x} &= 2 \frac{\cos 4x}{\sin 4x}, \\ \frac{\sin 2x \sin x + \cos 2x \cos x}{\cos x \sin 2x} &= \frac{\cos 4x}{\sin 2x \cos 2x}, \\ \frac{\cos x}{\cos x \sin 2x} &= \frac{\cos 4x}{\sin 2x \cos 2x}.\end{aligned}$$

Cancelling the left-hand side by $\cos x$ and multiplying both sides of the resulting equation by $\sin 2x \cos 2x$, we arrive at an equation

$$\cos 2x - \cos 4x = 0.$$

Out of all its solutions, only those will serve as solutions of the original equation for which

$$\cos x \neq 0, \sin 2x \neq 0, \cos 2x \neq 0,$$

or, which is the same,

$$\sin 4x \neq 0. \quad (1)$$

We reduce the resulting equation to the form $2 \sin x \sin 3x = 0$. If $\sin x = 0$, then $\sin 4x = 0$ as well, that is, condition (1) is not satisfied and such values of x cannot serve as solutions of the original equation. If $\sin 3x = 0$, then $x = \frac{\pi k}{3}$, $k \in \mathbb{Z}$. Only three cases, $k = 3l$, $k = 3l + 1$, and $k = 3l - 1$ are possible for any integral k , where $l \in \mathbb{Z}$. If $k = 3l$, then $x = \pi l$ and $\sin 4x = 0$; now if $k = 3l \pm 1$, then $x = \pm \frac{\pi}{3} + \pi l$ and $\sin 4x = \sin \left(\pm \frac{4\pi}{3} + 4\pi l \right) = \pm \frac{\sqrt{3}}{2} \neq 0$.

This means that only the values $x = \pm \frac{\pi}{3} + \pi l$ are solutions of the original equation.

$$\text{Answer: } \left\{ \pm \frac{\pi}{3} + \pi l \mid l \in \mathbb{Z} \right\}.$$

8. We write the equation in the form

$$\sqrt{1 + \sin x} = -\cos x$$

and square its both sides:

$$1 + \sin x = \cos^2 x. \quad (1)$$

All those and only those solutions of equation (1) will serve as solutions of the original equation for which

$$\cos x \leq 0. \quad (2)$$

Let us transform equation (1): $1 + \sin x = (1 + \sin x)(1 - \sin x)$, or $(1 + \sin x) \sin x = 0$, whence it follows that $\sin x = -1$, $x =$

$$\begin{aligned}& -\frac{\pi}{2} + 2\pi k, \quad k \in \mathbb{Z}; \quad \sin x = 0, \quad x = \pi l, \quad l \in \mathbb{Z}. \quad \text{All the values } x = \\ & -\frac{\pi}{2} + 2\pi k \text{ satisfy condition (2) and are, consequently solutions of}\end{aligned}$$

the original equation. If $x = \pi l$, then $\cos \pi l = (-1)^l$ and it can be seen that condition (2) is satisfied only for odd $l = 2m + 1$, $m \in \mathbb{Z}$.

Hence, out of the values of x being considered, only $x = \pi(2m + 1)$, $m \in \mathbb{Z}$, are solutions of the original equation.

$$\text{Answer: } \left\{ -\frac{\pi}{2} + 2\pi k, \pi(2k + 1) \mid k \in \mathbb{Z} \right\}.$$

9. Since

$$\begin{aligned} \cos x - 2 \sin 2x - \cos 3x &= (\cos x - \cos 3x) - 2 \sin 2x \\ &= 2 \sin x \sin 2x - 2 \sin 2x = 2 \sin 2x (\sin x - 1), \end{aligned}$$

and

$$\begin{aligned} 1 - 2 \sin x - \cos 2x &= (1 - \cos 2x) - 2 \sin x \\ &= 2 \sin^2 x - 2 \sin x = 2 \sin x (\sin x - 1), \end{aligned}$$

we have

$$|\sin 2x (\sin x - 1)| = \sin x (\sin x - 1).$$

Then, taking into account that $|\sin x - 1| = -(\sin x - 1)$ (because $\sin x - 1 \leq 0$), we get

$$-2 |\sin x| |\cos x| (\sin x - 1) = \sin x (\sin x - 1),$$

and hence, $\sin x - 1 = 0$, $x = -\frac{\pi}{2} + 2\pi k$, $k \in \mathbb{Z}$, or

$$2 |\sin x| |\cos x| = -\sin x.$$

Let us solve the last equation. If $\sin x \geq 0$, then $|\sin x| = \sin x$, and this means that

$$2 \sin x |\cos x| = -\sin x,$$

$$\sin x = 0, x = \pi k, k \in \mathbb{Z}, \text{ or } 2 |\cos x| = -1.$$

We see that the equation has no solutions. If $\sin x < 0$, then $|\sin x| = -\sin x$ and, therefore,

$$2 \sin x |\cos x| = \sin x.$$

Since $\sin x \neq 0$ ($\sin x < 0$), we have $|\cos x| = 1/2$, whence we have

$$\cos x = \pm 1/2, x = \pm \pi/3 + 2\pi k, \text{ or } \cos x = -1/2, x = \pm 2\pi/3 + 2\pi k, k \in \mathbb{Z}.$$

From the values of x obtained, we must choose only those for which $\sin x < 0$. In the first case we have $\sin\left(\pm \frac{\pi}{3} + 2\pi k\right) = \pm \sqrt{3}/2$ and this means that we must take only the values corresponding to the lower sign, i.e. $x = -\frac{\pi}{3} + 2\pi k$. In the second case we must also take the values corresponding to the lower sign, i.e. $x = -\frac{2\pi}{3} + 2\pi k$.

$$\text{Answer: } \left\{ -\frac{\pi}{3} + 2\pi k, -\frac{2\pi}{3} + 2\pi k \mid k \in \mathbb{Z} \right\}.$$

10. The function $\arccos y$ is the inverse of the function $y = \cos x$ for which the interval $[0, \pi]$ serves as the domain of definition. This means that for any $x \in [0; \pi]$ we have

$$\arccos(\cos x) = x$$

and for any $y \in [-1; 1]$ we have

$$\cos(\arccos y) = y.$$

If we designate $\arccos a = \alpha$ then we have $\cos \alpha = a$. Since $\alpha \in [0; \pi]$, it follows that $\pi - \alpha \in [0; \pi]$ as well. Bearing in mind that $\cos(\pi - \alpha) = -\cos \alpha = -a$, we get

$$\arccos(-a) = \arccos(\cos(\pi - \alpha)) = \pi - \alpha = \pi - \arccos a,$$

that is, $\arccos a + \arccos(-a) = \pi$, and that is what we had to prove.

11. Since

$$\int_0^x (t^2 - 8t + 13) dt = \left(\frac{1}{3} t^3 - 8 \cdot \frac{t^2}{2} + 13t \right) \Big|_0^x = \frac{x^3}{3} - 4x^2 + 13x,$$

we have $\frac{x^3}{3} - 4x^2 + 13x = x \sin \frac{a}{x}$. Hence, taking into account that $x \neq 0$ (for $x = 0$ the right-hand side of the equation is not defined), we get an equation

$$\frac{x^2}{3} - 4x + 13 = \sin \frac{a}{x}, \quad (1)$$

which is equivalent to the original equation. In the left-hand side we isolate a perfect square:

$$\frac{x^2}{3} - 4x + 13 = \frac{1}{3} (x^2 - 12x + 36 + 3) = \frac{1}{3} (x - 6)^2 + 1.$$

Now we see that $\frac{x^2}{3} - 4x + 13 \geq 1$ for all x . At the same time, $\sin(a/x) \leq 1$, and this means that equation (1) is possible if and only if

$$\begin{cases} \frac{1}{3}(x-6)^2 + 1 = 1, \\ \sin(a/x) = 1. \end{cases}$$

Hence it follows that $x = 6$, $\sin(a/6) = 1$. The last equation holds only in the case when $\frac{a}{6} = \frac{\pi}{2} + 2\pi k$, i.e. $a = 3(1 + 4k)\pi$, where $k \in \mathbb{Z}$. Thus, the original equation has a solution only for $a = 3(1 + 4k)\pi$, $k \in \mathbb{Z}$, and this solution is $x = 6$.

Answer: $a = 3(1 + 4k)\pi$, $k \in \mathbb{Z}$; $x = 6$.

12. From the second equation we obtain

$$\sin x - \sin\left(\frac{\pi}{2} - 2y\right) = 0,$$

$$2 \sin\left(\frac{x}{2} + y - \frac{\pi}{4}\right) \cos\left(\frac{x}{2} - y + \frac{\pi}{4}\right) = 0,$$

and, consequently, $\sin\left(\frac{x}{2} + y - \frac{\pi}{4}\right) = 0$, whence we have $x = \frac{\pi}{2} - 2y + 2\pi k$, $k \in \mathbb{Z}$, or $\cos\left(\frac{x}{2} - y + \frac{\pi}{4}\right) = 0$, whence we have $x = \frac{\pi}{2} + 2y + 2\pi k$, $k \in \mathbb{Z}$.

Substituting the values $x = \frac{\pi}{2} - 2y + 2\pi k$ into the first equation and taking into consideration that $\tan\left(\frac{\pi}{2} - 2y + 2\pi k\right) = \cot 2y$, we obtain

$$\cot 2y = \tan^3 y.$$

Expressing $\cot 2y$ in terms of $\tan y$, we get

$$\frac{\tan^2 y - 1}{2 \tan y} = \tan^3 y, \quad 2 \tan^4 y - \tan^2 y + 1 = 0.$$

This equation, quadratic with respect to $\tan^2 y$, has no solutions.

Substituting, by analogy, $x = \frac{\pi}{2} + 2y + 2\pi k$ into the first equation we get

$$-\cot 2y = \tan^3 y, \quad 2 \tan^4 y + \tan^2 y - 1 = 0,$$

whence it follows that $\tan^2 y = -1$ or $\tan^2 y = 1/2$. It is evident that only the second equation has solutions, and $y = \pm \arctan(1/\sqrt{2}) + \pi l$, $l \in \mathbb{Z}$. Hence $x = \frac{\pi}{2} \pm 2 \arctan \frac{1}{\sqrt{2}} + 2\pi(l+k)$, $l, k \in \mathbb{Z}$.

Note that whatever the value of l , the sum $l+k$ can be equal to any integral m because of the arbitrariness of k . In other words, each value $y = \pm \arctan \frac{1}{\sqrt{2}} + \pi l$ is associated with all the values $x = \frac{\pi}{2} \pm 2 \arctan \frac{1}{\sqrt{2}} + 2\pi m$, $m \in \mathbb{Z}$.

Answer: $\left\{ \frac{\pi}{2} \pm 2 \arctan \frac{1}{\sqrt{2}} + 2\pi m; \pm \arctan \frac{1}{\sqrt{2}} + \pi b \mid m, l \in \mathbb{Z} \right\}$.

13. Adding up the equations of the system term-by-term, we get

$$2(\cos x \cos y + \sin x \sin y) + \cos 2x + \cos 2y = 0,$$

whence we obtain $2 \cos(x-y) + 2 \cos(x+y) \cos(x-y) = 0$,

$$\cos(x-y)(1 + \cos(x+y)) = 0.$$

Together with any equation from the original system, say, with the first, this equation forms the system

$$\begin{cases} 2 \sin x \sin y + \cos 2y - \sqrt{2} = 0, \\ \cos(x-y)(1 + \cos(x+y)) = 0, \end{cases} \quad (1)$$

which is equivalent to the original system. It follows from the second equation of system (1) that

$$\cos(x-y)=0, \quad x=\frac{\pi}{2}+y+\pi k,$$

or

$$\cos(x+y)=-1, \quad x=\pi-y+2\pi k, \quad \text{where } k \in \mathbb{Z}.$$

Substituting $x=\frac{\pi}{2}+y+\pi k$ into the first equation of system (1) we get

$$(-1)^k 2 \cos y \sin y + \cos 2y - \sqrt{2} = 0, \quad \cos 2y + (-1)^k \sin 2y = \sqrt{2}.$$

(Introducing an auxiliary angle, we get $\cos\left(2y - (-1)^k \frac{\pi}{4}\right) = 1$,

whence we obtain $y = (-1)^k \frac{\pi}{8} + \pi l$, $l \in \mathbb{Z}$, and, hence,

$$x = \frac{\pi}{2} + (-1)^k \frac{\pi}{8} + \pi(k+l), \quad k, l \in \mathbb{Z}.$$

Substituting $x = \pi - y + 2\pi k$ into the first equation of system (1), we get $2 \sin^2 y + \cos 2y - \sqrt{2} = 0$,

$$1 - \cos 2y + \cos 2y - \sqrt{2} = 0, \quad 1 - \sqrt{2} = 0,$$

which is impossible.

$$\text{Answer: } \left\{ \left(\frac{\pi}{2} + (-1)^k \frac{\pi}{8} + \pi(k+l); (-1)^k \frac{\pi}{8} + \pi l \right) \mid k, l \in \mathbb{Z} \right\}.$$

14. After the substitution $x + \frac{\pi}{4} = s$, $y + \frac{\pi}{4} = t$, the system acquires the form

$$\begin{cases} \tan s = \sin 2t, \\ \sin 2s = \tan t. \end{cases} \quad (1)$$

Multiplying the equations of the system term-by-term, we get

$$\begin{aligned} \tan s \sin 2s &= \tan t \sin 2t, & 2 \sin^2 s &= 2 \sin^2 t, \\ 1 - \cos 2s &= 1 - \cos 2t, & \cos 2t - \cos 2s &= 0, \\ \sin(s-t) \sin(s+t) &= 0, \end{aligned}$$

whence we have $\sin(s-t) = 0$, $s = t + \pi k$, $k \in \mathbb{Z}$, or $\sin(s+t) = 0$, $s = -t + \pi l$, $l \in \mathbb{Z}$. The substitution $s = t + \pi k$ into system (1) leads to one equation $\tan t = \sin 2t$, which has the solutions

$t = \pi m$, $t = \frac{\pi}{4} + \frac{\pi m}{2}$, where $m \in \mathbb{Z}$. By the formula $s = t + \pi k$ we

find, respectively, $s = \pi(m+k)$, $s = \frac{\pi}{4} + \pi\left(\frac{m}{2} + k\right)$, $m, k \in \mathbb{Z}$. The set of solutions $s = \pi(m+k)$, $k \in \mathbb{Z}$, corresponding to $t = \pi m$, can be written as $s = \pi n$, $n \in \mathbb{Z}$, for the same reasons as in Example 12.

Substituting $s = -t + \pi l$ into system (1) we find, by analogy, that $t = \pi m$ and, hence, $s = (l-m)\pi$, $l, m \in \mathbb{Z}$. For any m , the

difference $l - m$ may be any integer because of the arbitrariness of l and, therefore, the set of values of s , corresponding to $t = \pi m$, can be written in the form $s = \pi n$, $n \in \mathbb{Z}$. It is evident that this set of values of s and t belongs to the set obtained earlier. Now we find

$$\begin{cases} x = -\frac{\pi}{4} + \pi n, & y = -\frac{\pi}{4} + \pi m, & m, n \in \mathbb{Z}; \\ x = \pi \left(\frac{m}{2} + k \right), & y = \frac{\pi m}{2}, & m, k \in \mathbb{Z}. \end{cases} \quad (2)$$

Answer: The solution of the system is given by formulas (2).

15. The given inequality can be easily reduced to the quadratic inequality

$$2 \sin^2 x + \sin x - 1 \geq 0$$

with respect to $\sin x$. Solving it, we obtain

$$\sin x \leq -1 \text{ or } \sin x \geq 1/2.$$

The first inequality can be reduced to the equation $\sin x = -1$ with solutions $x = -\frac{\pi}{2} + 2\pi k$, $k \in \mathbb{Z}$. The set of solutions of the second inequality is the union of the intervals $\frac{\pi}{6} + 2\pi k \leq x \leq \frac{5\pi}{6} + 2\pi k$, $k \in \mathbb{Z}$.

$$\text{Answer: } x = -\frac{\pi}{2} + 2\pi k, \frac{\pi}{6} + 2\pi k \leq x \leq \frac{5\pi}{6} + 2\pi k, k \in \mathbb{Z}.$$

16. Expressing $\tan 2x$ in terms of $\tan x$, we transform the inequality as follows:

$$\begin{aligned} \frac{4 \tan x}{1 - \tan^2 x} - 3 \tan x &\leq 0, \\ \frac{\tan x}{1 - \tan^2 x} (1 + 3 \tan^2 x) &\leq 0. \end{aligned}$$

Since $1 + 3 \tan^2 x > 0$, we have

$$\frac{\tan x}{1 - \tan^2 x} \leq 0.$$

This inequality is, evidently, equivalent to the original one. Solving it with respect to $\tan x$, we obtain $-1 < \tan x \leq 0$ and $1 < \tan x$.

From this we find $-\frac{\pi}{4} + \pi n < x \leq \pi n$, $\frac{\pi}{4} + \pi n < x < \frac{\pi}{2} + \pi n$, $n \in \mathbb{Z}$, respectively.

$$\text{Answer: } -\frac{\pi}{4} + \pi n < x \leq \pi n, \frac{\pi}{4} + \pi n < x < \frac{\pi}{2} + \pi n, n \in \mathbb{Z}.$$

17. Let us draw the graphs of the functions $\sin x$ and $|\cos x|$ (Fig. 204) and use the figure to make the reasoning more visual. On the interval $[\pi; 2\pi]$, $\sin x \leq 0$ and $|\cos x| \geq 0$, with $|\cos x| = 0$ only for $x = 3\pi/2$, and in that case $\sin x = -1$. This means that all

numbers belonging to the interval $[\pi; 2\pi]$ satisfy the given inequality. On the interval $[\pi/2; \pi]$, $|\cos x| = -\cos x$, and the equation $\sin x = \cos x$ has one solution $x = 3\pi/4$. Since $\sin x$ decreases monotonically on that interval and $|\cos x|$ monotonically increases, all the numbers belonging to the interval $[3\pi/4; \pi]$ satisfy the inequality $\sin x < |\cos x|$. On the interval $[2\pi; 5\pi/2]$, $|\cos x| = \cos x$ and the equation $\sin x = \cos x$ has one solution $9\pi/4$. Here the function

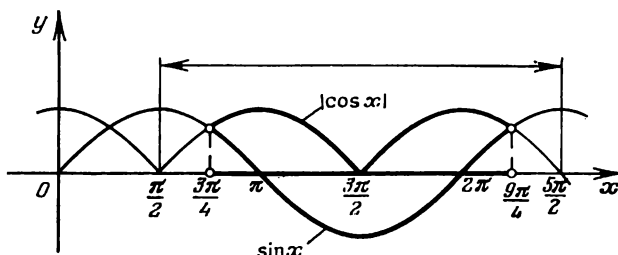


Fig. 204

$\sin x$ increases and $\cos x$ decreases and, therefore, all numbers belonging to the interval $[2\pi; 9\pi/4]$ satisfy the inequality $\sin x < |\cos x|$. Thus, on the interval $[\pi/2; 5\pi/2]$, which is 2π in length, all solutions of the original inequality form the interval $(3\pi/4; 9\pi/4)$. Since 2π is the common period of the functions $\sin x$ and $|\cos x|$, the set of all solutions of the given inequality is the union of the intervals $\left(\frac{3\pi}{4} + 2\pi n; \frac{9\pi}{4} + 2\pi n\right)$, $n \in \mathbb{Z}$.

$$\text{Answer: } \frac{3\pi}{4} + 2\pi n < x < \frac{9\pi}{4} + 2\pi n, n \in \mathbb{Z}.$$

CHAPTER 2

1. Passing to the base 2, we obtain, after obvious transformations, an equation

$$2^{2x} - 3 \cdot 2^{x + \sqrt{x}} - 4 \cdot 2^{\sqrt{x}} = 0,$$

which is a homogeneous equation of degree 2 with respect to 2^x and $2^{\sqrt{x}}$. Dividing it by $2^{2\sqrt{x}}$ and designating $t = 2^{x - \sqrt{x}}$, we obtain a quadratic equation $t^2 - 3t - 4 = 0$, which possesses the roots -1 and 4 . The equation $2^{x - \sqrt{x}} = -1$ has no solutions. The equation $2^{x - \sqrt{x}} = 4$ can be reduced to the equation $x - \sqrt{x} = 2$, which is quadratic with respect to $\sqrt{x}$. Its roots are 2 and -1 , but since $\sqrt{x} \geq 0$, it follows that $\sqrt{x} = 2$, whence we have $x = 4$.

Answer: 4.

2. Taking the logarithms to the base 5 (both sides of the equation being positive), we obtain an equation

$$\log_5 (8x) \cdot \log_5 x = \log_5 16 + \frac{4}{3} \log_5 x,$$

which is equivalent to the original one. We reduce it to the form $\log_5^2 x + \left(3 \log_5 2 - \frac{4}{3}\right) \log_5 x - 4 \log_5 2 = 0$ and, solving this equation quadratic with respect to $\log_5 x$, we find that $\log_5 x = 4/3$, $x = 5^{4/3}$; $\log_5 x = -3 \log_5 2$, $x = 1/8$.

Answer: $5^{4/3}$, $1/8$.

3. Taking antilogarithms, we get an equation

$$1 + \cos x = 2 \sin^2 x,$$

transforming it we find that $1 + \cos x = 2(1 + \cos x)(1 - \cos x)$, $(1 + \cos x)(2 \cos x - 1) = 0$. Consequently, $\cos x = -1$, $x =$

$\pi(2n + 1)$, $n \in \mathbb{Z}$; $\cos x = 1/2$, $x = \pm \frac{\pi}{3} + 2\pi n$, $n \in \mathbb{Z}$. The values $x = \pi(1 + 2n)$ and $x = -\frac{\pi}{3} + 2\pi n$ do not satisfy the condition

$\sin x > 0$. But, as is easy to verify, the values $x = \frac{\pi}{3} + 2\pi n$ are solutions of the original equation.

Answer: $\frac{\pi}{3} + 2\pi n$, where $n \in \mathbb{Z}$.

4. We pass to the base 3:

$$1 - \frac{1}{2} \log_3 (x + 1)^2 = \log_3 \frac{x + 5}{x + 3};$$

Taking into account that $\log_3 (x + 1)^2 = 2 \log_3 |x + 1|$, we get an equation

$$1 - \log_3 |x + 1| = \log_3 \frac{x + 5}{x + 3},$$

which yields

$$\frac{3}{|x + 1|} = \frac{x + 5}{x + 3}. \quad (1)$$

The left-hand side of the equation being positive, we can perform the inverse transformations and get equation (1), which is equivalent to the original equation. Then we solve equation (1). We have $3(x + 3) = |x + 1| \cdot (x + 5)$ and, if $x + 1 > 0$, we have $x^2 + 3x - 4 = 0$, whence it follows that $x = 1$, $x = -4$, the condition $x + 1 > 0$ being satisfied only by the value $x = 1$; if $x + 1 < 0$, then $x^2 + 9x + 14 = 0$, whence $x = -2$, $x = -7$, both these values satisfying the condition $x + 1 < 0$.

Answer: 1, -2, -7.

5. Passing to the base 10, we obtain

$$\frac{2}{\log(0.5 + \cos^2 x)} = \frac{1}{\log \sin 2x},$$

whence

$$\begin{aligned}\log \sin^2 2x &= \log (0.5 + \cos^2 x), \quad \sin^2 2x = 0.5 + \cos^2 x, \\ 1 - \cos^2 2x &= 1 + 0.5 \cos 2x, \quad \cos 2x (\cos 2x + 0.5) = 0.\end{aligned}$$

(a) $\cos 2x = 0$, $2x = \frac{\pi}{2} + \pi n$, and in this case we have $\sin 2x = (-1)^n$, that is, the right-hand side of the original equation is not defined.

(b) $\cos 2x = -0.5$, $2x = \pm \frac{2\pi}{3} + 2\pi n$, $x = \pm \frac{\pi}{3} + \pi n$. The values $x = -\frac{\pi}{3} + \pi n$ do not satisfy the condition $\sin 2x > 0$ whereas the values $x = \frac{\pi}{3} + \pi n$ are solutions of the original equation, which fact can be easily verified.

Answer: $\frac{\pi}{3} + \pi n$, $n \in \mathbb{Z}$.

6. Passing to the base 2, we obtain

$$\frac{1}{\log_2 \sin x} + \frac{1}{\log_2 \cos x} + \frac{1}{\log_2 \sin x \cdot \log_2 \cos x} = 0,$$

whence $\log_2 \cos x + \log_2 \sin x + 1 = 0$, $\log_2 \sin 2x = 0$, $\sin 2x = 1$, $2x = \frac{\pi}{2} + 2\pi n$. We can make sure by verification that only the values $x = \frac{\pi}{4} + 2\pi n$ can serve as solutions of the original equation (it is easy to see that it is sufficient to verify the conditions $0 < \sin x < 1$, $0 < \cos x < 1$).

Answer: $\frac{\pi}{4} + 2\pi n$, $n \in \mathbb{Z}$.

7. Designating $t = x - 1$, we have $x = t + 1$ and $\log_t (t + 1 + a) = 1/2$. If we also introduce the designation $b = 1 + a$, the original equation will assume the form

$$\log_t (t + b) = 1/2. \quad (1)$$

Taking antilogarithms to the base t , we get

$$t + b = \sqrt{t}. \quad (2)$$

Every solution of (1) is evidently a solution of (2). Conversely, if t is a solution of (2) and

$$t > 0, \quad t \neq 1, \quad (3)$$

then, taking logarithms in (2) to the base t , we find that (1) is a true equality.

Solving (2) as an equation quadratic with respect to $\sqrt{t}$, we find that

$$(1^\circ) \quad \sqrt{t} = \frac{1}{2} (1 + \sqrt{1-4b}) \quad \text{and}$$

$$(2^\circ) \quad \sqrt{t} = \frac{1}{2} (1 - \sqrt{1-4b}).$$

It is clear that (2) possesses solutions only if $1 - 4b \geq 0$, i.e. $b \leq 1/4$. Under this condition $1 + \sqrt{1-4b} \geq 1$ and, therefore, formula (1°) defines the solution $t_1 = \frac{1}{2} (1 - 2b + \sqrt{1-4b})$ of equation (2). The first condition of (3) is evidently satisfied for the solution. In addition, it is clear from (1°) that the equation $\sqrt{t_1} = 1$ (and, hence $t_1 = 1$) is possible only for $b = 0$. Thus, if $b \leq 1/4$ and $b \neq 0$, then (1°) gives the solution t_1 of equation (1). Let us consider formula (2°). The condition $\sqrt{t} > 0$ (and, hence, $t > 0$ from (3)) is satisfied if and only if $1 - \sqrt{1-4b} > 0$, whence it follows that $b > 0$.

In this case, $1 - \sqrt{1-4b} < 1$, and $\sqrt{t} < 1/2$ and $t < 1/4 < 1$, that is, the second condition of (3) is also satisfied. This means that

formula (2°) gives the solution $t_2 = \frac{1}{2} (1 - 2b - \sqrt{1-4b})$ of equation

(1) if $0 < b \leq 1/4$. As a result we find that equation (1) has one solution t_1 for $b < 0$, and for $0 < b \leq 1/4$ its solutions are t_1 and t_2 , with $t_1 = t_2$ only at $b = 1/4$. This means that (1), and the original equation together with it, have a unique solution only for $b < 0$ or $b = 1/4$ or, which is the same, for $a < -1$ or $a = -3/4$.

Answer: $a < -1$, $a = -3/4$.

8. In the first equation we pass to the base y :

$$\log_y x + \frac{1}{\log_y x} = 2.5.$$

After obvious transformations we arrive at an equation $\log_y^2 x - 2.5 \log_y x + 1 = 0$, whence $\log_y x = 2$, $\log_y x = 0.5$ and, hence, $x = y^2$, $x = \sqrt{y}$. In the first case, we find from the second equation of the system that $y^3 = 27$, $y = 3$, and then $x = 9$. In the second case $y^{1.5} = 27$, $y = 9$, and $x = 3$.

Answer: (3; 9), (9; 3).

9. If we designate $u = 2^{x+y}$, $v = 3^{x-y}$, we have

$$\begin{cases} u^2 = 27 + v^2, \\ u^3 - 21u = v^3 + 21v, \end{cases} \quad \begin{cases} u^2 - v^2 = 27, \\ u^3 - v^3 = 21(u + v), \end{cases}$$

$$\begin{cases} (u - v)(u + v) = 27, \\ (u - v)(u^2 + uv + v^2) = 21(u + v). \end{cases}$$

Taking into account that $u + v > 0$, we express $u - v = 27(u + v)^{-1}$ from the first equation of the last system, substitute it into the second equation and transform it as follows: $27(u^2 + uv + v^2) = 21(u + v)^2$, $2u^2 - 5uv + 2v^2 = 0$. Solving this equation as a quadratic equation with respect to u , we find that $u = 2v$, $u = v/2$. Substitution into the equation $u^2 - v^2 = 27$ yields $4v^2 - v^2 = 27$, $v = 3$ (since $v > 0$), and then, $u = 6$; $\frac{1}{4}v^2 - v^2 = 27$, $-\frac{3}{4}v^2 = 27$. The equation has no solutions. Thus we have $2^{x+v} = 6$, $3^{x-v} = 3$, whence it follows that $x + y = \log_2 6$, $x - y = 1$, and $x = \frac{1}{2} \log_2 12 = \log_4 12$, $y = \frac{1}{2} \log_2 3 = \log_4 3$.

Answer: $(\log_4 12; \log_4 3)$.

10. Taking logarithms to the base 10, we get a system

$$\begin{cases} \log 5 \cdot \log x = \log 3 \cdot \log y \\ \log 3 \cdot (\log 3 + \log x) = \log 5 \cdot (\log 5 + \log y), \end{cases}$$

which is equivalent to the original system. We find $\log y = \frac{\log 5}{\log 3} \log x$ from the first equation, substitute the result into the second equation and transform it as follows: $\log 3 \cdot (\log 3 + \log x) = \log 5 \cdot \left(\log 5 + \frac{\log 5}{\log 3} \log x \right)$, $(\log^2 3 - \log^2 5)(\log 3 + \log x) = 0$, whence we have $\log 3 + \log x = 0$ since $\log^2 3 - \log^2 5 \neq 0$. Hence, $x = 1/3$, and then $\log y = -\log 5$, $y = 1/5$.

Answer: $(1/3, 1/5)$.

11. As was noted earlier, if the base and the exponent contain a variable, then only those values of the variable are considered everywhere here for which the base is positive. In the present case, we should consider only the values $x > 0$, $y > 0$. Taking this into account and taking logarithms to the base 10, we get a system

$$\begin{cases} y \log x = x \log y, \\ x \log x = 9y \log y, \end{cases}$$

which is equivalent to the original system. We find $\log y = \frac{y}{x} \log x$ in the first equation and substitute the result into the second equation:

$x \log x = 9y \cdot \frac{y}{x} \log x$, $(x^2 - 9y^2) \log x = 0$. Hence we have

(1) $\log x = 0$, $x = 1$, and then $\log y = 0$, $y = 1$;

(2) $x^2 - 9y^2 = 0$, $x = 3y$ (since $x > 0$ and $y > 0$) and this means

that $\log y = \frac{1}{3} \log 3y$, $y^3 = 3y$, $y^2 = 3$, $y = \sqrt[3]{3}$, and then $x = 3\sqrt[3]{3}$.

Answer: $(1; 1)$ $(3\sqrt[3]{3}; \sqrt[3]{3})$.

12. We transform the system as follows:

$$\begin{cases} \log_2 \frac{x^2 + y^2}{x} = \log_2 \frac{x+3y}{2}, \\ \log_2 \frac{xy+1}{2y^2+y-x+2} = \log_2 \frac{x}{2y}, \end{cases} \quad \begin{cases} \frac{x^2 + y^2}{x} = \frac{x+3y}{2}, \\ \frac{xy+1}{2y^2+y-x+2} = \frac{x}{2y}, \end{cases} \quad \begin{cases} x^2 - 3xy + 2y^2 = 0, \\ x^2 - xy + 2y - 2x = 0. \end{cases} \quad (1)$$

Every solution of the original system is also a solution of system (1), the converse may not always be true. All solutions of system (1) must be verified by substitution into the original system. We write the second equation of system (1) in the form $(y-x)(x-2) = 0$. If $x = 2$, then we find from the first equation of system (1) that $y^2 - 3y + 2 = 0$, whence we have $y = 2$, $y = 1$. This means that $(2; 2)$, $(2; 1)$ are solutions of system (1). If $y = x$, then the first equation of system (1) is a true equality: $x^2 - 3x^2 + 2x^2 = 0$. This means that any pair of numbers of the form $(c; c)$ is also a solution of system (1). The solution $(2; 2)$ obtained before is contained in this set.

Let us verify which of the solutions $(2; 1)$, $(c; c)$, where $c \in \mathbb{R}$, of system (1) satisfy the original system. We can easily verify by means of substitution that $(2; 1)$ is a solution of that system. It follows from the first equation of the original system that $x > 0$ and, therefore, we should consider only the pairs $(c; c)$, where $c > 0$. After the substitution, we obtain

$$\begin{cases} \log_2 (2c^2) - \log_2 c = \log_2 (4c) - 1, \\ \log_2 \frac{c^2+1}{2c^2+2} = -1, \end{cases}$$

and now it is easy to see that these equalities are true for $c > 0$. This means that all pairs of the form $(c; c)$, where $c > 0$, are also solutions of the original system.

Answer: $(2; 1)$, $(c; c)$, where $c > 0$.

13. Multiplying the first equation of the system by $2^{-x+y} \cdot 3^{2y}$ term-by-term, we obtain $3^{2y} - 2 \cdot 3^{x+y} - 2^{-x-y} = 0$, whence we have $2^{-x-y} = 3^{2y} - 2 \cdot 3^{x+y}$. Substitution into the second equation of the system yields $3^{2y} - 4 \cdot 3^{x+y} + 3 \cdot 3^{2x} = 0$, or, after the multiplication by 3^{-2x} , $3^{2(y-x)} - 4 \cdot 3^{y-x} + 3 = 0$. Solving this equation, quadratic with respect to 3^{y-x} , we find that $3^{y-x} = 3$, $3^{y-x} = 1$. Hence we have $y - x = 1$, $y - x = 0$. If $y - x = 1$, then it follows from the first equation of the original system that $6^{-2y} = 1/6$, $y = 1/2$, and then we have $x = y - 1 = -1/2$. Now if $y - x = 0$, then we have, by analogy, $6^{-2y} = -1$, and in this case there are no solutions.

Answer: $(-1/2; 1/2)$.

14. We designate $t = 3^x$ and then the inequality has the form $\frac{7}{t^2-2} \geq \frac{2}{t-1}$. Transforming it, we get $\frac{2t^2-7t+3}{(t^2-2)(t-1)} \leq 0$. Fac-

toring the numerator as a quadratic trinomial and $t^2 - 2$ as the difference between the squares, we have $\frac{2(t-3)(t-0.5)}{(t+\sqrt{2})(t-\sqrt{2})(t-1)} \leq 0$, whence taking into account that $t > 0$ and, hence, $t + \sqrt{2} > \sqrt{2} > 0$, we arrive at an inequality $\frac{(t-3)(t-0.5)}{(t-\sqrt{2})(t-1)} \leq 0$. Solving it, say, by the method of intervals, we find that $1/2 \leq t < 1$, $\sqrt{2} < t \leq 3$. Hence it follows that $1/2 \leq 3^x < 1$, $\sqrt{2} < 3^x \leq 3$ and, correspondingly, $-\log_3 2 \leq x < 0$, $\frac{1}{2} \log_3 2 < x \leq 1$.

Answer: $[-\log_3 2; 0) \cup (\frac{1}{2} \log_3 2; 1]$.

15. We pass to the base 3: $\log_3(x+2) > \frac{4}{\log_3(x+2)}$ and, introducing the designation $t = \log_3(x+2)$, we reduce the inequality to the form $\frac{(t-2)(t+2)}{t} > 0$. Solving this inequality, we find that $-2 < t < 0$, $2 < t$. From this we get

$$\begin{aligned} -2 < \log_3(x+2) < 0, \quad 1/9 < x+2 < 1, \quad -17/9 < x < -1; \\ 2 < \log_3(x+2), \quad 9 < x+2, \quad 7 < x. \end{aligned}$$

Answer: $(-17/9; -1) \cup (7; +\infty)$.

16. We pass to the base 2 and designate $\log_2 x = t$ and get, as a result, the inequality

$$\frac{8(t^2+2t)}{3(t+1)(t+3)} < 5.$$

Transforming it, we obtain

$$\frac{7t^2+44t+45}{(t+1)(t+3)} > 0, \quad \frac{\left(t+\frac{9}{7}\right)(t+5)}{(t+1)(t+3)} > 0.$$

By the method of intervals we find that $t < -5$, $-3 < t < -9/7$, $-1 < t$. From this we find, respectively, that $\log_2 x < -5$, $0 < x < 2^{-5}$, $-3 < \log_2 x < -9/7$, $2^{-3} < x < 2^{-9/7}$, $-1 < \log_2 x$, $0.5 < x$.

Answer: $(0; 2^{-5}) \cup (2^{-3}; 2^{-9/7}) \cup (0.5; +\infty)$.

17. If $x > 1$, then $6x - 1 > 2x$, $x > 1/4$. This means that all the values $x > 1$ are solutions of the original equation. If $0 < x < 1$, then $0 < 6x - 1 < 2x$. The left-hand inequality $0 < 6x - 1$ has solutions $x > 1/6$ and the right-hand inequality has solutions $x < 1/4$. This means that for $0 < x < 1$ the original inequality has solutions $1/6 < x < 1/4$.

Answer: $(1/6; 1/4) \cup (1; +\infty)$.

18. Since $x > 0$, we have $\log_a x^2 = 2 \log_a x$. We pass to the base a and transform the inequality as follows:

$$\log_a x \cdot \frac{1 - 2 \log_a (x+a)}{\log_a (x+a)} \leq 0. \quad (1)$$

Suppose $a > 1$. Then $x + a > a > 1$, $\log_a (x+a) > 1 > 0$, $1 - 2 \log_a (x+a) < 0$ and, therefore, inequality (1) can be reduced to the inequality $\log_a x \geq 0$, whence we have $x \geq 1$. This means that for $a > 1$ the set of solutions of the original equation is the interval $[1; +\infty)$.

Assume that $0 < a < 1$. If $x > 1$, then $\log_a x < 0$, $\log_a (x+a) < 0$, and $1 - 2 \log_a (x+a) > 0$, and, hence, inequality (1) is not true. The number $x = 1$ is a solution of equation (1). If $0 < x < 1$, then $\log_a x > 0$ and equation (1) can be reduced to the inequality

$$\frac{1 - 2 \log_a (x+a)}{\log_a (x+a)} \leq 0.$$

Solving it for $\log_a (x+a)$, we find that $\log_a (x+a) \geq 1/2$, $\log_a (x+a) < 0$. From this we have, correspondingly, $x+a \leq \sqrt{a}$, $x \leq \sqrt{a} - a$; $x+a > 1$, $x > 1-a$.

Thus, for $0 < a < 1$ inequality (1) has solutions $0 < x \leq \sqrt{a} - a$, $1-a < x \leq 1$.

Answer: If $a > 1$, then $x \geq 1$; if $0 < a < 1$, then $0 < x \leq \sqrt{a} - a$, $1-a < x \leq 1$.

19. Setting $x > 1$, we transform the inequality:

$$2 + \log_3 x < \frac{6(x-1)}{2x-1}, \quad \log_3 x < \frac{2(x-2)}{2x-1} \quad (1)$$

In the case $0 < x < 1$ we have

$$2 + \log_3 x > \frac{6(x-1)}{2x-1}, \quad \log_3 x > \frac{2(x-2)}{2x-1}. \quad (2)$$

To make it more illustrative, we depict the graphs of the functions $\log_3 x$ and $2(x-2)/(2x-1)$ (Fig. 205). Suppose $x > 1$ (inequality (1)). If $1 < x \leq 2$, then $\log_3 x > 0$ but $2(x-2)/(2x-1) \leq 0$, and this means that there are no solutions here. If $x \geq 3$, then $\log_3 x \geq 1$, and $\frac{2(x-2)}{2x-1} = \frac{2x-1-3}{2x-1} = 1 - \frac{3}{2x-1} < 1$ and, therefore, there are no solutions here either. Suppose $2 < x < 3$. Here $\log_3 x >$

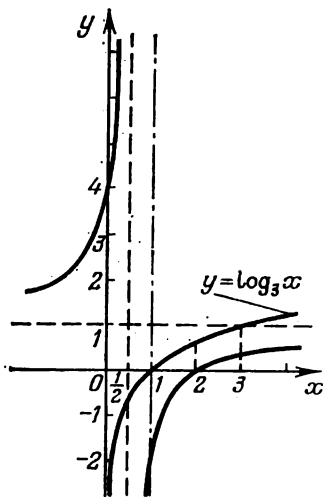


Fig. 205

$\log_3 2$, and the function $2(x-2)/(2x-1)$ increases monotonically (since $3/2x-1$ decreases monotonically and, hence, $1-\frac{3}{2x-1}$ increases) and this means that $2(x-2)/(2x-1) < 2 \cdot (3-2)/(2 \cdot 3-1) = 2/5$. Let us compare the numbers $\log_3 2$ and $2/5$. We have $\frac{2}{5} = \log_3 3^{2/5} = \log_3 9^{1/5} < \log_3 32^{1/5} = \log_3 2$. Hence it follows that

$$\log_3 x > \log_3 2 > 2/5 > 2(x-2)/(2x-1)$$

for $2 < x < 3$. Thus it follows that for $2 < x < 3$ as well and, hence, for all $x > 1$, inequality (1) has no solutions.

Suppose $0 < x < 1$ (inequality (2)). If $0 < x < 1/2$, then $\log_3 x < 0$ and $2(x-2)/(2x-1) > 0$, i.e. $\log_3 x < 2(x-2)/(2x-1)$, and all these values of x do not serve as solutions to inequality (2). For $1/2 < x < 1$ the function $2(x-2)/(2x-1) = 1 -$

$3/(2x-1)$ increases monotonically and, hence, $\frac{2(x-2)}{2x-1} < 2(1-2)/$

$(2 \cdot 1 - 1) = -2$. At the same time we have here $\log_3 x > \log_3 (1/2) = -\log_3 2 > -1$. Therefore, $\log_3 x > 2(x-2)/(2x-1)$, that is, all values $1/2 < x < 1$ are solutions of inequality (2).

Answer: $(1/2; 1)$.

CHAPTER 3

1. From the five-element set of all the workers we isolate three-element subsets consisting of the workers who leave for vacations. In case (a) we must consider ordered subsets (the first worker gets the first place in the sanatorium, the second worker, the second place, the third worker, the third place) and, therefore, their number is equal to $A_5^3 = 5 \cdot 4 \cdot 3 = 60$. In case (b) the order in which the workers are chosen is of no importance since all the places in the sanatorium are of equal value and, consequently, the number of ways of choosing is equal to $C_5^3 = C_5^2 = \frac{5 \cdot 4}{2} = 10$.

2. Let us first consider case (a). Suppose the white balls lie side-by-side, then we can arrange black balls so that we have one black ball for six places: 4 places between five white balls and 2 places in front and behind the white balls. But we can arrange four balls in six places by C_6^4 ways because there are as many four-element subsets in a six-element set. Consequently, in case (a) the number of ways is equal to $C_6^4 = C_6^2 = \frac{6 \cdot 5}{2} = 15$. In case (b) we can obtain $5! \cdot 4!$ choices from each arrangement of the balls considered in case (a) since the white balls can be permuted in $5!$ ways and the black balls in $4!$ ways. Therefore, in case (b) the number of ways is equal to $C_6^4 \cdot 5! \cdot 4! = 43200$.

3. Connecting every pair of vertices of the polygon, we get either a diagonal or a side of the polygon. The number of different pairs of vertices of an n -gon is equal to C_n^2 (this is the number of two-element subsets in an n -element set). Since there are n sides in an n -gon, the

number of diagonals is

$$C_n^2 - n = \frac{n(n-1)}{2} - n = \frac{n(n-3)}{2}.$$

4. There may be triangles of two kinds. In the triangles of the first kind one vertex lies on the first line and two vertices lie on the second line. We can use 10 ways of choosing the vertex on the first line and C_{20}^2 ways of choosing the two vertices on the second line. There are, evidently, $10 \cdot C_{20}^2$ triangles of the first kind. In the triangles of the second kind one vertex lies on the second line and two other vertices lie on the first line. The number of such triangles can be calculated by analogy. It is equal to $20 \cdot C_{10}^2$. Thus, the desired number of all triangles is

$$10 \cdot C_{20}^2 + 20 \cdot C_{10}^2 = 10 \frac{20 \cdot 19}{2} + 20 \frac{10 \cdot 9}{2} = 100(19 + 9) = 2800.$$

5. **First solution.** A set consisting of 17 elements contains C_{17}^5 five-element subsets. Therefore, we can choose chapters for the first writer in C_{17}^5 ways. Similarly, we can choose 4 chapters from the remaining 12 in C_{12}^4 ways. Five chapters for the third writer can be chosen in C_8^5 ways, and after that 3 chapters remain for the fourth writer. Consequently, the number of ways of distributing the chapters is equal to

$$C_{17}^5 C_{12}^4 C_8^5 = 171531360.$$

Second solution. Let us enumerate, for the sake of convenience, the chapters in Arabic numerals and the writers in Roman numerals. Then we can use the following method to distribute the chapters:

I	II	III	IV
2, 4, 8, 11, 14	1, 5, 7, 13	9, 10, 15, 16, 17	3, 6, 12

Any permutation of the numbers in the second row will lead to a new distribution, provided that it does not involve a permutation of the numbers in the columns. Since there are $17!$ ways of permutations of the numbers in the second row and any permutation of numbers in a column (there are $5!$, $4!$, $5!$, $3!$ of them respectively) does not give new ways, it follows that the desired number of ways of distributing the chapters is

$$\frac{17!}{(5!)^2 \cdot 4! \cdot 3!} = 171531360.$$

6. Let us calculate the maximum number of crocodiles with different sets of teeth. It is evident that their number is equal to the number of all subsets of a 68-element set, that is, 2^{68} . Since $16^{17} = 2^{68}$, it follows that 16^{17} crocodiles may have different sets of teeth. It is clear that if there are more than 16^{17} crocodiles, then there can always be found at least two with the same collection of teeth.

7. Let us consider a 10-element set of places on which the digits of the given number are written. We take its any subset and put the number 1 on the places belonging to that subset. The other places will be filled by the number 2. We have thus written the number. Having isolated another subset and carried out the same operations, we shall get some other number. It is clear that we can write as many numbers as there are subsets in the 10-element set, i.e. $2^{10} = 1024$.

8. Let us consider the set of places on which we put dots and dashes when writing a letter containing k signs. This set consists of k elements. Let us now consider its any subset, fill it by dots and put dashes on the other places. We have written the letter. There will evidently be as many different letters whose notation requires k signs as there are subsets in the k -element set, namely, 2^k . Since, in accordance with the hypothesis, we cannot use more than four signs to write the letter, we can write

$$\sum_{k=1}^4 2^k = 2 + 2^2 + 2^3 + 2^1 = 30 \text{ letters.}$$

9. The first place in a car registration number can be occupied by any one of the 30 letters. Consequently, the first letter can be chosen in 30 ways. The second place can also be occupied by any one of the 30 letters and, therefore, the first two letters can be chosen in 30^2 ways. It is clear now that three letters can be chosen in 30^3 ways. Reasoning by analogy, we find that four letters can be chosen in 10^4 ways. Thus, we can compose numbers for $30^3 \cdot 10^4 = 27 \cdot 10^7$ cars.

10. This problem can be solved with the aid of the formulas for the number of associations and the number of permutations. We can obtain the result quicker and with less difficulty, however, if we reason as follows. We designate the number of partitionings into pairs in the case of $2n$ elements by R_{2n} and select some element. With the use of that element we can form a pair in $2n - 1$ ways. Every time $2n - 2$ elements will remain after the formation of one pair, which can be partitioned into pairs in R_{2n-2} ways. Therefore, $R_{2n} = (2n - 1) R_{2n-2}$. Using this recurrence formula, we obtain

$$R_{2n} = (2n - 1) R_{2n-2} = (2n - 1) (2n - 3) R_{2n-4} = \dots$$

$$\dots = (2n - 1) (2n - 3) \dots 3 \cdot R_2 = (2n - 1) (2n - 3) \dots 3 \cdot 1.$$

Thus, the number of partitions is equal to the product of all odd natural numbers not exceeding $2n - 1$.

11. The fifth term of the expansion is its middle term. In the present case, the formula for the general term of the expansion yields

$$T_5 = C_{10}^5 (x^{-1/5}) (x^{1/3})^5 = C_{10}^5 x^{2/3} = 252 x^{2/3}.$$

12. We form a system of inequalities:

$$T_5 > T_4, \quad T_5 > T_6.$$

Using the formula for the general term of the expansion, we get

$$\begin{cases} C_9^5 (2x)^4 3^5 > C_9^4 (2x)^5 3^4, \\ C_9^5 (2x)^4 3^5 > C_9^6 (2x)^3 3^6, \end{cases}$$

After cancelling, we obtain

$$\begin{cases} 3x^4 > 2x^5 \\ x^4 > x^3 \end{cases} \quad \text{or} \quad \begin{cases} 3 - 2x > 0, \\ x(x-1) > 0. \end{cases}$$

Solving the system we get

$$x \in (-\infty, 0) \cup (1; 3/2).$$

13. Let us write the general term of the expansion:

$$T_k = C_9^k (\sqrt{3})^{9-k} (\sqrt[3]{2})^k = C_9^k 3^{(9-k)/2} 2^{k/3}, \quad k=0, 1, 2, \dots, 9.$$

The numbers T_k are integral if the exponents $(9-k)/2$ and $k/3$ are integers. From the fact that $k/3$ is an integer it follows that k can assume the values 0; 3; 6; 9. For these values of k the exponent $(9-k)/2$ will be an integer only for $k=3$ and $k=9$. Consequently, only the third and the ninth term of the expansion are integers, and

$$T_3 = C_9^3 3^3 2 = 4536, \quad T_9 = C_9^9 2^3 = 8.$$

14. Setting $n=10$, $a=1$, and $b=3x+2x^3$ in Newton's formula, we obtain

$$(1+3x+2x^3)^{10} = \sum_{k=0}^{10} C_{10}^k (3x+2x^3)^k.$$

It is easy to see that only for $k=2$ and $k=4$ we get summands containing x^4 . Replacing by dots the summands which obviously contain x not in the fourth degree, we obtain

$$\begin{aligned} (1+3x+2x^3)^{10} &= C_{10}^2 (3x+2x^3)^2 + C_{10}^4 (3x+2x^3)^4 + \dots \\ &= C_{10}^2 12x^4 + \dots + C_{10}^4 (3x)^4 + \dots = (12C_{10}^2 + 3^4 C_{10}^4) x^4 + \dots \end{aligned}$$

Thus, the required coefficient is equal to $12C_{10}^2 + 3^4 C_{10}^4 = 17\,550$.

15. The sum of the coefficients of any polynomial

$$P_n(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

is equal to the value of the polynomial for $x=1$. Indeed,

$$P_n(1) = a_n + a_{n-1} + \dots + a_1 + a_0.$$

For $x=1$ the given polynomial turns into unity and, therefore, the sum of all its coefficients is equal to unity.

16. Let us use Newton's formula to write the expansions for $(1+x)^{2n}$ and $(1-x)^{2n}$:

$$(1+x)^{2n} = C_{2n}^0 + C_{2n}^1 x + C_{2n}^2 x^2 + \dots + C_{2n}^{2n} x^{2n},$$

$$(1-x)^{2n} = C_{2n}^0 - C_{2n}^1 x + C_{2n}^2 x^2 - \dots + C_{2n}^{2n} x^{2n}.$$

We set $x=1$ in these identities and have

$$C_{2n}^0 + C_{2n}^1 + C_{2n}^2 + \dots + C_{2n}^{2n} = 2^{2n},$$

$$C_{2n}^0 - C_{2n}^1 + C_{2n}^2 - \dots + C_{2n}^{2n} = 0.$$

Adding up these equalities term-by-term, we obtain

$$C_{2n}^0 + C_{2n}^2 + C_{2n}^4 + \dots + C_{2n}^{2n} = 2^{2n-1}.$$

17. There are a total of C_{10}^5 ways to take 5 tickets out of 10. Consequently, the number of outcomes in cases (a) and (b) is the same and equals C_{10}^5 .

In case (a) the favourable outcome consists in the fact that one ticket is taken out of two winning tickets (there are two ways of doing this) and four tickets are chosen out of eight failing tickets (this can be done in C_8^4 ways). The number of favourable outcomes is $2 \cdot C_8^4$ and the probability in case (a) is equal to the ratio $2C_8^4/C_{10}^5 = 5/9$.

In case (b) the favourable outcome consists in the fact that there are two winning tickets among the chosen five (there is one way of choosing them) and three failing tickets (they can be chosen in C_8^3 ways). Consequently, the number of favourable outcomes is equal to C_8^3 and the probability in case (b) is equal to the ratio $C_8^3/C_{10}^5 = 2/9$.

18. Every person can leave the lift cabin in seven ways (at every floor beginning with the first). Therefore, 7^5 outcomes are possible here. A favourable outcome consists in the fact that all five people leave the lift cabin at different floors. Consequently, the first person can do it in seven ways, the second person in six ways, the third person in five ways, the fourth person in four ways and the fifth person can do it only in three ways. Thus, the number of favourable outcomes is equal to $A_7^5 = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3$ and the desired probability is equal to

$$\frac{A_7^5}{7^5} = \frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3}{7^5} = \frac{360}{2401} \approx 0.15.$$

CHAPTER 4

1. (a) One of the antiderivatives of the function x^5 is the function $\frac{1}{6}x^6$. Consequently, by virtue of the theorem in 4.1 any antiderivative has the form $F(x) = \frac{1}{6}x^6 + C$, where C is an arbitrary constant.

(b) One of the antiderivatives of the function $\frac{1}{2}\sin(3x+1)$ is the function $-\frac{1}{6}\cos(3x+1)$ and, consequently, any antiderivative has the form $F(x) = -\frac{1}{6}\cos(3x+1) + C$.

(c) It was proved in 4.1 that one of the antiderivatives of the function $1/x$ is the function $\ln|x|$. It is easy to see that the function $\frac{1}{5}\ln|5x+6|$ is one of the antiderivatives of the function $1/(5x+6)$. Consequently, any antiderivative of the given function has the form

$$F(x) = \frac{1}{5}\ln|5x+6| + C.$$

(d) We can verify by differentiation that the function $\frac{3}{2} \tan(2x+1)$ is an antiderivative of the function $3/\cos^2(2x+1)$. We can analogously verify that the antiderivative of the function $\cos 6x$ is the function $\frac{1}{6} \sin 6x$. Consequently, any antiderivative of the given function has the form

$$F(x) = \frac{3}{2} \tan(2x+1) + \frac{1}{6} \sin 6x + C.$$

2. Suppose $F(x)$ is an antiderivative of the given function. Then $F'(x) = x$ for $x > 0$ and $F'(x) = -x$ for $x < 0$. Consequently, $F(x) = \frac{1}{2}x^2 + C_1$ for $x > 0$ and $F(x) = -\frac{1}{2}x^2 + C_2$ for $x < 0$, where C_1 and C_2 are arbitrary constants. By the hypothesis, the function $F(x)$ has a derivative at each point and is, therefore, continuous throughout the real axis. It follows from the continuity of the function $F(x)$ at the point $x = 0$ that $C_1 = C_2$. Thus, the assumption that the function $f(x) = |x|$, $x \in \mathbb{R}$, possesses the antiderivative $F(x)$ signifies that $F(x)$ has the form

$$F(x) = \begin{cases} \frac{x^2}{2} + C, & x \geq 0, \\ -\frac{x^2}{2} + C, & x < 0, \end{cases}$$

where C is an arbitrary constant. It is easy to verify that every function of this kind is really an antiderivative of the given function $|x|$.

3. The desired antiderivative of the function $f(x) = x^3$ has the form

$$F(x) = \frac{1}{3}x^3 + C,$$

where C is some constant. To determine C , let us make use of the condition that the graph of the desired antiderivative passes through the point $(2; 3)$, i.e. $F(2) = 3$. Hence we get $C = 1/3$. Consequently, $F(x) = (x^3 + 1)/3$.

4. The desired antiderivative has the form

$$F(x) = \ln|x| + \frac{2^{3x}}{3 \ln 2} + C,$$

where C is some constant. From the condition $F(1) = 3$ we obtain $C = 3 - \frac{2^3}{3 \ln 2}$ and, consequently, the desired antiderivative is

$$F(x) = \ln|x| + \frac{2^{3x}}{3 \ln 2} + 3 - \frac{8}{3 \ln 2}.$$

5. (a) The indefinite integral taken of the sum of the functions is equal to the sum of the indefinite integrals of those functions and,

therefore,

$$\int (t^3 + 3e^{5t}) dt = \int t^3 dt + 3 \int e^{5t} dt,$$

where, in addition, the constant factor 3 is taken outside the sign of integral. The antiderivative of t^3 is the function $t^4/4$ and the antiderivative of e^{5t} is the function $e^{5t}/5$. Consequently,

$$\int (t^3 + 3e^{5t}) dt = \frac{1}{4} t^4 + \frac{1}{5} e^{5t} + C,$$

where C is an arbitrary constant.

(b) By analogy with the above case,

$$\begin{aligned} \int \frac{x^{1/3} + x^{1/5}}{x^2} dx &= \int x^{-5/3} dx + \int x^{-9/5} dx \\ &= -\frac{3}{2} x^{-2/3} - \frac{5}{4} x^{-4/5} + C. \end{aligned}$$

(c) Using the relation

$$\sin 2x \cos 4x = (\sin 6x - \sin 2x)/2,$$

we obtain

$$\begin{aligned} \int \sin 2x \cos 4x dx &= \frac{1}{2} \int \sin 6x dx - \frac{1}{2} \int \sin x dx \\ &= -\frac{1}{12} \cos 2x + \frac{1}{4} \cos 2x + C. \end{aligned}$$

$$\begin{aligned} \text{(d)} \quad \int \left(5^{3x} + \frac{1}{\sin^2(2x+1)} + \frac{1}{\sqrt[5]{4x-1}} \right) dx \\ = \int 5^{3x} dx + \int \frac{dx}{\sin^2(2x+1)} + \int \frac{dx}{\sqrt[5]{4x-1}} \\ = \frac{5^{3x}}{3 \ln 5} - \frac{1}{2} \cot(2x+1) + \frac{5}{16} \sqrt[5]{(4x-1)^4} + C. \end{aligned}$$

6. (a) We set $u = x$ and $v' = \cos x$. Then, $u' = 1$ and $v = \sin x$, and we find, by the formula for integration by parts, that

$$\int x \cos x dx = x \sin x - \int \sin x dx = x \sin x + \cos x + C.$$

(b) We set $u = \ln x$ and $v' = x$. Then, $u' = 1/x$, $v = x^2/2$ and

$$\int x \ln x dx = \frac{1}{2} x^2 \ln x - \frac{1}{2} \int x dx = \frac{1}{2} x^2 \ln x - \frac{1}{4} x^2 + C.$$

7. Let us assume that the function $f(x)$ possesses an antiderivative $F(x)$ on $\mathbb{R}$. Then, $F'(x) = 1$ for $x > 0$ and $F'(x) = -1$ for $x < 0$. Consequently, $F(x) = x + C_1$ for $x > 0$ and $F(x) = -x + C_2$ for $x < 0$. It follows from the continuity of the function $F(x)$ at $x = 0$

that $F(0) = C_1 = C_2$. Therefore, the function $F(x)$ has the form $F(x) = |x| + C$, where C is some constant. For $x = 0$ this function has no derivative and because of this fact cannot serve as an antiderivative on $\mathbb{R}$ of any function. And this contradicts the assumption. Consequently, our assumption is wrong and, therefore, the function $f(x) = \operatorname{sign} x$ does not possess an antiderivative on $\mathbb{R}$.

8. (a) Since the function $x^{4/3}$ is an antiderivative of x^3 , we find by the Newton-Leibniz formula, that

$$\int_0^1 x^3 dx = \frac{1}{4} x^4 \Big|_0^1 = \frac{1}{4}.$$

(b) The function $-\cos x$ is an antiderivative of $\sin x$ and, therefore,

$$\int_{-\pi/4}^{\pi/3} \sin x dx = -\cos x \Big|_{-\pi/4}^{\pi/3} = -\frac{1}{2} + \frac{\sqrt{2}}{2} = \frac{\sqrt{2}-1}{2}.$$

(c) We have

$$\int_4^9 \left(2\sqrt{x} - \frac{3}{\sqrt{x}} \right) dx = \left(\frac{4}{3} x^{3/2} - 6\sqrt{x} \right) \Big|_4^9 = \frac{58}{3}.$$

(d) The antiderivative of the function $1/\sqrt{2x+3}$ is the function $\sqrt{2x+3}$. Therefore,

$$\int_1^3 \frac{1}{\sqrt{2x+3}} dx = \sqrt{2x+3} \Big|_1^3 = 3 - \sqrt{5}.$$

(e) We pass to a new variable of integration by setting $3x^3 + 4 = y$. Then, $(3x^3 + 4)^{1/3} = y^{1/3}$, $y' = 9x^2$ and, consequently, $(3x^3 + 4)^{1/3} x^2 dx = y^{1/3} \frac{1}{9} dy$. We find the new limits of integration: $A = 3 \cdot 1^3 + 4 = 7$, $B = 3 \cdot 2^3 + 4 = 28$. By the formula for a change of variables, we obtain

$$\begin{aligned} \int_1^2 (3x^3 + 4)^{1/3} x^2 dx &= \int_7^{28} \frac{1}{9} y^{1/3} dy = \frac{1}{9} \cdot \frac{3}{4} y^{4/3} \Big|_7^{28} \\ &= \frac{1}{12} (28)^{4/3} - \frac{1}{12} \cdot 7^{4/3}. \end{aligned}$$

(f) We set $u = x$ and $v' = \sin x$. Then, $u' = 1$ and $v = -\cos x$. By the formula for integration by parts we obtain

$$\int_0^{\pi/2} x \sin x dx = -x \cos x \Big|_0^{\pi/2} + \int_0^{\pi/2} \cos x dx = \sin x \Big|_0^{\pi/2} = 1.$$

(g) We set $u = \ln x$ and $v' = 1/x^2$ and then $u' = 1/x$, $v = -1/x$. By the formula for integration by parts, we obtain

$$\int_1^e \frac{\ln x}{x^2} dx = -\frac{1}{x} \ln x \Big|_1^e + \int_1^e \frac{1}{x^2} dx = -\frac{1}{e} - \frac{1}{x} \Big|_1^e = 1 - \frac{2}{e}.$$

(h) The function $|x|$ is equal to x for $x > 0$ and to $-x$ for $x < 0$. Therefore, the integrand $f(x)$ is equal to $x(x-2) = x^2 - 2x$ for $x > 0$ and to $-x^2 + 2x$ for $x < 0$. Partitioning the integral into two integrals, we get

$$\begin{aligned} \int_{-2}^2 f(x) dx &= \int_{-2}^0 (-x^2 + 2x) dx + \int_0^2 (x^2 - 2x) dx \\ &= \left(-\frac{x^3}{3} + x^2 \right) \Big|_{-2}^0 + \left(\frac{x^3}{3} - x^2 \right) \Big|_0^2 = -\frac{8}{3} - 4 + \frac{8}{3} - 4 = -8. \end{aligned}$$

9. (a) Assume that the function $F(t)$ is an antiderivative of the function $f(t) = t^2 \sin t$. By the Newton-Leibniz formula we get $\Phi(x) = F(x) - F(0)$. It follows from the definition of the antiderivative that $F'(x) = x^2 \sin x$. Therefore, $\Phi'(x) = (F(x) - F(0))' = F'(x) = x^2 \sin x$.

(b) Assume that the function $F(t)$ is an antiderivative of the function $f(t) = 2t \cos(\sin t)$. Then $\Phi(x) = F(3x^2) - F(2x)$. Applying now the rule of differentiation of a composite function, we get

$$\begin{aligned} \Phi'(x) &= (F(3x^2) - F(2x))' = 2 \cdot 3x^2 \cos(\sin(3x^2)) \cdot 6x \\ &\quad - 2^{2x} \cos(\sin(2x)) \cdot 2. \end{aligned}$$

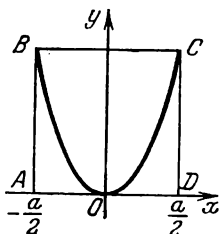


Fig. 206

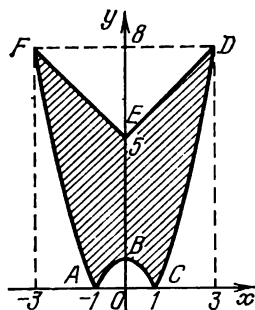


Fig. 207

10. We designate the length of the side of the square as a and choose the system of coordinates as is shown in Fig. 206. In this system of coordinates, the equation of the parabola has the form $y = kx^2$, where k is defined by the condition that the parabola passes through the point

($a/2$; a). Thus, $k = 4/a$. The parabola divides the square into two parts: $ABOCD$ and BOC . We calculate the area of the curvilinear trapezoid $ABOCD$:

$$S_{ABOCD} = \int_{-a/2}^{a/2} \frac{4}{a} x^2 dx = \frac{4}{a} \frac{x^3}{3} \Big|_{-a/2}^{a/2} = \frac{a^2}{3}.$$

Hence it follows that $S_{BOC} = a^2 - \frac{a^2}{3} = \frac{2}{3} a^2$. Consequently, the

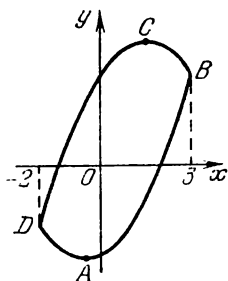


Fig. 208

parabola divides the area of the square in the ratio 1/2.

11. We construct the graph of the functions $y = |x^2 - 1|$ and $y = |x| + 5$ and see that we should find the area of the figure $ABCDEF$ (Fig. 207). From the system of equations

$$\begin{cases} y = |x^2 - 1| \\ y = |x| + 5 \end{cases}$$

we find the coordinates of the points D and F at which the given curves intersect, namely, $D(3; 8)$ and $F(-3; 8)$. It can be seen in the figure that $|x^2 - 1| < |x| + 5$

for any $x \in (-3; 3)$ and the given figure is symmetric about the x -axis. Consequently, $S = 2S_{BCDE}$ and by formula (2) in 4.3 we have

$$S_{BCDE} = \int_0^3 (|x| + 5 - |x^2 - 1|) dx.$$

It is easy to see that

$$\int_0^3 (|x| + 5) dx = \int_0^3 x dx + \int_0^3 5 dx = \frac{9}{2} + 15 = \frac{39}{2}.$$

Furthermore,

$$\begin{aligned} \int_0^3 (|x^2 - 1|) dx &= \int_0^1 (1 - x^2) dx + \int_1^3 (x^2 - 1) dx \\ &= \left(x - \frac{x^3}{3} \right) \Big|_0^1 + \left(\frac{x^3}{3} - x \right) \Big|_1^3 = \frac{22}{3}. \end{aligned}$$

Consequently, $S = 2 \cdot \left(\frac{39}{2} - \frac{22}{3} \right) = \frac{73}{3}$.

12. The curve $2y = x^3 + x - 6$ is a parabola with vertex at the point $A(-1/2; -25/8)$ (Fig. 208), and the curve $2y = -x^3 + 3x + 6$

is a parabola with vertex at the point $C(3/2; 33/8)$. It is required to find the area S of the figure $ABCD$, where B and D are the points of intersection of the parabolas. Solving the system

$$\begin{cases} 2y = x^2 + x - 6 \\ 2y = -x^2 + 3x + 6, \end{cases}$$

we find that the coordinates of the point D are the numbers $x = -2$, $y = -2$ and the coordinates of the point B are the numbers $x = 3$, $y = 3$. Using formula (3) in 4.3, we obtain

$$\begin{aligned} S &= \int_{-2}^3 \left(\frac{1}{2}(-x^2 + 3x + 6) - \frac{1}{2}(x^2 + x - 6) \right) dx \\ &= \int_{-2}^3 (-x^2 + x + 6) dx = \left(-\frac{x^3}{3} + \frac{x^2}{2} + 6x \right) \Big|_{-2}^3 = \frac{125}{6}. \end{aligned}$$

13. Since

$$\int_{-a}^a f(x) dx = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx,$$

we must prove that

$$\int_{-a}^0 f(x) dx = \int_0^a f(x) dx.$$

For that purpose we make change of variable $x = -t$ and then change the places of the limits of integration:

$$\int_{-a}^0 f(x) dx = - \int_a^0 f(-t) dt = \int_0^a f(-t) dt.$$

The function f being even, we have $f(-t) = f(t)$ and

$$\int_0^a f(-t) dt = \int_0^a f(t) dt.$$

Consequently,

$$\int_{-a}^0 f(x) dx = \int_0^a f(t) dt = \int_0^a f(x) dx.$$

14. We note first of all that

$$\int_0^{a+T} f(x) dx = \int_a^T f(x) dx + \int_T^{a+T} f(x) dx. \quad (1)$$

In the second integral on the right-hand side we make a change of variables $x = t + T$. Then

$$\int_T^{a+T} f(x) dx = \int_0^a f(t+T) dt,$$

and since $f(t+T) = f(t)$, we have

$$\int_T^{a+T} f(x) dx = \int_0^a f(t) dt. \quad (2)$$

In the last integral we again designate the variable of integration as x and find, from (1) and (2), that

$$\int_a^{a+T} f(x) dx = \int_a^T f(x) dx + \int_0^a f(x) dx = \int_0^T f(x) dx.$$

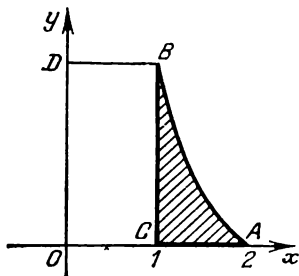


Fig. 209

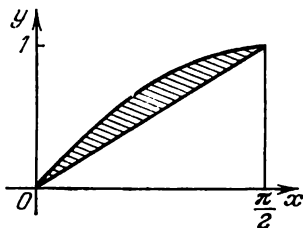


Fig. 210

15. Assume that A and B are the points of intersection of the hyperbola $x(y+2) = 4$ and the straight lines $y = 0$ and $x = 1$, and C is the point of intersection of those lines. Then $A(2; 0)$, $B(1; 2)$, and $C(1; 0)$ (Fig. 209). It is required to calculate the volume V of the solid generated by the rotation of the figure ABC about the y -axis. The desired volume is equal to the difference $V_1 - V_2$ between the volume of the solid generated by the rotation of the curvilinear trapezoid $OABD$ about the y -axis and that of the cylinder with radius 1 and altitude 2. The volume V_1 can be calculated by the formula for the

volume of a solid of revolution:

$$V_1 = \pi \int_0^2 \left(\frac{4}{x+2} \right)^2 dy = -16\pi \frac{1}{y+2} \Big|_0^2 = 4\pi.$$

The volume V_2 of the cylinder is equal to 2π . Consequently, $V = V_1 - V_2 = 2\pi$.

16. The volume V of the given solid (Fig. 210) is equal to the difference $V_1 - V_2$ of the volume of the body, generated by the rotation of the curvilinear trapezoid, bounded by the graph of the function $y = \sin x$, $x \in [0; \pi/2]$, and the segments of the curves $y = 0$ and $x = \pi/2$, and those of the cone with the radius of the base 1 and the altitude $\pi/2$. The volume V_1 can be calculated using integrals

$$\begin{aligned} V_1 &= \pi \int_0^{\pi/2} \sin^2 x \, dx = \pi \int_0^{\pi/2} \frac{1 - \cos 2x}{2} \, dx \\ &= \frac{\pi}{2} \left(x - \frac{1}{2} \sin 2x \right) \Big|_0^{\pi/2} = \frac{\pi^2}{4}. \end{aligned}$$

The volume V_2 of the cone is equal to $\pi^2/6$. Consequently, the sought-for volume is equal to $\frac{\pi^2}{4} - \frac{\pi^2}{6} = \frac{\pi^2}{12}$.

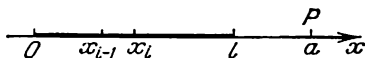


Fig. 211

17. Let us partition the interval $[0; l]$ of the x -axis into n equal parts (Fig. 211). In accordance with Newton's law, the part of the bar $[x_{i-1}; x_i]$ of mass $\delta \Delta x_i$ attracts the particle P with the force approximately equal to

$$-k \frac{m \delta \Delta x_i}{(a - x_i)^2},$$

where k is the proportionality factor (gravitational constant). Then the force F with which the homogeneous bar attracts the particle P is approximately equal to the sum

$$- \sum_{i=1}^n \frac{km \delta \Delta x_i}{(a - x_i)^2}.$$

Passing to the limit as $n \rightarrow \infty$, we obtain

$$F = -km\delta \int_0^l \frac{dx}{(a-x)^2}.$$

Consequently,

$$F = - \frac{km\delta}{a-x} \Big|_0^l = - \frac{km\delta}{a} - \frac{km\delta}{a-l} = - \frac{km\delta l}{a(a-l)}.$$

18. Let us choose a system of coordinates as shown in Fig. 212. We divide the interval $[0; R]$ of the x -axis into n equal parts. The kinetic energy E_i of the circular ring contained between the circles of

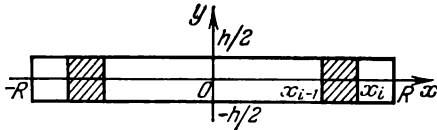


Fig. 212

the radii $r = x_{i-1}$ and $r = x_i$ is approximately equal to $m_i v_i^2/2$, where m_i is the mass of the ring and $v_i = x_i \omega$. Since

$$m_i = \rho h \pi (x_i^2 - x_{i-1}^2) = \rho h \pi (x_i + x_{i+1}) \Delta x_i,$$

m_i is approximately equal to $2\rho h \pi x_i \Delta x_i$ and the kinetic energy E_i is approximately equal to $\rho h \pi \omega^2 x_i^3 \Delta x_i$. Then the kinetic energy E of the whole body is approximately equal to the sum

$$\sum_{i=1}^n \pi \rho h \omega^2 x_i^3 \Delta x_i.$$

Passing to the limit as $n \rightarrow \infty$, we obtain

$$E = \int_0^R \pi \rho h \omega^2 x^3 dx = \frac{1}{4} \pi \rho h \omega^2 R^4.$$

CHAPTER 5

1. Let the diagonals of the trapezoid $ABCD$ meet at a point O (Fig. 213). The triangles ABD and ACD have a common base AD and equal altitude drawn to that base, and, therefore, their areas are equal. The areas of the triangles ABO and CDO are less than the areas of the triangles ABD and ACD by the value of the area of the triangle ADO and, consequently, they are equal.

2. Suppose $PM \parallel AC$ and $MQ \parallel AB$ (Fig. 214). We designate $b = |BC|$, $x = |BM|/|MC|$ and then $|BM| = \frac{x}{1+x}$ and

$$|MC| = \frac{1}{1+x} b.$$

It is easy to see that the triangles PBM and QMC are similar to the triangle ABC . Therefore, if S is the area of the triangle ABC , then

$$S_{PBM} = \left(\frac{x}{1+x} \right)^2 S \quad \text{and} \quad S_{QMC} = \frac{1}{(1+x)^2} S.$$

By the hypothesis, $S - S_{PBM} - S_{QMC} = \frac{5}{18} S$, whence we get

$$1 - \left(\frac{x}{1+x} \right)^2 - \frac{1}{(1+x)^2} = \frac{5}{18}, \quad \text{or}$$

$$5x^2 - 26x + 5 = 0.$$

Solving the equation, we find that $x = 5$ and $x = 1/5$. In both cases the point M divides the side BC into segments whose lengths are related as $5/1$.

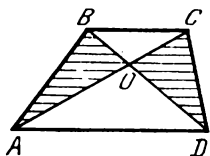


Fig. 213

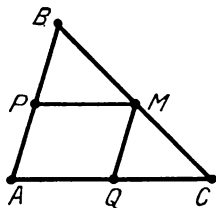


Fig. 214

3. Suppose S_0 , S_1 , S_2 , and S_3 are the areas of the triangles ABC , PEC , AMF , and DBN respectively (Fig. 215). It is easy to see that

$$S_0 = S_{AMC} + S_{ABN} + S_{PBC} + S_{FDE} - (S_1 + S_2 + S_3),$$

and since $S_{AMC} = S_{ABN} = S_{PBC} = \frac{1}{3} S_0$, $S_{FDE} = S$, we obtain

$$S_1 + S_2 + S_3 = S. \quad (1)$$

Let us find S_1 . We draw $MK \parallel BP$ and then $\triangle AMN \sim \triangle ABP$. From the similarity of the triangles it follows that $|AK| = \frac{1}{3} |AP|$, and since $|AP| = \frac{2}{3} |AC|$, we have $|AK| = \frac{2}{9} |AC|$.

Then we have $S_{AMK} = \frac{2}{9} S_{AMC} = \frac{2}{27} S_0$ and $S_{KMC} = \frac{7}{27} S_0$.

The triangles PEC and KMC are also similar and, therefore, $S_1 = (|PC|/|KC|)^2 S_{KMC}$. Taking into account that $|PC| = \frac{1}{3} |AC|$ and $|KC| = \frac{7}{9} |AC|$, we get $S_1 = \frac{1}{21} S_0$.

By analogy we establish that $S_2 = S_3 = \frac{1}{21} S_0$, and, therefore, it follows from (1) that $S = \frac{1}{7} S_0$, i.e. $S_0 = 7S$.

4. Upon rotation through the angle α every line rotates through the same angle and, consequently, $\angle AB'A' = \alpha$ (Fig. 216). Then it follows from the condition $A'B' \perp AC$ that $\angle A = \frac{\pi}{2} - \alpha$. Furthermore, since $|CB'| = |CB|$, the triangle CBB' is isosceles and $\angle BB'C = \angle B$. On the other hand, the angles ABC and $A'B'C$ are congruent, that is,

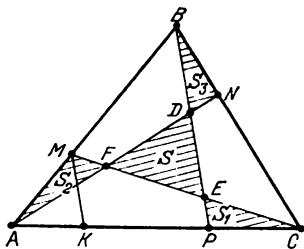


Fig. 215

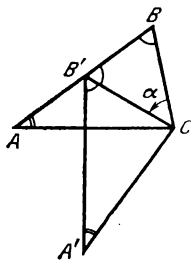


Fig. 216

$\angle A'B'C = \angle B$. The angles $AB'A'$, $A'B'C$, and $CB'B$ form a straight angle and, therefore, $\alpha + 2\angle B = \pi$, whence we find that $\angle B = \frac{\pi}{2} - \frac{\alpha}{2}$. The angle C can be found from the condition that the sum of the angles of a triangle is equal to π , and we obtain $\angle C = \frac{3}{2}\alpha$.

5. We draw straight lines through the vertices A , B , and C of the triangle parallel to its opposite sides (Fig. 217). These lines meet and bound the triangle $A'B'C'$. In the quadrangles $AC'BC$ and $ABA'B$ the opposite sides are pairwise parallel and, consequently, they are parallelograms and $|C'B| = |AC|$, $|BA'| = |AC|$. Thus, the point B is the midpoint of the side $A'C'$. By analogy we establish that the points A and C are the midpoints of the sides $C'B'$ and $A'B'$.

If BK is the altitude of the triangle ABC , then the line BK is the midline of the side $A'C'$ of the triangle $A'B'C'$. Similarly, if $AM \perp BC$ and $CN \perp AB$, then the lines AM and CN are the midlines of the sides $C'B'$ and $A'B'$ respectively. The midlines of the sides of a triangle meet at one point and, consequently, three straight lines which contain the altitudes of the triangle also meet at one point.

6. Assume that the extensions of the altitudes drawn from the vertices A , B , and C cut the circle at the points A' , B' , and C' (Fig. 218) and suppose that

$$\widehat{C'BA'} : \widehat{A'CB'} : \widehat{B'AC'} = p : q : r. \quad (1)$$

We designate the angles into which the altitudes divide the vertex angles by A_1 and A_2 , B_1 and B_2 and C_1 and C_2 respectively. Considering the inscribed angles subtended by the arcs $C'BA'$, $A'CB'$ and $B'AC'$ we obtain, from (1),

$$(\angle C_1 + \angle A_2) : (\angle A_1 + \angle B_2) : (\angle B_1 + \angle C_2) = p : q : r. \quad (2)$$

Suppose O is the point of intersection of the altitudes of the triangle ABC . It follows from the similitude of the right triangles AOK and

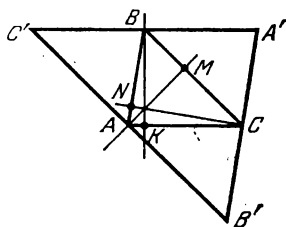


Fig. 217

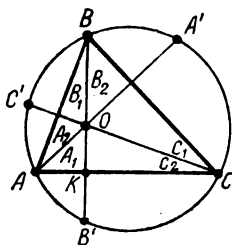


Fig. 218

BOM that $\angle A_1 = \angle B_2$. We establish by analogy that $\angle C_1 = \angle A_2$ and $\angle B_1 = \angle C_2$. Taking this into account, we obtain, from (2),

$$2\angle C_2 : 2\angle A_1 : 2\angle B_1 = p : q : r,$$

whence it follows that

$$\angle A_1 = \frac{q}{r} \angle B_1 \quad \text{and} \quad \angle C_1 = \frac{p}{r} \angle B_1.$$

Now, from the condition $\angle A + \angle B + \angle C = 2(\angle A_1 + \angle B_1 + \angle C_1) = \pi$ we easily find $\angle B_1$, $\angle C_1$, $\angle A_1$ and calculate the angles of the triangle:

$$\angle A = \frac{p+q}{p+q+r} \frac{\pi}{2}, \quad \angle B = \frac{q+r}{p+q+r} \frac{\pi}{2}, \quad \angle C = \frac{p+r}{p+q+r} \frac{\pi}{2}.$$

7. Suppose O_1 and O_2 are the centres of the major and the minor circle respectively, D is the point of intersection of the chord AB and the minor circle (Fig. 219). If $O_2M \perp AB$, then $|AM| = \frac{1}{2} |AD|$,

and if $O_1N \perp AB$, then $|AN| = \frac{1}{2} |AB|$. From the similitude of the right triangles O_2AM and O_1AN we obtain $|AO_2|/|AO_1| = |AM|/|AN|$, whence we have $|AD|/|AB| = r/R$. By the secant-tangent theorem we have

$$|BC|^2 = |AB| |BD| = |AB| (|AB| - |AD|)$$

or $a^2 = |AB|^2 \left(1 - \frac{r}{R}\right)$. We find $|AB| = a \sqrt{R/(R-r)}$.

8. Suppose P , Q , and N are the points of tangency of the circle and the sides of the triangle (Fig. 220). We designate $a = |BC|$ and $3x = |BM|$. It follows from the secant-tangent theorem that

$$|BP| = |BN| = |QM| = \sqrt{2}x.$$

But since $|CP| = |CQ|$, we have $|BC| = |CM| = a$. The point M is the midpoint of the side AC and, consequently, $|AC| = 2a$ and $|AM| = a$.

Furthermore, $|AN| = |AQ| = |AM| + |MQ| = a + \sqrt{2}x$ and, therefore, $|AB| = a + 2\sqrt{2}x$.

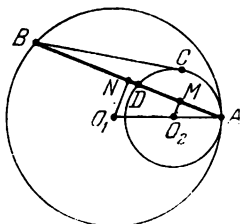


Fig. 219

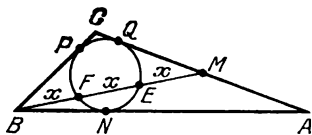


Fig. 220

Using the expression for the median in terms of the lengths of the sides of the triangle (see Problem 5 in 5.2), we can write

$$2|BM|^2 = |BC|^2 + |AB|^2 - \frac{|AC|^2}{2}$$

or $18x^2 = a^2 + (a + 2\sqrt{2}x)^2 - 2a^2$. This yields a quadratic equation $5x^2 - 2\sqrt{2}ax = 0$. Thus we have $x = \frac{2\sqrt{2}}{5}a$ and $|AB| = \frac{13}{5}a$. Thus we have $|BC| : |AC| : |AB| = 5 : 10 : 13$.

9. Suppose O is the point of intersection of the altitudes of the triangle ABC (Fig. 221). The right triangles BAM and BON contain equal acute angles at the vertex B and, consequently, $\angle BON = \angle A$. The angle BAC is subtended by the arc BC and the same arc subtends the angle BQC and, therefore, $\angle BQC = \angle A$, the triangle BOQ is isosceles and $|BO| = |BQ|$. We designate $x = |BO|$. The triangles QOP and BOC are similar ($\angle CQP = \angle CBP$ and $\angle QPB = \angle QCB$),

$$\frac{|QO|}{|BO|} = \frac{|QP|}{|BC|} = \frac{6}{5}, \text{ whence it follows that } |QO| = \frac{6}{5}x. \text{ In an isosceles triangle an altitude is also a median and, therefore, } |NO| =$$

$\frac{3}{5}x$ and $\cos A = \frac{|NO|}{|BO|} = \frac{3}{5}$. Then, $\sin A = \frac{4}{5}$ and the radius R of the circle circumscribed about the triangle ABC can be found by

formula (4) in 5.2, i.e. $R = \frac{a}{2 \sin A} = \frac{5}{8}a$.

10. Assume that $DF \perp AB$ (Fig. 222). Then the triangle AFE is right angled and FK is its median. Consequently, $|FK| = |AK|$, and since $\angle FAK = 60^\circ$, the triangle AFK is regular. Hence it follows that $FK \parallel BC$ and $|FB| = |KC|$, i.e. $FBCK$ is an isosceles trapezoid about which we can circumscribe a circle.

Furthermore, by the hypothesis, the opposite angles BFD and BCD in the quadrangle $FBCD$ are right angles. This means that we can

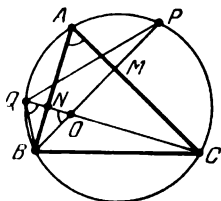


Fig. 221

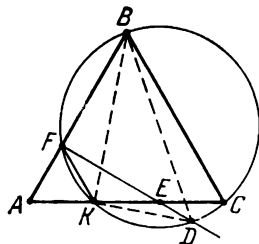


Fig. 222

circumscribe a circle about the quadrangle $FBCD$ too and BD is the diameter of that circle.

The two indicated circles pass through the points F, B , and C and, therefore, coincide. Hence it follows that the angle BKD rests on the

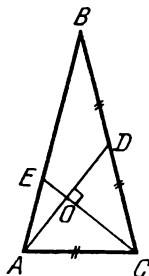


Fig. 223

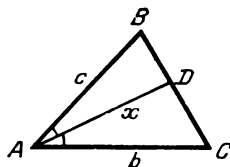


Fig. 224

diameter and is, therefore, a right angle: $\angle BKD = 90^\circ$ and $\angle KDB = \angle KCB = 60^\circ$. Then, $\angle KBD = 30^\circ$.

11. Assume that the median AD and the bisector CE meet at a point O (Fig. 223). The bisector CO of the triangle ACD is also its altitude and, therefore (see 5.1), the triangle ACD is isosceles: $|AC| = |CD|$. Then the condition that the point D is the midpoint of the segment BC implies that $|BC| = 2|AC|$ and $\cos ACB = \frac{1}{2} |AC|/|BC| = \frac{1}{4}$.

Thus we have $\angle ACB = \arccos(1/4)$.

12. Suppose AD is the bisector of the angle BAC (Fig. 224). We designate $x = |AD|$. It is evident that the area of the triangle ABC is equal to the sum of the areas of the triangles ABD and ADC and, therefore, using formula (2) in 5.2, we can write

$$\frac{1}{2} bc \sin \alpha = \frac{1}{2} cx \sin \frac{\alpha}{2} + \frac{1}{2} bx \sin \frac{\alpha}{2}.$$

From this, taking into account that $\sin \alpha = 2 \sin(\alpha/2) \cos(\alpha/2)$ and $\sin(\alpha/2) \neq 0$, we get $2bc \cos(\alpha/2) = x(b+c)$, whence it follows that $x = \frac{2bc}{b+c} \cos \frac{\alpha}{2}$.

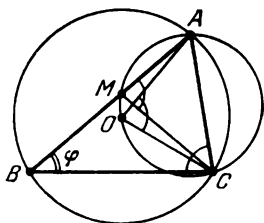


Fig. 225

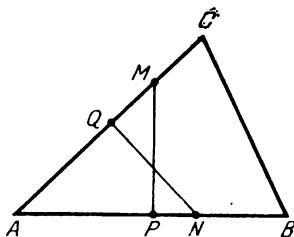


Fig. 226

13. Suppose O is the centre of the circle circumscribed about the triangle ABC (Fig. 225). The inscribed angle ABC rests on the arc AC which also subtends the central angle AOC and, consequently, $\angle AOC = 2\varphi$. In the circle circumscribed about the triangle AOC the arc AC subtends the inscribed angles AOC and AMC and, therefore, $\angle AMC = \angle AOC = 2\varphi$, and since the vertex angles A in the triangles ABC and ACM are equal, we have $\angle ACM = \angle ABC = \varphi$.

Applying the sine law to the triangles AMC and ABC , we get

$$\frac{|AM|}{\sin \varphi} = \frac{|AC|}{\sin 2\varphi} \quad \text{and} \quad \frac{|AC|}{\sin \varphi} = \frac{|AB|}{\sin 2\varphi},$$

whence we have $\frac{|AM|}{|AB|} = \left(\frac{\sin \varphi}{\sin 2\varphi} \right)^2 = \frac{1}{4 \cos^2 \varphi}$.

14. Assume that P in the triangle ABC is the midpoint of the side AB and Q is the midpoint of the side AC (Fig. 226). We set $a = |BC|$, $b = |AC|$, $c = |AB|$, and then $|AM| = \frac{3}{4}b$, $|MC| = \frac{1}{4}b$, $|AN| = \frac{2}{3}c$, and $|NB| = \frac{1}{3}c$. It follows from the right triangles AMP and AQN that

$$\cos A = \frac{|AP|}{|AM|} = \frac{2}{3} \frac{c}{b} \quad \text{and} \quad \cos A = \frac{|AQ|}{|AN|} = \frac{3}{4} \frac{b}{c}.$$

Hence we get $c = \frac{3}{2\sqrt{2}}b$ and $\cos A = \frac{\sqrt{2}}{2}$. Thus, $\angle A = \pi/4$.

By the cosine law, we find from the triangle ABC that

$$a = \sqrt{b^2 + c^2 - 2bc \cos A} = \frac{\sqrt{5}}{2\sqrt{2}}b.$$

Using now the sine law, we get

$$\sin B = \frac{b}{a} \sin A = \frac{2}{\sqrt{5}} \quad \text{and} \quad \sin C = \frac{c}{a} \sin A = \frac{3}{\sqrt{10}}.$$

Thus, $\angle A = \pi/4$, $\angle B = \arcsin(2/\sqrt{5})$, and $\angle C = \arcsin(3/\sqrt{10})$.

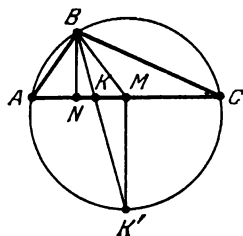


Fig. 227

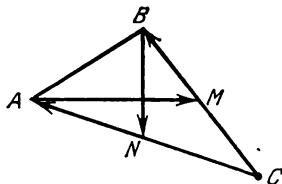


Fig. 228

15. Assume that in the triangle ABC the lengths of the sides adjacent to the vertex B are different (otherwise the triangle ABC would be isosceles and the bisector of the angle B would be the median and the altitude) and suppose that BK is a bisector of that triangle (Fig. 227). Let us circumscribe a circle about the triangle ABC and extend the line BK till it meets the circle at a point K' . The point K' is, evidently, the midpoint of the arc AC and, therefore, if we draw a straight line through the point K' at right angles to the chord AC (that is, parallel to the altitude BN of the triangle ABC), it will bisect the chord AB . Hence it follows that the point K lies between the point M , which is the midpoint of the side AB , and the point N , which is the foot of the altitude, that is, the bisector lies between the median and the altitude.

16. Let us apply vector algebra. Assume that $\vec{CB} = \mathbf{a}$, $\vec{CA} = \mathbf{b}$, $\vec{BN} = \mathbf{n}$, and $\vec{AM} = \mathbf{m}$ (Fig. 228). Then $\mathbf{n} = \frac{1}{2}\mathbf{b} - \mathbf{a}$ and $\mathbf{m} = \frac{1}{2}\mathbf{a} - \mathbf{b}$. Let us now find the scalar product of the vectors $\mathbf{m}$ and $\mathbf{n}$:

$$\mathbf{m} \cdot \mathbf{n} = \frac{5}{4} \mathbf{a} \cdot \mathbf{b} - \frac{1}{2} (|\mathbf{a}|^2 + |\mathbf{b}|^2). \quad (1)$$

If $\overrightarrow{AB} = \mathbf{c}$, then $\mathbf{c} = \mathbf{a} - \mathbf{b}$ and $|\mathbf{c}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2 - 2\mathbf{a} \cdot \mathbf{b}$, whence, using the condition of the problem $|\mathbf{a}|^2 + |\mathbf{b}|^2 = 5|\mathbf{c}|^2$, we find that $\mathbf{a} \cdot \mathbf{b} = \frac{2}{5}(|\mathbf{a}|^2 + |\mathbf{b}|^2)$. Substituting this into (1), we get $\mathbf{m} \cdot \mathbf{n} = 0$. Therefore, the vectors $\overrightarrow{AM}$ and $\overrightarrow{BN}$ and, consequently, the medians AM and BN are perpendicular.

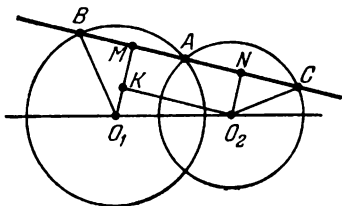


Fig. 229

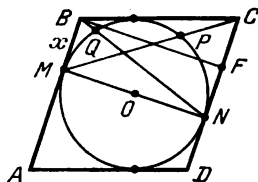


Fig. 230

17. Let O_1 and O_2 be the centres of the circles (Fig. 229), $O_1M \perp BC$ and $O_2N \perp BC$. Then, if $|AB| = x$, we have

$$|BM| = |MA| = |AN| = |NC| = x/2$$

and

$$|O_1M| = \sqrt{5 - \frac{x^2}{4}} \quad \text{and} \quad |O_2N| = \sqrt{2 - \frac{x^2}{4}}.$$

We draw $O_2K \parallel CB$ and then $O_2K \perp O_1M$ and $|O_1K|^2 = |O_1O_2|^2 - |O_2K|^2$, i.e.

$$\left(\sqrt{5 - \frac{x^2}{4}} - \sqrt{2 - \frac{x^2}{4}} \right)^2 = 9 - x^2.$$

Solving this equation, we find that $x = 6/\sqrt{5}$. Thus we have $|AB| = 6/\sqrt{5}$ cm.

18. Let us designate $|MC| = y$ (Fig. 230). Then, from the condition $|CP|/|PM| = 9/16$ we find $|CP| = \frac{9}{25}y$. By the secant-tangent theorem we have $|CN|^2 = |CP| \cdot |MC|$, whence we find $|CN| = \frac{3}{5}y$.

Suppose O is the centre of the circle. Then, $OM \perp AB$ and $ON \perp DC$, and it follows from the condition $AB \parallel DC$ that the points M , O , and N lie on the same straight line. We find from the right triangle CMN that $|MN| = \frac{4}{5}y$. We designate $|MB| = x$ and then have $|DN| =$

$|MB| = x$ and $|BC| = |DC| = x + \frac{3}{5}y$. We draw $BF \parallel MN$ and see that $|BF| = |MN| = \frac{4}{5}y$, and $|CF| = \frac{3}{5}y - x$. By the Pythagorean theorem we have

$$\left(\frac{3}{5}y + x\right)^2 = \left(\frac{3}{5}y - x\right)^2 + \left(\frac{4}{5}y\right)^2,$$

whence we find that $x = \frac{4}{15}y$.

Furthermore, from the right triangle MBN we get $|BN| = \frac{4}{3}\sqrt{\frac{2}{5}}y$, and by the secant-tangent theorem we have $|MB|^2 = |BN| \cdot |BQ|$, whence $|BQ| = \frac{2}{15}\sqrt{\frac{2}{5}}y$. Then we find

$$|BQ|/|QN| = |BQ|/(|BN| - |BQ|) = 1/9.$$

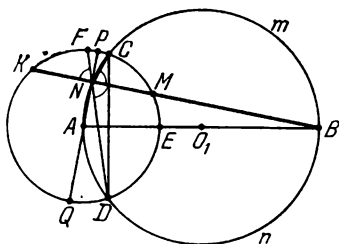


Fig. 231

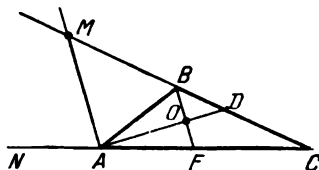


Fig. 232

19. Let the line BM cut the second circle at a point K (Fig. 231). The angle ANB rests on the diameter and is equal to 90° , and therefore the diameter QP is perpendicular to the chord KM and bisects it: $|KN| = |NM|$. Furthermore, the chord CD is perpendicular to the diameter AB and, therefore, $\widehat{CmB} = \widehat{DnB}$ and $\angle CNB = \angle DNB$. Suppose the line DN cuts the second circle at a point F . Then $\angle KNF = \angle DNB = \angle CNB$. In addition, being symmetric about the diameter PQ , the second circle passes into itself, and since $KN \perp PQ$ and $|KN| = |NM|$, the point M is projected onto the point K . The angles MNC and KNF are congruent and, therefore, the point C is projected onto the point F . Hence it follows that $|FN| = |NC|$. By the property of intersecting chords (sec 5.3) we have $|KM| |NM| = |FN| |ND|$, whence we get $|MN| = \sqrt{|NC| |ND|} = \sqrt{ab}$.

20. Suppose $BF \parallel AM$ (Fig. 232); cutting the sides of the angle MCA , the lines BF and AM intercept proportional segments:

$$\frac{|MB|}{|MC|} = \frac{|AF|}{|AC|}. \quad (1)$$

Assume that AD is the bisector of the triangle ABC . Then $\angle BAD = \frac{1}{2} \angle BAC$, but $\angle MAB = \frac{1}{2} (180^\circ - \angle BAC)$ and, therefore, $\angle MAD = \angle MAB + \angle BAD = 90^\circ$, that is, the bisectors of the exterior and interior angles at the same vertex of the triangle are perpendicular. In accordance with the construction, $BF \parallel AM$ and then $BF \perp AD$. Thus, in the triangle ABF the bisector AO is an altitude and, consequently, the triangle ABF is isosceles: $|AF| = |AB|$. Bearing this

in mind, we get, from (1), $\frac{|MB|}{|MC|} = \frac{|AB|}{|AC|}$.

The case when the line BC is cut by the extension of the bisector AM can be treated analogously.

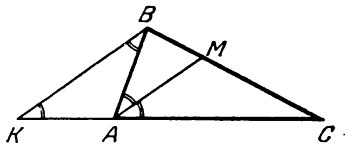


Fig. 233

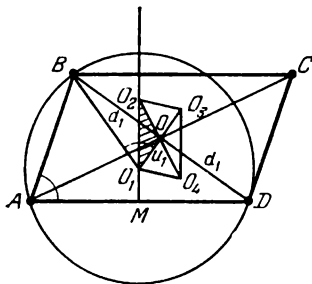


Fig. 234

21. Only two cases are possible here: either the point M lies within the segment BC or outside it. In the first case (Fig. 233) we draw $BK \parallel AM$ and then $|KA|/|AC| = |BM|/|MC|$.

By the hypothesis, $|BM|/|MC| = |AB|/|AC|$ and, consequently, $|KA|/|AC| = |AB|/|AC|$, whence it follows that $|KA| = |AB|$. In the isosceles triangle KAB the base angles are equal, but $\angle BKA = \angle MAC$ and $\angle KBA = \angle BAM$ and, therefore, $\angle BAM = \angle MAC$, that is, AM is a bisector of the triangle ABC .

The reasoning in the second case is quite analogous (see Fig. 232).

22. Let us consider the parallelogram $ABCD$ (Fig. 234). The centres O_1 and O_3 of the circles circumscribed about the triangles DAB and DBC lie on the midline of the segment BD on different sides of BD and at the same distance from BD . Similarly, the centres O_2 and O_4 of the circles circumscribed about the triangles DAC and ABC lie on the midline of the segment AC on different sides of AC and at the same distance from AC . Hence it follows that in the quadrangle

$O_1O_2O_3O_4$ the diagonals O_1O_3 and O_2O_4 meet at a point O (at the point of intersection of the diagonals BD and AC of the parallelogram $ABCD$) and are divided in half. Consequently, $O_1O_2O_3O_4$ is a parallelogram.

Let us introduce the designations $|BD| = 2d_1$, $|AC| = 2d_2$, $|O_1O_3| = 2u_1$, and $|O_2O_4| = 2u_2$. Since $O_1O_3 \perp BD$ and $O_2O_4 \perp AC$, the angle between the diagonals BD and AC of the parallelogram $ABCD$ is equal to the angle between the diagonals O_1O_3 and O_2O_4 of the parallelogram $O_1O_2O_3O_4$ and, therefore,

$$S_{ABCD}/S_{O_1O_2O_3O_4} = d_1d_2/u_1u_2.$$

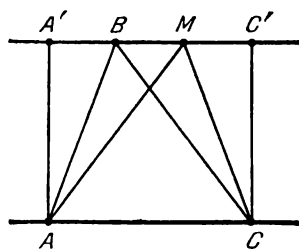
Let M be the midpoint of the side AD . Then the points M , O_1 , and O_2 lie on the same straight line and, as can be easily seen, $\angle ADO = \angle O_2O_1O$ (being the angles with perpendicular respective sides). As we have established above, $\angle O_1OO_2 = \angle DOA$ and, therefore, the triangles O_1O_2O and DAO are similar and $|OO_1|/|OO_2| = |OD|/|OA|$, i.e. $u_1/u_2 = d_1/d_2$. Hence it follows that

$$S_{ABCD}/S_{O_1O_2O_3O_4} = (d_1/u_1)^2.$$

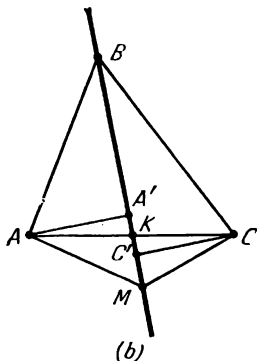
Note that O_1 is the centre of the circle circumscribed about the triangle DAB and, therefore, the inscribed angle DAB and the central angle DO_1B are subtended by the same arc and $\angle DO_1B = 2\angle DAB = 2\alpha$. Furthermore, $O_1O \perp DB$ and $|DO| = |OB|$ and, consequently, $\angle OO_1B = \alpha$ and $d_1/u_1 = \tan \alpha$. Thus, $S_{ABCD}/S_{O_1O_2O_3O_4} = \tan^2 \alpha$.

CHAPTER 6

1. Note that neither of the points M belonging to the line AB or to the line BC can be contained in the desired set. Indeed, if $M \in (AB)$, then the triangle AMB does not exist; if $M \in (BC)$, then the triangle BMC does not exist.



(a)



(b)

Fig. 235

Suppose M belongs to the desired set. This means that the areas of the triangles AMB and BMC are equal (Fig. 235). And since the triangles have the side BM in common, their altitudes AA' and CC' are equal, i.e. $|AA'| = |CC'|$. Hence it follows that the point M belongs to the line which passes through the point B and is equidistant from the points A and C . There are, evidently, two lines of this kind: one of them is parallel to the segment AC (Fig. 235a) and the other passes through the midpoint K of the segment AC (Fig. 235b) since in that case the equality $|AA'| = |CC'|$ yields the equality of the areas of the triangles ABK and KBC from which it follows that $|AK| = |KC|$. As was noted earlier, the point B at which the lines meet does not belong to the desired set. It is clear from the argument that if the point M does not belong to the indicated lines, then the areas of the triangles AMB and BMC are not equal.

Thus, the desired set is formed by two lines intersecting at the vertex B (the point of intersection exclusive): the line parallel to the side AC and the line passing through the midpoint of the side AC .

2. Let us designate as Q the area of the triangle ABC and assume, for definiteness, that $s > Q$. Note that the desired set cannot contain points lying on the lines AB and AC since in those cases either the triangle AMB , or the triangle AMC , does not exist.

Suppose the point M belongs to the desired set. Let us first consider the case when M is in the interior of the angle EAF (Fig. 236). By the hypothesis, we have $s_{AMB} + s_{AMC} = s$. Taking into consideration the obvious equality

$$s_{AMB} + s_{AMC} = s_{ABC} + s_{BMC},$$

we obtain $s_{BMC} = s - Q$. Consequently, the desired set contains those points M for which the area of the triangle BMC is constant and equal

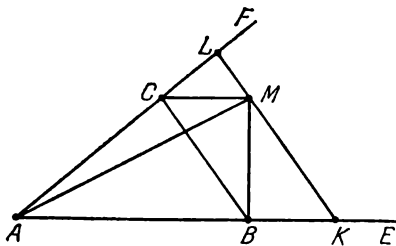


Fig. 236

to $s - Q$. It is clear that the area of the triangle BMC is constant if the point M lies on the line KL parallel to the side BC of the given triangle. It is equal to $s - Q$ if the distance between the side BC and the line KL is equal to $2\frac{s-Q}{|BC|}$. Since M lies in the interior of the angle EAF , the desired set contains all the points from the interval KL . If the point M is in the interior of the angle EAF but does not belong to the interval KL , then, evidently, $s_{BMC} \neq s - Q$.

exclusive, as the desired set of points, but now one side of the parallelogram is in the interior ($s < Q$) or on the boundary ($s = Q$) of the triangle ABC .

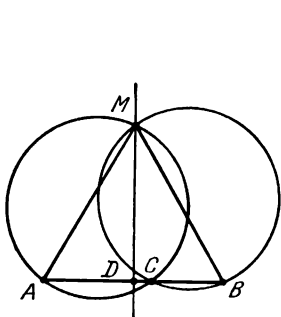


Fig. 239

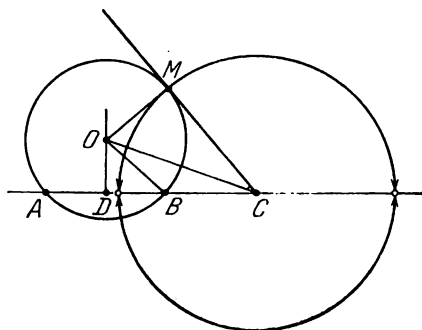


Fig. 240

3. It is evident that the point C belongs to the required set. Suppose M is an arbitrary point of the required set not coinciding with the point C (Fig. 239). Let us consider the angles MAB and MBA . These angles are subtended by the congruent arcs CM in the congruent circles and are, therefore, congruent. Consequently, $\angle MAB = \angle MBA$, that is, the triangle AMB is isosceles, with $|MA| = |MB|$. Thus, if the point M belongs to the required set and does not coincide with the point C , then it is equidistant from the given points A and B , that is, lies on the line perpendicular to the segment AB and passing through its midpoint.

Assume that the point M belongs to the line perpendicular to the segment AB and passing through its midpoint D . If M does not coincide with D , then M is the point of intersection of two congruent circles. In fact, let us draw one circle through the points A , C , and M and another circle through the points B , C , and M . These circles satisfy all the conditions of the problem: the first circle passes through the points A and C , the other through the points B and C ; the circles intersect and are congruent since, by the sine law, the radius of the first circle is equal to $\frac{|CM|}{2\sin MAC}$, and the radius of the second circle is equal to $\frac{|CM|}{2\sin MBC}$. The radii are equal because of the equality of the angles $\angle MAC = \angle MBC$.

Thus, the required set consists of the point C and all the points of the line perpendicular to the segment AB and passing through its midpoint, the midpoint of the segment AB exclusive.

It is easy to see that in the case when the point C is the midpoint of the segment AB , the required set is the line perpendicular to the segment AB and passing through its midpoint.

4. Let the point M belong to the required set, that is, be the point of tangency of some circle, passing through the points A and B , and the straight line passing through the point C (Fig. 240). The square of a tangent is known to be equal to the product of the secant by its external part. Consequently, $|CM^2| = |AC| |BC|$, that is, the point M lies on the circle of radius $R = \sqrt{|AC| |BC|}$ with centre at the point C .

It is clear that the point M cannot lie on the given line. Suppose now that M is a point of the indicated circle (Fig. 240) not lying on the given line.

Let us draw, through M , a line perpendicular to the segment CM . Suppose O is the point of intersection of that line and the perpendicular to the segment AB of the line passing through its middle. Let us consider the circle with centre at O passing through the point M . The line CM is, evidently, a tangent to that circle. It remains to show that the circle passes through the points A and B . Applying the Pythagorean theorem three times, we obtain

$$\begin{aligned} |OB|^2 &= |OD|^2 + |BD|^2 = |OC|^2 - |DC|^2 + |BD|^2 \\ &= |OM|^2 + |CM|^2 - |DC|^2 + |BD|^2, \end{aligned}$$

and transform the right-hand side of the equations obtained, using the formula for the difference between the squares of two numbers:

$$\begin{aligned} |OB|^2 &= |OM|^2 + |CM|^2 - (|DC| - |BD|) (|BD| + |DC|)^{\frac{1}{2}} \\ &= |OM|^2 + |CM|^2 - |BC| |AC|. \end{aligned}$$

Since $|CM|^2 = |BC| |AC|$, it follows that $|OB| = |OM|$.

Thus, the required set is a circle of radius $\sqrt{|AC| |BC|}$ with centre at C , without two points lying on the given line.

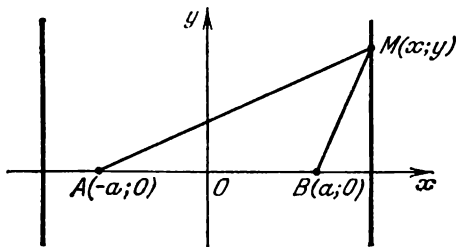


Fig. 241

5. Suppose the distance between the given points A and B is equal to $2a$. We choose a system of coordinates as shown in Fig. 241. Assume that the point $M(x; y)$ belongs to the required set. Then

$$|AM|^2 |AM|^2 - |BM|^2 |^2 = s.$$

By the Pythagorean theorem we find that

$$|AM|^2 = (x + a)^2 + y^2, \quad |BM|^2 = (x - a)^2 + y^2.$$

Thus, the required set includes the points whose coordinates x and y satisfy the equation

$$|(x+a)^2 + y^2 - (x-a)^2 - y^2| = s.$$

Simplifying this equation, we get $|4xa| = s$. Consequently, the point $M(x; y)$ belongs either to the line $x = s/4a$ or to the line $x = -s/4a$. The transformations carried out being equivalent, the converse assertion is also true: every point belonging to the lines $x = s/4a$ and $x = -s/4a$ belongs to the required set. Figure 241 shows the positions of the lines for the case $s = 6a^2$.

6. Let us designate the first two points as A and B and the third point as C . We choose a system of coordinates such that the points A

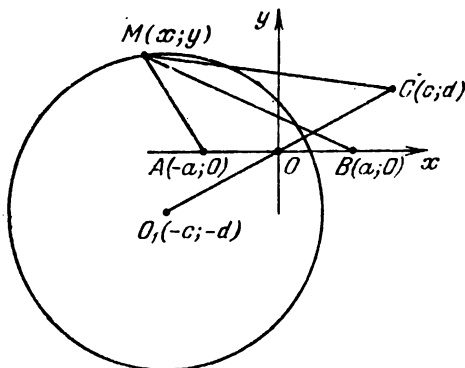


Fig. 242

and B have the coordinates $(-a; 0)$ and $(a; 0)$ (Fig. 242). The coordinates of the point C in that system will be designated as c and d respectively. Let the point $M(x; y)$ belong to the required set. Then

$$|MA|^2 + |MB|^2 = |MC|^2 \quad (1)$$

or

$$(x+a)^2 + y^2 + (x-a)^2 + y^2 = (x-c)^2 + (y-d)^2.$$

Simplifying this equation, we get

$$\begin{aligned} x^2 + y^2 + 2cx + 2dy &= c^2 + d^2 - 2a^2, \\ (x+c)^2 + (y+d)^2 &= 2c^2 + 2d^2 - 2a^2. \end{aligned} \quad (2)$$

Thus, the coordinates of the point M must satisfy equation (2). The converse assertion is also true: if the coordinates of the point M satisfy equation (2), then condition (1) is fulfilled for the point M , that is, the point M belongs to the required set.

Let us consider equation (2). If $c^2 + d^2 > a^2$, then (2) is an equation of a circle of radius

$$R = \sqrt{2c^2 + 2d^2 - 2a^2}$$

with centre at the point $O_1(-c; -d)$. If $c^2 + d^2 = a^2$, then equation (2) is satisfied only by the coordinates of the point $O_1(-c; -d)$. If $c^2 + d^2 < a^2$, then equation (2) is not satisfied by the coordinates of any point of the plane.

Thus we obtain the final result. If the point C is located farther from the midpoint of the segment AB than the points A and B , then the required set is a circle of radius $R = \sqrt{2c^2 + 2d^2 - a^2}$ with centre at the point symmetric with respect to the point C about the midpoint of the segment AB . If the point C is at the same distance from the midpoint of the segment AB as the points A and B , then the required set is a point symmetric with respect to the point C about the midpoint of the segment AB . If the point C lies nearer to the midpoint of the segment AB than the points A and B , then the required set is empty.

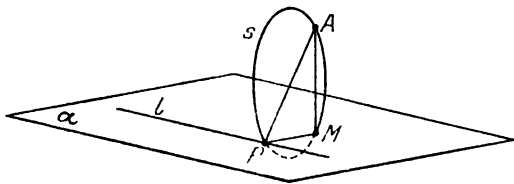


Fig. 243

7. Assume that the point M belongs to the required set, that is, is the foot of the perpendicular drawn through the given point A to some plane α passing through the given line l (Fig. 243).

Let us drop a perpendicular MP from the point M to the line l and connect the point A with the point P . By the theorem on three perpendiculars, $[AP] \perp l$, and, consequently, the point M belongs to the plane β such that $\beta \perp l$ and $A \in \beta$. Furthermore, since the angle AMP in the triangle AMP is a right angle, the point M belongs to the circle s with diameter AP lying in the plane β .

Let us prove the converse assertion. Let M be an arbitrary point of the circle s . We draw a plane α through the point M and the line l . That plane is perpendicular to the plane β which contains the circle s (because $\beta \perp l$). Furthermore, since AP is a diameter, $[AM] \perp [MP]$ and, consequently, $[AM] \perp \alpha$, that is, the point M is the foot of the perpendicular drawn through the point A to the plane passing through the given line l .

The proof carried out does not hold for two points of the circle s : the point A and the point P which is its projection onto the line l .

Notwithstanding this fact, both points belong to the required set. Indeed, the point A is the foot of the perpendicular drawn through it to the plane passing through the line l and the point A . The point P is the foot of the perpendicular drawn through the point A to the plane passing through the line l at right angles to the segment AP .

Thus, the required set is a circle which lies in the plane drawn through the given point A at right angles to the line l and whose diameter is the segment AP , where P is the projection of the point A onto the line l .

8. Suppose the point M belongs to the required set, that is, is the point of intersection of the medians of a certain triangle ABC (Fig. 244). We draw the medians CK and SK of the triangles ABC and ASB . Assume that O is the point of intersection of the medians of the triangle ASB . We connect the point M with the point O . Since

$$\frac{|CM|}{|MK|} = \frac{|SO|}{|OK|} = 2,$$

we have $[OM] \parallel [SC]$, that is, the point M belongs to the line which is parallel to the third edge of the trihedral angle and passes through the point O .

It is clear that the points of that line which lie in the exterior of that trihedral angle do not belong to the required set. The point O belongs to the set since it

is the point of intersection of the medians of the triangle ASB (C coincides with S).

Let us prove that all points of the ray ON belong to the required set. Suppose M is an arbitrary point of that ray. We draw a plane AMB . It cuts the third edge at some point C .

Let us consider the plane SMC . Since $[OM] \parallel [SC]$, the plane SMC cuts the segment AB at its midpoint K . Consequently, the lines CM and SO meet at the point K , with

$$\frac{|MC|}{|KM|} = \frac{|SO|}{|OK|}, \quad \text{but} \quad \frac{|SO|}{|OK|} = 2$$

and, consequently, M is the point of intersection of the medians of the triangle ABC . Thus, the required set is the ray ON whose direction is similar to that of the third edge of the trihedral angle and which begins at O , the point of intersection of the medians of the triangle ASB (S being the vertex of the trihedral angle).

9. Suppose the point M belongs to the required set, that is, is the point of tangency of some sphere, passing through the end points A and B of the given segment, and the given plane (Fig. 245). Let us extend the segment AB till its intersection with the plane α at the point O and connect the point O with the point M . We draw a plane through the rays OB and OM . It will cut the sphere along the circle touching the ray OM at the point M . By the secant-tangent theorem, the tangent and the secant being drawn from the same point to the circle, we find

$$|OM| = \sqrt{|OB| \cdot |OA|}, \quad (1)$$

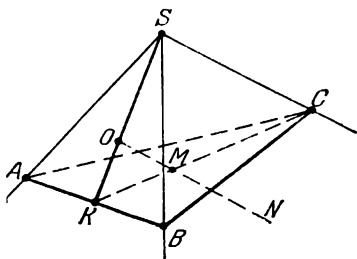


Fig. 244

that is, the point M lies on the circle of radius $R = \sqrt{|OB| |OA|}$ with centre at the point O .

Let us prove the converse assertion. Assume that the point M lies on the circle obtained. Then equation (1) holds true.

Let us draw an auxiliary circle through three points A , B , and M and designate its centre as S_1 . By virtue of condition (1), the segment OM is a tangent to that circle, and, consequently, $[OM] \perp [MS_1]$.

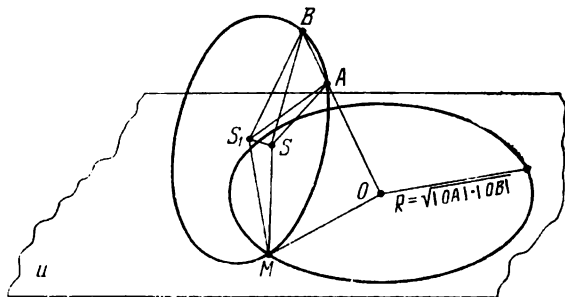


Fig. 245

Let us now draw a perpendicular through the point M to the given plane and a perpendicular through the point S_1 to the plane of the auxiliary circle. These two perpendiculars lie in the plane perpendicular to the segment OM and passing through the point M . It is easy to see that they cannot be parallel. We designate by S their point of intersection, and connect the point S with the points A and B . The projections of the segments SA , SB , and SM onto the plane of the auxiliary circle are the radii S_1A , S_1B , and S_1M respectively. Hence we infer that $|SA| = |SB| = |SM|$. Consequently, the sphere with centre at S and with the radius $|SM|$ touches the given plane at the point M and passes through the end points of the segment AB , that is, the point M belongs to the required set.

Thus we have found the set. It is a circle lying in the given plane with centre at O (the point of intersection of the extension of the given segment with the plane) and radius $R = \sqrt{|OB| |OA|}$.

10. Let us draw, through the given line l_1 , a plane α_1 parallel to the line l_2 and, through the line l_2 , a plane α_2 parallel to the line l_1 (Fig. 246). We shall prove that the required set is the set of all points of the planes α_1 and α_2 which do not belong to the lines l_1 and l_2 . If the point M_1 belongs to the plane α_2 but does not lie in the line l_2 , and a line passes through it and cuts the line l_2 , then that line lies in the plane α_2 and, therefore, cannot cut the line l_1 which lies in the plane α_1 . We can prove, by analogy, that every point of the plane α_1 , which does not lie in the line l_1 , belongs to the required set. We shall also prove that no other points of space possess the required property, that is, we shall show that through any other point of space we can

draw a line cutting l_1 and l_2 . If the point M lies on the lines l_1 and l_2 , then it is obvious. Suppose the point M does not lie either in the plane α_1 or in the plane α_2 (Fig. 247). We draw a plane α through the point M and a line l_1 and a plane β through the point M and the line l_2 .

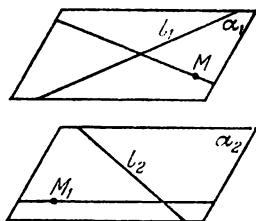


Fig. 246

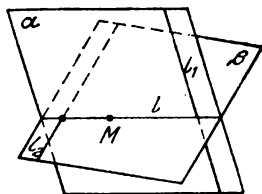


Fig. 247

The planes α and β intersect along some line l , which necessarily cuts both the line l_1 and the line l_2 . Indeed, the line l lies in the same plane α as the line l_1 and is not parallel to l_1 , since otherwise the point M would lie in the plane α_2 . Similarly, the line l lies in the same plane β as the line l_2 and is not parallel to l_2 , since otherwise the point M would lie in the plane α_1 .

11. Let us apply the method of similitude. We remove the condition that the point E lies on the side BC . We lay off an arbitrary seg-

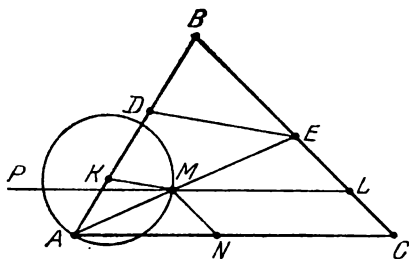


Fig. 248

ment AK on the side AB (Fig. 248) and an interval $[CL]$ on the side BC so that $|CL| = |AK|$, and a straight line LP , parallel to the side AC , through the point L . Next, we draw a circle of radius $|KA|$ from the point K as the centre. Suppose M is the point of intersection of the circle and the line LP and lies in the interior of the triangle. We connect the point M with the point K and draw, through the point M , a line parallel to $[BC]$. Assume that N is the point of intersection of that line and the side AC . We get a quadrangle $AKMN$ whose three sides are equal $|AK| = |KM| = |MN|$.

Let us extend the segment AM till it meets the side BC at the point E and draw a line, through the point E , parallel to the segment KM and intersecting the side AB at the point D . The quadrangle $ADEC$ is similar to the quadrangle $AKMN$ and, consequently, $|AD| = |DE| = |EC|$.

The problem has one and only one solution.

12. Suppose the channel is h wide. We draw a straight line, at right angles to the channel, through the point A , and lay off on it a segment AA_1 of length h as shown in Fig. 249. We connect the points A_1

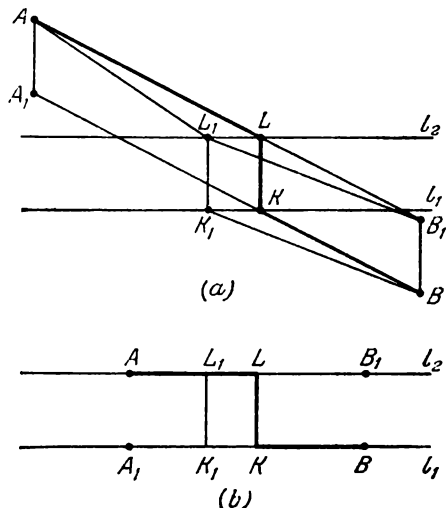


Fig. 249

and B . Assume that K is the point of intersection of the line A_1B and the line l_1 (Fig. 249a). The bridge must be constructed at the point K since the way $ALKB$ is the shortest. Indeed, for any other way AL_1K_1B we obtain

$$\begin{aligned} |AL_1| + |L_1K_1| + |K_1B| &= |AL_1| + |LK| + |L_1B_1| \\ &\geq |AL| + |LK| + |LB_1| = |AL| + |LK| + |KB|. \end{aligned}$$

We can put the equality sign only if the points A and B are located on the banks of the channel: $A \in l_2$, $B \in l_1$ (Fig. 249b). In that special case the bridge can be constructed at any point belonging to the segment A_1B .

13. Let us construct a point B_1 , which is symmetric with respect to the point B about the line l (Fig. 250). Through the points A and B_1 we draw a line p . Suppose $C = l \cap p$. Let us prove that C is the desired point. We consider the arbitrary point $C_1 \in l$, not coinciding

of the angle. Assume that $O = l_1 \cap l_2$. We construct a circle with centre at O of radius $r = |OB| = |OC|$. From the given point M we draw a tangent MT to that circle. The triangle KLA , where K and L are the points of intersection of the tangent with the sides of the angle, has a perimeter $2p$, that is, is the desired one. Let us prove this:

$$\begin{aligned} |AK| + |KL| + |AL| &= |AK| + |KN| + |NL| + |AL| \\ &= |AK| + |KB| + |LC| + |AL| = |AB| + |AC| = 2p. \end{aligned}$$

CHAPTER 7

1. (a) We designate the midpoints of the edges of the tetrahedron as $S_1, S_2, \dots, S_6$ (Fig. 253) and the points of intersection of the medians of the faces as M_1, M_2, M_3 , and M_4 . The segments DM_1 and

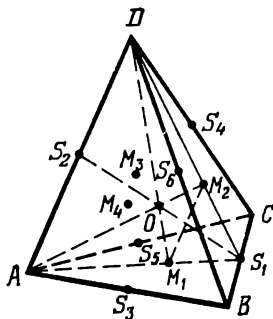


Fig. 253

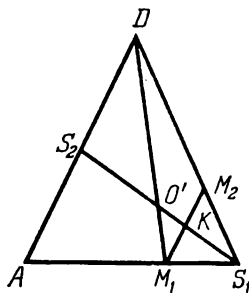


Fig. 254

AM_2 lie in the plane ADS_1 and, evidently, intersect. We designate their point of intersection as O . Since $[M_1M_2] \parallel [DA]$ and $|M_1M_2|/|DA| = 1/3$, it follows from the similarity of the triangles M_1OM_2 and DOA that $|DO|/|OM_1| = |AO|/|OM_2| = |DA|/|M_1M_2| = 3/1$. We can prove by analogy (considering the triangles BS_3D and CS_3D) that the segments BM_3 and CM_4 , as well, intersect the segment DM_1 at the point which divides it in the ratio $3/1$, reckoning from the vertex, that is, at the same point O . The segments BM_3 and CM_4 are divided by that point in the same ratio.

(b) We shall prove that the point O is the midpoint of the segment S_1S_2 . Let us consider the triangle ADS_1 (Figs. 253, 254). We shall designate $O' = [DM_1] \cap [S_1S_2]$ and prove that $O' = O$. The segment S_1S_2 is a median in the triangle DS_1A and, since $[M_1M_2] \parallel [AD]$, the segment S_1S_2 cuts the segment M_1M_2 at its midpoint K (Fig. 254). Taking into account that $|KM_1|/|S_2D| = |M_1M_2|/|DA| = 1/3$, we get, from the similarity of the triangles KM_1O' and S_2DO' that $|KO'|/|S_2O'| = |M_1O'|/|DO'| = 1/3$. The last equality yields $O' = O$. In addition, we find that $|O'S_2| = \frac{3}{4} |KS_2|$, and since

$|KS_2| = \frac{2}{3} |S_1S_2|$, we have $|O'S_2| = \frac{1}{2} |S_1S_2|$. Thus, the point O ($O = O'$) is the midpoint of the segment S_1S_2 . Considering the triangles CS_3D and BS_5D , we can prove, by analogy, that O is the midpoint of the segments S_3S_4 and S_5S_6 .

2. The construction of the section is shown in Fig. 255: (1) $S = (MN) \cap (AC)$, and (2) $Q = (SP) \cap [AB]$.

Designating $|BQ| = xa$, we draw $[QQ_1] \parallel [AC]$. Then $|QQ_1| = |BQ_1| = xa$ and, since $|BP| = \frac{4}{5}a$, we have $|Q_1P| = \left(\frac{4}{5} - x\right)a$. It follows from the similarity of the triangles SPC and QPQ_1 that $|SC|/|QQ_1| = |CP|/|Q_1P|$, whence we have $|SC| = \frac{x}{4-5x}a$. We draw $[MM_1] \parallel [AC]$. Reasoning by analogy, we find that $|SC| = a/3$. This means that $x/(4-5x) = 1/3$, whence we get $x = 1/2$, $|BQ| = a/2$.

Answer: $a/2$.

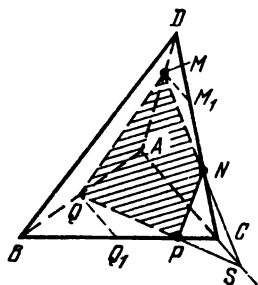


Fig. 255

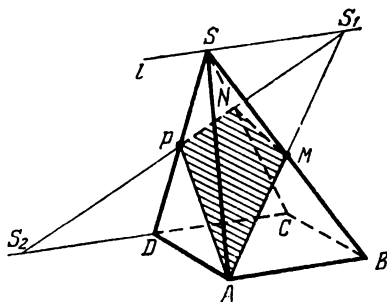


Fig. 256

3. We construct the point of intersection of the line AM and the plane SCD . Since $(AB) \parallel (CD)$, we have $(AB) \parallel (SCD)$ and this means that the planes SAB and SCD intersect along a straight line parallel to (AB) (in Fig. 256, $l \parallel (AB)$). We find that $S_1 = (AM) \cap l = (AM) \cap (SCD)$, $N = (S_1P) \cap [SC]$. The section is a quadrangle $AMNP$.

Suppose $S_2 = (S_1P) \cap (CD)$. M and P being the midpoints of the segments SB and SD , we have $|S_1S| = |AB|$ and $|S_2D| = |S_1S|$. And since $|CD| = |AB|$, we have $|S_2D| = |CD|$, and $|S_2C| = 2|CD|$. Now, from the similarity of the triangles SNS_1 and CNS_2 we get $|SN|/|NC| = |SS_1|/|CS_2| = 1/2$.

Answer: $1/2$, reckoning from the vertex.

4. Let us consider the plane NQP . It intersects the face BCC_1B_1 along the segment CQ . The plane of the section cuts the plane NQP along the line parallel to the line NQ . We draw a straight line $(PT) \parallel (NQ)$ in the plane NQP (Fig. 257). The point $T = (PT) \cap (CQ)$ belongs to the plane of the section. Note that since $CP| = |PN|$, we

have $|CT| = |TQ|$. Then we perform the following constructions: $S_1 = (MP) \cap (BC)$, $S_2 = (S_1T) \cap [CC_1]$, $S_3 = (S_1T) \cap [B_1C_1]$, $S_4 = (S_1T) \cap (BB_1)$, $S_5 = (MP) \cap (AB)$, $S_6 = (S_4S_5) \cap [A_1B_1]$, and $S_7 = (S_4S_5) \cap [AA_1]$. The section is a hexagon $MPS_2S_3S_6S_7$.

Let us now find the area of the section. We can easily see that $|S_1C| = |S_5A| = a/3$. T being the midpoint of the segment CQ , we have $|QS_3| = |S_1C| = a/3$ and $|S_3B_1| = a/3$. The triangles S_1S_2C and $S_3S_4B_1$ being similar to the triangle $S_3S_5C_1$, it follows that $|S_2C| = |S_4B_1| = a/3$. Thus we have $|BS_1| = |BS_4| = |BS_5| = 4a/3$ and, hence, the triangle $S_1S_4S_5$ is regular and its side is equal to $4\sqrt{2}a/3$ and its area is equal to $8\sqrt{3}a^2/9$. Each of the triangles S_1S_2P , $S_4S_6S_3$, and S_5MS_7 is similar to the triangle $S_1S_4S_5$ with a ratio $1/4$. This means that the area of each of these triangles is equal to $\frac{1}{16} \cdot \frac{8\sqrt{3}}{9} a^2 = \frac{\sqrt{3}}{18} a^2$. And now we can find the area of the

section: $\frac{8\sqrt{3}}{9} a^2 - 3 \cdot \frac{\sqrt{3}}{18} a^2 = \frac{13\sqrt{3}}{18} a^2$.

Answer: $\frac{13\sqrt{3}}{18} a^2$.

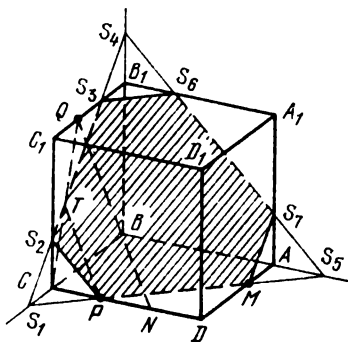


Fig. 257

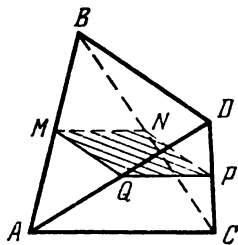


Fig. 258

5. Let $MNPQ$ be a section of the tetrahedron formed by a plane parallel to the lines AC and BD (in Fig. 258, $[MN] \parallel [QP] \parallel [AC]$, $[MQ] \parallel [NP] \parallel [BD]$). Then $(MN, MQ) = (AC, BD) = \varphi$, and this means that $\angle QMN = \varphi$ or $\angle QMN = \pi - \varphi$. In both cases, $\sin QMN = \sin \varphi$. Designating $|AM|/|AB| = x$, we get $|MQ| = x \times |BD| = xb$ and, since $|MB| = (1-x)|AB|$, it follows that $|MN| = (1-x)|AC| = (1-x)a$. Then we find the area of the section: $S = |MQ||MN| \sin \varphi = x(1-x)ab \sin \varphi$, where $0 < x < 1$. The function $x(1-x)$ (being a quadratic trinomial $-x^2 + x$) has the greatest value for $x = 1/2$ and it is equal to $1/4$. This

means that the section passing through the midpoints of the edges AB , BC , CD , and AD has the greatest area and that area is equal to $\frac{1}{4} ab \sin \alpha$.

Answer: $\frac{1}{4} ab \sin \alpha$.

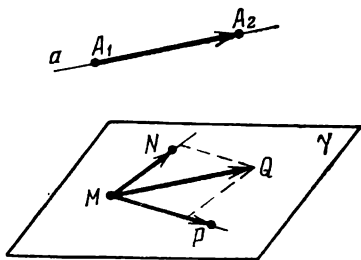


Fig. 259

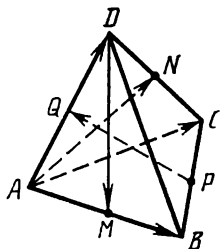


Fig. 260

6. Necessity. Suppose $a \parallel \gamma$, then the rays $[A_1A_2]$, $[MN]$, and $[MP]$ (Fig. 259) lie on the straight lines parallel to the plane γ . This means that the vectors $\overrightarrow{A_1A_2}$, $\overrightarrow{MN}$, and $\overrightarrow{MP}$ are coplanar. Since the vectors $\overrightarrow{MN}$ and $\overrightarrow{MP}$ are not collinear (the points M , N , and P do not lie on the same line), the vector $\overrightarrow{A_1A_2}$ can be represented in terms of these vectors: $\overrightarrow{A_1A_2} = \alpha \overrightarrow{MN} + \beta \overrightarrow{MP}$.

Sufficiency. Assume $\overrightarrow{A_1A_2} = \alpha \overrightarrow{MN} + \beta \overrightarrow{MP}$. We lay off the vector $\overrightarrow{MQ} = \alpha \overrightarrow{MN} + \beta \overrightarrow{MP}$ from the point M . If $\beta = 0$, then $\overrightarrow{MQ} = \alpha \overrightarrow{MN}$ and the point Q lies on the line MN . Since $\overrightarrow{A_1A_2} = \overrightarrow{MQ}$, the directions of these vectors coincide and this means that the rays $[A_1A_2]$ and $[MQ]$, defining these directions, lie on the parallel lines, i.e. $a \parallel (MN)$. Hence it follows that $a \parallel \gamma$. Similarly, if $\alpha = 0$, then $a \parallel (MP)$, i.e. $a \parallel \gamma$. If $\alpha \neq 0$ and $\beta \neq 0$, then, in accordance with the parallelogram rule, the point Q belongs to the plane γ . Hence, $(MQ) \subset \gamma$ as well. From the fact that $\overrightarrow{A_1A_2} = \overrightarrow{MQ}$ we get, as before, $a \parallel (MQ)$, i.e. $a \parallel \gamma$.

7. We suppose that $Q \in [AD]$ and designate $|AQ|/|AD| = x$. (Fig. 260). The lines DM and AN being parallel to the plane drawn through the line PQ , the vectors $\overrightarrow{DM}$, $\overrightarrow{AN}$, and $\overrightarrow{PQ}$ are coplanar. And since the vectors $\overrightarrow{DM}$ and $\overrightarrow{AN}$ are, evidently, not collinear, the vector $\overrightarrow{PQ}$ can be represented in terms of these vectors, that is, there

are numbers α and β such that

$$\overrightarrow{PQ} = \alpha \overrightarrow{DM} + \beta \overrightarrow{AN}. \quad (1)$$

Let us represent the vectors $\overrightarrow{PQ}$, $\overrightarrow{DM}$, and $\overrightarrow{AN}$ in terms of the non-complanar vectors $\overrightarrow{AB}$, $\overrightarrow{AC}$, and $\overrightarrow{AD}$. We have $\overrightarrow{DM} = \overrightarrow{DA} + \overrightarrow{AM} = -\overrightarrow{AD} + \frac{1}{2}\overrightarrow{AB}$, $\overrightarrow{AN} = \frac{1}{2}(\overrightarrow{AD} + \overrightarrow{AC})$, and $\overrightarrow{PQ} = \overrightarrow{PB} + \overrightarrow{BA} + \overrightarrow{AQ} = \frac{1}{2}\overrightarrow{CB} - \overrightarrow{AB} + x\overrightarrow{AD} = \frac{1}{2}(\overrightarrow{AB} - \overrightarrow{AC}) - \overrightarrow{AB} + x\overrightarrow{AD} = \frac{1}{2}\overrightarrow{AB} - \frac{1}{2}\overrightarrow{AC} + x\overrightarrow{AD}$. Substituting these resolutions into (1), we obtain $-\frac{1}{2}\overrightarrow{AB} - \frac{1}{2}\overrightarrow{AC} + x\overrightarrow{AD} = \frac{\alpha}{2}\overrightarrow{AB} + \frac{\beta}{2}\overrightarrow{AC} + \left(\frac{\beta}{2} - \alpha\right)\overrightarrow{AD}$, whence we find that $\alpha = -1$, $\beta = -1$, and $x = 1/2$. Hence, $|\overrightarrow{AQ}| = |\overrightarrow{AD}|/2$, and $|\overrightarrow{AQ}|/|\overrightarrow{QD}| = 1/1$.

Answer: 1/1.

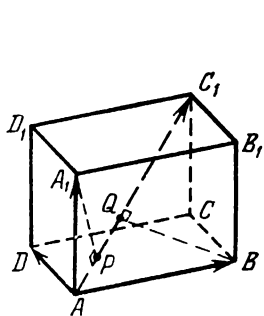


Fig. 261

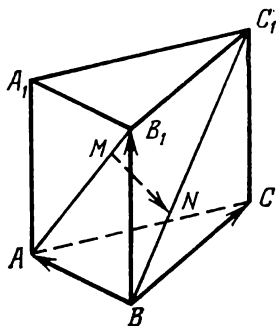


Fig. 262

8. The vectors $\overrightarrow{AP}$ and $\overrightarrow{AQ}$ are collinear with the vector $\overrightarrow{AC_1}$ (Fig. 261) and, therefore, $\overrightarrow{AP} = x\overrightarrow{AC_1}$, and $\overrightarrow{AQ} = y\overrightarrow{AC_1}$. Hence it follows that $\overrightarrow{A_1P} = -\overrightarrow{AA_1} + x\overrightarrow{AC_1}$, and $\overrightarrow{BQ} = -\overrightarrow{AB} + y\overrightarrow{AC_1}$. Since $(A_1P) \perp (AC_1)$, we have $\overrightarrow{A_1P} \cdot \overrightarrow{AC_1} = 0$, whence it follows that $x = \frac{\overrightarrow{AA_1} \cdot \overrightarrow{AC_1}}{|\overrightarrow{AC_1}|^2}$. Since $\overrightarrow{AC_1} = \overrightarrow{AB} + \overrightarrow{AD} + \overrightarrow{AA_1}$, we have $|\overrightarrow{AC_1}|^2 = a^2 + b^2 + c^2$ and $\overrightarrow{AA_1} \cdot \overrightarrow{AC_1} = c^2$. This means that $x = c^2/(a^2 + b^2 + c^2)$. Similarly, we find that $y = a^2/(a^2 + b^2 + c^2)$. Further-

more, $\overrightarrow{PQ} = \overrightarrow{AQ} - \overrightarrow{AP} = (y - x) \overrightarrow{AC_1}$ and this means that $|PQ| = |y - x| \cdot |AC_1| = |c^2 - a^2| / \sqrt{a^2 + b^2 + c^2}$.

Answer: $|c^2 - a^2| / \sqrt{a^2 + b^2 + c^2}$.

9. Since $(MN) \perp (AB)$, it follows that $\overrightarrow{MN} \cdot \overrightarrow{BA} = 0$. We represent $\overrightarrow{MN}$ in terms of the noncoplanar vectors $\overrightarrow{BA}$, $\overrightarrow{BC}$, and $\overrightarrow{BB_1}$ (Fig. 262). If we designate $|\overrightarrow{MB_1}|/|\overrightarrow{AB_1}| = x$, and $|\overrightarrow{BN}|/|\overrightarrow{BC_1}| = y$, we have $\overrightarrow{MN} = \overrightarrow{MB_1} + \overrightarrow{B_1B} + \overrightarrow{BN} - x\overrightarrow{AB_1} - \overrightarrow{BB_1} + y\overrightarrow{BC_1} = x(\overrightarrow{BB_1} - \overrightarrow{BA}) - \overrightarrow{BB_1} + y(\overrightarrow{BB_1} + \overrightarrow{BC}) = -x\overrightarrow{BA} + y\overrightarrow{BC} + (x + y - 1)\overrightarrow{BB_1}$. Taking into account that $|\overrightarrow{BA}| = |\overrightarrow{BC}| = |\overrightarrow{BB_1}| = a$, $\overrightarrow{BA} \cdot \overrightarrow{BC} = a^2/2$, and $\overrightarrow{BA} \cdot \overrightarrow{BB_1} = \overrightarrow{BC} \cdot \overrightarrow{BB_1} = 0$, we obtain, from the fact that $\overrightarrow{MN} \cdot \overrightarrow{BA} = 0$ and $|\overrightarrow{MN}|^2 = a^2/3$, the following system:

$$\begin{cases} -xa^2 + \frac{y}{2}a^2 = 0, \\ x^2a^2 + y^2a^2 + (x + y - 1)^2a^2 - xy a^2 = a^2/3. \end{cases}$$

It follows from the first equation that $y = 2x$ and, therefore, from the second equation we get $36x^2 - 18x + 2 = 0$, whence we have $x_1 = 1/3$ and $x_2 = 1/6$. Correspondingly, we find that $y_1 = 2/3$ and $y_2 = 1/3$. Hence it is easy to find that either $|\overrightarrow{AM}|/|\overrightarrow{MB_1}| = |\overrightarrow{BN}|/|\overrightarrow{NC_1}| = 2/1$ or $|\overrightarrow{AM}|/|\overrightarrow{MB_1}| = |\overrightarrow{BN}|/|\overrightarrow{NC_1}| = 1/2$.

Answer: $|\overrightarrow{AM}|/|\overrightarrow{MB_1}| = |\overrightarrow{BN}|/|\overrightarrow{NC_1}| = 2/1$, or $|\overrightarrow{AM}|/|\overrightarrow{MB_1}| = 5/1$, $|\overrightarrow{BN}|/|\overrightarrow{NC_1}| = 1/2$.

CHAPTER 8

1. Assume $\gamma \perp \alpha$, $\gamma \perp \beta$, $\alpha \cap \beta = c$, $\gamma \cap \alpha = a$, $\gamma \cap \beta = b$. It is required to prove that $\gamma \perp c$.

Let us take a point M on the line c and draw through it a perpendicular c_1 to α in the plane α and a perpendicular c_2 to β in the plane β . Then, $c_1 \perp \gamma$ and $c_2 \perp \gamma$, and, since only one perpendicular to γ can be drawn through one point M , we have $c_1 = c_2$. Since $c_1 \subset \alpha$, and $c_2 \subset \beta$, we have $c_1 = c_2 = \alpha \cap \beta = c$. This means that $c \perp \gamma$ and that is what we had to prove.

2. We shall prove that (a) yields (b). Suppose that $|\overrightarrow{SA_1}| = |\overrightarrow{SA_2}| = \dots = |\overrightarrow{SA_n}| = l$ (Fig. 263), SO is the altitude of the pyramid, and $|\overrightarrow{SO}| = H$. We have $\sin \angle SA_1O = \sin \angle SA_2O = \dots = \sin \angle SA_nO = H/l$, whence we find that $\angle SA_1O = \angle SA_2O = \dots = \angle SA_nO$, that is, the lateral edges are at the same angle to the plane of the base.

Let us now prove that (b) yields (c). Suppose $\angle SA_1O = \angle SA_2O = \dots = \angle SA_nO = \varphi$. Then, $|\overrightarrow{OA_1}| = |\overrightarrow{OA_2}| = \dots = |\overrightarrow{OA_n}| =$

$H \cot \varphi$ and this means that a circle with centre at O and radius $R = H \cot \varphi$ is circumscribed about the base of the pyramid.

Finally, we shall prove that (c) yields (a). We assume that O is the centre of the circle circumscribed about the base and then $|OA_1| = |OA_2| = \dots = |OA_n| = R$ is the radius of the circle. Since SO is the altitude of the pyramid, we have $|SA_1| = |SA_2| = \dots = |SA_n| = \sqrt{R^2 + H^2}$, that is, the lateral edges are of the same length. Thus we have proved that (a) leads to (b), (b) to (c), and (c) to (a). Hence it follows that any two of these three assertions are equivalent.

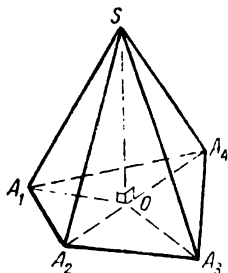


Fig. 263

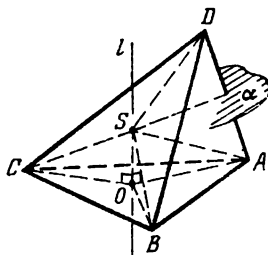


Fig. 264

3. A plane which passes through the midpoint of a line segment and is perpendicular to it is the set of all points equidistant from the end points of that segment.

Suppose O is the centre of the circle circumscribed about the face ABC of the tetrahedron (Fig. 264) and l is a straight line passing through the point O at right angles to the plane ABC . Every point of the line l is equidistant from the points A, B , and C . Indeed, $|OA| = |OB| = |OC|$, and if $S \in l, S \neq O$, then we find, from the right triangles SOA, SOB , and SOC , that $|SA| = |SB| = |SC|$. Assume that the plane α passes through the midpoint of the edge AD and is perpendicular to it. We shall prove that α and l intersect. Assume that $\alpha \parallel l$. From the fact that $(AD) \perp \alpha$ and $l \parallel \alpha$, it follows that $(AD) \perp l$. Since, in addition, $l \perp (AB)$, we have $l \perp (ABD)$. We have found that two distinct planes ABC and ABD are drawn through the point A at right angles to the line l . But that is impossible and, consequently, the statement that $l \parallel \alpha$ is incorrect, i.e. $l \nparallel \alpha$. Suppose $S = l \cap \alpha$. Then, $|SD| = |SA|$ since $S \in \alpha$ and $|SB| = |SC| = |SA|$ because $S \in l$. This means that the point S is equidistant from all the points A, B, C , and D and, therefore, belongs to every plane passing through the midpoint of the edge of the tetrahedron at right angles to that edge.

4. A perpendicular drawn in the plane ASC to the edge SC through the midpoint M of that edge lies in the plane of the section. We designate the point of intersection of that perpendicular and AC as K

and find $|KC|$. From the triangle SOC (Fig. 265a, b) we have $\cos \varphi = |OC|/|CS| = a/\sqrt{2}l$. Consequently, (a) $\cos \varphi = 1/\sqrt{3}$ and (b) $\cos \varphi = 1/\sqrt{5}$. Now, from the triangle KMC we find: (a) $|KC| = |MC|/\cos \varphi = \frac{3}{4} a \sqrt{2} = \frac{3}{4} |AC|$ and (b) $|KC| = \frac{5}{4} a \sqrt{2} = \frac{5}{4} |AC|$. Proceeding from the point K we have found, we construct the section. The plane α of the section and the line BD are perpendicular to the edge SC and, therefore, they are parallel. This means that

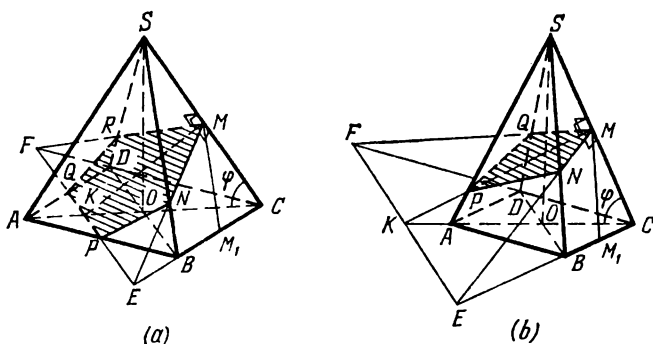


Fig. 265

the line of intersection of the planes α and $ABCD$ is also parallel to (BD) . We draw $(EF) \parallel (BD)$ and then (EM) and (FM) . As a result we obtain the required sections: (a) a pentagon $MNPQR$ and (b) a quadrangle $MNPQ$.

Let us now find the areas of the sections.

(a) We have $|MK| = |MC| \cdot \tan \varphi = \frac{\sqrt{3}}{2} a$, $|EF| = 2 |KC| = \frac{3}{\sqrt{2}} a$, $S_{EFM} = |EF| \cdot |MK| = \frac{3\sqrt{6}}{8} a^2$. It is easy to see that $|CE| = \frac{3}{2} |BC|$, and $|BE| = \frac{1}{3} |CE|$. Hence $|EP| = \frac{1}{3} |EF|$. We draw $[MM_1] \parallel [SB]$, and then we have $|EB| = |BM_1|$, and this means that $|EN| = \frac{1}{2} |EM|$. From the fact that $|EP| = \frac{1}{3} |EF|$ and $|EN| = \frac{1}{2} |EM|$ we find that $S_{EPN} = \frac{1}{6} S_{EFM}$. It follows from the symmetry of all the figures about the plane SAC that $S_{FQR} = S_{EPN} = \frac{1}{6} S_{EFM}$. Consequently, the area of the section is equal to $S_{EFM} - S_{EPN} - S_{FQR} = \frac{2}{3} S_{EFM} = \frac{\sqrt{6}}{4} a^2$.

(b) By analogy we find that $|KM| = \sqrt{\frac{5}{2}} a$, $|EF| = \frac{5}{\sqrt{2}} a$, $S_{EFM} = \frac{5\sqrt{5}}{4} a^2$. In $\triangle PSM$ we have $\angle PSM = \pi - 2\varphi$, $\tan PSM = -\tan 2\varphi = 4/3$, $|PM| = |MS| \cdot \tan PSM = \frac{2}{3} \sqrt{\frac{5}{2}} a$. Hence, $|PM| = \frac{2}{3} |KM|$. We draw $[MM_1] \parallel [SB]$. Then we have $|EC| = \frac{5}{2} |BC|$, $|EM_1| = 2 |BC|$, $|BM_1| = \frac{1}{2} |BC|$ and, consequently, $|MN|/|ME| = |M_1B|/|M_1E| = 1/4$. From the fact that $|MP| = \frac{2}{3} |MK|$, $|MN| = \frac{1}{4} |ME|$, it follows that $S_{MNP} = \frac{1}{6} S_{MEK}$, i.e. $S_{MNPQ} = \frac{1}{6} S_{MEF} = \frac{5\sqrt{5}}{24} a^2$.

Answer: (a) $\frac{\sqrt{6}}{4} a^2$, (b) $\frac{5\sqrt{5}}{24} a^2$.

5. The diagonal A_1C of the cube is perpendicular to the plane BDC_1 (Fig. 266) and, therefore, the plane of the section is parallel to

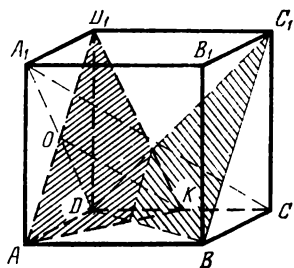


Fig. 266

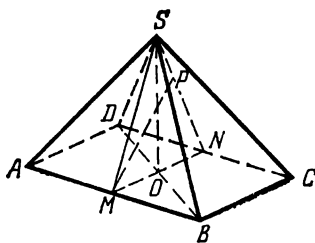


Fig. 267

the line A_1C . Proceeding from this fact, we construct the section as follows: in the plane A_1DCB_1 , through the point O of its intersection with the diagonal AD_1 , we draw a straight line, parallel to A_1C , till it meets the edge CD at a point K and, as a result, we obtain a section, which is a triangle AD_1K . Since $|DO| = |OA_1|$, we have $|DK| = |KC|$. We designate the length of the edge of the cube as a . The volume of the pyramid D_1ADK is equal to $\frac{1}{3} \cdot \frac{1}{2} \cdot |AD| \cdot |DK| \times |DD_1| = \frac{1}{12} a^3$. Hence it follows that the ratio between the volumes of the parts of the cube into which it is divided by the section is $1/11$.

Answer: $1/11$.

6. We designate $|SB| = a$ (Fig. 267). Next we find the length h of the perpendicular dropped from the point B to the plane SCD , that is, the distance from that point to the plane SCD . Since $(AB) \parallel (SCD)$, the distance from the midpoint M of the edge AB to the plane SCD is also h . Let N be the midpoint of the edge CD . Then $(SMN) \perp (SCD)$ and if $[MP] \perp (SN)$, then $[MP] \perp (SCD)$ and, hence, $|MP| = h$. Let us find $|MP|$ as an altitude of the triangle SMN . Since $\angle SBO = \pi/4$, we have $|SO| = |OB| = a/\sqrt{2}$ and this means that $|MO| = a/2$; $|MN| = a$, $|SN| = \sqrt{|SO|^2 + |ON|^2} = a\sqrt{3}/2$. Now we can find $h = |MP| = |SO| \cdot |MN| / |SN| = \sqrt{2}/3a$. Hence it follows that if α is the angle between (SB) and (SCD) , then $\sin \alpha = h/a = \sqrt{2}/3$.

Answer: $\arcsin \sqrt{2}/3$.

7. If $M_0 \in \alpha$, then $\rho = 0$ and $ax_0 + by_0 + cz_0 + d = 0$ and this means that the formula is true in this case. Suppose $M_0 \notin \alpha$, $M_1(x_1; y_1; z_1)$ is the foot of the perpendicular dropped from the point

M_0 to α . We designate $\vec{r}_0 = \vec{OM}_0 = (x_0; y_0; z_0)$ and $\vec{r}_1 = \vec{OM}_1 = (x_1; y_1; z_1)$, where O is the origin. The equation $ax_1 + by_1 + cz_1 + d = 0$ can be rewritten in the form $\vec{n} \cdot \vec{r}_1 + d = 0$, where $\vec{n} = (a; b; c)$ is a vector perpendicular to the plane. The vectors $\vec{M}_0M_1 = \vec{r}_1 - \vec{r}_0$ and $\vec{n}$ are collinear and, therefore, $\vec{r}_1 - \vec{r}_0 = \lambda \vec{n}$, whence we have $\vec{r}_1 = \vec{r}_0 + \lambda \vec{n}$. Then $(\vec{r}_0 + \lambda \vec{n}) \cdot \vec{n} + d = 0$ and, hence, $\lambda = -\frac{\vec{n} \cdot \vec{r}_0 + d}{|\vec{n}|^2}$. Now we have

$$\rho = |\vec{M}_0M_1| = |\lambda| \cdot |\vec{n}| = \frac{|\vec{n} \cdot \vec{r}_0 + d|}{|\vec{n}|} = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

8. If M is the centre of the base ABC (Fig. 268), then $\vec{OM} = \frac{1}{3}(\vec{OA} + \vec{OB} + \vec{OC}) = (4; -1; 0)$. The point S has the coordinates $S(0; y; z)$ and, therefore, $\vec{MS} = \vec{OS} - \vec{OM} = (-4; y + 1; z)$. We find that $\vec{AB} = (0; -3; 3)$ and $\vec{AC} = (-3; -3; 0)$. Since $(MS) \perp (ABC)$, we have $\vec{MS} \cdot \vec{AB} = 0$ and $\vec{MS} \cdot \vec{AC} = 0$. Writing these equalities in the coordinates, we get $-3(y + 1) + 3z = 0$ and $12 - 3(y + 1) = 0$, whence we have $y = 3, z = 4$.

Suppose P is the midpoint of the edge AC , and PQ is a perpendicular to (SB) in the plane SBP . Since $(AC) \perp (SBP)$, we have $(AC) \perp (PQ)$ and this means that PQ is a common perpendicular to the lines AC and SB , and $|PQ|$ is the distance between these lines. We have $\vec{OP} = \frac{1}{2}(\vec{OA} + \vec{OC}) = (7/2; -1/2; -1)$, $\vec{BP} = \vec{OP} - \vec{OB} = (-3/2; 3/2;$

$-3)$, $\overrightarrow{BS} = \overrightarrow{OS} - \overrightarrow{OB} = (-5; 5; 2)$, and $\overrightarrow{MS} = (-4; 4; 4)$. Hence, $|BP| = \sqrt{27/2}$, $|BS| = \sqrt{54}$, $|MS| = \sqrt{48}$, and $|PQ| = |BP| \cdot |MS|/|BS| = 2\sqrt{3}$.

9. Let l be an edge of the dihedral angle shown in Fig. 269. We draw the planes $(APM) \perp l$ and $(BQN) \perp l$ and then we have $\angle APM = \angle BQN = \pi/3$. In addition, we draw perpendiculars AA_1 and BB_1 to the planes of the faces γ_2 and γ_1 of the angle. The dihedral angle being acute, the points A_1 and B_1 lie on the sides PM and QN of the plane angles respectively. By the hypothesis, $\angle BAB_1 = \pi/6$, $\angle ABA_1 = \arcsin 0.7$. If $(AC) \parallel l$, then $(AC) \perp (BQN)$ and, hence, $\angle BCA = \pi/2$. The angle BAC is acute and is equal to the angle between (AB) and l . We designate $AB = a$.

We find that $|BB_1| = a \cdot \sin BAB_1 = a/2$, $|QB_1| = |BB_1| \times \cot |QBB_1| = \frac{a}{2\sqrt{3}}$, $|AA_1| = a \cdot \sin ABA_1 = 0.7a$, $|PA| = |AA_1|/\sin APA_1 = 7a/5\sqrt{3}$. Since $|QC| = |PA|$, we have

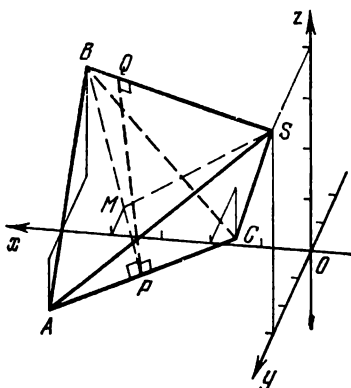


Fig. 268

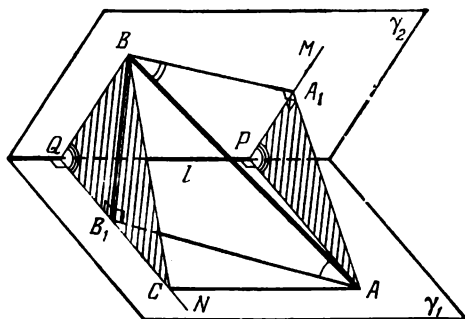


Fig. 269

$|B_1C| = |PA| - |QB_1| = 9a/10\sqrt{3}$. In the triangle AB_1C we have $|AB_1| = a \cdot \cos(\pi/6) = a\sqrt{3}/2$, $|AC| = \sqrt{|AB_1|^2 - |B_1C|^2} = \frac{2\sqrt{3}}{5}a$. Consequently, $\cos BAC = |AC|/|AB| = 2\sqrt{3}/5$.

Answer: $\arccos(2\sqrt{3}/5)$.

10. We construct a section $ABMN$ (in Fig. 270, $[MN] \parallel [AB] \parallel [CD]$). We designate the volume of the pyramid $SABCD$ as V and the volume of the pyramid $SABMN$ as V_1 . The volume V_1 is equal to the sum of the volumes V_2 and V_3 of the pyramids $SBMN$ and $SBAN$. Let us compare the volumes of the pyramids $SBMN$ and $SBCD$. We take the faces SMN and SCD as the bases of these pyramids, the point B being their common vertex. Since $|SM| = \frac{1}{2}|SC|$, we have

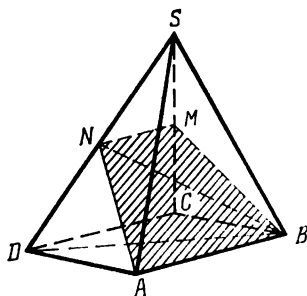


Fig. 270

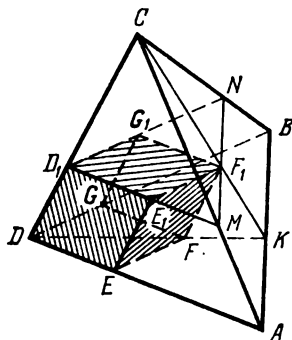


Fig. 271

$S_{SMN} = \frac{1}{4} S_{SCD}$ and this means that $V_2 = \frac{1}{4} V_{SBCD}$. It is evident that $V_{SBCD} = \frac{1}{2} V$, and, therefore, $V_2 = \frac{1}{8} V$.

Let us consider the pyramids $SBAN$ and $SBAD$. We take the faces SAN and SAD as their bases, the point B being their common vertex. Since N is the midpoint of the edge SD , we have $S_{SAN} = \frac{1}{2} S_{SAD}$ and this means that $V_3 = \frac{1}{2} V_{SBAD} = \frac{1}{4} V$. Now we can find the volume $V_1 = V_2 + V_3 = \frac{3}{8} V$. The volume of the other cut-off part is equal to $\frac{5}{8} V$ and the ratio of the volumes of the parts is $3/5$.

11. It is clear that the tetrahedron and the parallelepiped have a common vertex which we designate as D (Fig. 271). The edges of the parallelepiped issuing from that vertex lie on the edges of the tetrahedron. The face ABC of the tetrahedron can only contain the vertex of the parallelepiped not lying in the same face with the vertex D . There is only one vertex answering the requirement, we designate it as F_1 . Let us consider the sections of the tetrahedron formed by the

planes $D_1E_1F_1G_1$ and DD_1F_1F . We introduce the designations $|CF_1|/|CK| = y$, $|MF_1|/|MN| = x$, H is the altitude of the tetrahedron drawn from the vertex C , and h is the altitude of the parallelepiped dropped to the base $DEFG$. The altitude of the tetrahedron CD_1MN drawn from the vertex C is equal to $H - h$. This tetrahedron is homothetic to the tetrahedron $CDAB$ with centre at C and a ratio y . Therefore, $S_{D_1MN} = y^2 \cdot S_{DAB}$, $H - h = yH$, i.e. $h = (1 - y)H$. The triangles ME_1F_1 and MD_1N are similar, with a ratio x , and, therefore, $S_{ME_1F_1} = x^2 \cdot S_{MD_1N}$ and the triangles NG_1F_1 and ND_1M are similar, with a ratio of similitude $1 - x$, and, therefore, $S_{NG_1F_1} = (1 - x)^2 S_{ND_1M}$. Hence it follows that $S_{D_1E_1F_1G_1} = S_{D_1MN} - S_{ME_1F_1} - S_{NG_1F_1} = 2x(1 - x) S_{D_1MN} = 2x(1 - x) \times y^2 S_{DAB}$. We find the volume of the parallelepiped: $V_p = S_{D_1E_1F_1G_1} \cdot h = 2x(1 - x) y^2 (1 - y) S_{DAB} \cdot H$. Since $S_{DAB} \times H = 3V$, we have $V_p = 6x(1 - x) y^2 (1 - y) V$, where $0 < x < 1$, $0 < y < 1$. The function $x(1 - x)$ has the greatest value for $x = 1/2$ and it is equal to $1/4$. It is easy to establish that the function $y^2(1 - y)$ assumes the greatest value on the interval $(0; 1)$ for $y = 2/3$ and it is equal to $4/27$. This means that $x(1 - x) y^2 (1 - y) \leq (1/4) \cdot (4/27) = 1/27$ for $0 < x < 1$, $0 < y < 1$, the equality taking place at $x = 1/2$, $y = 2/3$. Thus, the greatest value of the volume of the parallelepiped is equal to $6 \cdot \frac{1}{27} V = \frac{2}{9} V$. The parallelepiped whose vertex F_1 coincides with the point of intersection of the medians of the face ABC has the greatest volume.

Answer: $\frac{2}{9} V$.

CHAPTER 9

1. Suppose R is the radius of the circle and r is the radius of the base of the cone. Then $2\pi r = R\alpha$, whence $r = R\alpha/2\pi$. We find the altitude of the cone and its volume: $H = \sqrt{R^2 - r^2} = R \sqrt{1 - \frac{\alpha^2}{4\pi^2}}$ and $V = \frac{1}{3} \pi r^2 H = \frac{\pi R^3}{3} \cdot \frac{\alpha^2}{4\pi^2} \sqrt{1 - \frac{\alpha^2}{4\pi^2}}$. We designate $\alpha^2/4\pi^2 = x$, $0 < x < 1$, and $V_0 = \pi R^3/3$. Then, $V = V_0 x \sqrt{1 - x}$. We find the derivative $V' = V_0 \frac{2-3x}{2\sqrt{1-x}}$ and the critical point $x_0 = 2/3$. It is easy to see that at that point the function V has the greatest value.

Answer: For $\alpha = 2\pi \sqrt{2/3}$.

2. When we rotate the rectangular trapezoid $MNBC$ (Fig. 272), we obtain a truncated cone with the altitude $H = |BD| = a\sqrt{3}/2$ and the radii of the bases $R = |MC| = b + (a/2)$ and $r = |NB| = b$. The volume of the cone is equal to

$$V_1 = \frac{1}{3} \pi H (R^2 + Rr + r^2) = \frac{\pi H}{3} \left(3b^2 + \frac{3}{2} ab + \frac{a^2}{4} \right).$$

By analogy we find the volume of the truncated cone, resulting from the rotation of the rectangular trapezoid $MNBA$:

$$V_2 = \frac{\pi H}{3} \left(b^2 + b \left(b - \frac{a}{2} \right) + \left(b - \frac{a}{2} \right)^2 \right) = \frac{\pi H}{3} \left(3b^2 - \frac{3}{2} ab + \frac{a^2}{4} \right).$$

The volume of the given solid of revolution is equal to the difference between the volumes of the cones: $V = V_1 - V_2 = \pi H ab = \frac{\sqrt{3}\pi}{2} ba^2$.

Answer: $\frac{\sqrt{3}\pi}{2} ba^2$.

3. It follows from the hypothesis that one of the lateral faces of the prism is inscribed into the base of the cone (Fig. 273). That face

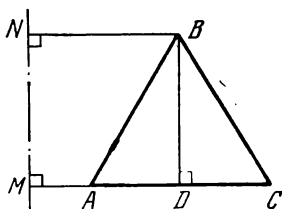


Fig. 272

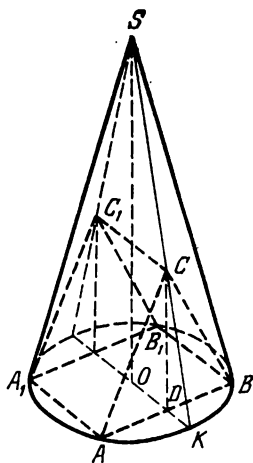


Fig. 273

is a rectangle with the sides of the lengths 1.6 dm and 1.2 dm. Consequently, the radius of the base of the cone is equal to $R = \frac{1}{2} \sqrt{1.6^2 + 1.2^2} = 1$ dm. The right triangles CDK and SOK are similar and, therefore, $|SO|/|CD| = |OK|/|DK|$. Hence, taking into account that $|CD| = 0.8\sqrt{3}$ dm, $|OK| = 1$ dm, and $|DK| = |OK| - |OD| = 0.4$ dm, we obtain $|SO| = 2\sqrt{3}$ dm. Now we can find the volume of the cone: $V = \frac{\pi}{3} R^2 \cdot |SO| = 2\pi/\sqrt{3}$ dm³.

4. We introduce a system of coordinates as shown in Fig. 274. The points B , C_1 , and M (the midpoint of the edge AD) and N (the centre of the face CC_1D_1D) have the coordinates $B(a; a; 0)$, $C_1(0; a; a)$, $N(0; a/2; a/2)$ and $M(a/2; 0; 0)$. Substituting these coordinates into

the equation for the sphere

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = R^2$$

we obtain a system of equations

$$\begin{cases} (a - x_0)^2 + (a - y_0)^2 + z_0^2 = R^2, \\ x_0^2 + (a - y_0)^2 + (a - z_0)^2 = R^2, \\ \left(\frac{a}{2} - x_0\right)^2 + y_0^2 + z_0^2 = R^2, \\ x_0^2 + \left(\frac{a}{2} - y_0\right)^2 + \left(\frac{a}{2} - z_0\right)^2 = R^2 \end{cases}$$

with respect to x_0, y_0, z_0 (the coordinates of the centre of the sphere) and R (the radius of the sphere). Subtracting the second equation

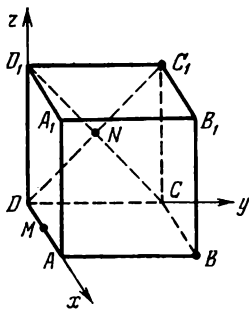


Fig. 274

from the first, and the fourth equation from the second and the third, we obtain, after simplification,

$$\begin{cases} z_0 = x_0, \\ y_0 + z_0 = \frac{3}{2} a, \\ x_0 = y_0 + z_0 - \frac{a}{4}. \end{cases}$$

Hence we find $x_0 = z_0 = 5a/4$, and $y_0 = a/4$. Substituting these values into any equation of the original system, we find that $R = a\sqrt{35}/4$.

5. Suppose O is the centre of the sphere (Fig. 275). From the fact that $|OC| = |OA| = r$ and $|AC| = \sqrt{2}r$, it follows that $\angle AOC = \pi/2$. We draw a plane $\beta \perp (CD)$ through the point B and suppose that $N = \beta \cap (CD)$. We also draw a straight line $(AP) \parallel (CD)$, $P \in \alpha$. Then $(CD) \perp (PN)$ and this means that $(PN) \parallel (AO)$ and, therefore,

$|ON| = |AP|$. Since $\angle APB = \pi/2$ and $\angle BAP = \widehat{AB, CD} = \pi/3$, we have $|AP| = |AB| \cos(\pi/3) = r/2$. This means that the plane β passes through the midpoint of one of the radii OC or OD . If we assume that the plane β passes through the midpoint M of the radius OC , then, from the fact that in the triangle OBC the altitude BM is also a median, it follows that $|BC| = |OB| = r$, and this contradicts the condition $|BC| > |AC|$. Hence, the plane β passes through the midpoint N of the radius OD and then $|BD| = r$.

Answer: r .

6. The sphere touches the bases of the cone at their centres and, consequently, the altitude of the cone is equal to $2R$. We designate

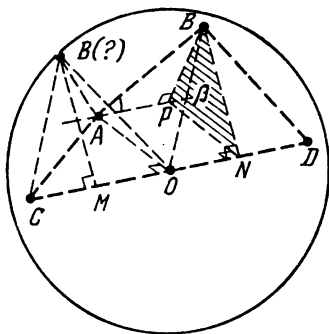


Fig. 275

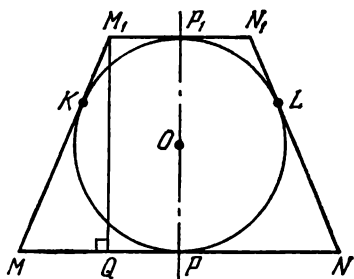


Fig. 276

the radii of the bases of the cone as r and r_1 ($r > r_1$). Let us consider the section of the given solids formed by the plane passing through the axis PP_1 of the cone (Fig. 276). The section of the sphere is the major circle, the section of the cone is the isosceles trapezoid MM_1N_1N circumscribed about the circle. Assume that K is the point of tangency. Then, by the property of tangents, we have $|MK| = |MP| = r$, $|M_1K| = |M_1P_1| = r_1$. But $|MK| + |KM_1| = a$ and, hence,

$$a = r + r_1 \quad (1)$$

In the right triangle MM_1Q we have $|M_1O| = 2R$, $|MQ| = r - r_1 = \sqrt{|MM_1|^2 - |M_1Q|^2} = \sqrt{a^2 - 4R^2}$. Hence, we find, from (1), that

$$r = \frac{1}{2} (a + \sqrt{a^2 - 4R^2}), \quad r_1 = \frac{1}{2} (a - \sqrt{a^2 - 4R^2}).$$

The volume of the cone is equal to $\frac{\pi}{3} \cdot 2R (r^2 + rr_1 + r_1^2) = \frac{2}{3} \pi R (a^2 - R^2)$.

7. It is easy to establish that the centre of the sphere touching the lateral faces of the pyramid lies on the ray SP (Fig. 277), where P

is the centre of the base $ABCD$. From the fact that the second sphere touches not only the lateral faces but also the sphere circumscribed about the pyramid, it follows that the centre of the second sphere lies on the diameter SQ of the circumscribed sphere and Q is the point of tangency of those spheres. Assume that O is the centre of the second

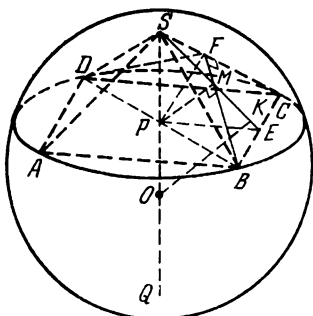


Fig. 277

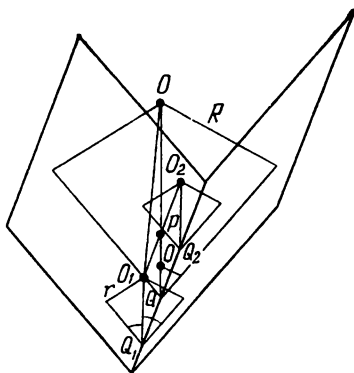


Fig. 278

sphere, K is its point of tangency with the face SBC , and r is the radius of that sphere. We designate $\angle OSK = \alpha$ and have

$$|OQ| = |OK| = r, |SO| = 2R - r, |OK| = |SO| \sin \alpha,$$

that is, $r = (2R - r) \sin \alpha$, whence $r = 2R \sin \alpha / (1 + \sin \alpha)$.

Let us find $\sin \alpha$. Assuming $[BF] \perp (SC)$, we have $\angle BFD = 120^\circ$, $\angle PFB = 60^\circ$ and $\angle PBF = 30^\circ$. Supposing $[SE] \perp [BC]$, $M = (BF) \cap (SE)$, we find that $[PM] \perp (SBC)$ since $(BFD) \perp (SBC)$, $(SPE) \perp (SBC)$, and $(PM) = (BFD) \cap (SPE)$. Thus we have $[PM] \perp [BF]$, and from the right triangle PMB we find that $|PM| = \frac{1}{2} |PB|$. Taking into account that $|PE| = \frac{1}{\sqrt{2}} |PB|$, we obtain

$\sin |PEM| = \frac{1}{\sqrt{2}}$. Hence it follows that $\angle PEM = 45^\circ$ and this

means that $\alpha = \angle PSE = 45^\circ$ as well. Then we easily find that $r = 2(\sqrt{2} - 1)R$.

Answer: $2(\sqrt{2} - 1)R$.

8. Let us designate the centres of the given spheres as O_1 and O_2 , the centre of the third sphere as O , and the radius of that sphere as R (Fig. 278). The point O belongs to the bisector of the given angle and to the plane passing through the midpoint P of the line segment O_1O_2 at right angles to it. Consequently, the point O belongs to the intersection of the bisector and that plane, that is, to the ray QP . We have

$|OQ| = R/\sin \alpha$, $|PQ| = |O_1Q_1| = r/\sin \alpha$, $|OP| = \left| \frac{R}{\sin \alpha} - \frac{r}{\sin \alpha} \right|$, $|O_1P| = r$, and $|OO_1| = R + r$. By the Pythagorean theorem, we get, from the right triangle O_1OP , $|OP^2| = |O_1O|^2 - |O_1P|^2$, whence we find $R^2 \cos^2 \alpha - 2r(1 + \sin^2 \alpha)R + r^2 = 0$ and

$$R = \frac{1 + \sin^2 \alpha \pm \sqrt{3 \sin^2 \alpha + \sin^4 \alpha}}{\cos^2 \alpha} r. \quad (1)$$

Answer: R is given by formula (1).

9. *Necessity.* If the prism is inscribed into the sphere, then its every face and, in particular, the base, is inscribed into the circle, which is the section of the sphere formed by the plane of that face. Every lateral face of such a prism is a parallelogram, inscribed into the circle and, hence, the lateral faces are rectangles. It follows that the lateral edges of the prism are perpendicular to the planes of the bases, that is, the prism is right-angled.

Sufficiency. Suppose AA_1 is a lateral edge of the prism, and P and P_1 are the centres of the circles circumscribed about the bases (Fig. 184 in Ch. 9). The parallel

displacement of $\overrightarrow{AA_1}$ maps one base of the prism onto the other, and under this mapping the point P_1 is the image of the point P .

This means that $\overrightarrow{PP_1} = \overrightarrow{AA_1}$ and, therefore, the line PP_1 is perpendicular to the base of the prism. Hence it follows that the midpoint O of the segment PP_1 is equidistant from all vertices of the prism, that is, the given prism is inscribed into the sphere with centre O and radius $R = |OA|$.

10. Suppose the sphere with centre O touches the edge SB and SC of the pyramid (Fig. 279) at points K_1 and K_2 , the extension of the edge SA beyond the vertex S at a point Q , and the plane ABC at a

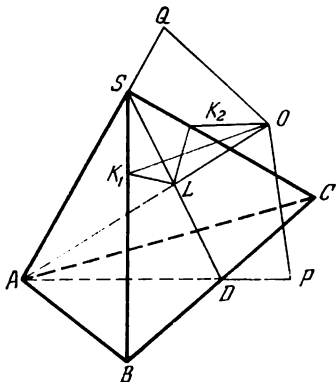


Fig. 279

point P . Assume $|OL| \perp (SBC)$. Then, from the fact that $|OK_1| \perp [SB]$ and $|OK_2| \perp [SC]$, it follows that $|LK_1| \perp [SB]$ and $|LK_2| \perp [SC]$. The equalities $|OK_1| = |OK_2| = R$, where R is the radius of the sphere, yield $|LK_1| = |LK_2|$. From this equation and from the fact that the feet of the perpendiculars LK_1 and LK_2 to the lines SB and SC belong to the segments SB and SC , it follows that the point L lies on the bisector SD of the angle BSC . The plane SAD and the line OL are perpendicular to the plane SBC and, hence, the point O lies in the plane ASD and the point P lies on the ray AD . It follows from

the equation $|OP| = |OQ|$ that AO is the bisector of the angle SAD . The triangle SAD is isosceles and, therefore, the bisector AO passes through the midpoint of the side SD and is perpendicular to it. Taking into account that $(OL) \perp (SD)$, we find that the lines OL and AO coincide and this means that L is the midpoint of the side SD .

Let us find $|LK_1|$. In the triangle SBD we have

$$|BD| = a/\sqrt{3}, \quad |SD| = \sqrt{2/3}a, \quad |SL| = a/\sqrt{6},$$

$$|LK_1| = |BD| \frac{|SL|}{|SB|} = \frac{a}{3\sqrt{2}}.$$

From the triangle DAL we find $\sin DAL = 1/\sqrt{6}$, $|AL| = \sqrt{5/6}a$, and from the triangle AOP we find that $|AO| = |OP|/\sin DAL = \sqrt{6}R$ and then $|LO| = \sqrt{6}R - \sqrt{5/6}a$. Now we find, from the triangle OLK_1 that

$$\sqrt{|OK_1|^2 - |LK_1|^2} = |LO|, \quad \sqrt{R^2 - \frac{a^2}{18}} = \sqrt{6}R - \sqrt{\frac{5}{6}}a.$$

This equation has a unique solution $R = 4a/(3\sqrt{5})$.

Answer: $4a/(3\sqrt{5})$.

11. Assume that O_1 is the centre of the circle inscribed into the triangle ASB , D and K are the points of tangency of the circle and the sides AB and SA , and r is the radius of the circle (Fig. 280). From the similarity of the triangles SBD and SO_1K we find that $r = a/(2\sqrt{7})$. Let SP be the altitude of the pyramid. From the triangle SDP we find that $|SP| = a/3$. The radius R of the sphere touching the edges of the pyramid can be found by the same method as was used in Problem 9 of 9.3. Carrying out the calculations, we get $R = a/(2\sqrt{3})$. The radius of the circle inscribed into the base ABC of the pyramid is also equal to $a/(2\sqrt{3})$ and, therefore, the centre of the sphere coincides with the centre P of the base. In the exterior of the pyramid there are three spheric segments cut off by the lateral faces and a hemisphere cut off by the base. The area of the semi-sphere is equal to $2\pi R^2 = \pi a^2/6$. We find the area of the segment by the formula $S_{\text{segm}} = 2\pi R h$, where $h = R - |PO_1|$ is the altitude of the segment. We have $|PO_1| = \sqrt{R^2 - r^2} = \frac{a}{\sqrt{21}}$, $h = \frac{\sqrt{7}-2}{2\sqrt{21}}a$.

$S_{\text{segm}} = \frac{\pi(\sqrt{7}-2)}{6\sqrt{7}}a^2$. The area of the part of the sphere lying

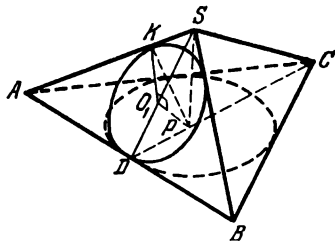


Fig. 280

13. We designate the centres of the spheres as O_1 , O_2 , and O_3 (Fig. 282a), the points of tangency of two spheres with the lower base of the cylinder as K_1 and K_2 , and the point of tangency of the third sphere with the upper base as K_3 . We have $|O_1K_1| = |O_2K_2| = |O_3K_3| = r$, $|O_1O_2| = |O_1O_3| = |O_2O_3| = 2r$. From the point O_3 we drop a perpendicular O_3L to the plane of the lower base of the

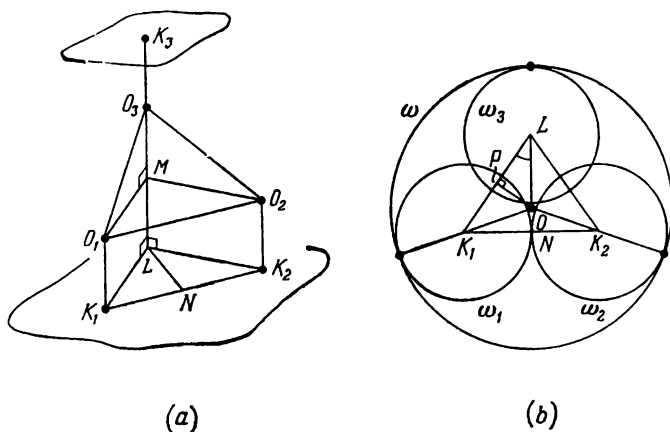


Fig. 282

cylinder, and then we have $|O_3L| = 2r$. Suppose $|O_1M| \perp |O_3L|$, then $|ML| = |O_3M| = r$, and we find, from the right triangle O_1O_3M , that $|O_1M| = \sqrt{3}r$. It is evident that $|MO_2| = \sqrt{3}r$ as well.

Let us consider the orthogonal projections of the spheres and the lateral surface of the cylinder onto the plane of its lower base. The projections of the spheres are circles of radius r and centres at K_1 , K_2 , and L (Fig. 282b). We designate the circumferences of these circles as ω_1 , ω_2 , and ω_3 . The projection of the lateral surface of the cylinder is the circumference of its lower base, which we designate as ω . The point of tangency of the lateral surface of the cylinder with the first sphere will be projected into a point belonging both to ω and to ω_1 , and it will be a *single* common point of the two circumferences. Hence, there is an internal tangency between the circumferences ω and ω_1 . In just the same way ω touches the circumferences ω_2 and ω_3 . Thus, the centre O of the circumference ω is at the distance $R - r$ from each of the points K_1 , K_2 , and L , where R is the radius of ω . This means that O is the centre of the circle circumscribed about the triangle K_1LK_2 . From the triangle K_1LN ($[LN] \perp [K_1K_2]$) we find that $|LN| = \sqrt{2}r$, $\cos K_1LN = \sqrt{2/3}$, $|LO| = \frac{1}{2} |K_1L|/\cos K_1LN = 3r/2\sqrt{2}$. Hence $R = |LO| + r = (3\sqrt{2} + 4)r/4$.

Answers to Problems of Section II and of the Appendix

CHAPTER 1 *

1. $\pi(1+2n)/8, \pi(1+2n)/4$. 2. $\pi(1+2n)/2, (-1)^n \frac{\pi}{6} + \pi n$.
3. $\pi(1+2n)/8, \pi(1+2n)/4$. 4. $\pi(8n-1)/16, \pi(8n+1)/24$.
5. $\pi(1+4n)/6, 2\pi n$. 6. $\pi n/2, \pi(1+2n)/8$.
7. $\pi(12n-1)/12, \pi(12n-5)/12$.
8. $\pi(8n \pm 3)/4$. 9. $\pm \arctan \frac{1}{\sqrt{2}} + \pi n$.
10. $\pm \frac{1}{2} \arccos \frac{\sqrt{21}-1}{4} + \pi n$.
11. $\pi(1+2n)/2$. 12. $(-1)^{n+1} \frac{\pi}{12} + \frac{\pi n}{2}$.
13. $\pm \frac{\pi}{12} + 2\pi n, \pm \frac{5\pi}{12} + 2\pi n$.
14. $-\frac{\pi}{2} + 2\pi n, (-1)^n \frac{\pi}{6} + \pi n$. 15. $\pm \arccos(-1/2\sqrt{2}) + 2\pi n$.
16. $(-1)^{n+1} \frac{\pi}{6} + \pi n$. 17. $\pm \arccos(-1/4) + 2\pi n$. 18. $\pi, 4\pi/3$.
19. $\pm \frac{\pi}{6} + \frac{\pi n}{2}$. 20. $\frac{\pi}{4}(1+2n), \frac{(-1)^n}{2} \arcsin(2\sqrt{2}-2) + \frac{\pi n}{2}$.
21. $\pi n, \pi(4n-1)/4$. 22. $\pi n, \pi(3n-1)/3$. 23. $|p| \leq 1/\sqrt{3}$.
24. $\arccos(3/5) + \pi(2n+1)$ 25. $\pi(2n+1)/2$. 26. $3/4$.
27. $\frac{\pi}{4} + 2\pi n, \frac{\pi}{4} \pm \frac{2\pi}{3} + 2\pi n$. 28. $-\frac{\pi}{4} \pm \arccos \frac{1}{2\sqrt{2}} + 2\pi n$.
30. $\pi(1+2n)/2$. 31. $\pi n, \pi(3n-1)/3$.

* In these answers, for the sake of brevity, we have indicated only the values of the variable, the integral parameters are designated here as $m, n, \dots$. For instance, the notation " $1. \frac{\pi(1+2n)}{8}, \frac{\pi(1+2n)}{4}$ " should be understood as " $1. \left\{ \frac{\pi(1+2n)}{8}, \frac{\pi(1+2n)}{4} \mid n \in \mathbb{Z} \right\}$ ".

32. $\frac{\pi}{4}(1+2n)$, $(-1)^{n+1}\frac{\pi}{12} + \frac{\pi n}{2}$.
 33. $\pi(1+2n)/2$, $\pi(1+2n)/4$.
 34. $\pi(4n+1)/4$, $\pi(24n+1)/12$, $\pi(24n+5)/12$.
 35. $\frac{\pi}{4} + \pi n$, $-\frac{\pi}{4} + (-1)^n \arcsin \frac{\sqrt{5}-2}{\sqrt{2}} + \pi n$.
 36. $-\frac{\pi}{4} + \pi n$, $(-1)^{n+1}\frac{\pi}{12} + \frac{\pi n}{2}$. 37. $\pm \frac{\pi}{4} + 2\pi n$, $\pm \frac{\pi}{6} + \pi n$.
 38. If $|a| \geq 1$, then $x = \frac{\pi}{4} + \pi n$, $x = \pm \frac{1}{2} \arccos \frac{1}{a} + \pi n$; if $|a| < 1$, then $x = \frac{\pi}{4} + \pi n$.
 39. $\frac{\pi}{4} + \pi n$, $(-1)^{n+1}\frac{\pi}{8} + \frac{\pi n}{2}$. 40. $\pi(1+4n)/4$. 41. $\pi n/4$.
 42. $(-1)^n \frac{\pi}{6} + \pi n$, $-\frac{\pi}{2} + 2\pi n$, πn . 43. $\frac{\pi n}{4}$, $\pm \frac{2\pi}{3} + 2\pi n$.
 44. $\pi n/3$. 45. $\pi(1+2n)/12$, $\pi(1+2n)/2$.
 46. $\pi(1+2n)$, $\pi(6n \pm 1)/3$. 47. $\frac{\pi n}{3}$, $\frac{\pi}{2} + \pi n$, $\pm \frac{1}{2} \arccos \frac{3}{4} + \pi n$.
 48. $2\pi(3n \pm 1)/3$. 49. $\pi(1+2n)$.
 50. $\pi(1+6n)/6$. 51. $\pi n/3$, $\pi(1+2n)/7$.
 52. $\frac{1}{2} \left(\frac{\pi}{2} + \pi n \pm \sqrt{\left(\frac{\pi}{2} + \pi n \right)^2 - 4} \right)$, $n \neq -1, 0$.
 53. -2 , $-\frac{1}{2}$, 1 , $\frac{1}{2}(-7 \pm 3\sqrt{5})$.
 54. $\pi(16n \pm 1)/8$. 55. $\pi(1+2n)/6$. 56. $\pi(1+4n)/4$.
 57. πn , $\pi(6n \pm 1)/9$. 58. $\pi(1+2n)/4$, $\pi(3n \pm 1)/3$.
 59. $\pi n/2$. 60. πn , $\pi(6n \pm 1)/6$.
 61. $\pi(1+2n)/2$, $\pi(6n \pm 1)/3$. 62. $\pi(6n \pm 1)/3$, $4\pi n$.
 63. $\pi(1+2n)/2$, $\pi(1+4n)/18$. 64. $\pi n/2$, $2\pi(3n \pm 1)/3$.
 65. $\pi(1+2n)/14$, $\pi(1+2n)/4$.
 66. (a) $\arccos(1/\sqrt{3}) + \pi(1+4n)/2$; (b) $\pi(2n+1) - \arctan 3$.
 67. $\pi n/2$. 68. $(-1)^n \frac{\pi}{6} + \pi n$. 69. $(-1)^{n+1} \frac{\pi}{18} + \frac{\pi n}{3}$.
 70. πn , $\pi(1+2n)/16$. 71. $\pi(1+2n)/7$, $n \neq 7m+3$, $m \in \mathbb{Z}$.
 72. $\pi(6n \pm 1)/6$. 73. $\pi(1+2n)/5$, $n \neq 5m+2$, $m \in \mathbb{Z}$.
 74. $\pi(1+2n)/2$. 75. $\pi(16n+3 \pm 2)/8$, $\pi(8n-3)/4$,
 $\pi(24n-3 \pm 2)/12$.
 76. $2\pi n$, $\pi(1+2n)/2$, $\pi(6n-1)/6$. 77. $|a| \geq 10$.
 78. If $|a| \geq 1/2$, then $x = (\arctan(4a \pm 2\sqrt{4a^2-1}) + \pi n)/2$;
 if $|a| < 1/2$, then there are no solutions.

79. If $a \in \left(\frac{1}{3}; 1\right)$, then $x = \pi(1+2n)/2$; if $a \notin \left(\frac{1}{3}; 1\right)$, then $x = \pi(1+2n)/2$, $x = \pm \frac{1}{2} \arccos \frac{a}{2a-1} + \pi n$.
80. If $|a| > \sqrt{5}$, then $x = \pi n/2$; if $|a| \leq \sqrt{5}$, then $x = \pi n/2 + \frac{1}{2} \left(\arcsin \frac{1}{\sqrt{5}} + (-1)^{n+1} \arcsin \frac{a}{\sqrt{5}} + \pi n \right)$.
81. πn , $\pi(8n-3)/4$. 82. $2\pi n$, $\pi(4n-1)/2$.
 83. $\pi(12n-5)/6$. 84. $\pi(3n \pm 1)/6$. 85. $\pi(1+2n)/4$.
86. $\frac{\pi}{6} + 2\pi n$, $-\arcsin \frac{\sqrt{17}-1}{4} + 2\pi n$. 87. $|a| \leq 1$.
88. (a) $1 \leq a \leq \sqrt{4+2\sqrt{2}}$; (b) $(\sqrt{6} + \sqrt{3}-1)/2 \leq a < (\sqrt{6} + \sqrt{3}+1)/2$, $a = \sqrt{4+2\sqrt{2}}$.
89. $\pi n/3$. 90. $\pi(6n+1)/18$.
91. $\pi(4n-1)/2$, $\frac{\pi}{4} \pm \arccos \frac{1}{2\sqrt{2}} + 2\pi n$.
92. $-\frac{\pi}{4} + \pi n$, $\frac{\pi}{4} + (-1)^n \arcsin \frac{1}{2\sqrt{2}} + \pi n$, $2\pi n$, $-\frac{\pi}{2} + 2\pi n$.
93. $2\pi n/3$. 94. $\pi n/2$. 95. 0. 96. $2n - \frac{1}{2}$. 97. $\pi(1+2n)/2$.
- 98*. $\left(\pm \frac{\pi}{3} + \pi n; \left(\frac{5}{2} \mp \frac{2}{3} \right) \pi - 2\pi n \right)$.
99. $\left(\frac{3}{2} \pi - 2\pi n; \pi n \right)$, $\left(\left(\frac{3}{2} \mp \frac{1}{3} - 2n \right) \pi; \pm \frac{\pi}{6} + \pi n \right)$.
100. If $a = 2\pi n$, then $x = \pm \frac{\pi}{3} + \pi(n+m)$, $n = \mp \frac{\pi}{3} + \pi(n-m)$; if $a = (2n+1)\pi$, then $x = \pm \frac{\pi}{6} + \pi\left(n+m+\frac{1}{2}\right)$, $y = \mp \frac{\pi}{6} + \pi\left(n-m+\frac{1}{2}\right)$; if $a \neq \pi n$, then there are no solutions.
101. $(\mp 0.5; \pm 0.5)$, $(\pm 1; 0)$, $(0; \pm 1)$.
102. $(\arctan(2 \pm 0.8\sqrt{5}) + \pi m; \arctan(2 \mp 0.8\sqrt{5}) + \pi n)$.
103. $(\pi(8m \pm 1)/2; 2\pi(2n+1))$, $(2\pi(2m+1); \pi(8n \pm 1)/2)$.

* If there are double signs in the notation of the solutions x , y of the system, then, unless otherwise specified, we must take, simultaneously, either only the upper signs or only the lower signs for both x and y .

$$104. \left(-\frac{\pi}{4} \pm \arccos \frac{1}{2\sqrt{2}} + 2\pi m; -\frac{\pi}{4} \pm \arccos \left(-\frac{1}{2\sqrt{2}} \right) + 2\pi n \right).$$

$$105. (\pi(2m-1)/4; \pi(2m+8n+1)/4).$$

$$106. \left((-1)^{n+1} \frac{\pi}{3} + \frac{\pi}{2} + \pi m; (-1)^n \frac{\pi}{6} + \frac{\pi n}{2} \right).$$

$$107. \left(\pm \frac{\pi}{3} + 2\pi(m+n); \pm \frac{\pi}{3} + 2\pi(m-n) \right), \text{ the signs are taken arbitrarily.}$$

$$108. x \text{ is any number not equal to } \pi(1+2m)/4, y = \pi(1+2n)/2.$$

$$109. \left(\frac{\pi}{2}(1+2m); \frac{\pi}{2}(1+2n) \right), \left(\frac{\pi}{4}(1+2m); 2\pi n - \frac{\pi}{4}(1+2m) \right).$$

$$110. \left(\pm \frac{\pi}{3} + \pi m; \mp (-1)^n \frac{\pi}{3} + \pi n \right), \left(\frac{\pi m}{5}; \pi n \right).$$

$$111. ((-1)^n \arcsin(7/8) + \pi(n+2m); (-1)^n \arcsin(1/4) + \pi n).$$

$$112. \left(\frac{\pi}{8} + \frac{\pi m}{2}; \frac{\pi}{2} + \pi(2n+m) \right), \left(\frac{\pi}{2} + \pi m; (-1)^{m+1} \frac{\pi}{4} + \pi(2n-m) \right), \left(\frac{\pi}{4} + \pi m; \pi(2n-m) \right).$$

$$113. \left((-1)^n \frac{\pi}{4} + \pi m; \frac{\pi}{2} + \pi n \right), \left((-1)^{n+1} \arctan \sqrt{2} + \pi m; \frac{\pi}{4} + \pi n \right).$$

$$114. \left(-\frac{\pi}{4} + \pi m; (-1)^{m+n+1} \arcsin \frac{\sqrt{10}-\sqrt{2}}{4} + \pi n \right), \left(\pi m; (-1)^m \frac{\pi}{2} + 2\pi n \right), \left(\frac{\pi}{2} + \pi m; (-1)^{m+1} \frac{\pi}{2} + 2\pi n \right).$$

$$115. (\pi m; \pi(1+2n)/4), (\pi(1+2m)/4; \pi n).$$

$$116. (\pi m; \pi n), \left(\pi m; \pm \frac{\pi}{4} + \pi(m+2n+1) \right).$$

$$117. \left(\mp \frac{\pi}{3} + 2\pi m; \pm \frac{\pi}{3} + 2\pi n \right).$$

$$118. (\pi m; \arctan 2 + \pi n), \left(\frac{\pi}{2} - (-1)^n \frac{\pi}{6} + \pi m; (-1)^n \frac{\pi}{12} + \frac{\pi n}{2} \right).$$

$$119. \left(\frac{\pi}{2} + \pi m; \pi n \right), \left(-\frac{\pi}{4} + \pi m; \frac{3\pi}{4} + \pi(2n-m) \right), \left(\frac{\pi}{2} - \arctan 2 + \pi(2m+n); -\arctan 2 + \pi n \right).$$

$$120. (\pi m; 2\pi n), \left(\mp \arccos \frac{1}{7} + 2\pi m; \pm \frac{2\pi}{3} + 2\pi n \right).$$

$$121. \left(\pi (2m+n); \frac{\pi}{6} + \pi n \right), \left(\frac{\pi}{2} + \pi (2m+n); \frac{\pi}{6} + \pi n \right).$$

$$122. (\pi m; \pi n; \pi (2p-m-n)), \left(\pm \frac{\pi}{3} + \pi (2m-n-p); \pm \frac{\pi}{4} + \pi n; \pm \frac{\pi}{4} + \pi p \right).$$

$$123. \left(\frac{\pi}{4} + \pi m; \pi (2n+m); \frac{3\pi}{4} + \pi (2p+m) \right), \\ \left(-\frac{\pi}{4} + \pi m; \frac{\pi}{2} + \pi (2n+m); \frac{\pi}{4} + \pi (2p+m) \right), \left(-\frac{1}{2} \arctan \times \right. \\ \left. \frac{1}{2} + \frac{\pi m}{2}; \frac{3\pi}{4} + \frac{1}{2} \arctan \frac{1}{2} + \pi \left(2n - \frac{m}{2} \right); -\frac{\pi}{2} - \frac{1}{2} \arctan \times \right. \\ \left. \frac{1}{2} + \pi \left(2p + \frac{m}{2} \right) \right).$$

$$124. \left(\frac{\pi}{6} + \pi m; \frac{\pi}{3} + \pi n; -\frac{\pi}{3} + \pi (2p-m-n) \right), \left(-\frac{\pi}{6} + \pi m; \right. \\ \left. -\frac{\pi}{3} + \pi n; \frac{\pi}{3} + \pi (2p-m-n+1) \right).$$

$$125. -\frac{\pi}{6} + \pi n < x < \frac{\pi}{6} + \pi n. \quad 126. \frac{\pi}{4} + 2\pi n < x < \frac{5\pi}{4} + 2\pi n.$$

$$127. 2 \arctan 2 + 2\pi n < x < 2\pi (n+1).$$

$$128. -\frac{\pi}{6} + 2\pi n < x < \frac{7\pi}{6} + 2\pi n.$$

$$129. x = -\frac{\pi}{2} + 2\pi n, \quad 2\pi n \leq x \leq \pi (2n+1).$$

$$130. \arctan \sqrt{2} + \pi n \leq x < \frac{\pi}{2} + \pi n, \quad -\arctan \sqrt{2} + \pi n \leq x < \pi n.$$

$$131. n - \frac{1}{2} < x \leq n - \frac{1}{3}, \quad n - \frac{1}{4} < x \leq n, \quad n + \frac{1}{4} < x \leq n + \frac{1}{3}.$$

$$132. -\frac{2\pi}{3} + 2\pi n < x < \frac{\pi}{4} + 2\pi n, \quad \frac{2\pi}{3} + 2\pi n < x < \frac{3\pi}{4} + 2\pi n.$$

$$133. 2\pi n < x < \frac{\pi}{8} + 2\pi n, \quad \frac{3\pi}{8} + 2\pi n < x < \frac{\pi}{2} + 2\pi n, \quad \frac{\pi}{2} + \\ 2\pi n < x < \frac{5\pi}{8} + 2\pi n, \quad \frac{7\pi}{8} + 2\pi n < x < \pi + 2\pi n, \quad \frac{9\pi}{8} + 2\pi n < x < \\ \frac{11\pi}{8} + 2\pi n, \quad \frac{13\pi}{8} + 2\pi n < x < \frac{15\pi}{8} + 2\pi n.$$

$$134. \frac{\pi}{4} + \pi n < x < \frac{3\pi}{4} + \pi n.$$

$$135. -\frac{\pi}{3} + 2\pi n \leq x \leq \frac{\pi}{3} + 2\pi n, \quad \frac{\pi}{2} + 2\pi n \leq x \leq \frac{3\pi}{2} + 2\pi n.$$

$$136. \frac{\pi}{4} + \pi n \leq x \leq \arctan 3 + \pi n.$$

$$137. \frac{\pi}{6} + 2\pi n \leq x \leq \frac{5\pi}{8} - \frac{1}{2} \arcsin \frac{1}{2\sqrt{2}} + 2\pi n, \quad \frac{1}{2} \arcsin \times \\ \frac{1}{2\sqrt{2}} - \frac{7\pi}{8} + 2\pi n \leq x \leq -\frac{\pi}{6} + 2\pi n.$$

$$138. 2\pi n \leq x < \arcsin(\sqrt{5} - \sqrt{2}) - \frac{\pi}{4} + 2\pi n, \quad \frac{3\pi}{4} - \arcsin \times \\ (\sqrt{5} - \sqrt{2}) + 2\pi n < x \leq \frac{\pi}{2} + 2\pi n.$$

$$140. \sin^2(\alpha + \beta). \quad 144. (a) a^3; (b) 225/128. \quad 145. (a+b)^{-3}.$$

$$146. (a) 4/5; (b) -1/8; (c) 2/\sqrt{5}; (d) -2/\sqrt{5}; (e) 9/25. \quad 148. \pi.$$

$$149. a=4, a>5. \quad 150. n \pm \frac{1}{6}.$$

$$151. \pi(1+2n)/2, \pm \arctan(1/2) + \pi n.$$

$$152. 6; \frac{4\pi}{3}; \frac{7\pi}{3}; \pm \frac{\pi}{3} + 2\pi n, n \geq 2; \pm \frac{2\pi}{3} + 2\pi n, n \leq 0.$$

$$153. -\frac{2}{3}; \frac{1}{3}; \pm \frac{1}{3} + 2n, n \geq 1; \pm \frac{2}{3} + 2n, n \leq -1.$$

$$154. -2\pi/3; -\pi/3; 4\pi/3; 5\pi/3. \quad 156. 35.$$

$$157. -15/2; -3/10; 3/2.$$

$$158. a_n = \pm \left(2\pi n - \frac{\pi}{2}\right)^{-1/2}, \quad x_n = 2\pi n - \frac{\pi}{2}, \quad n \geq 1.$$

$$159. 12\pi - 1. \quad 160. -\frac{\pi}{4} + 2\pi n; \arctan \frac{1}{2} + 2\pi n.$$

$$161. (-1)^{n+1} \arcsin \frac{\sqrt{5}-1}{2} + \pi n. \quad 162. \sqrt{2\pi}, (\sqrt{1+8\pi}-1)/2.$$

$$163. 1/2. \quad 164. -1. \quad 165. -\arctan(1/2), -\arctan(1/3).$$

$$166. a = \frac{\pi^3}{32}, \quad \frac{\pi^3}{8} < a \leq \frac{7\pi^3}{8}.$$

$$167. (a) x \text{ is any number, } y = 2\pi n; (b) x = \pi m, y = \pi n;$$

$$(c) x = \pm \frac{\pi}{3} + 2\pi m, \quad y = \pm \frac{\pi}{3} + 2\pi n; \quad (d) x + \pm \frac{\pi}{3} + \pi m, y = \\ \pm \frac{\pi}{3} + \pi(2n+m), \text{ the combination of signs is arbitrary.}$$

$$168. \left(0; \frac{\pi}{2} + 2\pi n\right), \left(\frac{1}{2}; \frac{5\pi}{4} + 2\pi n\right).$$

$$170. \text{For } a = \frac{\tan^2 1 + 2}{2(2 \tan 1 - 1)} \text{ there are three solutions: } x=0,$$

$x = \pm 2\pi$; for $a = -1/2$ there are four solutions:

$x = \pm \sqrt{3\pi^2 \pm 2\pi \arccos \frac{\pi}{4} - \left(\arccos \frac{\pi}{4}\right)^2}$, the combination of signs being arbitrary.

CHAPTER 2

1. 0.6. 2. 1.5. 3. 2; $\log_5 0.1$. 4. 2. 5. 3.
 6. (a) $\frac{b+1}{2-ab}$; (b) $\frac{2b-1}{3(2a+2b-ab-3)}$. 7. $\pi(2n+1)/2$.
 8. $\pi(2n+1)/4$. 9. $\pi(2n+1)/4$. 10. $1/2$. 11. $-2 - \sqrt{10}$. 12. 2.
 13. $-2 - \sqrt{10}$.
 14. -1 . 15. 1. 16. $(-1)^n \frac{\pi}{6} + \pi n$. 17. 7; 14. 18. 9; $1/9$.
 19. 3. 20. 3. 21. 6. 22. $\pm \frac{\pi}{3} +$
 23. $(-1)^n \frac{\pi}{6} + \pi n$. 24. -5 . 25. $5\pi/4$. 26. $\pi/4$.
 27. (a) $\log_2 11$; (b) $\sqrt{6}$. 28. πn ; $(-1)^n \frac{\pi}{6} + \pi n$. 29. $\frac{\pi}{4} + \pi n$.
 30. 5. 31. -0.5 . 32. $(-1)^n \frac{\pi}{8} + \pi n$. $(-1)^n \frac{3\pi}{8} + \pi n$.
 33. $\frac{\pi}{4} + 2\pi n$. 34. 3; $\sqrt{3/2}$. 35. $\sqrt{2}$. 36. 2^{12} . 37. $2 - \sqrt{2}$.
 38. 0.25. 39. 9; $1/9$. 40. -3 ; -8 . 41. 15; 1.
 42. If $a \neq 1$, $n \neq 0$, then $x=1$, $x=a^{(2-n)/(n-1)}$; if $a \neq 1$, $n=0$, then $x=1$; if $a=1$, then there are no solutions.
 44. $\sqrt{6}$; $\sqrt{2}$. 45. 20. 46. 3; $\sqrt{2}$. 47. 2. 48. 4.
 49. $1/9$. 50. 8. 51. $5/12$. 52. $\frac{\pi}{4} + \pi n$. 53. $2\pi n$. 54. 2; -2 .
 55. $a = -7/4$, $a < -2$. 56. Two solutions. 57. 2. 58. 9.
 59. (9; 7), (7; 9). 60. (10; 0.1). 61. (4; 1), (1; 9). 62. (3; 2).
 63. (5; -1). 64. (3; $\log_3 2$). 65. (1; 1), $(1/\sqrt[3]{3}; \sqrt[3]{9})$.
 66. $\left((-1)^n \frac{\pi}{6} + \pi n; \pm \frac{\pi}{3} + 2\pi m\right)$. 67. (1/2; 1/4), (1/8; 6/4).
 68. ($\sqrt{2}$; $1/\sqrt{2}$), (2; 1). 69. (9; 27), (1/9; 1/3).
 70. (3; 2). 71. (1; 2). 72. (5.5; 2.5). 73. (16; 4).
 74. $\left(\sqrt{\frac{\sqrt{5}+1}{2}}; \sqrt{\frac{\sqrt{5}-1}{2}}\right)$. 75. (2; 3), (c ; 1), where
 $1 < c < 3$.

76. (25; 13). 77. $(10^{-1}; 10^{-4})$, $(10^{-4}; 10^{-1})$.
 78. (1.5; 0.5), (0.5; 1.5). 79. $(-0.5; -0.5)$.
 80. (0; 0), (4; 4), $(2 \pm \sqrt{7}; 3)$, $(2 \pm \sqrt{7}; 1)$.
 81. (0; 1; 2), (1; 0; 2), (0; 2; 1), (2; 0; 1), (2; 1; 0), (1; 2; 0).
 82. $(\pm 3; \pm 2; \pm 1)$.
 83. $(9; 3; \sqrt{3})$, $(9; 1/3)$, $1/\sqrt{3}$; $(1/9; 1/3; \sqrt{3})$, $(1/9; 3; 1/\sqrt{3})$.
 84. $\left(\frac{2}{\sqrt{21}}; \frac{2}{\sqrt{35}}; \frac{2}{\sqrt{15}}\right)$. 85. (1; -1). 86. $a > -1$.
 87. $a = -1$. 88. (a) $\log_{\log_2 2} 0.5 > 0$; (b) $\log_3 2 > \frac{5}{8}$.
 89. $(0; 0.5) \cup (4; +\infty)$.
 90. $(-\infty; -1) \cup (2; +\infty)$. 91. $(-\infty; -2) \cup (5/8; +\infty)$.
 92. $(-1; -0.5) \cup (1; 2)$. 93. $(1; +\infty)$. 94. $(6/5; 4/3)$.
 95. $(2 - \sqrt{2}; 3/4) \cup [13/4; 2 + \sqrt{2}]$. 96. $(-19/4; -3) \cup (4; 23/4)$.
 97. If $a > 1$, then $|x| > \sqrt{a}$ or $0 < |x| < 1/\sqrt{a}$; if $0 < a < 1$, then $0 < |x| < \sqrt{a}$ or $|x| > 1/\sqrt{a}$.
 98. $(-\infty; 0) \cup (1; +\infty)$. 99. $[0; 1]$. 100. $(-\infty; -1)$.
 101. $[0.5; 2]$. 102. $(0; 10) \cup [10^5; +\infty)$.
 103. $(0; \log_3 28 - 3) \cup (\log_3 4; +\infty)$.
 104. $(-\infty; \log_3 \log_3 2) \cup (0; +\infty)$. It is not true.
 105. $(3; +\infty)$. 106. $(1 - \sqrt{5}; -1/3) \cup (1 + \sqrt{5}; +\infty)$.
 107. $\frac{\pi}{24} + 2\pi n < x < \frac{5\pi}{24} + 2\pi n$. 108. $(-2; 4)$.
 109. $(1; 1 + \sqrt{2}]$. 110. $(1; 2) \cup [3; (5 + \sqrt{7})/2)$.
 111. $(-\infty; -2) \cup (1; 2)$. 112. $[\log_3 0.9; 2)$.
 113. $(-\infty; 0) \cup (2; +\infty)$. 114. $(0; 1/\sqrt{6}) \cup [1; +\infty)$.
 115. $(-1; -0.5) \cup (0; 1)$. 116. $(1; 2\sqrt{2}) \cup (2^{-\sqrt{2}}; 0.5)$.
 117. If $0 < a < 1$, then $a^2 < x < 1$, $a^{-3} < x$; if $a > 1$, then $1 < x < a^2$, $0 < x < a^{-3}$.
 118. $(2; 3) \cup (\sqrt{10}; +\infty)$. 119. $[0.2; 5]$. 120. $[0; 1)$.
 121. If $0 < a < 1$, then $0 < x < a$, $a^{-2} < x$; if $1 < a$, then $a^{-2} < x < a$.
 122. $(-1; 0) \cup (1.5; 4)$. 123. $(-0.5; -0.25) \cup (0; 0.5]$.
 124. $(3; (5 + \sqrt{7})/2) \cup (4; 4.5]$. 125. $(1; 2) \cup [4; +\infty)$.
 126. $(1; \sqrt{(\sqrt{69} - 3)/2})$. 127. $(1; \sqrt{2}) \cup (2; +\infty)$.
 128. $[7; +\infty)$. 129. $(1; 2) \cup [16; +\infty)$.
 130. $\frac{1}{2} \arcsin \frac{1}{2} (\sqrt{3} - 1) + 2\pi n < x < \frac{\pi}{2} - \frac{1}{2} \arcsin \frac{1}{2} \times (\sqrt{3} - 1) + 2\pi n$.

131. If $0 < a < 1$, then $a^{-a}\sqrt{2-a^2} < x \leq a^{-1}$; if $1 < a$, then there are no solutions.

132. If $a < 0$, then $1 < x < (1 + \sqrt{1-4a})/2$; if $a = 0$, then there are no solutions; if $0 < a \leq 1/4$, then $a < x < (1 - \sqrt{1-4a})/2$, $(1 + \sqrt{1-4a})/2 < x < 1$; if $1/4 < a < 1$, then $a < x < 1$; if $1 \leq a$, then there are no solutions. *Hint.* To make the solution more illustrative, construct the graphs of the functions x^2 and $x - a$.

133. (1; 1.5). 134. $a < -2$. 135. $|a| > 2$.

137. (a) $\log_3 30 > \log_4 60$; (b) $\log_3 75 > \log_2 11$.

139. $\{-2; -1; 0; 1\}$.

144. $x_1 = \pi$, $y_1 = 0$; $x_2 = \arccos(1/3)$; $y_2 = (\pi - \arccos(1/3))^2$, $x_3 = \pi - \arccos(1/3)$, $y_3 = (\arccos(1/3))^2$.

145. (2; 2). 146. (1; 2).

147. (a) An open ray emanating from the point $(8/7; 9/7)$, passing through the point $(5; 0)$. (b) The interior points of the quadrangle $ABCD$ with vertices $A(-2; 3)$, $B(0; 1)$, $C(1; 2)$, $D(1; 3)$ and the points belonging to the interval BC .

148. (0; 1). 149. 5.

CHAPTER 3

1. (a) $1/15$; (b) $k(k+1)$, if $k \geq n$.

2. (a) 27; (b) 2; (c) 3; 14; (d) 10, if $k \leq 10$; (e) 3; (f) $\{0; 1; 2\}$; (g) $\{0; 1; 2; \dots; 27\}$.

3. (a) $\{1; 2; 3\}$; (b) $\{1; 9; 14; 15; 35\}$. 4. (a) 870; (b) 435.

5. 12. 6. 16. 7. 210. 8. 576. 9. 18; 10. 18. 11. 4060.

12. 72. 13. (a) 120; (b) 60; (c) 420. 14. $\frac{28!}{(7!)^4}$.

15. 5040. 16. 15 625. 17. 12. 18. 16. 19. $(n+1)! m!$.

20. 945. 21. 330. 22. They may not be enough.

23. If $n < 7$, then there are less numbers with the digit 9; if $n \geq 7$, then there are more numbers with the digit 9.

24. (a) $(x+1)^7 = x^7 + 7x^6 + 21x^5 + 35x^4 + 35x^3 + 21x^2 + 7x + 1$; (b) $\left(\sqrt{\frac{a}{b}} - \sqrt{\frac{b}{a}}\right)^5 = a^{5/2}b^{-5/2} - 5a^{3/2}b^{-3/2} + 10a^{1/2}b^{-1/2} - 10a^{-1/2}b^{1/2} + 5a^{-3/2}b^{3/2} - a^{-5/2}b^{5/2}$.

25. (a) $89\sqrt{3} - 109\sqrt{2}$; (b) 16.

26. (a) $1716a^3\sqrt{ab^2}$; (b) 5005; (c) $153\sqrt{2}x^8$; (d) $C_{1980}^{196}x^{1980}$.

27. (a) 60; (b) 625; 7000; 7000; 1120; 16. 28. 32.

29. (a) $27/64$; (b) $\sqrt[10]{50^3 C_{70}^3}$. 30. (a) 378; (b) 245.

31. (a) -1 ; (b) $1/16$. 33. (a) $1/4$; (b) $1/3$; (c) 0; (d) $5/12$.

34. $5/36$. 35. $93/168$. 36. $120/343$. 37. $29/52$.

38. $164/1081$. 39. (a) $9/17$; (b) $8/17$.

CHAPTER 4

1. (a) $\frac{1}{3} e^{3x+2} + \frac{2}{\pi} \sin \frac{\pi x}{4} - 2x^{3/2} + C$; (b) $\frac{1}{2 \ln 2} \cdot 2^{2x} - \frac{3}{\pi} \times \tan \frac{\pi x}{3} + \frac{3}{4} (2x-1)^{2/3} + C$; (c) $\frac{3}{2} \ln |2x+5| + 4 \cot \frac{x}{2} + \frac{4}{3} \times \sin 3x + C$.

2. (a) $2\sqrt{x} - \frac{1}{2} \cos(2x+1) + \frac{1}{2} \cos 3 - 1$; (b) $\ln |x-3| + \frac{1}{2} e^{2x+1} + 2 - \ln 2 - \frac{1}{2} e^3$.

3. $F(x) = \frac{2}{3} \sin 3x + \frac{3}{4} \cos x + \frac{5}{3}$. 4. $f(x) = \frac{5}{3} e^{3x} + \frac{7}{3}$.

5. In the interval $(-3; 1)$ we have $F(x) = -\frac{2}{3} x^3 + \frac{7}{2} x^2 - 6x + C$; in the interval $(1; 3)$ we have $F(x) = -\varphi(x) + C$ for $x < 2$ and $F(x) = \varphi(x) - \frac{20}{3} + C$ for $x \geq 2$, where $\varphi(x) = \frac{2}{3} x^3 - \frac{7}{2} x^2 + 6x$.

6. 2. 7. Not always.

8. Not always, for instance, $f(x) = \cos x$, $F(x) = \sin x + 3$.

9. Yes.

10. (a) $\frac{1}{3} x^3 - \frac{2}{3} \cos 3x + \frac{1}{2} e^{2x} + C$; (b) $-\frac{1}{14} \cos 7t - \frac{1}{10} \cos 5t + C$; (c) $\frac{2}{7} x^{7/2} + 2x + \frac{1}{2} x^2 + C$; (d) $\frac{1}{2 \ln 3} 3^{2x} + \frac{12}{\pi} \tan \frac{\pi x}{3} + C$; (e) $\frac{1}{2} \ln |2u-1| + \frac{9}{8} (4u+1)^{2/3} + C$; (f) $\frac{x^2}{2} - x + C$, if $x > 1$, and $-\frac{x^2}{2} + x - 1 + C$, if $x \leq 1$; (g) $-\ln |1 + \cos x| + C$.

11. (a) $-x \cos x + \sin x + C$; (b) $x^2 \sin x + 2x \cos x - 2 \sin x + C$; (c) $(x^2 - 2x + 3) e^x + C$; (d) $\left(\frac{x}{3 \ln 2} - \frac{1}{(3 \ln 2)^2} \right) 2^{3x} + C$.

12. (a) 3.5; (b) 1; (c) $\frac{56}{3 \ln 2} - \frac{1}{2} (\sin 4 - \sin 2)$; (d) $\frac{1}{2} \ln \frac{5}{3}$; (e) $\frac{5}{6} (8^{1/5} - 2^{1/5})$; (f) $\frac{1}{2} (e^{-1} - e^{-4})$; (g) $\sin 1 + \cos 1 - 1$; (h) $1/(2 \ln 2)$; (i) 1.

13. 1. 14. $A = \frac{1}{\ln 2}$, $B = \frac{7}{3} - \frac{7}{3 (\ln 2)^2}$. 15. $\sqrt{2\pi} \cdot \frac{-1 + \sqrt{1+8\pi}}{2}$.

16. (a) $(x+1)e^x$; (b) $-2t^4 2t \ln(1+t^4)$; (c) $-\sin u \cdot e^{\cos u} \times \sin^3(2 \cos u + 5) - \cos u \cdot e^{\sin u} \cdot \sin^3(2 \sin u + 5)$.

17. (a) 1; (b) $5/12$; (c) $e + \frac{1}{e} - 2$; (d) $\frac{\pi}{2} + 2$; (e) $8/3$; (f) $3 - \frac{3}{4 \ln 2}$; (g) $7 \ln 3$; (h) $\frac{37\sqrt{37}-55}{54}$; (i) $9 - 4 \ln 4$; (j) $\frac{15}{4} - \ln 2$.

18. $\ln 2 - \frac{5}{8}$. 19. $0 < a < 1/6$. 20. $200/243$.

21. $1/8$. 23. $\frac{\pi}{7} (3^7 - 15)$. 24. $\frac{3}{8} \pi^2$.

25. (a) $\frac{\pi}{2\alpha+1}$; (b) $\frac{2\pi}{2+\alpha}$; (c) $\pi \frac{3\alpha+1}{(\alpha+1)(2\alpha+1)}$; (d) $\frac{2\pi}{(\alpha+1)(\alpha+2)}$; (e) $\frac{3\pi\alpha\sqrt{2}}{2(2+\alpha)(1+2\alpha)}$, if $\alpha > 1$, $\frac{\pi}{3\sqrt{2}}$, if $0 < \alpha \leq 1$.

CHAPTER 5

2. $\frac{ah}{\sqrt{4h^2+a^2}}$. 3. $2a^2$. 4. $\frac{3\sqrt{3}}{4}a$. 5. $\frac{\pi}{4}ab$. 6. $1/4$.

7. $\frac{\sqrt{3}-1}{2}a$; $(2-\sqrt{3})a$; $\frac{\sqrt{3}-1}{2}a$. 8. $\frac{\sqrt{mn}}{2}b$. 9. $3/4$; $1/3$.

10. $7/1$. 11. $\sqrt{n+2}$. 12. 20 cm^2 . 13. $2/1$.

14. $\frac{S_1 S_3 (S_1 + S_2) (S_2 + S_3)}{S_2 (S_2^2 - S_1 S_3)}$. 15. $1/2$.

16. $\sqrt{\frac{na^2+mb^2}{m+n}}$. 17. $\frac{1}{2}\sqrt{a^2+b^2+6ab}$. 19. $\frac{2rh}{\sqrt{h(h-r)}}$.

20. $m(p+1)+p(n+1)+n(m+1)+1$. 21. $\frac{1}{9}S$.

22. $\frac{m^2-mn+n^2}{(m+n)^2}$. 23. $5/8$; $4/1$. 24. $\frac{6}{5}a$.

25. $\frac{5\pi}{12}$; $\frac{7\pi}{12}$. 26. $\frac{4\sqrt{5}}{3}R^2$.

27. *Hint.* Draw a straight line, through the points D and A , till it meets the line BM . Show that DM is a median of the resulting triangle.

28. $\frac{b\sqrt{a^2+b^2}}{4a}$. 29. $\frac{a^2}{\sqrt{4a^2-b^2}}$. 30. $R/3$; $3R$.

31. $\frac{2\sqrt{5}}{5}R$. 32. $\frac{3\sqrt{3}}{4}R^2$. 33. $a/2$; $a^2/2b$.

34. $\sqrt{\frac{4k+3}{k+1}} R$. 35. 6 cm^2 . 36. $\sqrt{3}/2$. 37. $3/2$.
 38. $(n-m)/2m$. 39. $\arctan(1/\sqrt{2})$. 40. 144 cm^2 .
 41. $\sqrt{m^2+mn}$; $\sqrt{n^2+mn}$. 42. $\frac{35}{2} (5\sqrt{2} \pm 4\sqrt{3})$. 43. $(\sqrt{41}-4)/4$.
 44. Use vector algebra. 45. Use the result of Problem 44.
 46. $a\sqrt{(R+r)/R}$. 47. 2. 48. $20/17$. 49. 8 cm . 50. 12 cm .
 51. $\sqrt{7/2} \text{ cm}$. 52. $3/2$. 53. $8:7$. 54. $(2n+1):1$.
 55. $\pi/2$; $\pi/3$; $\pi/6$. 56. $\frac{b(b+m-n)}{b-n} \sqrt{mn}$.
 57. $\pi/4$. 58. 3. 59. $\sqrt{4\sqrt{3}-3}$. 60. $\pi/3$; $\pi/12$; $7\pi/12$.
 61. $\frac{b}{8(1-\cos\alpha)}$. 62. $\frac{5}{24} R$. 63. $\frac{3}{10} S$; $\frac{3}{2} S$. 64. $2 \arctan(6/7)$.
 65. $2 \sin^2 \alpha$. 66. $2 \cot^2 \alpha$. 67. $2 \sin^2 \alpha$.
 68. $\frac{4a \cos^2 \alpha}{1+4 \cos^2 \alpha}$. 69. $\pi/3$. 70. $\frac{\sin 2\alpha}{4} a^2$.
 71. $\frac{4}{3} R$. 72. $\frac{\pi}{2}$; $\frac{\pi}{8}$; $\frac{3\pi}{8}$ (see solution of Problem 15 in Section I).

CHAPTER 6

1. A circle with diameter AO , where A is the given point and O is the centre of the given circle.

2. A circle with centre at the midpoint of the segment AO (A being the given point and O , the centre of the given circle) and radius half that of the given circle.

3. A circle with diameter AB .

4. A circle with radius $|AB|$ and centre at the point B .

5. A triangle OM_1M_2 , where M_1 and M_2 are points lying on the sides of the given angle at the distance l from the vertex O .

6. A segment.

8. A circle with radius $\frac{\sqrt{2s-|AB|^2}}{2}$ and centre at the midpoint of the segment AB if $2s > |AB|^2$; a point (the midpoint of the segment AB) if $2s = |AB|^2$; an empty set if $2s < |AB|^2$.

9. A circle of diameter AB , the points A and B exclusive.

10. A segment connecting the middle of the base with the middle of the altitude dropped to the base.

11. Four points: the centre of the circle inscribed into the triangle formed by the given lines and three points which are the centres of the circles each of which touches one side of the indicated triangle and the extensions of the other two sides of that triangle.

12. A circle with centre at the point C and radius $|AC|$, the point A exclusive.

13. The line containing the hypotenuse.
 14. An octagon whose four sides are congruent and parallel to the sides of the square and are $\frac{3}{2}a$ distant from the centre of the square.
 15. The line of intersection of the planes which are perpendicular to the sides of the trapezoid and pass through their midpoints.
 16. Four straight lines perpendicular to the plane α and passing, respectively, through the four points of that plane which are indicated in the answer to Problem 11.
 17. If $A \notin l$, then it is the line lying in the plane passing through l and A , parallel to l , not passing through A , and lying at a distance from l equal to the distance from A to l . If $A \in l$, then it is the line l .
 18. A straight line which is perpendicular to l and passes through A .
 19. If $A \notin l$ then it is a circle with diameter AA_1 , where A_1 is a point symmetric with respect to the point A about the line l . If $A \in l$, then it is the point A .
 20. A sphere with centre at the point B and radius $|AB|$.
 21. A sphere with centre at the point B and radius $|AB|$.
 22. An arc of a circle with diameter OA (O being the centre of the ball and A being the point of intersection of the given line l and the plane α which is perpendicular to l and passes through O), lying in the plane α and contained inside the ball.
 23. A sphere of radius $\frac{\sqrt{6}}{2}R$ whose centre coincides with the centre of the given sphere.
 24. Let h be the distance between l_1 and l_2 . If $h = 0$, then it is a circle with centre at the point of intersection of the lines l_1 and l_2

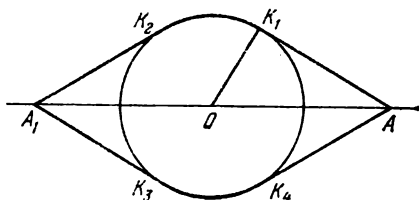


Fig. 283

and with radius $a/2$ (Problem 2 in 14.2). If $0 < h \leq a$, then it is a circle lying in the plane equidistant from the parallel planes containing l_1 and l_2 , with centre at the midpoint of the common perpendicular of the lines l_1 and l_2 and with radius $\sqrt{a^2 - h^2}/2$ (for $h = a$ the circle degenerates into a point). If $h > a$, then it is an empty set.

40. The point of tangency of the circle passing through the points A and B and the side of the angle.

42. *Hint.* Consider the points symmetric with respect to the point M about the sides of the angle.

44. The pedestrian can be at any point of the figure bounded by two arcs K_1K_2 and K_3K_4 (Fig. 283) and four segments AK_1 , K_2A_1 , A_1K_3 , K_4A . At the initial moment the pedestrian was at the point O ; $|OA| = 6$ km, $|OK_1| = 3$ km.

CHAPTER 7

2. It does not.

6. If $\alpha \parallel \beta$, then γ must be drawn so that $\gamma \cap \alpha \neq \emptyset$; if $\alpha \nparallel \beta$, then γ should be drawn parallel to the line of intersection of the planes α and β .

7. If the given lines are not parallel.

9. Suppose $\alpha \supset a$, $\alpha \parallel b$; $\beta \supset b$, $\beta \parallel a$. The problem has no solutions if $M \in \alpha$ but $M \notin a$, or if $M \in \beta$ but $M \notin b$.

10. It is not unique. 11. (a) They can; (b) they cannot.

12. (a) It can; (b) it can; (c) it cannot. 13. (a) It cannot; (b) it cannot; (c) it can.

$$19. \frac{\sqrt{11}}{12} a^2. \quad 20. 1 : 2. \quad 21. m - n + p.$$

22. 2 : 1, reckoning from the point M . 23. 1 : 5.

24. 2 : 3; 1 : 1. 25. 2 : 9. 26. 3 : 1, reckoning from the point K .

27. 2 : 3, reckoning from the vertex S . 28. 2 : 1, reckoning from the vertex S . 29. 3 : 7, reckoning from the vertex A_1 . 30. 5 : 1, reckoning from the vertex B .

$$31. \frac{7\sqrt{6}}{16} a^2. \quad 32. \frac{a^2\sqrt{6}}{4}.$$

33. A section passing through the midpoint of the edge AB .

$$34. mp : n. \quad 35. \frac{al\sqrt{2}}{3}. \quad 36. 1. \frac{\sqrt{5}}{3} a; 2. \frac{a}{\sqrt{5}}.$$

$$37. 3/4. \quad 38. 1/3. \quad 39. \frac{2}{3} a. \quad 40. \frac{3}{4} a. \quad 41. \frac{ma}{3}. \quad 45. \vec{AP} =$$

$$a + \frac{5}{2} b + c, \quad \vec{AQ} = \frac{1}{9} (4a + 5b + 4c). \quad 46. \vec{MN} = (1 - \lambda) \vec{AB} + \lambda \vec{CD}.$$

$$49. 1/1; 3/1. \quad 50. \frac{a\sqrt{6}}{2}. \quad 51. 3:5.$$

54. $\arccos(2/3)$; $\arccos(2/3)$; $\arccos(1/3)$.

55. (a) $\arccos(1/\sqrt{10})$; (b) $\arccos(1/\sqrt{10})$; (c) $\arccos \sqrt{2/5}$.

56. $\arccos(1/2\sqrt{3})$. 57. $\alpha/2$; $(\pi - \alpha)/2$. 58. (a) $\arccos(1/6)$; (b) $\arccos(5/6)$.

$$59. \frac{(\sqrt{3}-1)}{2} \quad 60. a \sqrt{2} \left(\frac{1}{2} - \lambda \right).$$

$$61. \text{It does not exist.} \quad 64. \arccos \left(\frac{1}{4} + \cos^2 \alpha \right).$$

$$65. \arccos(5/(3\sqrt{6})). \quad 67. \pi/3,$$

68. $\sqrt{b^2 - 2a^2(1 \pm \cos \alpha)}$; $b > \sqrt{2}a$. 69. $a/\sqrt{2}$.
 70. $1/\sqrt{2}$. 71. $\frac{\sqrt{2}}{3}a$. 72. $\frac{\sqrt{10}}{3}a$. 73. (a) $\frac{\sqrt{38}}{4}a$;
 (b) $\frac{\sqrt{11}}{6}a^2$.
 74. $n^2/4$, reckoning from the vertices A and D .
 75. $3/2$, reckoning from the vertices D and B .
 76. $c/\sqrt{10}$. 77. $\frac{\sqrt{23}}{6}a$. 78. a ; $3a$. 79. a ; $\frac{a}{2\sqrt{2}}$.
 81. $\frac{5\sqrt{3}}{36}a^2$. 82. $\arccos(\cos \beta - \cos \alpha)$.
 83. $\arccos\left(-\frac{1}{3}\right)$. 84. $-3a^2$.
 86. Assume that M and N are the midpoints of the segments AC and BD , $p = \frac{1}{2}(|AC|^2 - |BD|^2)$. If $M = N$, then the desired set is:
 (a) the entire space for $y_0 = p$; (b) an empty set for $y_0 \neq p$. If $M \neq N$, then the desired set is a plane perpendicular to the line MN .
 87. $8b^2 + 4a^2(2 + \sqrt{2})$.
 88. The midpoint of the segment PQ , where P and Q are the midpoints of the segments AB and CD (if the segment degenerates into a point, then this point is considered to be its middle).

CHAPTER 8

1. $B\left(\frac{-1 \pm 2\sqrt{3}}{3}; \frac{7 \mp 2\sqrt{3}}{3}; \frac{7 \pm \sqrt{3}}{3}\right)$,
 $C\left(\frac{-3 \mp 2\sqrt{3}}{9}; \frac{21 \pm 2\sqrt{3}}{9}; \frac{21 \mp \sqrt{3}}{9}\right)$.
 6. $\pi - 2 \arcsin \frac{1}{\sqrt{3}}$. 7. $a\sqrt{1 - 2\cos \alpha}$.
 8. $\frac{\sqrt{5}}{4}a$. 9. $\vec{SH} = \vec{SA} - \vec{SB} + \vec{SC}$. 10. a .
 13. (a) $\frac{1}{\sqrt{3}}S \cot(\alpha/2)$; (b) $\sqrt{3}S \tan \frac{\alpha}{2}$.
 14. $\frac{\sqrt{14}}{3}a$. 15. $\frac{2}{9}a^3$. 17. $\frac{1}{2}a^2$.
 18. $\sqrt{b^2 - 7a^2}$, $\sqrt{b^2 - 3a^2}$. 19. $\frac{2}{3}a$.
 20. (a) $\frac{2}{5}\sqrt{10}a$; (b) $\frac{1}{8}\sqrt{10}a$. 21. $\sqrt{\frac{7}{3}}a$.

22. (a) $\frac{3\sqrt{3}}{32}a^2$; (b) $\frac{41}{96}a^2$. 23. $\frac{3\sqrt{3}}{4}a^2$.
24. (a) $\frac{\sqrt{6}}{8}a^2$; (b) $\frac{3\sqrt{15}}{16}a^2$. 25. $\frac{1}{6}a^3\sin^2\alpha$.
26. $1/2$, reckoning from the vertex S . 27. $a^3/6$.
28. $2\sqrt{2}a$. 30. $2a\sqrt{1-\cos^2\alpha\cos^2\frac{\beta}{2}}$.
31. $\pi/6$. 32. (a) $\alpha_x = \alpha_y = \alpha_z = \arccos(1/\sqrt{3})$;
(b) $\arcsin(1/\sqrt{3})$. 33. $\frac{a\sqrt{3}\sin\alpha}{4\sin\beta}$.
34. $a/\sqrt{6}$; $a/2\sqrt{6}$. 35. (a) $2/1$; (b) $14/1$; $2/3$, reckoning from the vertex A .
36. $\arcsin(1/3)$. 38. 7. 39. $37/30$; 0.
40. (0; 0; 3), (0; 0; -2.5).
41. $D(1; 4; 1)$, $A_1(1; 3; 3)$, $B_1(3; 4; 2)$, $D_1(1; 6; 3)$; $\sqrt{3}$.
42. $\frac{a}{4\sqrt{2}}$. 43. $\frac{a\sqrt{3b^2-a^2}}{4b}$. 44. $\frac{2}{\sqrt{5}}H$.
45. $\frac{4}{9}a^3$. 46. $2/\sqrt{3}$. 47. 4.5 cm^2 . 48. $0.3a$.
49. $\sqrt{10}/6\text{ cm}$. 50. $1.5a$.
51. If $0 < l \leq a\sqrt{3}$, then $\rho = \sqrt{\frac{3}{4}a^2 + \frac{\sqrt{3}}{2}al + l^2}$; if $l > a\sqrt{3}$, then $\rho = \sqrt{\frac{9}{4}a^2 + l^2}$.
52. $\arcsin\left(\sin\varphi \cdot \cos\frac{\psi}{2}\right)$.
53. $\arcsin\left(\frac{\sqrt{5}}{2}\sin\varphi\right)$. 54. $\arccos\sqrt{-\cos\alpha}$.
55. (a) $\arctan\sqrt{2}$; (b) $\frac{\pi}{3}$; (c) $\arccos\frac{8}{9}$; (d) $\arccos\frac{3}{\sqrt{14}}$.
56. $\arctan\sqrt{5}$. 57. (a) $\arccos\frac{1}{3}$; (b) $\arccos\frac{2}{3}$.
58. (a) $\frac{b\sqrt{4a^2-b^2}}{4\cos\alpha}$; (b) 6 cm^3 . 59. $\arccos(\pm\cot\varphi \cdot \cot\psi)$.
60. $\arcsin\left(\frac{2}{\sqrt{3}}\sin\varphi \cdot \sin\psi\right)$. 61. $3a$. 63. $\frac{\sqrt{2}(1+\cos\alpha)^{3/2}}{(1+2\cos\alpha)^2}a^2$.
64. $\frac{a^3}{3}$. 65. $3a^3$. 66. $\arccos\frac{1}{6}$.

67. $2 \arccos \frac{1}{2 \sin \frac{\alpha}{2}}$; $\frac{\pi}{3} < \alpha < \pi$. 69. $3 \sqrt{3} a$.
70. $a \sqrt{\frac{1-2 \cos \varphi}{3}}$. 71. $2 \arcsin \frac{1}{\sqrt{2} \cos \frac{\alpha}{2}}$.
72. $\frac{a^3}{12}$ in all cases. 73. $\frac{2}{3} a^3 \sin^2 \frac{\alpha}{2} \sqrt{\cos \alpha}$.
74. $\frac{7 \sqrt{3}}{24} a^3$. 75. $\frac{b^3}{8} \sin 2\alpha$. 76. $\frac{b^3}{3} \sin^3 \frac{\alpha}{2} \tan \varphi$.
77. $\frac{1}{6} S \sqrt{S \sin \alpha} \tan \beta$. 79. $3:1$. 80. $\frac{a^3}{24}$. 81. $mnp:1$.
82. (a) $13:23$; (b) $5:19$; (c) $19:41$.
85. $9:5$. 86. $q^2(3+q):(1+3q)$. 87. $\frac{a}{3} \sqrt{(c^2-b^2)(b^2-a^2)}$.
88. $\frac{27}{4} \text{ cm}^3$; $\frac{25 \sqrt{5}}{4} \text{ cm}^3$. 89. $\frac{7}{18} V$. 90. $\frac{4}{3} a^3$.
91. $\frac{a^3}{12} \cdot (1 + \sqrt{3} \cot \varphi)^3 \tan \varphi$; $\varphi = \arctan(3 - \sqrt{3})$.
92. $4V$. 93. $\frac{125}{27}$. 94. $\frac{1}{2} V$. 95. $\frac{3 \sqrt{2}-2}{21} V$.
96. $\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{\sqrt{2}c}\right)^{-1}$, and also $\left(\frac{1}{a} + \frac{1}{b} - \frac{1}{\sqrt{2}c}\right)^{-1}$,
if $\frac{1}{a} + \frac{1}{b} - \frac{1}{\sqrt{2}c} > 0$.
97. $(20-14 \sqrt{2}):1$. 98. $\frac{125}{12} V$. 99. $\frac{27 \sqrt{3}}{4} a^3$.
100. (a) $2a$; (b) $\frac{2}{\sqrt{15}} a$. 101. $(4+3 \sqrt{6}):19$.
102. $\frac{6}{5} (2 \sqrt{2} - \sqrt{3}) \text{ cm}$.

CHAPTER 9

1. $r=h=\sqrt[3]{V/\pi}$. 2. $2a^3/\sqrt{3}$. 3. $\frac{\pi}{18} V$. 4. $a \sqrt{3}$.
5. $\pi a^3 \sqrt{23}/9$. 6. $7a/8 \sqrt{3}$. 8. $20\pi/9 \sqrt{3}$. 9. $\sqrt{11} R/3$, $\sqrt{3} R$.
10. One cylinder of radius $(aC/3\sqrt{2})$ and three cylinders of radius $a/\sqrt{2}$.

$$11. 2 \arctan (2/\sqrt{3}). \quad 12. \frac{\pi}{3} \sqrt{\frac{7}{3}}.$$

13. If $2\alpha \leq \pi/2$, then the axial section has the greatest area equal to $\frac{1}{2} l^2 \sin 2\alpha$; if $2\alpha > \pi/2$, then the section formed by the plane, making an angle of $\arccos (\sqrt{2} \cos \alpha)$ with the axis, has the greatest area equal to $\frac{1}{2} l^2$.

$$14. S(5/2; 5/2; 1), S_{\text{sec}} = 5\pi/18. \quad 15. 22.5\pi.$$

$$16. \frac{9}{8} \pi a^3. \quad 17. \frac{32}{15} a^3. \quad 18. \arccos(-1/71).$$

$$19. 2 \arctan \sqrt{\sin(\alpha/2)}. \quad 20. 25 \sqrt{5/3} \pi \text{ cm}^3. \quad 21. 7\pi/6 \sqrt{3}.$$

$$22. 2(\sqrt{2}-1)R. \quad 23. 16 \sqrt{3} r^3. \quad 24. 1:2.$$

$$27. \frac{64}{81} R^3. \quad 28. \frac{6 \sqrt{3}}{\pi} V. \quad 29. \pi/5 \sqrt{3}. \quad 30. \frac{6 \sqrt{3} + 4\pi}{27} a^2.$$

$$31. \frac{\pi(4\sqrt{2}-5)}{24} a^3. \quad 32. \frac{18\sqrt{2}-20}{31} \pi.$$

$$34. \frac{\sqrt{41}}{8} a. \quad 35. \frac{4 \sin^2 \alpha \cos \alpha}{3(1+\sin^2 \alpha)^{3/2}} R^3.$$

$$36. a. \quad 37. \frac{5 \pm \sqrt{7}}{4} a. \quad 38. \frac{a \sqrt{3} \sin \alpha}{4(\sin \alpha + \sin \beta)}.$$

$$39. \frac{3a}{4\sqrt{2}}. \quad 40. \sqrt{3} r. \quad 41. \frac{2}{3} \text{ cm}. \quad 42. \frac{4}{5} \text{ cm}.$$

$$43. (a) r:R=2\sqrt{3}-3; (b) 30^\circ. \quad 47. \frac{15}{2\sqrt{2}} r.$$

$$48. \sqrt{1.5} a. \quad 49. \frac{6-\sqrt{3}}{16} a^2. \quad 50. \frac{3\sqrt{17}}{16} a^2.$$

$$51. \frac{8}{375} \text{ cm}^3. \quad 52. \frac{\sqrt{3} \pm 1}{4\sqrt{2}} a. \quad 53. \frac{a}{2}; 2a.$$

$$54. \frac{40}{7} \text{ cm}; \frac{104}{7} \text{ cm}. \quad 55. \frac{4r}{\sqrt{3}}. \quad 56. \frac{2\sqrt{2}a}{5 \pm \sqrt{11}}.$$

$$57. \arccos \frac{3}{5}. \quad 58. \text{It is true.} \quad 59. \frac{16\pi}{27}.$$

$$60. \frac{Rr}{R \pm r}. \quad 61. (a) 8a^2; (b) 2a^2; (c) 8a^2.$$

$$62. R^2. \quad 63. \frac{\sqrt{3} \pm \sqrt{2}}{2\sqrt{3}} r. \quad 64. \frac{5}{16}.$$

$$65. \frac{2\sqrt{3}-3}{6} \pi a^2. \quad \text{Hint. Consider the tetrahedron homothetic}$$

to the given tetrahedron, with centre at A and the coefficient 2.

67. $4.5\pi a^3$.

68. The length of the base is $\frac{1}{4}P$, the length of the lateral side is $\frac{3}{8}P$.

69. $\frac{(n+1)(m-1)}{(n-1)(m+1)}$. 70. $V/\sqrt{3}$. 71. $\frac{8}{27}V$.

72. $2S$. 73. $\arccos(\sqrt{5}-2)$. 75. $\arcsin \sqrt{\frac{S_2}{S_1}}$.

76. $\arccos\left(-\frac{1}{6}\right)$. 77. $1:9$.

78. $\frac{3\sqrt{2}+1 \pm \sqrt{10+6\sqrt{2}}}{9}R$ (external tangency),

$$\frac{3\sqrt{2}-1 \pm \sqrt{10-6\sqrt{2}}}{9}R$$
 (internal tangency).

79. $ab/2c$, $bc/2a$, $ca/2b$.

80. $d_A = d_B = r \cos \alpha \sqrt{1 + \sin^2 \alpha}$, $|AB| = 2r \cos \alpha$.

81. $0.5 \sqrt{n(n+2)}a$.

82. $1:(2\sqrt{2}+2):1$; $17:(6\sqrt{2}+2):17$.

83. $0.5r$; $1.5r$. 84. $r(\sqrt{1.5} \pm 1)$. 85. $1:2:1$.

86. $\frac{\pi}{3}$. 87. $\frac{5}{\sqrt{3}}r$. 88. $(6+3\sqrt{2})V$.

89. $2 \arctan \sin \alpha$. 90. $2 \arctan(\sqrt{3/2})$.

APPENDIX

1. -10 . 2. $\left\{1, \frac{\sqrt{8\pi+1}-1}{2}\right\}$. 3. $9/2$. 4. $a < -1$.

5. 3. 6. $\{-1\} \cup [2; +\infty)$. 7. $-\frac{4\pi}{3} + 2\pi k$, $k \in \mathbb{Z}$.

8. $y = -x + \frac{5}{2}$. 9. $\{6, 10, 14, 30, 42, 70, 105, 210\}$.

10. $(3-\sqrt{5})/4$. 11. $\pm \frac{\pi}{3} + \pi k$, $k \in \mathbb{Z}$. 12. $114\frac{1}{3}$.

13. $x_1 = -63/4$, $x_2 = -23/8$. 14. $\pi/4$, $\pi/4$, $3\pi/4$, $3\pi/4$.

15. $11/\sqrt{170}$. 16. $(-\infty; -2] \cup [2; +\infty)$. 17. $\sqrt{7}$ m.

18. The point of minimum is e ; the intervals of decrease are $(0; 1)$ and $(1; e]$, the interval of increase is $[e; +\infty)$.

19. 60 km/h. 20. $\frac{\pi}{12} + 2\pi k$, $-\frac{7\pi}{12} + 2\pi l$ ($k, l \in \mathbb{Z}$).

21. The point $x = 0$ is a point of minimum, the intervals of decrease are $(-\infty; 0]$ and $(1; +\infty)$, the interval of increase is $[0; 1)$.

22. $310 \frac{2}{7} \pi$. 23. $\pm \frac{\pi}{12} + \frac{\pi}{2} k$, $k \in \mathbb{Z}$.

24. $(-\infty; -14/3) \cup (4; +\infty)$. 25. $\{1/16, 1\}$.

26. 40 kopecks, 60 kopecks, 80 kopecks, and 1 ruble.

27. -62 . 28. -1 . 29. $-\frac{\pi}{4} + \pi k$, $\arctan 7 + \pi l$ ($k, l \in \mathbb{Z}$).

30. $13 \frac{1}{3}$. 31. 164850.

32. For $\alpha \in (0; \pi/2]$, $\beta \in (0; \pi/2]$, $\alpha \neq \beta$ there are two solutions:
 $\frac{V}{\sin(\alpha+\beta)}$ and $\frac{V}{\sin|\alpha-\beta|}$; for $\alpha \in (0; \pi/2]$, $\beta \in (0; \pi/2]$, $\alpha = \beta$,
 there is one solution: $\frac{V}{\sin(\alpha+\beta)}$; here $V = \frac{4a^3}{75} \sin^2 \varphi \cdot \cos \varphi \times$
 $\sin \alpha \cdot \sin \beta$.

33. $\{(15; -12)\}$. 34. $\left\{-6\pi, -\frac{9\pi}{2}, 0\right\}$.

35. 100. 36. $\frac{\pi}{4} + \frac{\pi k}{2}$, $\pm \frac{\pi}{6} + \pi k$, $k \in \mathbb{Z}$.

37. $(0; 8/5) \cup (5/2; +\infty)$. 38. $z = 3$.

39. $-135/4$. 40. $(-\infty; -4) \cup [-3; 3) \cup [6; +\infty)$.

41. 24; 0. 42. $64/3$. 43. $3\pi R^3$. 44. 100 tables and 400 chairs.

45. $\{0, 4/3\}$. 46. $[0; 1] \cup [4; 16]$.

47. $\frac{\pi}{4} + k\pi$, $\arctan 2 + k\pi$, $k \in \mathbb{Z}$. 48. -24 .

49. $\pi/3$. 50. $\pm \frac{\pi}{6} + \pi k$, $k \in \mathbb{Z}$.

51. $A' = (6; 3)$, $M' = M$, $C' = (7; 9)$.

52. $C_{11}^6 < C_{11}^3 + C_{11}^4$, $\{6, 7, 8, 9, 10, 11\}$.

53. $e + \frac{2}{e} + 4$, $\frac{10 - 7 \ln 2}{2}$. 54. (b) $[10/3; +\infty)$.

55. $\pi + 2\pi k$, $\pm \frac{\pi}{3} + 2\pi l$ ($k, l \in \mathbb{Z}$).

56. $\{10\}$. 57. 9. 58. $2a$. 59. 0. 60. $12 - 5 \ln 5$.

61. *Hint.* Under the homothetic transformation with centre $[AC] \cap [BD]$ and the coefficient $-|CD|/|AB|$, the square constructed on $[AB]$ passes into a square constructed on $[CD]$.

62. $(-3; -1)$. 63. $\frac{1}{2} \arctan 2 + 2\pi k$, $\frac{1}{2} \arctan 2 - \frac{\pi}{2} + 2\pi l$ ($k, l \in \mathbb{Z}$).

64. The ratio between the volumes of the "upper" and "lower" polyhedra is $3/5$.

65. $(-1; +\infty)$. 66. $\{(1/2; 1/4), (1/8; 64)\}$.

67. $5\pi/12$, $5\pi/12$, $7\pi/12$, $7\pi/12$. 68. $\frac{\pi k}{3}$, $2\pi k - \frac{\pi}{2}$, $k \in \mathbb{Z}$.

69. If $V_1/V_2 = 2$, then $t_1/t_2 = 2/3$.

70. $\arcsin 2(\sqrt{2}-1)$. 71. $-\frac{1}{8}$.

72. $\{(\sqrt[4]{3}; 1), (\sqrt[4]{3}; -1)\}$.

73. $\sqrt{3}-1$. 74. $\left\{-\frac{\pi}{2}, \frac{\pi}{2}, \frac{3}{2}\pi\right\}$.

75. If the base of the isosceles triangle obtained in the section lies on the plane $ABCD$, then $S = \frac{a^2 \cos \alpha}{\sin^2 \alpha}$; if it lies on the plane CDD_1C_1 (or on the plane BCC_1B_1), then $S = \frac{a^2}{2\sqrt{-\cos 2\alpha}}$.

76. $b > 0$, $a \in \left(-\sqrt[4]{\frac{128}{3}}b; -\sqrt[4]{b}\right) \cup \left(\sqrt[4]{b}; \sqrt[4]{\frac{128}{3}}b\right)$ or $b < 0$, a being any number.

77. 294840. 78. For $a \in [0; 4)$, $x = \pi k$, for $a \notin [0; 4)$, $x = \pi k$ and $x = \pi k \pm \frac{1}{2} \arccos \frac{a+2}{2(a-1)}$, $k \in \mathbb{Z}$.

79. $\frac{a \sin^2(a/2)}{\sqrt{2} \cos(\alpha/2) (1 + \sqrt{-\cos \alpha})}$. 80. $a = (2; -2; -2)$. 81. 0.

82. $2\pi k$, $\frac{\pi}{2} + 2\pi k$, $k \in \mathbb{Z}$. 83. $[1; 3]$.

85. 6+6. 86. Three sons and 2 daughters.

87. $\frac{\pi}{6} + \frac{\pi k}{3}$, $k \in \mathbb{Z}$. 88. $\{20\}$. 89. 2, -10.

90. $\frac{np(np+1)}{2}$. 91. $\frac{4a}{t}$ m/s, $\frac{a}{t}$ m/s.

92. πk , $\frac{\pi}{4} + \frac{\pi k}{2}$, $k \in \mathbb{Z}$. 93. $\frac{\sqrt{30}}{5}$ cm, $\frac{3\sqrt{30}}{10}$ cm.

94. $\{1/2\} \cup (-\infty; 0)$. 95. $\frac{\sqrt{6}}{2}a^3$, $\frac{\sqrt{15}}{4}a^3$.

96. $\frac{\pi}{4} + \frac{\pi}{2}k$, $k \in \mathbb{Z}$. 97. $(0; +\infty)$. 98. $b = 2a/3$. $V = \frac{4}{27}a^2H$.

99. 13. 100. $V = \frac{4}{3}r^3 \frac{(\tan \varphi + 1)^3}{\tan^2 \varphi}$, $\varphi = \arctan 2$.

101. 32. 102. $(0; 1/2] \cup (8; 16)$. 103. 576. 105. $8\sqrt{2}$.

106. $\left[\frac{-7-2\sqrt{11}}{5}; \frac{-7+2\sqrt{11}}{5}\right]$.

107. $(-1/6; \pi/3] \cup [5\pi/3; 6)$. 108. $\frac{4}{3}R$.

109. 720. 110. $(1; \sqrt{2}) \cup (2; +\infty)$. 111. $\pi + 2\pi k$, $k \in \mathbb{Z}$.

List of Formulas

TRIGONOMETRIC FORMULAS

1°. Formulas for solving equations:

$$\sin x = a, \quad x = (-1)^n \arcsin a + \pi n, \quad n \in \mathbb{Z} \quad (|a| \leq 1);$$

$$\cos x = a, \quad x = \pm \arccos a + 2\pi n, \quad n \in \mathbb{Z} \quad (|a| \leq 1);$$

$$\tan x = a, \quad x = \arctan a + \pi n, \quad n \in \mathbb{Z} \quad (a \in \mathbb{R});$$

$$\cot x = a, \quad x = \operatorname{arccot} a + \pi n, \quad n \in \mathbb{Z} \quad (a \in \mathbb{R});$$

$$\sin x = 0, \quad x = \pi n;$$

$$\sin x = 1, \quad x = \frac{\pi}{2} + 2\pi n;$$

$$\sin x = -1, \quad x = -\frac{\pi}{2} + 2\pi n;$$

$$\cos x = 0, \quad x = \frac{\pi}{2} + \pi n;$$

$$\cos x = 1, \quad x = 2\pi n;$$

$$\cos x = -1, \quad x = \pi + 2\pi n,$$

where $n \in \mathbb{Z}$.

2°. Some values of inverse trigonometric functions:

a	$\arcsin a$	$\arccos a$
$\frac{\sqrt{3}}{2}$	$\frac{\pi}{3}$	$\frac{\pi}{6}$
$\frac{\sqrt{2}}{2}$	$\frac{\pi}{4}$	$\frac{\pi}{4}$
$\frac{1}{2}$	$\frac{\pi}{6}$	$\frac{\pi}{3}$
$-\frac{1}{2}$	$-\frac{\pi}{6}$	$\frac{2\pi}{3}$
$-\frac{\sqrt{2}}{2}$	$-\frac{\pi}{4}$	$\frac{3\pi}{4}$
$\frac{\sqrt{3}}{2}$	$-\frac{\pi}{3}$	$\frac{5\pi}{6}$

a	$\arctan a$	$\operatorname{arccot} a$
$\sqrt{3}$	$\frac{\pi}{3}$	$\frac{\pi}{6}$
1	$\frac{\pi}{4}$	$\frac{\pi}{4}$
$\frac{1}{\sqrt{3}}$	$\frac{\pi}{6}$	$\frac{\pi}{3}$
$-\frac{1}{\sqrt{3}}$	$-\frac{\pi}{6}$	$\frac{2\pi}{3}$
-1	$-\frac{\pi}{4}$	$\frac{3\pi}{4}$
$-\sqrt{3}$	$-\frac{\pi}{3}$	$\frac{5\pi}{6}$

3°. Recursion formulas:

$$\begin{aligned}\sin(-\alpha) &= -\sin \alpha; \\ \sin\left(\frac{\pi}{2} - \alpha\right) &= \sin\left(\frac{\pi}{2} + \alpha\right) = \cos \alpha, \\ \sin(\pi - \alpha) &= \sin \alpha; \quad \sin(\pi + \alpha) = -\sin \alpha; \\ \cos(-\alpha) &= \cos \alpha; \\ \cos\left(\frac{\pi}{2} - \alpha\right) &= \sin \alpha; \quad \cos\left(\frac{\pi}{2} + \alpha\right) = -\sin \alpha, \\ \cos(\pi - \alpha) &= \cos(\pi + \alpha) = -\cos \alpha; \\ \tan(-\alpha) &= -\tan \alpha; \\ \tan\left(\frac{\pi}{2} - \alpha\right) &= \cot \alpha; \quad \tan\left(\frac{\pi}{2} + \alpha\right) = -\cot \alpha; \\ \cot(-\alpha) &= -\cot \alpha; \\ \cot\left(\frac{\pi}{2} - \alpha\right) &= \tan \alpha; \quad \cot\left(\frac{\pi}{2} + \alpha\right) = -\tan \alpha; \\ \sin(\alpha + \pi n) &= (-1)^n \sin \alpha; \quad \cos(\alpha + \pi n) = (-1)^n \cos \alpha,\end{aligned}$$

where $n \in \mathbb{Z}$.

4°. Formulas which are true equalities for all values of the variables:

$$\begin{aligned}\sin^2 \alpha + \cos^2 \alpha &= 1; \\ \sin 2\alpha &= 2 \sin \alpha \cos \alpha; \\ \cos 2\alpha &= \cos^2 \alpha - \sin^2 \alpha = 2 \cos^2 \alpha - 1 = 1 - 2 \sin^2 \alpha; \\ \sin^2(\alpha/2) &= (1 - \cos \alpha)/2; \quad \cos^2(\alpha/2) = (1 + \cos \alpha)/2; \\ \sin(\alpha + \beta) &= \sin \alpha \cos \beta + \cos \alpha \sin \beta; \\ \sin(\alpha - \beta) &= \sin \alpha \cos \beta - \cos \alpha \sin \beta; \\ \cos(\alpha + \beta) &= \cos \alpha \cos \beta - \sin \alpha \sin \beta; \\ \cos(\alpha - \beta) &= \cos \alpha \cos \beta + \sin \alpha \sin \beta; \\ \sin 3\alpha &= \sin \alpha (2 \cos 2\alpha + 1) = \sin \alpha (3 - 4 \sin^2 \alpha); \\ \cos 3\alpha &= \cos \alpha (2 \cos 2\alpha - 1) = \cos \alpha (4 \cos^2 \alpha - 3); \\ \sin \alpha \cos \beta &= \frac{1}{2} (\sin(\alpha + \beta) + \sin(\alpha - \beta)); \\ \cos \alpha \cos \beta &= \frac{1}{2} (\cos(\alpha + \beta) + \cos(\alpha - \beta)); \\ \sin \alpha \sin \beta &= \frac{1}{2} (\cos(\alpha - \beta) - \cos(\alpha + \beta));\end{aligned}$$

$$\sin \alpha + \sin \beta = 2 \sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2};$$

$$\sin \alpha - \sin \beta = 2 \sin \frac{\alpha - \beta}{2} \cos \frac{\alpha + \beta}{2};$$

$$\cos \alpha + \cos \beta = 2 \cos \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2};$$

$$\cos \alpha - \cos \beta = 2 \sin \frac{\beta - \alpha}{2} \sin \frac{\beta + \alpha}{2};$$

for all $\alpha \in \mathbb{R}$ and $\beta \in \mathbb{R}$.

5°. Formulas which are true equalities *not for all* values of the variables:

$$\tan \alpha = \frac{1}{\cot \alpha}, \quad \cot \alpha = \frac{1}{\tan \alpha}$$

for $\alpha \neq \frac{\pi n}{2}$, $n \in \mathbb{Z}$.

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

for $\alpha \neq \frac{\pi}{4} + \frac{\pi n}{2}$ and $\alpha \neq \frac{\pi}{2} + \pi n$, $n \in \mathbb{Z}$;

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

for $\alpha + \beta \neq \frac{\pi}{2} + \pi k$, $\alpha \neq \frac{\pi}{2} + \pi l$, $\beta \neq \frac{\pi}{2} + \pi m$, $k, l, m \in \mathbb{Z}$;

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

for $\alpha - \beta \neq \frac{\pi}{2} + \pi k$, $\alpha \neq \frac{\pi}{2} + \pi l$, $\beta \neq \frac{\pi}{2} + \pi m$, $k, l, m \in \mathbb{Z}$;

$$\sin^2 \alpha = \frac{\tan^2 \alpha}{1 + \tan^2 \alpha}; \quad \cos^2 \alpha = \frac{1}{1 + \tan^2 \alpha};$$

$$\sin 2\alpha = \frac{2 \tan \alpha}{1 + \tan^2 \alpha}; \quad \cos 2\alpha = \frac{1 - \tan^2 \alpha}{1 + \tan^2 \alpha}$$

for $\alpha \neq \frac{\pi}{2} + \pi n$, $n \in \mathbb{Z}$;

$$\tan \alpha = \frac{\sin 2\alpha}{1 + \cos 2\alpha}; \quad \tan^2 \alpha = \frac{1 - \cos 2\alpha}{1 + \cos 2\alpha}$$

for $\alpha \neq \frac{\pi}{2} + \pi n$, $n \in \mathbb{Z}$.

FORMULAS FOR LOGARITHMIC FUNCTIONS

(in the indicated formulas, except for the last one, $a > 0$, $a \neq 1$):

$$a^{\log_a b} = b \text{ for } b > 0;$$

$$\log_a a^b = b \text{ for } b \in \mathbb{R};$$

$$\log_a b^c = c \log_a b \text{ for } b > 0, c \in \mathbb{R};$$

$$\log_a b^{2n} = 2n \log_a (-b) \text{ for } b < 0, n \in \mathbb{Z};$$

$$\log_a bc = \log_a b + \log_a c \text{ for } b > 0, c > 0;$$

$$\log_a bc = \log_a (-b) + \log_a (-c) \text{ for } b < 0, c < 0;$$

$$\log_a \frac{b}{c} = \log_a b - \log_a c \text{ for } b > 0, c > 0;$$

$$\log_a \frac{b}{c} = \log_a (-b) - \log_a (-c) \text{ for } b < 0, c < 0;$$

$$\log_a \frac{1}{b} = -\log_a b \text{ for } b > 0;$$

$$\log_a b = \frac{\log_c b}{\log_c a} \text{ for } b > 0, c > 0, c \neq 1;$$

$$\log_a b = \frac{1}{\log_b a} \text{ for } b > 0, b \neq 1;$$

$$\log_{a^c} b = \frac{1}{c} \log_a b \text{ for } c \neq 0, b > 0;$$

$$\log_{a^{2n}} b = \frac{1}{2n} \log_{(-a)} b \text{ for } n \in \mathbb{Z}, n \neq 0,$$

$$a < 0, a \neq -1, b > 0.$$

FORMULAS FOR THE VOLUME OF SOLIDS

The volume of a rectangular parallelepiped with the dimensions a , b , c :

$$V = abc;$$

the volume of a prism:

$$V = HS,$$

where H is the altitude, S is the area of the base of the prism;

$$V = lS_{\perp},$$

where l is the length of the lateral edge, $S_{\perp}$ is the area of the perpendicular section of the prism;

the volume of a pyramid:

$$V = \frac{1}{3} HS,$$

where H is the altitude, S is the area of the base of the pyramid;
the volume of a truncated pyramid:

$$V = \frac{1}{3} H (S_1 + \sqrt{S_1 S_2} + S_2),$$

where H is the altitude, S_1 and S_2 are the areas of the bases of the truncated pyramid;
the volume of a cylinder:

$$V = \pi R^2 H,$$

where R is the radius of the base, H is the altitude of the cylinder;
the volume of a cone:

$$V = \frac{1}{3} \pi R^2 H,$$

where R is the radius of the base, H is the altitude of the cone;
the volume of a truncated cone:

$$V = \frac{1}{3} \pi H (R^2 + \sqrt{Rr} + r^2),$$

where H is the altitude, R and r are the radii of the bases of the truncated cone;
the volume of a ball:

$$V = \frac{4}{3} \pi R^3,$$

where R is the radius of the ball;
the volume of a spherical segment:

$$V = \pi H^2 \left(R - \frac{H}{3} \right),$$

where H is the altitude of the segment, R is the radius of the ball;
the volume of a spherical sector:

$$V = \frac{2}{3} \pi R^3 H,$$

where R is the radius of the ball, H is the altitude of the spherical part of the sector;

the volume of a solid of revolution: assume that a curvilinear trapezoid is bounded by the graph of the continuous nonnegative function $f(x)$, $x \in [a; b]$, the lines $x = a$ and $x = b$, and by the abscissa axis. Then the volume of the solid resulting from the rotation of that trapezoid about the abscissa axis can be calculated by the formula

$$V = \pi \int_a^b f^2(x) dx.$$

FORMULAS FOR THE AREA OF A SURFACE

The area of the orthogonal projection of a polygon:

$$S_{pr} = S \cos \varphi,$$

where S is the area of the polygon, S_{pr} is the area of the projection, φ is the angle between the plane of the polygon and the plane of the projection;

the area of the lateral surface of a prism:

$$S = P_{\perp} \cdot l,$$

where $P_{\perp}$ is the perimeter of the perpendicular section, l is the length of the lateral edge (for the right prism, $P_{\perp}$ is the perimeter of the base);

the area of the lateral surface of a regular pyramid:

$$S = \frac{1}{2} P \cdot h,$$

where P is the perimeter of the base, h is the apothem;

the area of the lateral surface of a regular truncated pyramid:

$$S = \frac{1}{2} (P_1 + P_2) h,$$

where P_1 and P_2 are the perimeters of the bases, h is the apothem;

the area of the lateral surface of a cylinder:

$$S = 2\pi RH,$$

where R is the radius of the base, H is the altitude of the cylinder;

the area of the lateral surface of a cone:

$$S = \pi Rl,$$

where R is the radius of the base, l is the length of the generating line;

the area of the lateral surface of a truncated cone:

$$S = \pi (R_1 + R_2) \cdot l,$$

where R_1 and R_2 are the radii of the bases, l is the length of the generating line;

the area of a sphere:

$$S = 4\pi R^2,$$

where R is the radius;

the area of a spherical segment:

$$S = 2\pi RH,$$

where R is the radius of the sphere, H is the altitude of the segment,

TABLES OF TRIGONOMETRIC FUNCTIONS

α°	$\sin \alpha$	$\tan \alpha$	$\cot \alpha$	$\cos \alpha$	
0	0.0000	0.0000	—	1.000	90
1	0175	0175	57.3	1.000	89
2	0349	0349	28.6	0.999	88
3	0523	0524	19.1	999	87
4	0698	0699	14.3	998	86
5	0.0872	0.0875	11.4	0.996	85
6	1045	1051	9.51	995	84
7	1219	1228	8.14	993	83
8	139	141	7.11	990	82
9	156	158	6.31	988	81
10	0.174	0.176	5.67	0.985	80
11	191	194	5.145	982	79
12	208	213	4.705	978	78
13	225	231	4.331	974	77
14	242	249	4.011	970	76
15	0.259	0.268	3.732	0.966	75
16	276	287	487	961	74
17	292	306	271	956	73
18	309	325	3.078	951	72
19	326	344	2.904	946	71
20	0.342	0.364	2.747	0.940	70
21	358	384	605	934	69
22	375	404	475	927	68
23	391	424	356	921	67
24	407	445	246	914	66
25	0.423	0.466	2.145	0.906	65
26	438	488	2.050	899	64
27	454	510	1.963	891	63
28	469	532	881	883	62
29	485	554	804	875	61
30	0.500	0.577	1.732	0.866	60
31	515	601	664	857	59
32	530	625	600	848	58
33	545	649	540	839	57
34	559	675	483	829	56
35	0.574	0.700	1.428	0.819	55
36	588	727	376	809	54
37	602	754	327	799	53
38	616	781	280	788	52
39	629	810	235	777	51
40	0.643	0.839	1.192	0.766	50
41	656	869	150	755	49
42	669	900	111	743	48
43	682	933	072	731	47
44	695	966	036	719	46
45	0.707	1.000	1.000	0.707	45
	$\cos \alpha$	$\cot \alpha$	$\tan \alpha$	$\sin \alpha$	α°

α°	α radians	$\sin \alpha$	$\tan \alpha$	α°	α radians	$\sin \alpha$	$\tan \alpha$
0	0	0.000	0.000	91.7	1.6	1.000	-34.23
5.73	0.1	0.100	+0.100	97.4	1.7	0.992	-7.697
11.5	0.2	0.199	+0.203	103.1	1.8	0.974	-4.286
17.2	0.3	0.296	+0.309	108.9	1.9	0.946	-2.927
22.9	0.4	0.389	+0.423	114.6	2.0	0.909	-2.185
28.7	0.5	0.479	+0.546	120.3	2.1	0.863	-1.710
34.4	0.6	0.565	+0.684	126.1	2.2	0.808	-1.374
40.1	0.7	0.644	+0.842	131.8	2.3	0.746	-1.119
45.0	$\frac{\pi}{4}$	0.707	+1.000	135.0	$\frac{3\pi}{4}$	0.707	-1.000
45.8	0.8	0.717	+1.030	137.5	2.4	0.675	-0.916
51.6	0.9	0.783	+1.260	143.2	2.5	0.598	-0.747
57.3	1.0	0.841	+1.557	149.0	2.6	0.516	-0.602
63.0	1.1	0.891	+1.965	154.7	2.7	0.427	-0.473
68.8	1.2	0.932	+2.572	160.4	2.8	0.335	-0.356
74.5	1.3	0.964	+3.602	166.2	2.9	0.239	-0.246
80.2	1.4	0.985	+5.798	171.9	3.0	0.141	-0.143
86.0	1.5	0.997	+14.10	177.6	3.1	0.042	-0.042
90.0	$\frac{\pi}{2}$	1.000	—	180.0	π	0.000	0.000

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